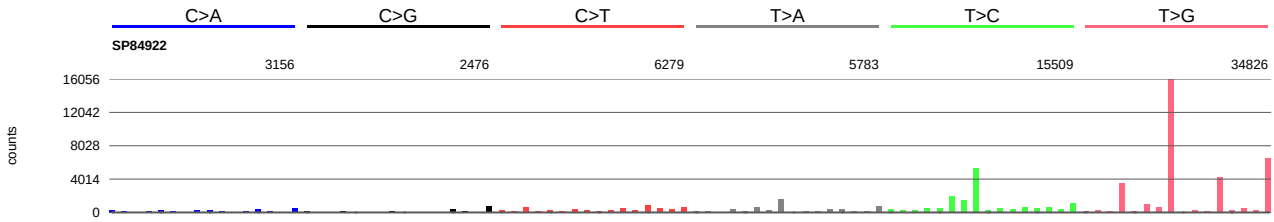
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SP84922
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6279
5783
15509
34826
16056
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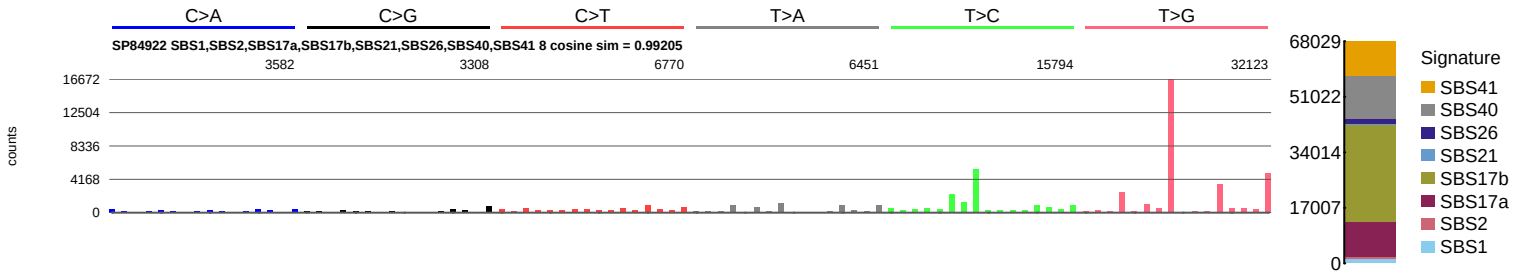
8028
4014
0
counts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SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.99205
3582
3308
6770
6451
15794
32123
16672
12504
8336
4168
0
counts
68029
51022
34014
17007
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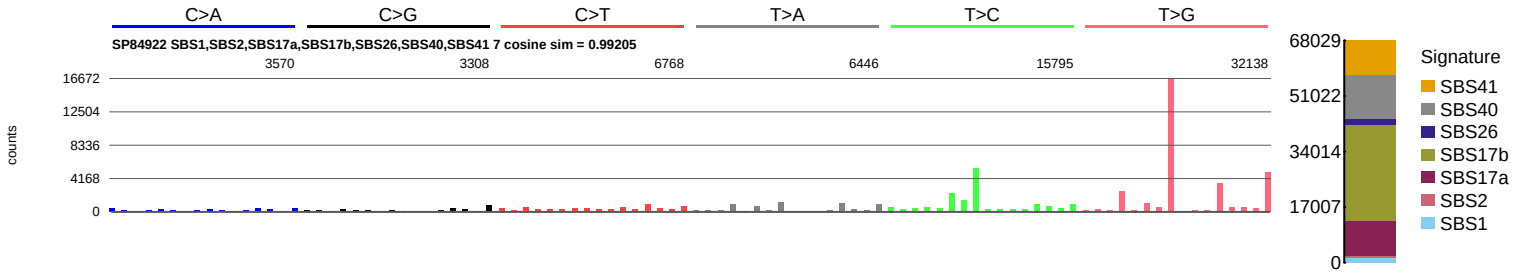
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS2
SBS1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C>G
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SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.99205
3570
3308
6768
6446
15795
32138
16672
12504
8336
4168
0
counts
68029
51022
34014
17007
0
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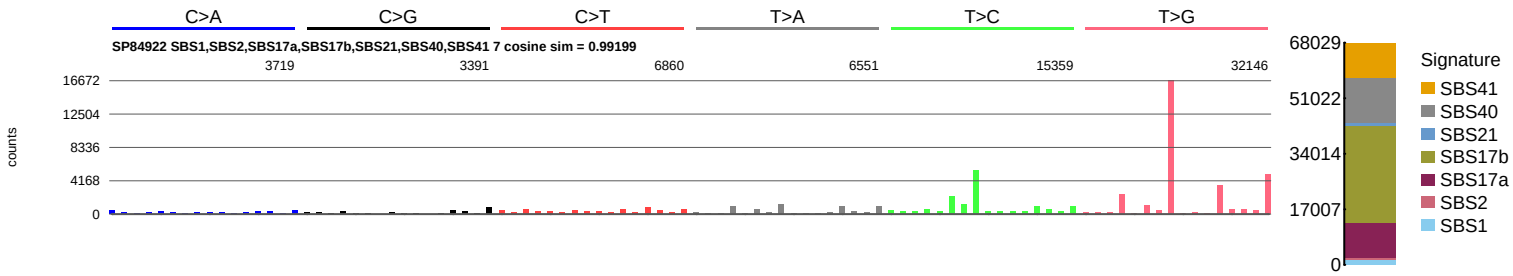
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1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SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.99199
3719
3391
6860
6551
15359
32146
16672
12504
8336
4168
0
counts
68029
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34014
17007
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SBS40
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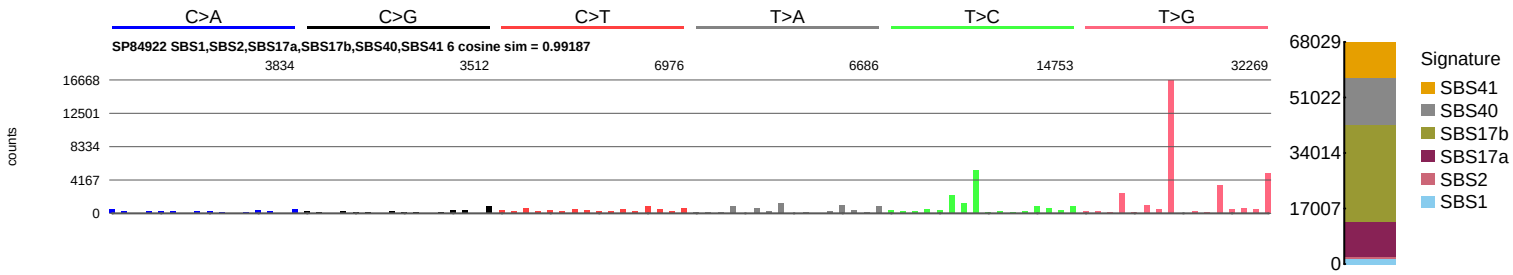
SBS17b
SBS17a
SBS2
SBS1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SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.99187
3834
3512
6976
6686
14753
32269
16668
12501
8334
4167
0
counts
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34014
17007
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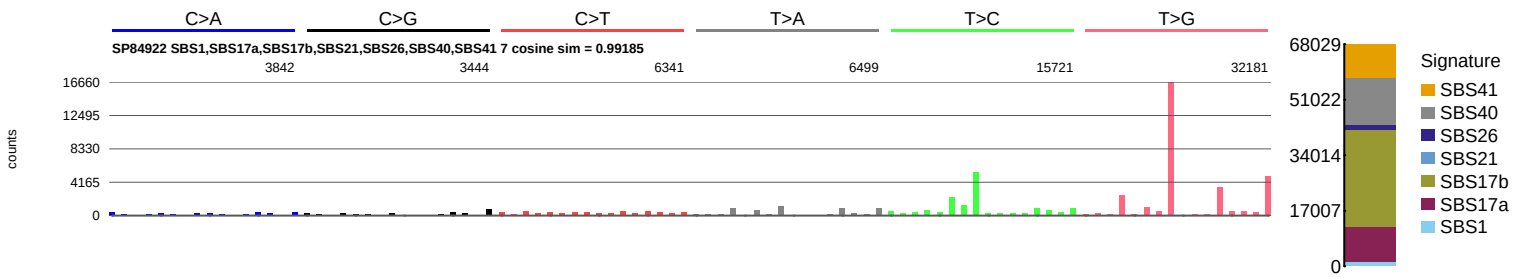
SBS41
SBS40
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.99185
3842
3444
6341
6499
15721
32181
16660
12495
8330
4165
0
counts
68029
51022
34014
17007
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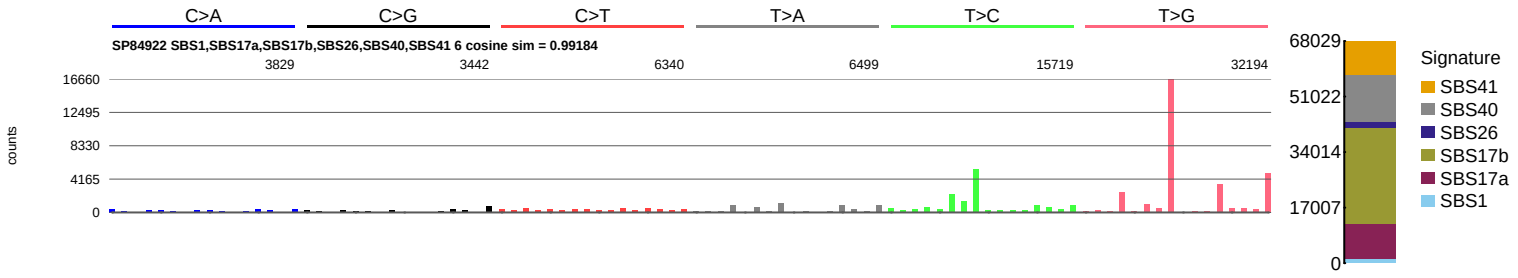
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS1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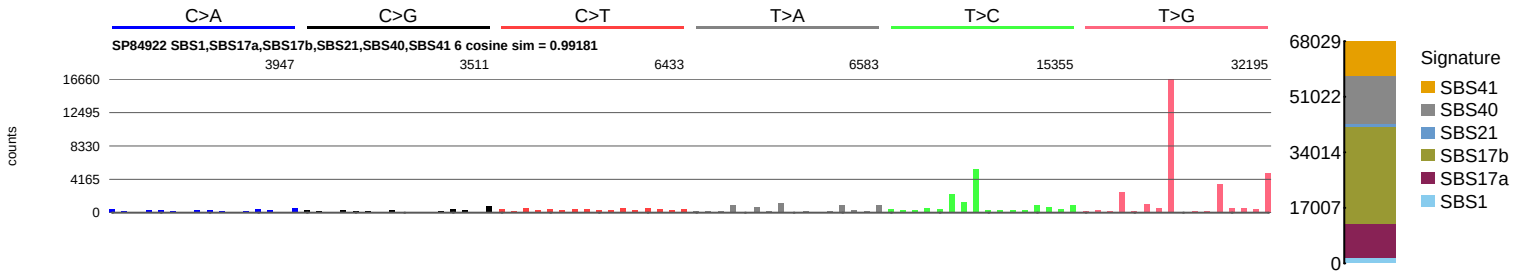
T>A
T>C
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.99184
3829
3442
6340
6499
15719
32194
counts
16660
12495
8330
4165
0
68029
51022
34014
17007
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS1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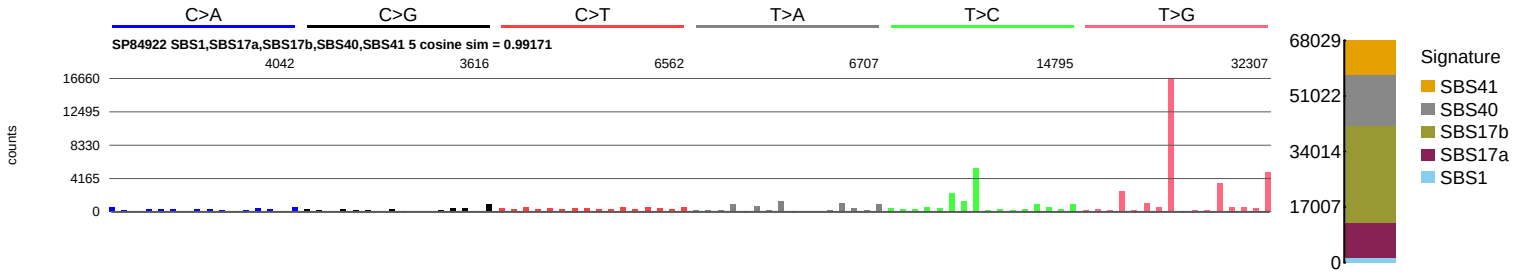
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.99181
3947
3511
6433
6583
15355
32195
counts
16660
12495
8330
4165
0
68029
51022
34014
17007
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.99171
4042
3616
6562
6707
14795
32307
counts
16660
12495
8330
4165
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68029
51022
34014
17007
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Signature
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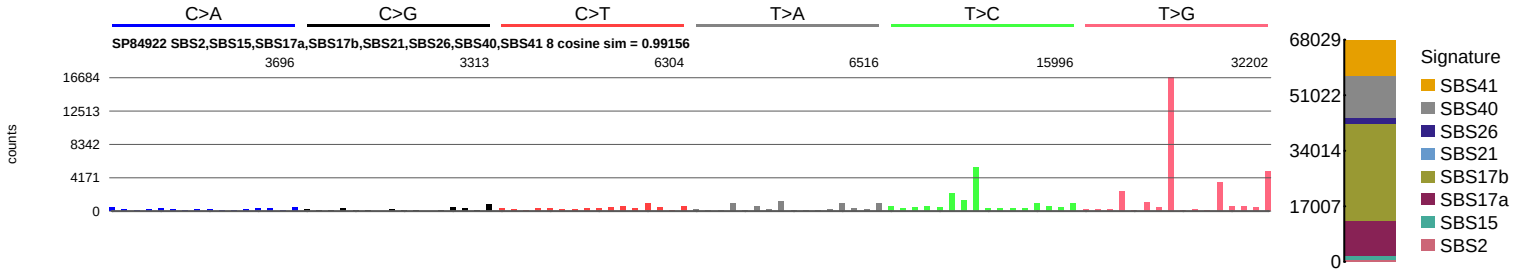
SBS40
SBS17b
SBS17a
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.99156
3696
3313
6304
6516
15996
32202
counts
16684
12513
8342
4171
0
68029
51022
34014
17007
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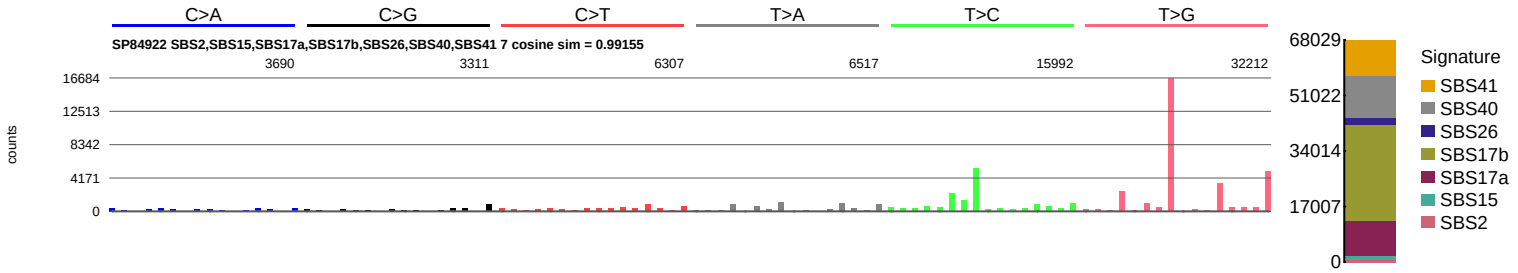
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2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C>G
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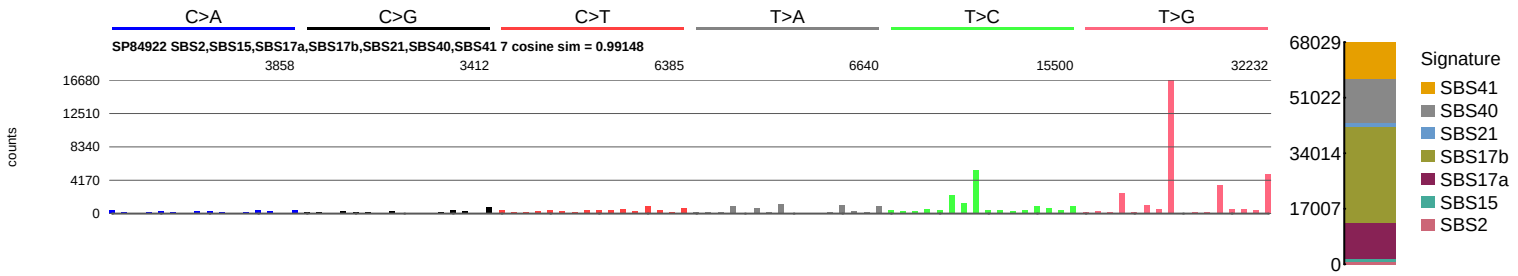
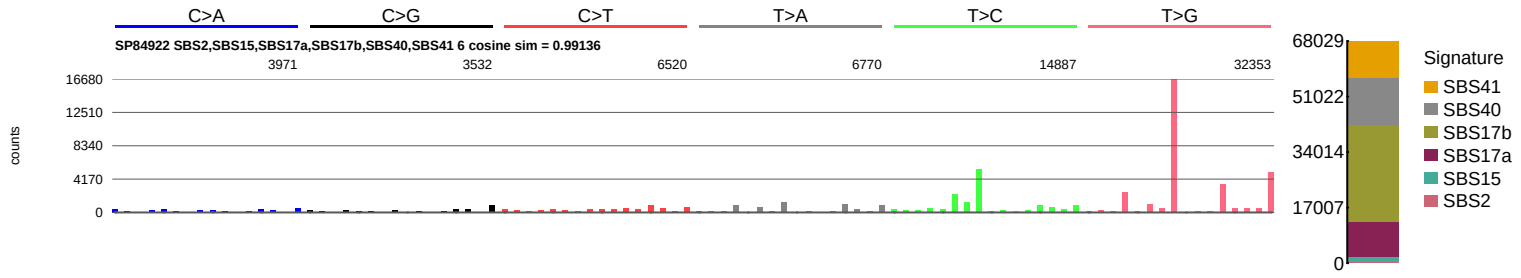
T>A
T>C
T>G
SP84922 SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.99155
3690
3311
6307
6517
15992
32212
counts
16684
12513
8342
4171
0
68029
51022
34014
17007
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.99148
3858
3412
6385
6640
15500
32232
counts
16680
12510
8340
4170
0
68029
51022
34014
17007
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2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C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.99136
3971
3532
6520
6770
14887
32353
counts
16680
12510
8340
4170
0
68029
51022
34014
17007
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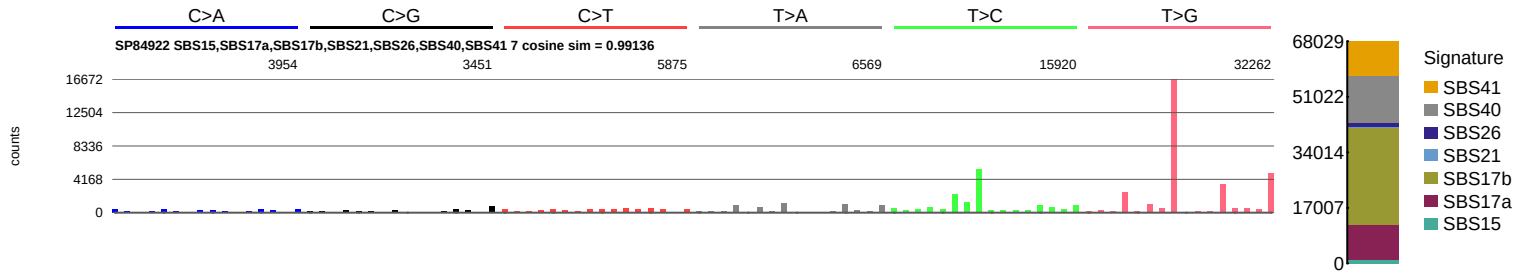
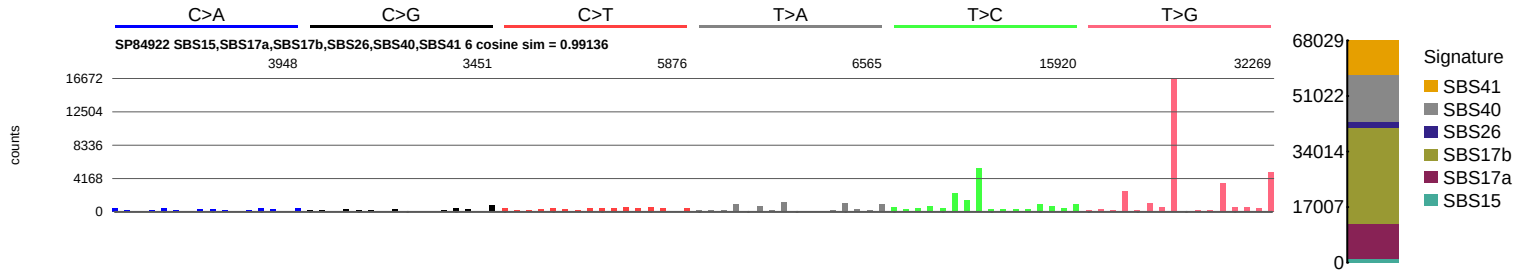
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.99136
3954
3451
5875
6569
15920
32262
counts
16672
12504
8336
4168
0
68029
51022
34014
17007
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15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C>G
C>T
T>A
T>C
T>G
SP84922 SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.99136
3948
3451
5876
6565
15920
32269
counts
16672
12504
8336
4168
0
68029
51022
34014
17007
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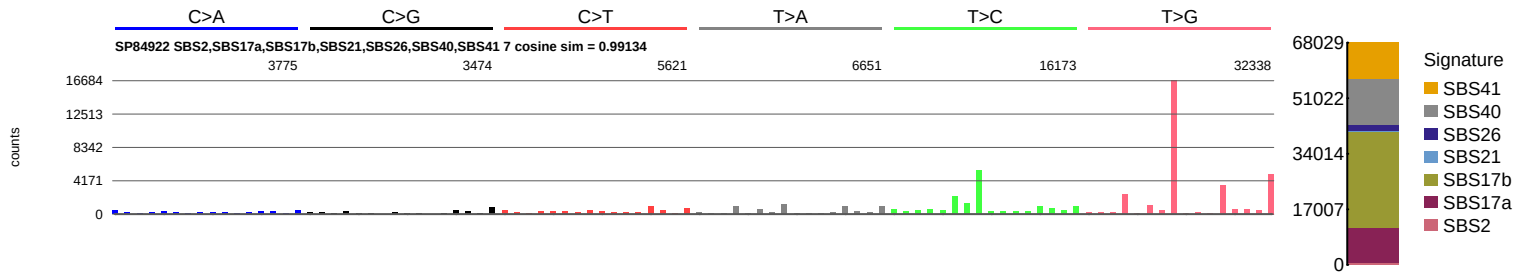
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15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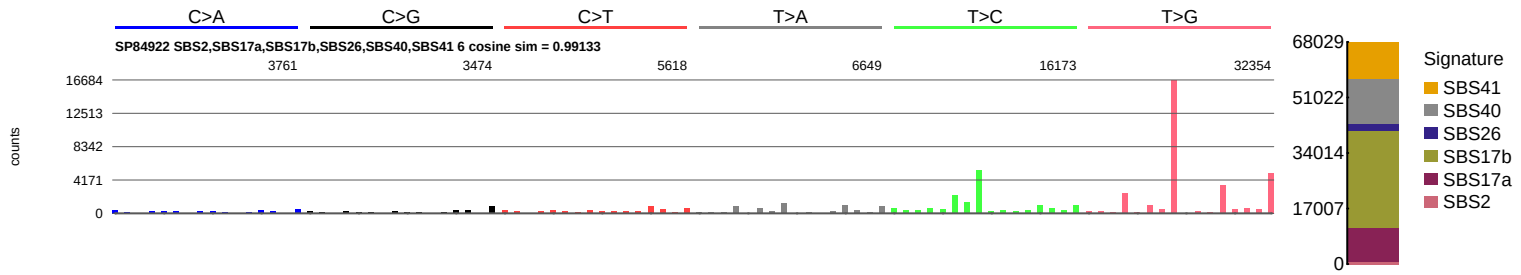
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.99134
3775
3474
5621
6651
16173
32338
counts
16684
12513
8342
4171
0
68029
51022
34014
17007
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS2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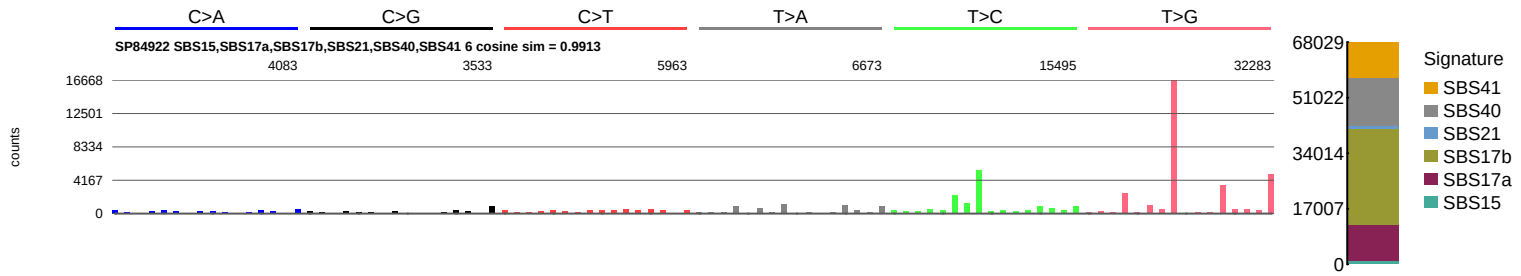
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.99133
3761
3474
5618
6649
16173
32354
counts
16684
12513
8342
4171
0
68029
51022
34014
17007
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS2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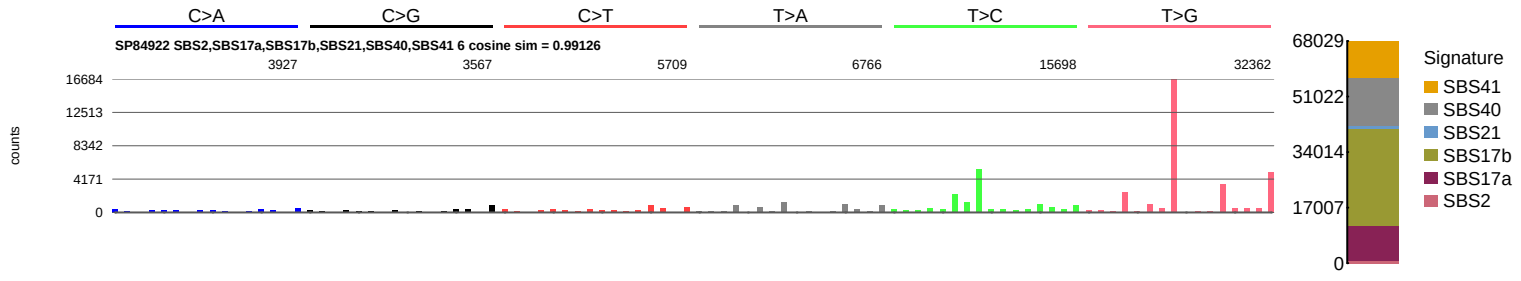
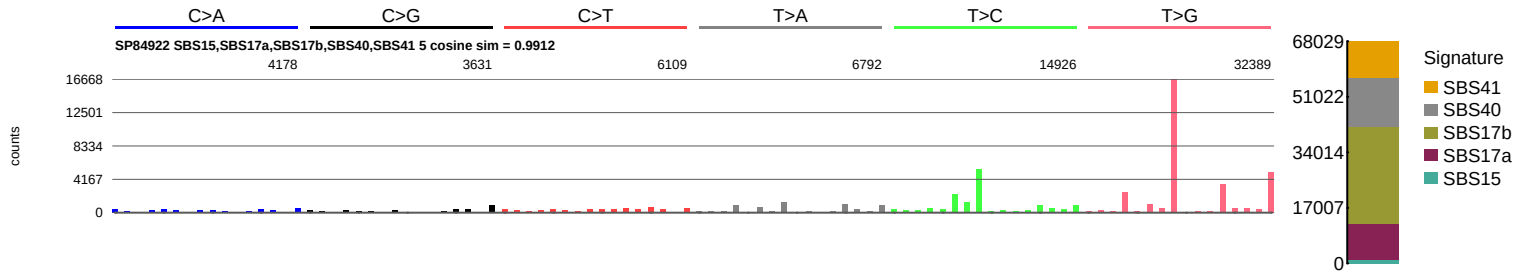
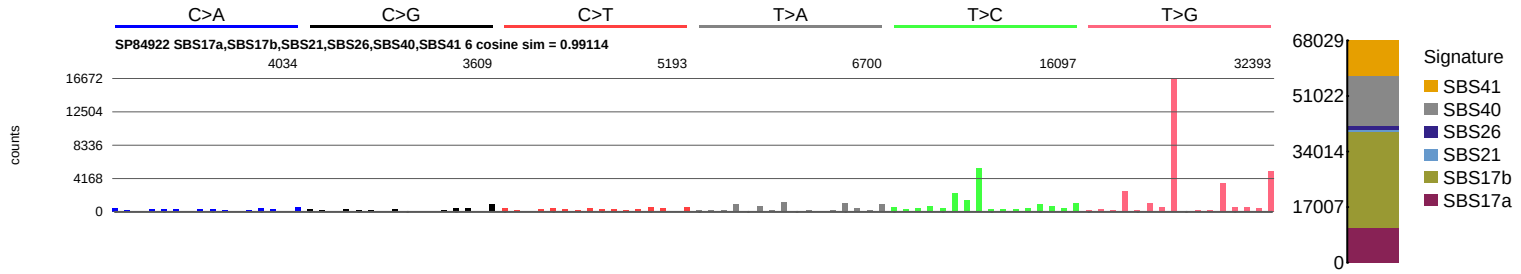
C>G
C>T
T>A
T>C
T>G
SP84922 SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.9913
4083
3533
5963
6673
15495
32283
counts
16668
12501
8334
4167
0
68029
51022
34014
17007
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.99126
3927
3567
5709
6766
15698
32362
16684
12513
8342
4171
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS15,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.9912
4178
3631
6109
6792
14926
32389
16668
12501
8334
4167
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15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C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 6 cosine sim = 0.99114
4034
3609
5193
6700
16097
32393
16672
12504
8336
4168
0
counts
68029
51022
34014
17007
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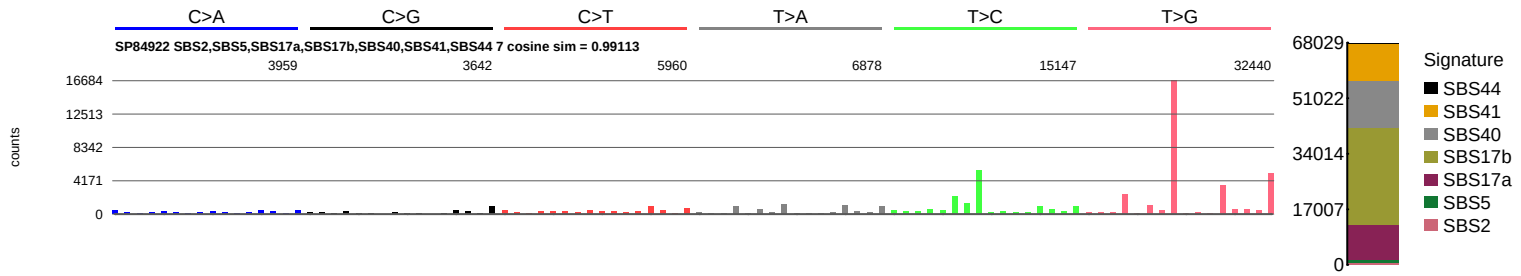
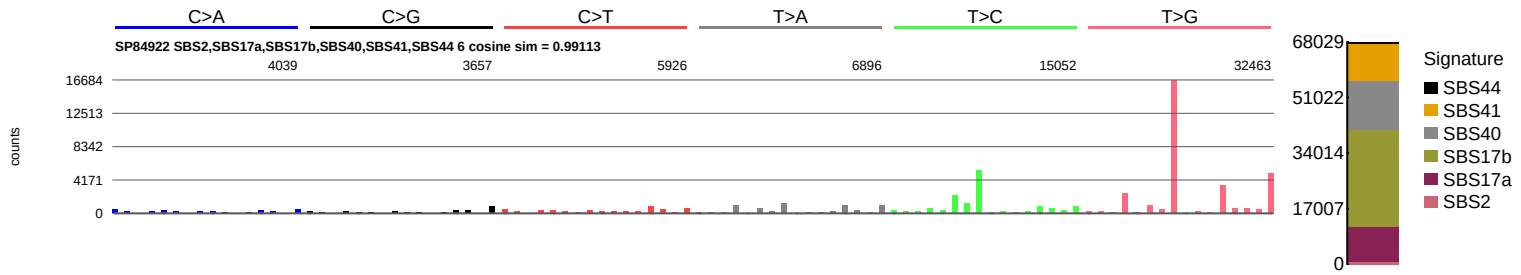
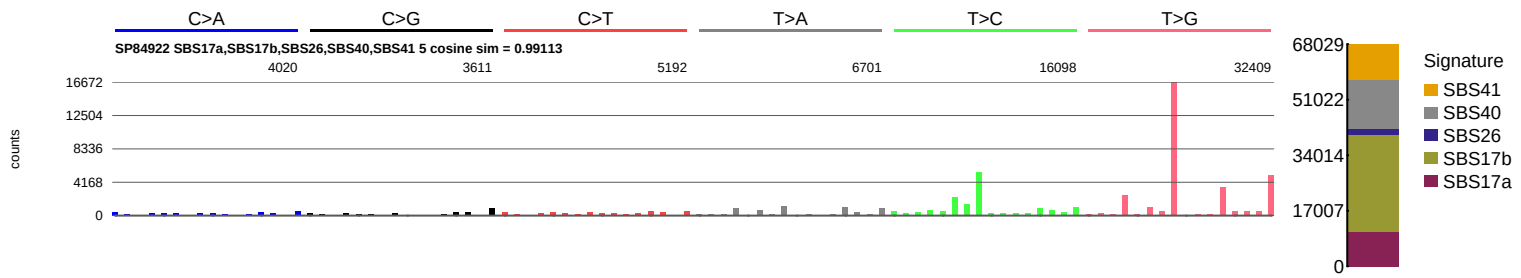
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.99113
3959
3642
5960
6878
15147
32440
16684
12513
8342
4171
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS40,SBS41,SBS44 6 cosine sim = 0.99113
4039
3657
5926
6896
15052
32463
16684
12513
8342
4171
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS26,SBS40,SBS41 5 cosine sim = 0.99113
4020
3611
5192
6701
16098
32409
16672
12504
8336
4168
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counts
68029
51022
34014
17007
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Signature
SBS41
SBS40
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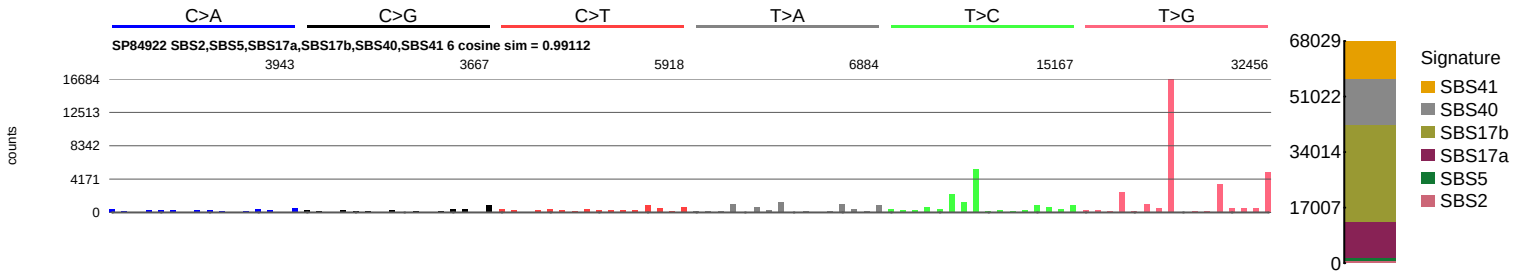
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.99112
3943
3667
5918
6884
15167
32456
counts
16684
12513
8342
4171
0
68029
51022
34014
17007
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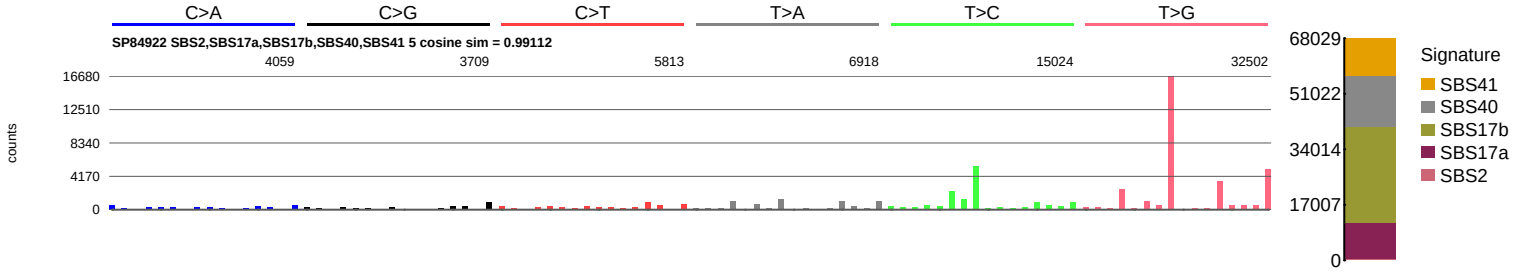
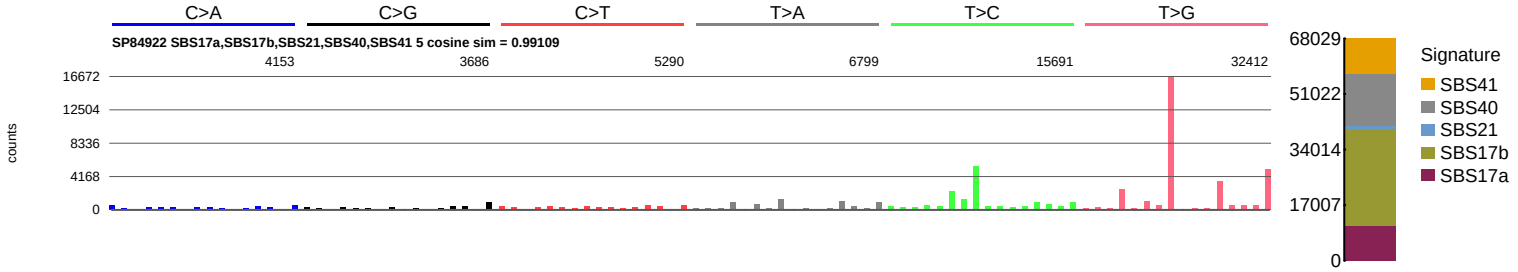
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.99112
4059
3709
5813
6918
15024
32502
counts
16680
12510
8340
4170
0
68029
51022
34014
17007
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS2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C>G
C>T
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T>G
SP84922 SBS17a,SBS17b,SBS21,SBS40,SBS41 5 cosine sim = 0.99109
4153
3686
5290
6799
15691
32412
counts
16672
12504
8336
4168
0
68029
51022
34014
17007
0
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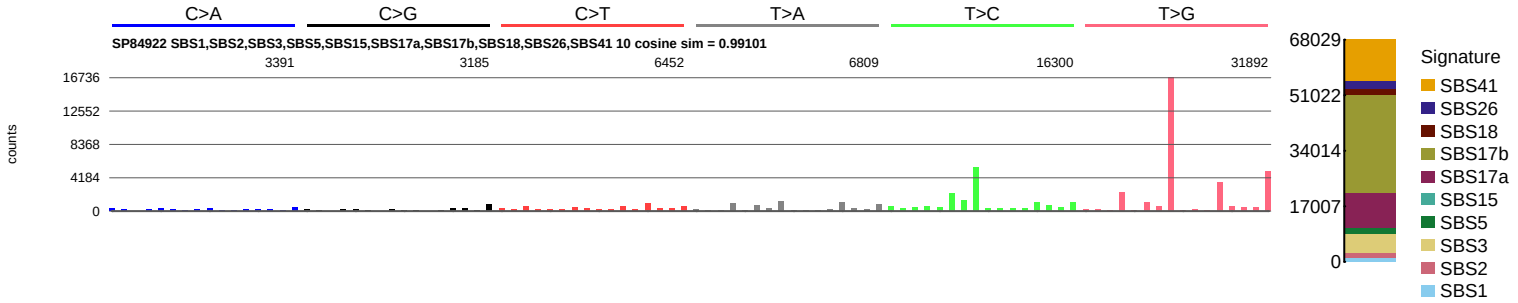
SBS41
SBS40
SBS21
SBS17b
SBS17a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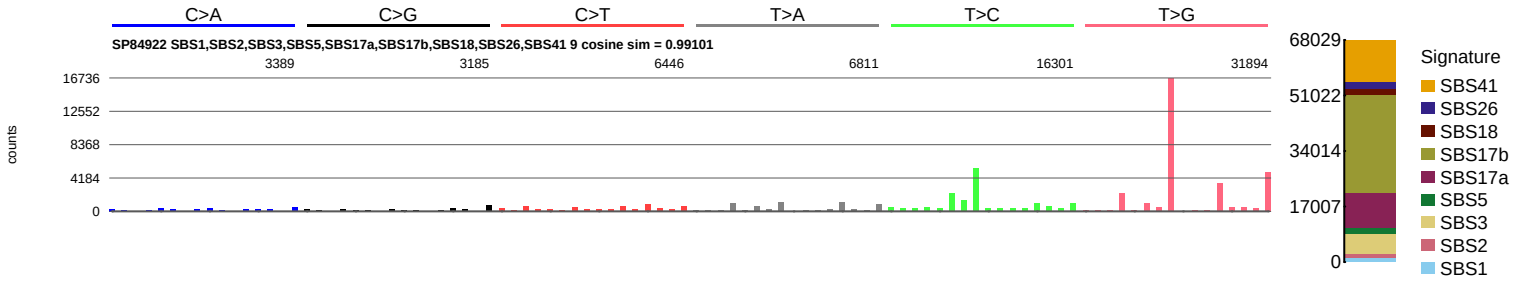
SP84922 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.99101
3391
3185
6452
6809
16300
31892
counts
16736
12552
8368
4184
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99101
3389
3185
6446
6811
16301
31894
counts
16736
12552
8368
4184
0
68029
51022
34014
17007
0
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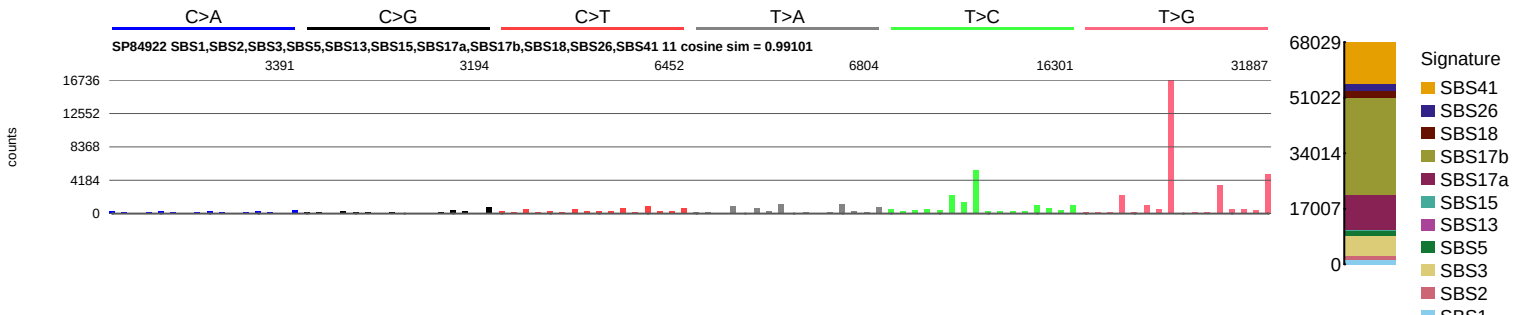
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 11 cosine sim = 0.99101
3391
3194
6452
6804
16301
31887
counts
16736
12552
8368
4184
0
68029
51022
34014
17007
0
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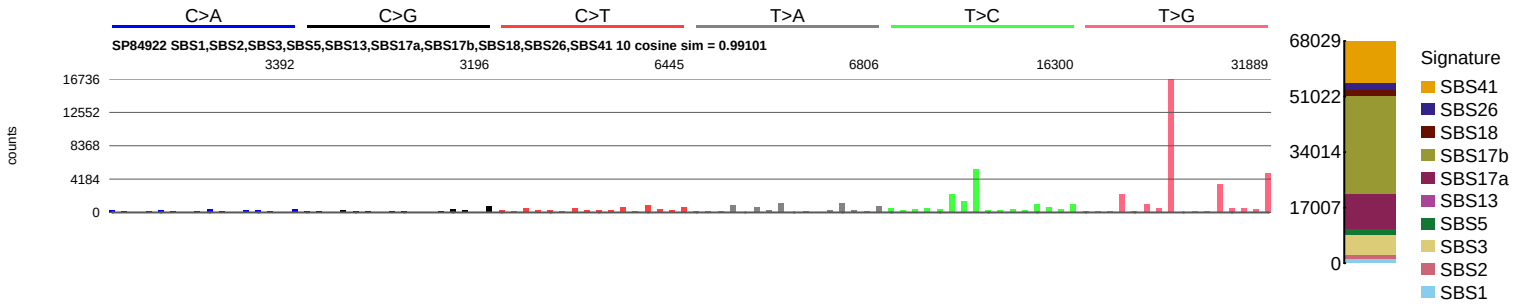
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2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C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.99101
3392
3196
6445
6806
16300
31889
counts
16736
12552
8368
4184
0
68029
51022
34014
17007
0
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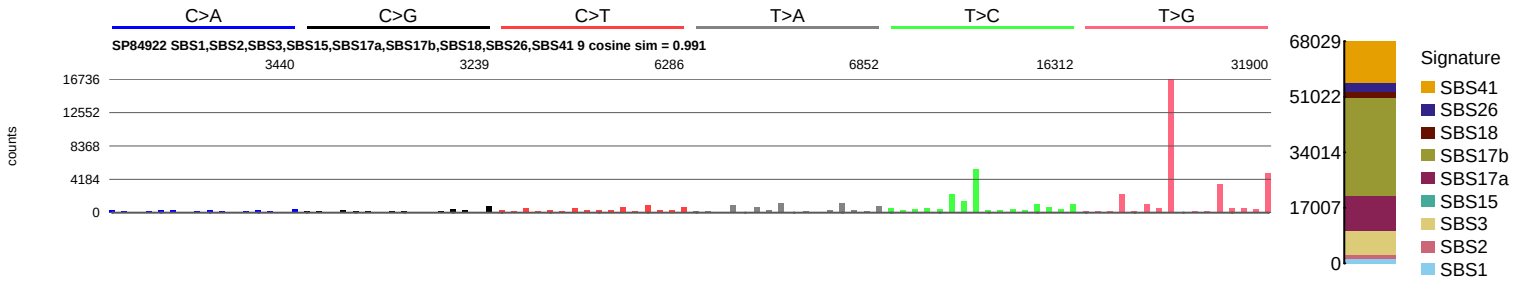
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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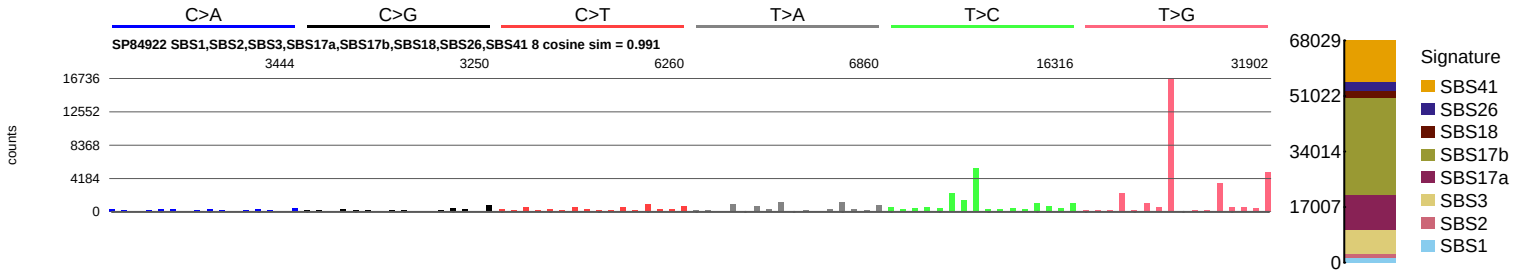
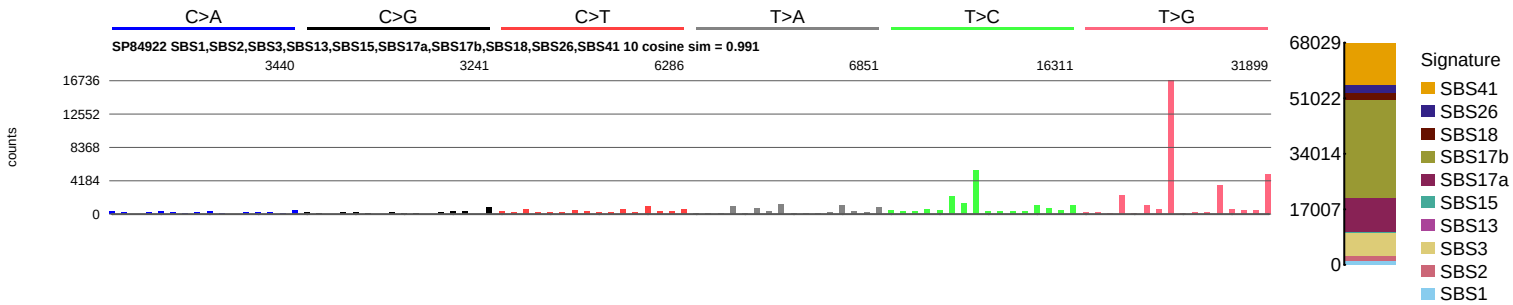
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.991
3440
3239
6286
6852
16312
31900
counts
16736
12552
8368
4184
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.991
3444
3250
6260
6860
16316
31902
counts
16736
12552
8368
4184
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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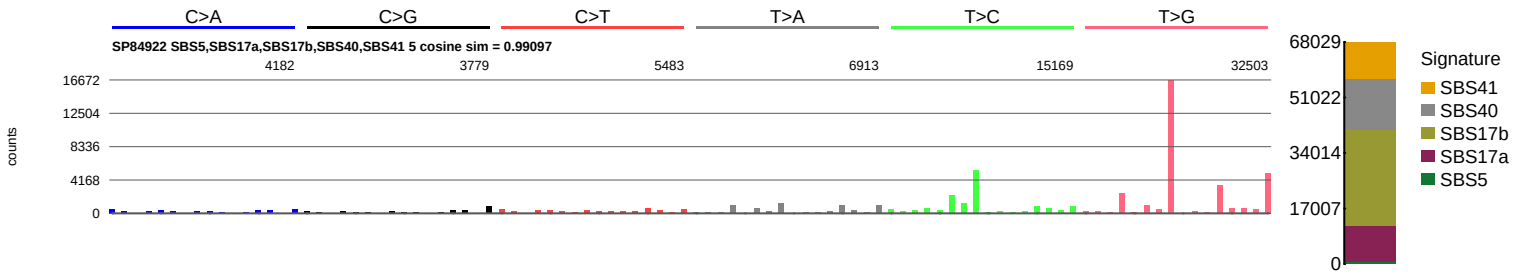
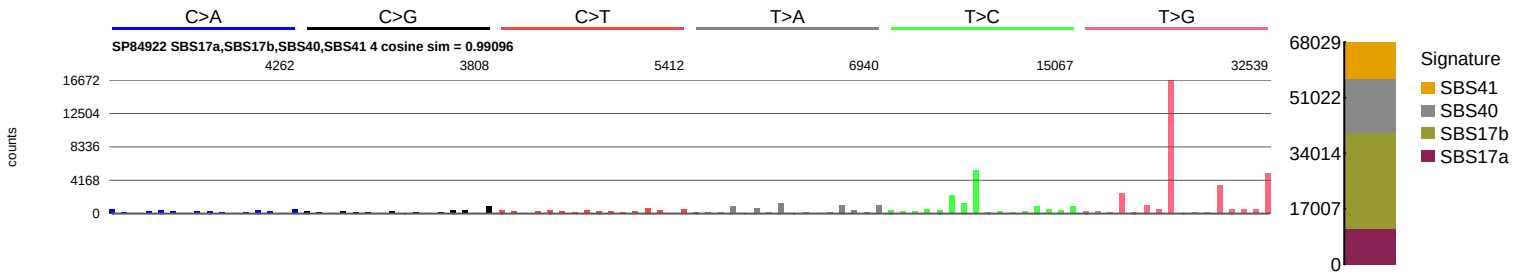
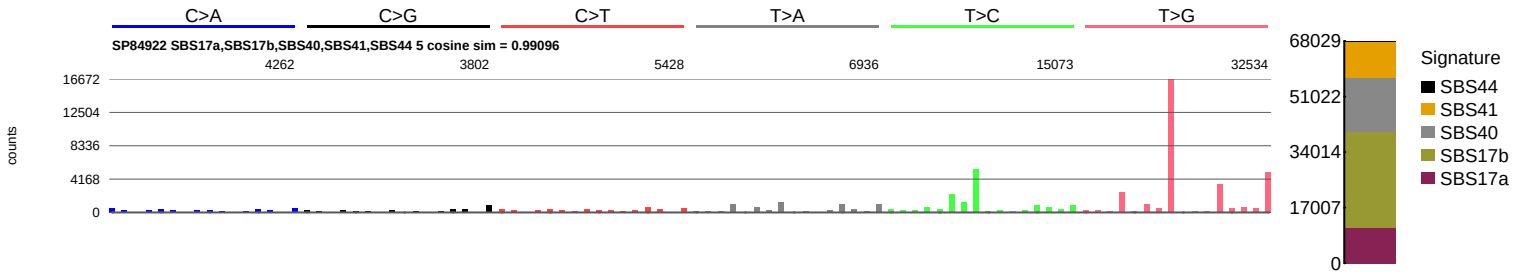
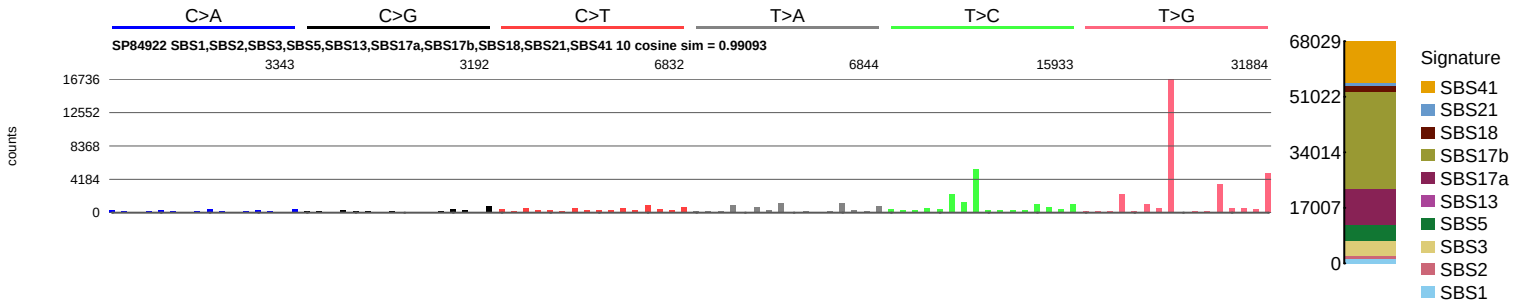
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.991
3440
3241
6286
6851
16311
31899
counts
16736
12552
8368
4184
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.99097
4182
3779
5483
6913
15169
32503
counts
16672
12504
8336
4168
0
68029
51022
34014
17007
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS40,SBS41 4 cosine sim = 0.99096
4262
3808
5412
6940
15067
32539
counts
16672
12504
8336
4168
0
68029
51022
34014
17007
0
Signature
SBS41
SBS40
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS40,SBS41,SBS44 5 cosine sim = 0.99096
4262
3802
5428
6936
15073
32534
counts
16672
12504
8336
4168
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a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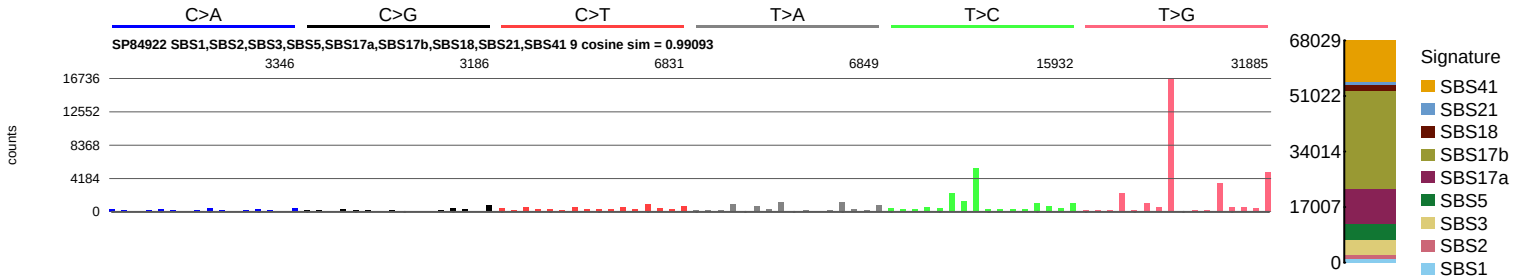
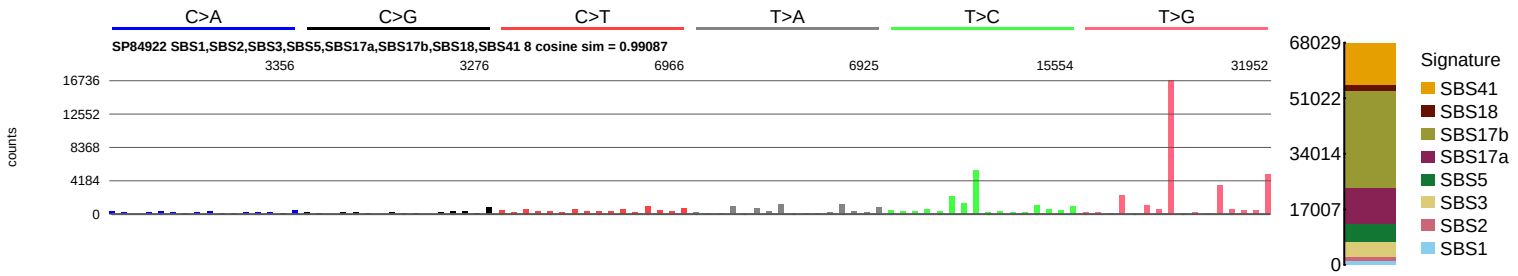
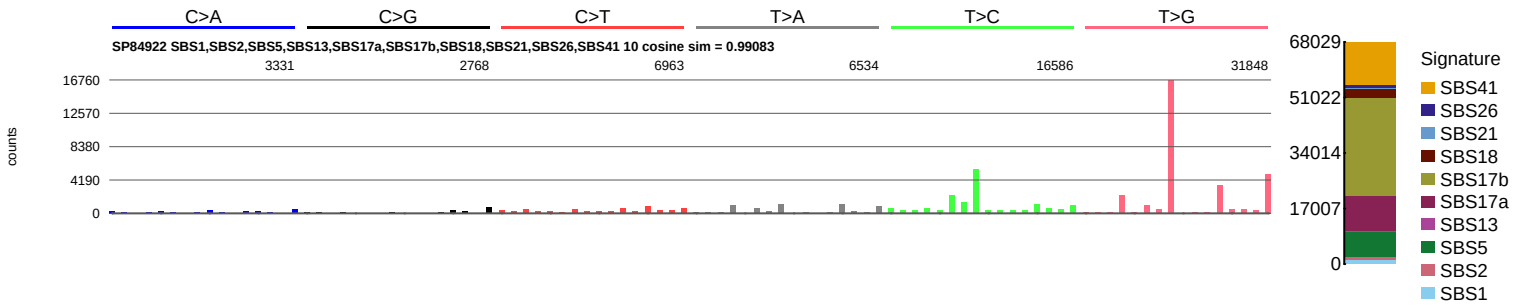
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99093
3343
3192
6832
6844
15933
31884
counts
16736
12552
8368
4184
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99093
3346
3186
6831
6849
15932
31885
counts
16736
12552
8368
4184
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99087
3356
3276
6966
6925
15554
31952
counts
16736
12552
8368
4184
0
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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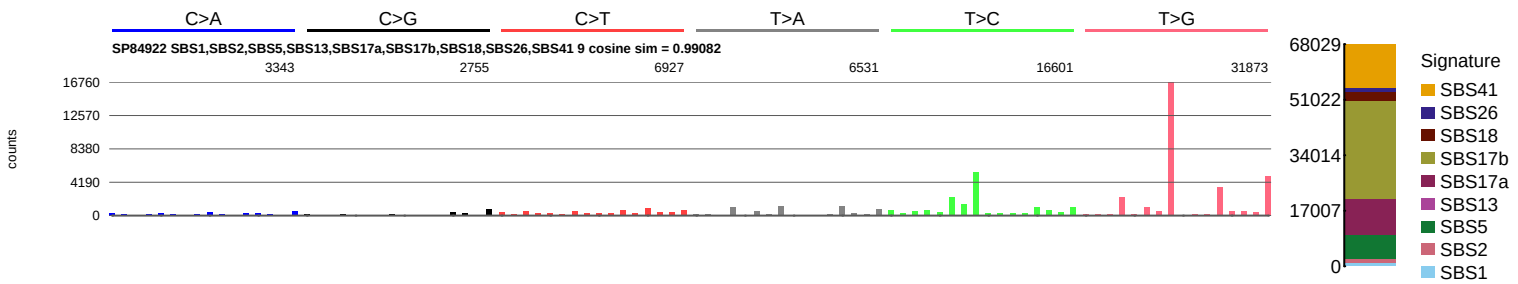
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.99083
3331
2768
6963
6534
16586
31848
counts
16760
12570
8380
4190
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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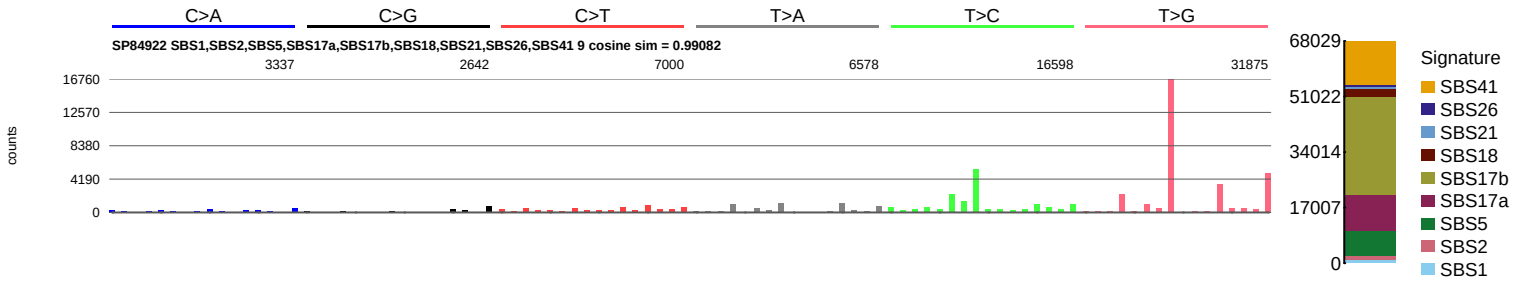
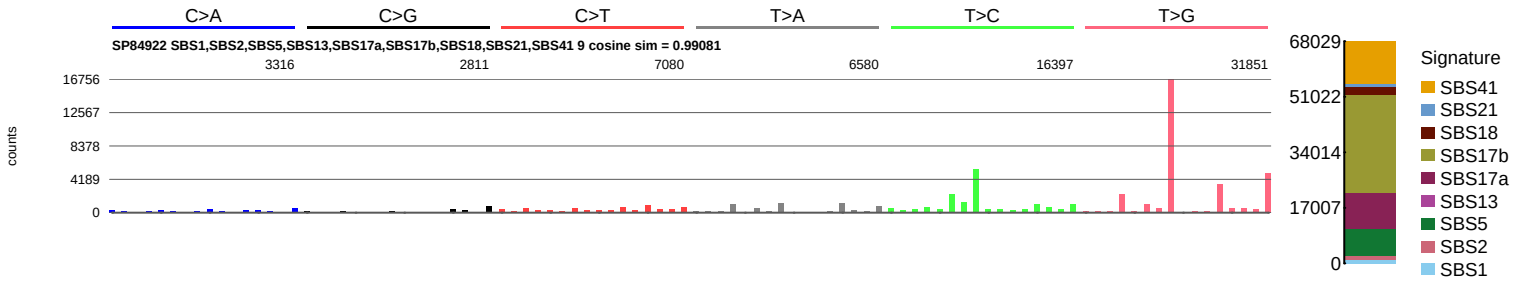
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99082
3343
2755
6927
6531
16601
31873
counts
16760
12570
8380
4190
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.99082
3337
2642
7000
6578
16598
31875
16760
12570
8380
4190
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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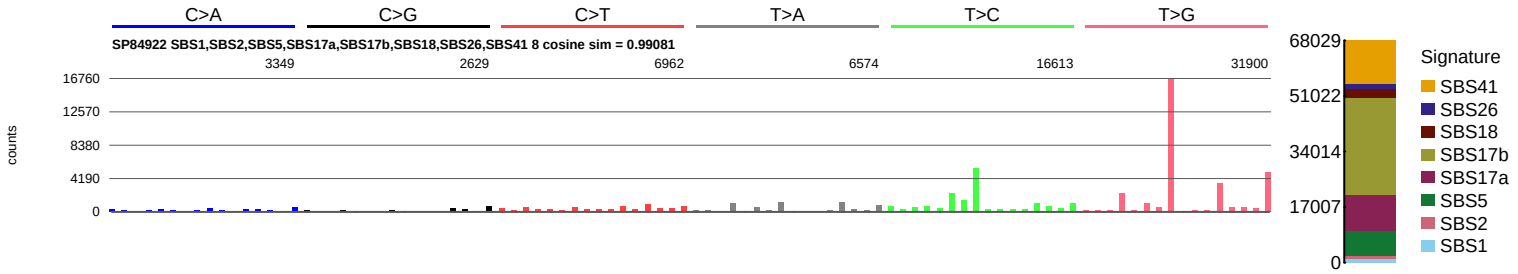
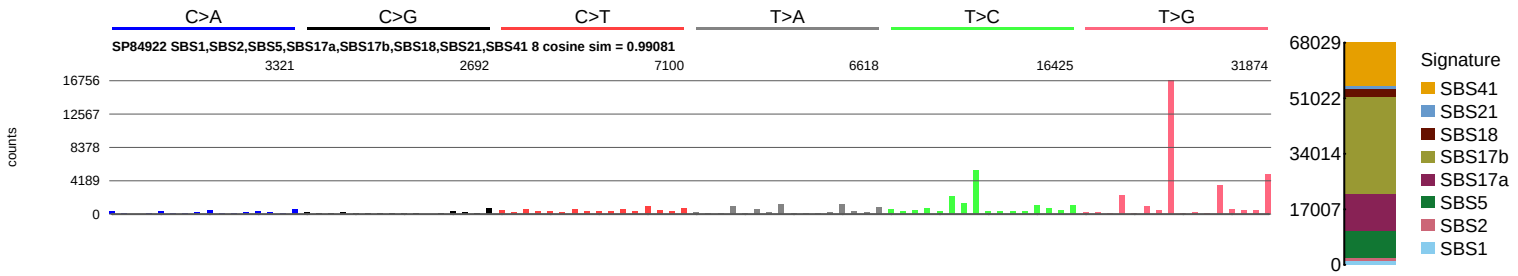
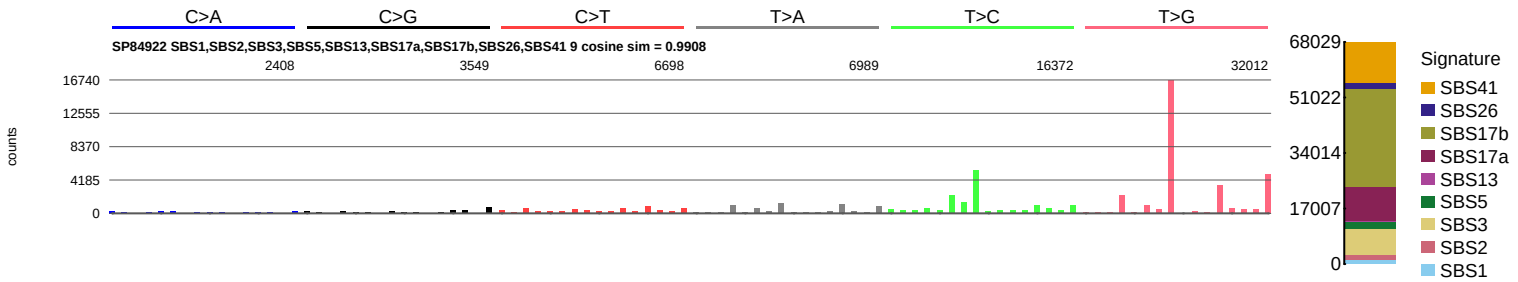
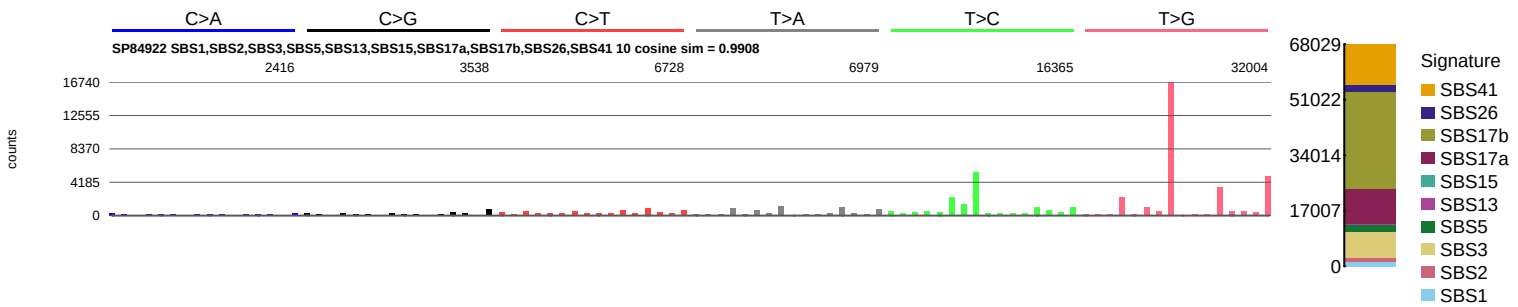
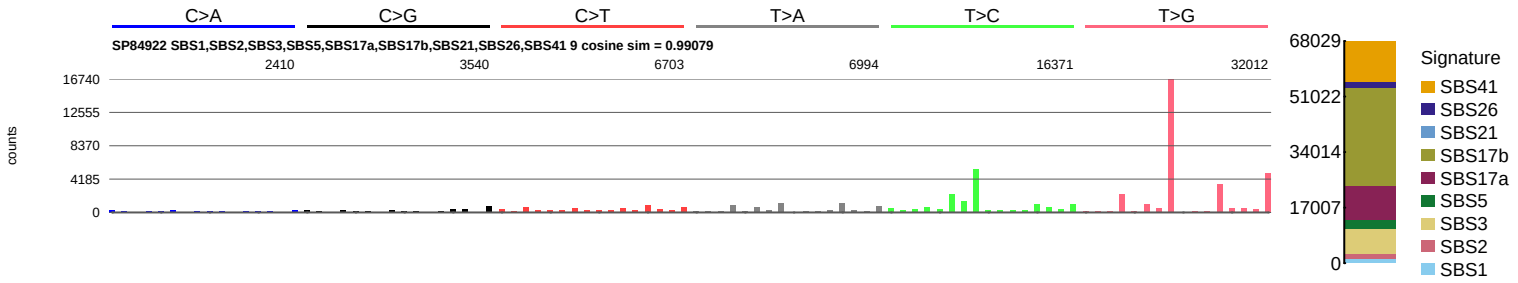
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99081
3316
2811
7080
6580
16397
31851
16756
12567
8378
4189
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

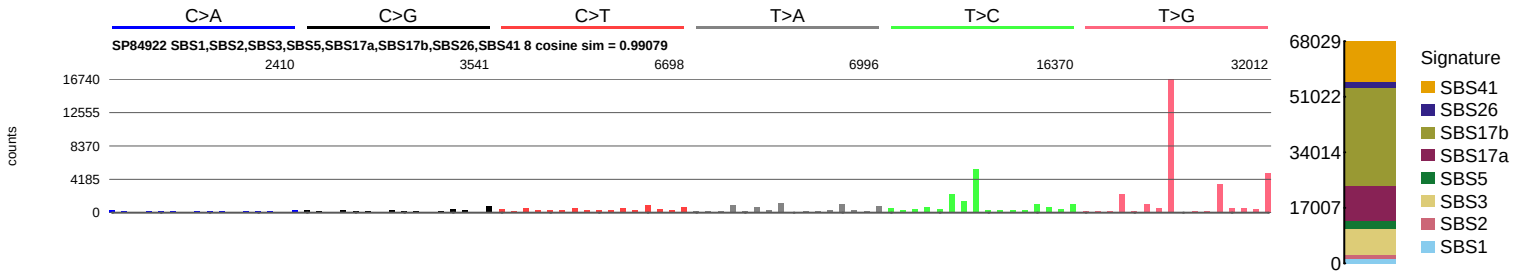

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99081
3349
2629
6962
6574
16613
31900
16760
12570
8380
4190
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

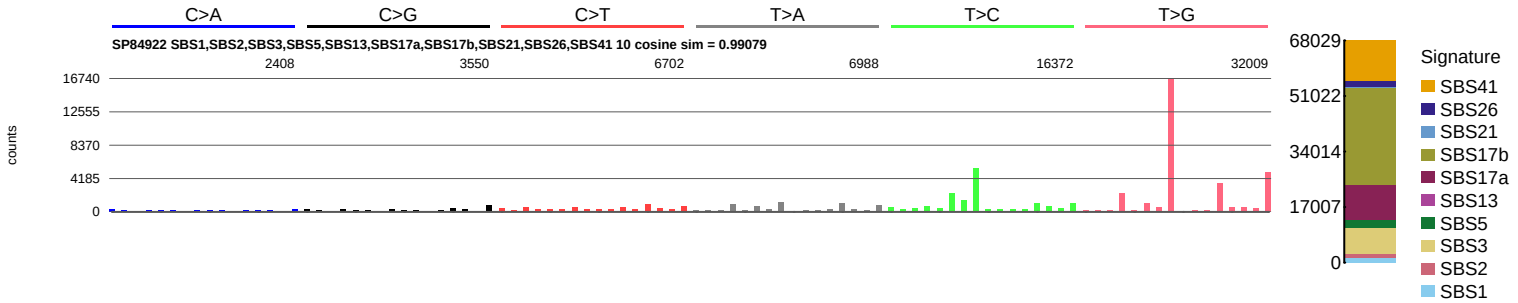

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99081
3321
2692
7100
6618
16425
31874
16756
12567
8378
4189
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS41 9 cosine sim = 0.9908
2408
3549
6698
6989
16372
32012
16740
12555
8370
4185
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 10 cosine sim = 0.9908
2416
3538
6728
6979
16365
32004
16740
12555
8370
4185
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS26,SBS41 9 cosine sim = 0.99079
2410
3540
6703
6994
16371
32012
16740
12555
8370
4185
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.99079
2410
3541
6698
6996
16370
32012
16740
12555
8370
4185
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41 10 cosine sim = 0.99079
2408
3550
6702
6988
16372
32009
16740
12555
8370
4185
0
counts
68029
51022
34014
17007
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SBS26
SBS21
SBS17b
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SBS13
SBS5
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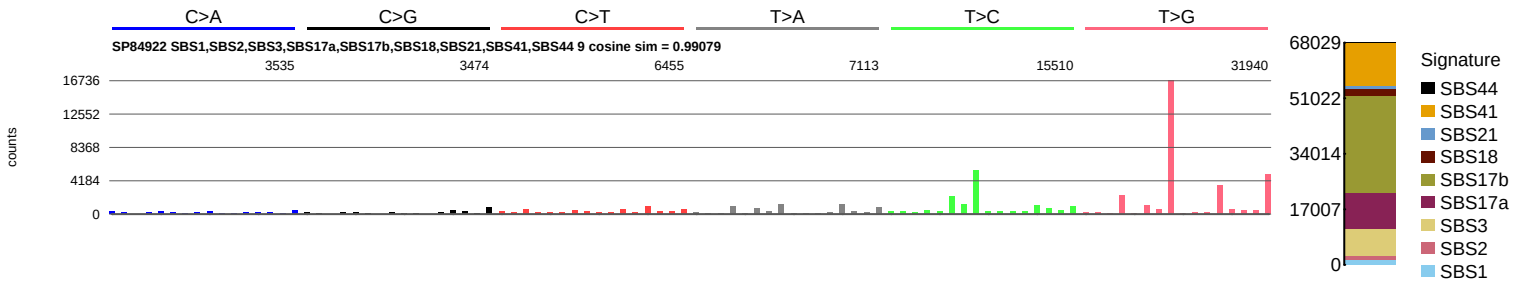
SBS2
SBS1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3535
3474
6455
7113
15510
31940
16736
12552
8368
4184
0
counts
68029
51022
34014
17007
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SBS41
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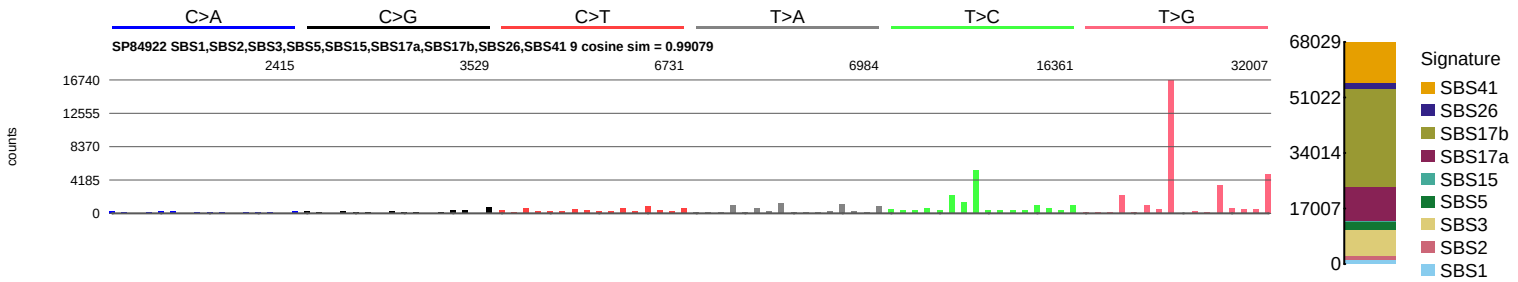
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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2415
3529
6731
6984
16361
32007
16740
12555
8370
4185
0
counts
68029
51022
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SBS17a
SBS15
SBS5
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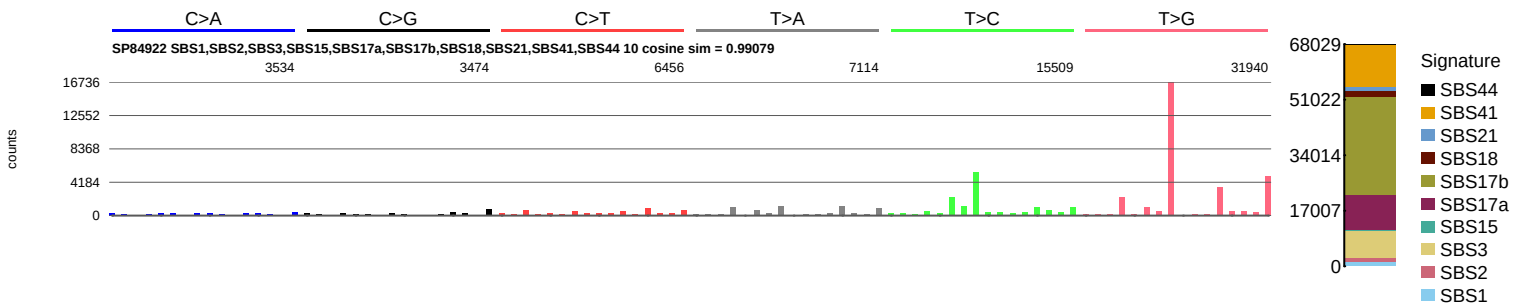
SBS2
SBS1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SP84922 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.99079
3534
3474
6456
7114
15509
31940
16736
12552
8368
4184
0
counts
68029
51022
34014
17007
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SBS18
SBS17b
SBS17a
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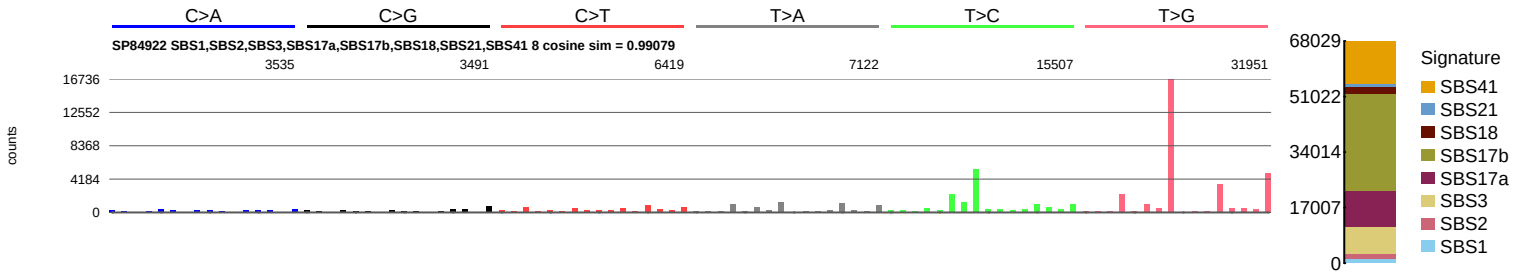
SBS3
SBS2
SBS1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C>T
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T>G
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3535
3491
6419
7122
15507
31951
counts
16736
12552
8368
4184
0
68029
51022
34014
17007
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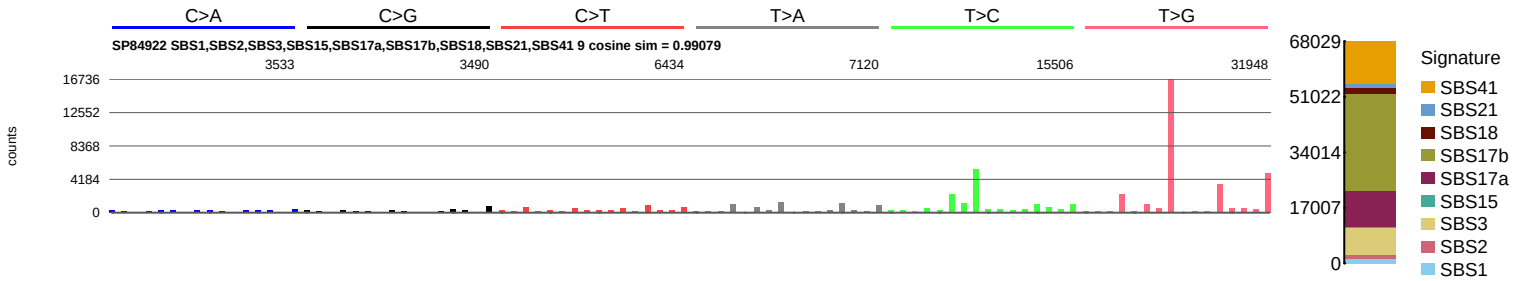
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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T>G
SP84922 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99079
3533
3490
6434
7120
15506
31948
counts
16736
12552
8368
4184
0
68029
51022
34014
17007
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SBS18
SBS17b
SBS17a
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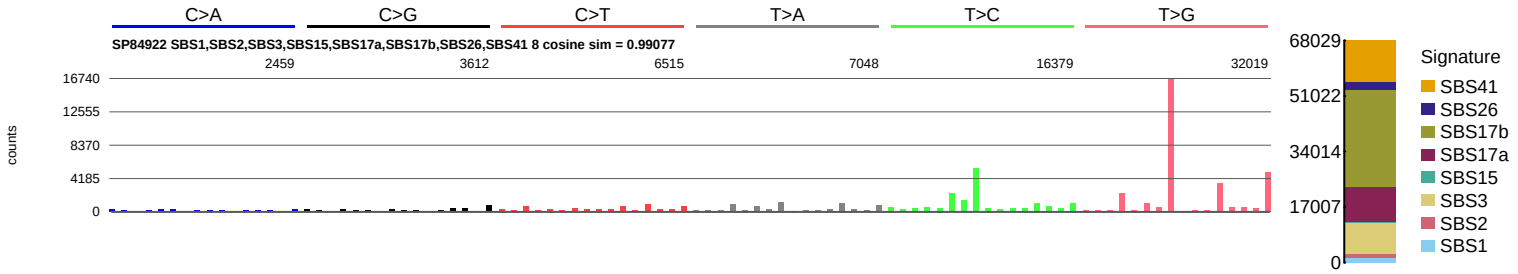
SBS3
SBS2
SBS1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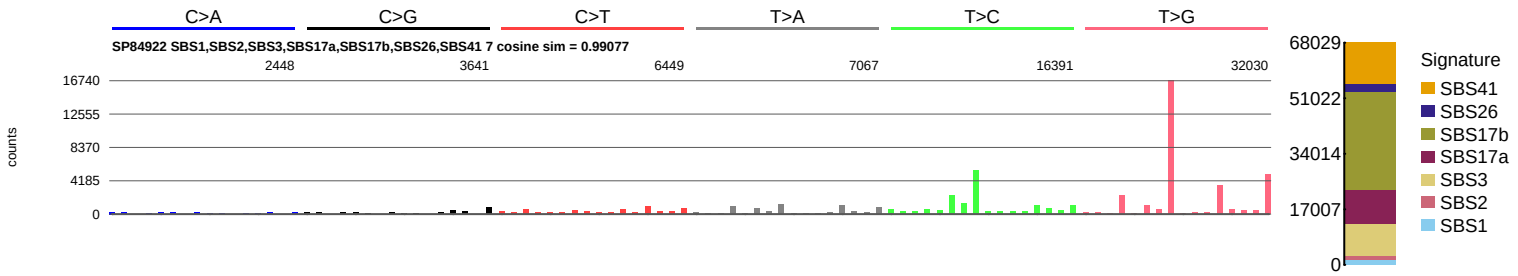
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.99077
2459
3612
6515
7048
16379
32019
counts
16740
12555
8370
4185
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.99077
2448
3641
6449
7067
16391
32030
counts
16740
12555
8370
4185
0
68029
51022
34014
17007
0
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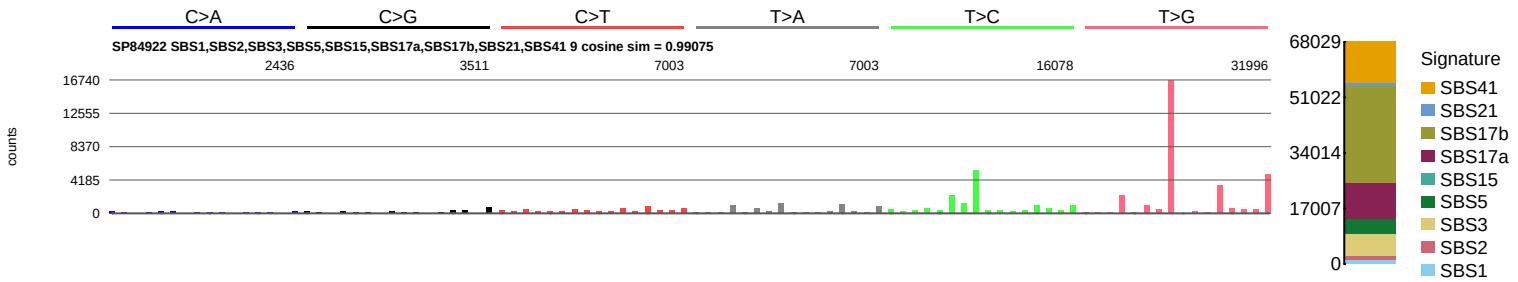
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.99075
2436
3511
7003
7003
16078
31996
counts
16740
12555
8370
4185
0
68029
51022
34014
17007
0
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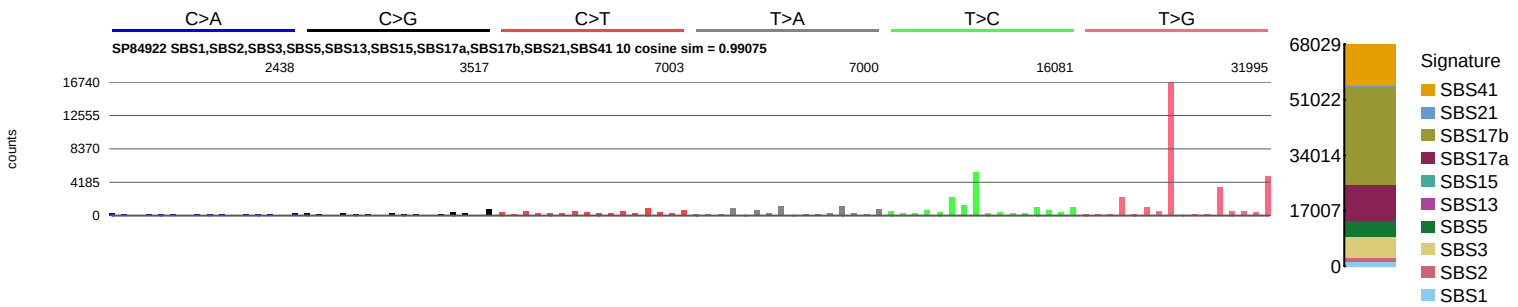
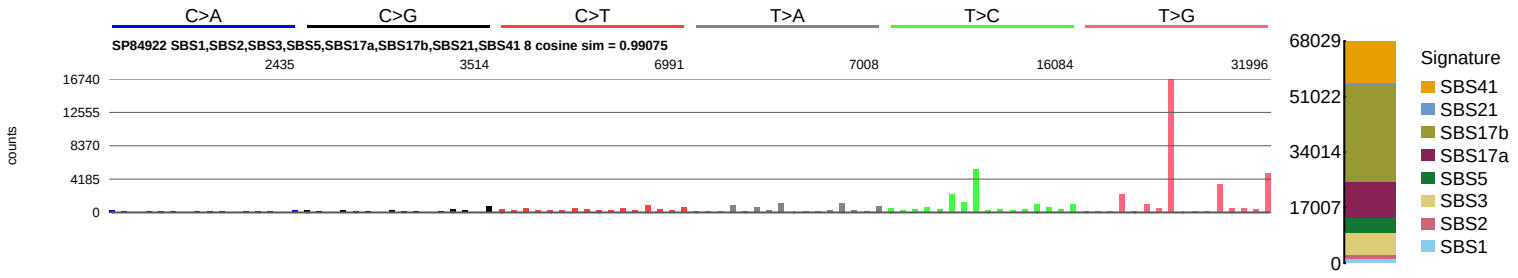
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 10 cosine sim = 0.99075
2438
3517
7003
7000
16081
31995
counts
16740
12555
8370
4185
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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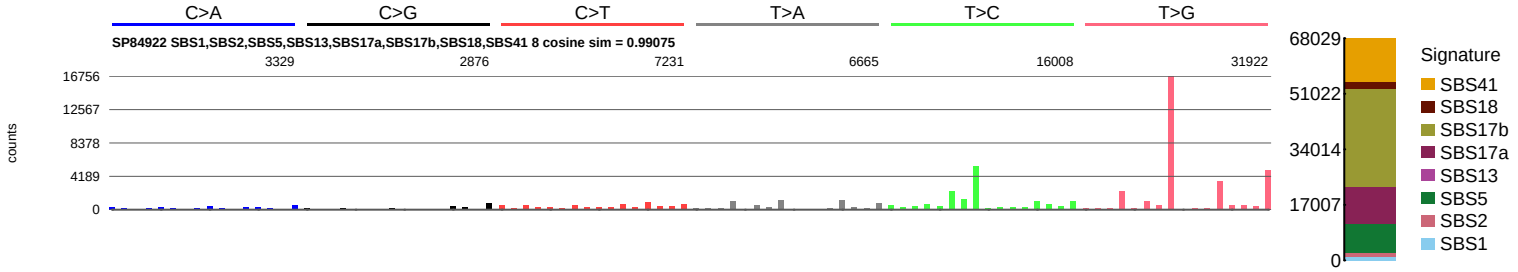
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.99075
2435
3514
6991
7008
16084
31996
16740
12555
8370
4185
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99075
3329
2876
7231
6665
16008
31922
16756
12567
8378
4189
0
counts
68029
51022
34014
17007
0
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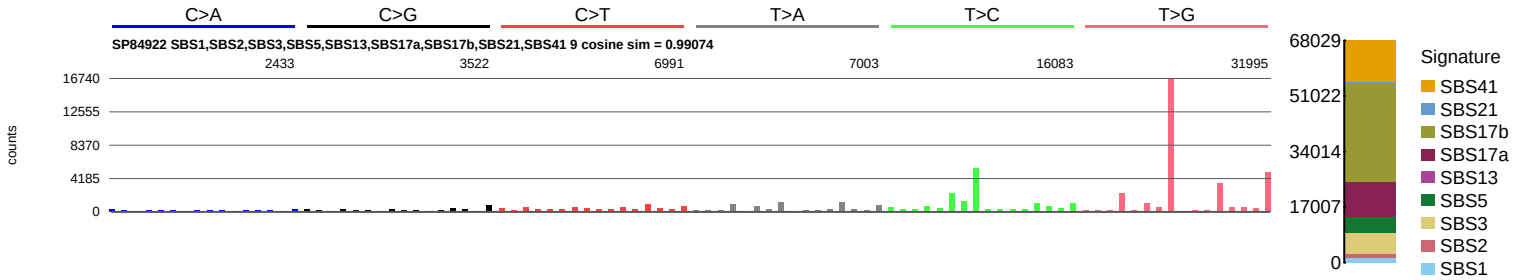
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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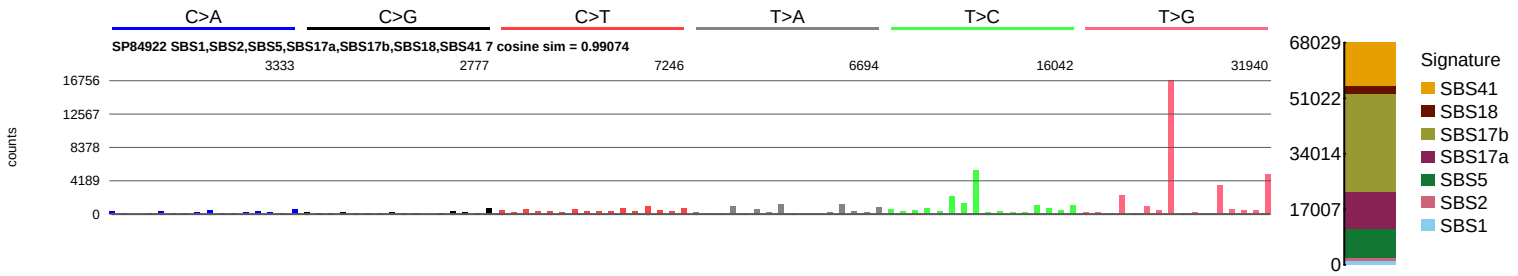
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.99074
2433
3522
6991
7003
16083
31995
16740
12555
8370
4185
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99074
3333
2777
7246
6694
16042
31940
16756
12567
8378
4189
0
counts
68029
51022
34014
17007
0
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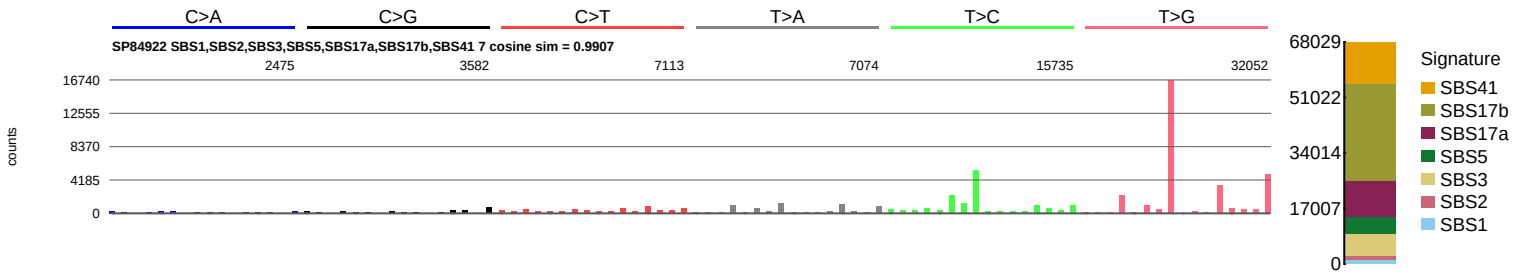
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41 7 cosine sim = 0.9907
2475
3582
7113
7074
15735
32052
16740
12555
8370
4185
0
counts
68029
51022
34014
17007
0
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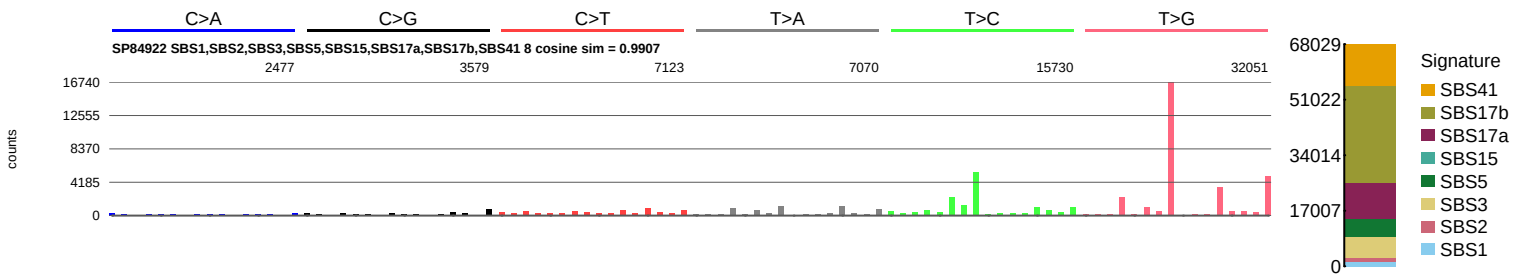
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.9907
2477
3579
7123
7070
15730
32051
16740
12555
8370
4185
0
counts
68029
51022
34014
17007
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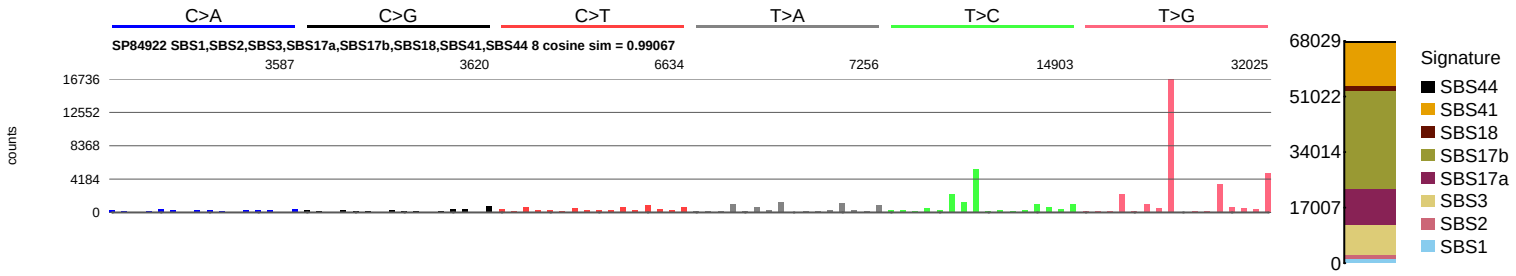
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.99067
3587
3620
6634
7256
14903
32025
counts
16736
12552
8368
4184
0
68029
51022
34014
17007
0
Signature
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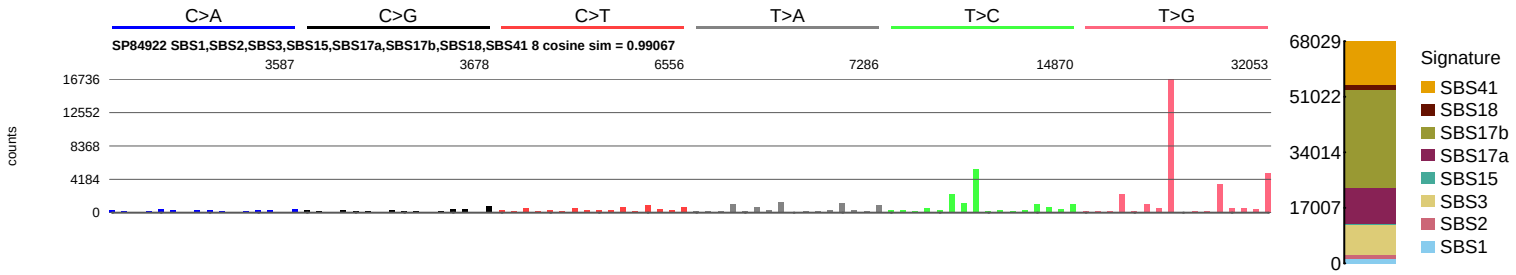
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99067
3587
3678
6556
7286
14870
32053
counts
16736
12552
8368
4184
0
68029
51022
34014
17007
0
Signature
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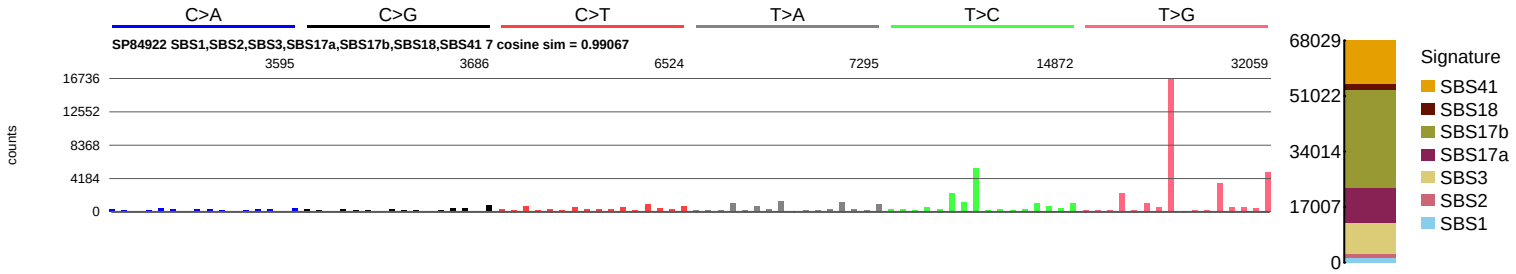
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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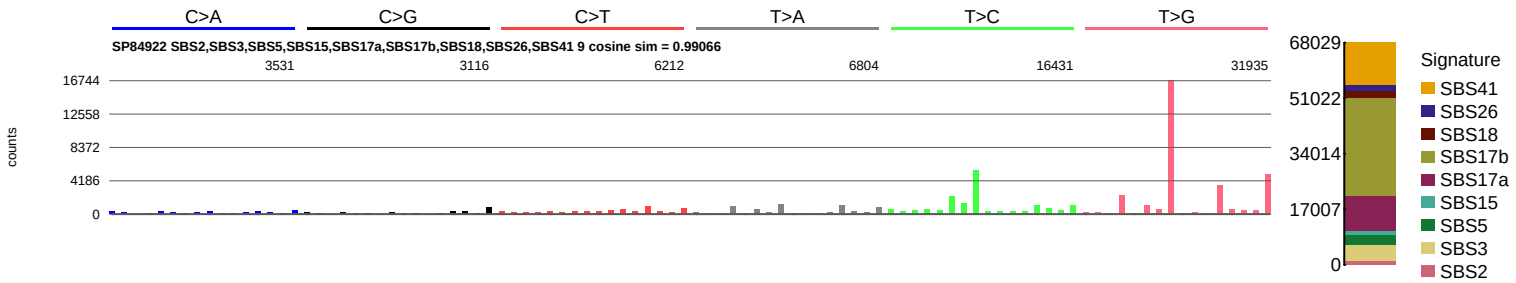
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99067
3595
3686
6524
7295
14872
32059
counts
16736
12552
8368
4184
0
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99066
3531
3116
6212
6804
16431
31935
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
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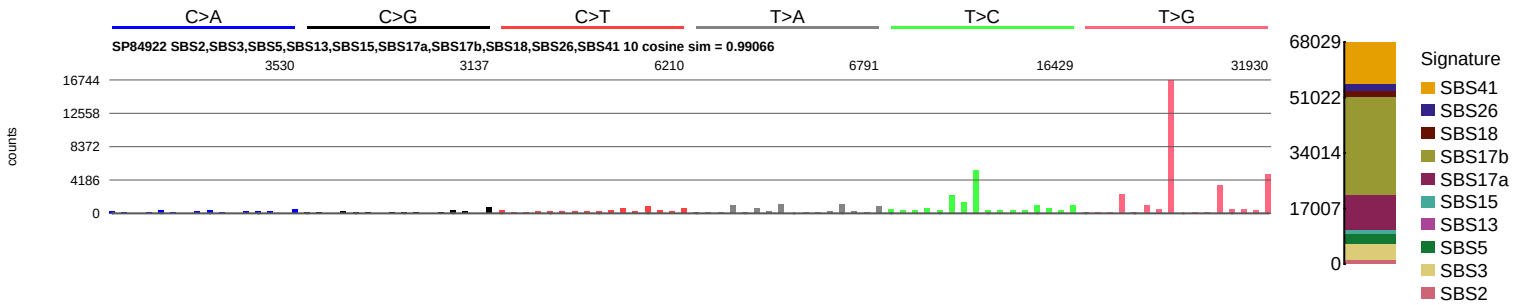
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2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C>G
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T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.99066
3530
3137
6210
6791
16429
31930
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
Signature
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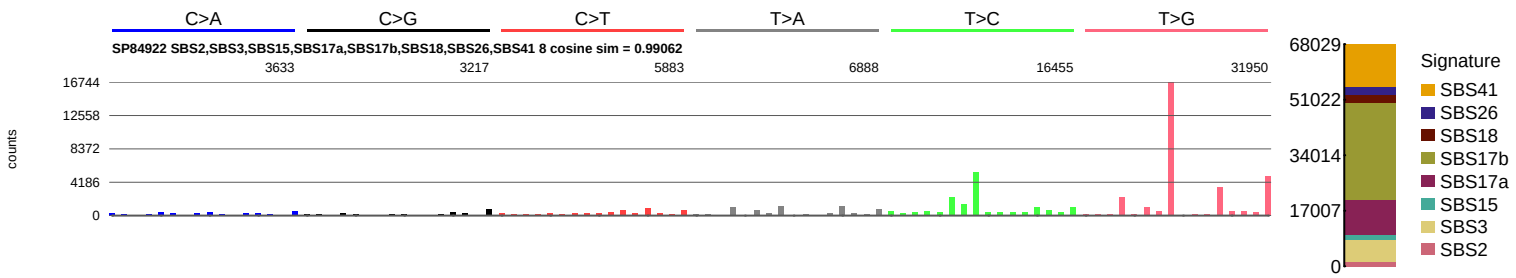
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99062
3633
3217
5883
6888
16455
31950
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
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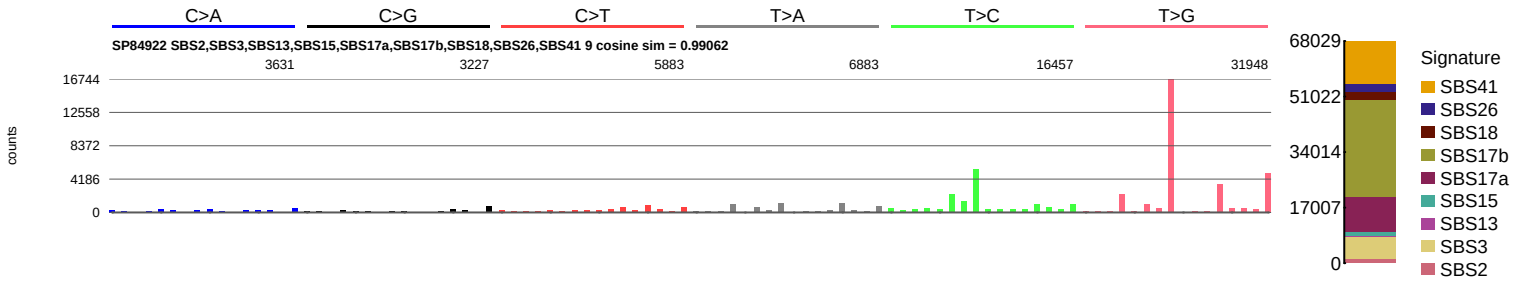
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2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C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99062
3631
3227
5883
6883
16457
31948
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
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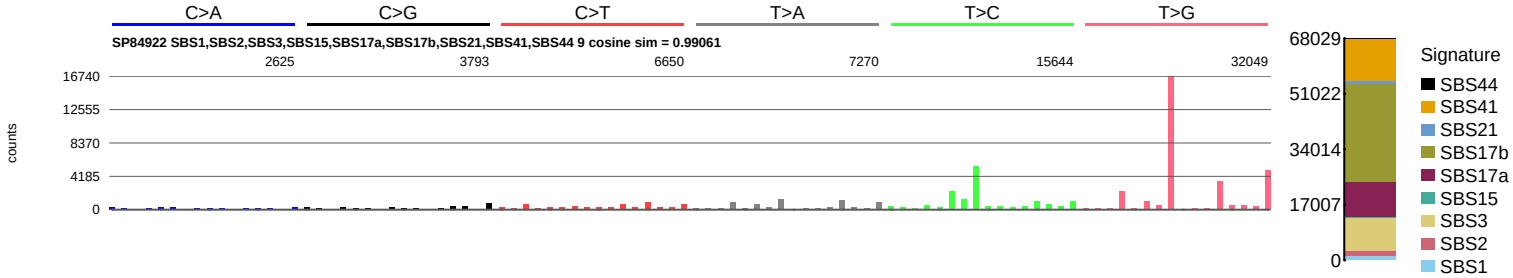
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
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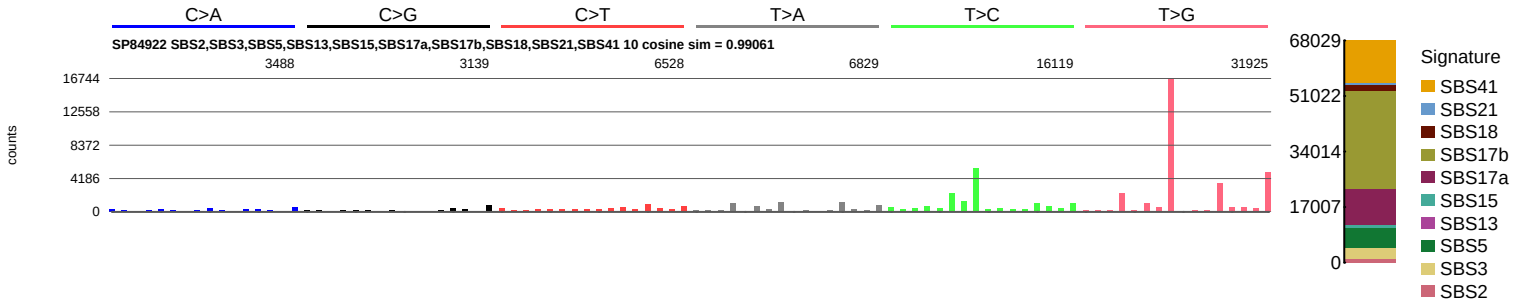
2625
3793
6650
7270
15644
32049
counts
16740
12555
8370
4185
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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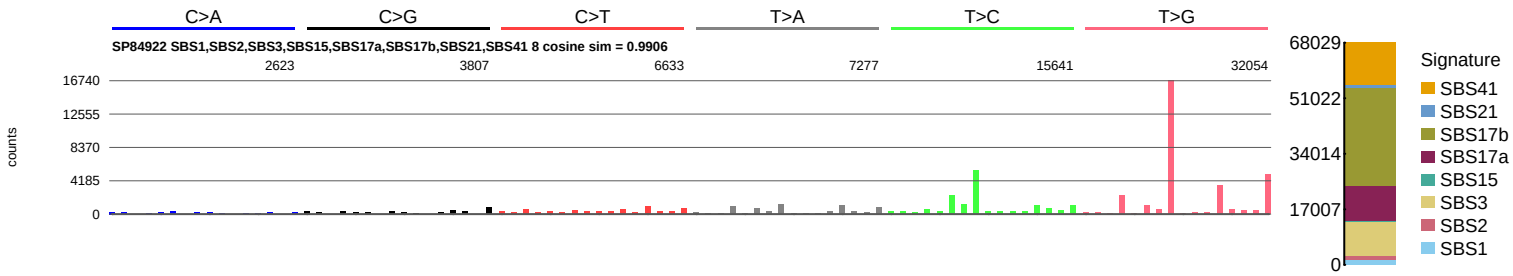
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99061
3488
3139
6528
6829
16119
31925
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2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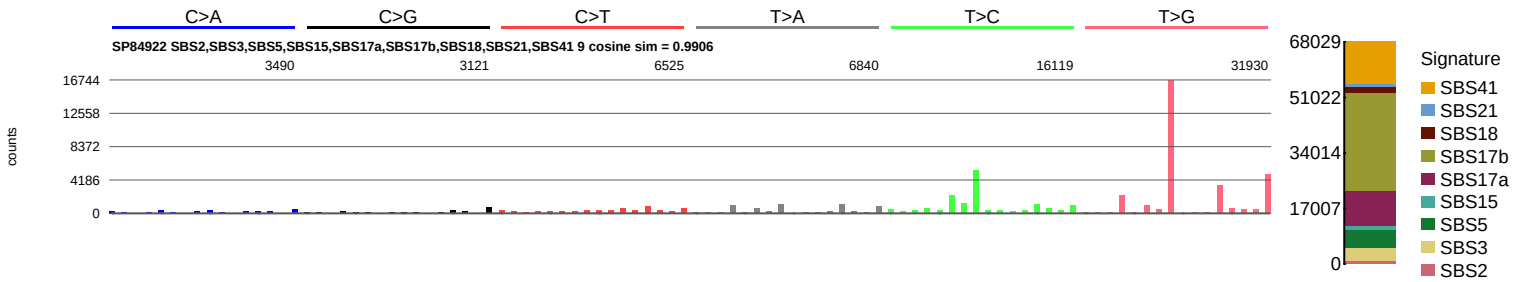
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.9906
2623
3807
6633
7277
15641
32054
counts
16740
12555
8370
4185
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.9906
3490
3121
6525
6840
16119
31930
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
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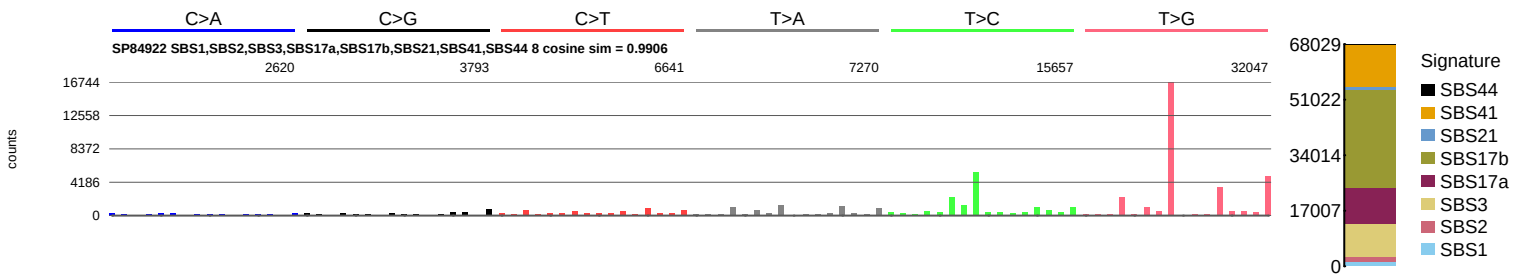
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2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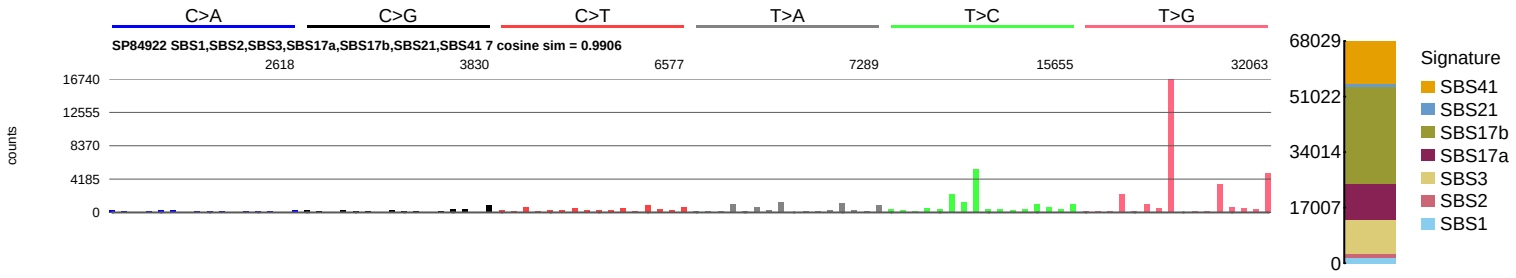
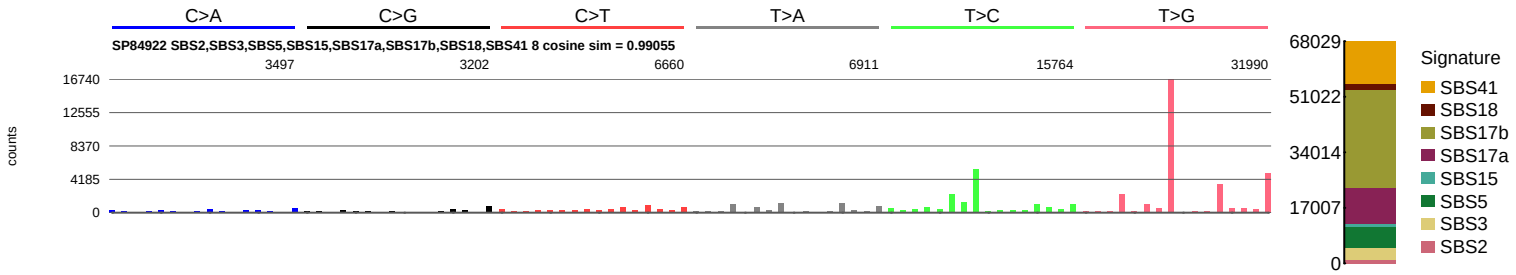
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.9906
2620
3793
6641
7270
15657
32047
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.9906
2618
3830
6577
7289
15655
32063
16740
12555
8370
4185
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1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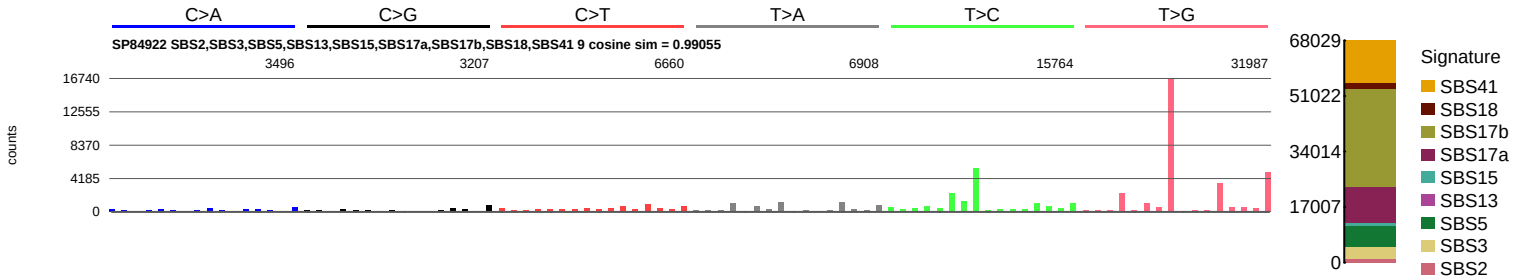
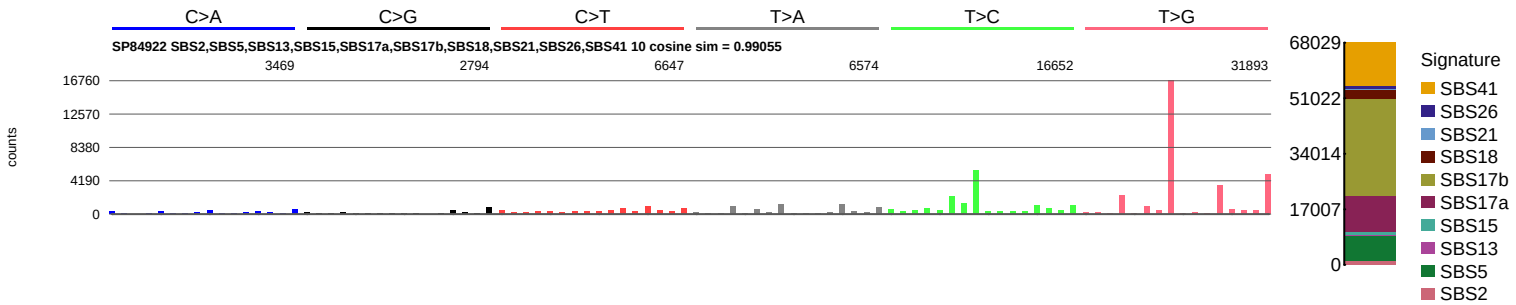
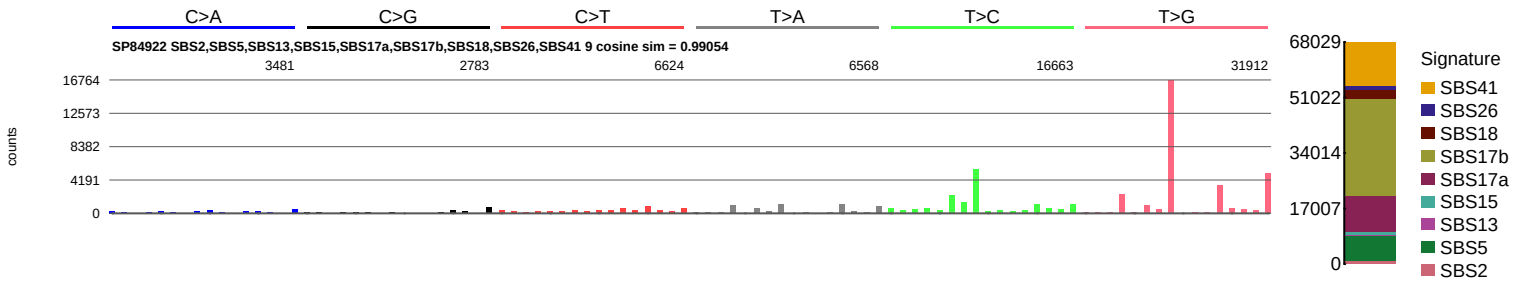
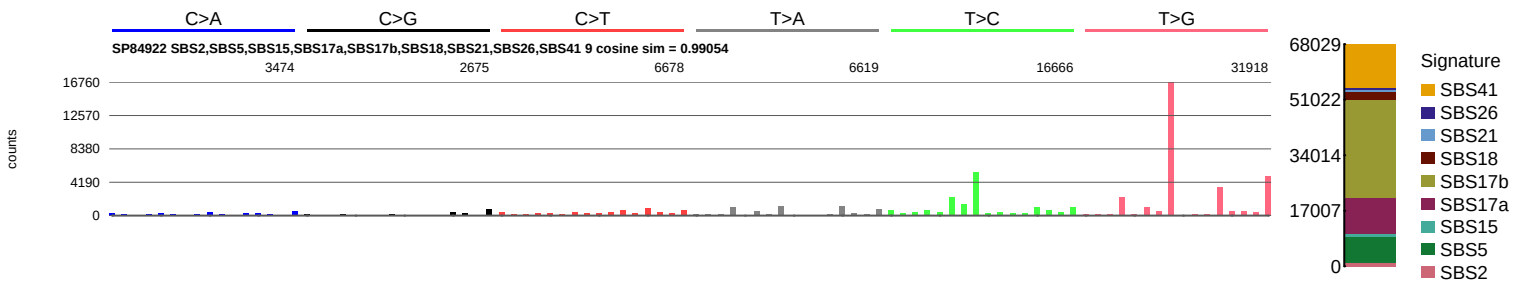
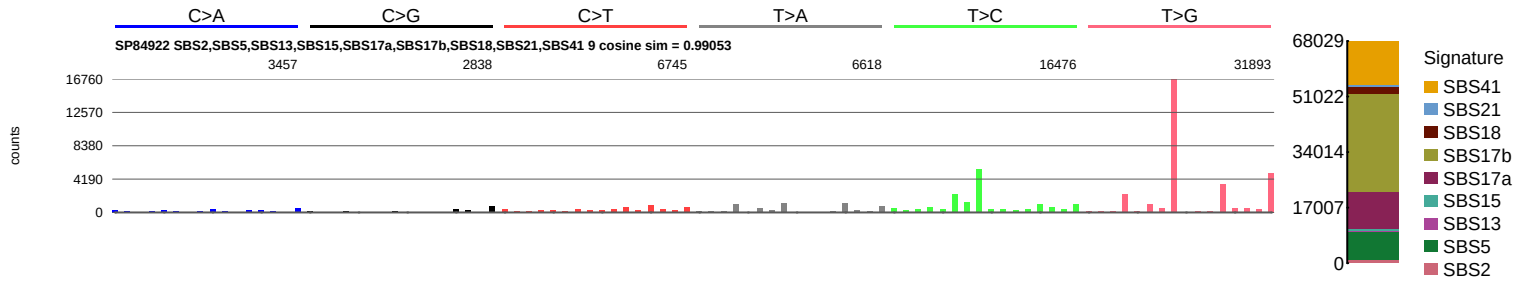
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99055
3497
3202
6660
6911
15764
31990
16740
12555
8370
4185
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

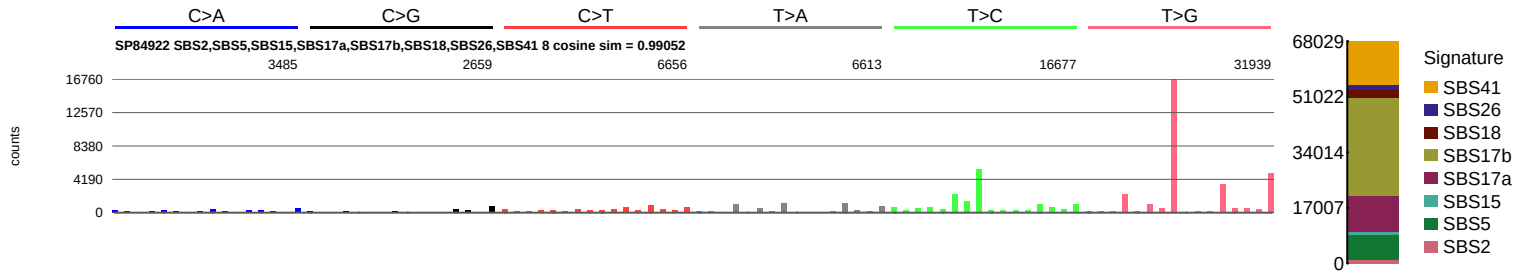

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99055
3496
3207
6660
6908
15764
31987
16740
12555
8370
4185
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

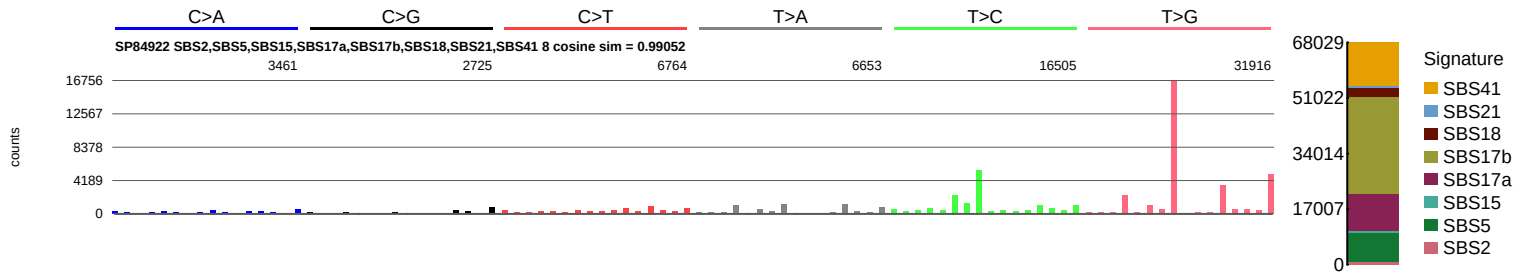

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.99055
3469
2794
6647
6574
16652
31893
16760
12570
8380
4190
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2

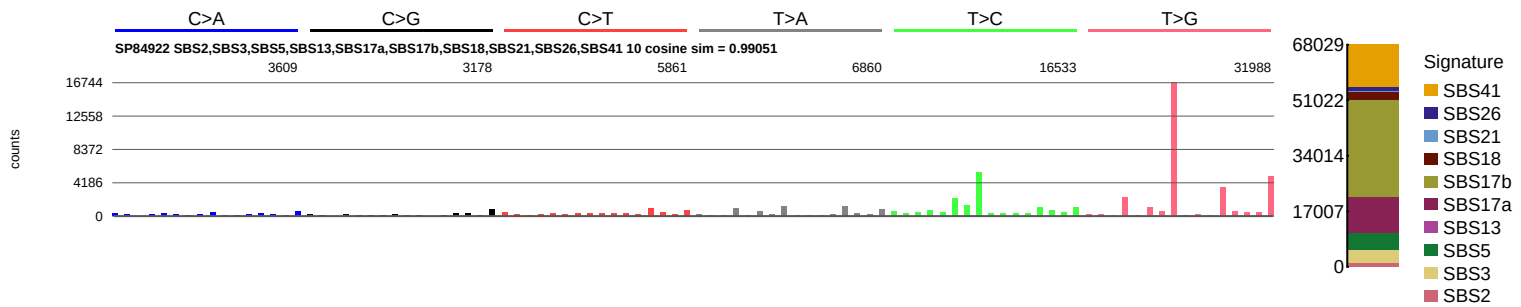

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99054
3481
2783
6624
6568
16663
31912
16764
12573
8382
4191
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.99054
3474
2675
6678
6619
16666
31918
16760
12570
8380
4190
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99053
3457
2838
6745
6618
16476
31893
16760
12570
8380
4190
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99052
3485
2659
6656
6613
16677
31939
16760
12570
8380
4190
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99052
3461
2725
6764
6653
16505
31916
16756
12567
8378
4189
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2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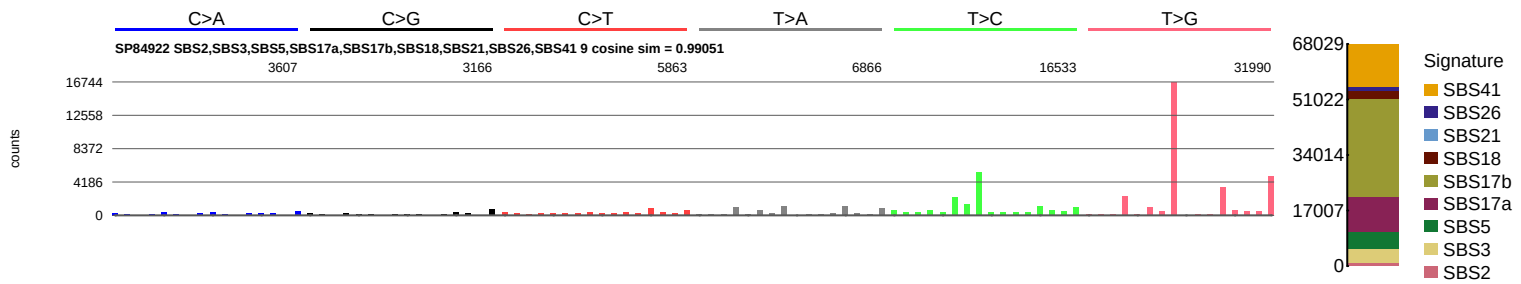
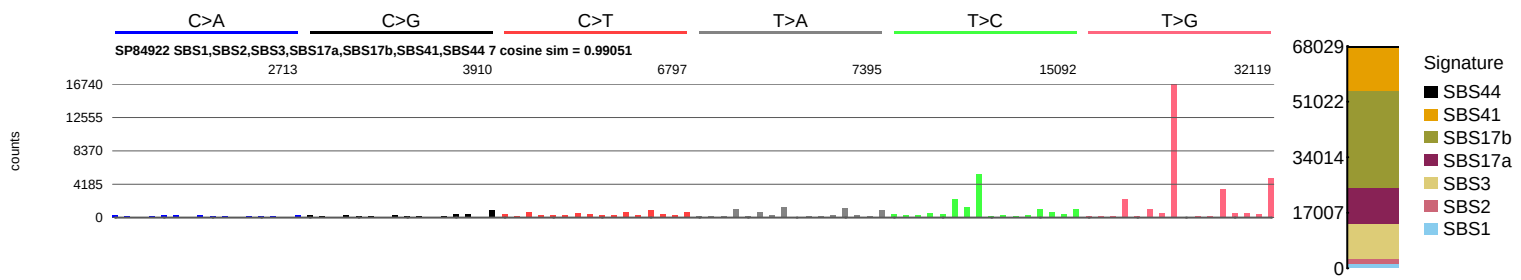
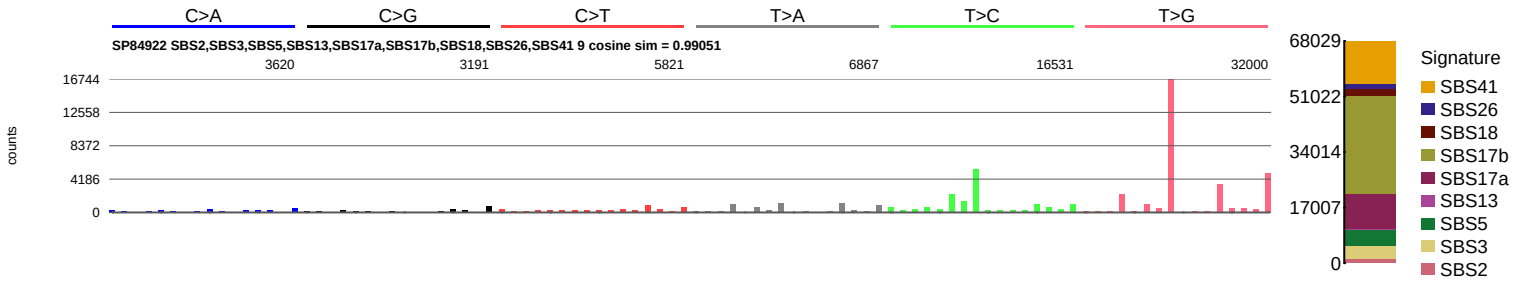
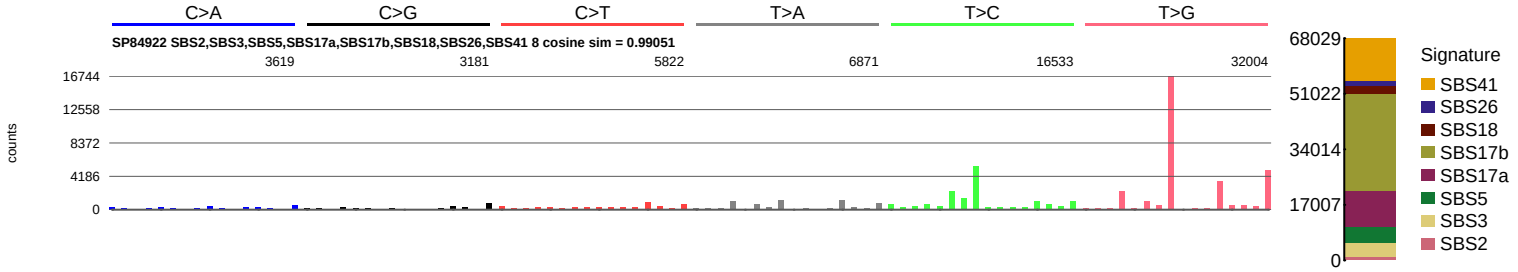
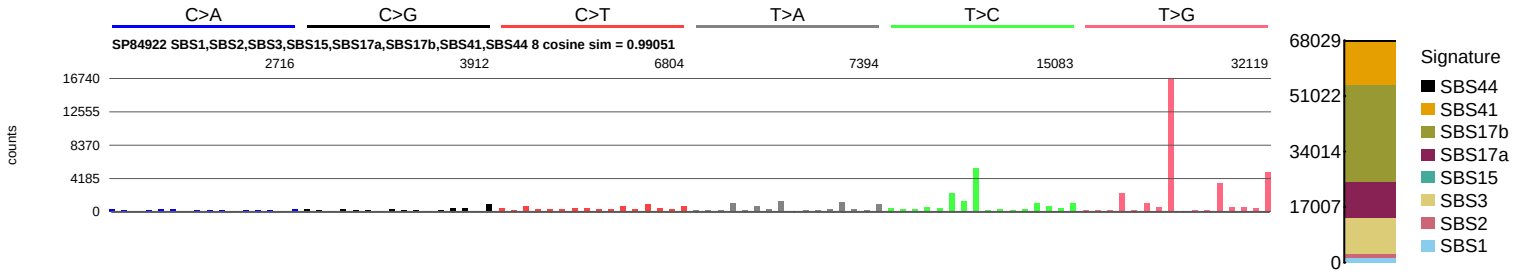
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.99051
3609
3178
5861
6860
16533
31988
16744
12558
8372
4186
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

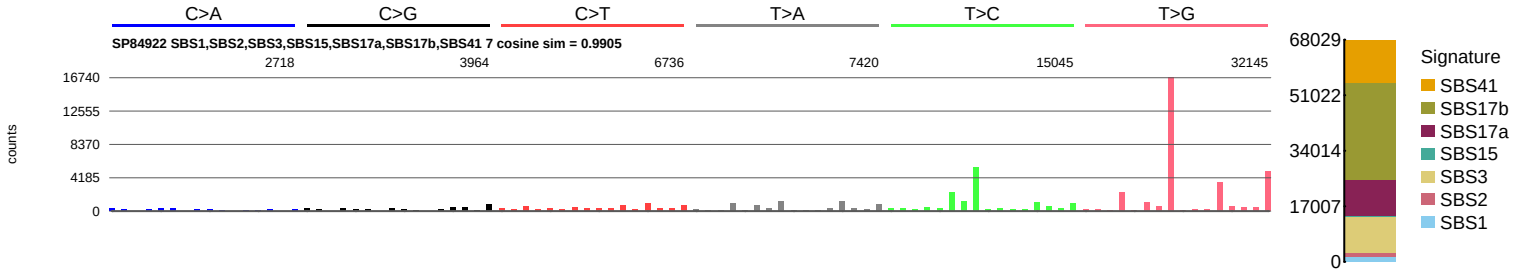

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.99051
3607
3166
5863
6866
16533
31990
16744
12558
8372
4186
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2

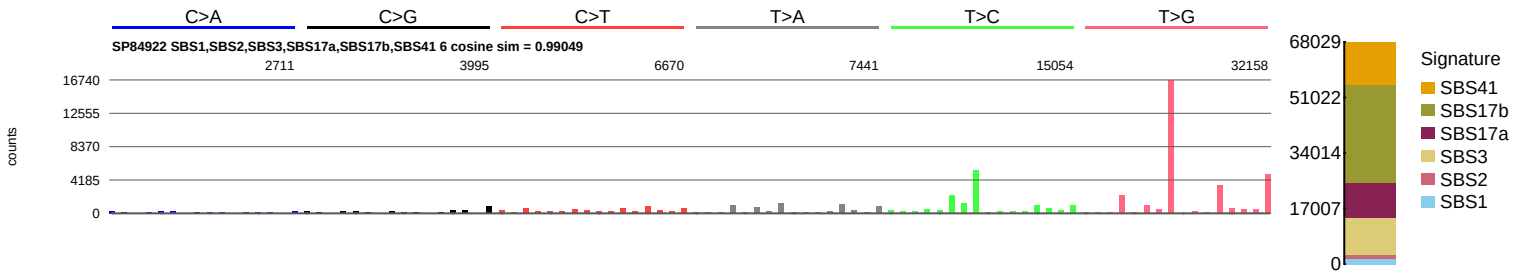

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.99051
2713
3910
6797
7395
15092
32119
16740
12555
8370
4185
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS3
SBS2
SBS1

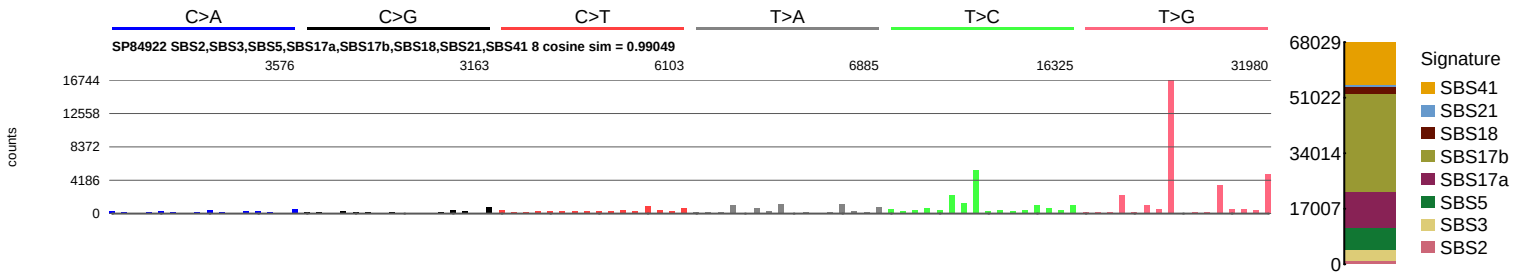

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99051
3620
3191
5821
6867
16531
32000
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

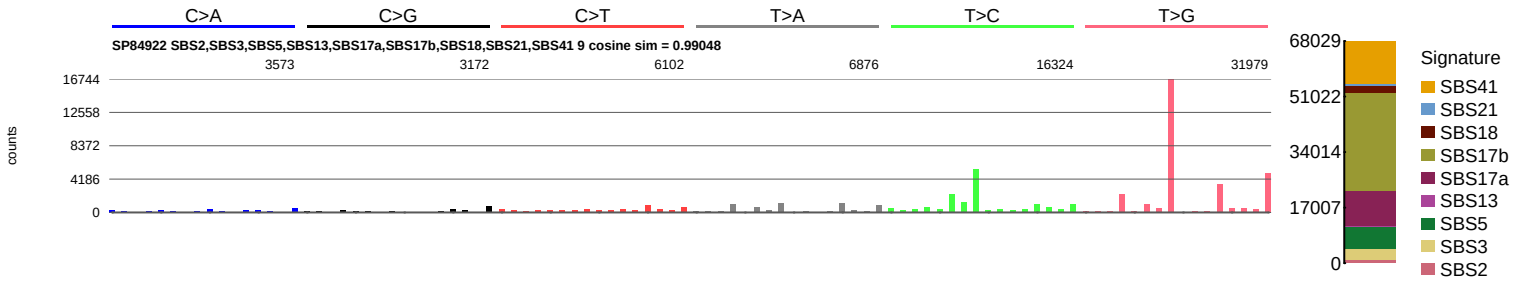

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99051
3619
3181
5822
6871
16533
32004
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2

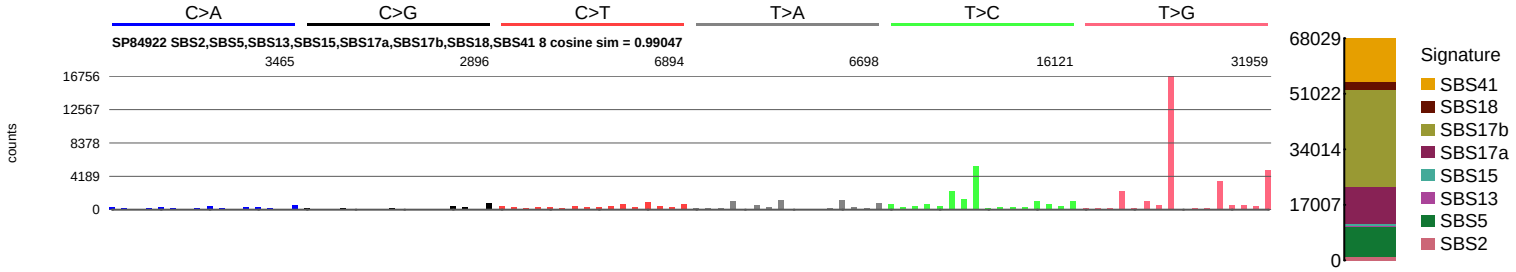

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.99051
2716
3912
6804
7394
15083
32119
counts
16740
12555
8370
4185
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

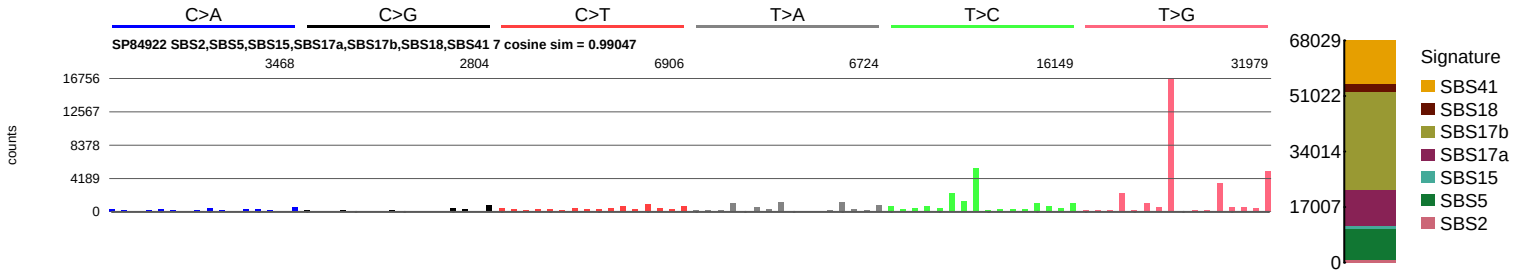

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.9905
2718
3964
6736
7420
15045
32145
counts
16740
12555
8370
4185
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS41 6 cosine sim = 0.99049
2711
3995
6670
7441
15054
32158
counts
16740
12555
8370
4185
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99049
3576
3163
6103
6885
16325
31980
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99048
3573
3172
6102
6876
16324
31979
16744
12558
8372
4186
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99047
3465
2896
6894
6698
16121
31959
16756
12567
8378
4189
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2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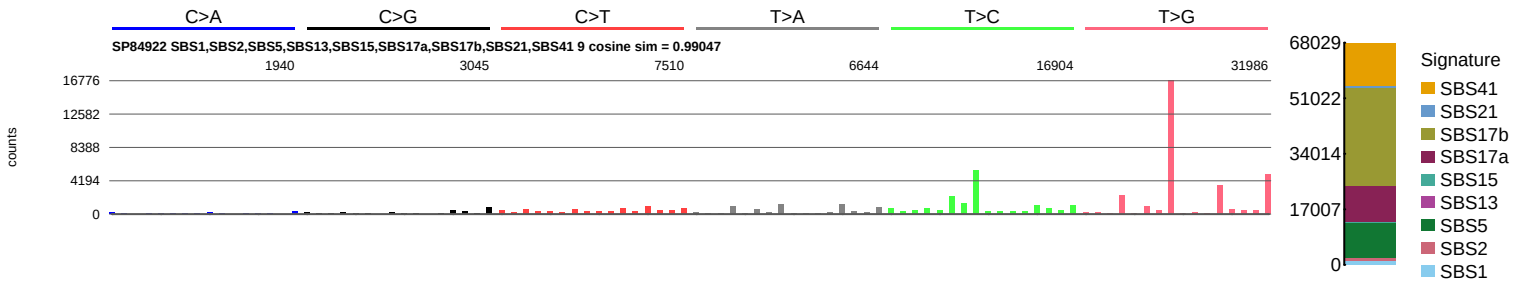
SP84922 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99047
3468
2804
6906
6724
16149
31979
16756
12567
8378
4189
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2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C>T
T>A
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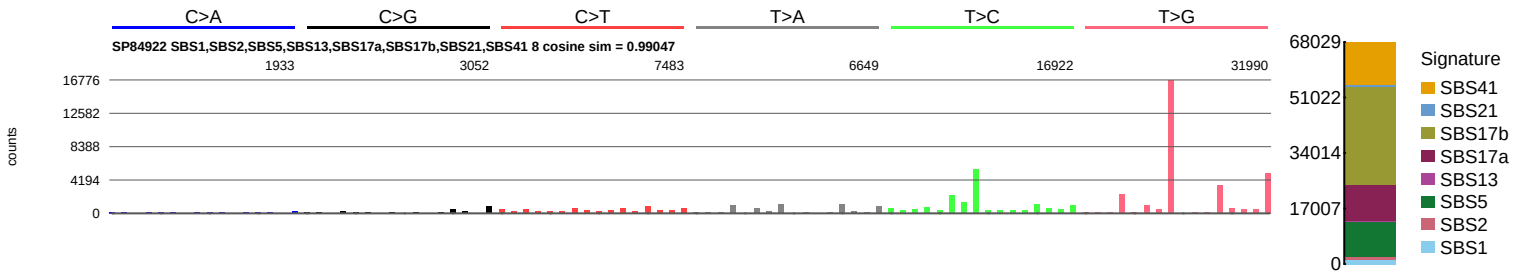
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.99047
1940
3045
7510
6644
16904
31986
16776
12582
8388
4194
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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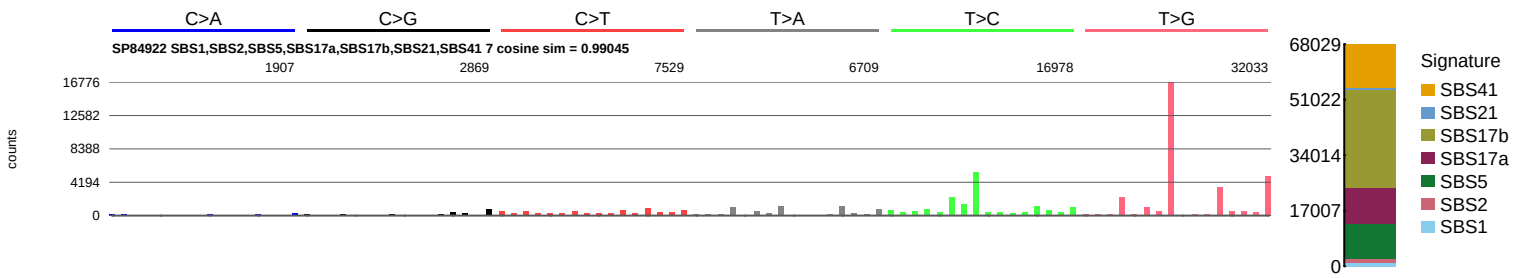
SP84922 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.99047
1933
3052
7483
6649
16922
31990
16776
12582
8388
4194
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>T
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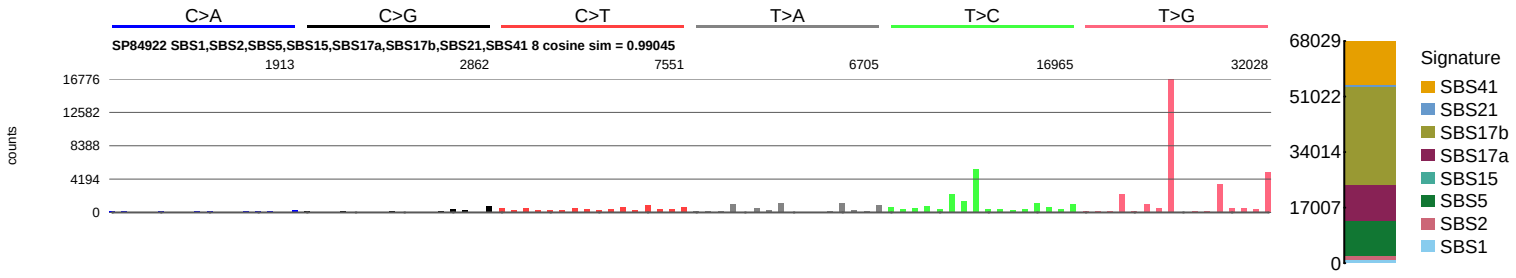
T>G
SP84922 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.99045
1907
2869
7529
6709
16978
32033
16776
12582
8388
4194
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1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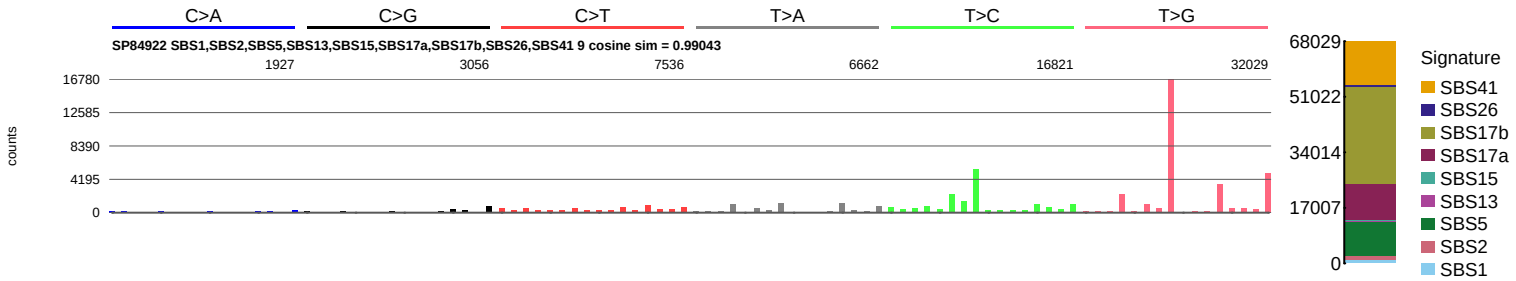
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.99045
1913
2862
7551
6705
16965
32028
16776
12582
8388
4194
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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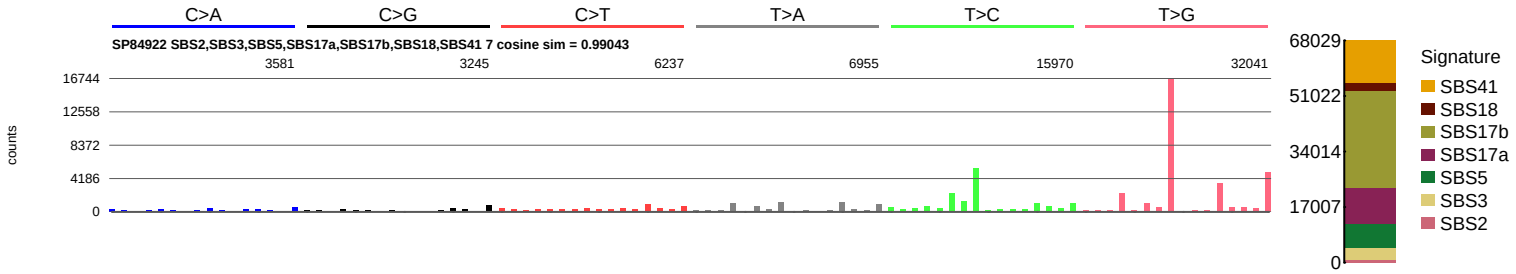
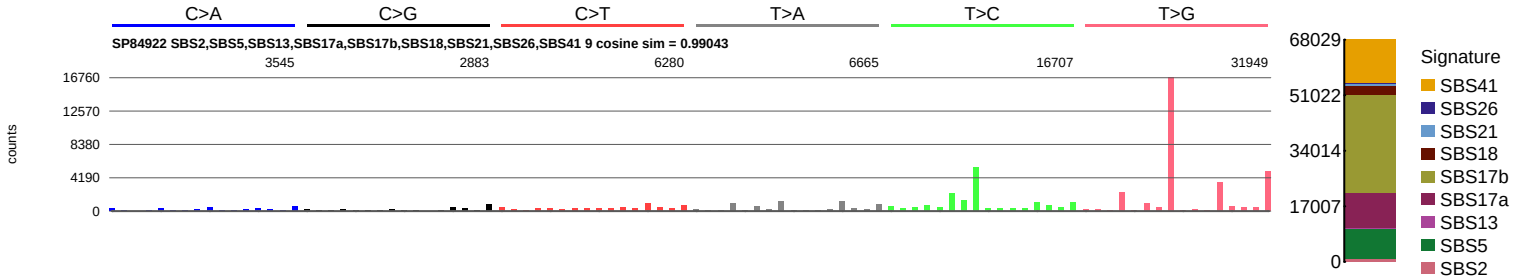
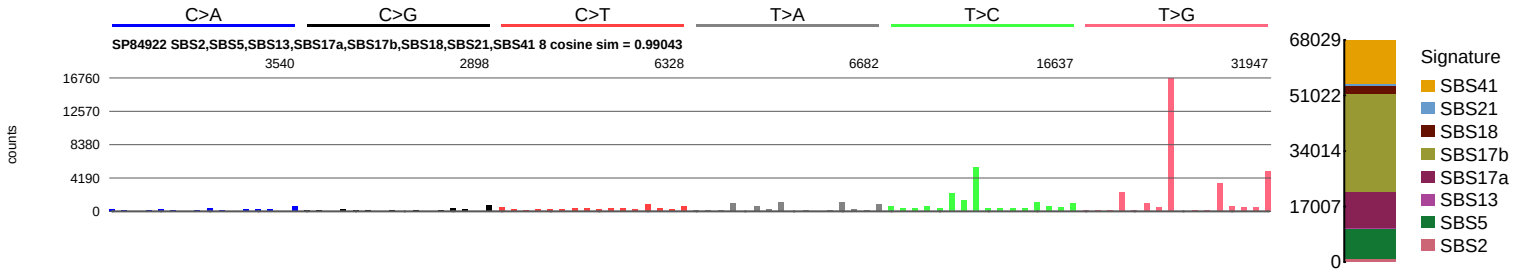
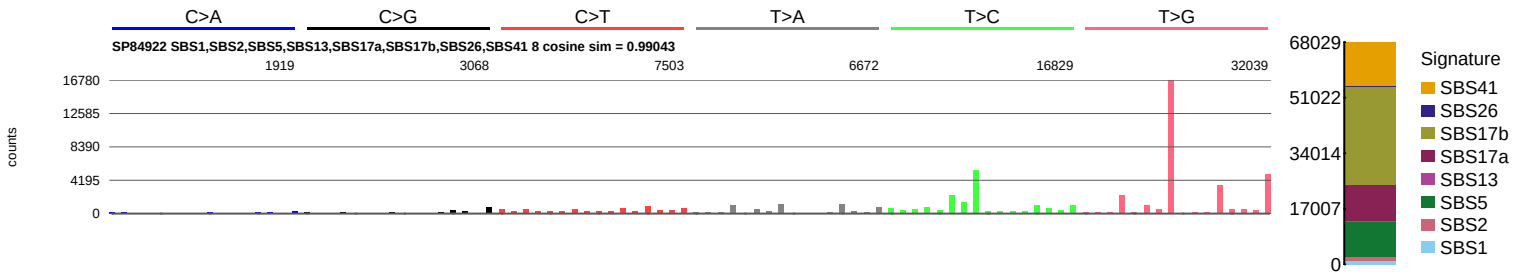
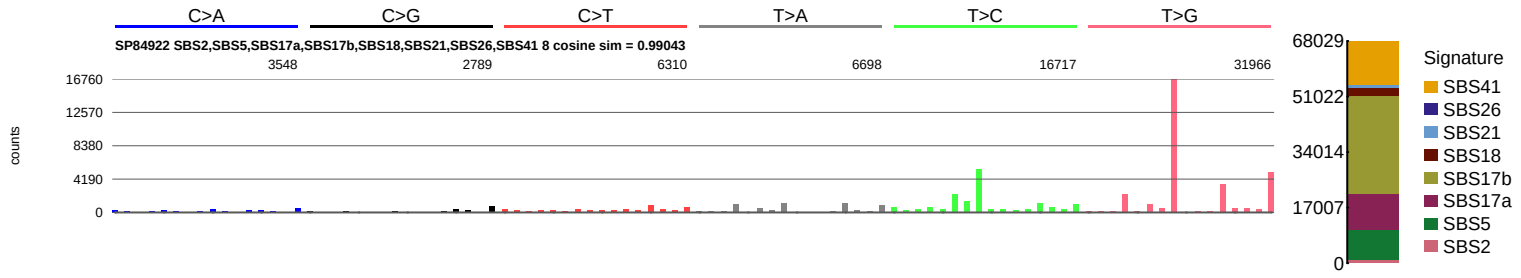
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 9 cosine sim = 0.99043
1927
3056
7536
6662
16821
32029
16780
12585
8390
4195
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

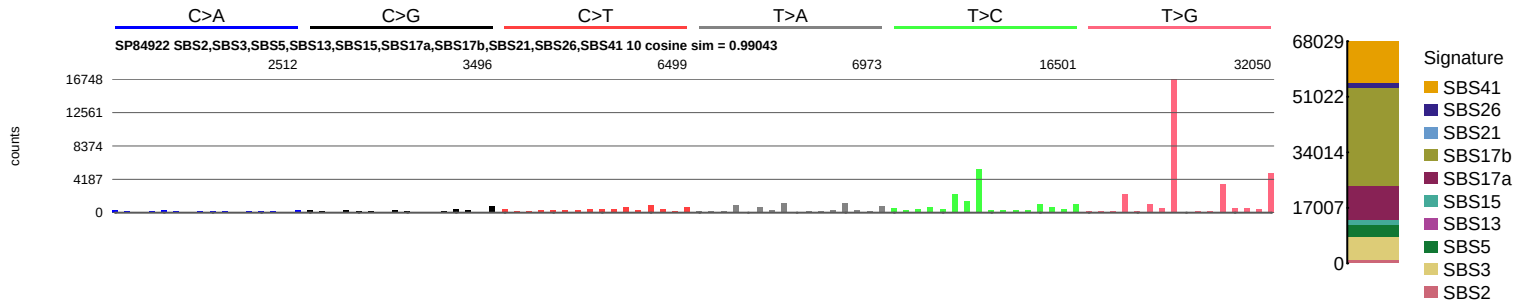

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99043
3581
3245
6237
6955
15970
32041
16744
12558
8372
4186
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2

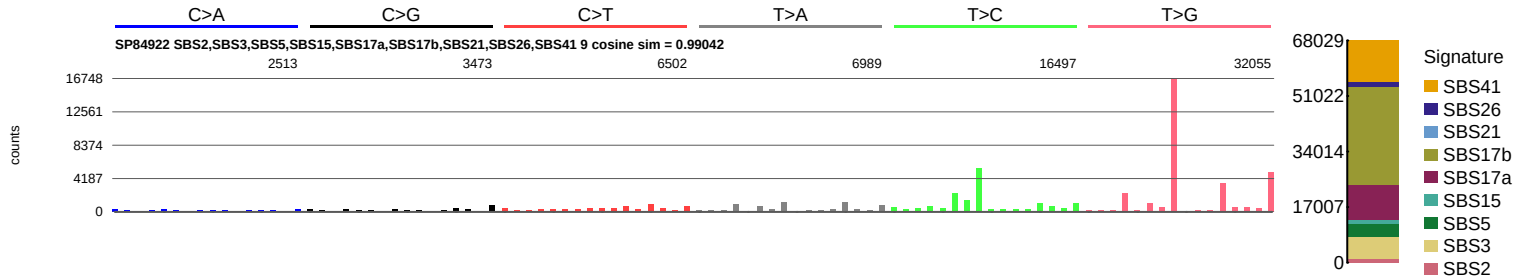

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.99043
3545
2883
6280
6665
16707
31949
16760
12570
8380
4190
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

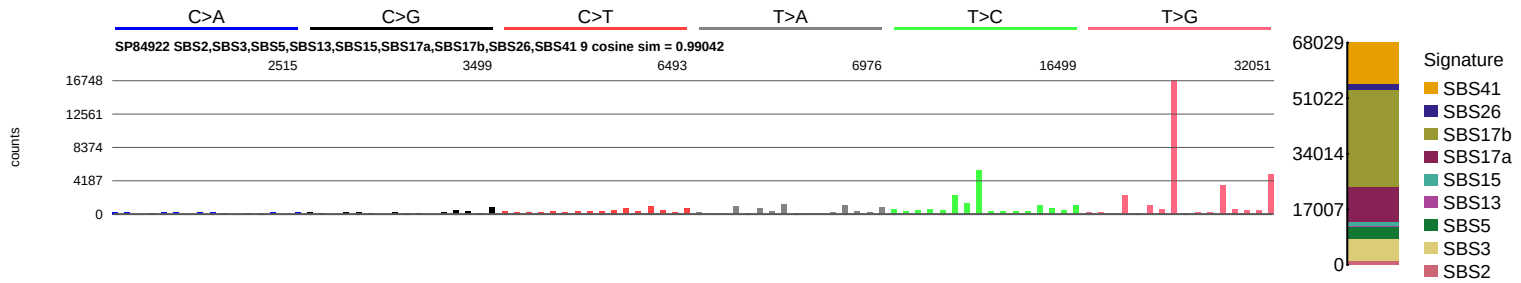

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99043
3540
2898
6328
6682
16637
31947
16760
12570
8380
4190
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.99043
1919
3068
7503
6672
16829
32039
16780
12585
8390
4195
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.99043
3548
2789
6310
6698
16717
31966
counts
16760
12570
8380
4190
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 10 cosine sim = 0.99043
2512
3496
6499
6973
16501
32050
counts
16748
12561
8374
4187
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 9 cosine sim = 0.99042
2513
3473
6502
6989
16497
32055
counts
16748
12561
8374
4187
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2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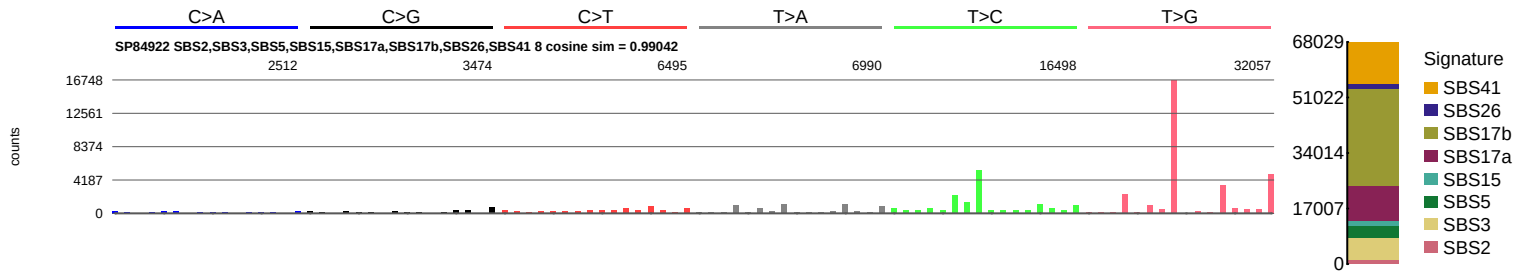
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 9 cosine sim = 0.99042
2515
3499
6493
6976
16499
32051
counts
16748
12561
8374
4187
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2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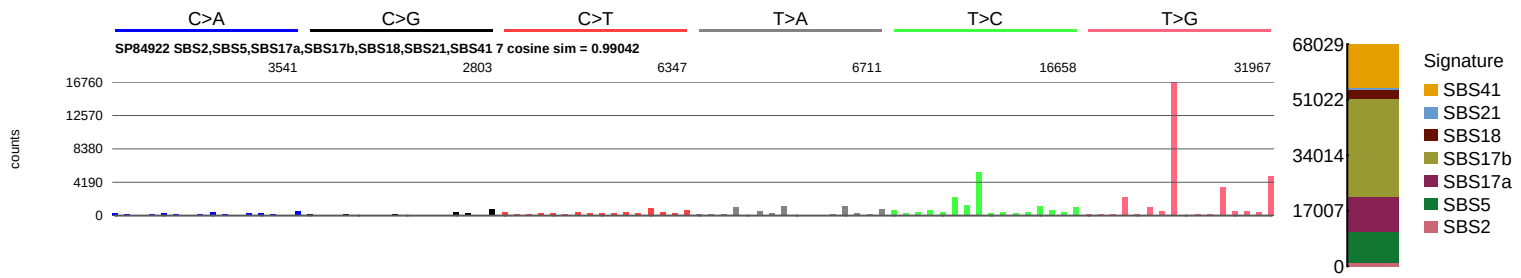
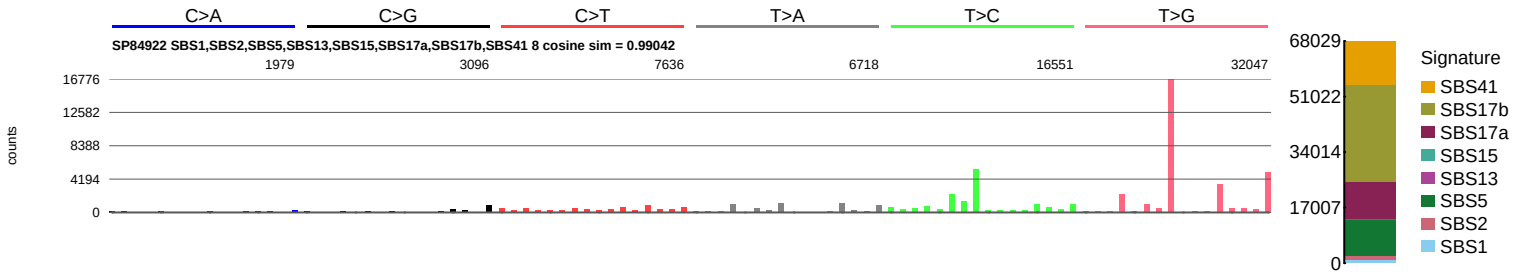
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.99042
2512
3474
6495
6990
16498
32057
counts
16748
12561
8374
4187
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.99042
3541
2803
6347
6711
16658
31967
counts
16760
12570
8380
4190
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2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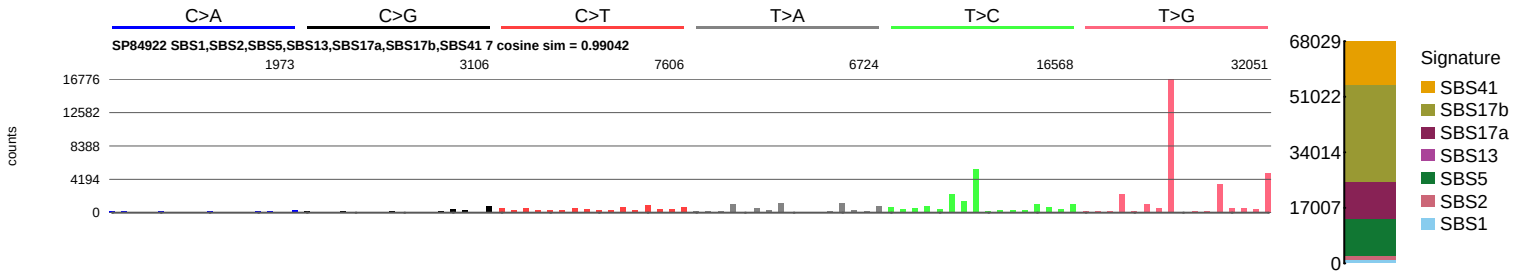
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.99042
1979
3096
7636
6718
16551
32047
counts
16776
12582
8388
4194
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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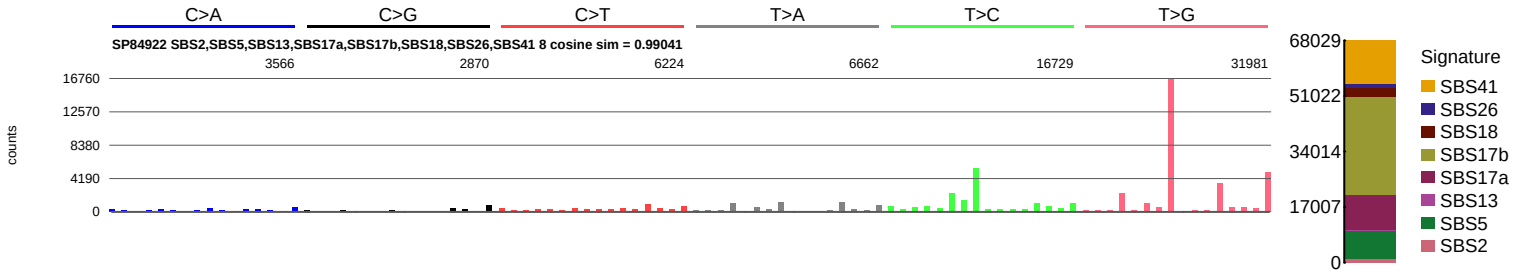
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS41 7 cosine sim = 0.99042
1973
3106
7606
6724
16568
32051
counts
16776
12582
8388
4194
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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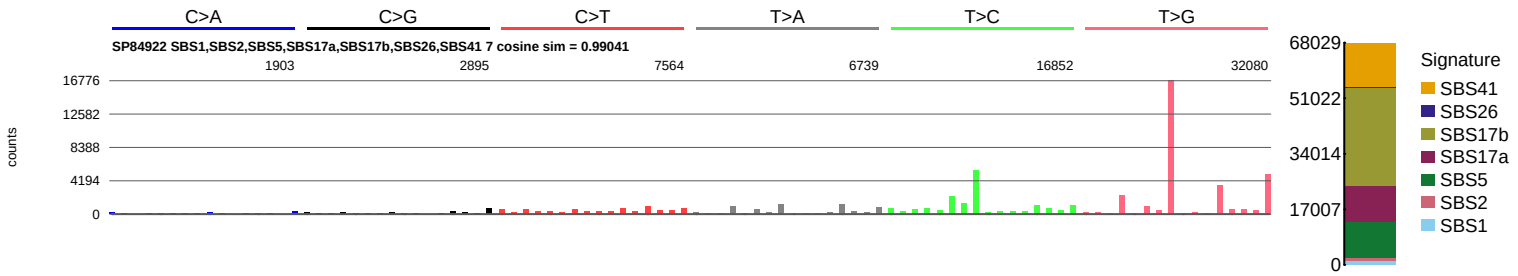
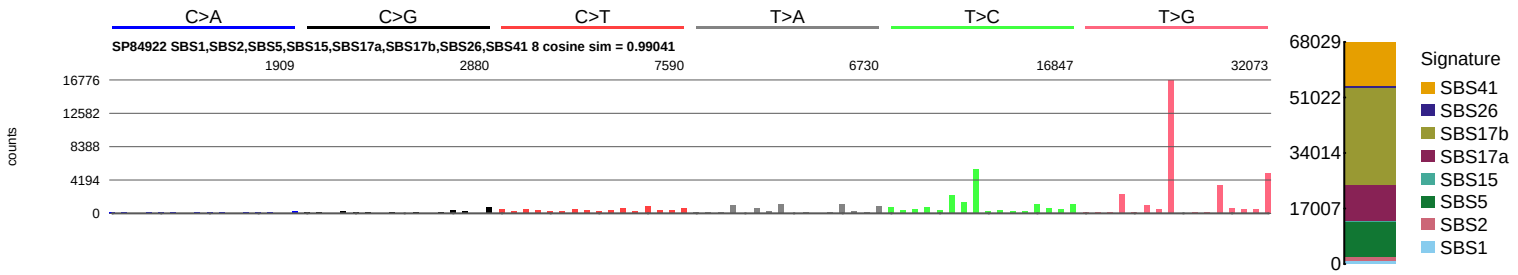
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99041
3566
2870
6224
6662
16729
31981
counts
16760
12570
8380
4190
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.99041
1903
2895
7564
6739
16852
32080
counts
16776
12582
8388
4194
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS5
SBS2
SBS1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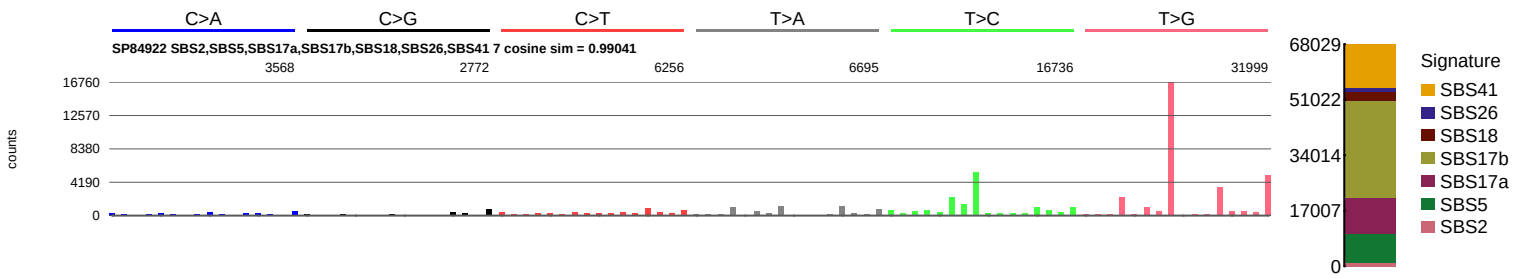
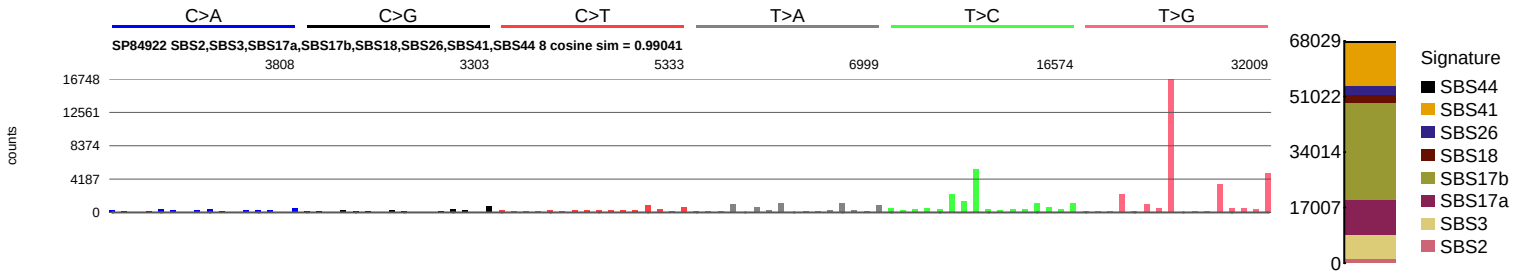
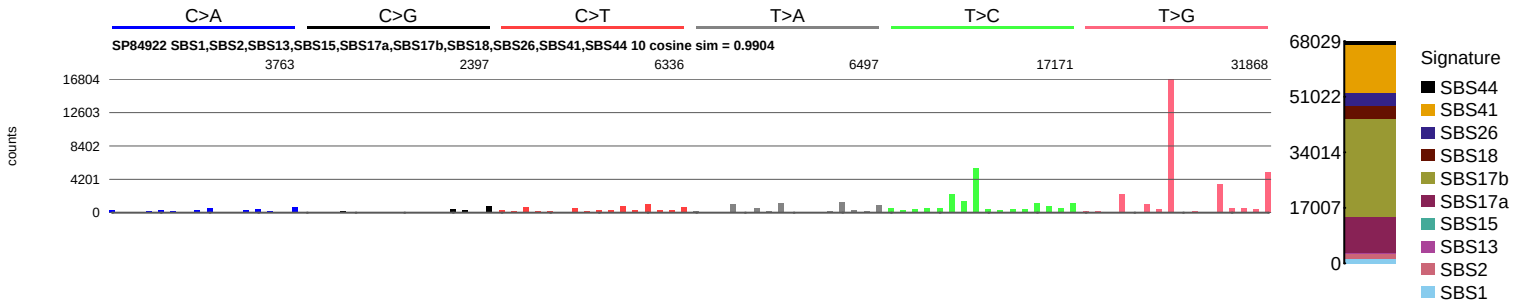
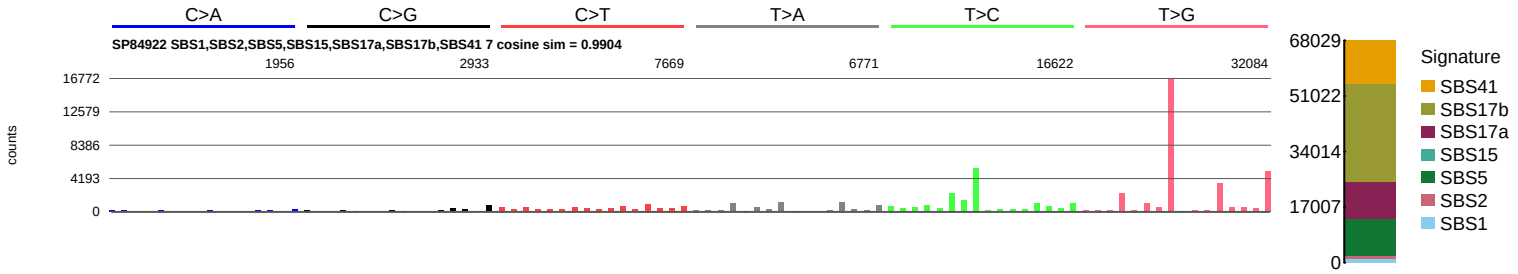
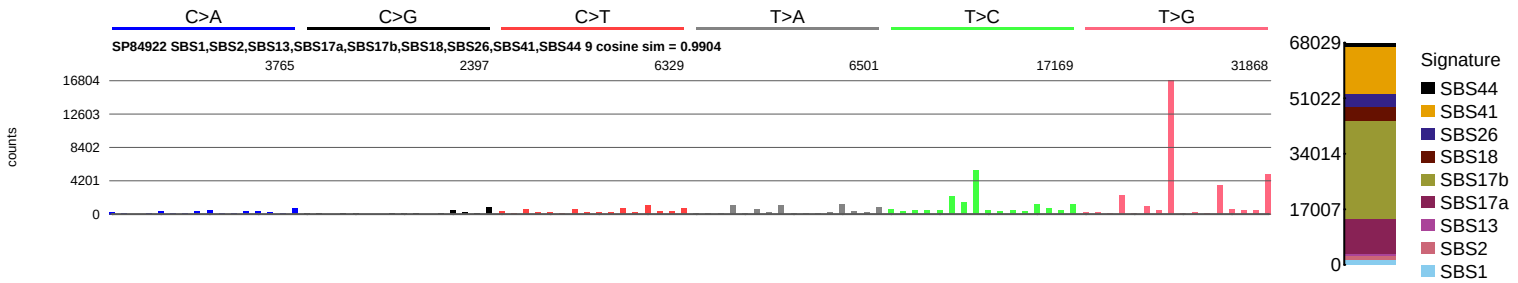
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.99041
1909
2880
7590
6730
16847
32073
counts
16776
12582
8388
4194
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

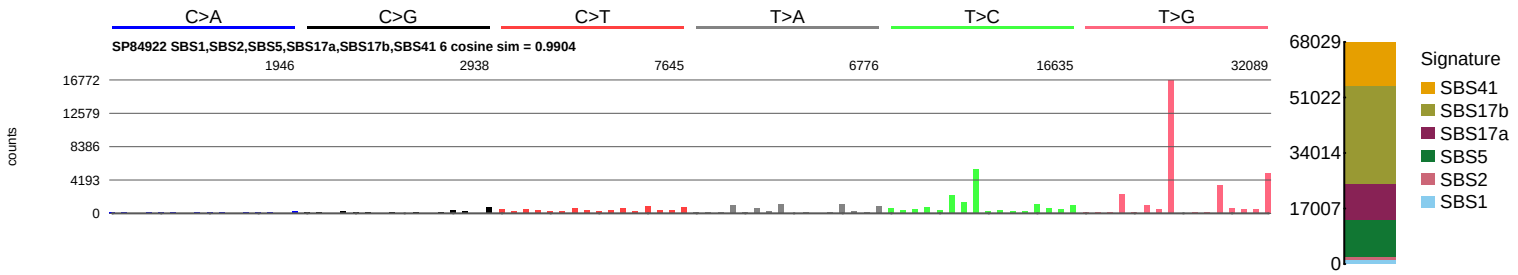

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.99041
3568
2772
6256
6695
16736
31999
counts
16760
12570
8380
4190
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2

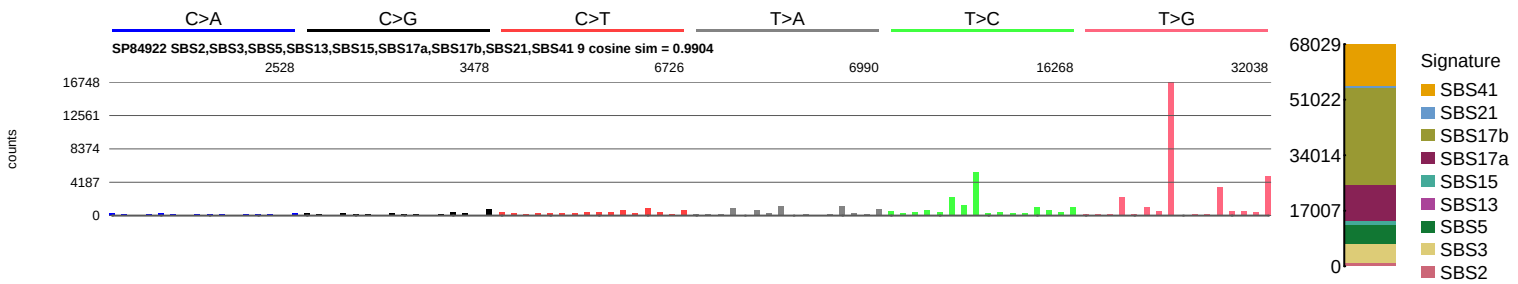

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.99041
3808
3303
5333
6999
16574
32009
counts
16748
12561
8374
4187
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2

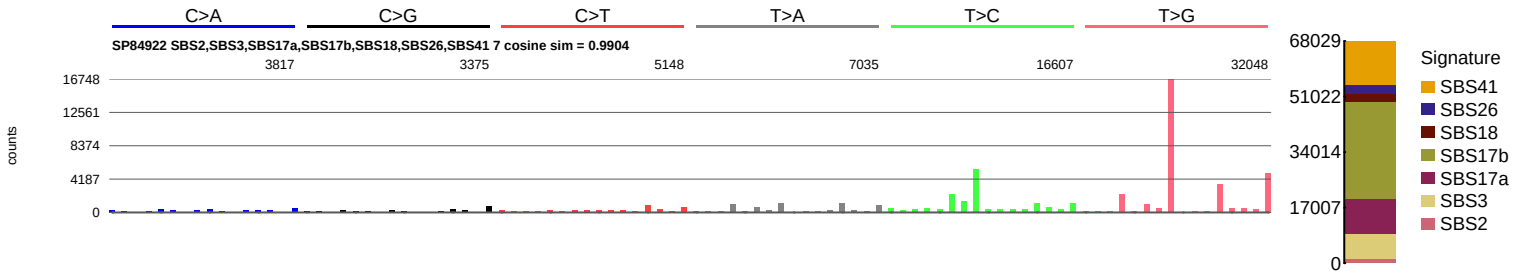

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.9904
3763
2397
6336
6497
17171
31868
counts
16804
12603
8402
4201
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

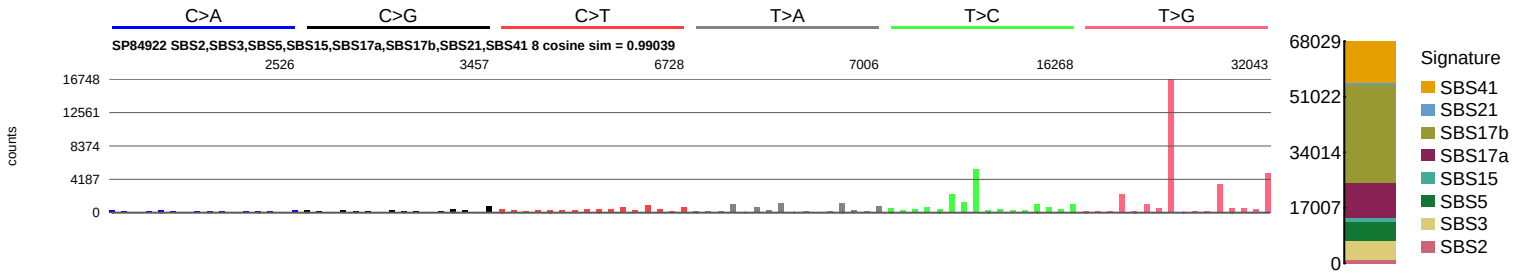

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.9904
1956
2933
7669
6771
16622
32084
counts
16772
12579
8386
4193
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

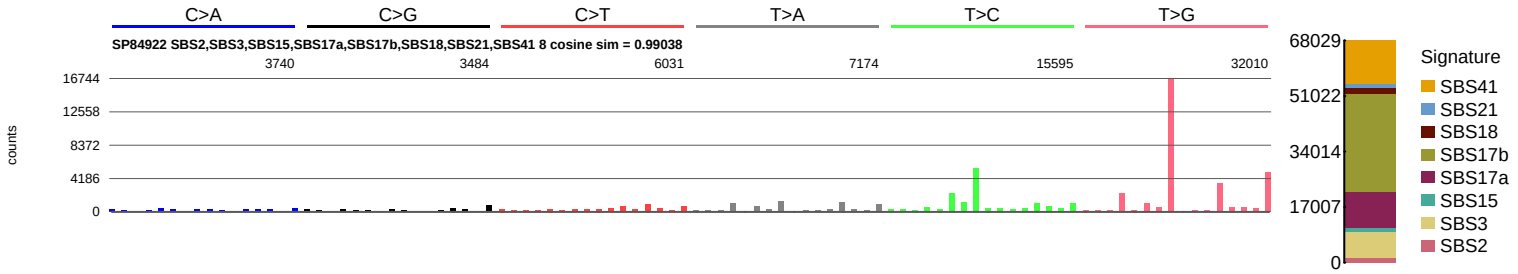

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.9904
3765
2397
6329
6501
17169
31868
counts
16804
12603
8402
4201
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

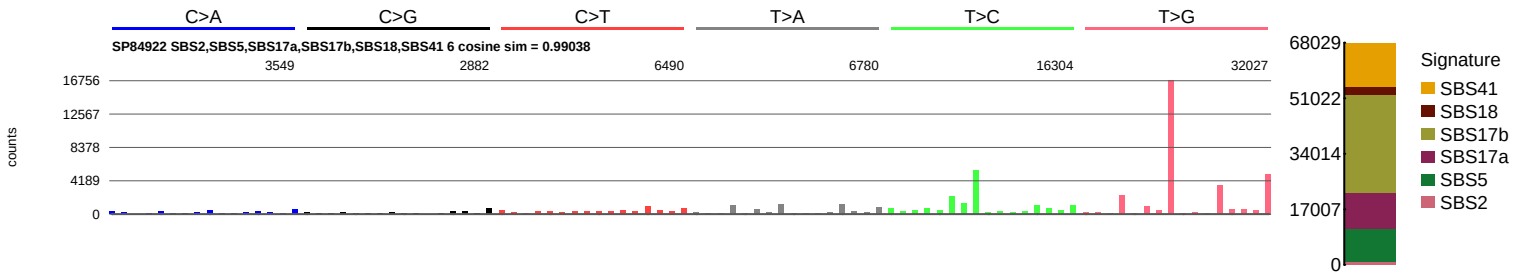

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.9904
1946
2938
7645
6776
16635
32089
counts
16772
12579
8386
4193
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS2
SBS1

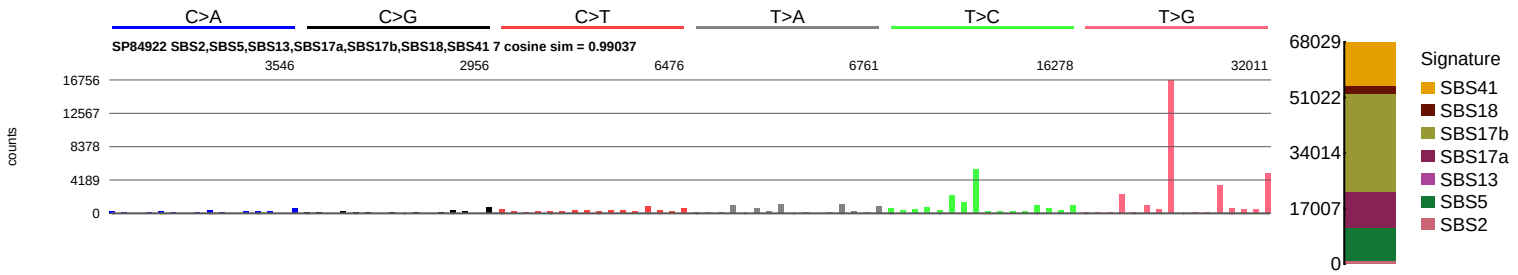

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.9904
2528
3478
6726
6990
16268
32038
counts
16748
12561
8374
4187
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

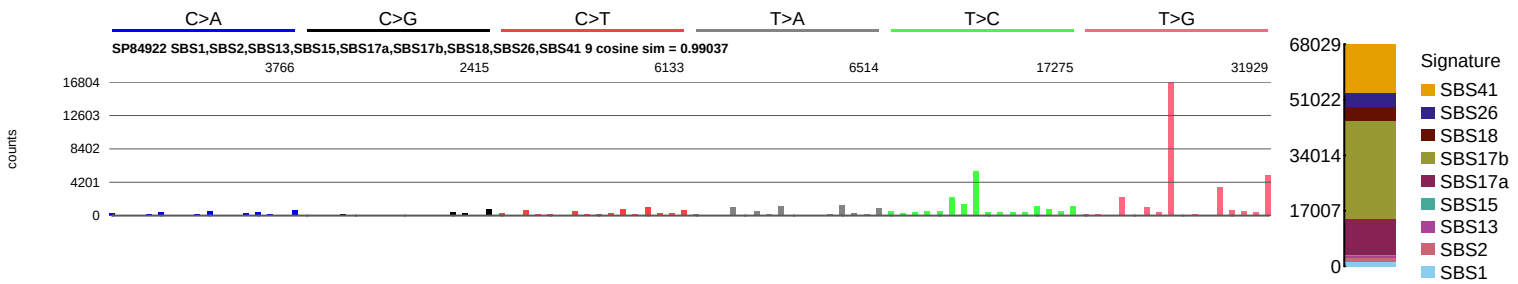

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.9904
3817
3375
5148
7035
16607
32048
16748
12561
8374
4187
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2

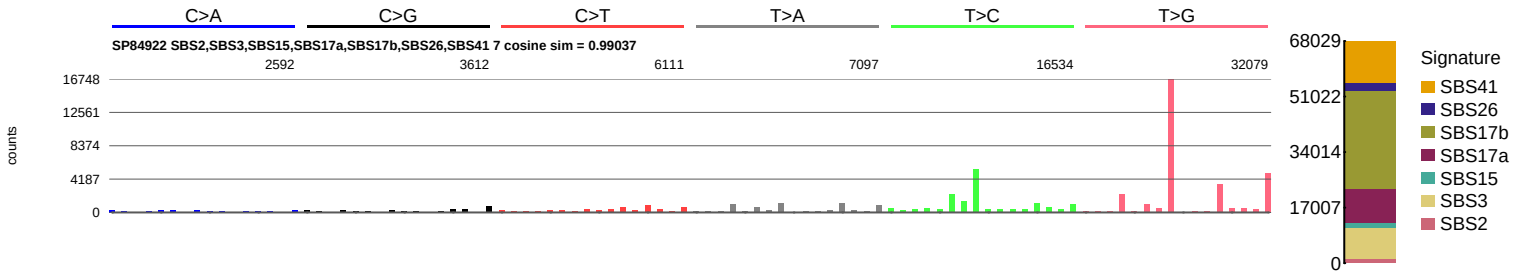

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.99039
2526
3457
6728
7006
16268
32043
16748
12561
8374
4187
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

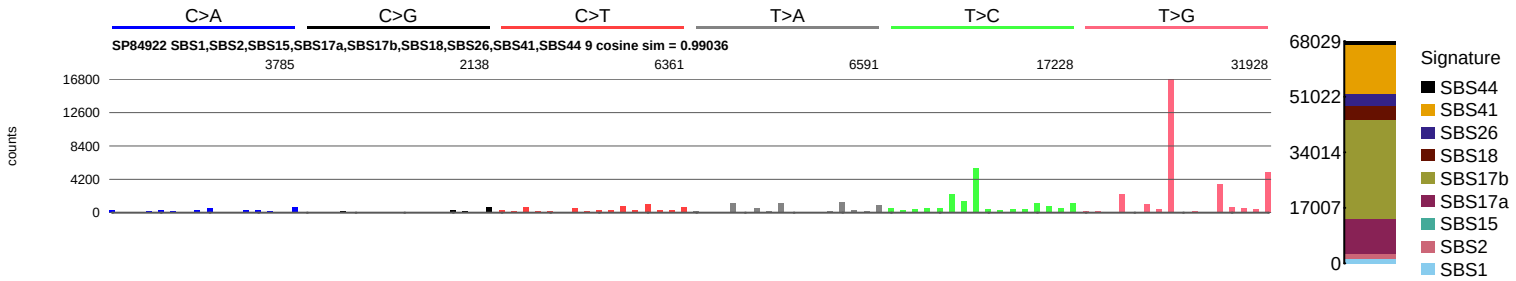

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99038
3740
3484
6031
7174
15595
32010
16744
12558
8372
4186
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

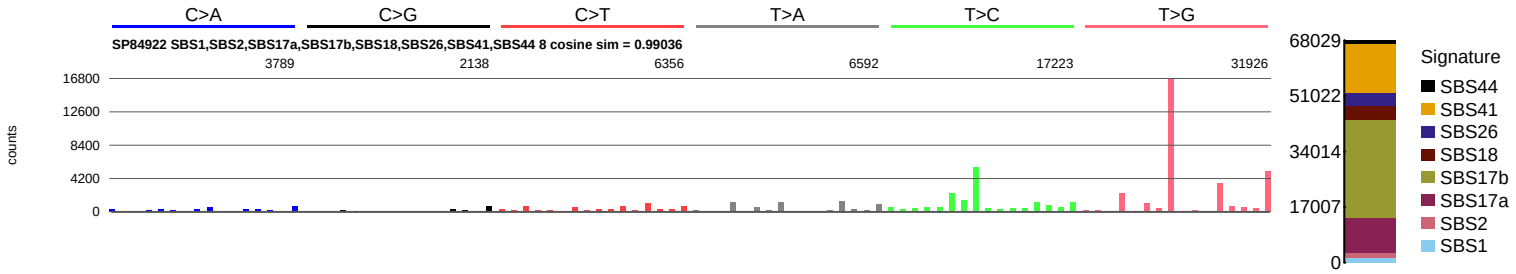

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.99038
3549
2882
6490
6780
16304
32027
16756
12567
8378
4189
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2

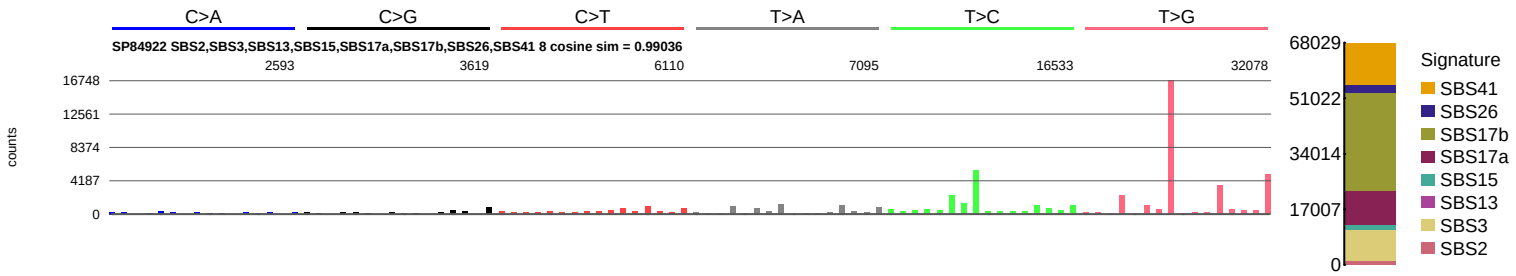

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99037
3546
2956
6476
6761
16278
32011
16756
12567
8378
4189
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

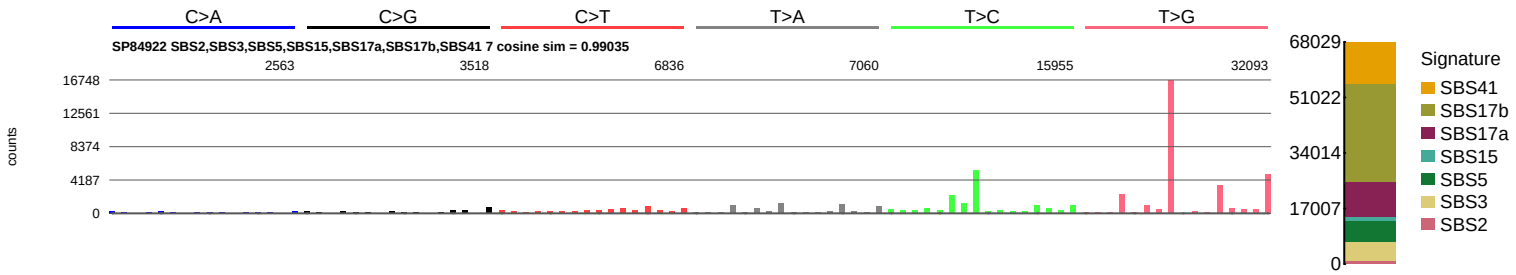

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99037
3766
2415
6133
6514
17275
31929
16804
12603
8402
4201
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

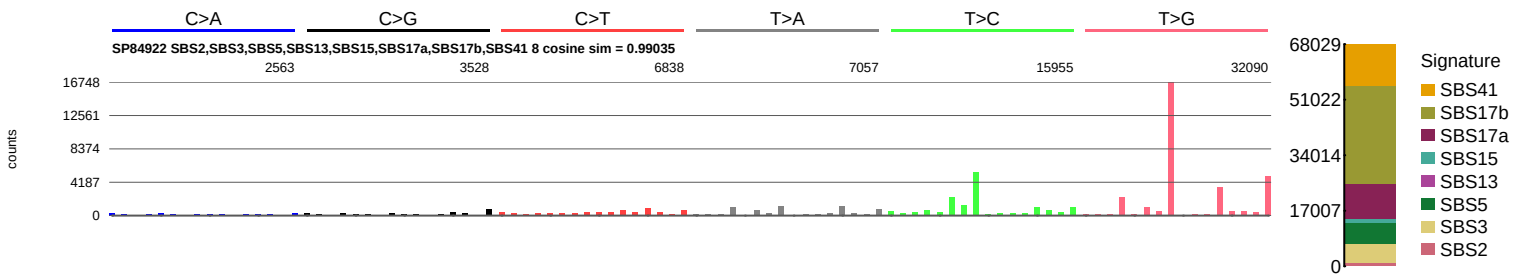

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.99037
2592
3612
6111
7097
16534
32079
counts
16748
12561
8374
4187
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2

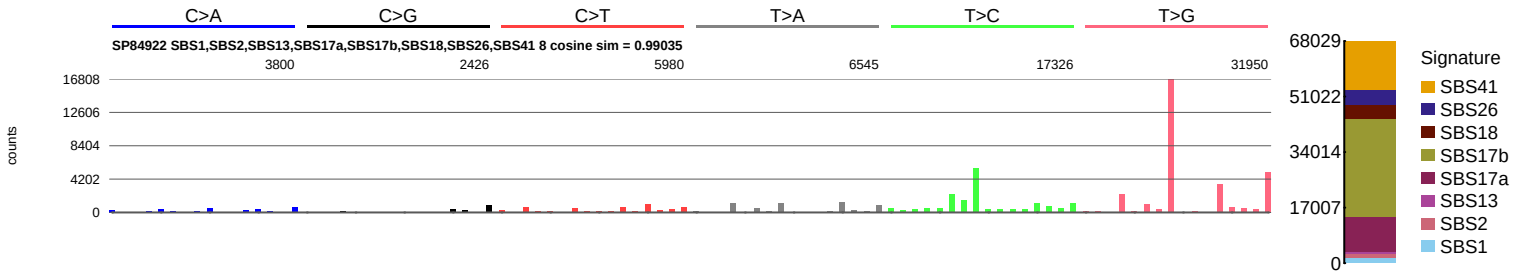

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.99036
3785
2138
6361
6591
17228
31928
counts
16800
12600
8400
4200
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

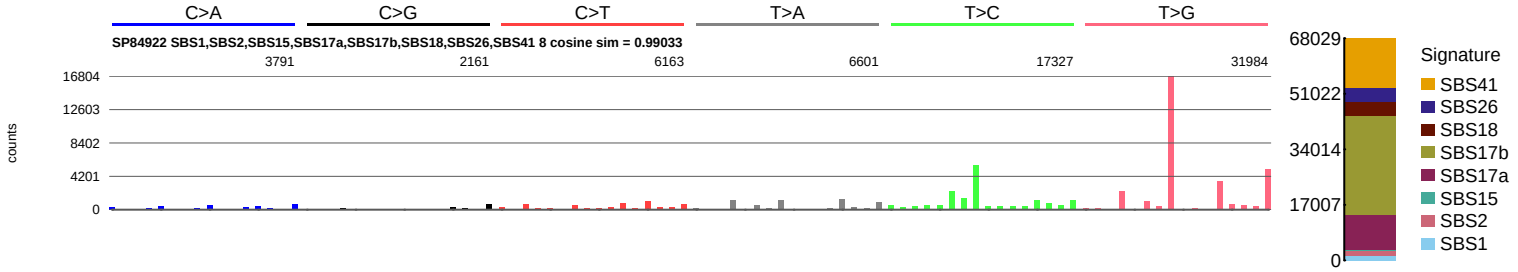

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.99036
3789
2138
6356
6592
17223
31926
counts
16800
12600
8400
4200
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

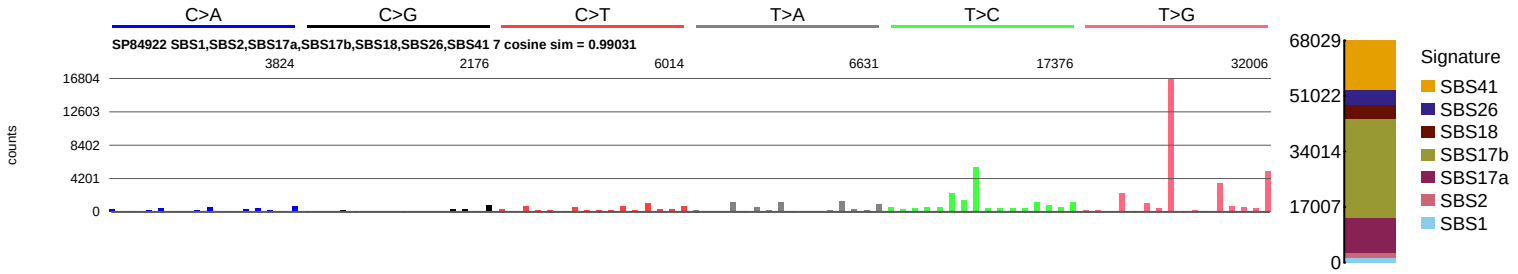

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.99036
2593
3619
6110
7095
16533
32078
counts
16748
12561
8374
4187
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

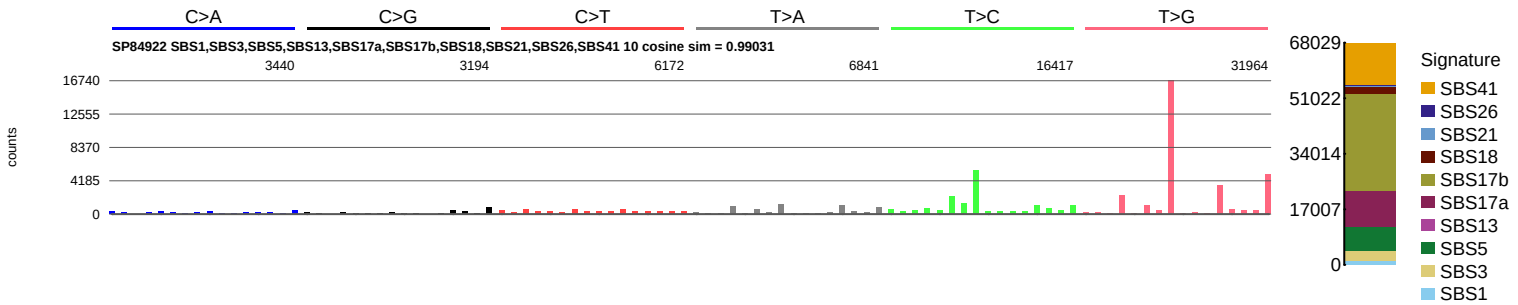

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.99035
2563
3518
6836
7060
15955
32093
counts
16748
12561
8374
4187
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

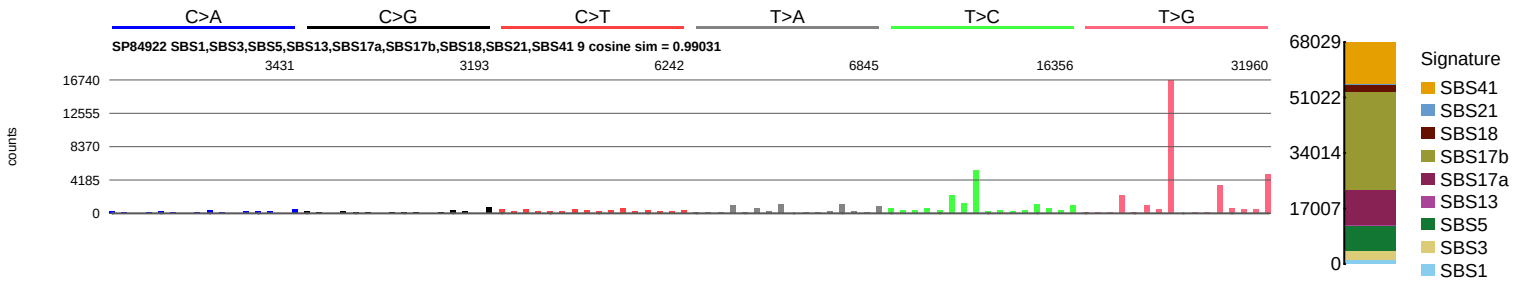

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.99035
2563
3528
6838
7057
15955
32090
counts
16748
12561
8374
4187
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

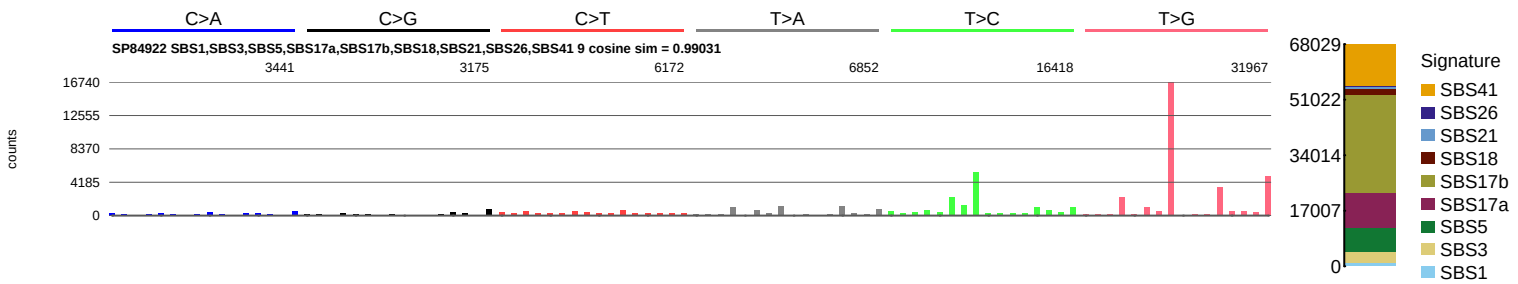

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99035
3800
2426
5980
6545
17326
31950
counts
16808
12606
8404
4202
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

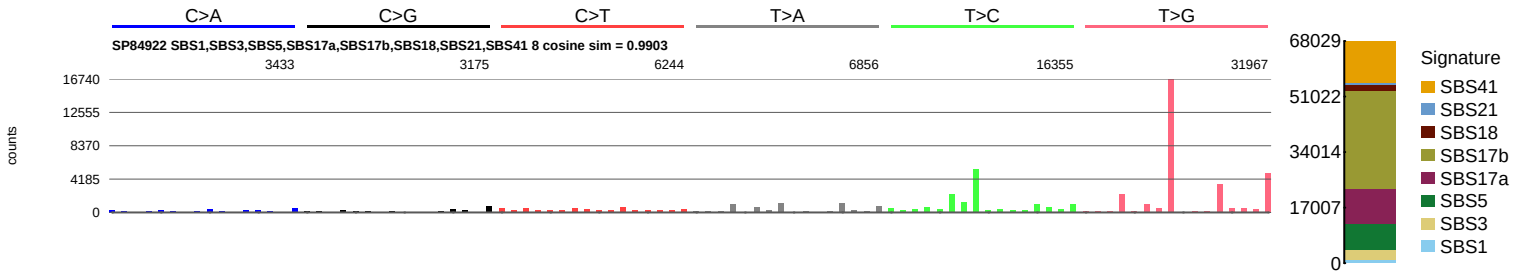

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99033
3791
2161
6163
6601
17327
31984
counts
16804
12603
8402
4201
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

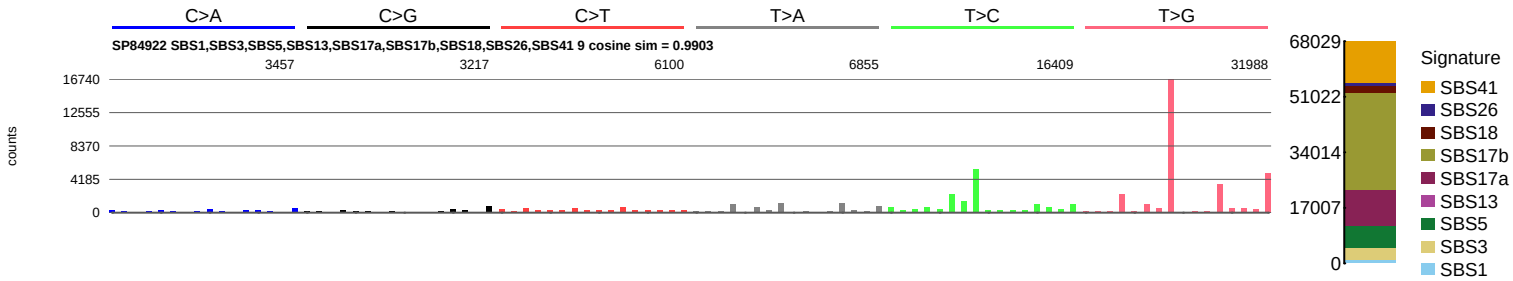

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.99031
3824
2176
6014
6631
17376
32006
counts
16804
12603
8402
4201
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

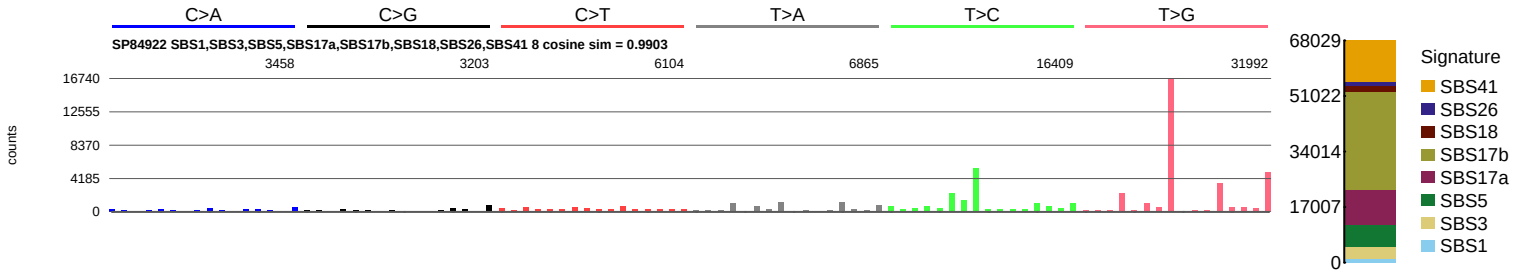

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.99031
3440
3194
6172
6841
16417
31964
counts
16740
12555
8370
4185
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

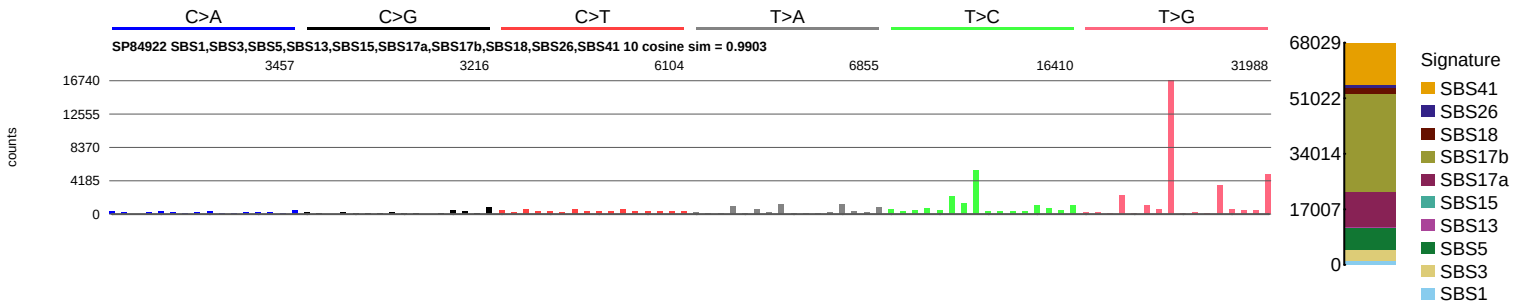

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99031
3431
3193
6242
6845
16356
31960
counts
16740
12555
8370
4185
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.99031
3441
3175
6172
6852
16418
31967
counts
16740
12555
8370
4185
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.9903
3433
3175
6244
6856
16355
31967
counts
16740
12555
8370
4185
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.9903
3457
3217
6100
6855
16409
31988
counts
16740
12555
8370
4185
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.9903
3458
3203
6104
6865
16409
31992
counts
16740
12555
8370
4185
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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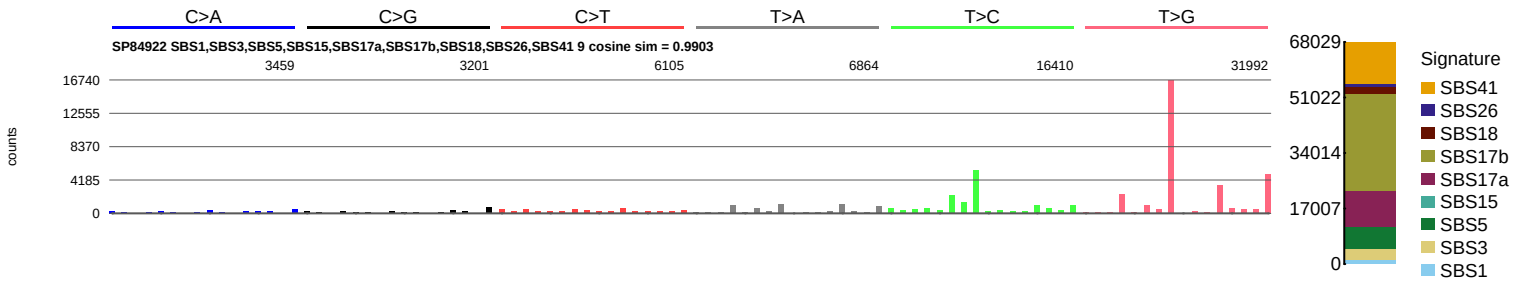
SP84922 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.9903
3457
3216
6104
6855
16410
31988
counts
16740
12555
8370
4185
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1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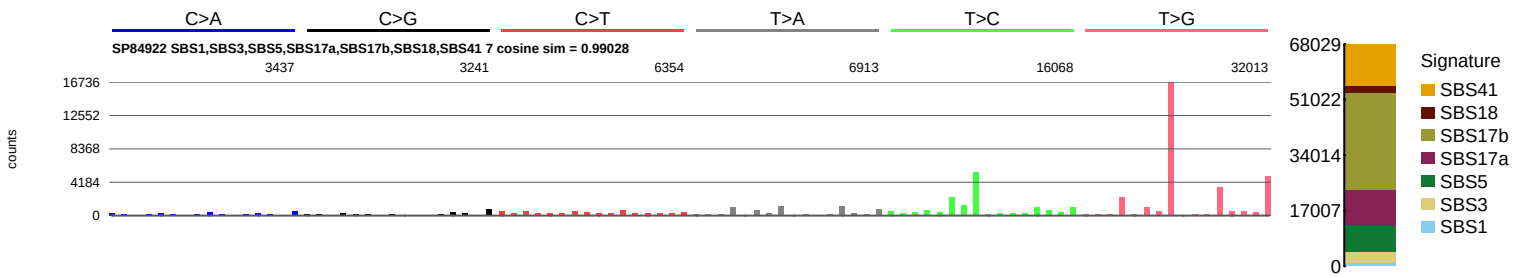
SP84922 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.9903
3459
3201
6105
6864
16410
31992
counts
16740
12555
8370
4185
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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C>G
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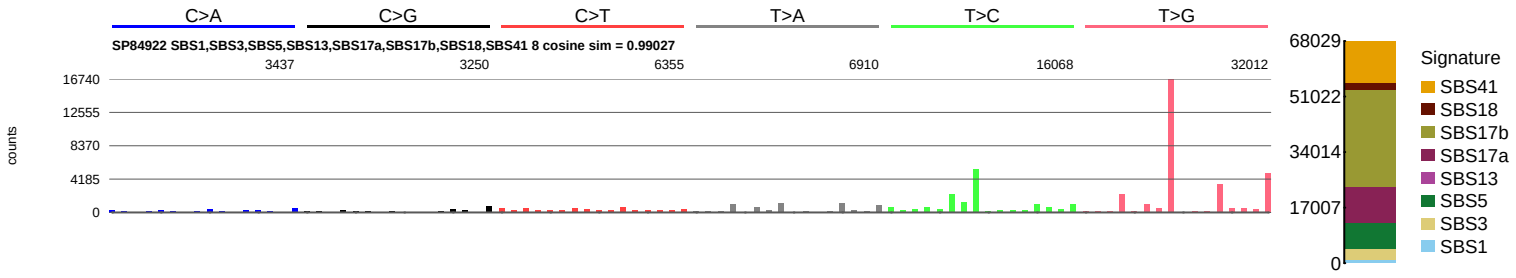
SP84922 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99028
3437
3241
6354
6913
16068
32013
counts
16736
12552
8368
4184
0
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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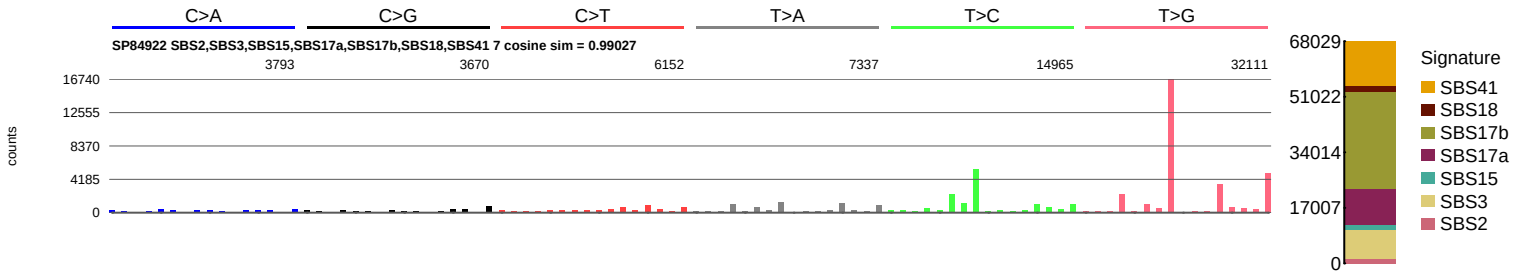
SP84922 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99027
3437
3250
6355
6910
16068
32012
counts
16740
12555
8370
4185
0
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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C>G
C>T
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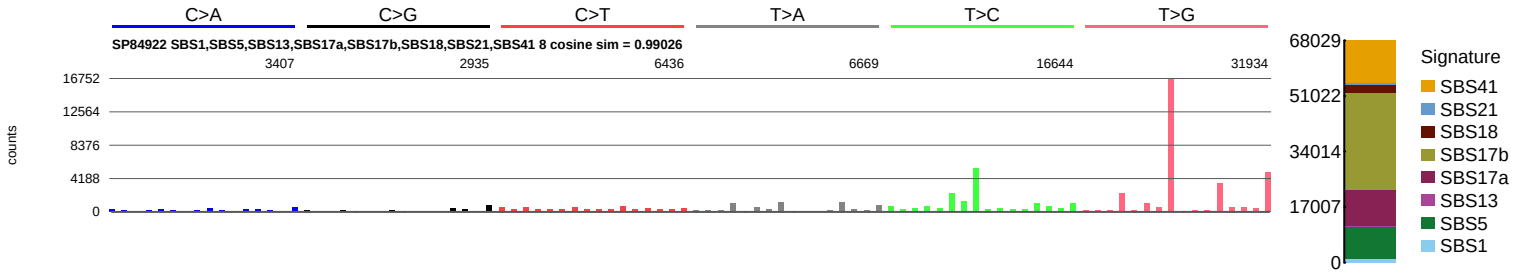
SP84922 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99027
3793
3670
6152
7337
14965
32111
counts
16740
12555
8370
4185
0
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2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C>G
C>T
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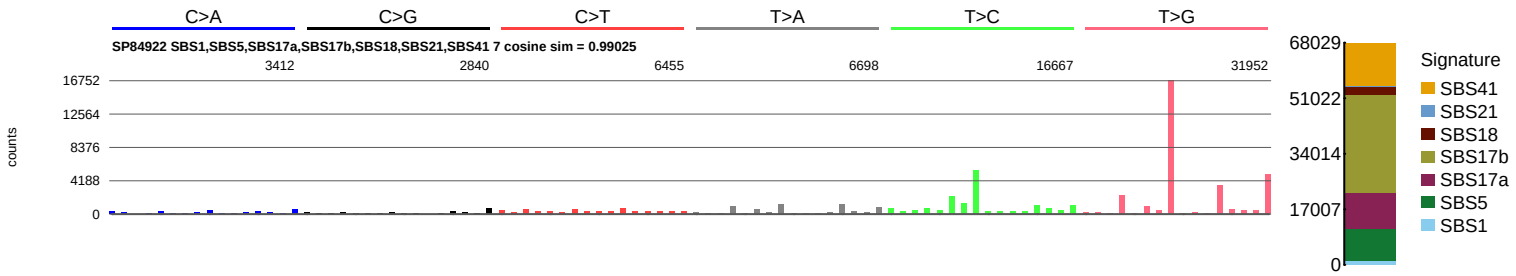
T>G
SP84922 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99026
3407
2935
6436
6669
16644
31934
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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C>G
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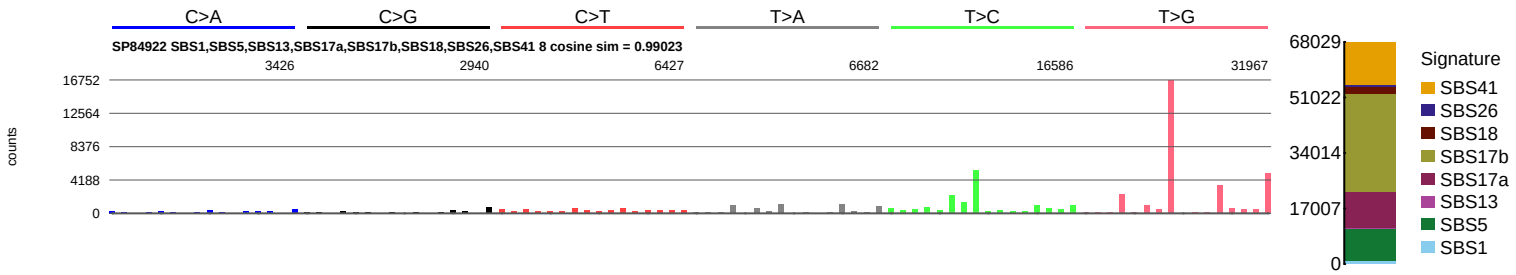
T>C
T>G
SP84922 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.99025
3412
2840
6455
6698
16667
31952
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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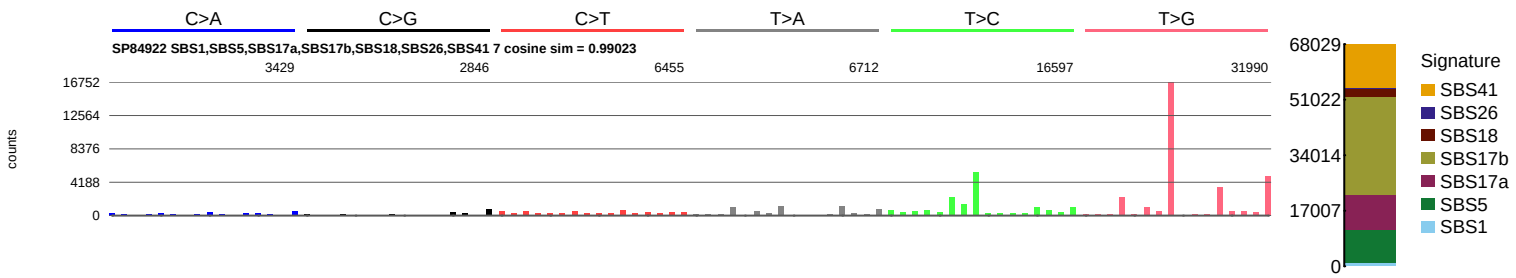
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99023
3426
2940
6427
6682
16586
31967
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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C>G
C>T
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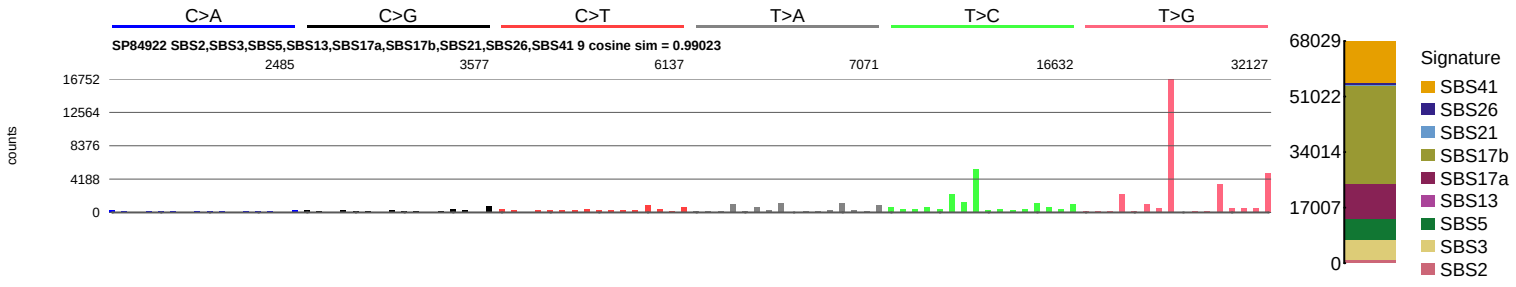
T>C
T>G
SP84922 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.99023
3429
2846
6455
6712
16597
31990
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS1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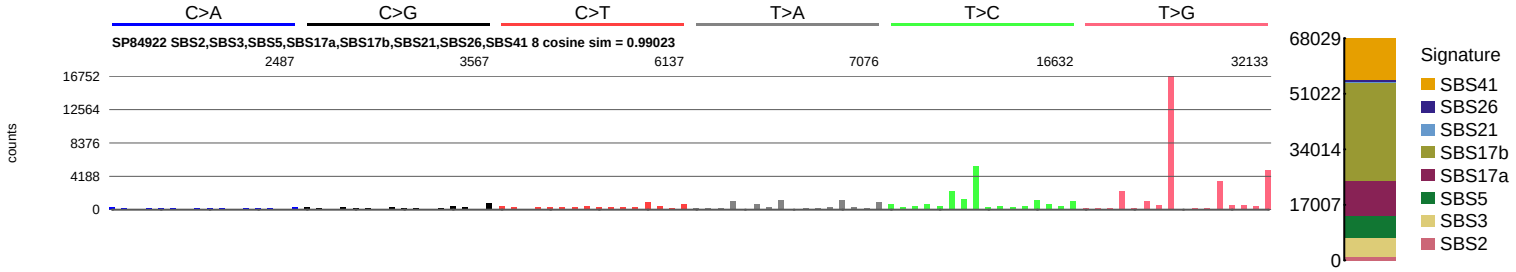
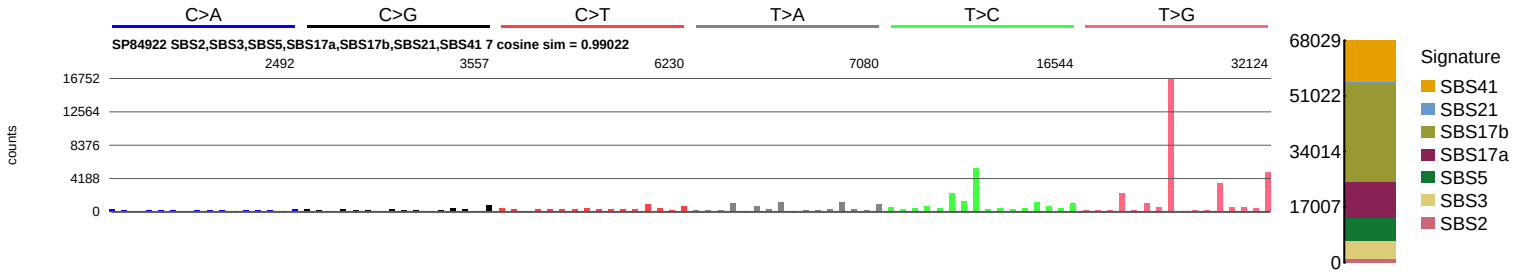
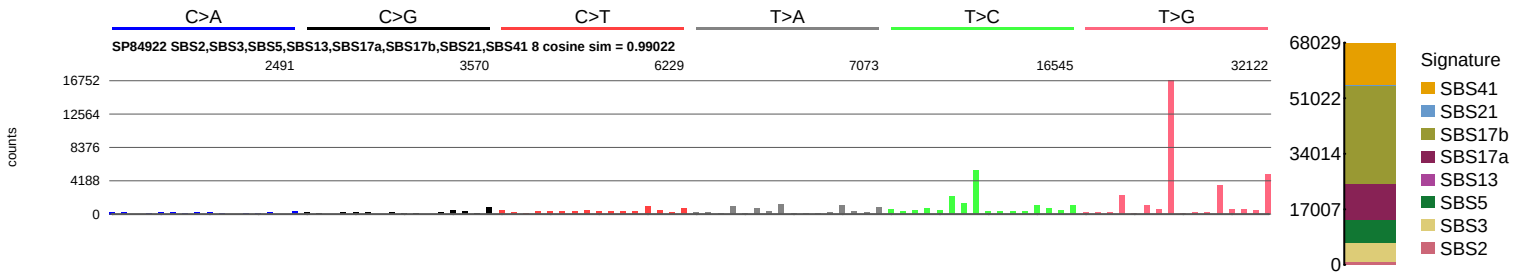
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41 9 cosine sim = 0.99023
2485
3577
6137
7071
16632
32127
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS26,SBS41 8 cosine sim = 0.99023
2487
3567
6137
7076
16632
32133
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.99022
2492
3557
6230
7080
16544
32124
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2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C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.99022
2491
3570
6229
7073
16545
32122
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
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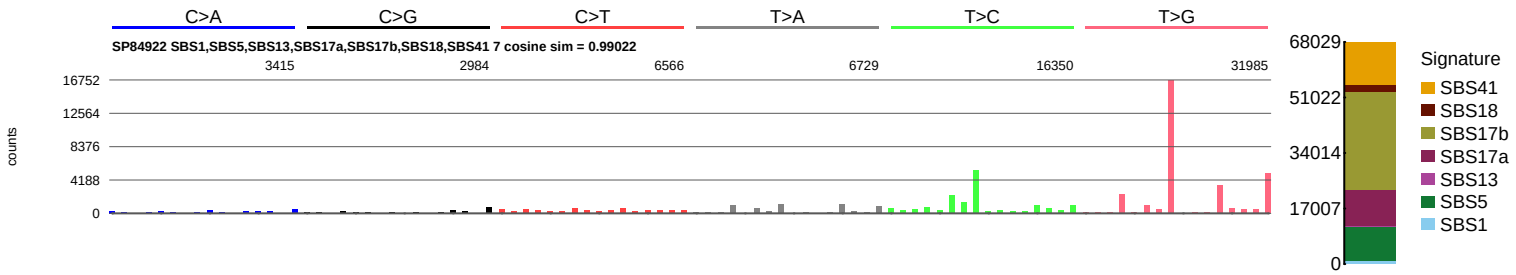
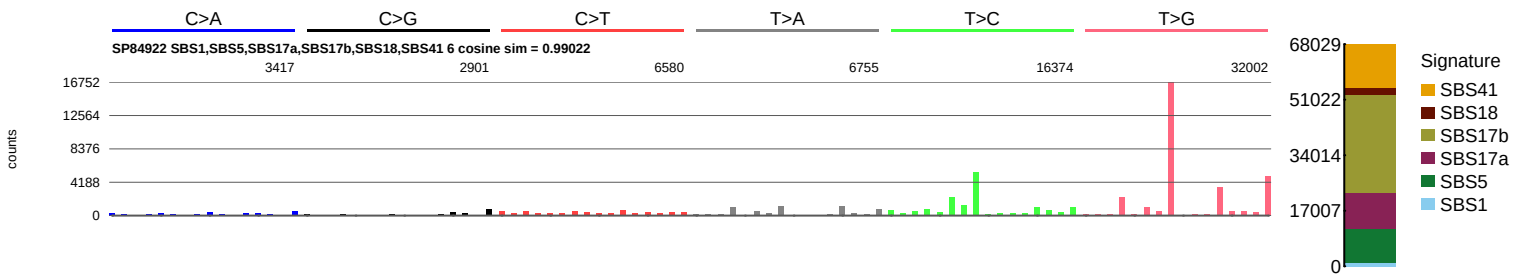
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99022
3415
2984
6566
6729
16350
31985
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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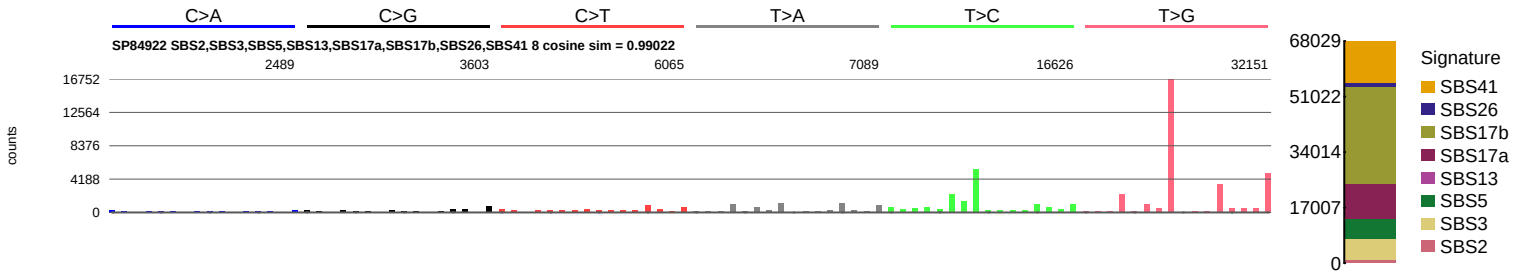
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.99022
3417
2901
6580
6755
16374
32002
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS1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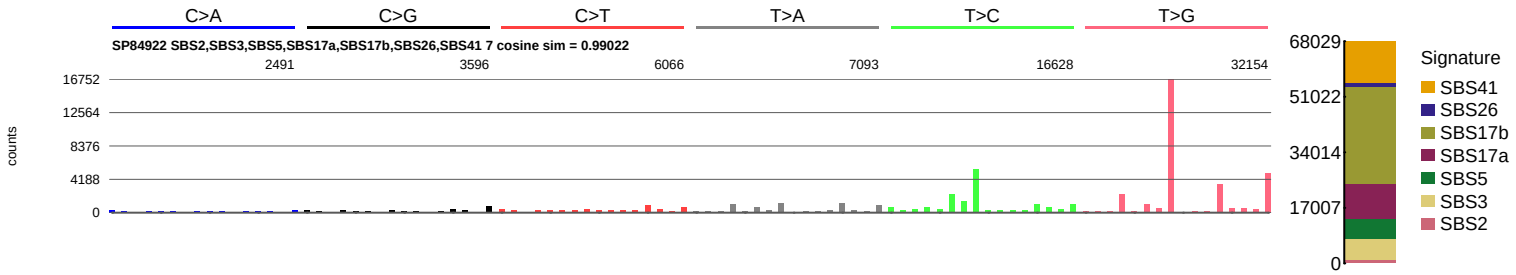
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.99022
2489
3603
6065
7089
16626
32151
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2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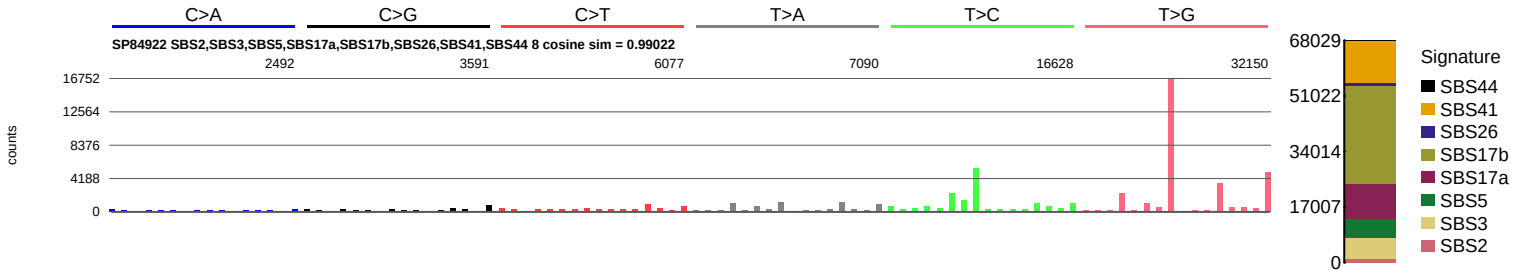
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.99022
2491
3596
6066
7093
16628
32154
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2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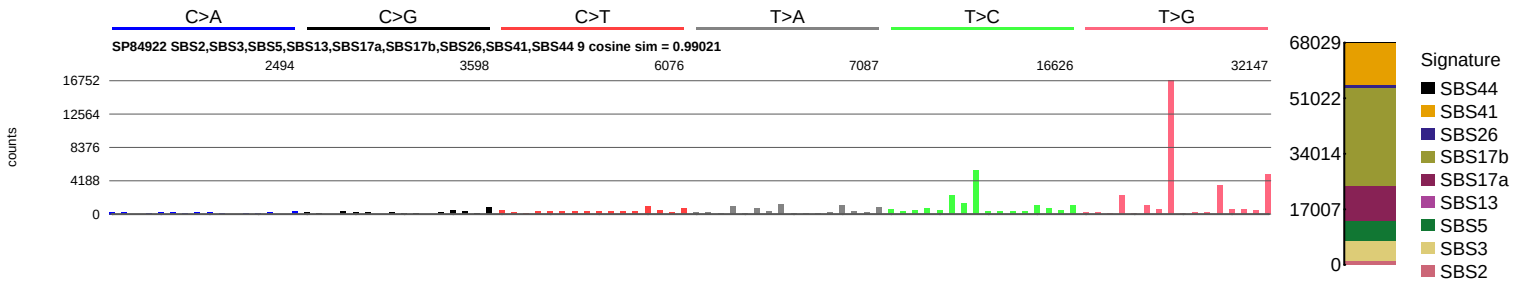
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.99022
2492
3591
6077
7090
16628
32150
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2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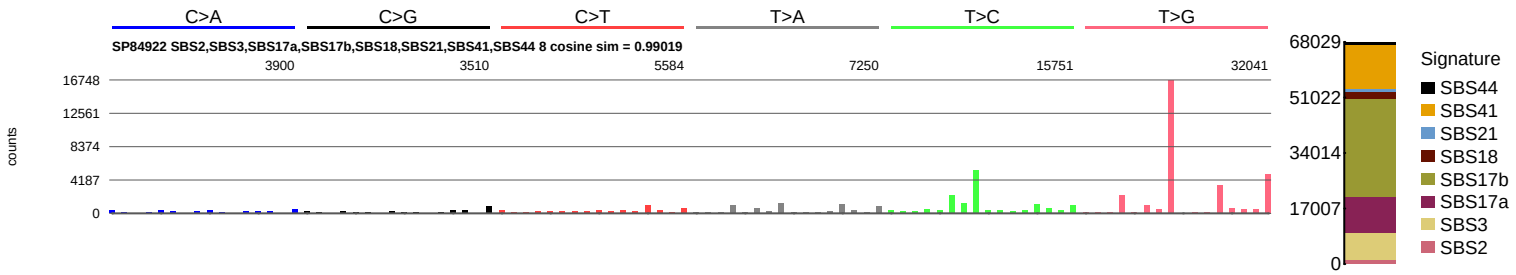
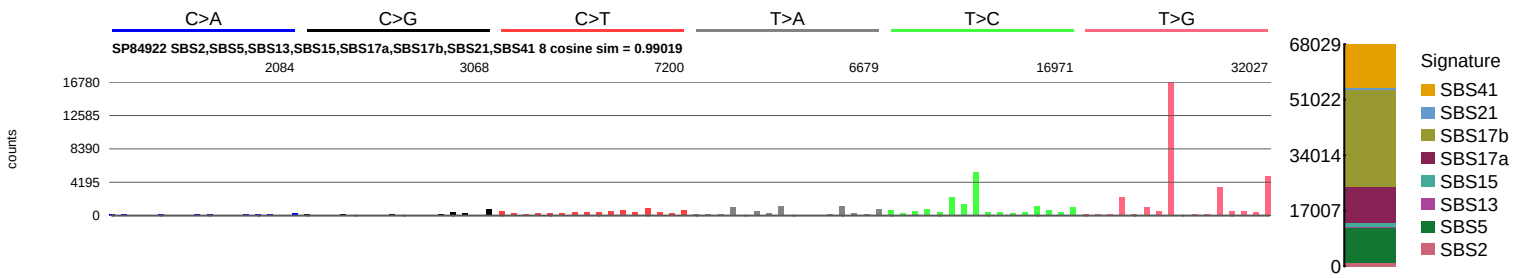
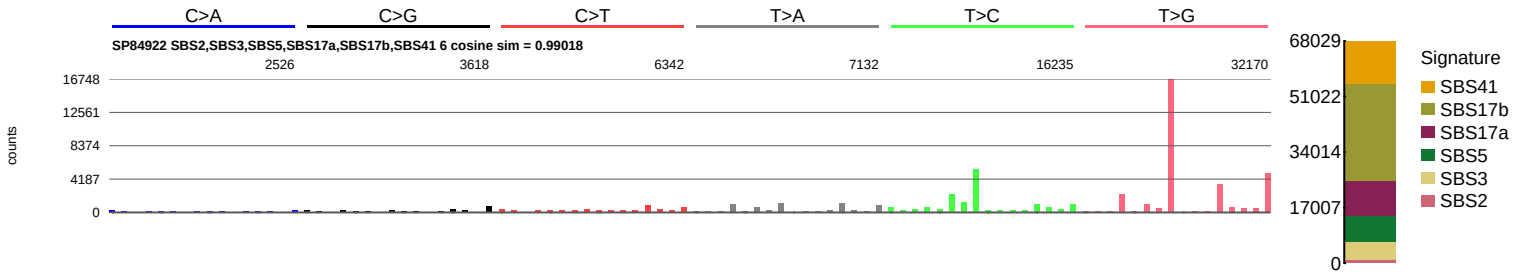
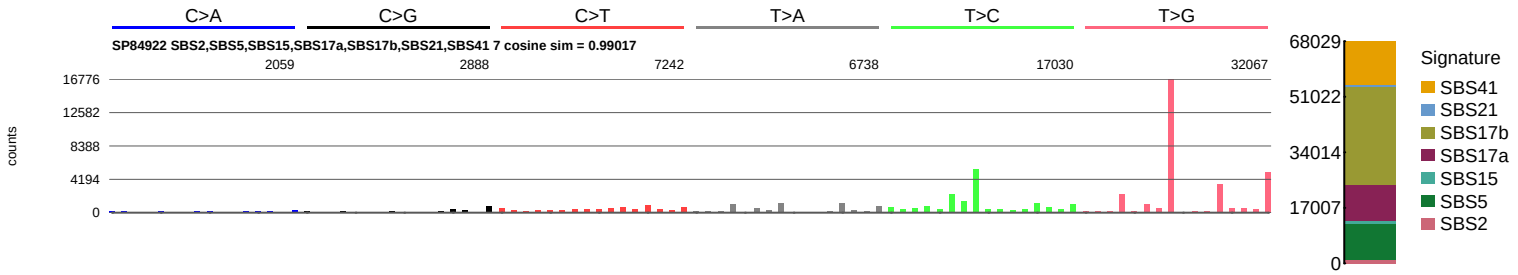
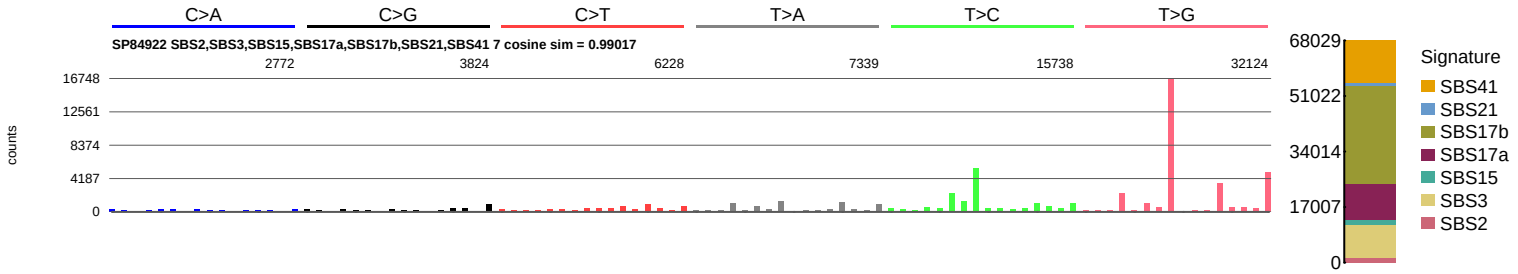
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.99021
2494
3598
6076
7087
16626
32147
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

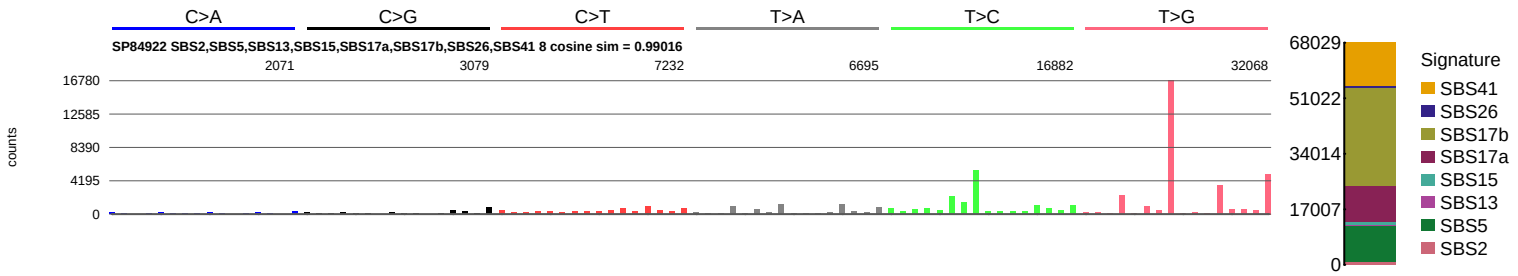

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.99019
3900
3510
5584
7250
15751
32041
counts
16748
12561
8374
4187
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.99019
2084
3068
7200
6679
16971
32027
counts
16780
12585
8390
4195
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.99018
2526
3618
6342
7132
16235
32170
16748
12561
8374
4187
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.99017
2059
2888
7242
6738
17030
32067
16776
12582
8388
4194
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.99017
2772
3824
6228
7339
15738
32124
16748
12561
8374
4187
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.99016
2071
3079
7232
6695
16882
32068
16780
12585
8390
4195
0
counts
68029
51022
34014
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SBS26
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SBS17a
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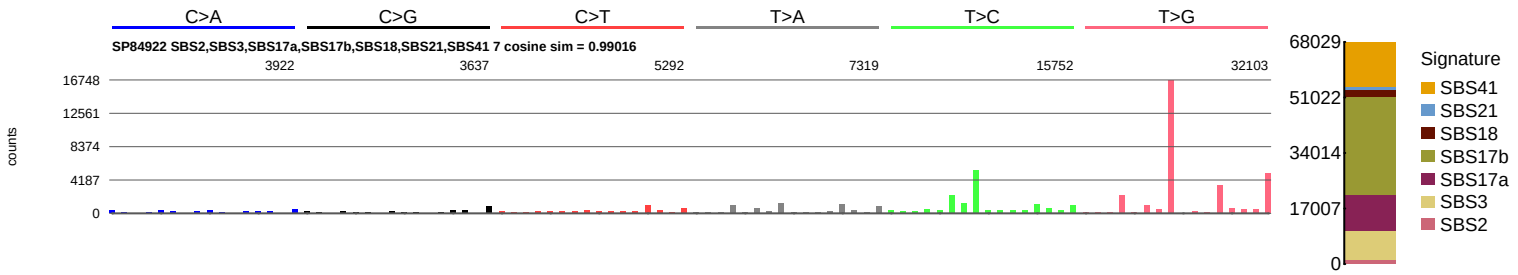
SBS5
SBS2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T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.99016
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3637
5292
7319
15752
32103
16748
12561
8374
4187
0
counts
68029
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SBS21
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SBS17b
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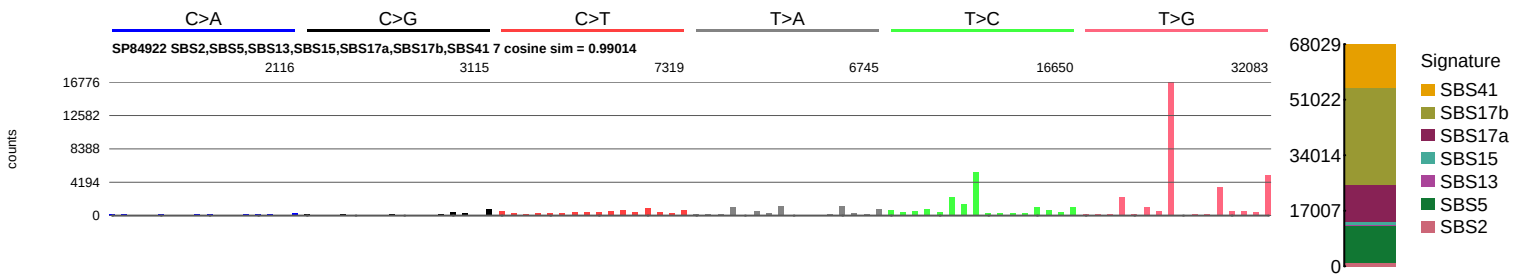
SBS3
SBS2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C>G
C>T
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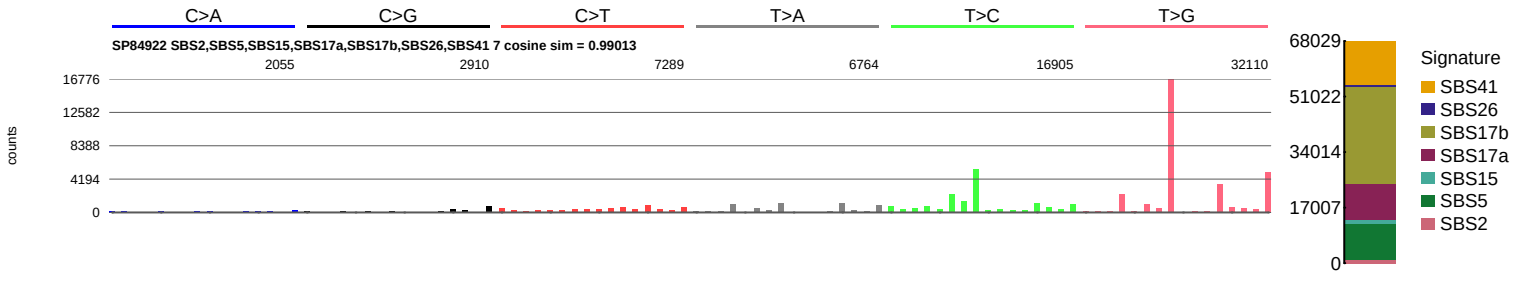
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.99014
2116
3115
7319
6745
16650
32083
16776
12582
8388
4194
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2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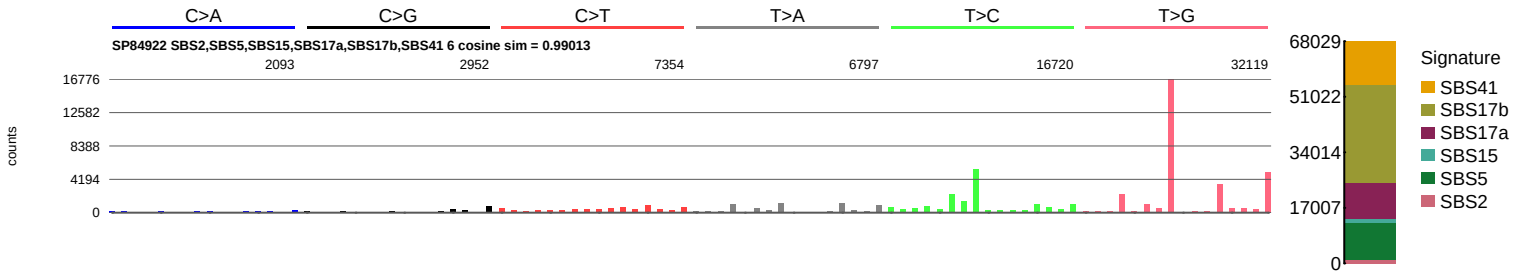
SP84922 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.99013
2055
2910
7289
6764
16905
32110
counts
16776
12582
8388
4194
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS2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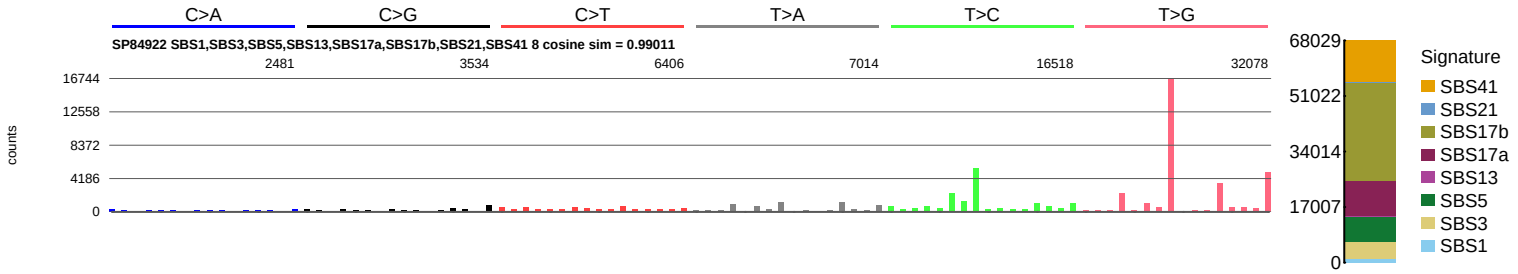
T>C
T>G
SP84922 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.99013
2093
2952
7354
6797
16720
32119
counts
16776
12582
8388
4194
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS2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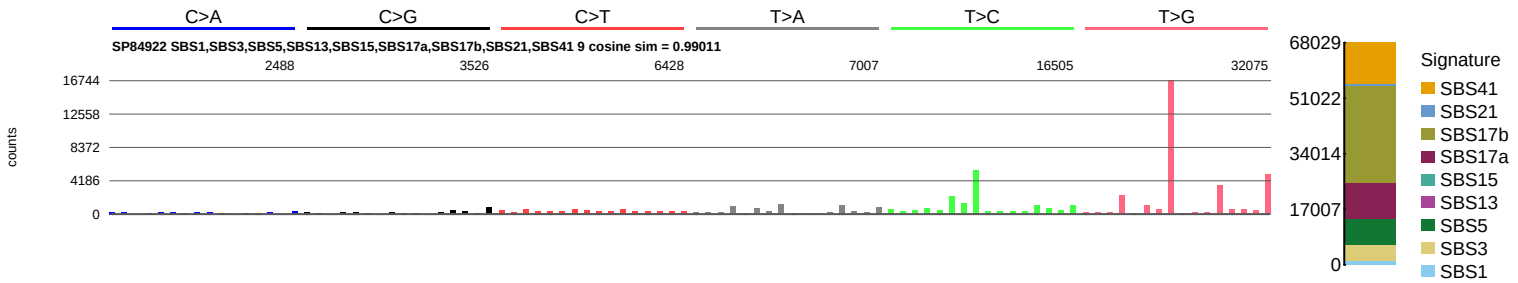
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.99011
2481
3534
6406
7014
16518
32078
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.99011
2488
3526
6428
7007
16505
32075
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
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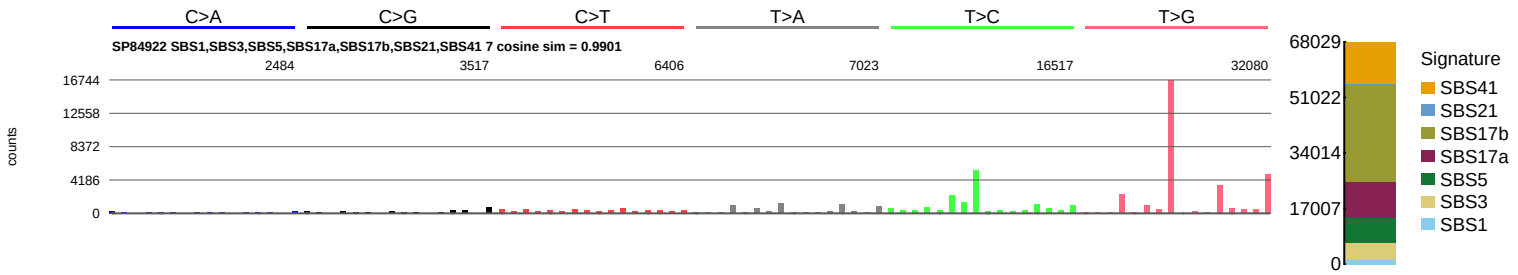
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1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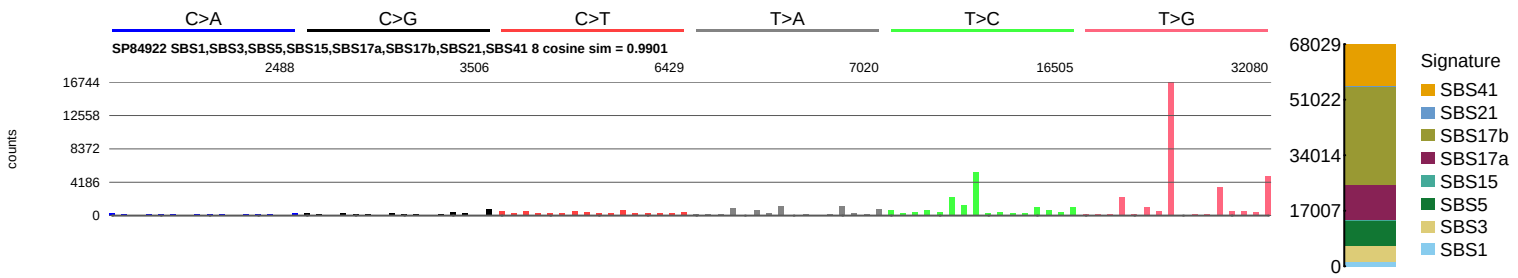
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.9901
2484
3517
6406
7023
16517
32080
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.9901
2488
3506
6429
7020
16505
32080
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
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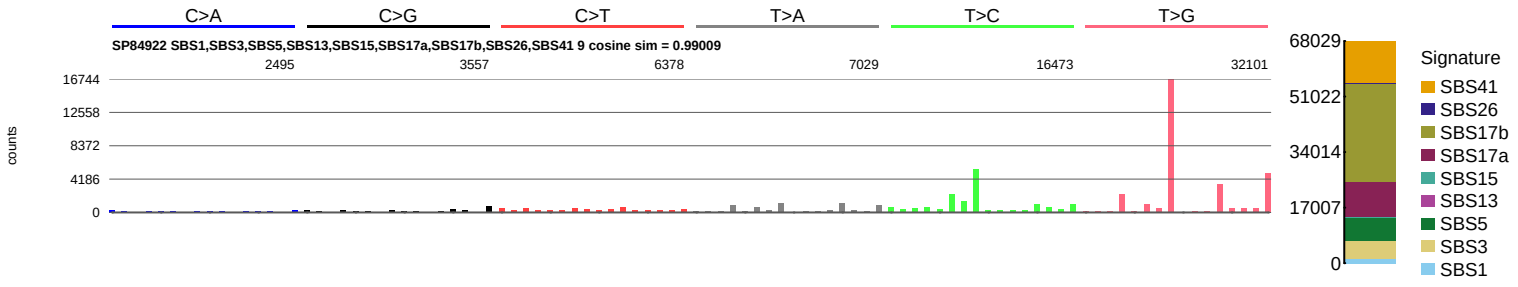
SBS3
SBS1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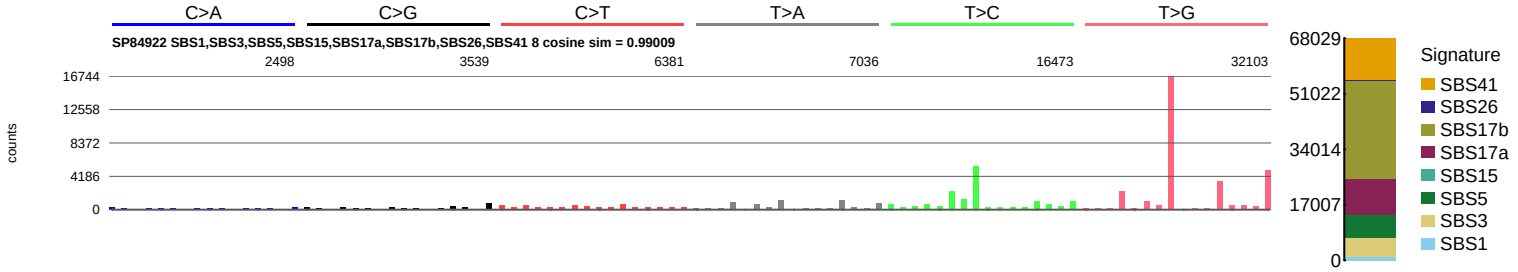
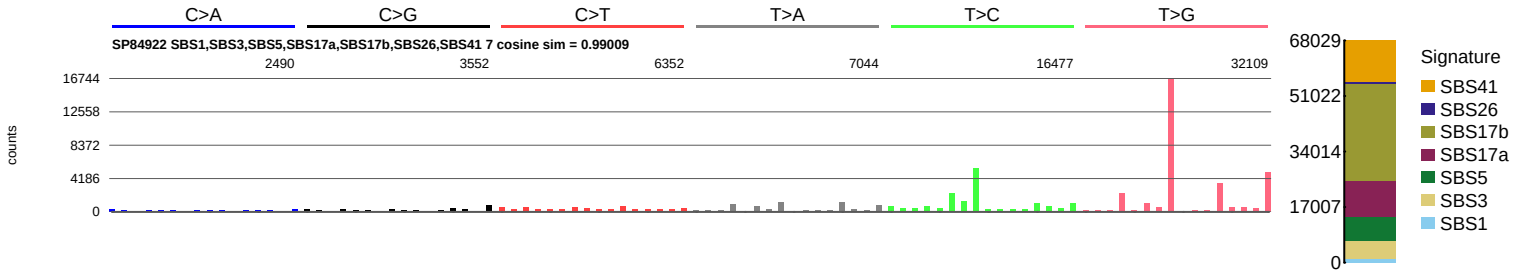
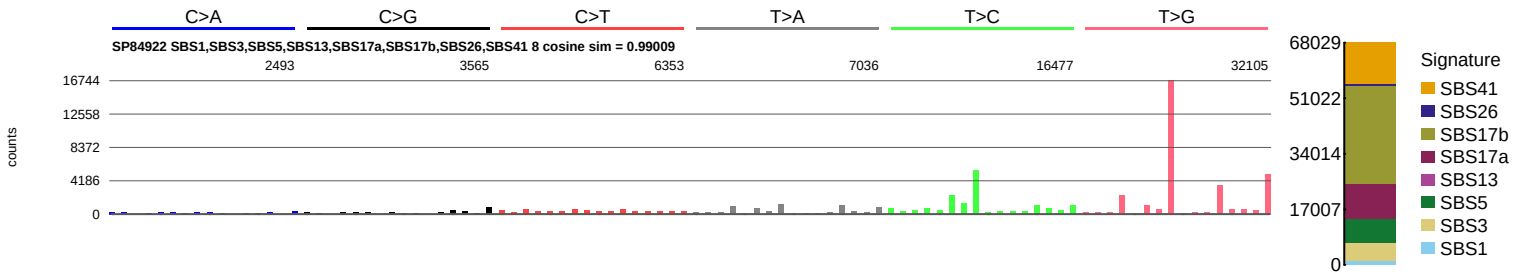
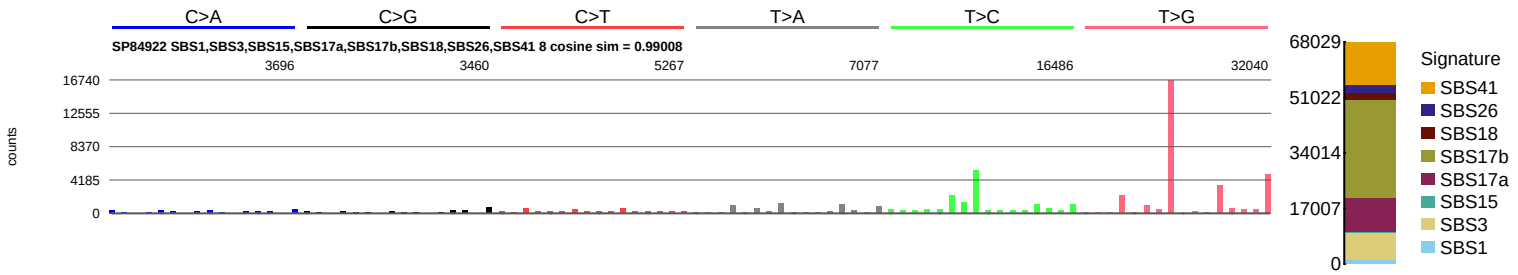
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 9 cosine sim = 0.99009
2495
3557
6378
7029
16473
32101
16744
12558
8372
4186
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.99009
2498
3539
6381
7036
16473
32103
16744
12558
8372
4186
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.99009
2490
3552
6352
7044
16477
32109
16744
12558
8372
4186
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.99009
2493
3565
6353
7036
16477
32105
16744
12558
8372
4186
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99008
3696
3460
5267
7077
16486
32040
16740
12555
8370
4185
0
counts
68029
51022
34014
17007
0
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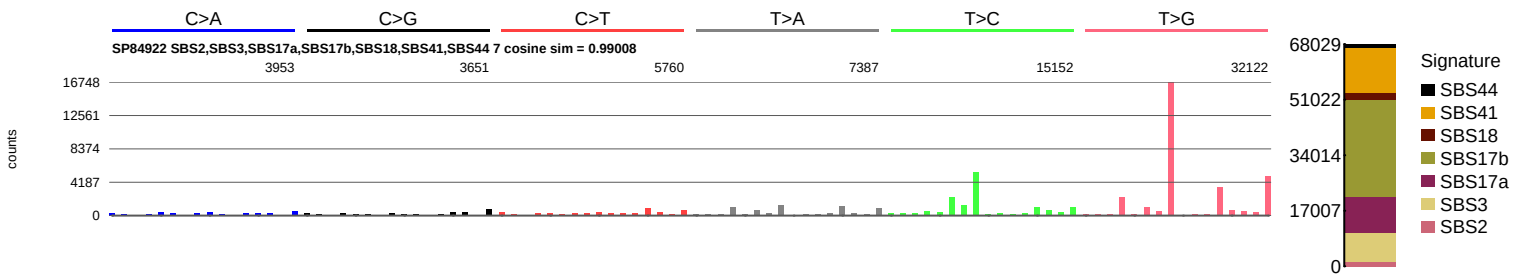
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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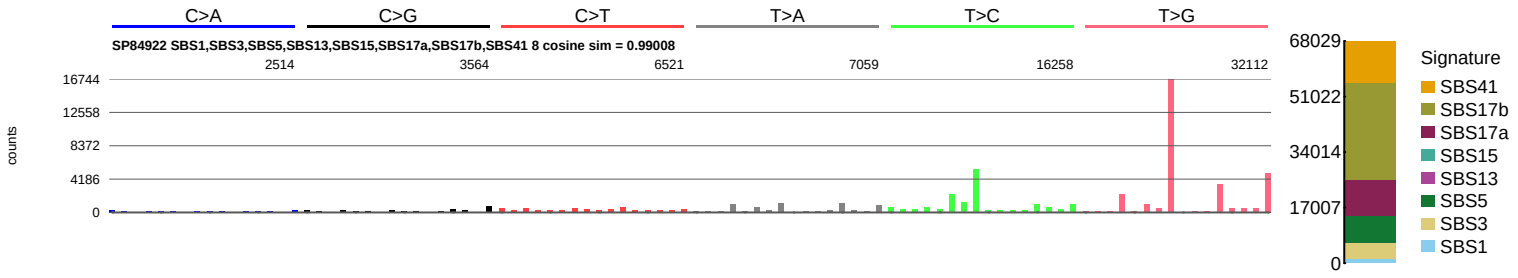
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.99008
3953
3651
5760
7387
15152
32122
16748
12561
8374
4187
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2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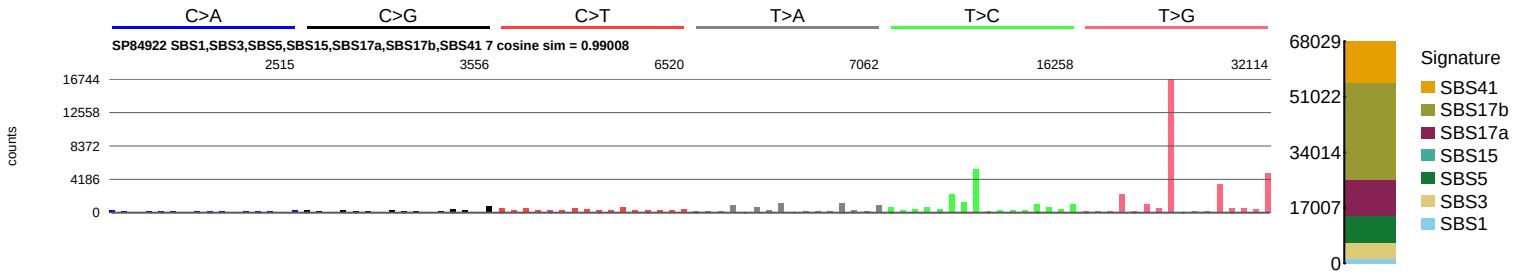
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.99008
2514
3564
6521
7059
16258
32112
16744
12558
8372
4186
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.99008
2515
3556
6520
7062
16258
32114
16744
12558
8372
4186
0
counts
68029
51022
34014
17007
0
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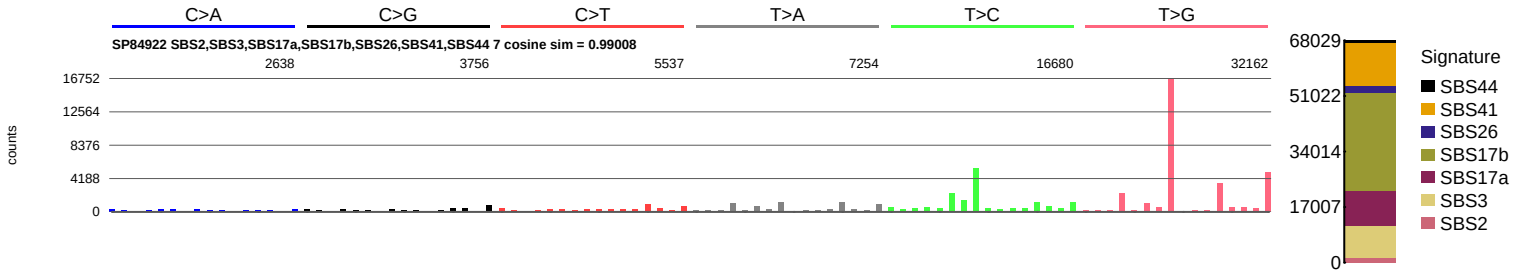
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.99008
2638
3756
5537
7254
16680
32162
16752
12564
8376
4188
0
counts
68029
51022
34014
17007
0
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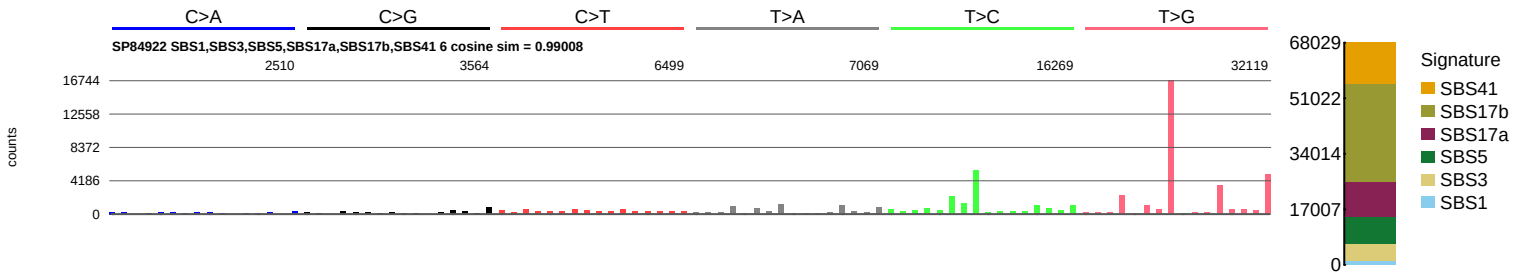
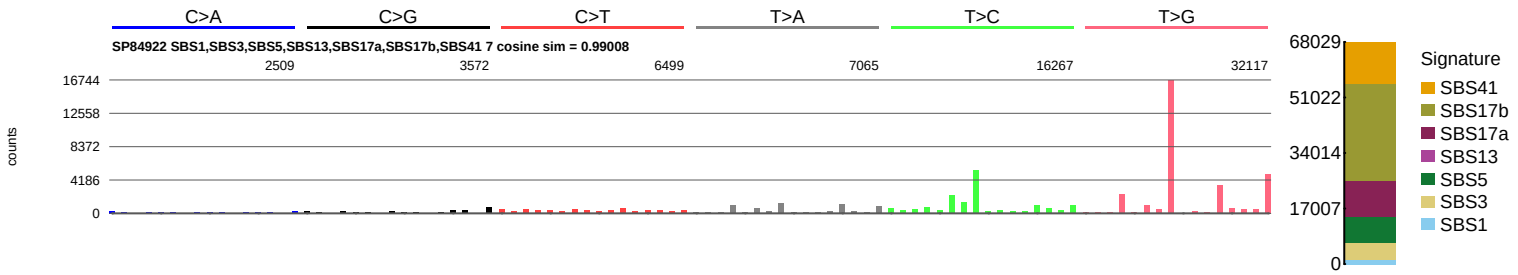
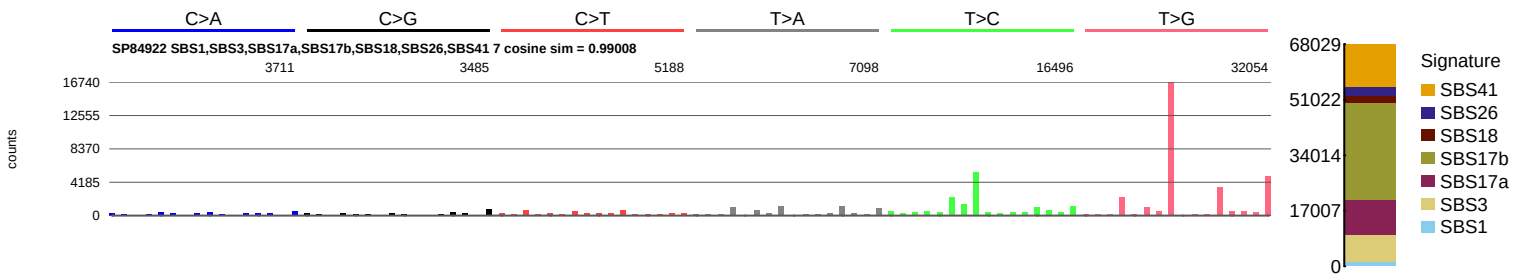
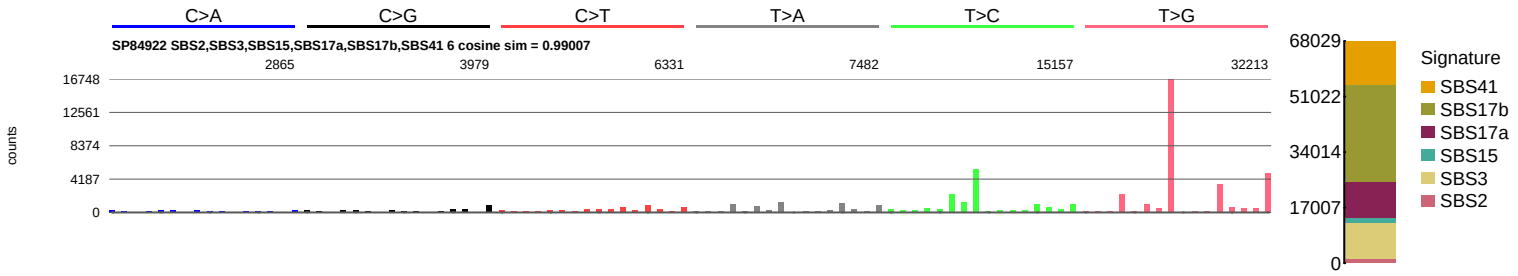
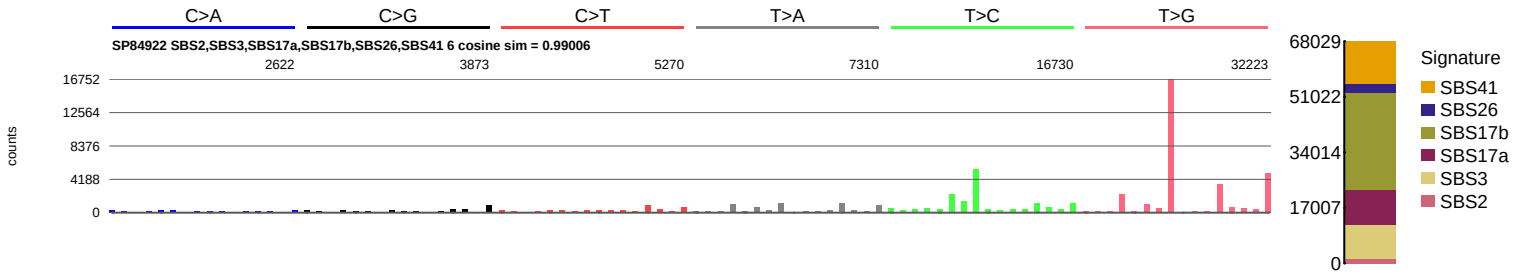
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2

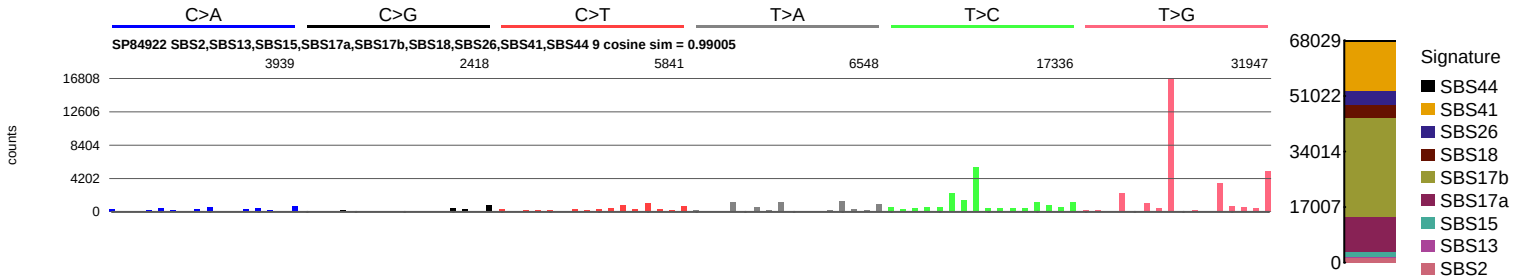

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.99008
2510
3564
6499
7069
16269
32119
16744
12558
8372
4186
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS41 7 cosine sim = 0.99008
2509
3572
6499
7065
16267
32117
16744
12558
8372
4186
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.99008
3711
3485
5188
7098
16496
32054
16740
12555
8370
4185
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.99007
2865
3979
6331
7482
15157
32213
16748
12561
8374
4187
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.99006
2622
3873
5270
7310
16730
32223
16752
12564
8376
4188
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.99005
3939
2418
5841
6548
17336
31947
16808
12606
8404
4202
0
counts
68029
51022
34014
17007
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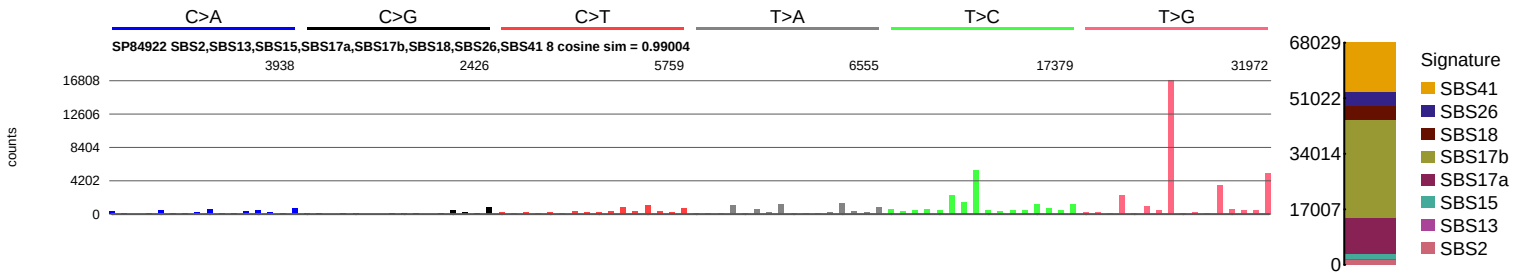
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2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C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99004
3938
2426
5759
6555
17379
31972
16808
12606
8404
4202
0
counts
68029
51022
34014
17007
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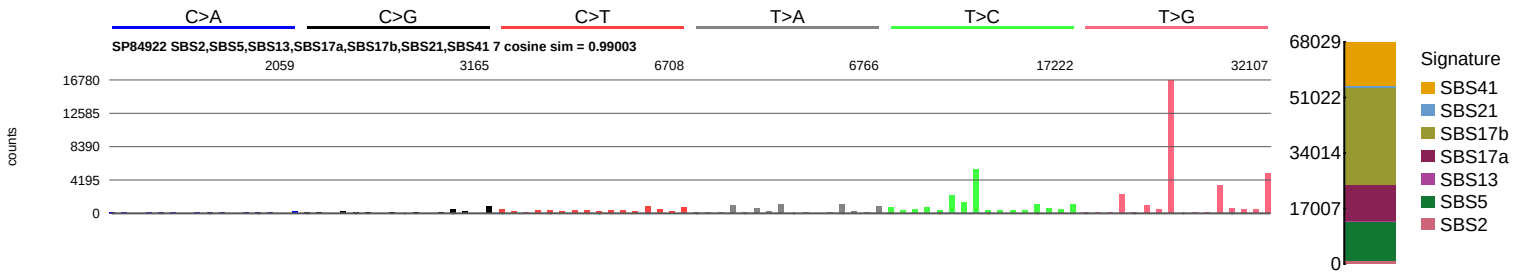
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.99003
2059
3165
6708
6766
17222
32107
16780
12585
8390
4195
0
counts
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SBS41
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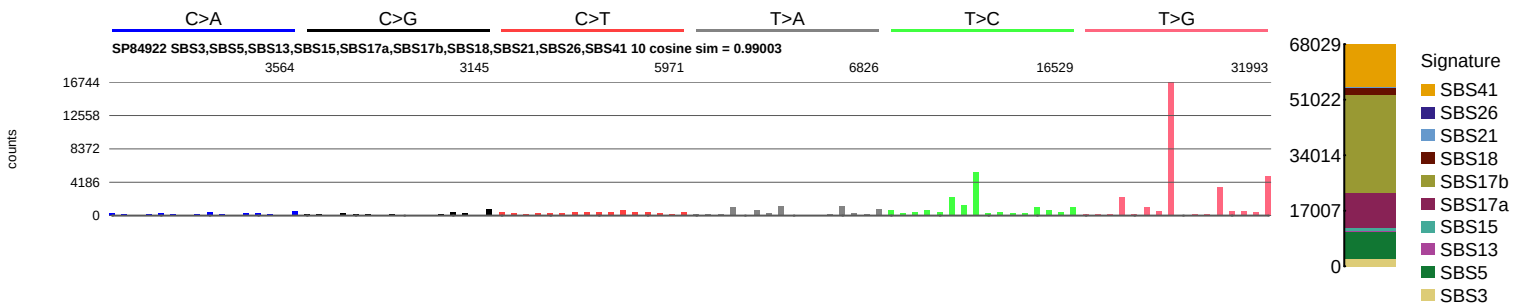
SBS2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SP84922 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.99003
3564
3145
5971
6826
16529
31993
16744
12558
8372
4186
0
counts
68029
51022
34014
17007
0
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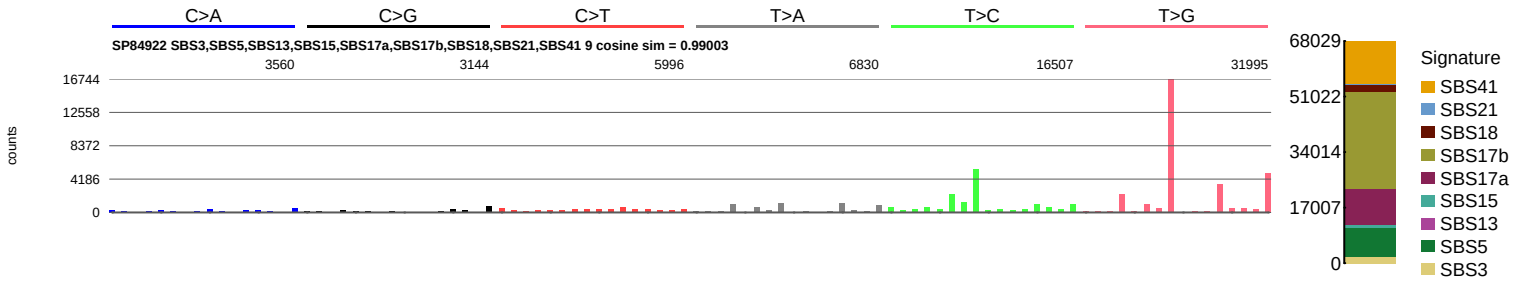
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3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SP84922 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99003
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3144
5996
6830
16507
31995
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
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SBS21
SBS18
SBS17b
SBS17a
SBS15
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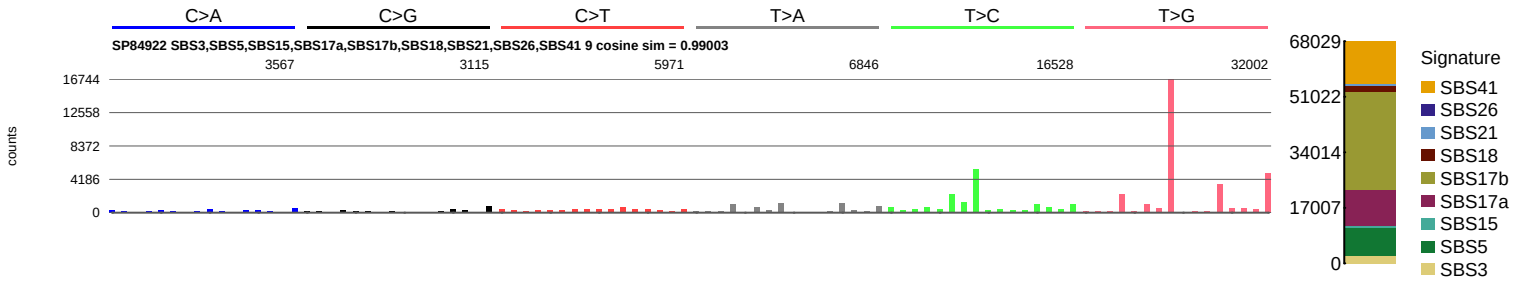
SBS5
SBS3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SP84922 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.99003
3567
3115
5971
6846
16528
32002
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
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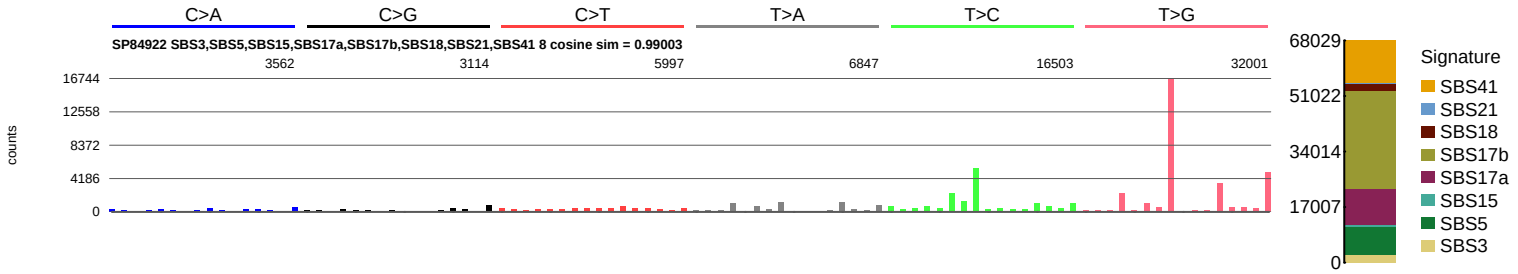
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3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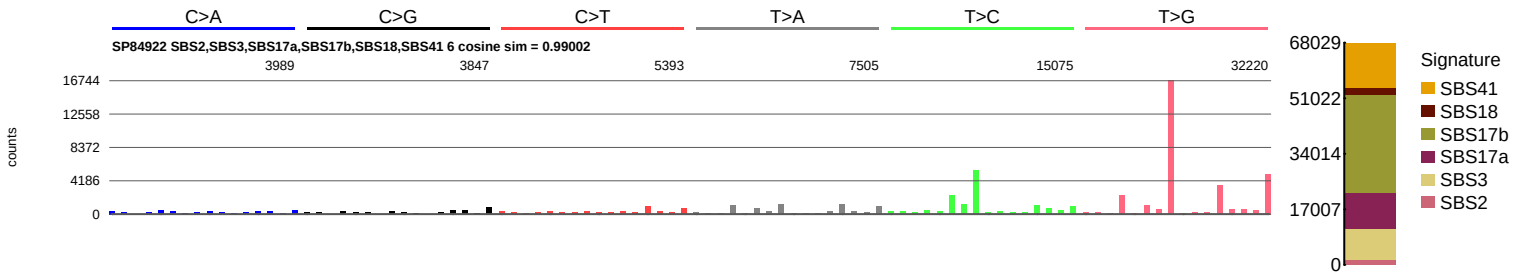
SP84922 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99003
3562
3114
5997
6847
16503
32001
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3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C>G
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SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.99002
3989
3847
5393
7505
15075
32220
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
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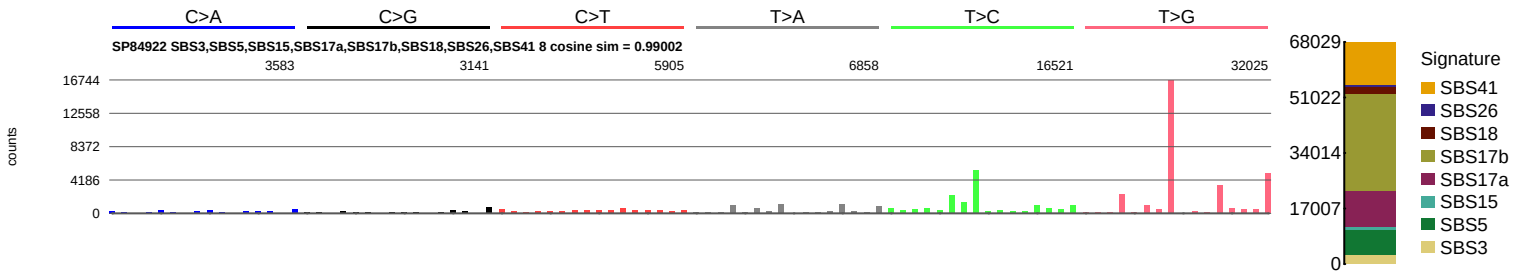
SBS17b
SBS17a
SBS3
SBS2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C>T
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T>C
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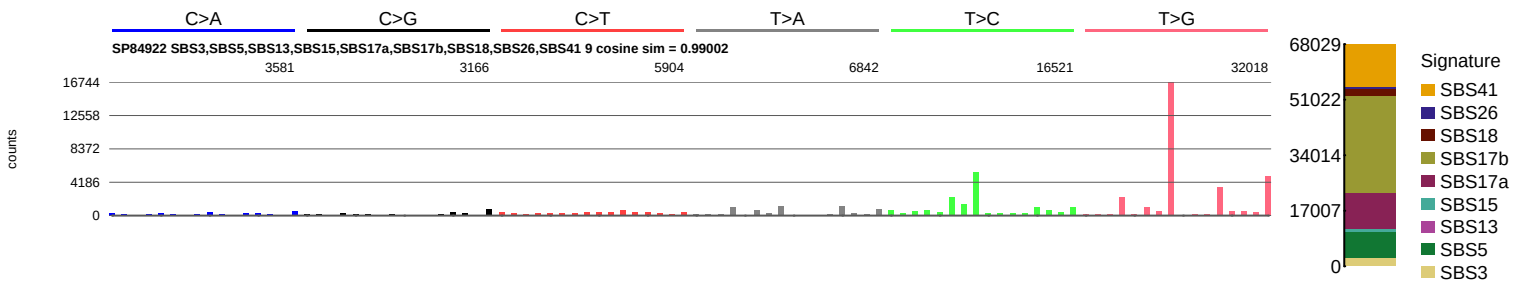
SP84922 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99002
3583
3141
5905
6858
16521
32025
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3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SP84922 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99002
3581
3166
5904
6842
16521
32018
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
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Signature
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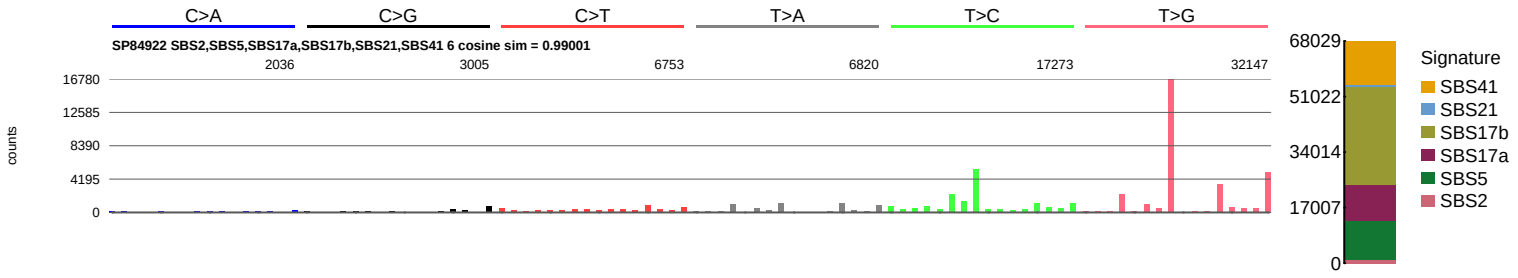
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3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C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.99001
2036
3005
6753
6820
17273
32147
counts
16780
12585
8390
4195
0
68029
51022
34014
17007
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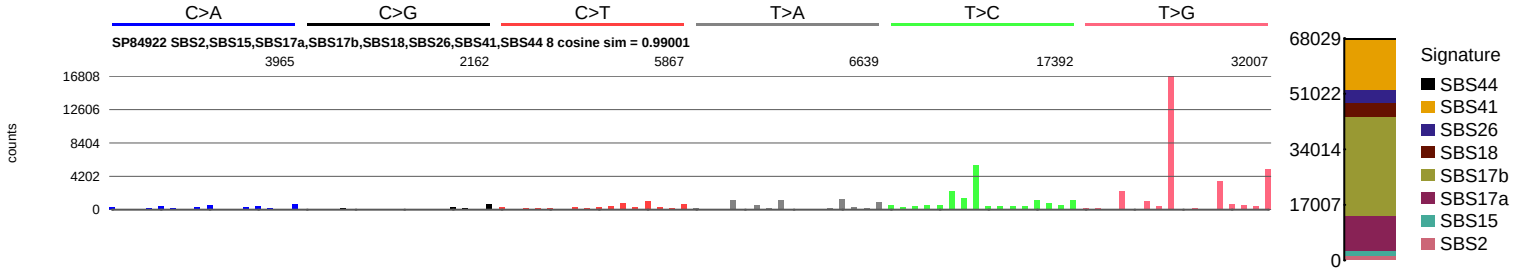
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS2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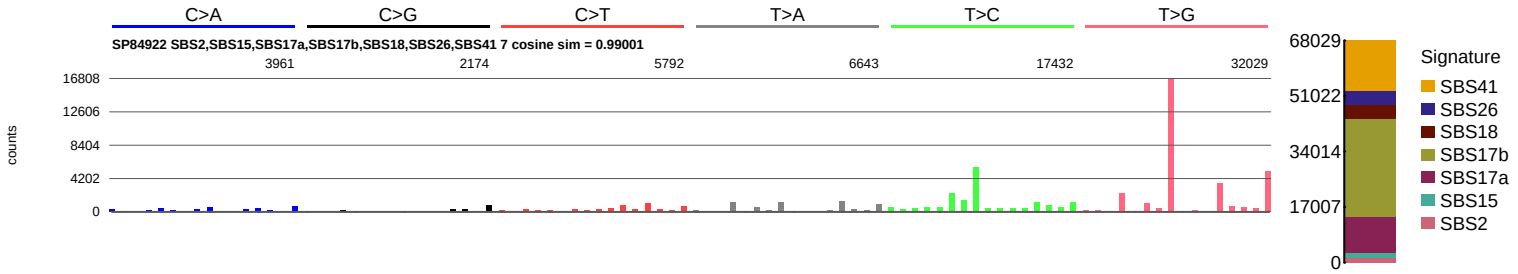
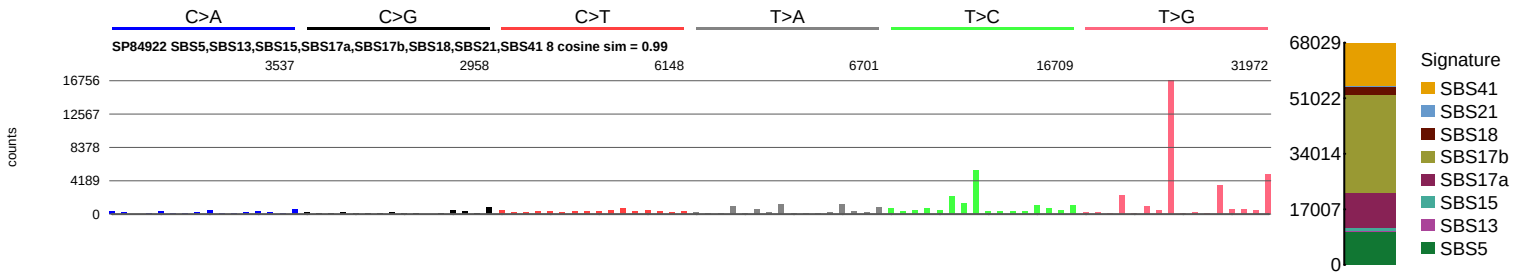
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.99001
3965
2162
5867
6639
17392
32007
counts
16808
12606
8404
4202
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.99001
3961
2174
5792
6643
17432
32029
counts
16808
12606
8404
4202
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2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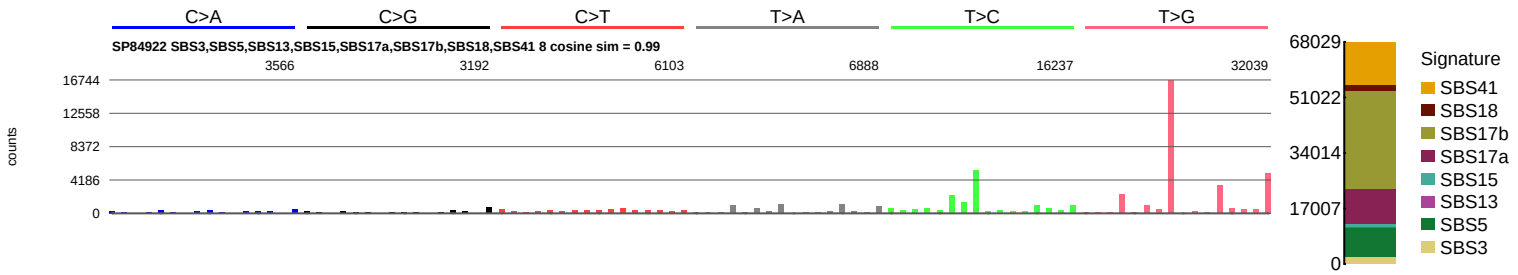
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99
3537
2958
6148
6701
16709
31972
counts
16756
12567
8378
4189
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5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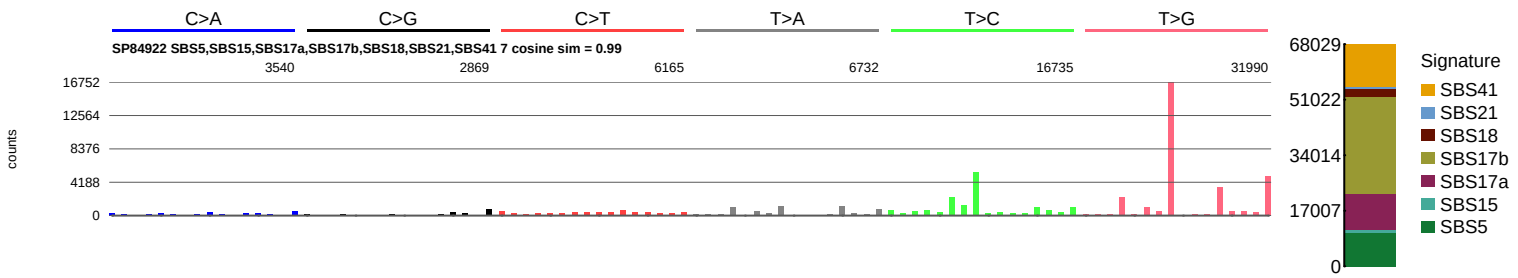
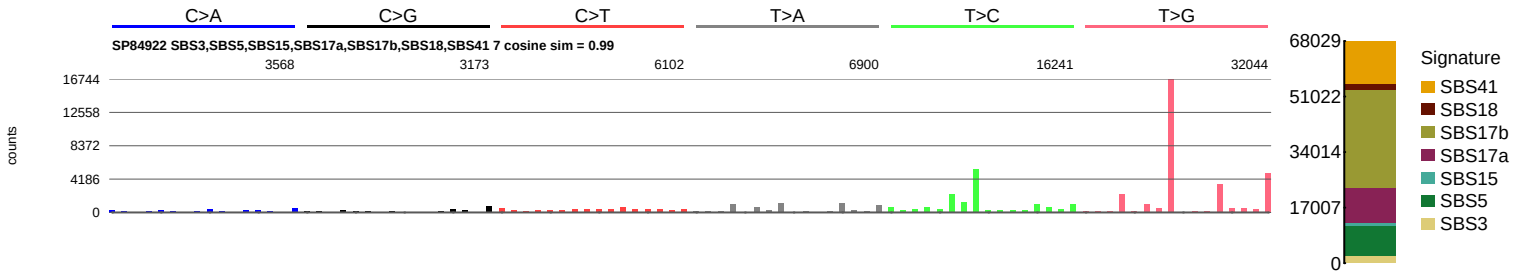
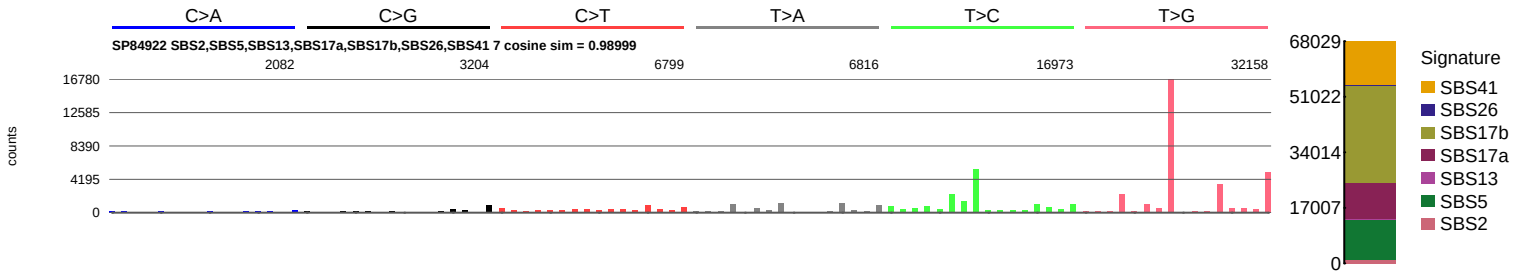
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99
3566
3192
6103
6888
16237
32039
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.99
3540
2869
6165
6732
16735
31990
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99
3568
3173
6102
6900
16241
32044
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3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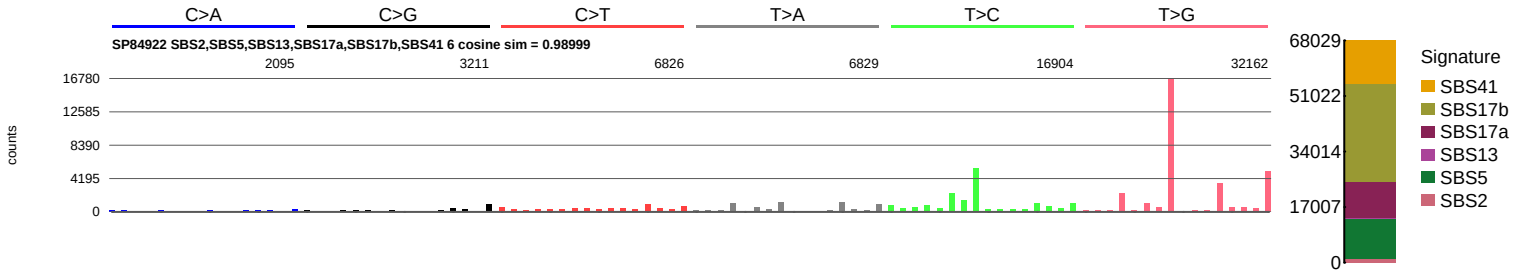
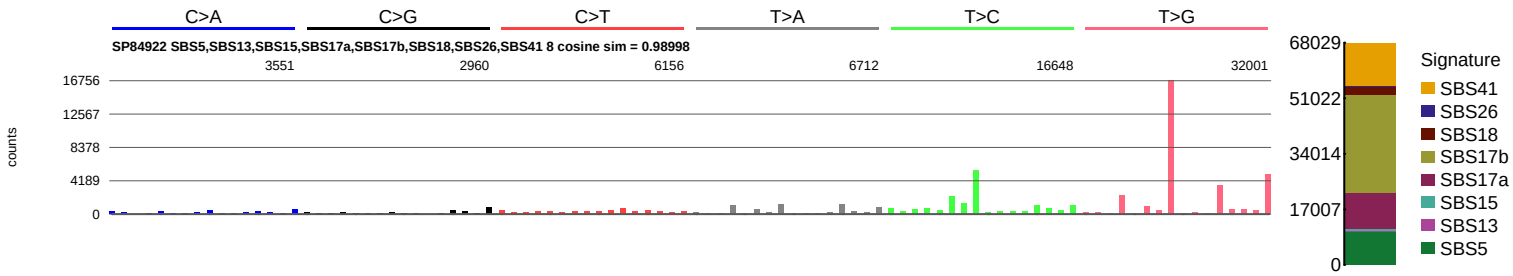
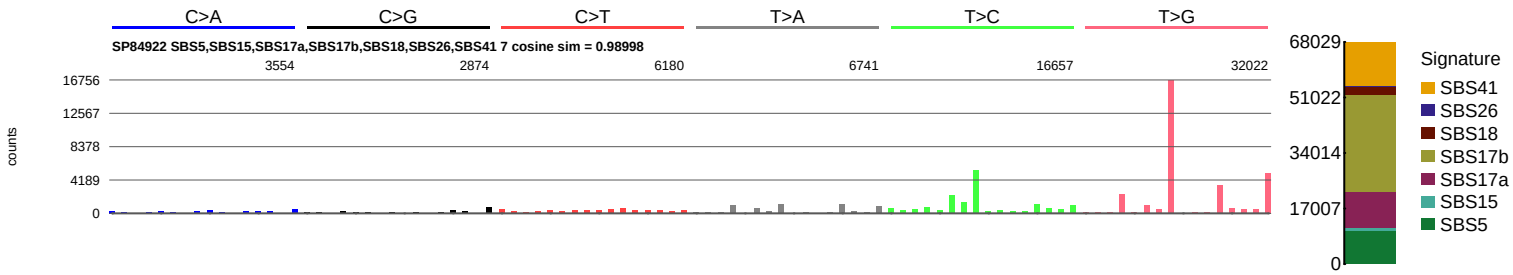
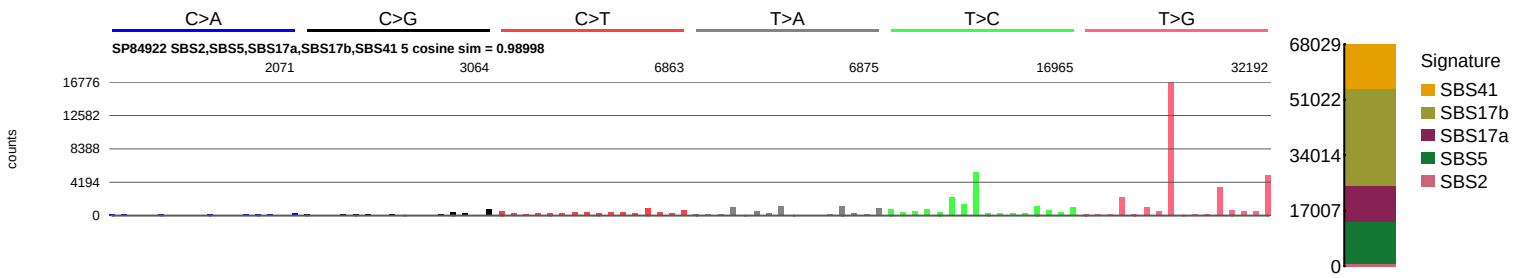
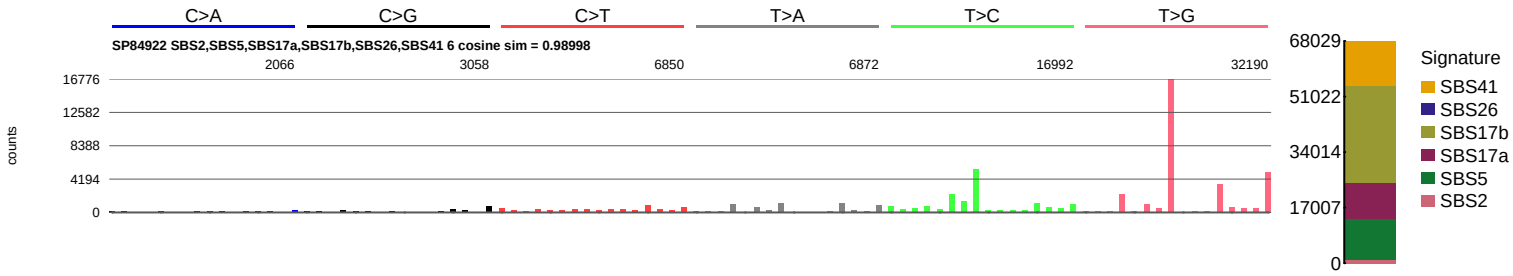
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98999
2082
3204
6799
6816
16973
32158
counts
16780
12585
8390
4195
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS2

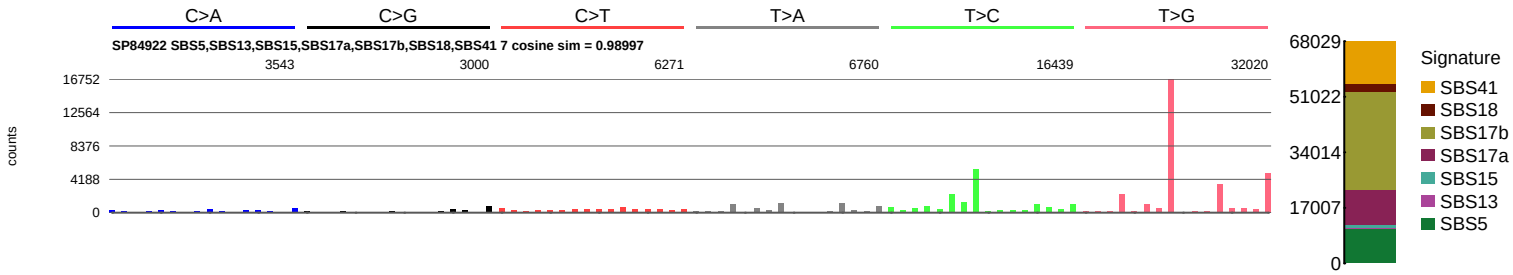

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98999
2095
3211
6826
6829
16904
32162
counts
16780
12585
8390
4195
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98998
3551
2960
6156
6712
16648
32001
counts
16756
12567
8378
4189
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98998
3554
2874
6180
6741
16657
32022
counts
16756
12567
8378
4189
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98998
2071
3064
6863
6875
16965
32192
counts
16776
12582
8388
4194
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98998
2066
3058
6850
6872
16992
32190
16776
12582
8388
4194
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98997
3543
3000
6271
6760
16439
32020
16752
12564
8376
4188
0
counts
68029
51022
34014
17007
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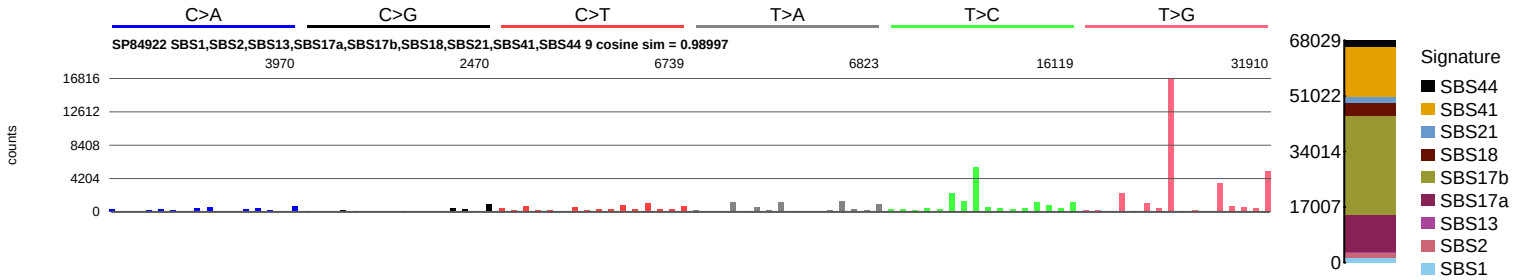
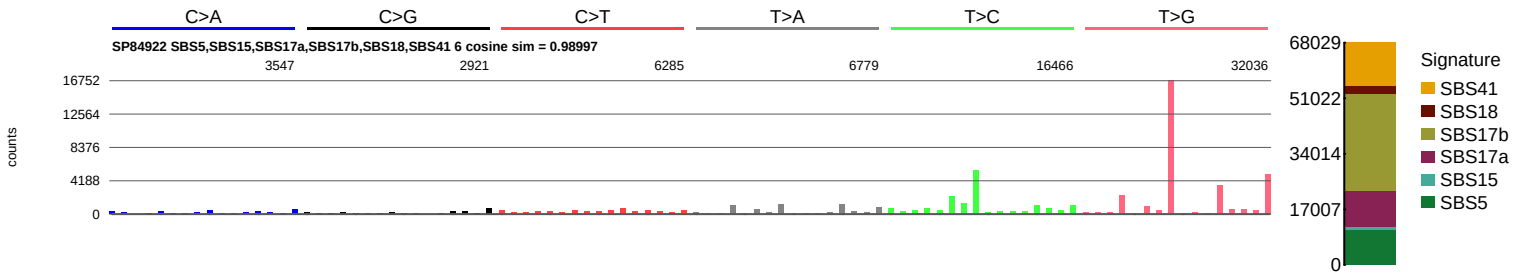
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98997
3970
2470
6739
6823
16119
31910
16816
12612
8408
4204
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98997
3547
2921
6285
6779
16466
32036
16752
12564
8376
4188
0
counts
68029
51022
34014
17007
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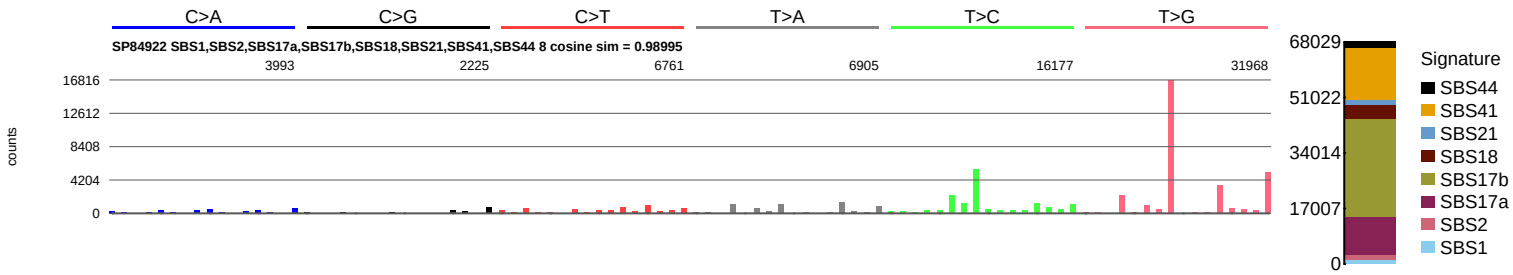
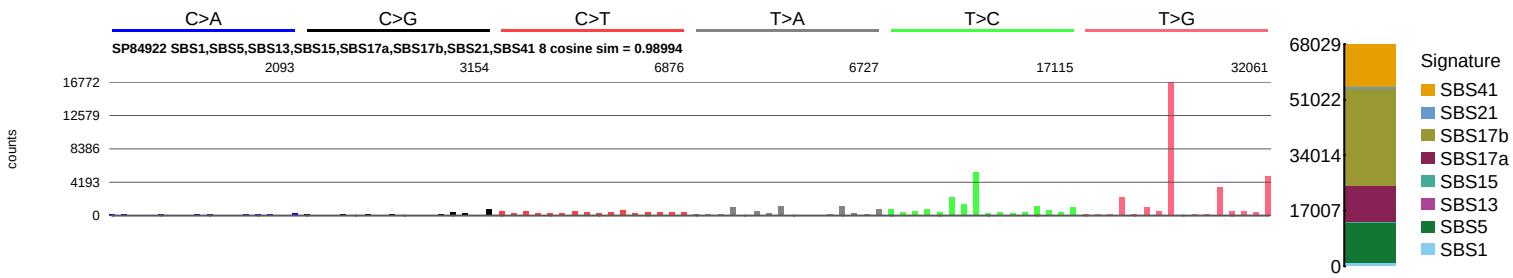
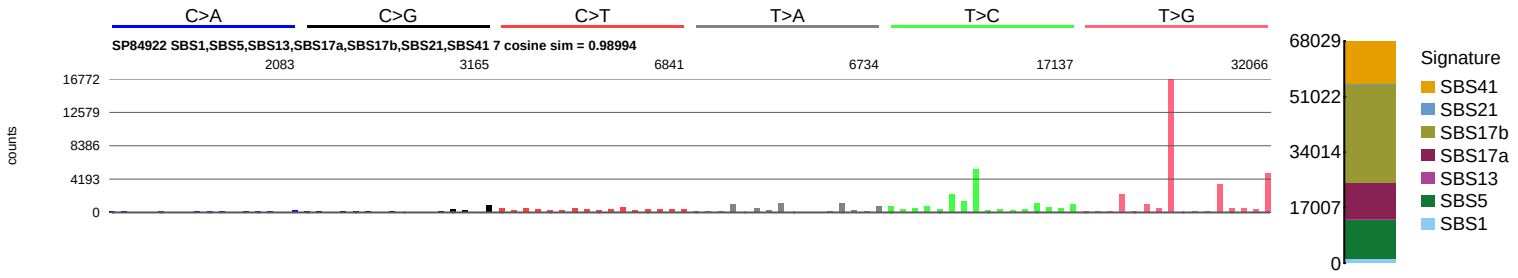
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98995
3993
2225
6761
6905
16177
31968
16816
12612
8408
4204
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98994
2093
3154
6876
6727
17115
32061
16772
12579
8386
4193
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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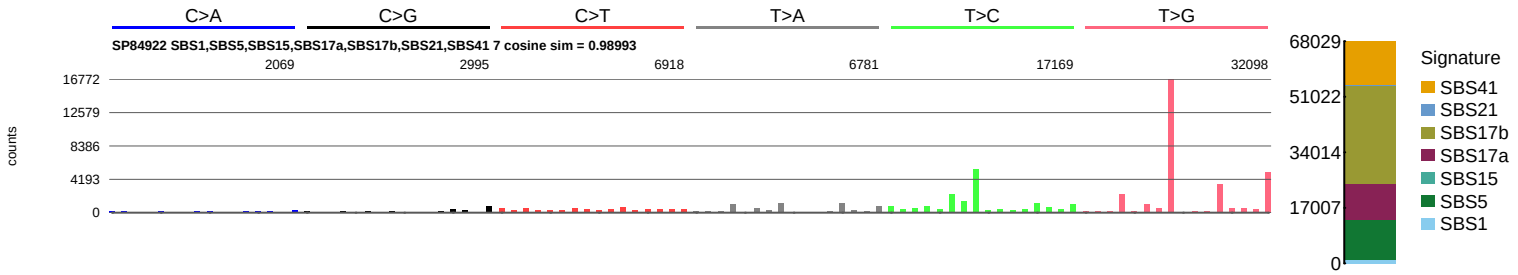
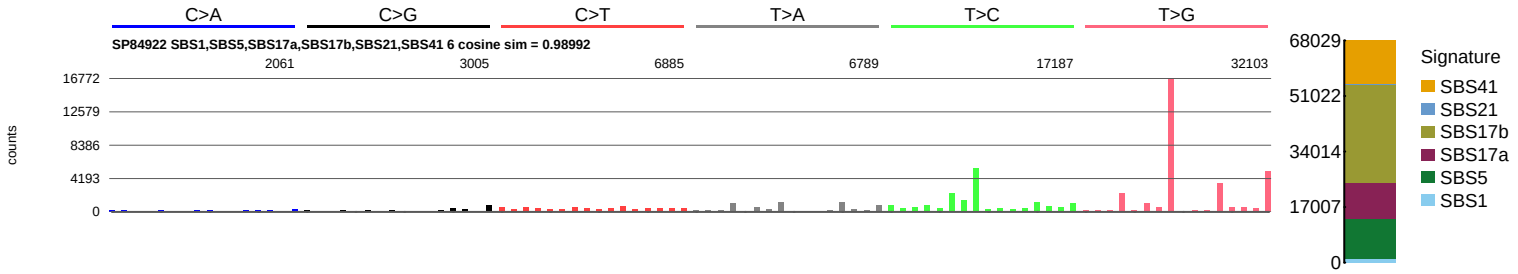
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98994
2083
3165
6841
6734
17137
32066
counts
16772
12579
8386
4193
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98993
2069
2995
6918
6781
17169
32098
counts
16772
12579
8386
4193
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1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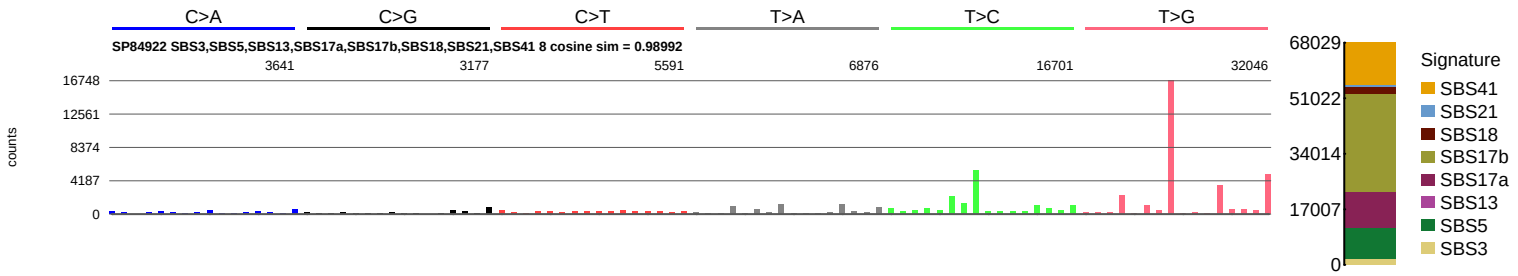
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98992
2061
3005
6885
6789
17187
32103
counts
16772
12579
8386
4193
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS1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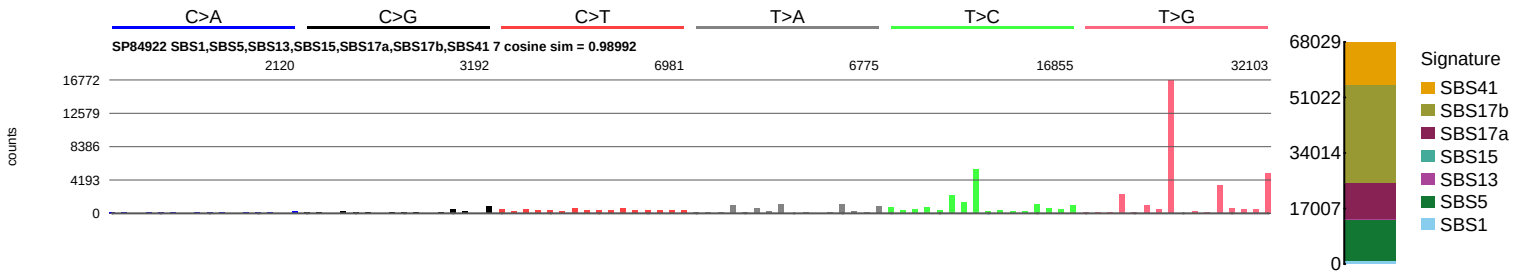
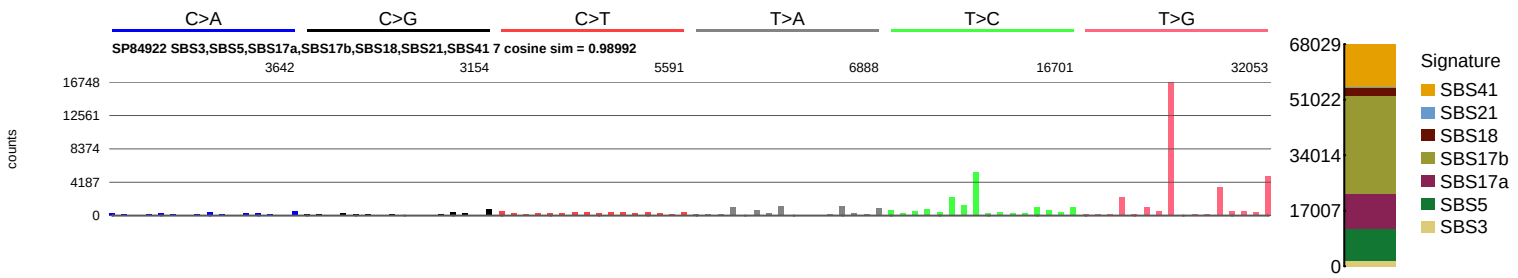
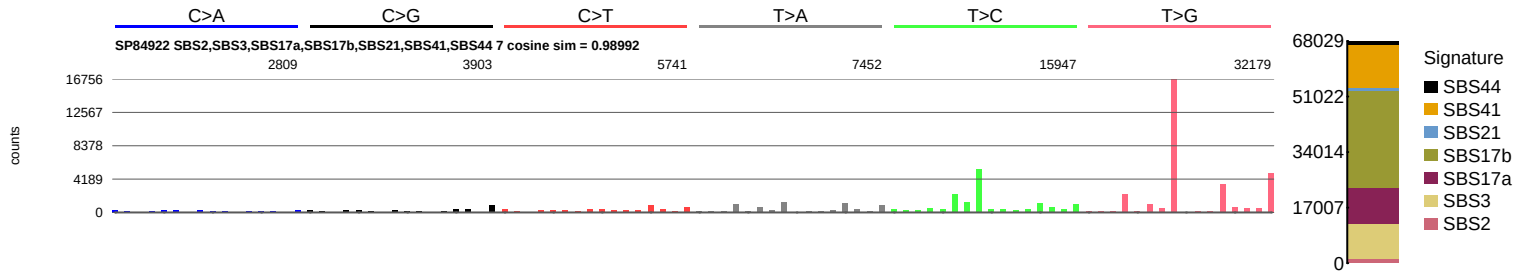
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98992
3641
3177
5591
6876
16701
32046
counts
16748
12561
8374
4187
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98992
2120
3192
6981
6775
16855
32103
counts
16772
12579
8386
4193
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98992
3642
3154
5591
6888
16701
32053
counts
16748
12561
8374
4187
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3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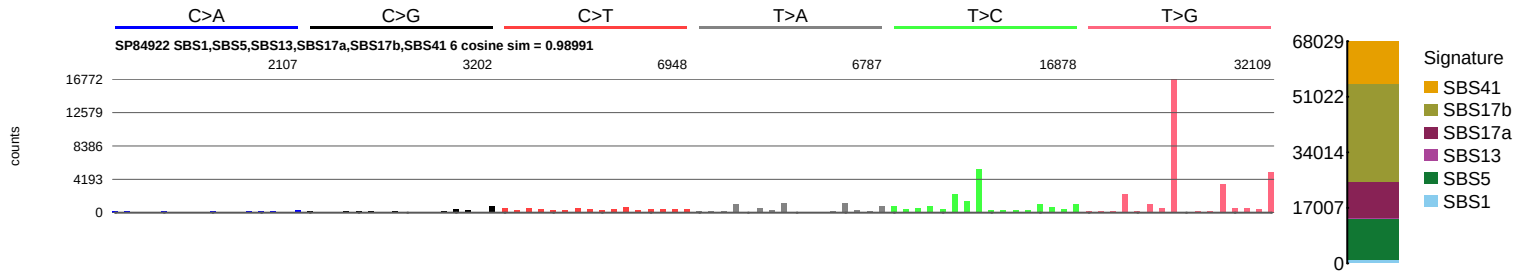
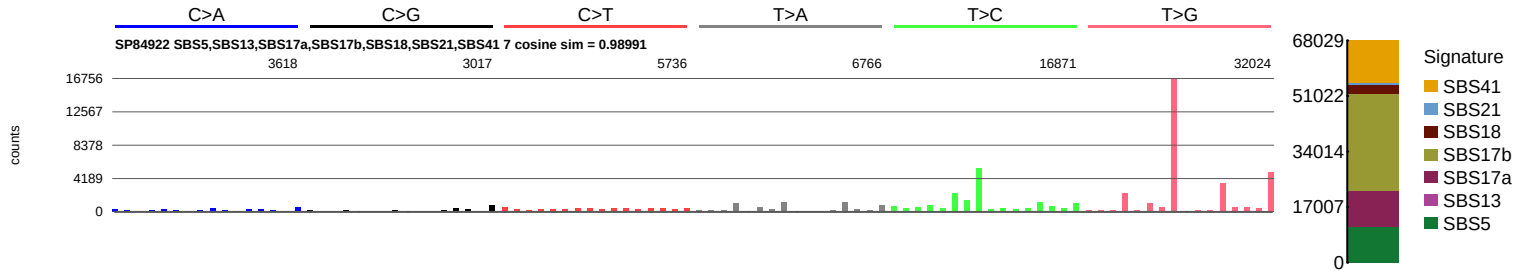
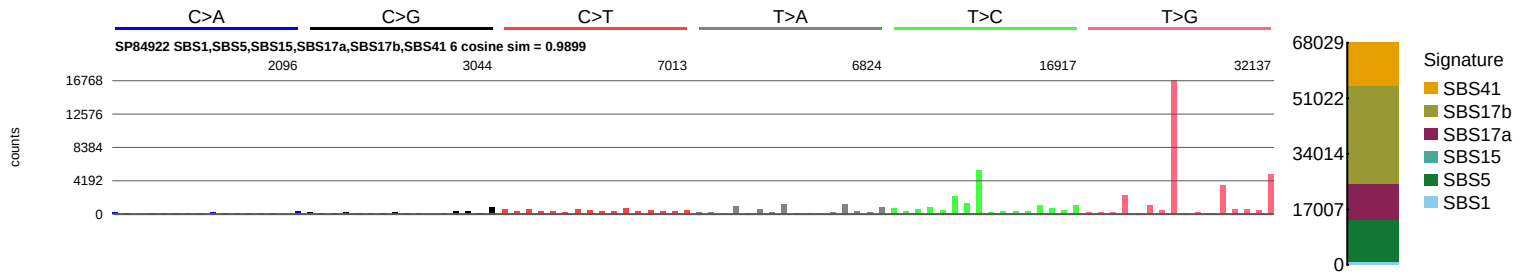
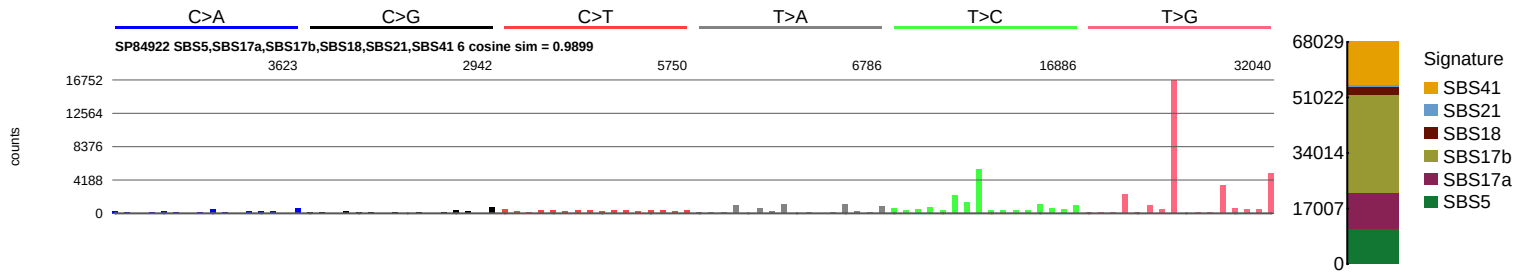
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98992
2809
3903
5741
7452
15947
32179
16756
12567
8378
4189
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98991
2107
3202
6948
6787
16878
32109
16772
12579
8386
4193
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98991
3618
3017
5736
6766
16871
32024
16756
12567
8378
4189
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.9899
2096
3044
7013
6824
16917
32137
16768
12576
8384
4192
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.9899
3623
2942
5750
6786
16886
32040
16752
12564
8376
4188
0
counts
68029
51022
34014
17007
0
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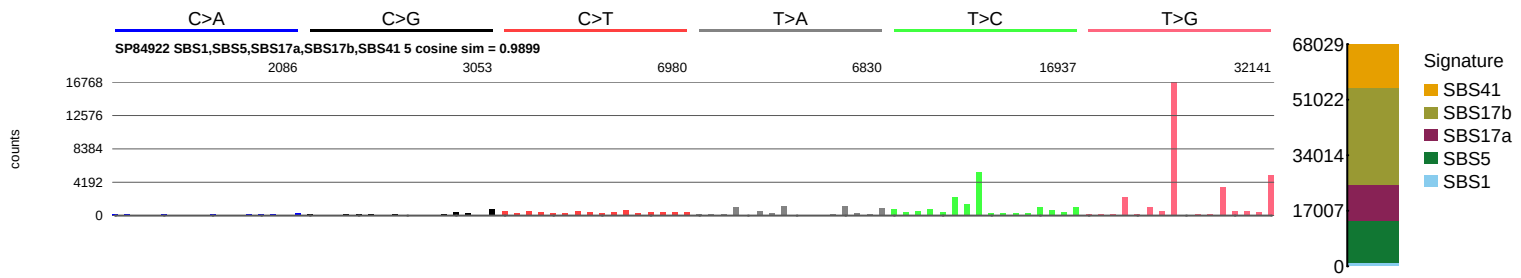
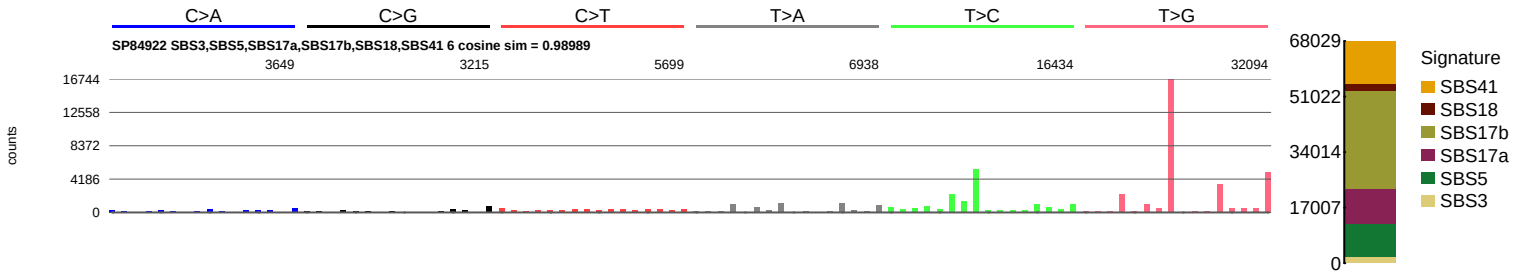
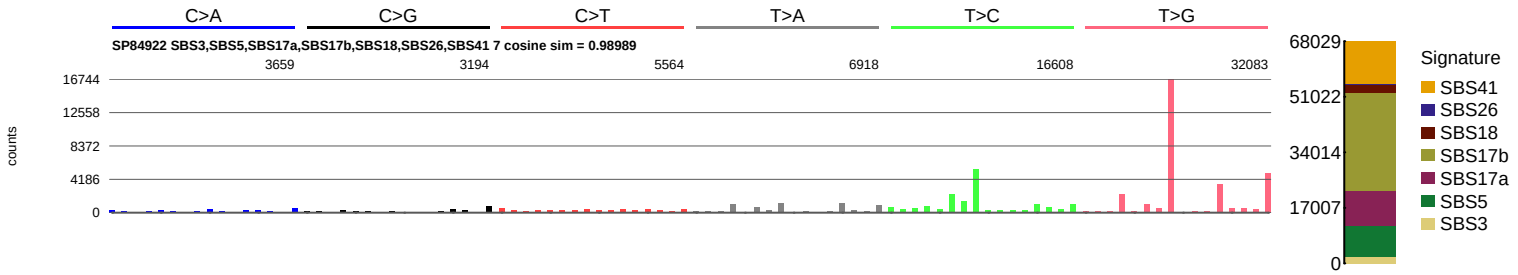
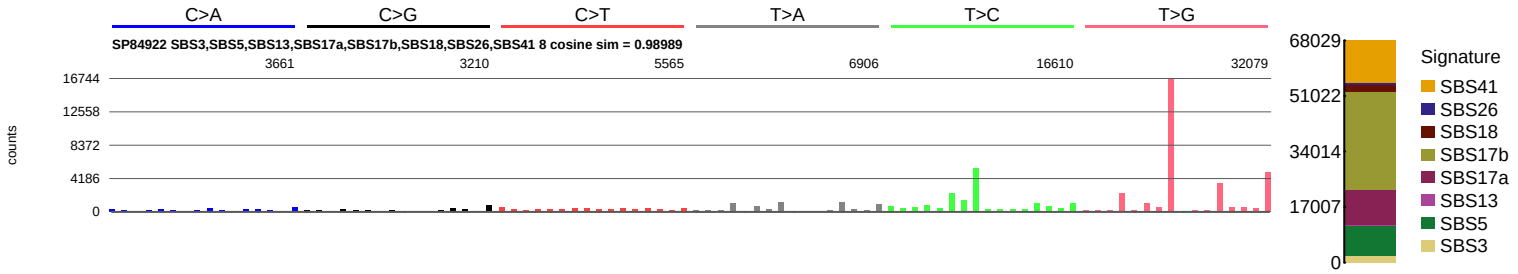
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS17a,SBS17b,SBS41 5 cosine sim = 0.9899
2086
3053
6980
6830
16937
32141
16768
12576
8384
4192
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98989
3649
3215
5699
6938
16434
32094
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98989
3659
3194
5564
6918
16608
32083
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3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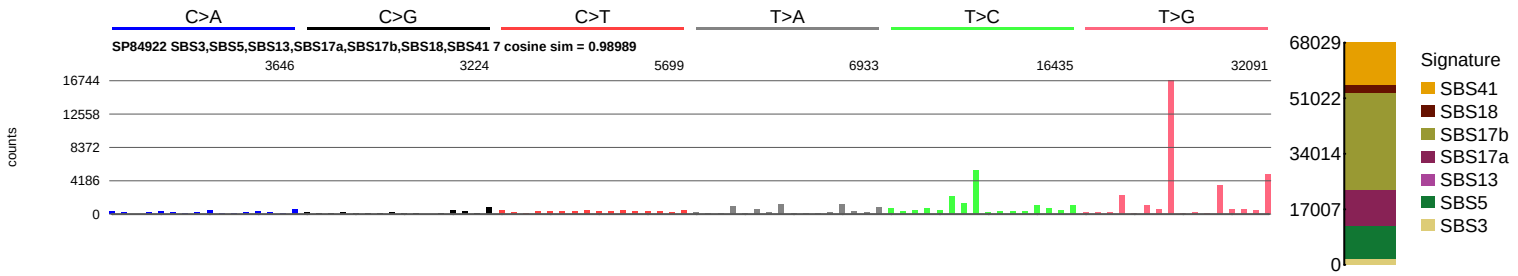
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98989
3661
3210
5565
6906
16610
32079
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98989
3646
3224
5699
6933
16435
32091
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
Signature
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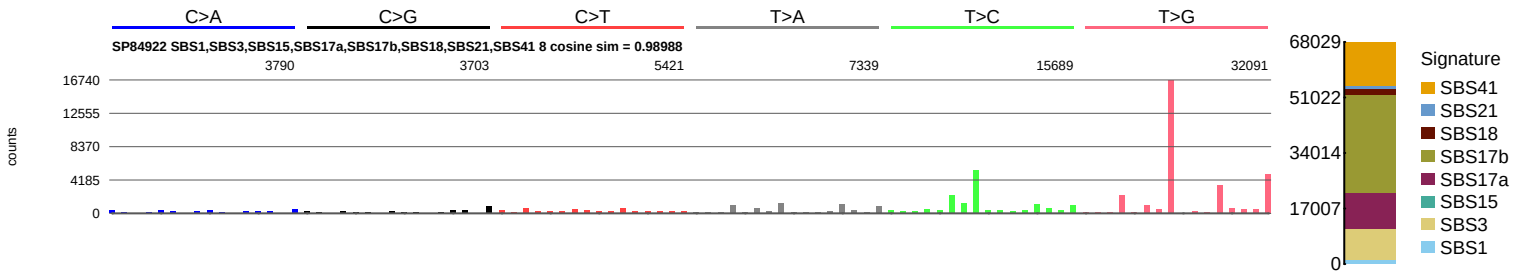
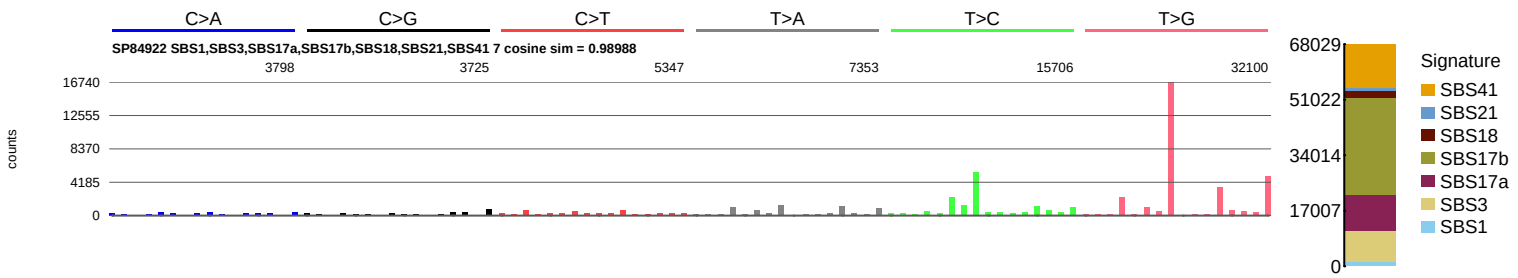
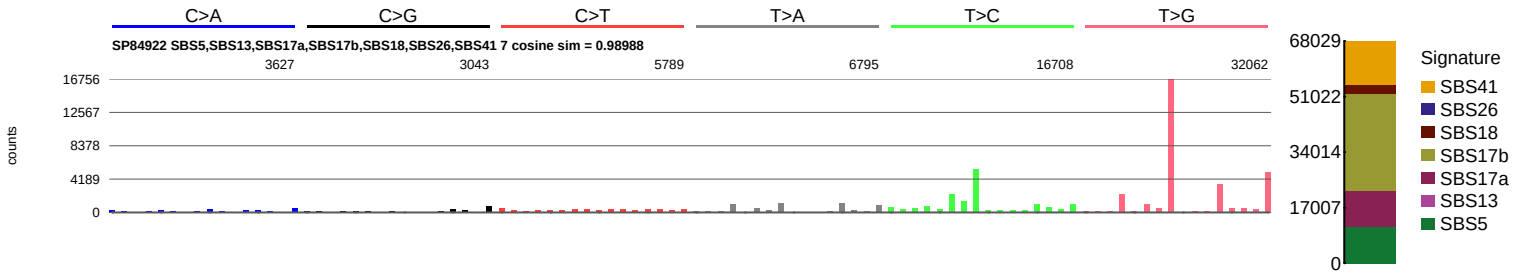
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98988
3790
3703
5421
7339
15689
32091
counts
16740
12555
8370
4185
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98988
3798
3725
5347
7353
15706
32100
counts
16740
12555
8370
4185
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98988
3627
3043
5789
6795
16708
32062
counts
16756
12567
8378
4189
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
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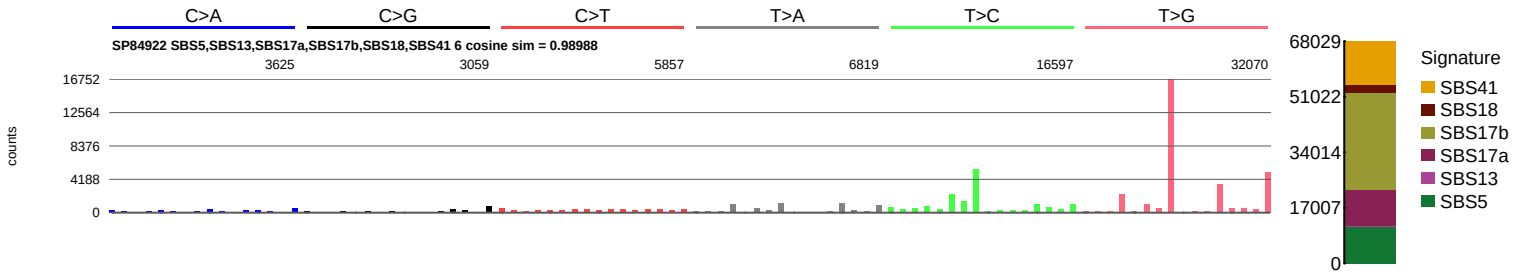
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98988
3625
3059
5857
6819
16597
32070
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
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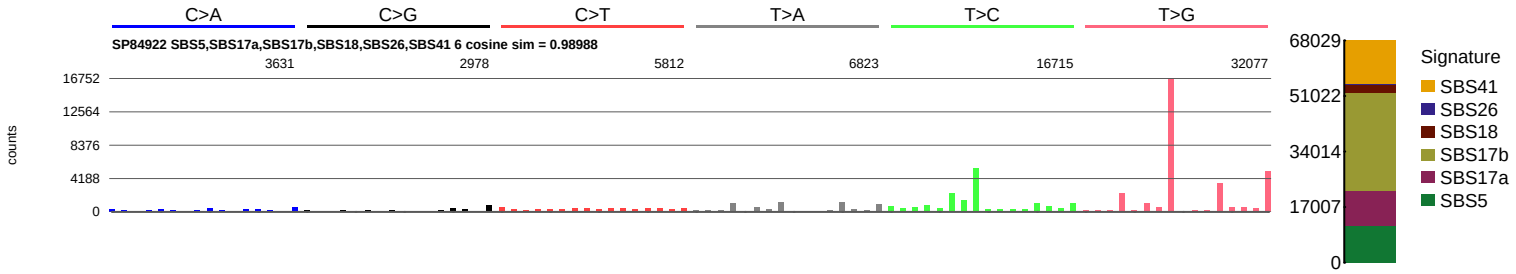
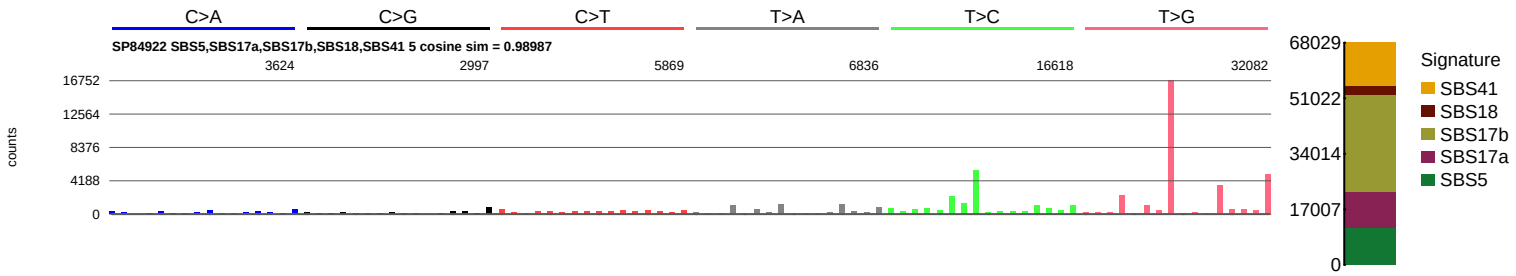
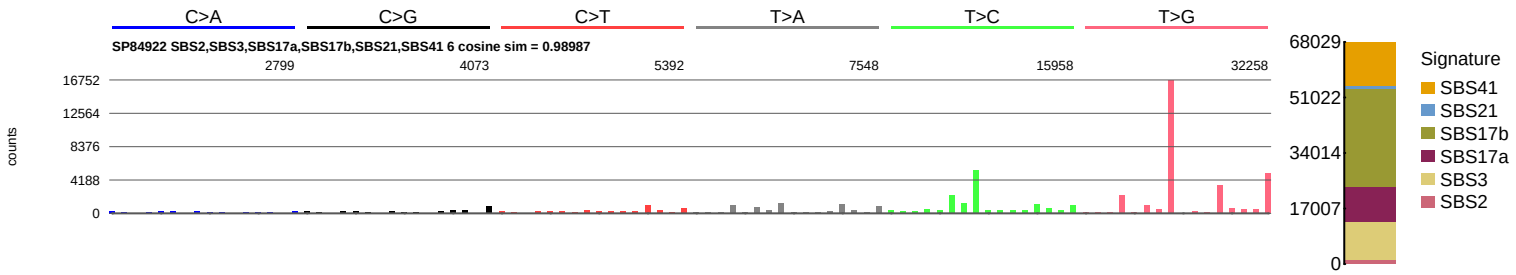
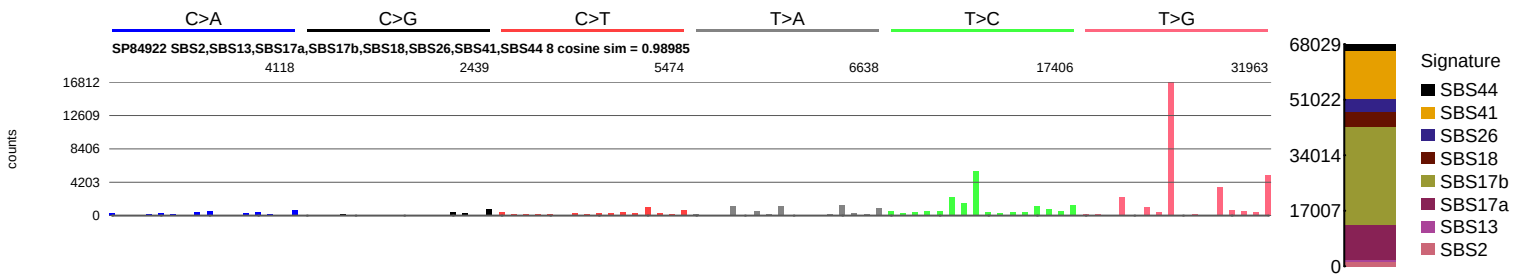
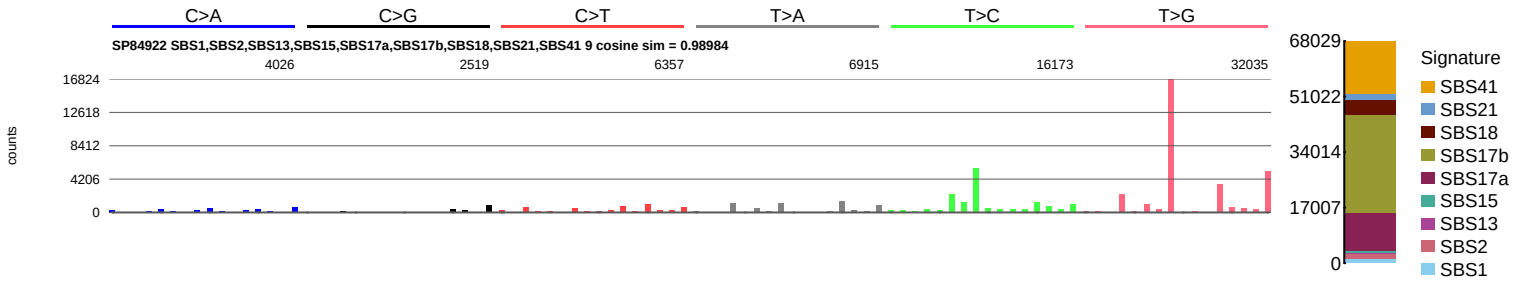
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.98988
3631
2978
5812
6823
16715
32077
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98987
3624
2997
5869
6836
16618
32082
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98987
2799
4073
5392
7548
15958
32258
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98985
4118
2439
5474
6638
17406
31963
counts
16812
12609
8406
4203
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98984
4026
2519
6357
6915
16173
32035
16824
12618
8412
4206
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
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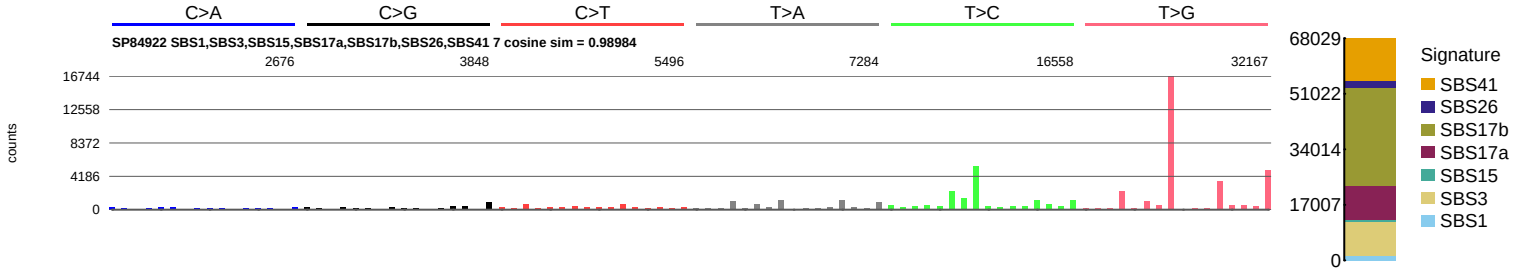
SBS13
SBS2
SBS1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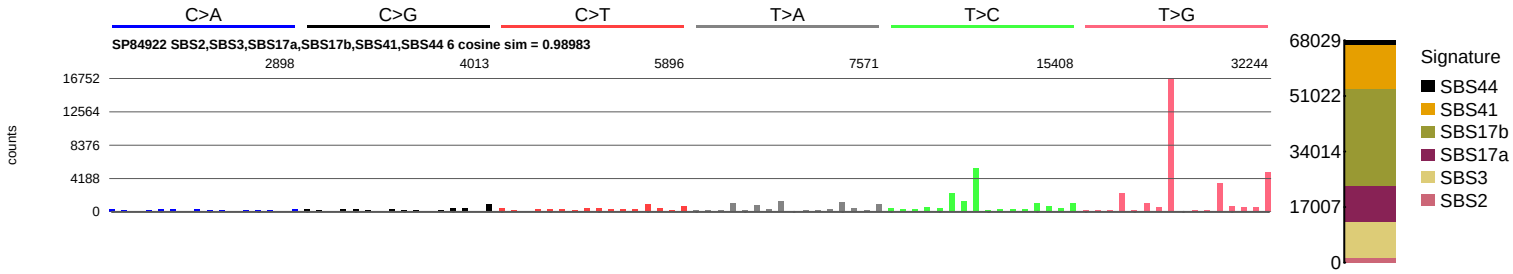
SP84922 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98984
2676
3848
5496
7284
16558
32167
16744
12558
8372
4186
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1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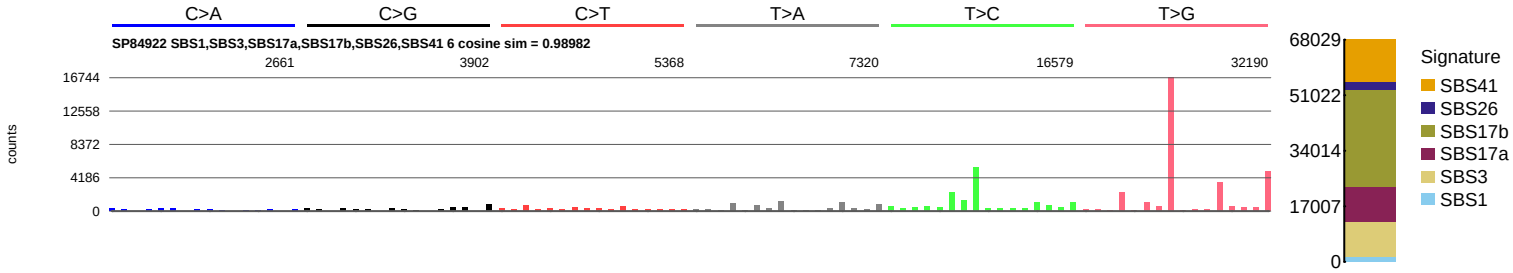
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98983
2898
4013
5896
7571
15408
32244
16752
12564
8376
4188
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS3
SBS2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T>C
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SP84922 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98982
2661
3902
5368
7320
16579
32190
16744
12558
8372
4186
0
counts
68029
51022
34014
17007
0
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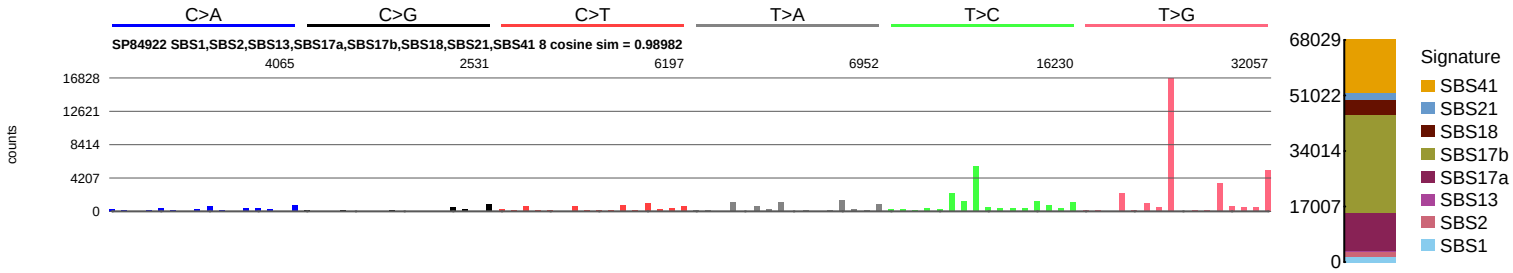
SBS26
SBS17b
SBS17a
SBS3
SBS1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C>T
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T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98982
4065
2531
6197
6952
16230
32057
16828
12621
8414
4207
0
counts
68029
51022
34014
17007
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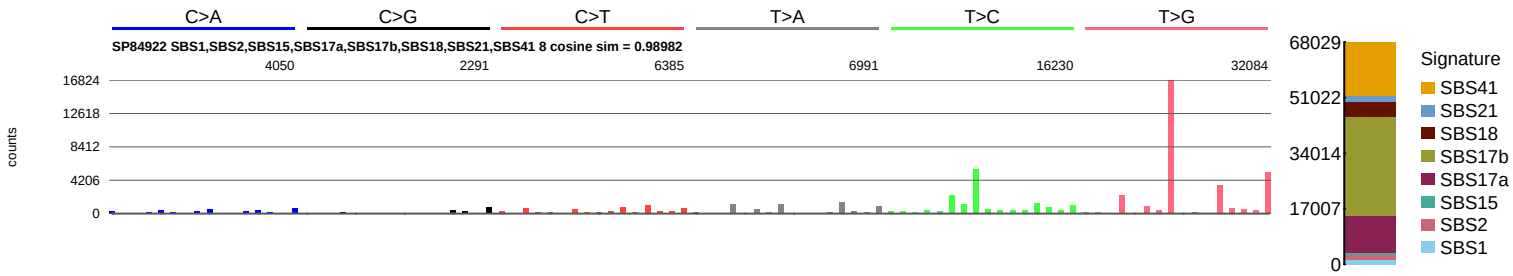
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84922 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98982
4050
2291
6385
6991
16230
32084
16824
12618
8412
4206
0
counts
68029
51022
34014
17007
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SBS41
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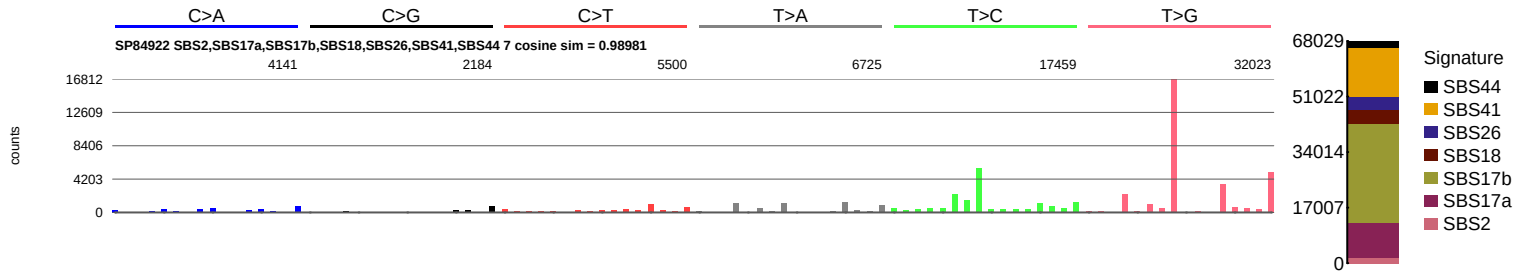
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98981
4141
2184
5500
6725
17459
32023
16812
12609
8406
4203
0
counts
68029
51022
34014
17007
0
Signature
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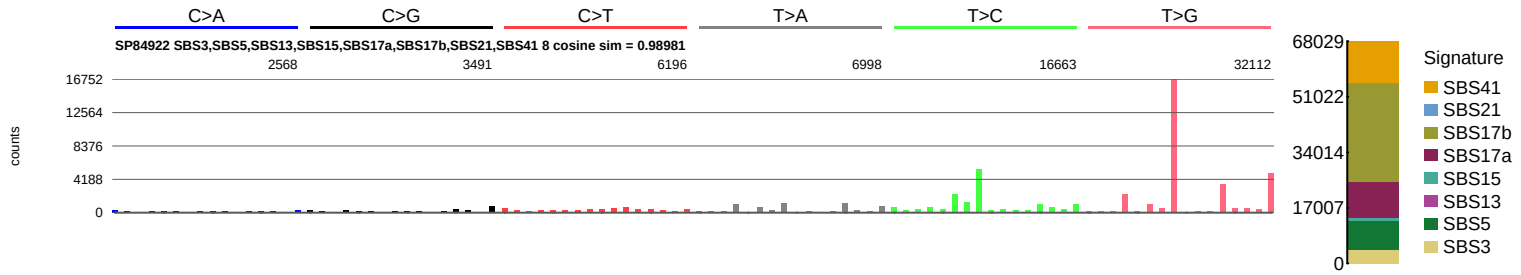
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2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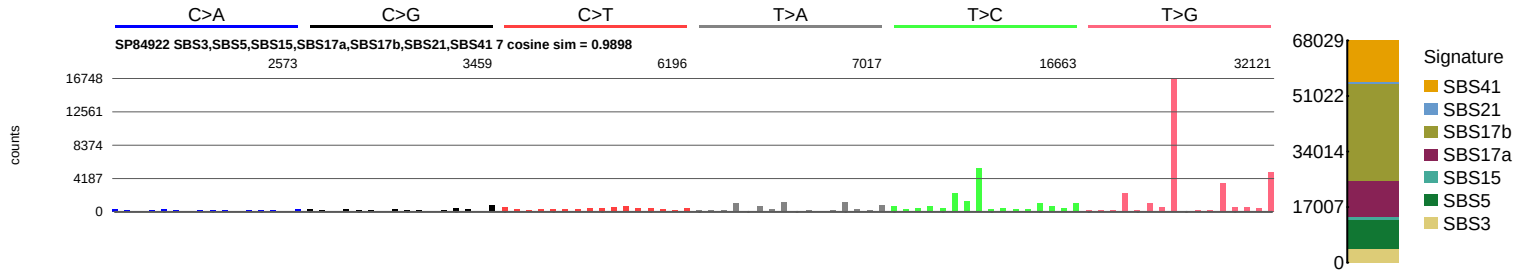
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98981
2568
3491
6196
6998
16663
32112
16752
12564
8376
4188
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3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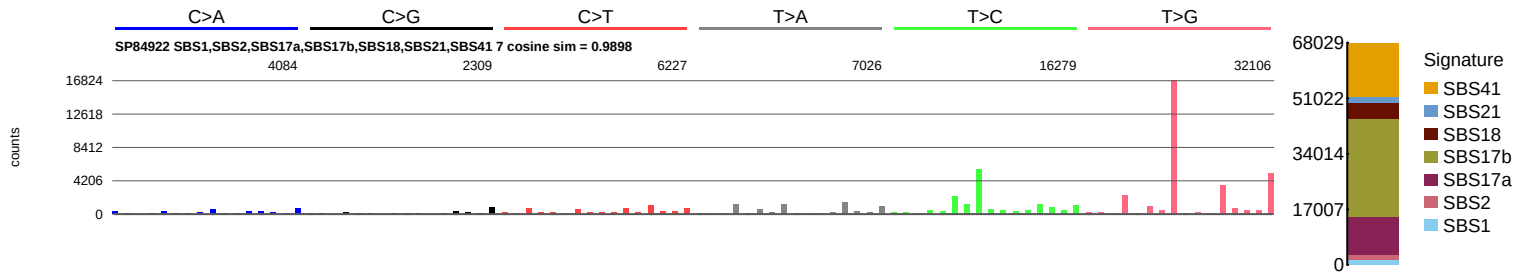
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.9898
2573
3459
6196
7017
16663
32121
16748
12561
8374
4187
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3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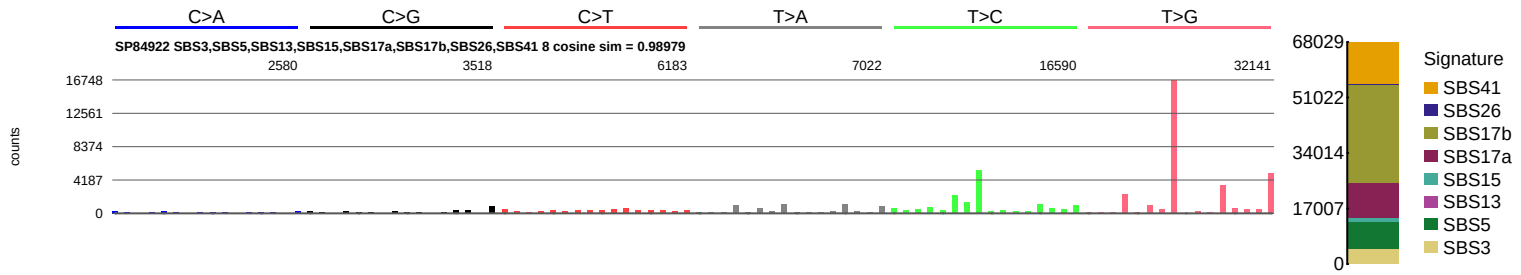
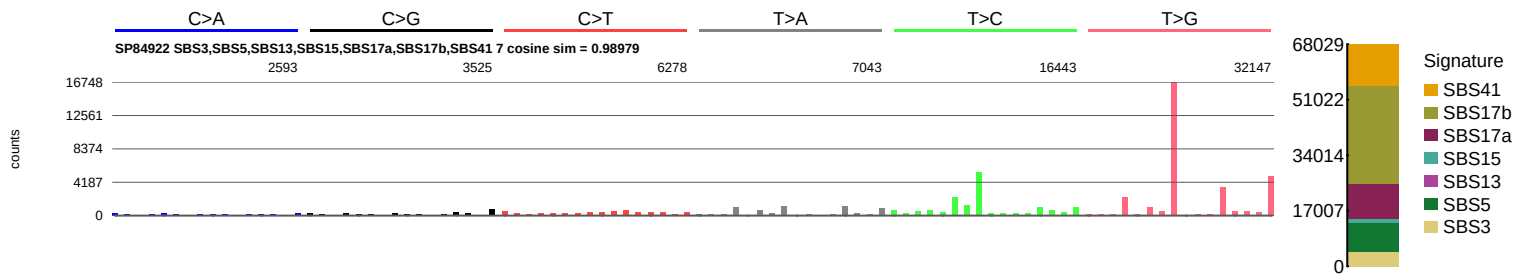
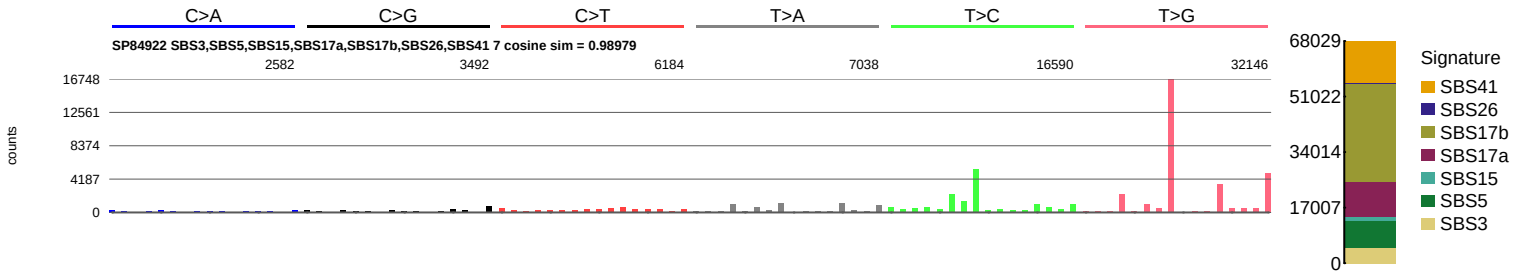
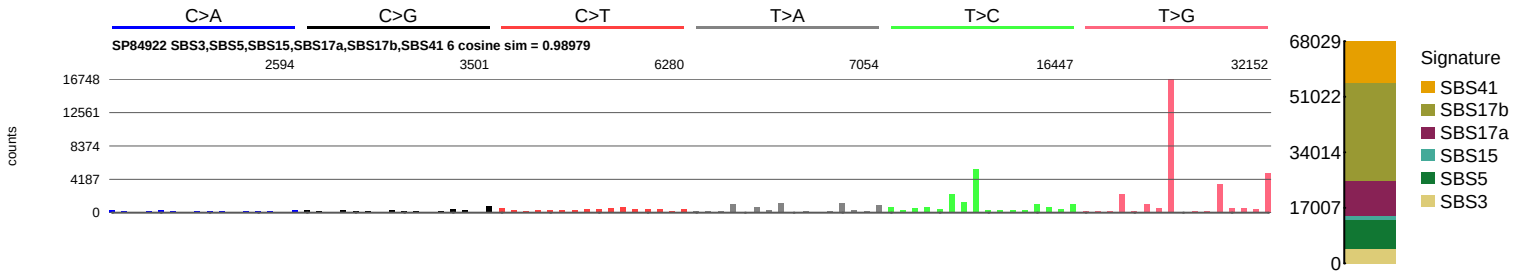
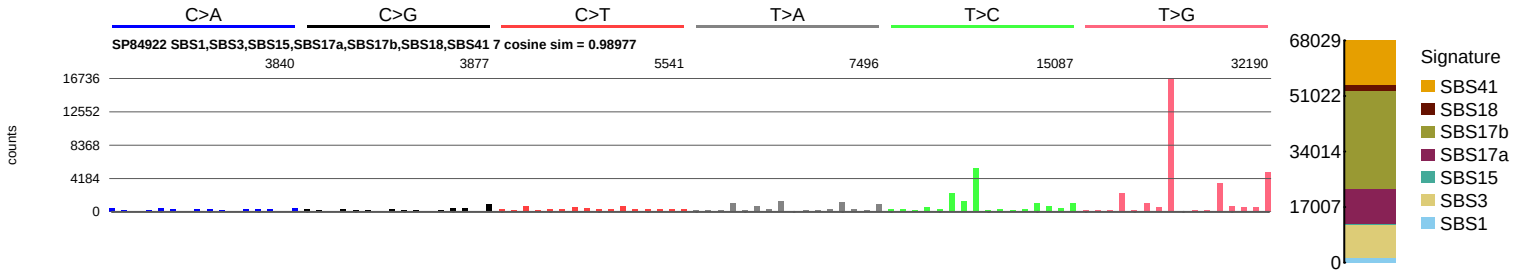
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.9898
4084
2309
6227
7026
16279
32106
16824
12618
8412
4206
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

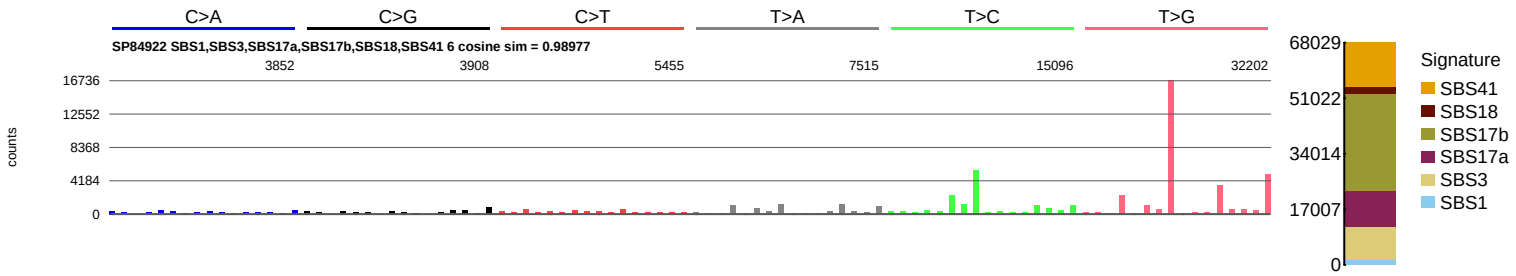

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.98979
2580
3518
6183
7022
16590
32141
16748
12561
8374
4187
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

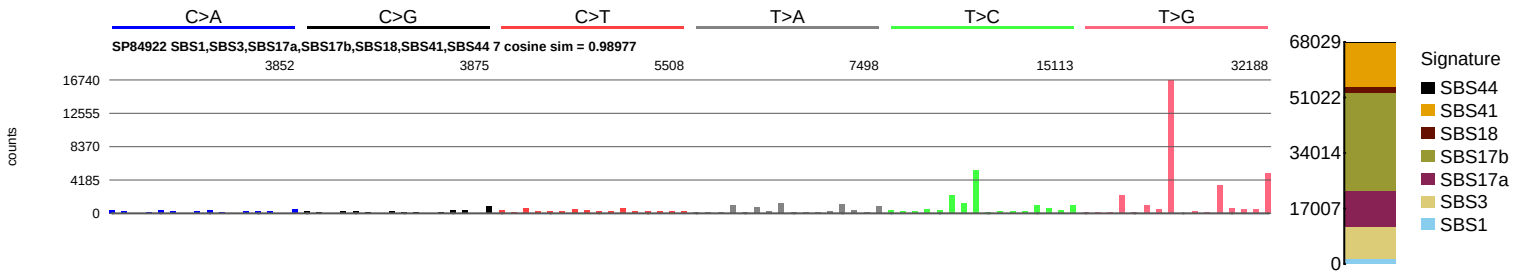

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98979
2593
3525
6278
7043
16443
32147
16748
12561
8374
4187
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

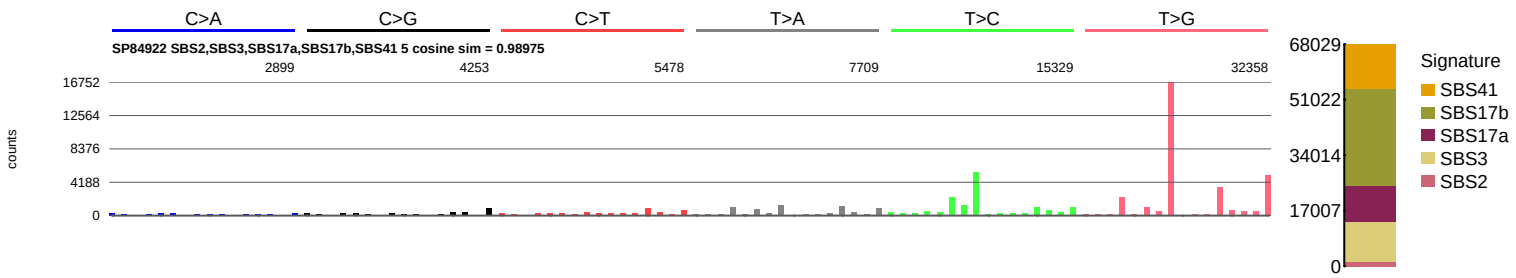

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98979
2582
3492
6184
7038
16590
32146
16748
12561
8374
4187
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3

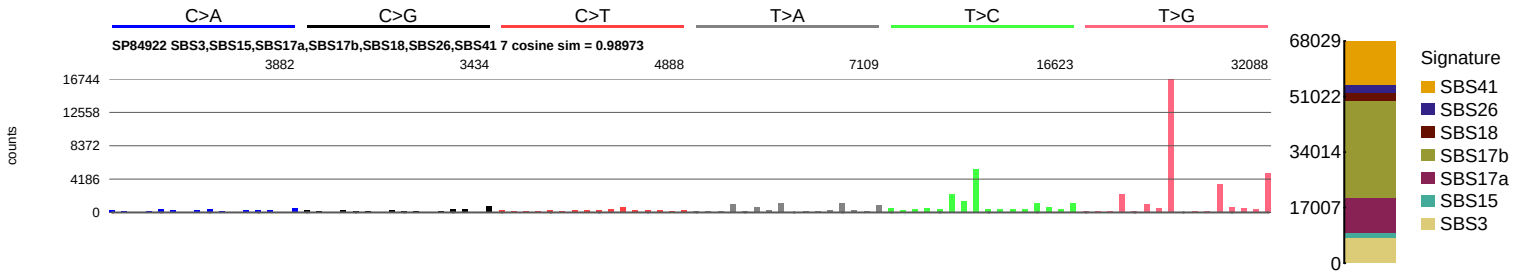

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98979
2594
3501
6280
7054
16447
32152
16748
12561
8374
4187
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3

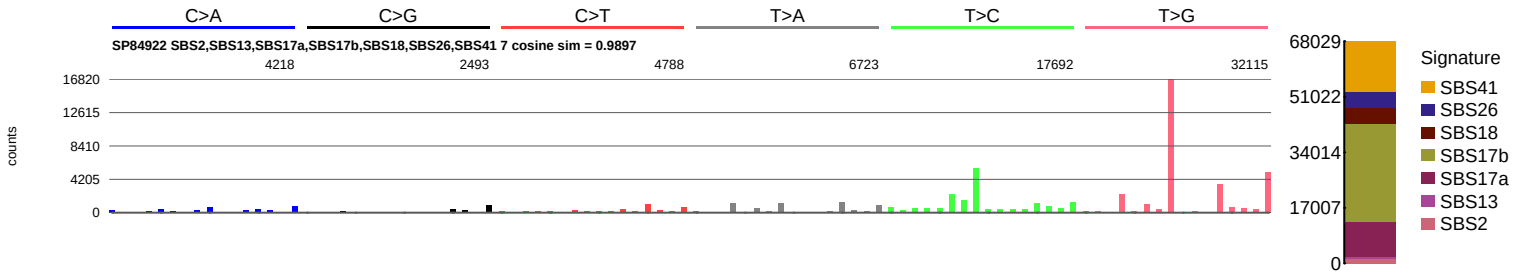

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98977
3840
3877
5541
7496
15087
32190
16736
12552
8368
4184
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98977
3852
3908
5455
7515
15096
32202
16736
12552
8368
4184
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98977
3852
3875
5508
7498
15113
32188
16740
12555
8370
4185
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98975
2899
4253
5478
7709
15329
32358
16752
12564
8376
4188
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98973
3882
3434
4888
7109
16623
32088
16744
12558
8372
4186
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.9897
4218
2493
4788
6723
17692
32115
16820
12615
8410
4205
0
counts
68029
51022
34014
17007
0
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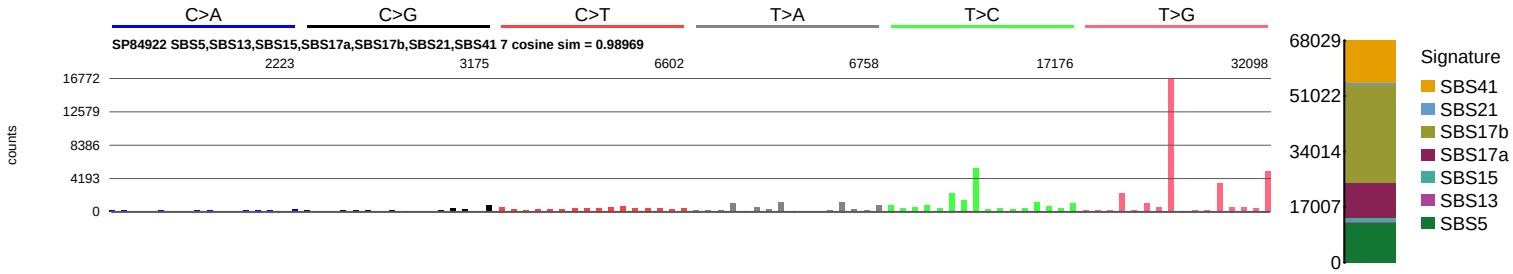
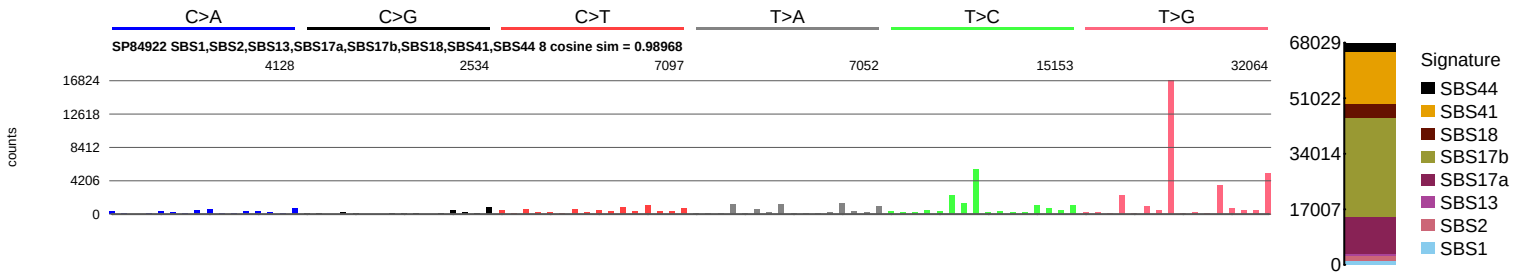
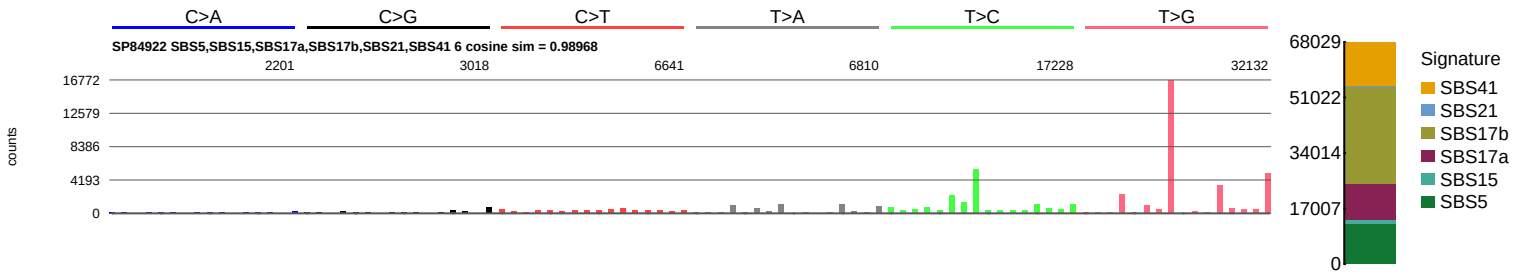
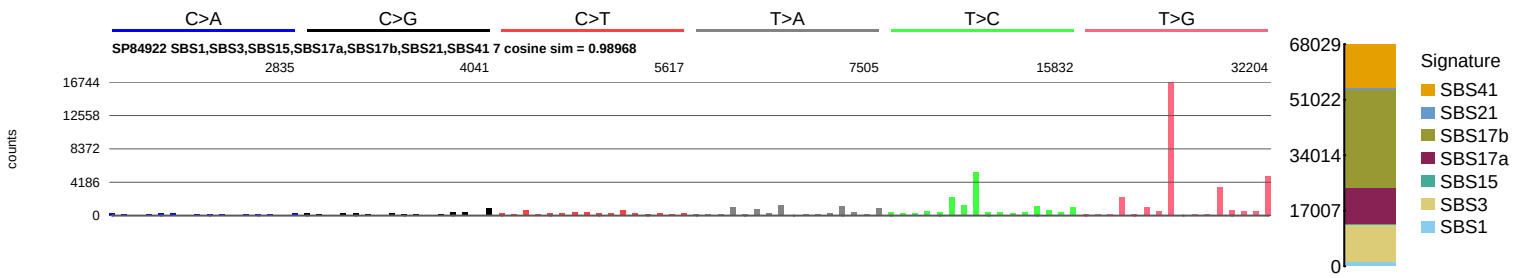
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98969
2223
3175
6602
6758
17176
32098
16772
12579
8386
4193
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98968
4128
2534
7097
7052
15153
32064
16824
12618
8412
4206
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98968
2201
3018
6641
6810
17228
32132
16772
12579
8386
4193
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98968
2835
4041
5617
7505
15832
32204
16744
12558
8372
4186
0
counts
68029
51022
34014
17007
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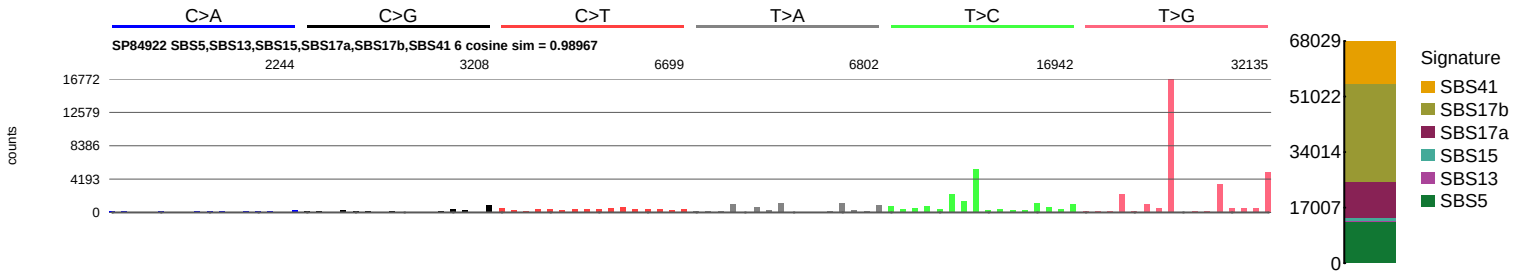
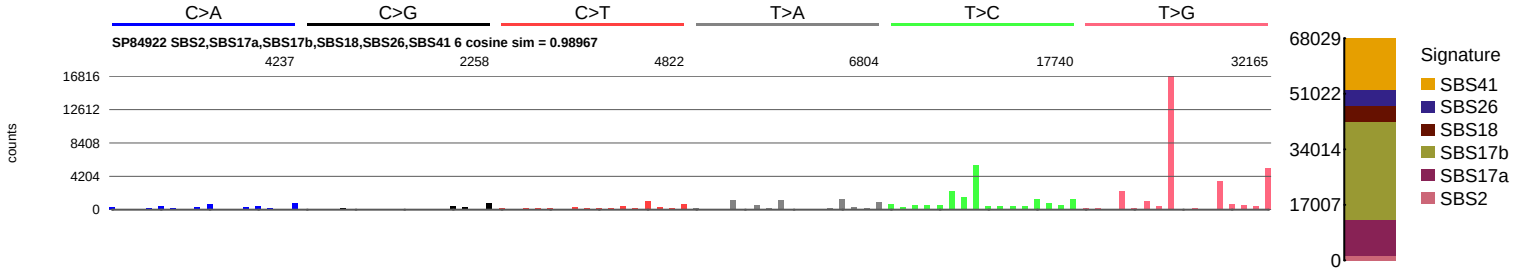
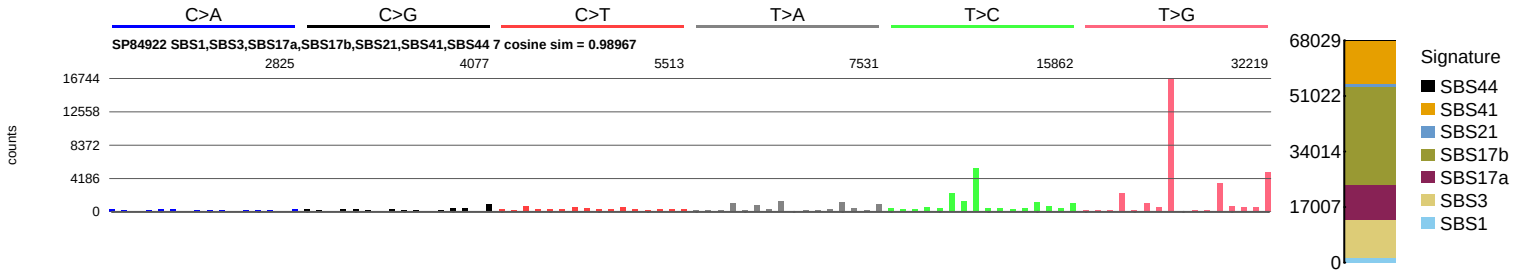
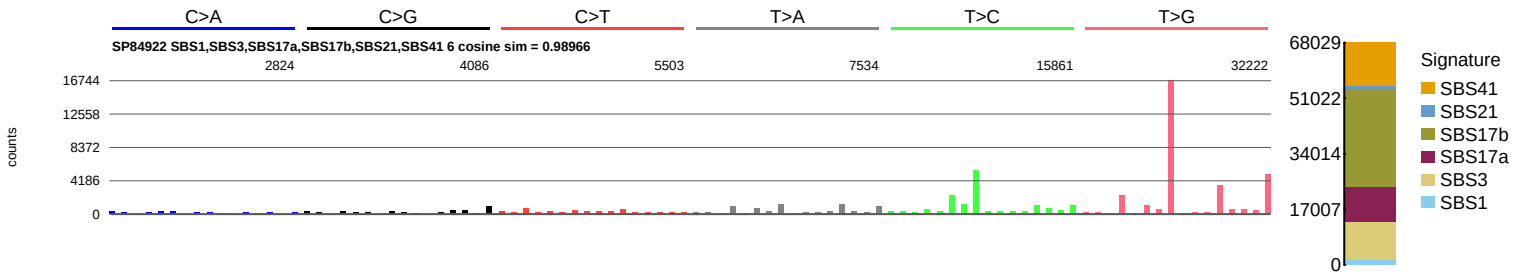
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98967
2244
3208
6699
6802
16942
32135
counts
16772
12579
8386
4193
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.98967
4237
2258
4822
6804
17740
32165
counts
16816
12612
8408
4204
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98967
2825
4077
5513
7531
15862
32219
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS1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C>G
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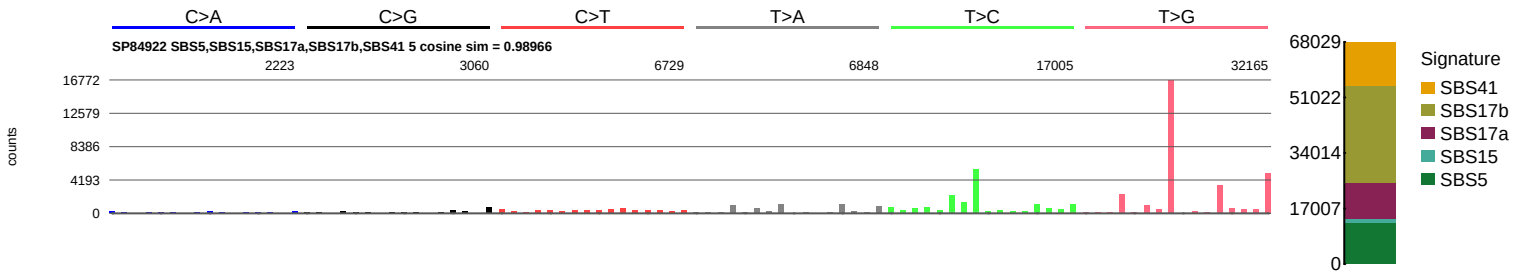
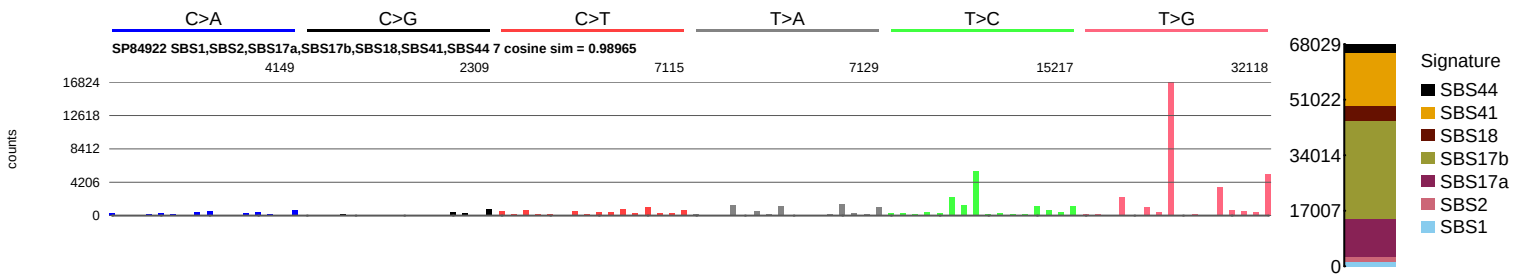
T>C
T>G
SP84922 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98966
2824
4086
5503
7534
15861
32222
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS15,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98966
2223
3060
6729
6848
17005
32165
counts
16772
12579
8386
4193
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5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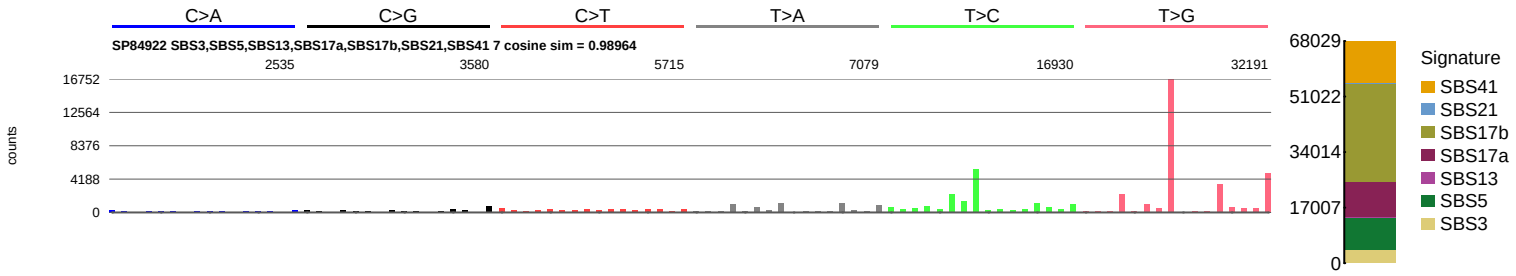
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98965
4149
2309
7115
7129
15217
32118
counts
16824
12618
8412
4206
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>T
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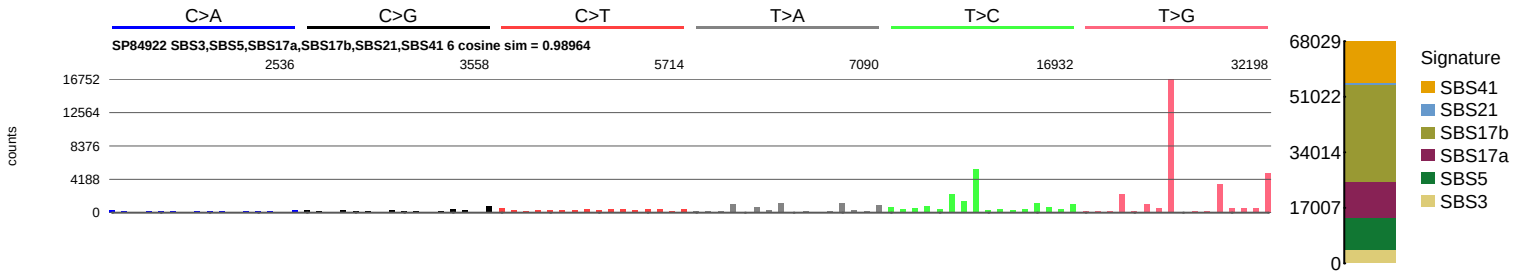
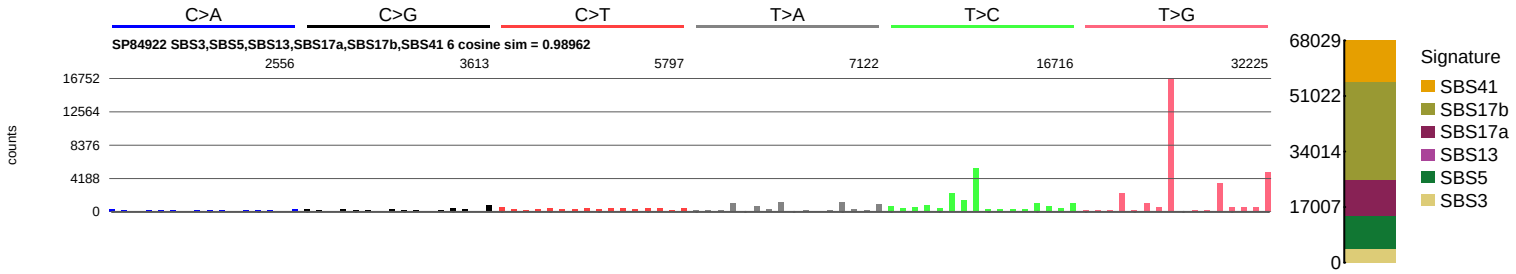
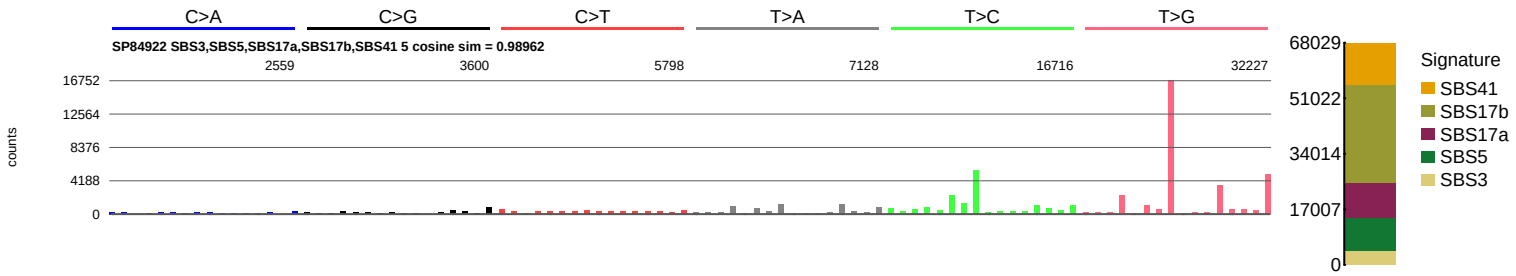
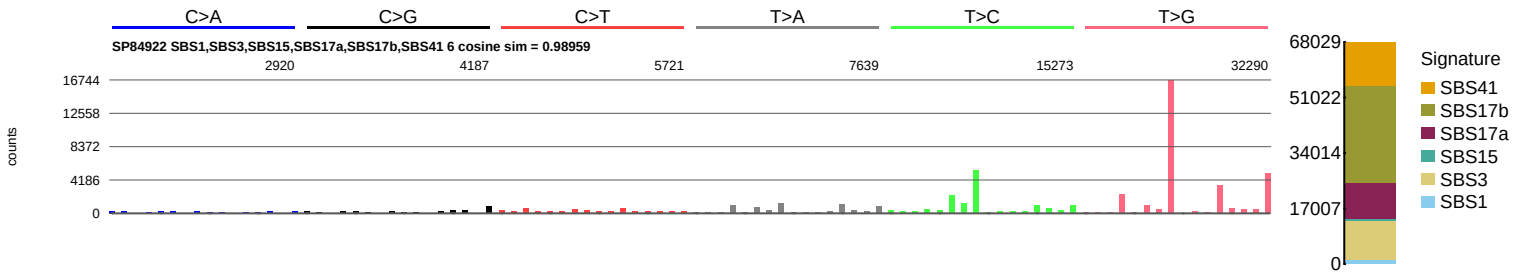
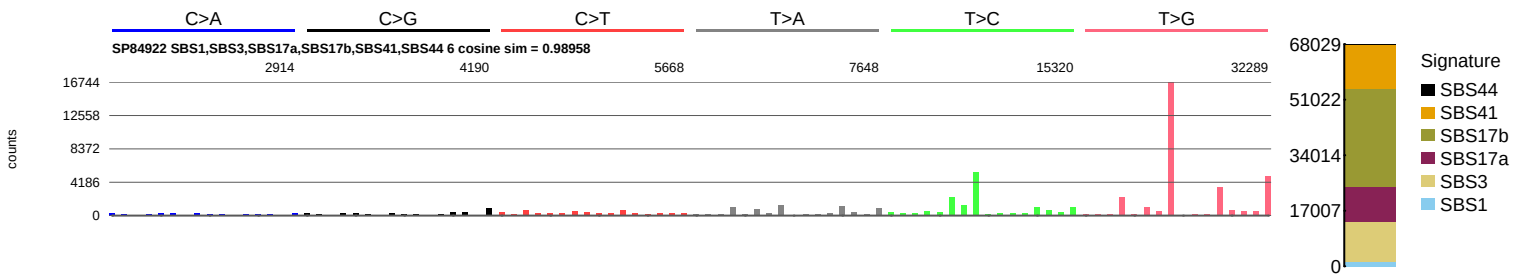
T>G
SP84922 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98964
2535
3580
5715
7079
16930
32191
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3

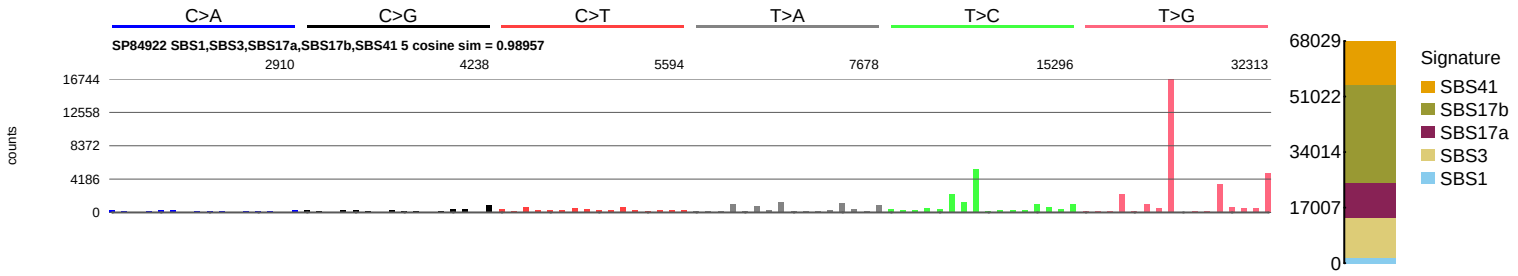

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98964
2536
3558
5714
7090
16932
32198
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3

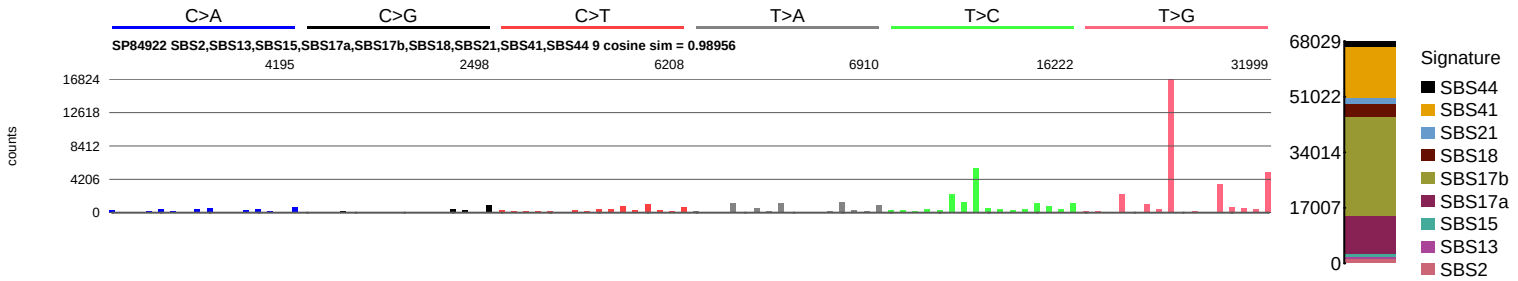

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98962
2556
3613
5797
7122
16716
32225
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS3

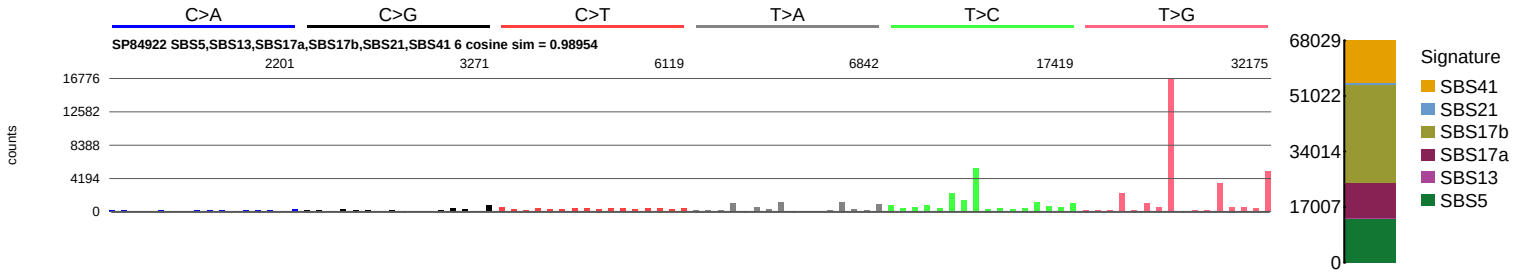

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98962
2559
3600
5798
7128
16716
32227
counts
16752
12564
8376
4188
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS3

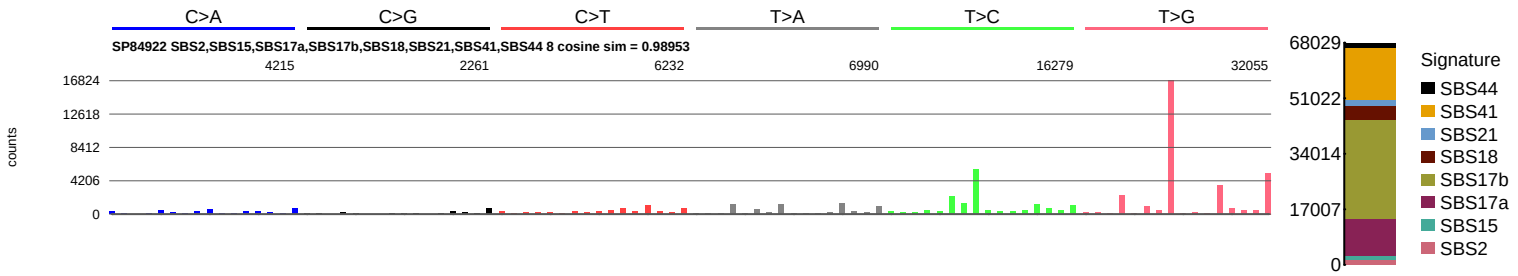

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98959
2920
4187
5721
7639
15273
32290
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS1

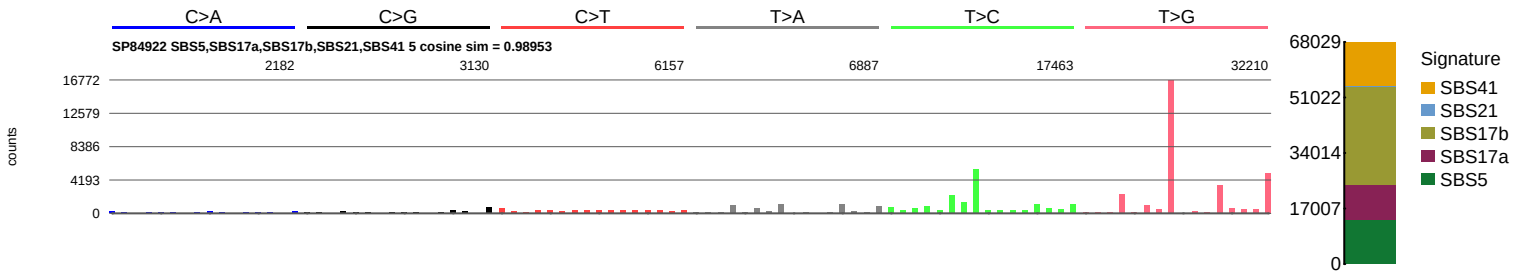

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98958
2914
4190
5668
7648
15320
32289
counts
16744
12558
8372
4186
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS3
SBS1

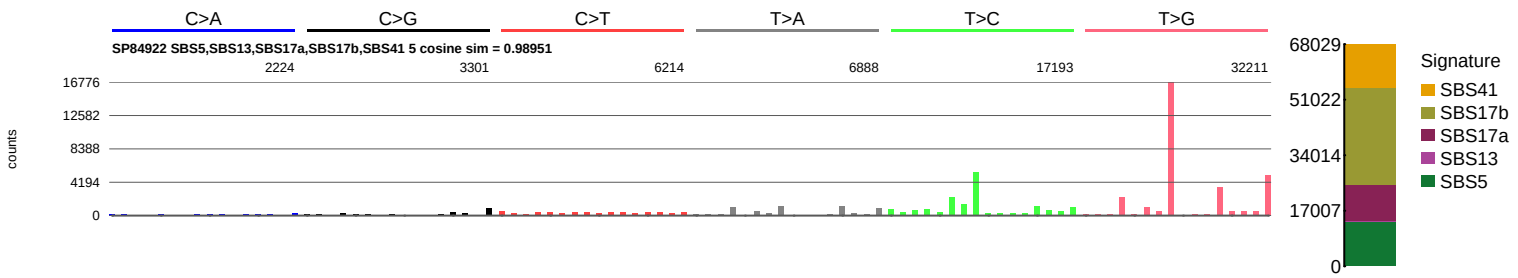

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98957
2910
4238
5594
7678
15296
32313
16744
12558
8372
4186
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS3
SBS1

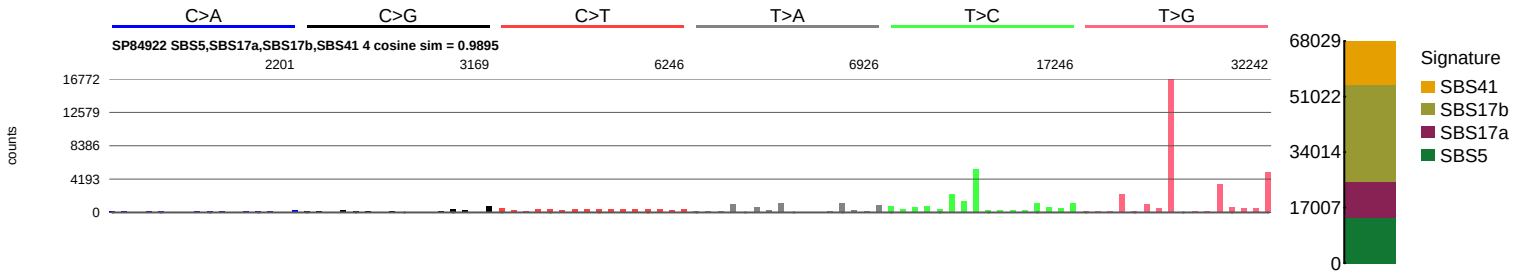

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98956
4195
2498
6208
6910
16222
31999
16824
12618
8412
4206
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2

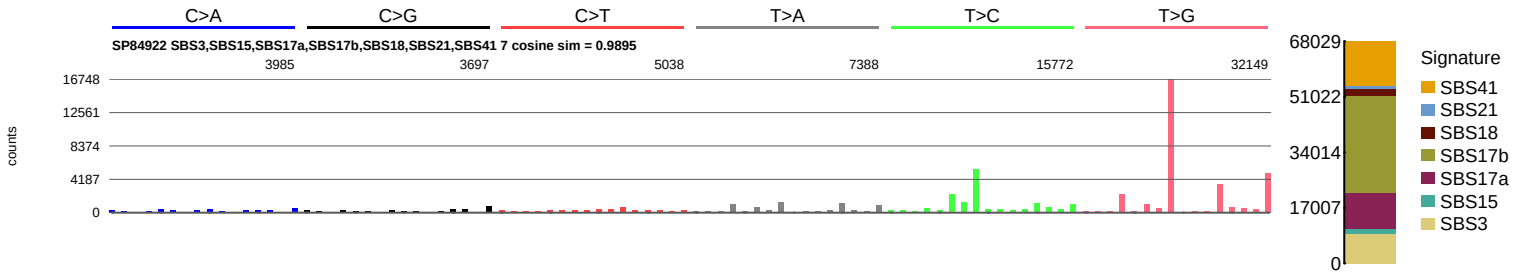

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98954
2201
3271
6119
6842
17419
32175
16776
12582
8388
4194
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5

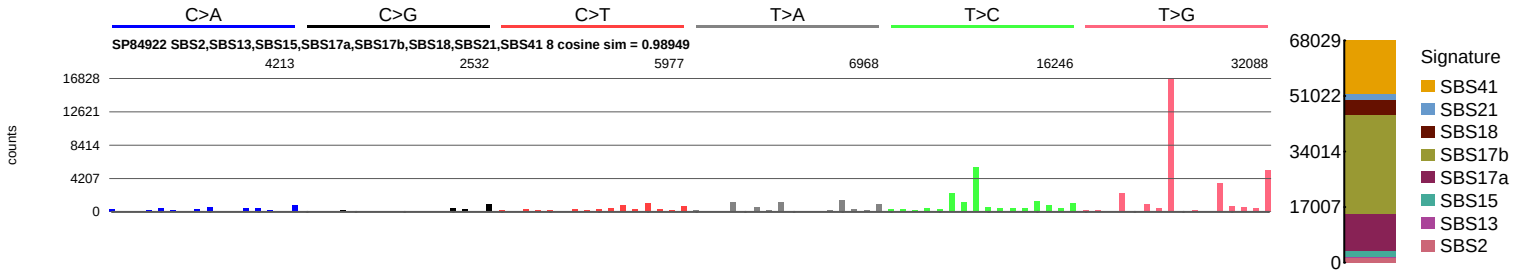

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98953
4215
2261
6232
6990
16279
32055
16824
12618
8412
4206
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2

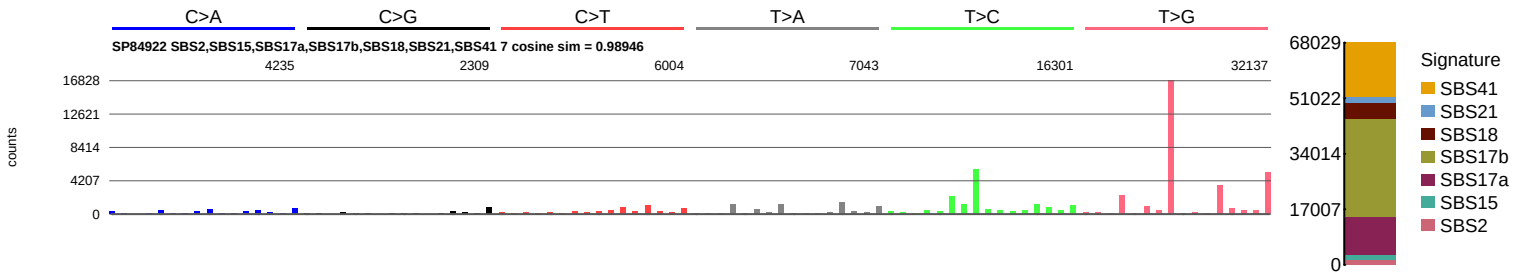

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS17a,SBS17b,SBS21,SBS41 5 cosine sim = 0.98953
2182
3130
6157
6887
17463
32210
16772
12579
8386
4193
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5

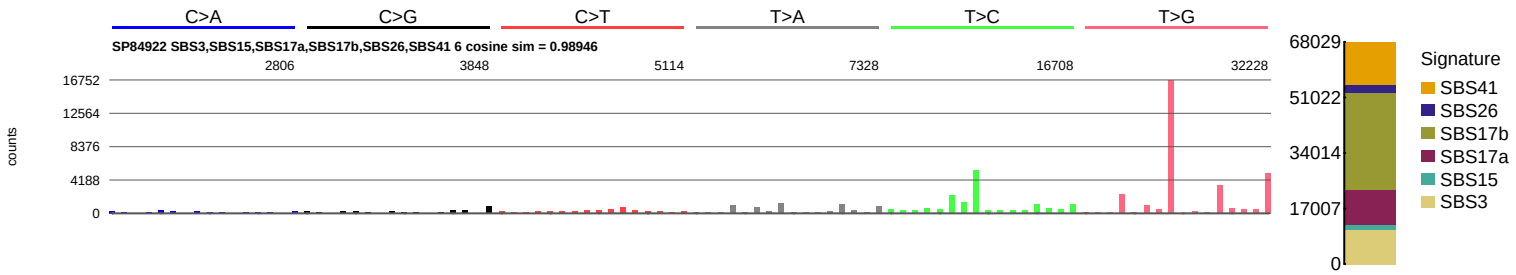

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98951
2224
3301
6214
6888
17193
32211
16776
12582
8388
4194
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5

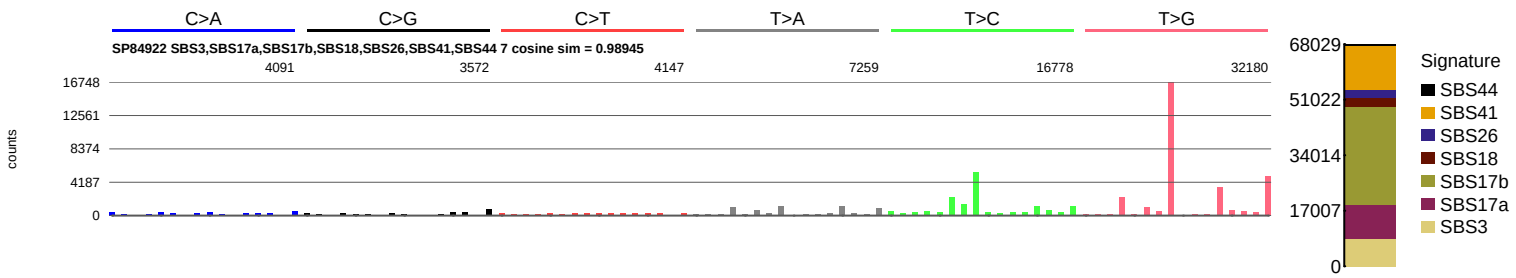

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS17a,SBS17b,SBS41 4 cosine sim = 0.9895
2201
3169
6246
6926
17246
32242
16772
12579
8386
4193
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS5

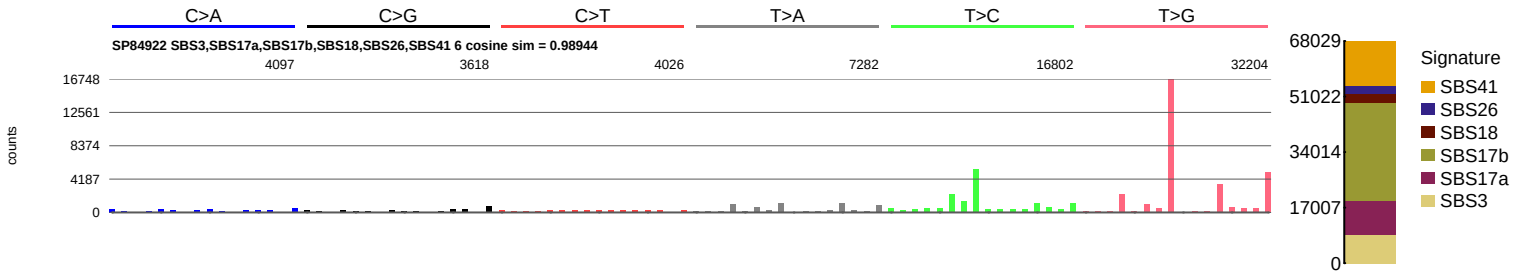

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.9895
3985
3697
5038
7388
15772
32149
16748
12561
8374
4187
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3

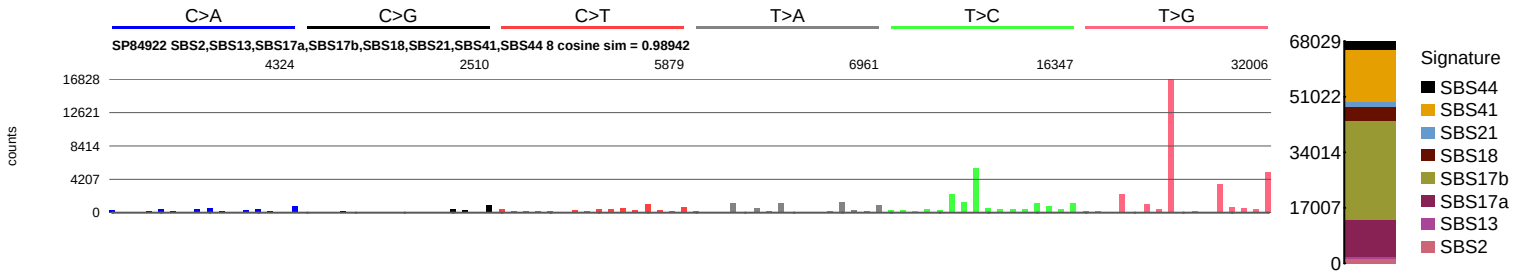

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98949
4213
2532
5977
6968
16246
32088
16828
12621
8414
4207
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2

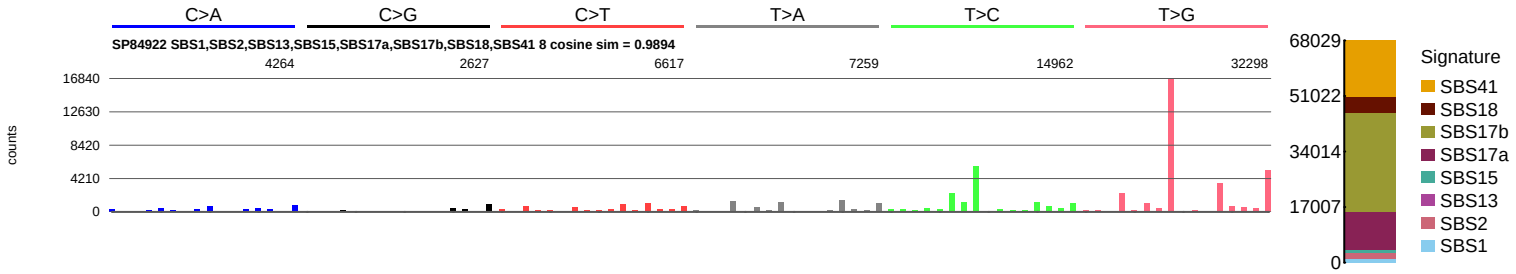

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98946
4235
2309
6004
7043
16301
32137
16828
12621
8414
4207
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98946
2806
3848
5114
7328
16708
32228
16752
12564
8376
4188
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98945
4091
3572
4147
7259
16778
32180
16748
12561
8374
4187
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.98944
4097
3618
4026
7282
16802
32204
16748
12561
8374
4187
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98942
4324
2510
5879
6961
16347
32006
16828
12621
8414
4207
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.9894
4264
2627
6617
7259
14962
32298
16840
12630
8420
4210
0
counts
68029
51022
34014
17007
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Signature
SBS41
SBS18
SBS17b
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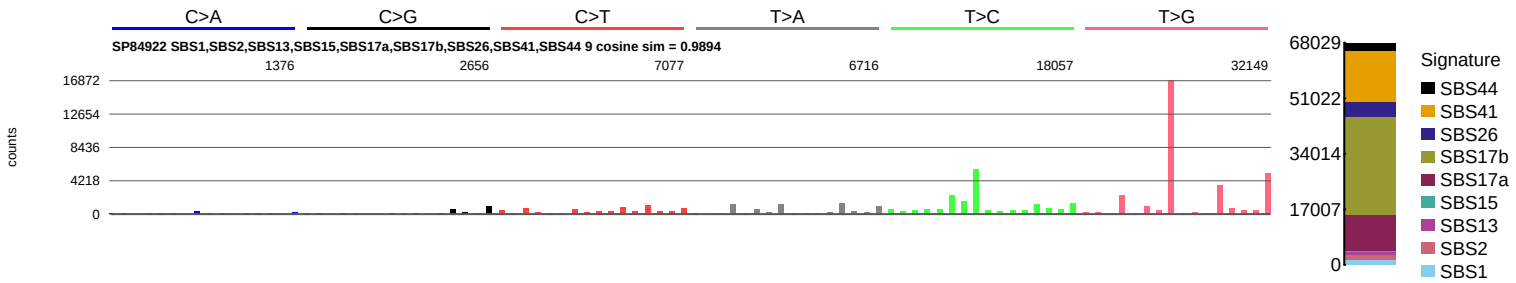
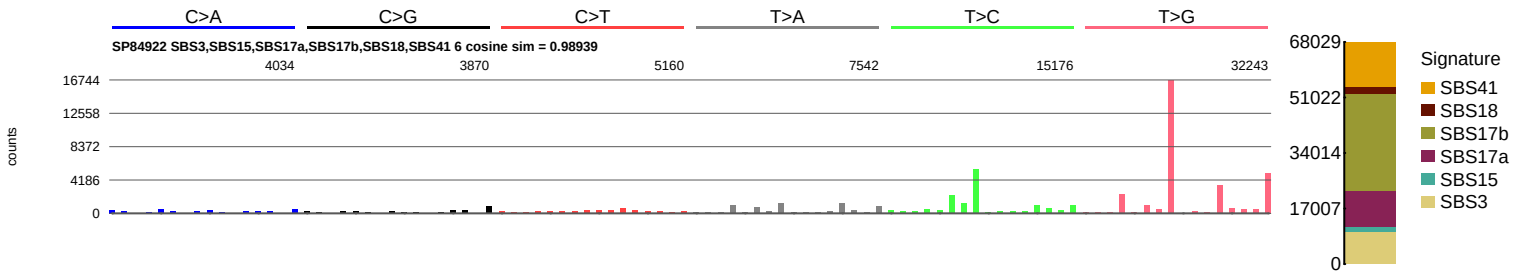
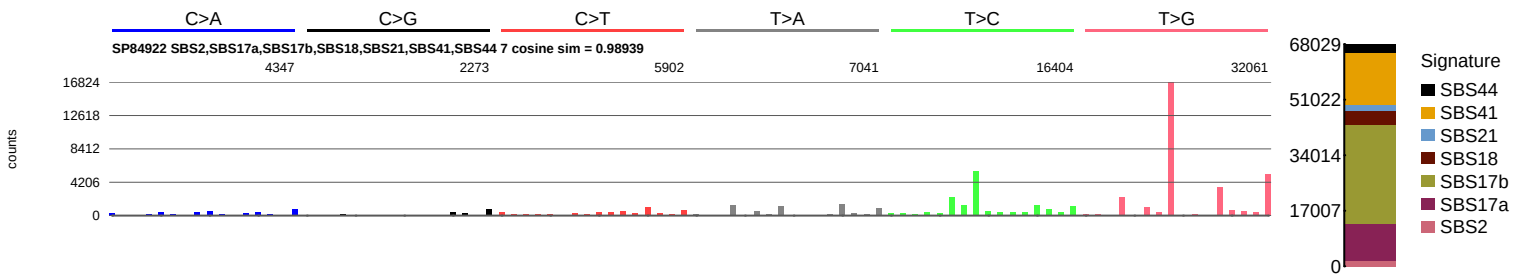
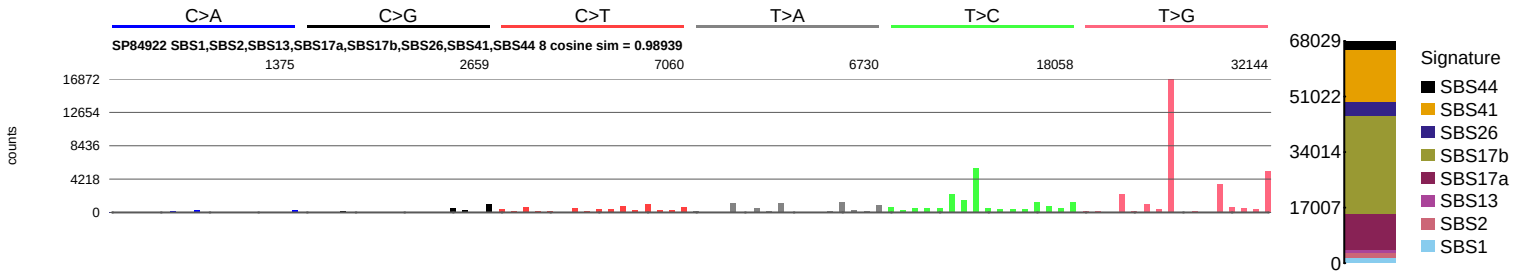
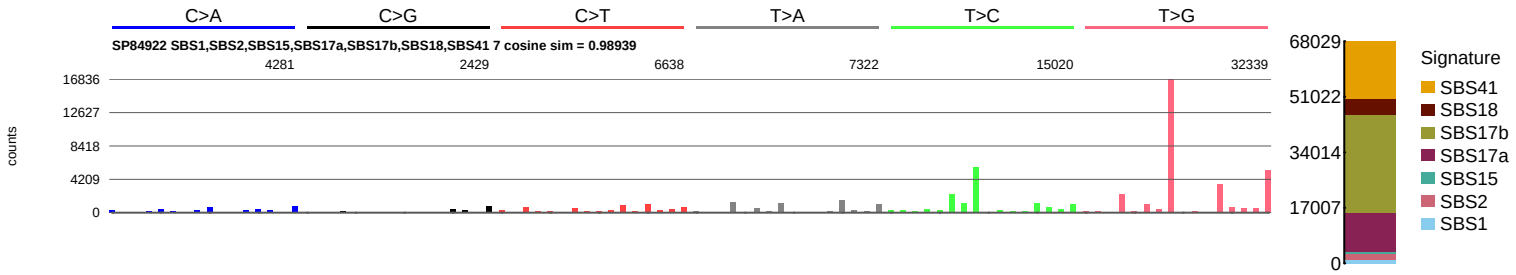
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.9894
1376
2656
7077
6716
18057
32149
16872
12654
8436
4218
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98939
4034
3870
5160
7542
15176
32243
16744
12558
8372
4186
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98939
4347
2273
5902
7041
16404
32061
16824
12618
8412
4206
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98939
1375
2659
7060
6730
18058
32144
16872
12654
8436
4218
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1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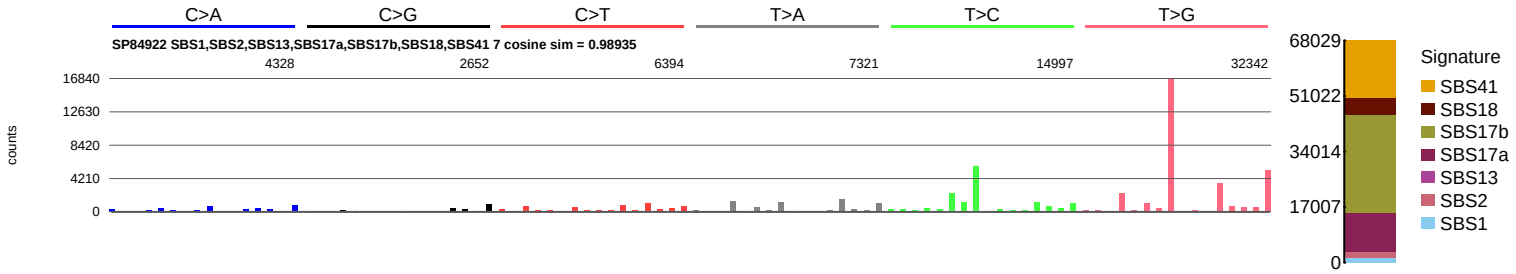
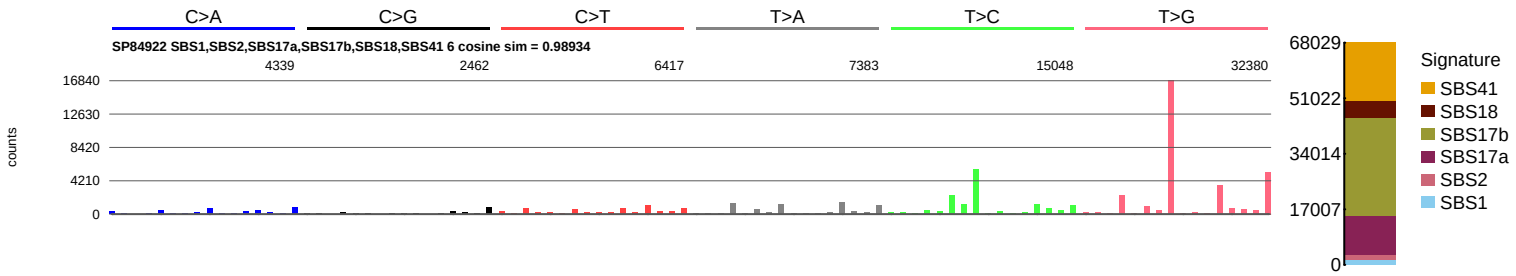
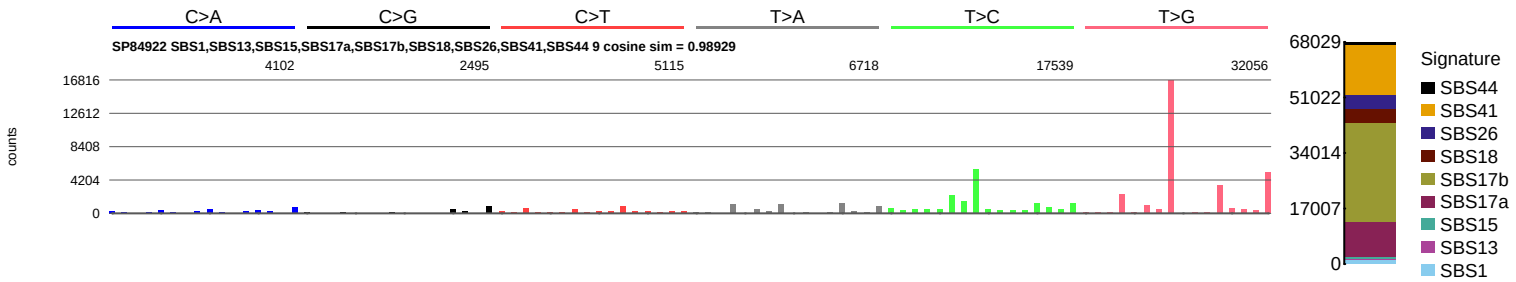
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98939
4281
2429
6638
7322
15020
32339
16836
12627
8418
4209
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98935
4328
2652
6394
7321
14997
32342
16840
12630
8420
4210
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98934
4339
2462
6417
7383
15048
32380
16840
12630
8420
4210
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS2
SBS1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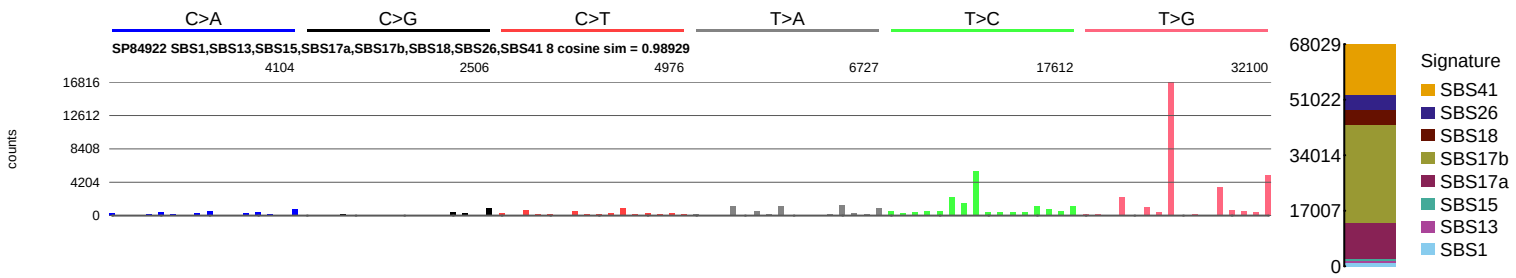
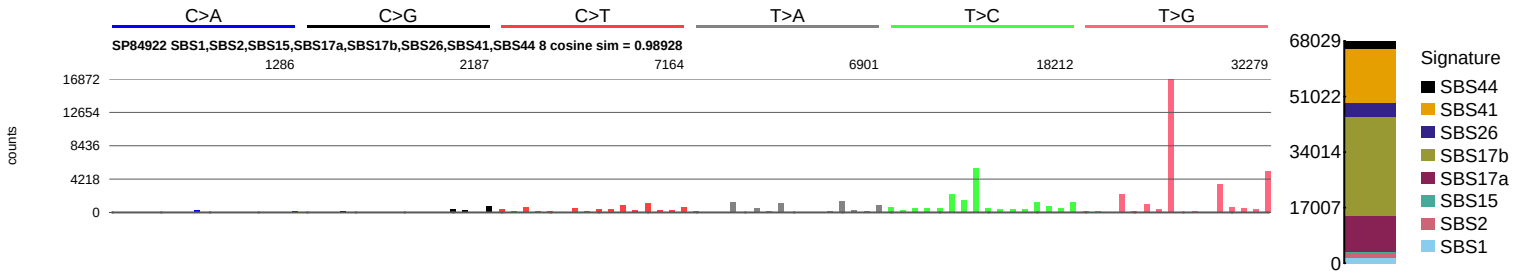
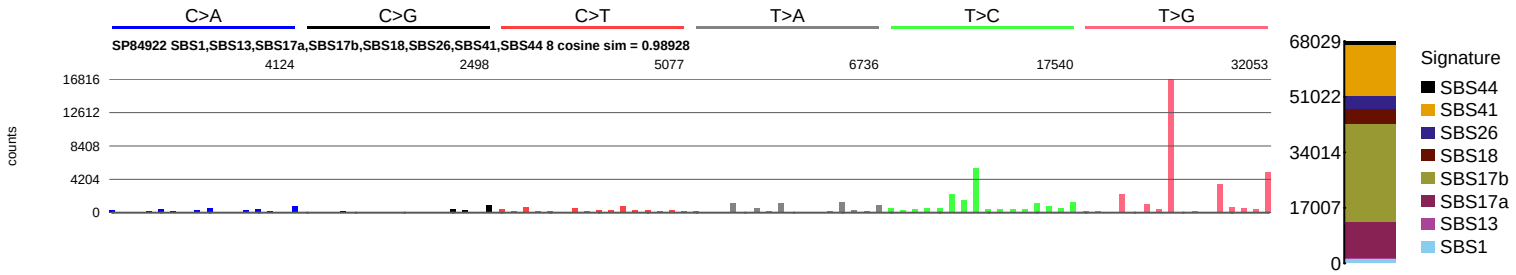
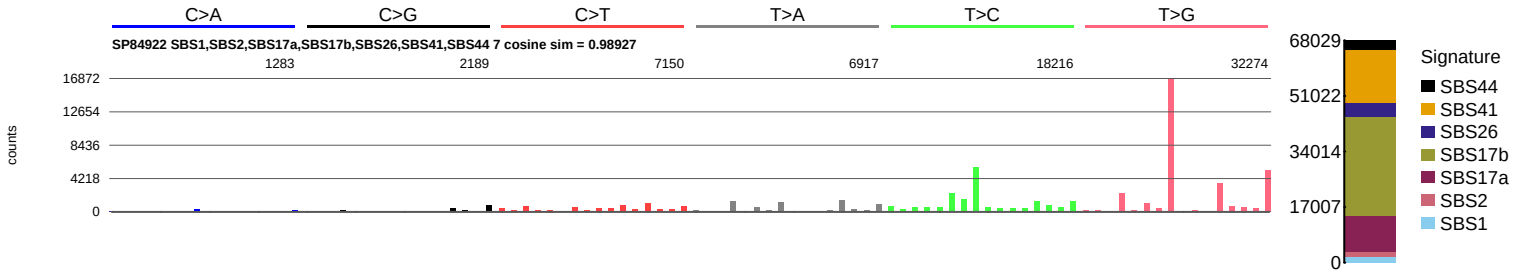
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98929
4102
2495
5115
6718
17539
32056
16816
12612
8408
4204
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98929
4104
2506
4976
6727
17612
32100
16816
12612
8408
4204
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98928
1286
2187
7164
6901
18212
32279
16872
12654
8436
4218
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98928
4124
2498
5077
6736
17540
32053
16816
12612
8408
4204
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98927
1283
2189
7150
6917
18216
32274
16872
12654
8436
4218
0
counts
68029
51022
34014
17007
0
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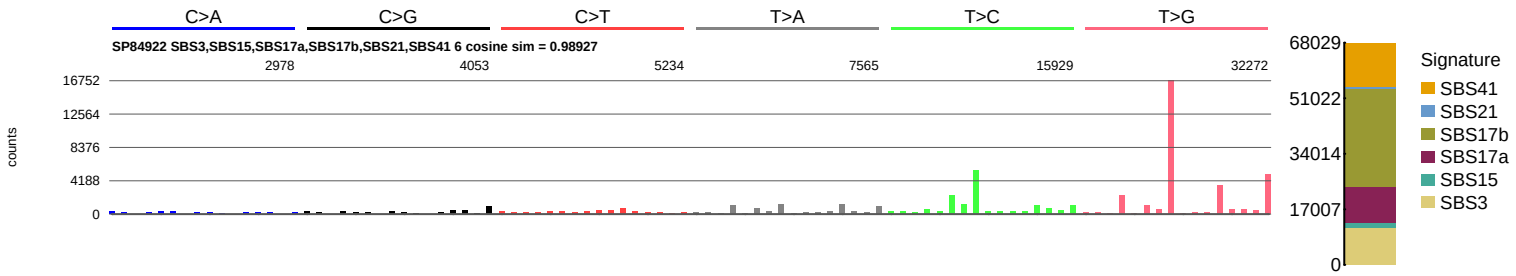
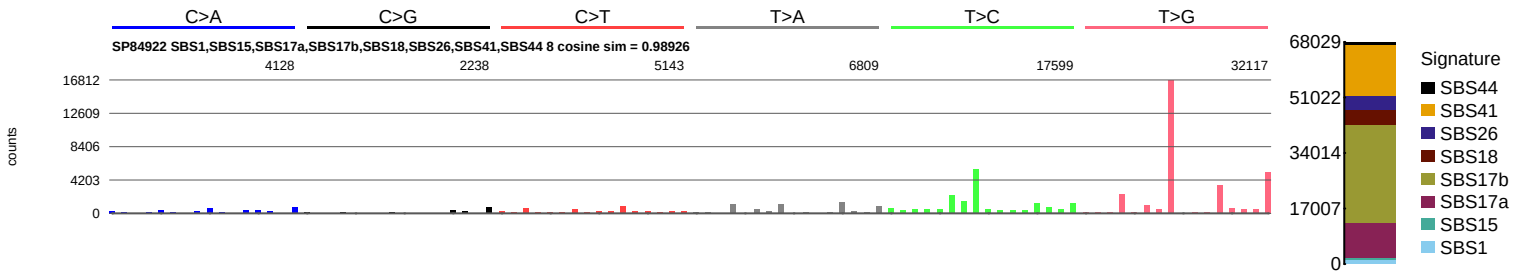
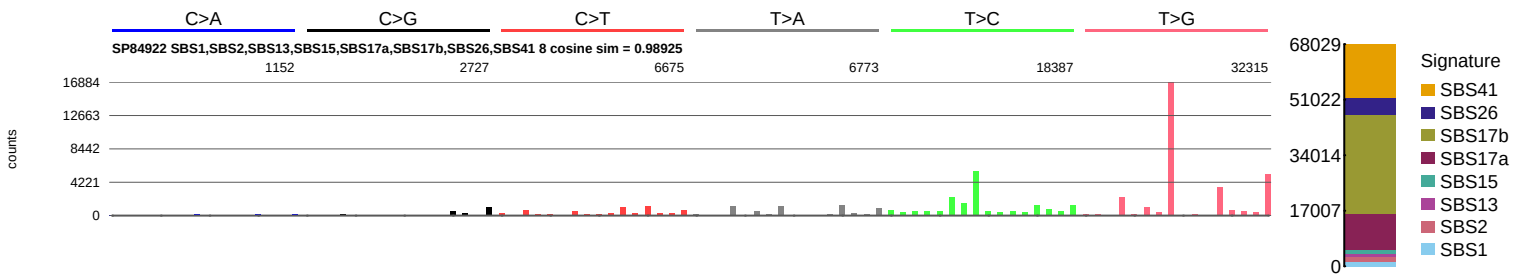
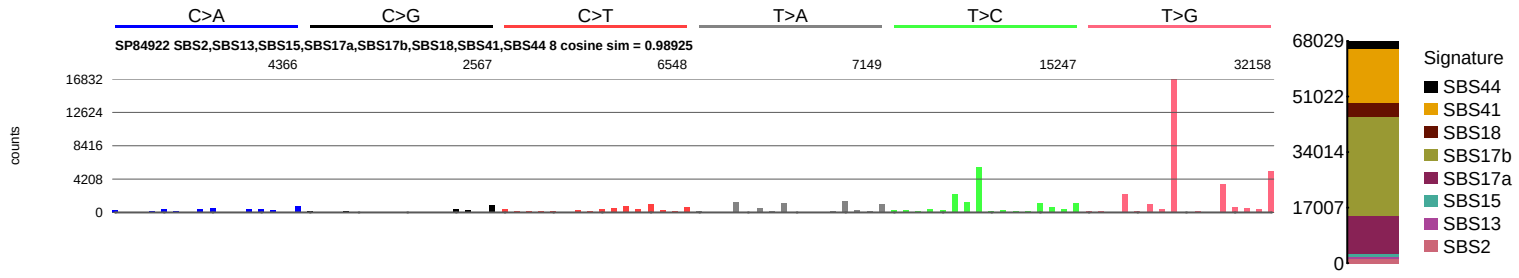
SBS41
SBS26
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98927
2978
4053
5234
7565
15929
32272
16752
12564
8376
4188
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98926
4128
2238
5143
6809
17599
32117
16812
12609
8406
4203
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.98925
1152
2727
6675
6773
18387
32315
16884
12663
8442
4221
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98925
4366
2567
6548
7149
15247
32158
16832
12624
8416
4208
0
counts
68029
51022
34014
17007
0
Signature
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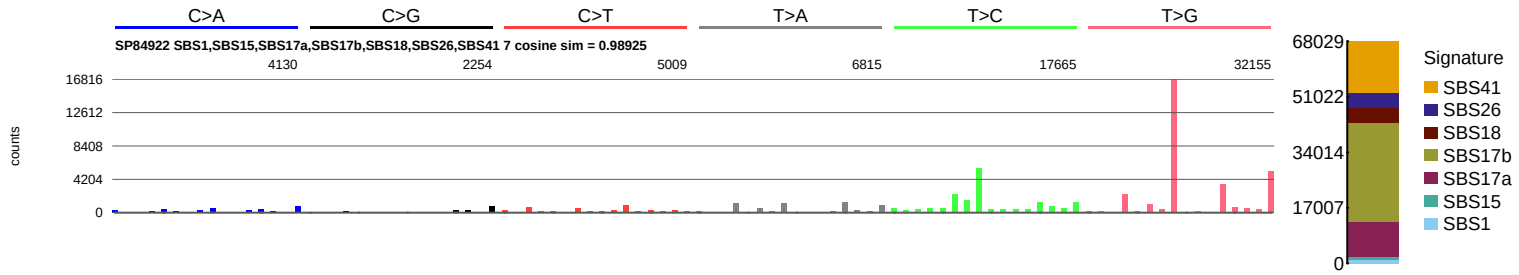
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98925
4130
2254
5009
6815
17665
32155
16816
12612
8408
4204
0
counts
68029
51022
34014
17007
0
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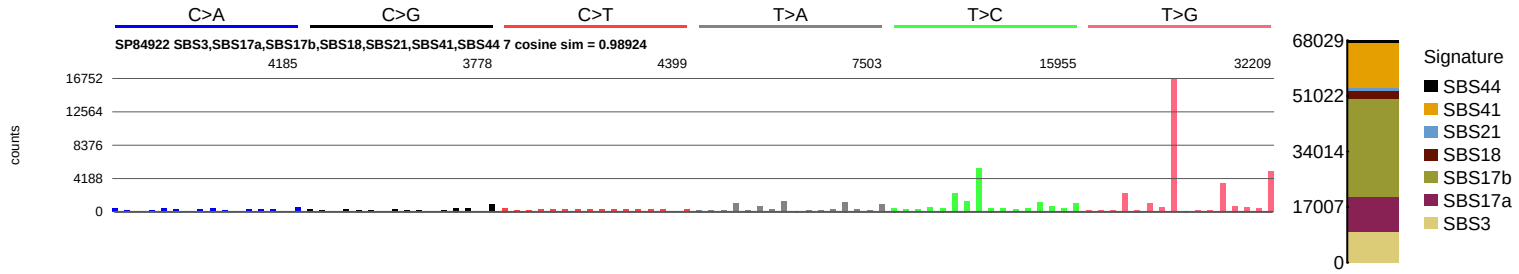
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98924
4185
3778
4399
7503
15955
32209
16752
12564
8376
4188
0
counts
68029
51022
34014
17007
0
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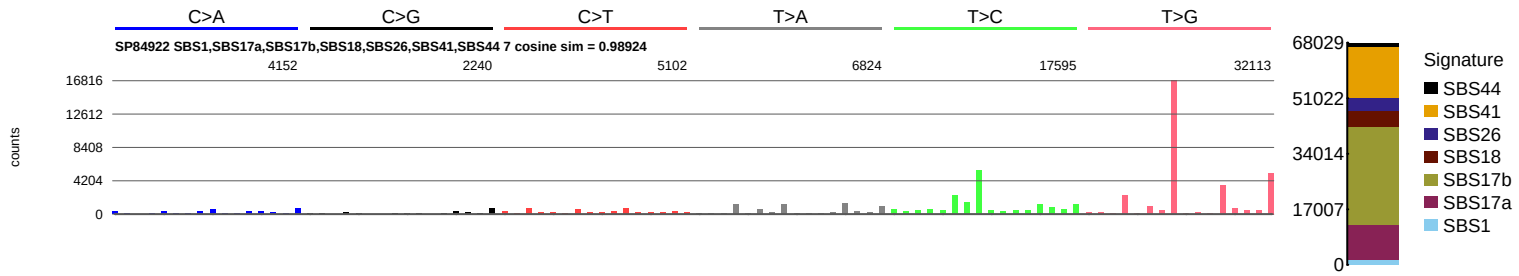
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98924
4152
2240
5102
6824
17595
32113
16816
12612
8408
4204
0
counts
68029
51022
34014
17007
0
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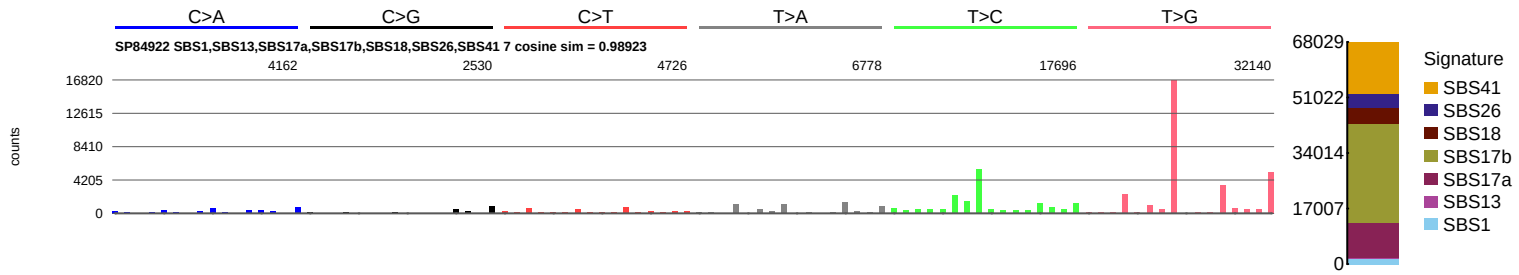
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98923
4162
2530
4726
6778
17696
32140
16820
12615
8410
4205
0
counts
68029
51022
34014
17007
0
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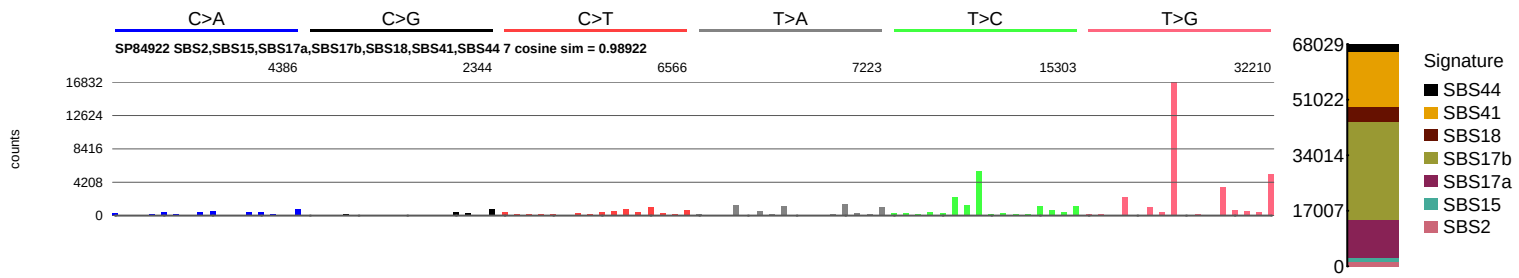
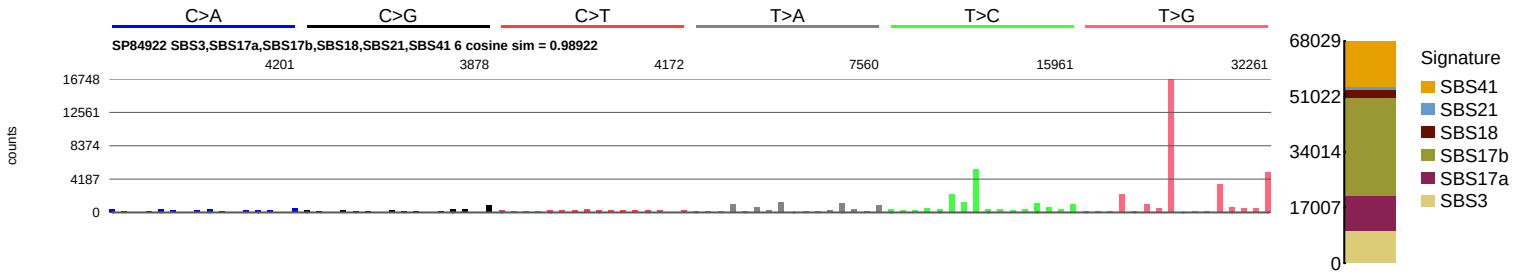
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98922
4386
2344
6566
7223
15303
32210
16832
12624
8416
4208
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98922
4201
3878
4172
7560
15961
32261
counts
16748
12561
8374
4187
0
68029
51022
34014
17007
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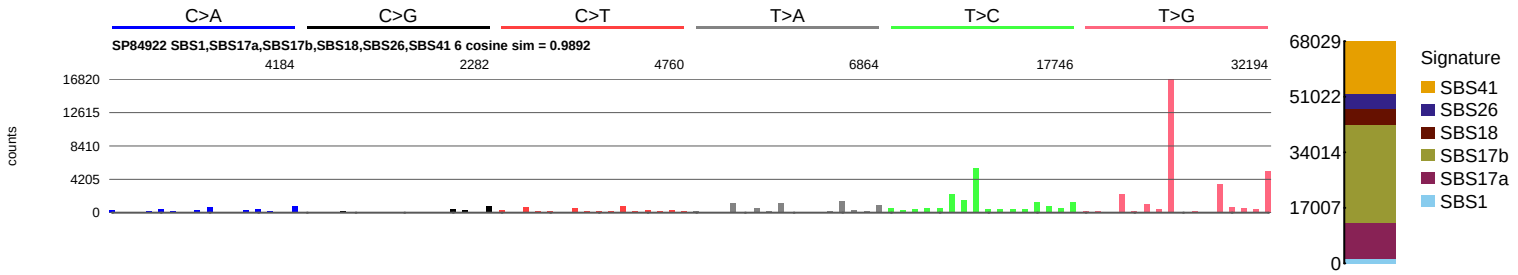
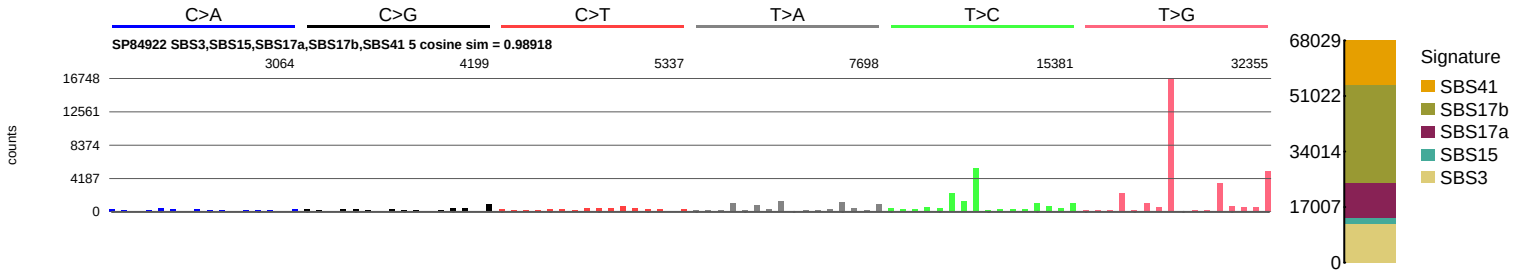
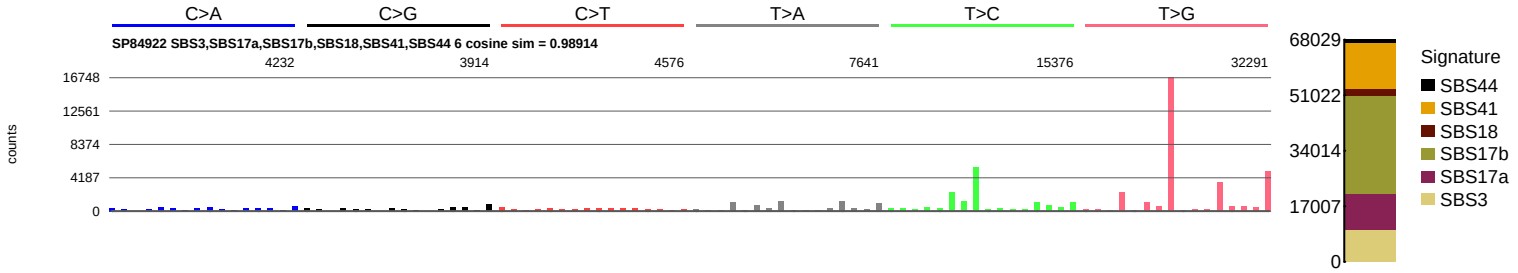
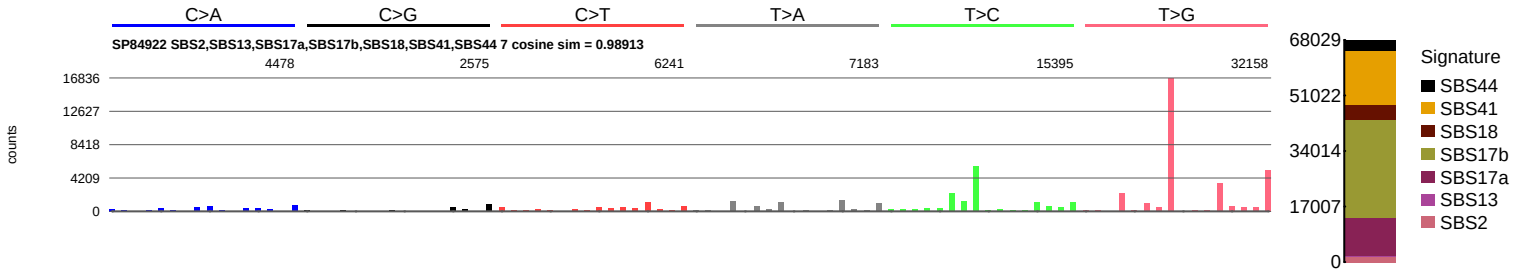
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.9892
4184
2282
4760
6864
17746
32194
counts
16820
12615
8410
4205
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS15,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98918
3064
4199
5337
7698
15381
32355
counts
16748
12561
8374
4187
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98914
4232
3914
4576
7641
15376
32291
counts
16748
12561
8374
4187
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98913
4478
2575
6241
7183
15395
32158
counts
16836
12627
8418
4209
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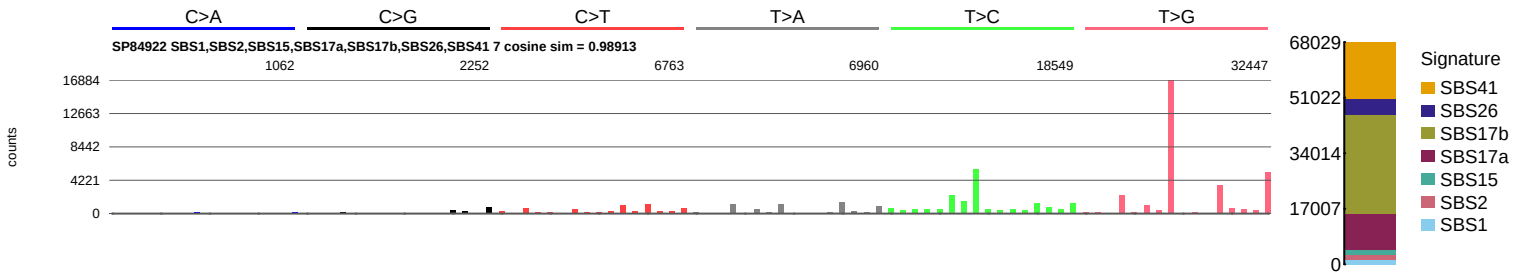
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2252
6763
6960
18549
32447
counts
16884
12663
8442
4221
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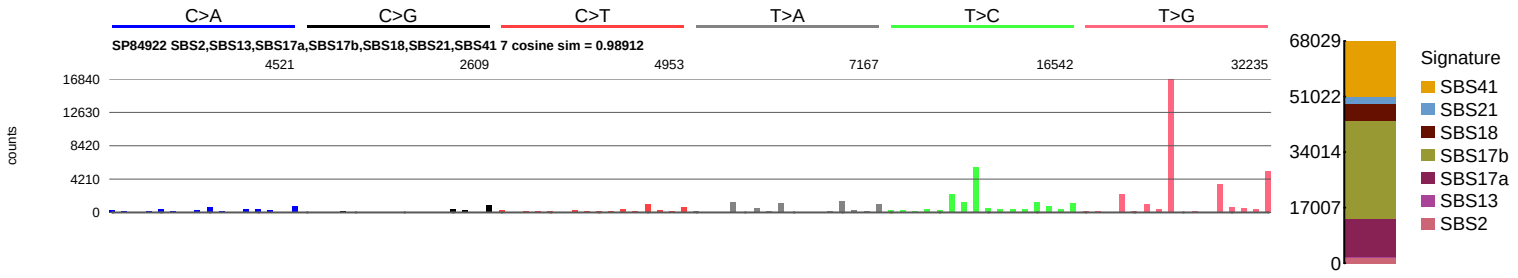
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SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98912
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2609
4953
7167
16542
32235
16840
12630
8420
4210
0
counts
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SBS17b
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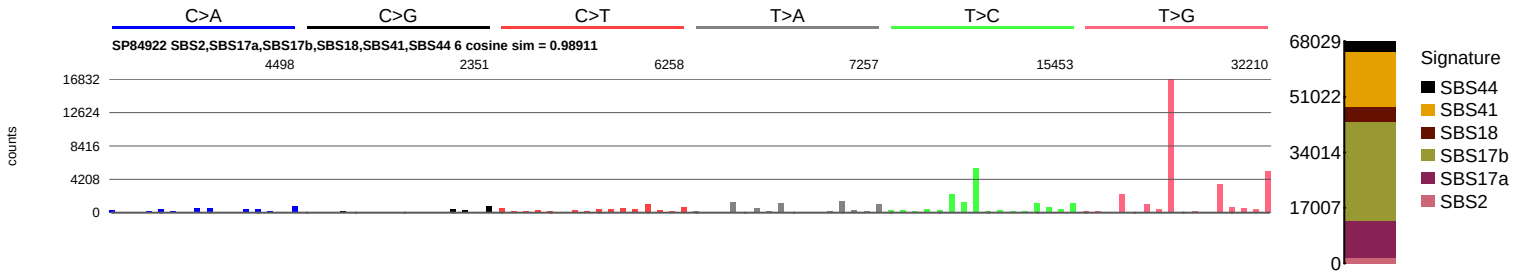
SBS13
SBS2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SP84922 SBS2,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98911
4498
2351
6258
7257
15453
32210
16832
12624
8416
4208
0
counts
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17007
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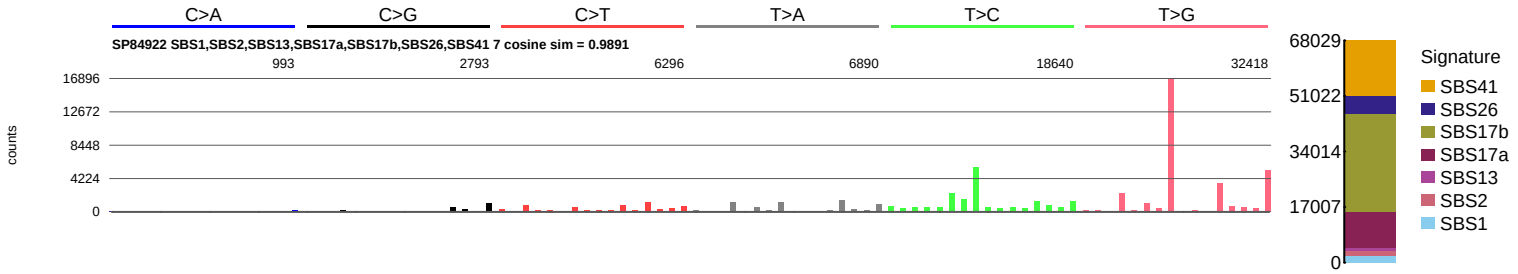
SBS2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C>T
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SP84922 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.9891
993
2793
6296
6890
18640
32418
16896
12672
8448
4224
0
counts
68029
51022
34014
17007
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SBS17a
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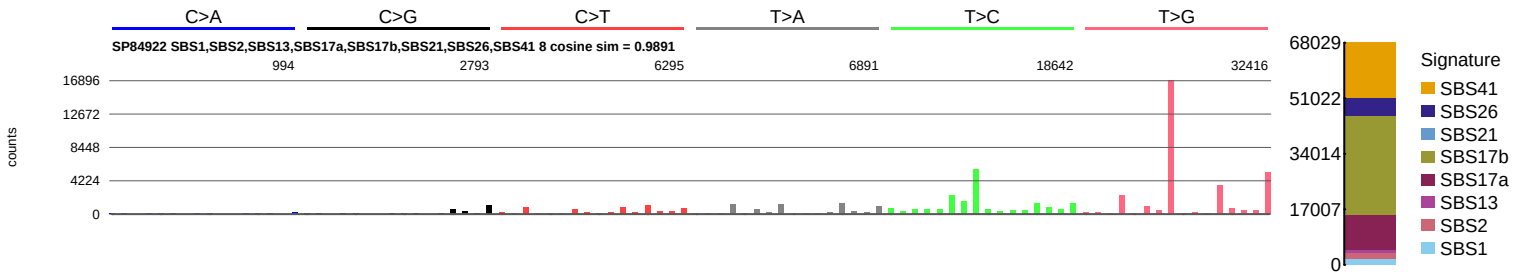
SBS2
SBS1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SP84922 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41 8 cosine sim = 0.9891
994
2793
6295
6891
18642
32416
16896
12672
8448
4224
0
counts
68029
51022
34014
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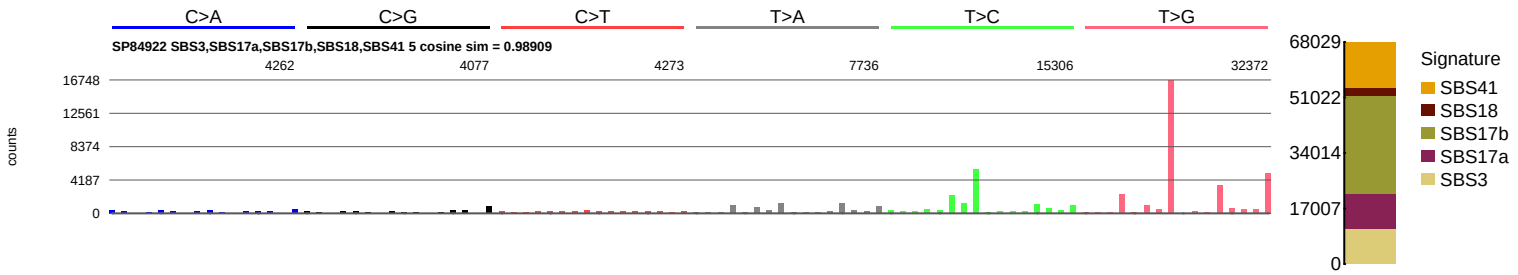
SBS2
SBS1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SP84922 SBS3,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98909
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4077
4273
7736
15306
32372
16748
12561
8374
4187
0
counts
68029
51022
34014
17007
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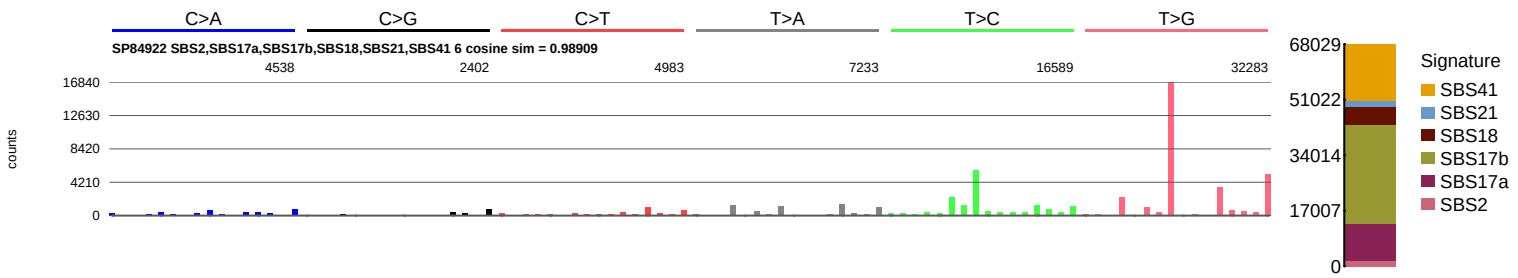
SBS18
SBS17b
SBS17a
SBS3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SP84922 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98909
4538
2402
4983
7233
16589
32283
16840
12630
8420
4210
0
counts
68029
51022
34014
17007
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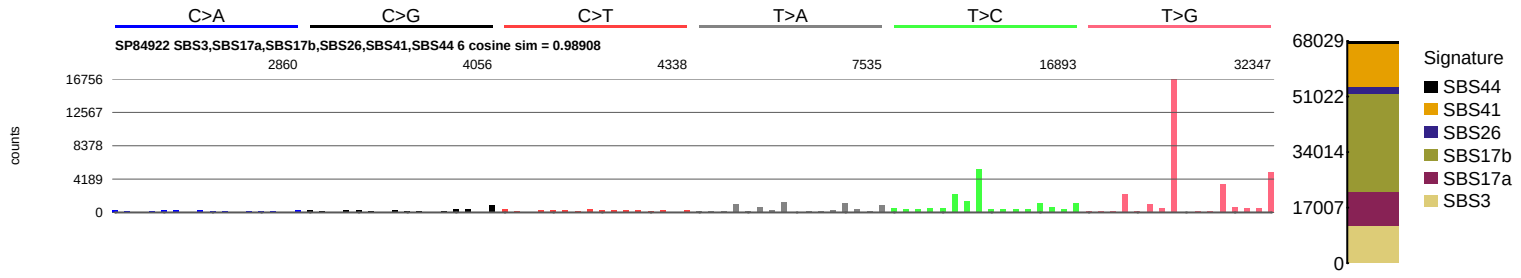
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
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SP84922 SBS3,SBS17a,SBS17b,SBS26,SBS41,SBS44 6 cosine sim = 0.98908
2860
4056
4338
7535
16893
32347
counts
16756
12567
8378
4189
0
68029
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34014
17007
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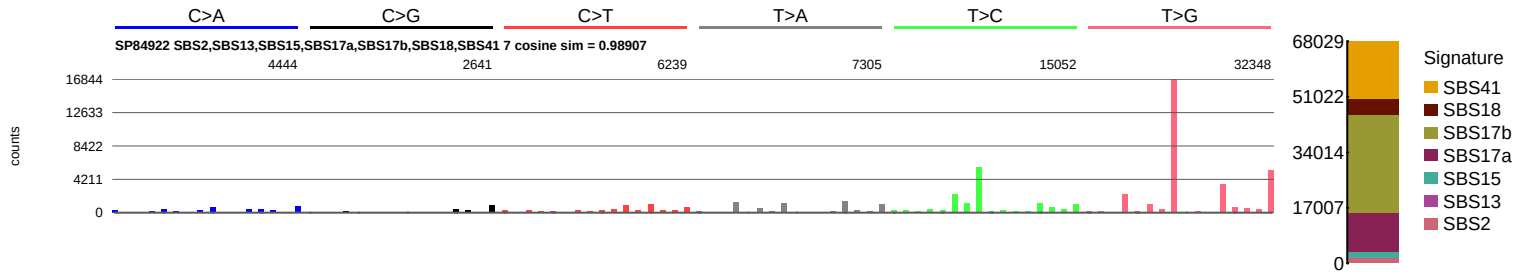
SBS26
SBS17b
SBS17a
SBS3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T>C
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SP84922 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98907
4444
2641
6239
7305
15052
32348
counts
16844
12633
8422
4211
0
68029
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17007
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SBS17b
SBS17a
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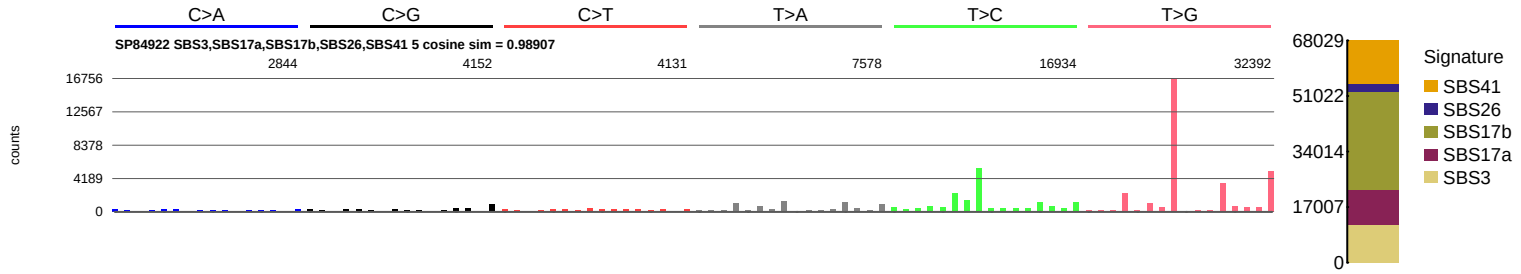
SBS13
SBS2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C>G
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T>A
T>C
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SP84922 SBS3,SBS17a,SBS17b,SBS26,SBS41 5 cosine sim = 0.98907
2844
4152
4131
7578
16934
32392
counts
16756
12567
8378
4189
0
68029
51022
34014
17007
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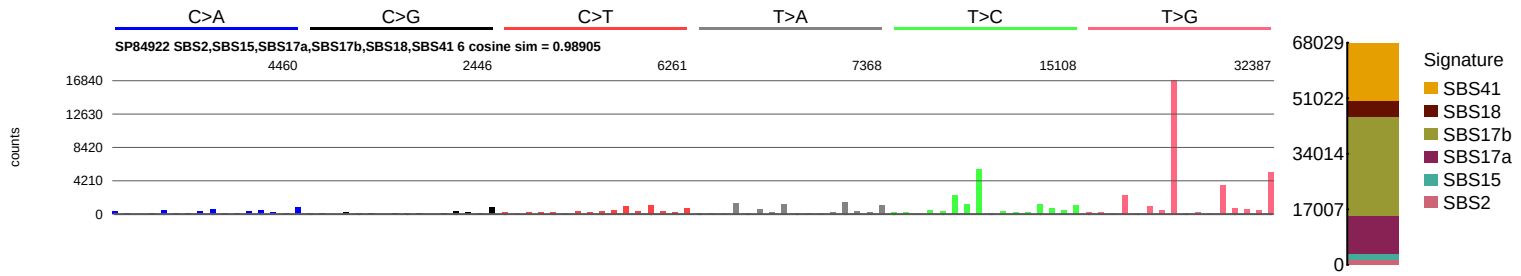
SBS26
SBS17b
SBS17a
SBS3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T>C
T>G
SP84922 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98905
4460
2446
6261
7368
15108
32387
counts
16840
12630
8420
4210
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68029
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34014
17007
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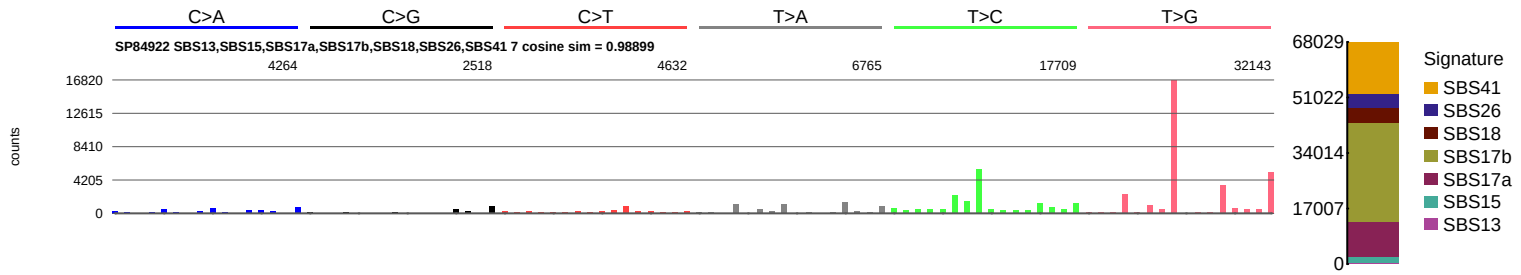
SBS17b
SBS17a
SBS15
SBS2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C>T
T>A
T>C
T>G
SP84922 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98899
4264
2518
4632
6765
17709
32143
counts
16820
12615
8410
4205
0
68029
51022
34014
17007
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SBS26
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SBS17b
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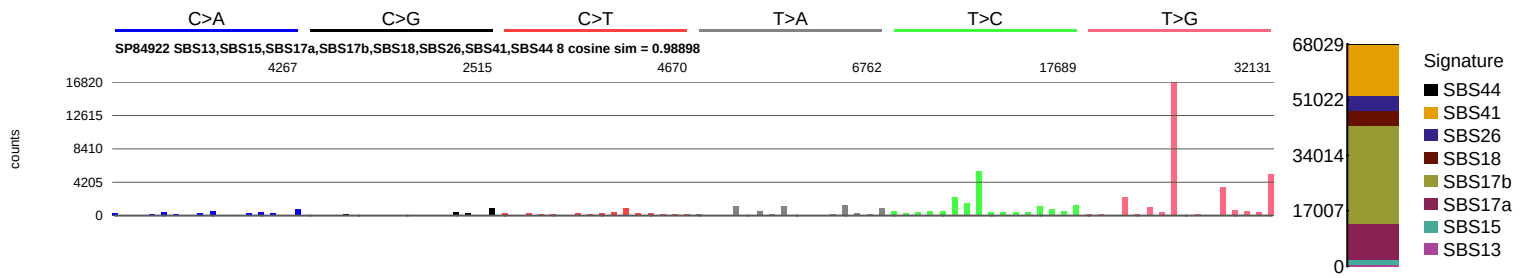
SBS15
SBS13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T>C
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SP84922 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98898
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2515
4670
6762
17689
32131
counts
16820
12615
8410
4205
0
68029
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34014
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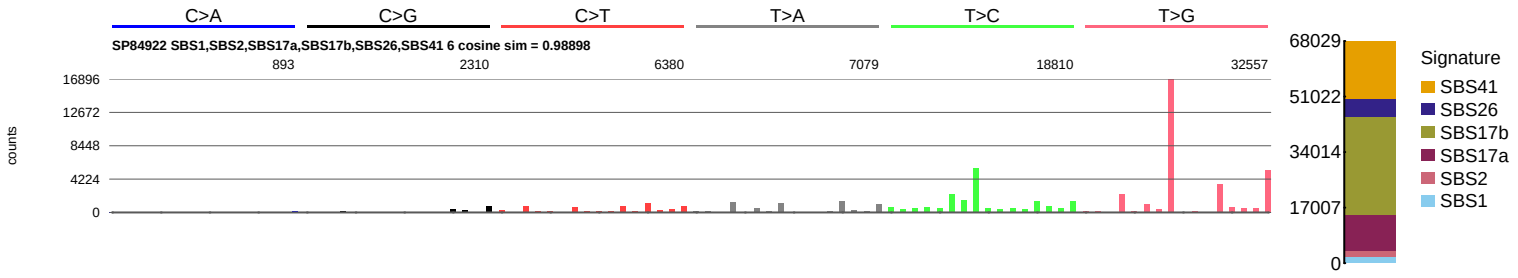
SBS15
SBS13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T>C
T>G
SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98898
893
2310
6380
7079
18810
32557
counts
16896
12672
8448
4224
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68029
51022
34014
17007
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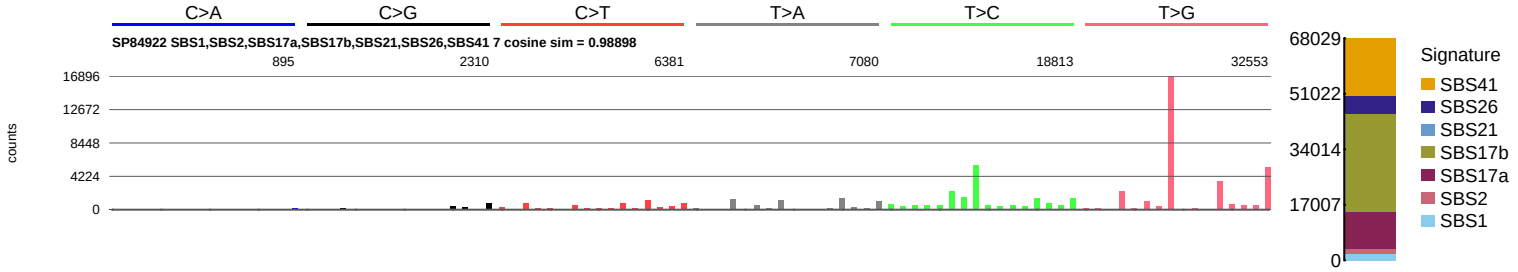
SBS17a
SBS2
SBS1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T>C
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SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS41 7 cosine sim = 0.98898
895
2310
6381
7080
18813
32553
counts
16896
12672
8448
4224
0
68029
51022
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17007
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SBS26
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SBS17b
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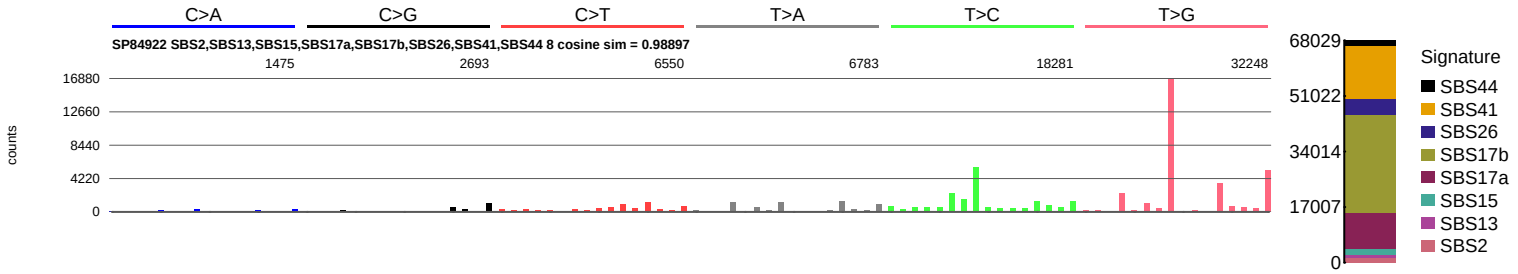
SBS2
SBS1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SP84922 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98897
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2693
6550
6783
18281
32248
counts
16880
12660
8440
4220
0
68029
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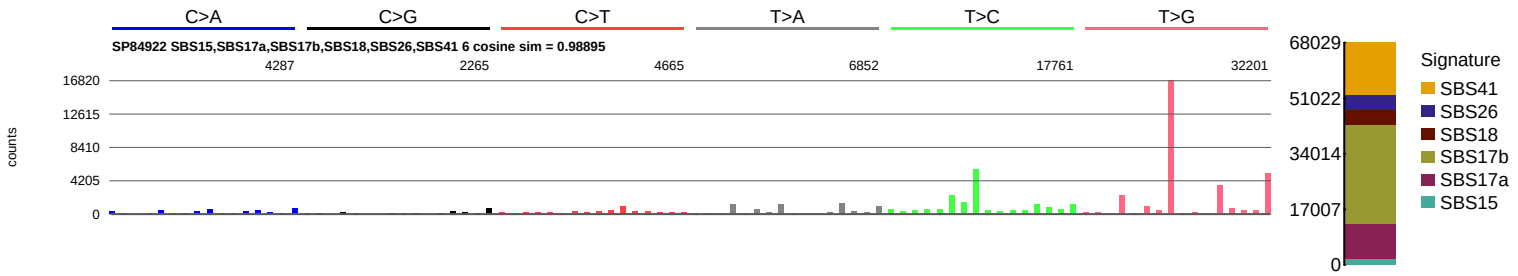
SBS13
SBS2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SP84922 SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.98895
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2265
4665
6852
17761
32201
counts
16820
12615
8410
4205
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68029
51022
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17007
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SBS41
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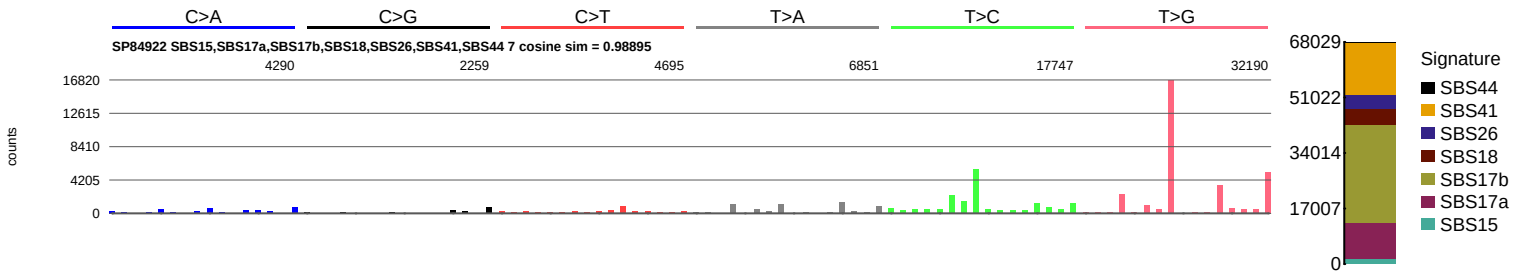
SBS17b
SBS17a
SBS15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T>C
T>G
SP84922 SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98895
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2259
4695
6851
17747
32190
counts
16820
12615
8410
4205
0
68029
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17007
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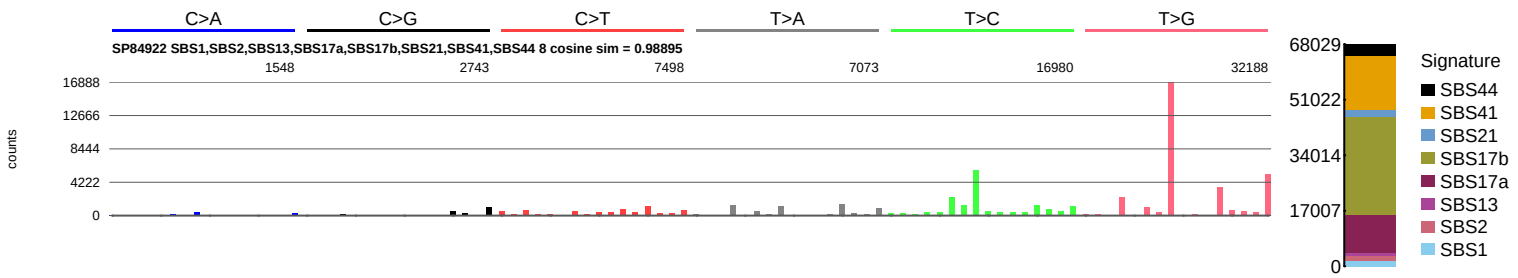
SBS15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T>C
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SP84922 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98895
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2743
7498
7073
16980
32188
counts
16888
12666
8444
4222
0
68029
51022
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17007
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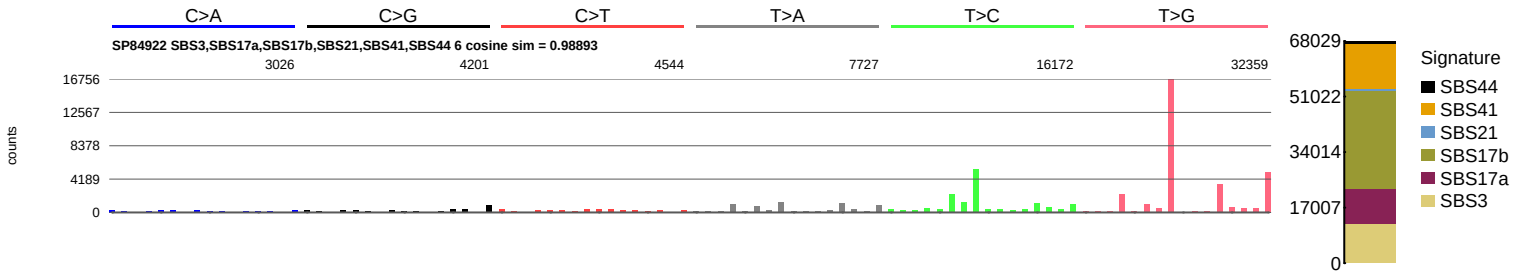
SBS2
SBS1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T>C
T>G
SP84922 SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 6 cosine sim = 0.98893
3026
4201
4544
7727
16172
32359
counts
16756
12567
8378
4189
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68029
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17007
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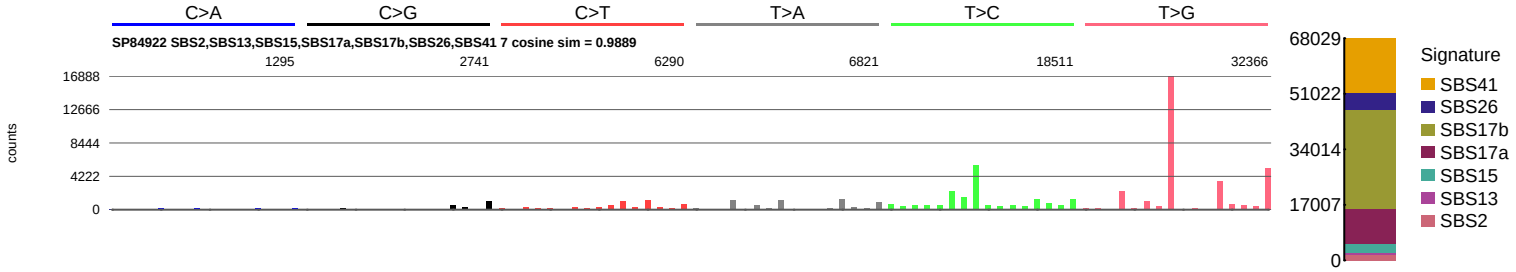
SBS3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T>C
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SP84922 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.9889
1295
2741
6290
6821
18511
32366
counts
16888
12666
8444
4222
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68029
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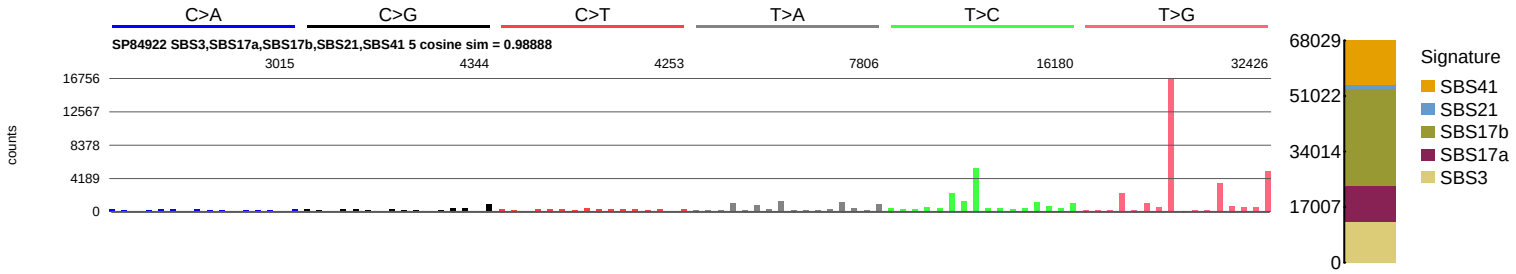
SBS2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T>C
T>G
SP84922 SBS3,SBS17a,SBS17b,SBS21,SBS41 5 cosine sim = 0.98888
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4344
4253
7806
16180
32426
counts
16756
12567
8378
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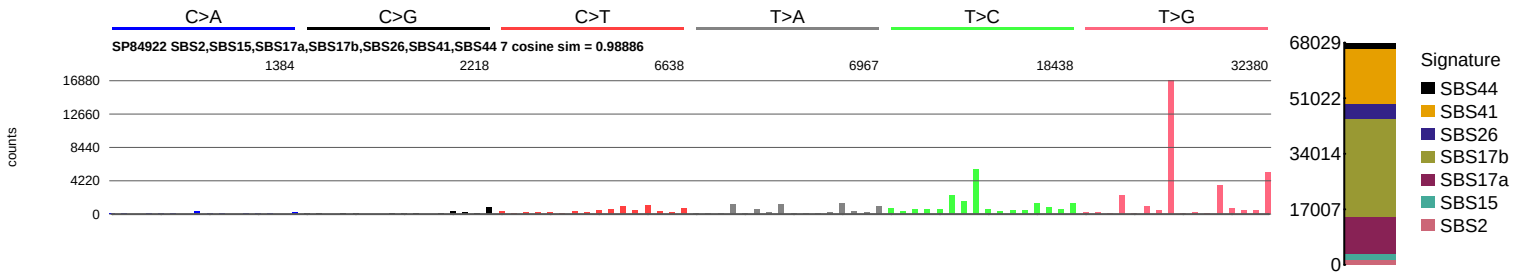
SBS3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T>C
T>G
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1384
2218
6638
6967
18438
32380
counts
16880
12660
8440
4220
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68029
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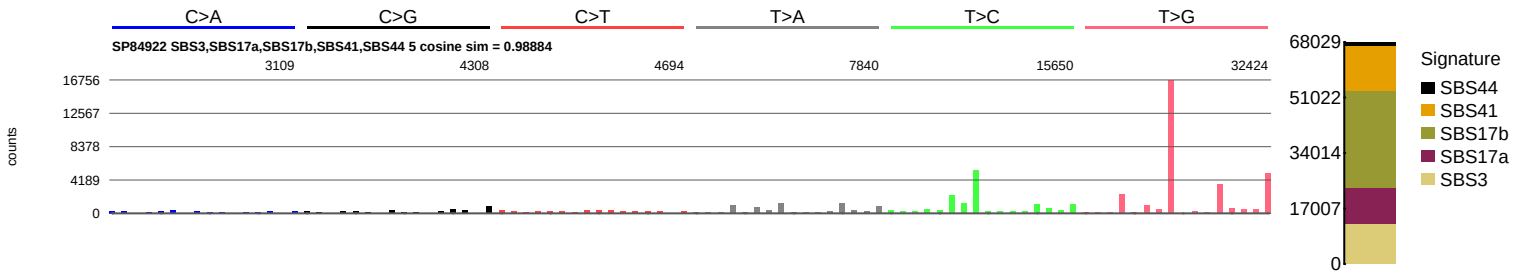
SBS2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T>C
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SP84922 SBS3,SBS17a,SBS17b,SBS41,SBS44 5 cosine sim = 0.98884
3109
4308
4694
7840
15650
32424
counts
16756
12567
8378
4189
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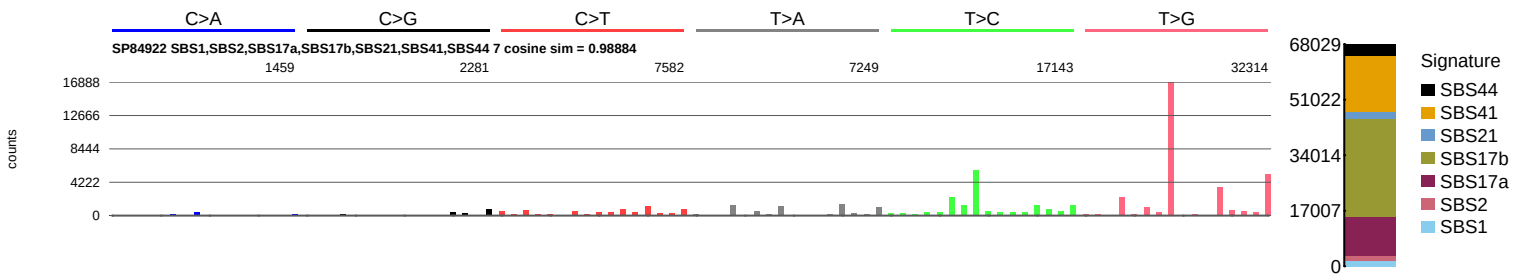
SBS3


C>A
C>G
C>T
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T>C
T>G
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1459
2281
7582
7249
17143
32314
counts
16888
12666
8444
4222
0
68029
51022
34014
17007
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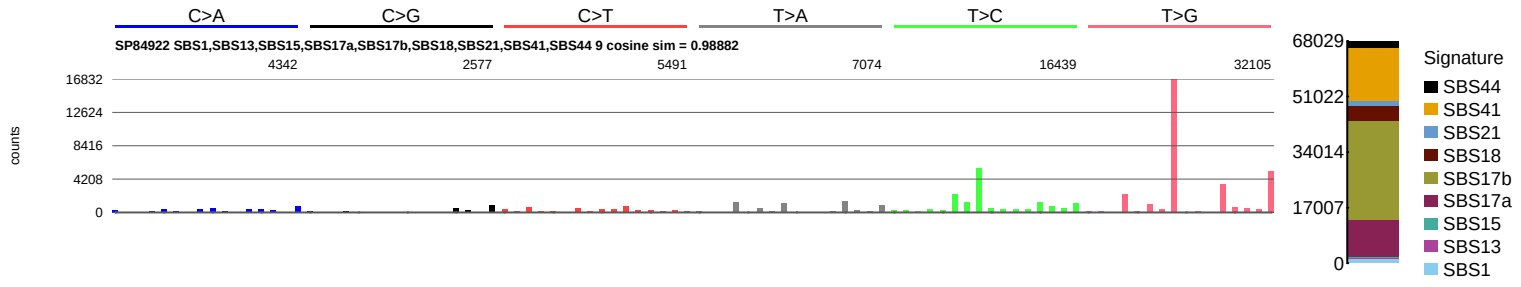
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS2
SBS1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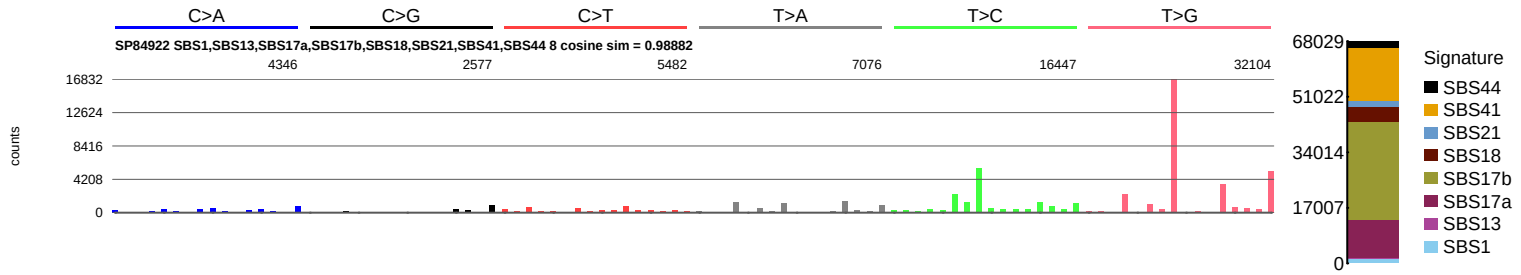
T>C
T>G
SP84922 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98882
4342
2577
5491
7074
16439
32105
16832
12624
8416
4208
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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C>G
C>T
T>A
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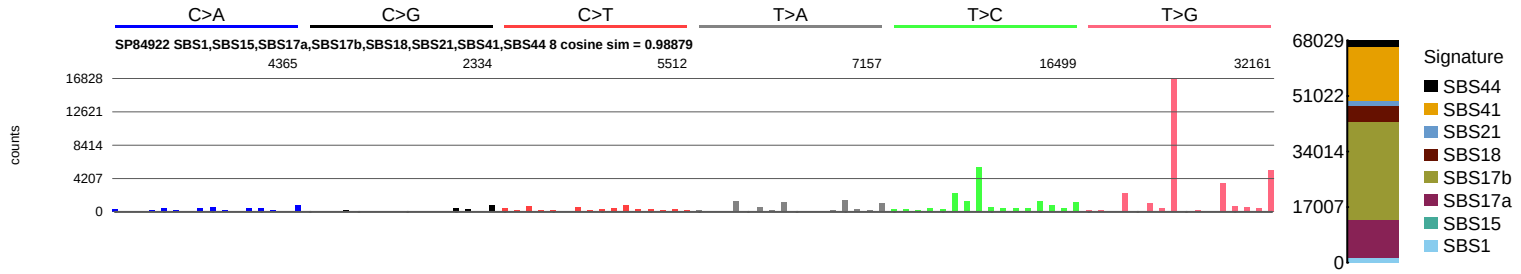
T>G
SP84922 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98882
4346
2577
5482
7076
16447
32104
16832
12624
8416
4208
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1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C>G
C>T
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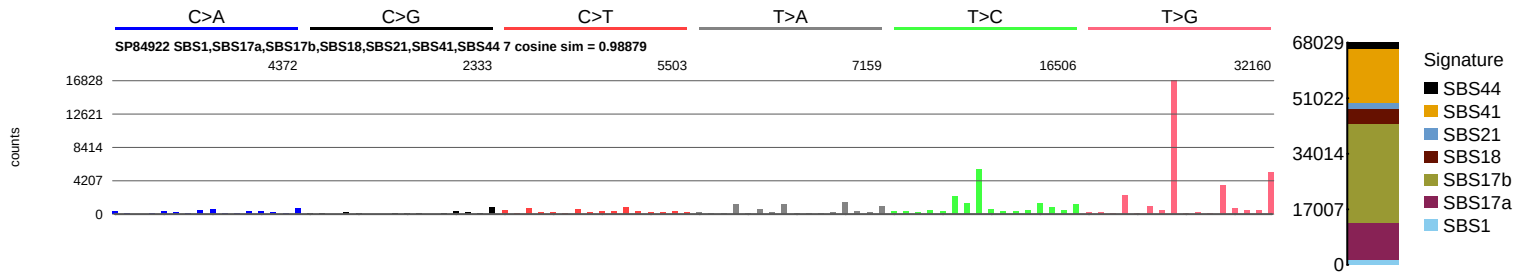
T>C
T>G
SP84922 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98879
4365
2334
5512
7157
16499
32161
16828
12621
8414
4207
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
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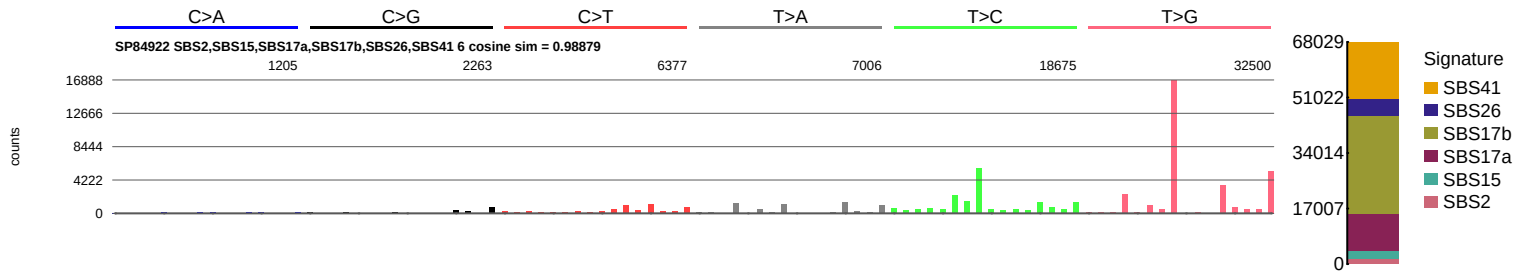
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98879
4372
2333
5503
7159
16506
32160
16828
12621
8414
4207
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
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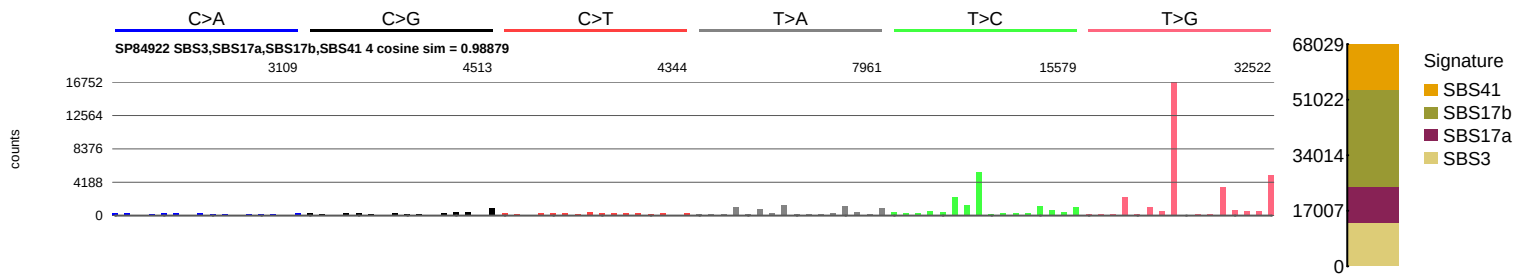
T>C
T>G
SP84922 SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98879
1205
2263
6377
7006
18675
32500
16888
12666
8444
4222
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS2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T>C
T>G
SP84922 SBS3,SBS17a,SBS17b,SBS41 4 cosine sim = 0.98879
3109
4513
4344
7961
15579
32522
16752
12564
8376
4188
0
counts
68029
51022
34014
17007
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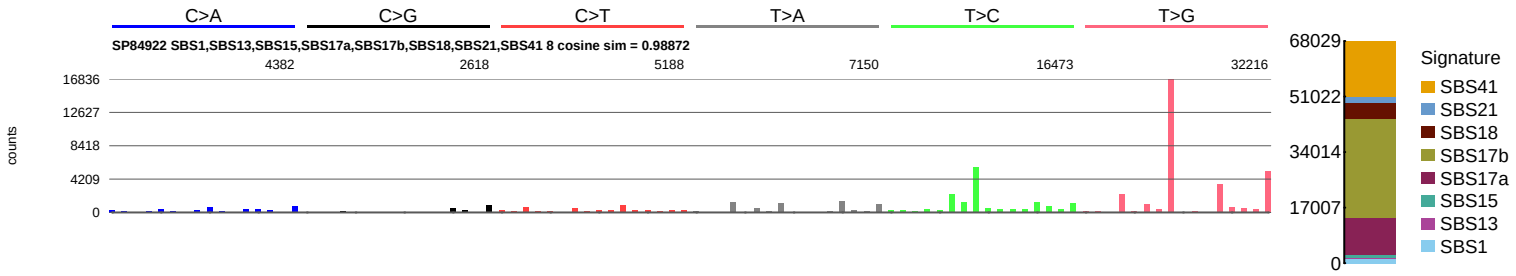
Signature
SBS41
SBS17b
SBS17a
SBS3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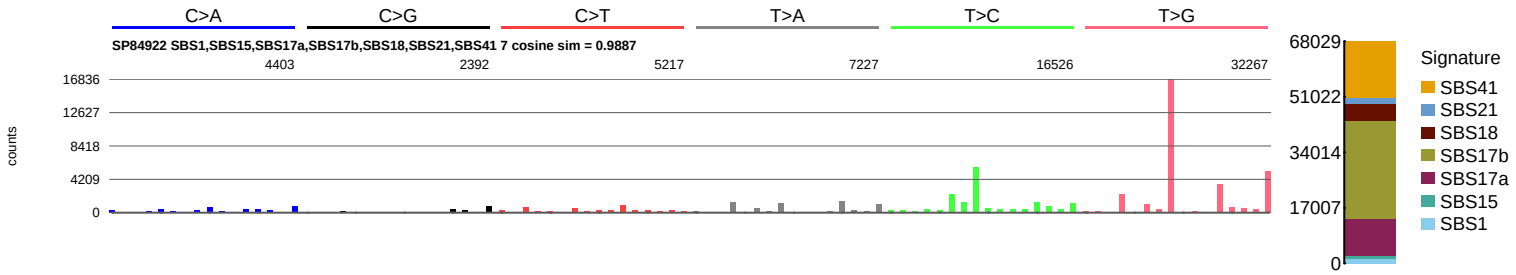
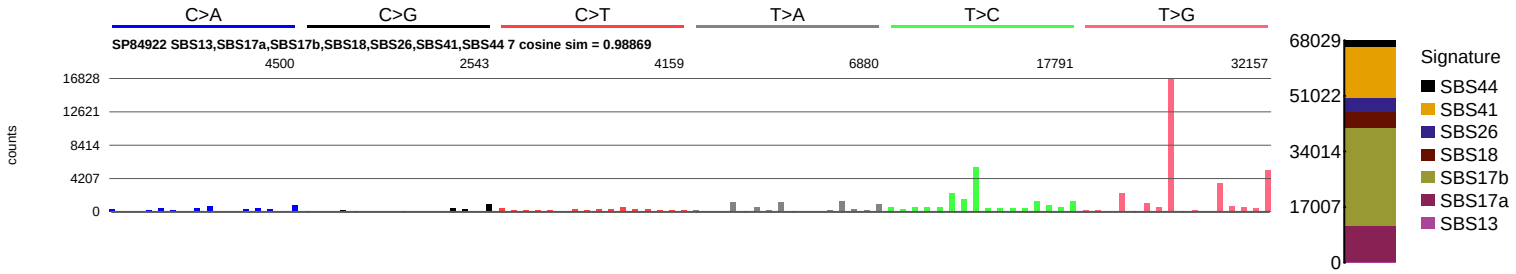
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98872
4382
2618
5188
7150
16473
32216
16836
12627
8418
4209
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.9887
4403
2392
5217
7227
16526
32267
16836
12627
8418
4209
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
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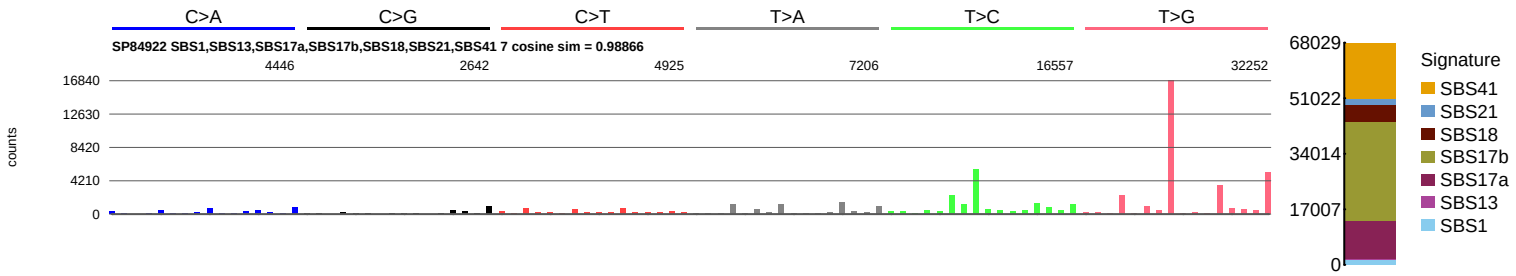
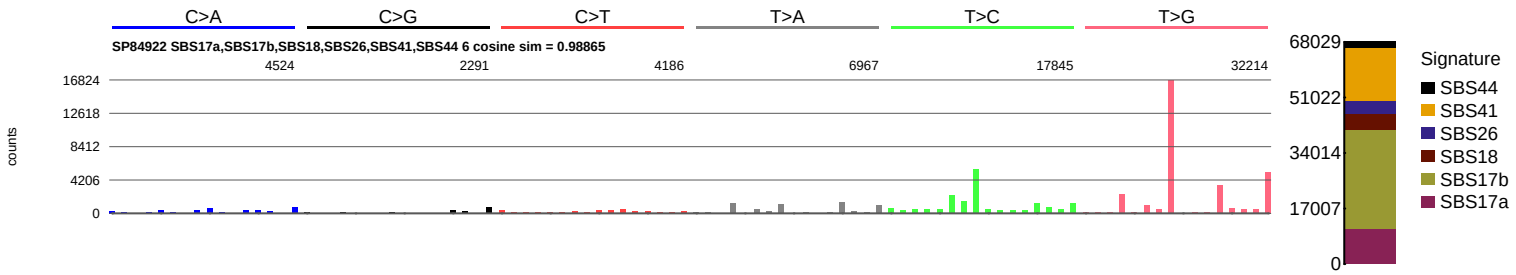
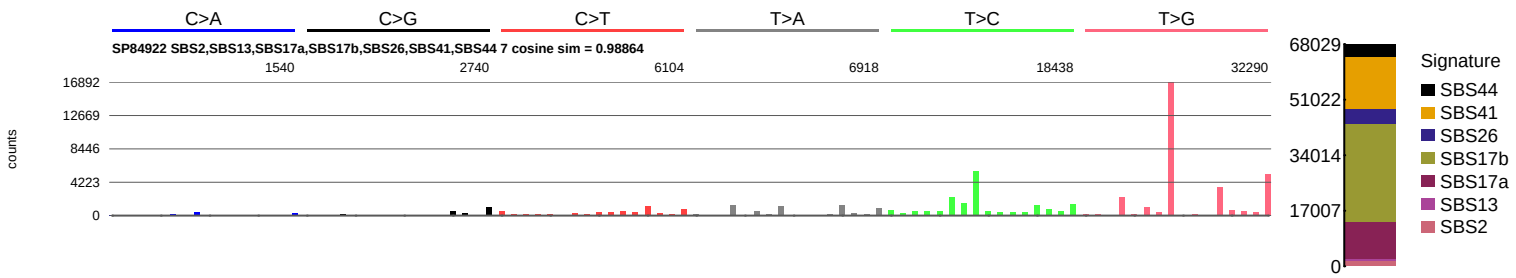
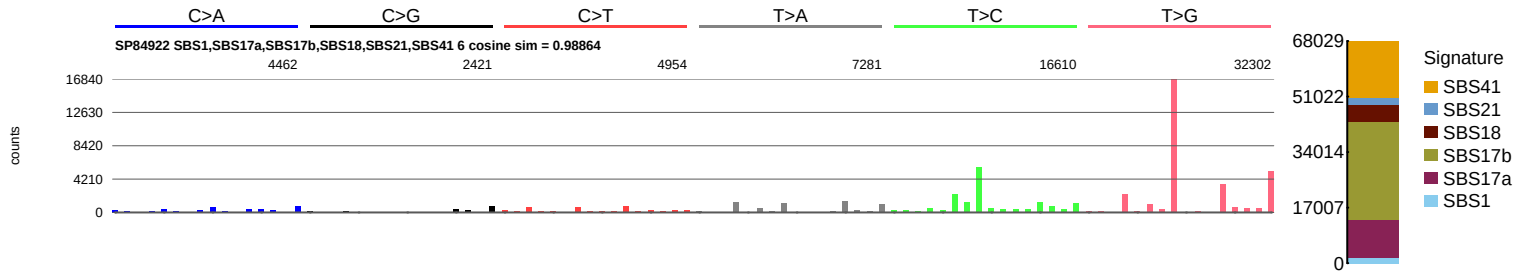
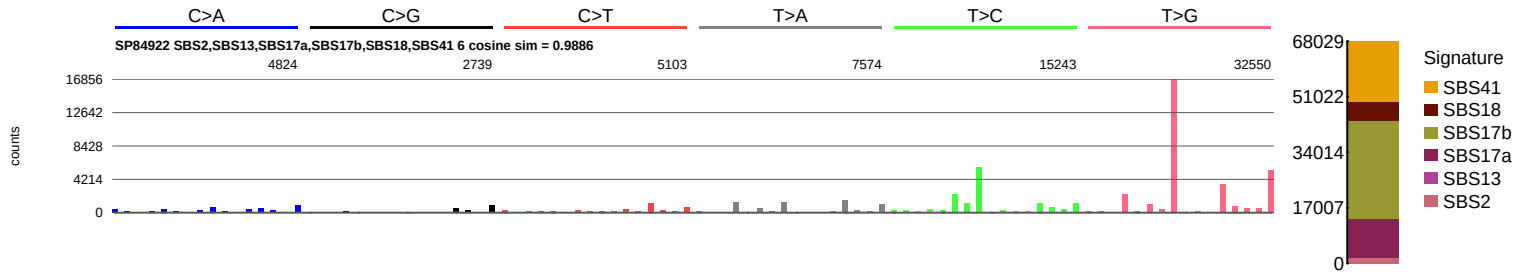
T>A
T>C
T>G
SP84922 SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98869
4500
2543
4159
6880
17791
32157
16828
12621
8414
4207
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98866
4446
2642
4925
7206
16557
32252
16840
12630
8420
4210
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 6 cosine sim = 0.98865
4524
2291
4186
6967
17845
32214
16824
12618
8412
4206
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98864
1540
2740
6104
6918
18438
32290
16892
12669
8446
4223
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98864
4462
2421
4954
7281
16610
32302
16840
12630
8420
4210
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS1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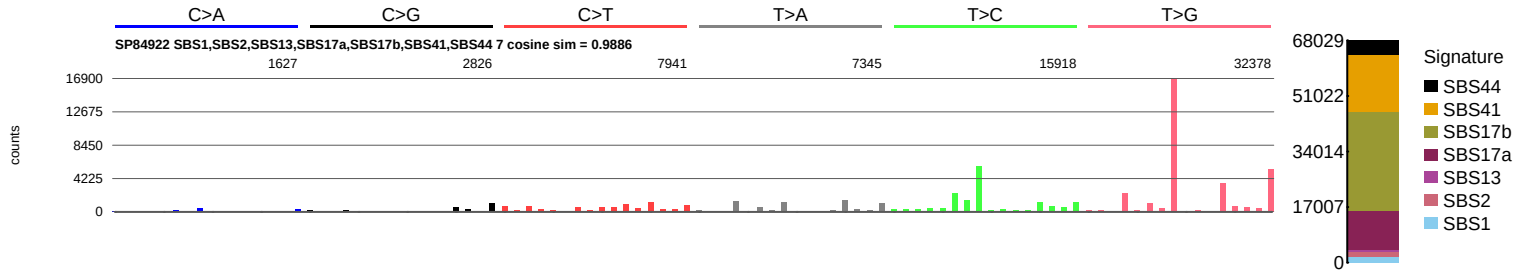
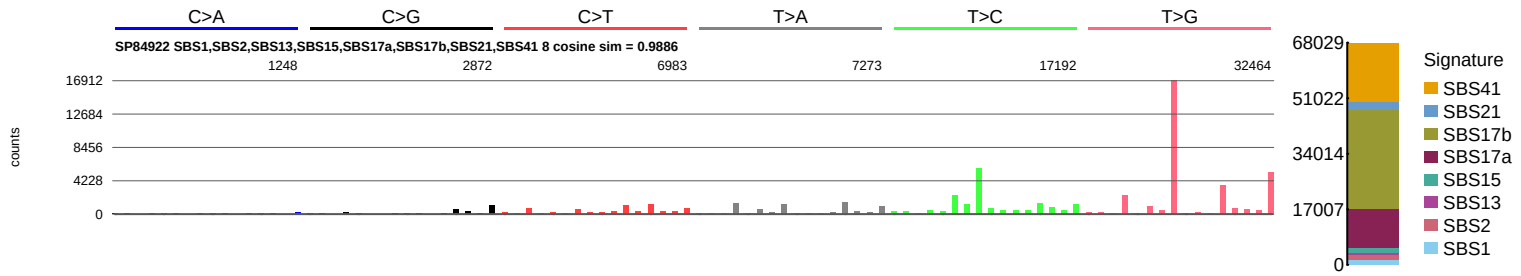
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.9886
4824
2739
5103
7574
15243
32550
16856
12642
8428
4214
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.9886
1627
2826
7941
7345
15918
32378
16900
12675
8450
4225
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS2
SBS1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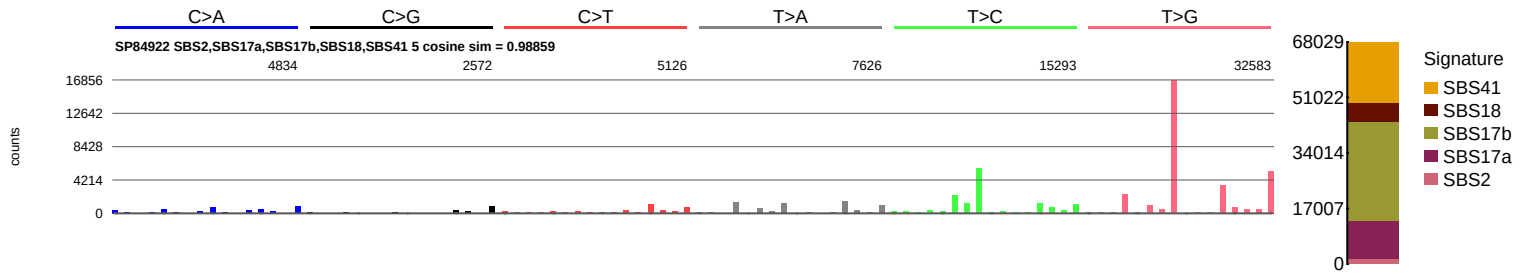
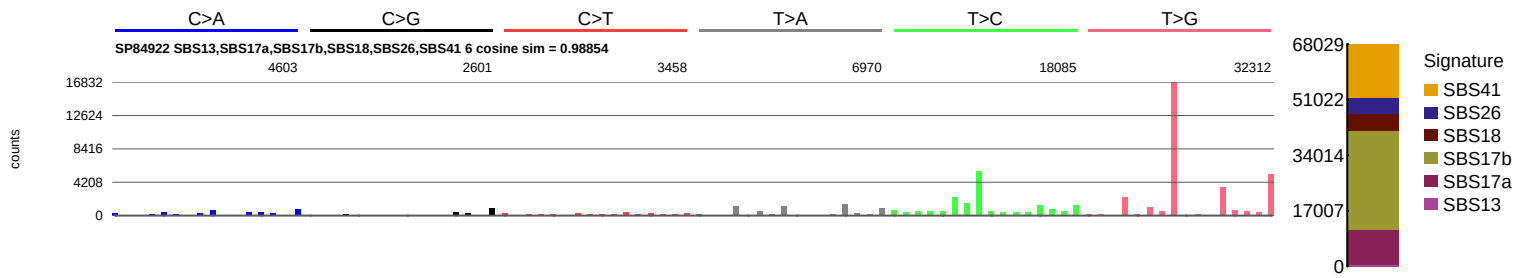
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.9886
1248
2872
6983
7273
17192
32464
16912
12684
8456
4228
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98859
4834
2572
5126
7626
15293
32583
16856
12642
8428
4214
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.98854
4603
2601
3458
6970
18085
32312
16832
12624
8416
4208
0
counts
68029
51022
34014
17007
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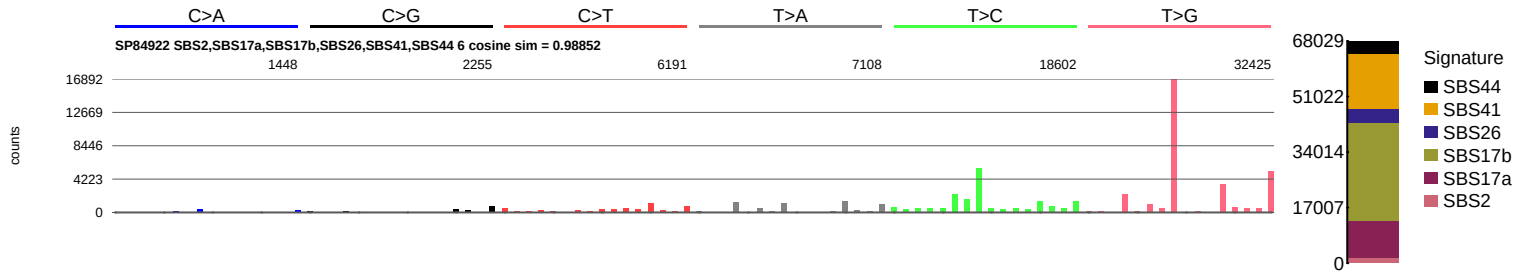
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS26,SBS41,SBS44 6 cosine sim = 0.98852
1448
2255
6191
7108
18602
32425
16892
12669
8446
4223
0
counts
68029
51022
34014
17007
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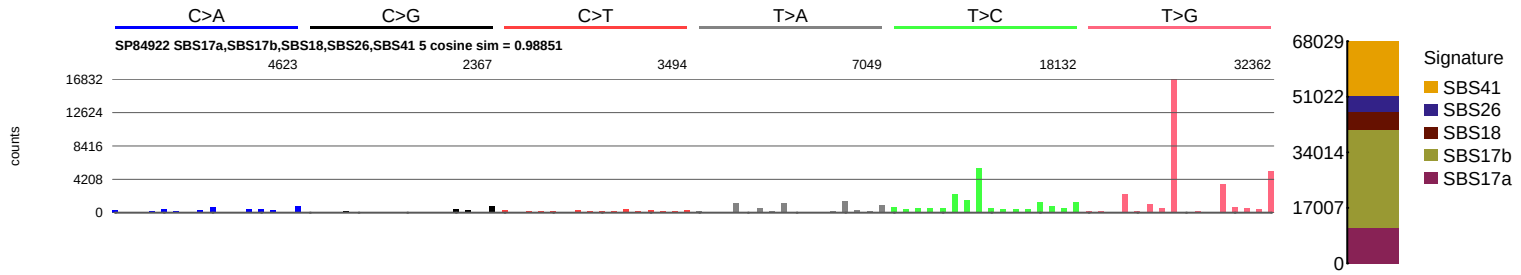
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS2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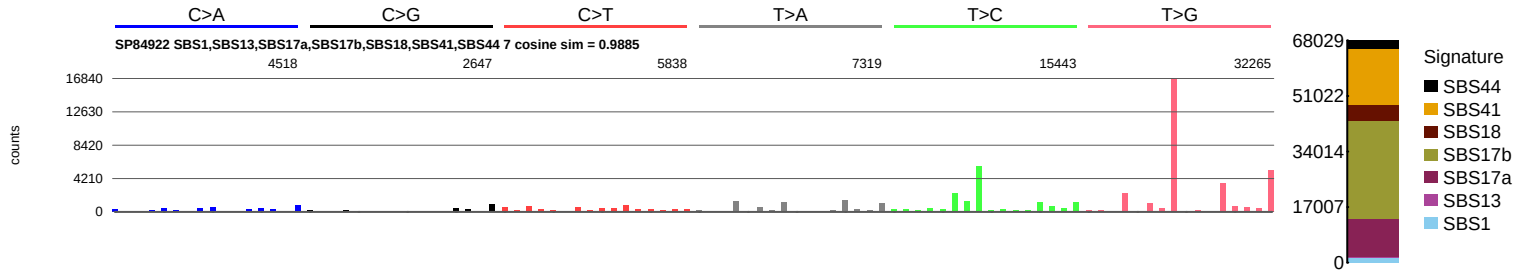
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS18,SBS26,SBS41 5 cosine sim = 0.98851
4623
2367
3494
7049
18132
32362
16832
12624
8416
4208
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.9885
4518
2647
5838
7319
15443
32265
16840
12630
8420
4210
0
counts
68029
51022
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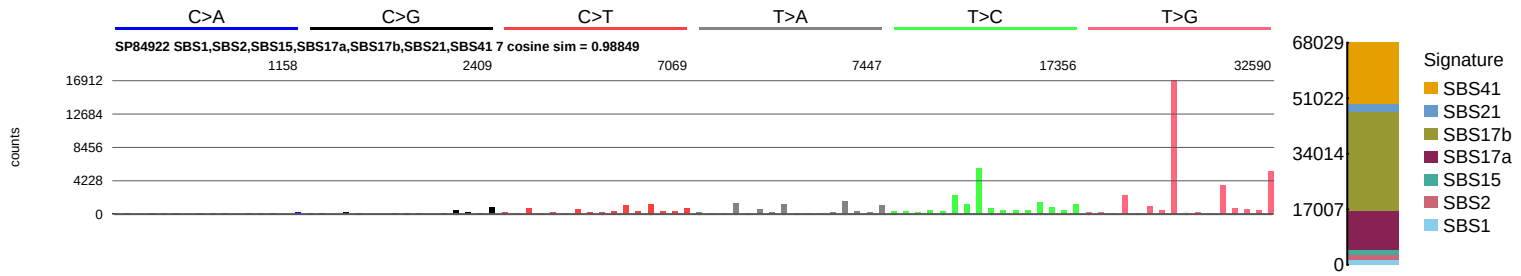
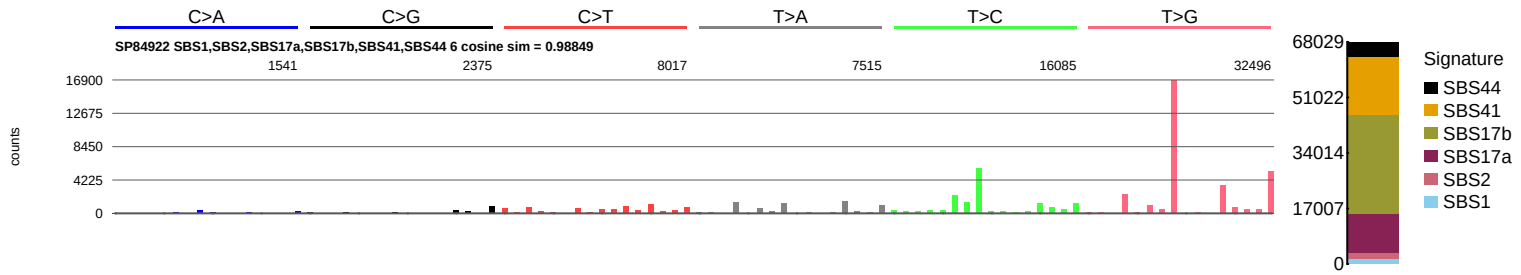
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98849
1158
2409
7069
7447
17356
32590
16912
12684
8456
4228
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98849
1541
2375
8017
7515
16085
32496
16900
12675
8450
4225
0
counts
68029
51022
34014
17007
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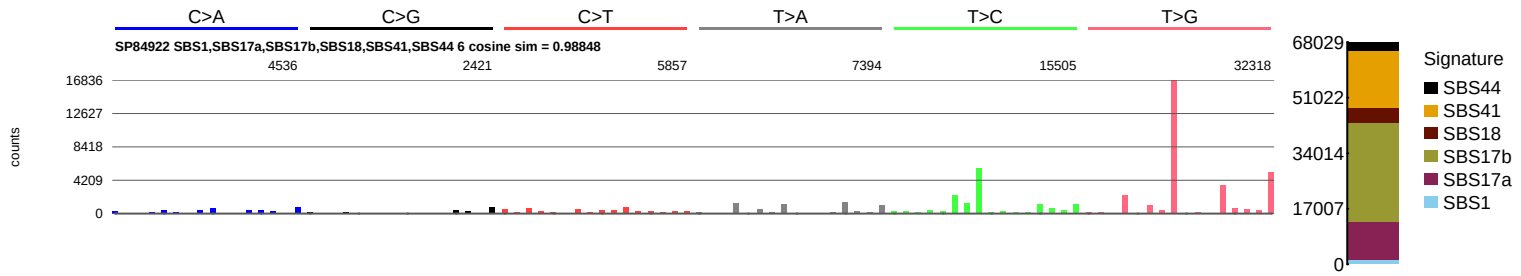
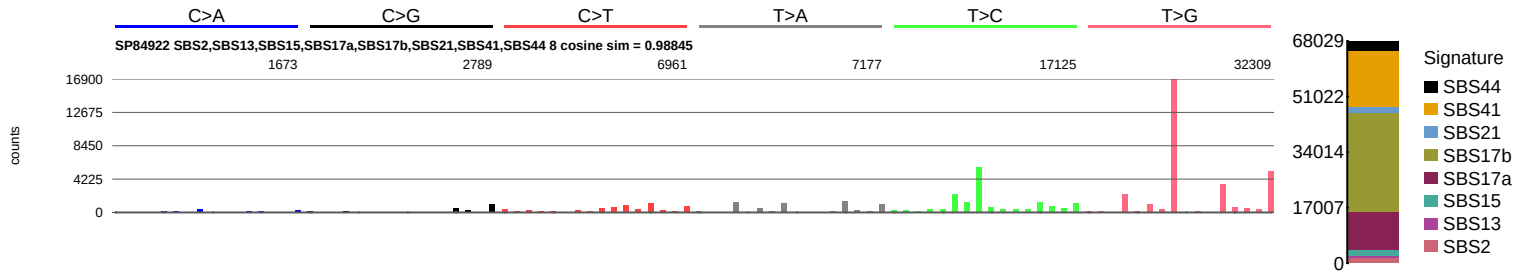
SBS44
SBS41
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98848
4536
2421
5857
7394
15505
32318
16836
12627
8418
4209
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98845
1673
2789
6961
7177
17125
32309
counts
16900
12675
8450
4225
0
68029
51022
34014
17007
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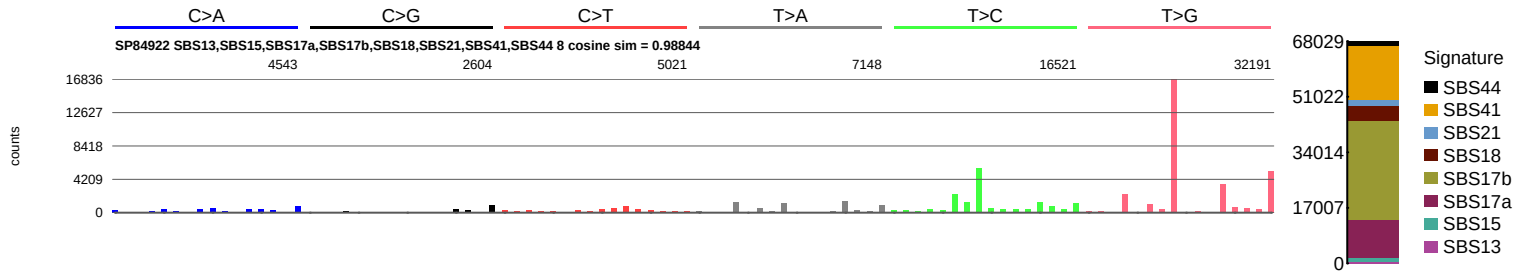
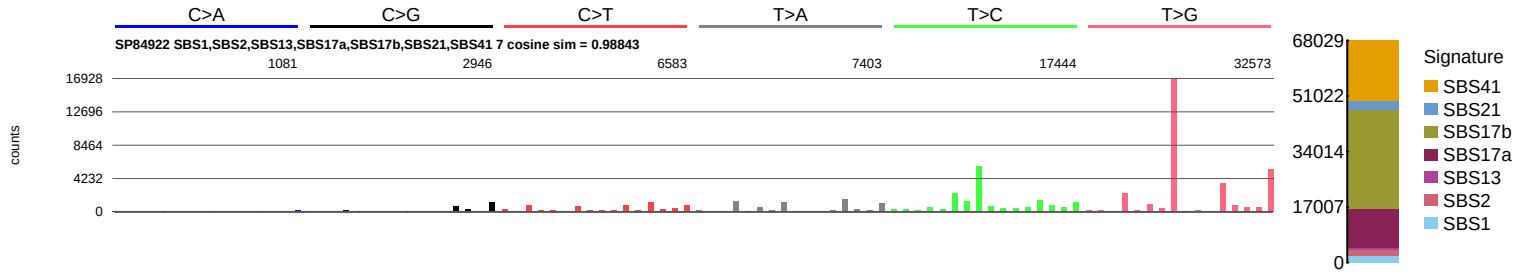
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98844
4543
2604
5021
7148
16521
32191
counts
16836
12627
8418
4209
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13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C>G
C>T
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T>C
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SP84922 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98843
1081
2946
6583
7403
17444
32573
counts
16928
12696
8464
4232
0
68029
51022
34014
17007
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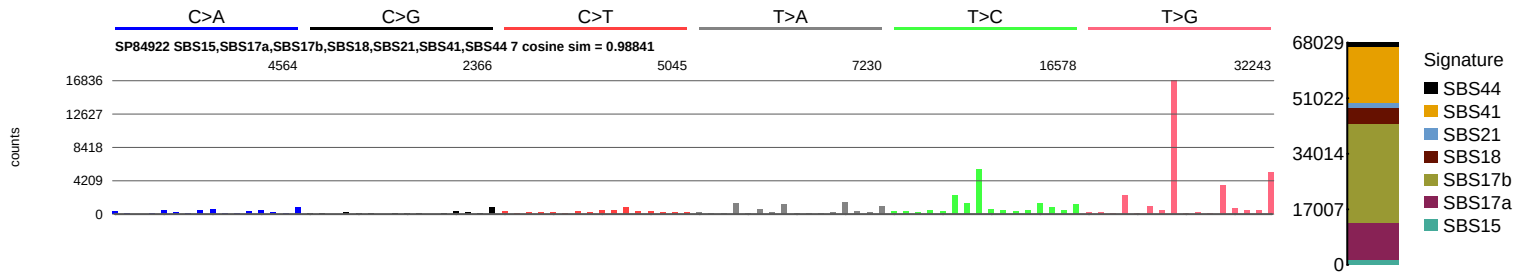
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>T
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T>C
T>G
SP84922 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98841
4564
2366
5045
7230
16578
32243
counts
16836
12627
8418
4209
0
68029
51022
34014
17007
0
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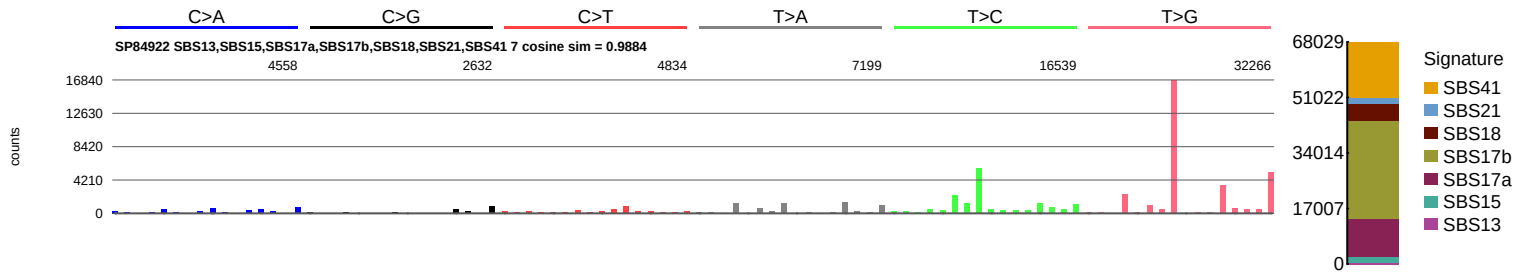
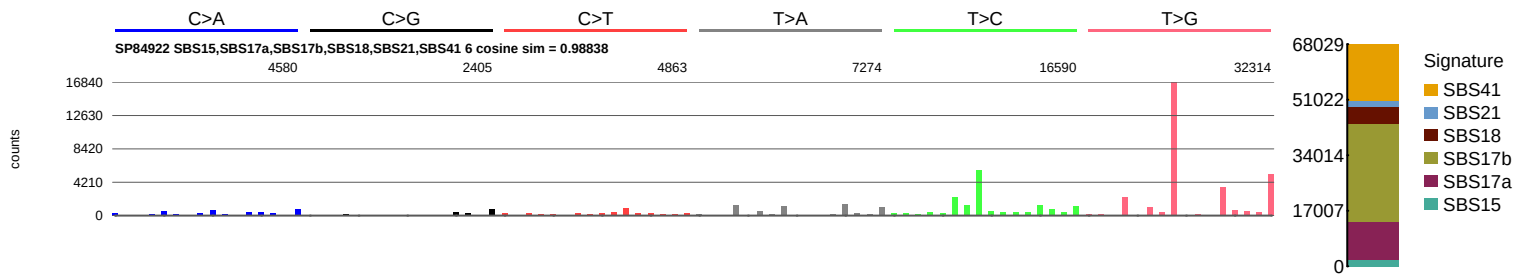
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.9884
4558
2632
4834
7199
16539
32266
counts
16840
12630
8420
4210
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13


C>A
C>G
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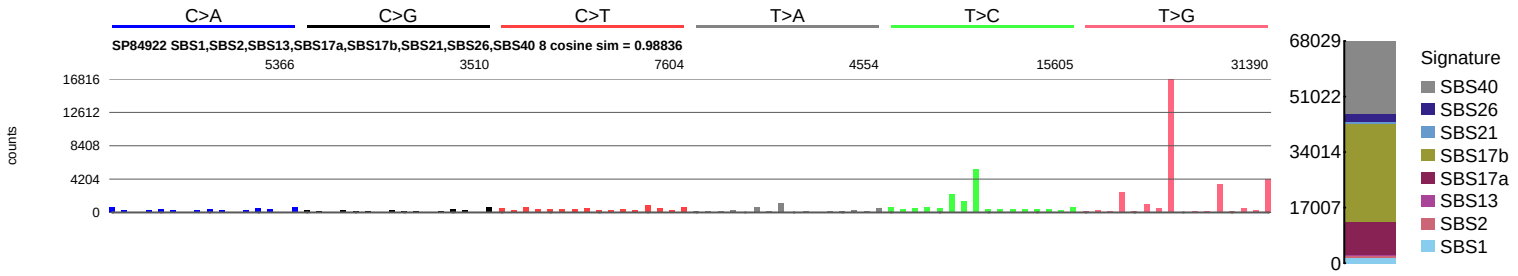
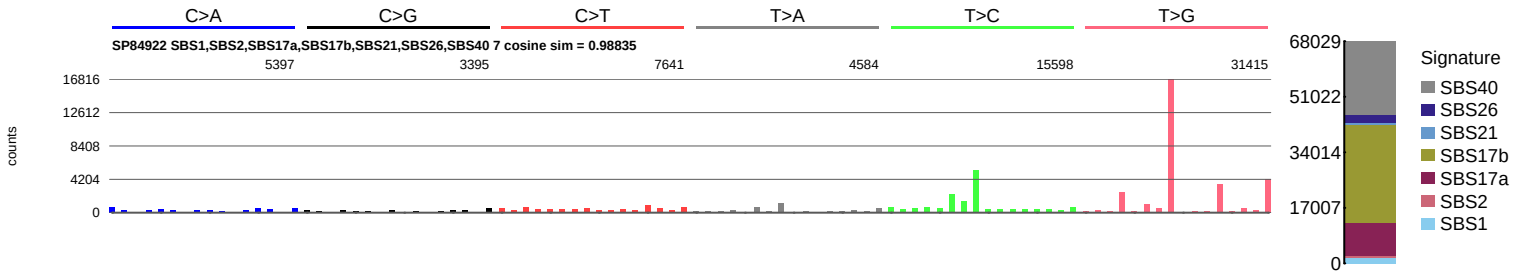
T>C
T>G
SP84922 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98838
4580
2405
4863
7274
16590
32314
counts
16840
12630
8420
4210
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40 8 cosine sim = 0.98836
5366
3510
7604
4554
15605
31390
counts
16816
12612
8408
4204
0
68029
51022
34014
17007
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.98835
5397
3395
7641
4584
15598
31415
counts
16816
12612
8408
4204
0
68029
51022
34014
17007
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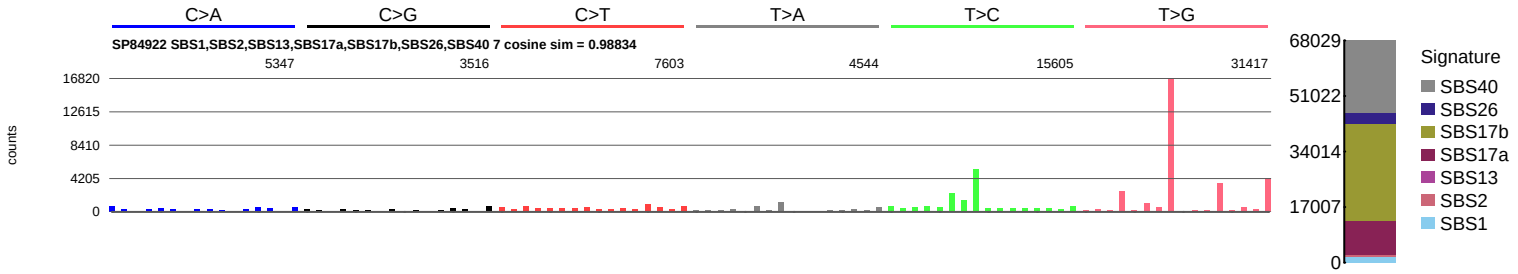
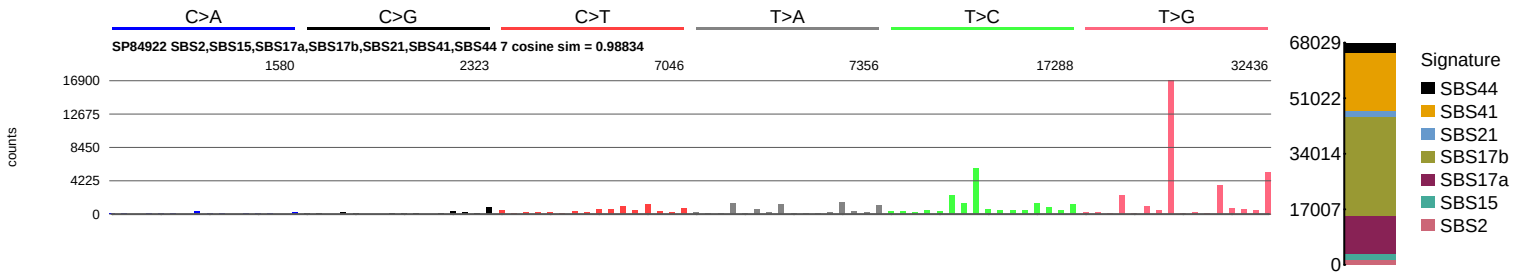
SBS26
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98834
5347
3516
7603
4544
15605
31417
counts
16820
12615
8410
4205
0
68029
51022
34014
17007
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98834
1580
2323
7046
7356
17288
32436
counts
16900
12675
8450
4225
0
68029
51022
34014
17007
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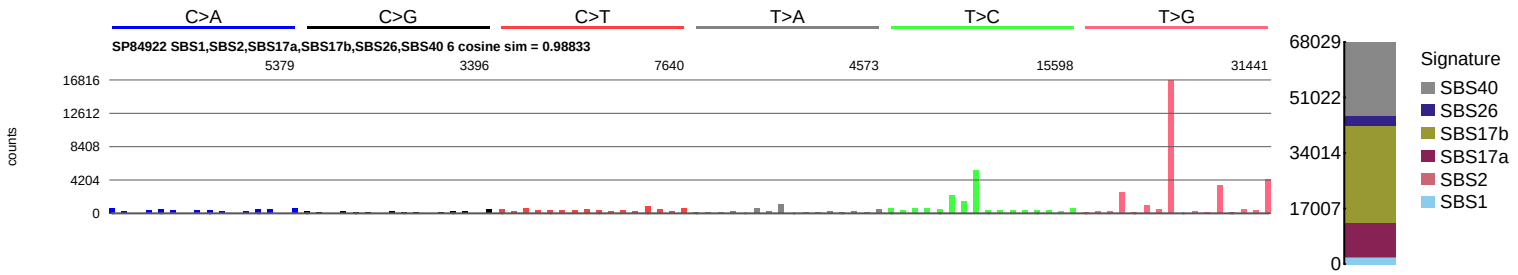
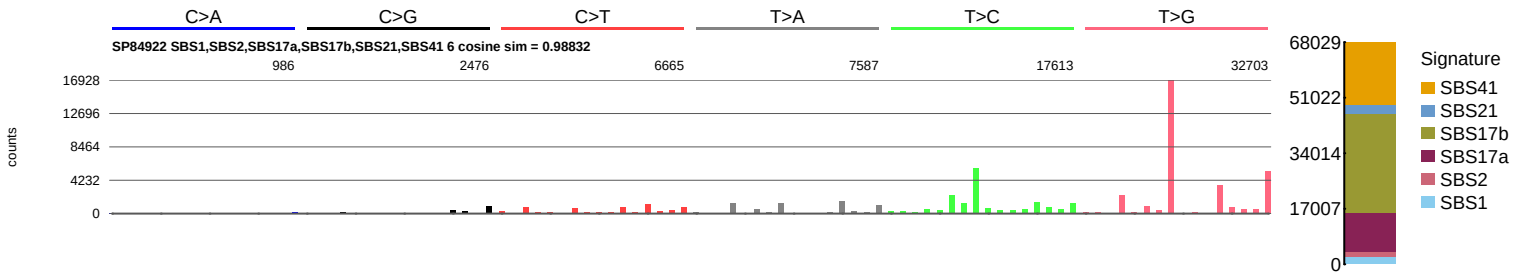
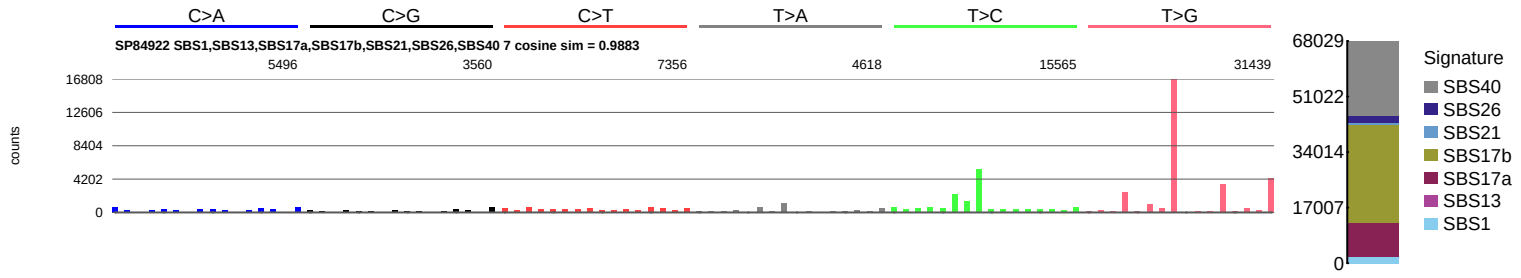
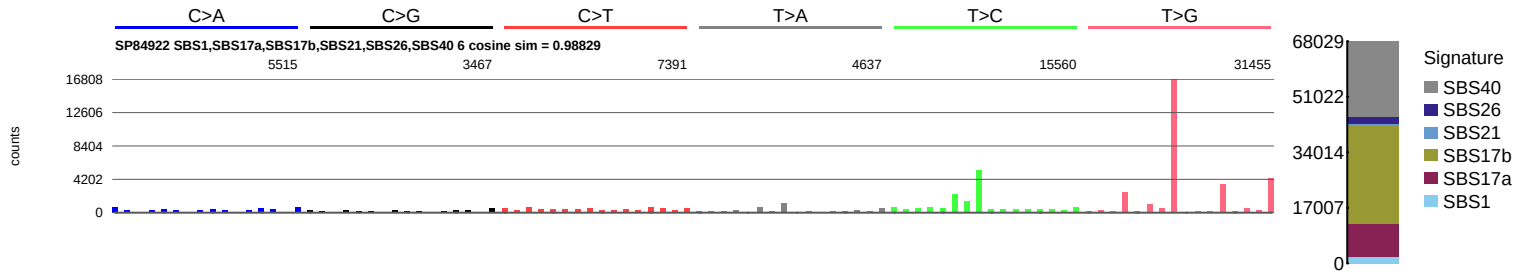
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98833
5379
3396
7640
4573
15598
31441
counts
16816
12612
8408
4204
0
68029
51022
34014
17007
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98832
986
2476
6665
7587
17613
32703
counts
16928
12696
8464
4232
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.9883
5496
3560
7356
4618
15565
31439
counts
16808
12606
8404
4202
0
68029
51022
34014
17007
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS21,SBS26,SBS40 6 cosine sim = 0.98829
5515
3467
7391
4637
15560
31455
counts
16808
12606
8404
4202
0
68029
51022
34014
17007
0
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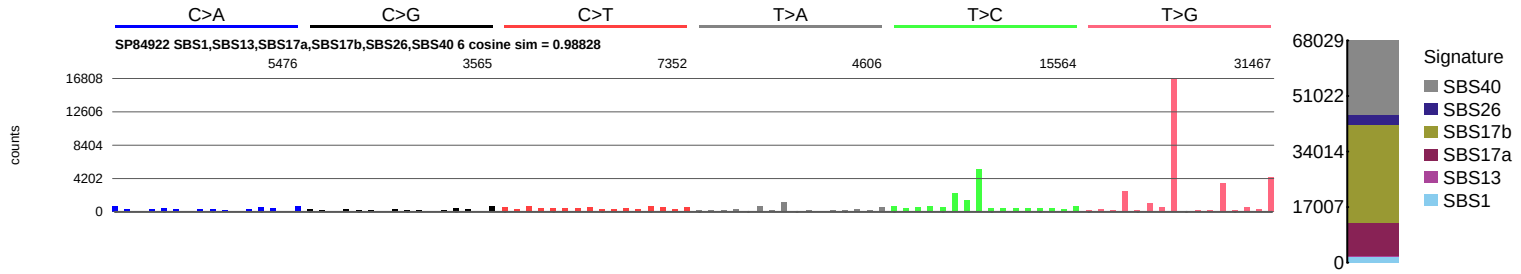
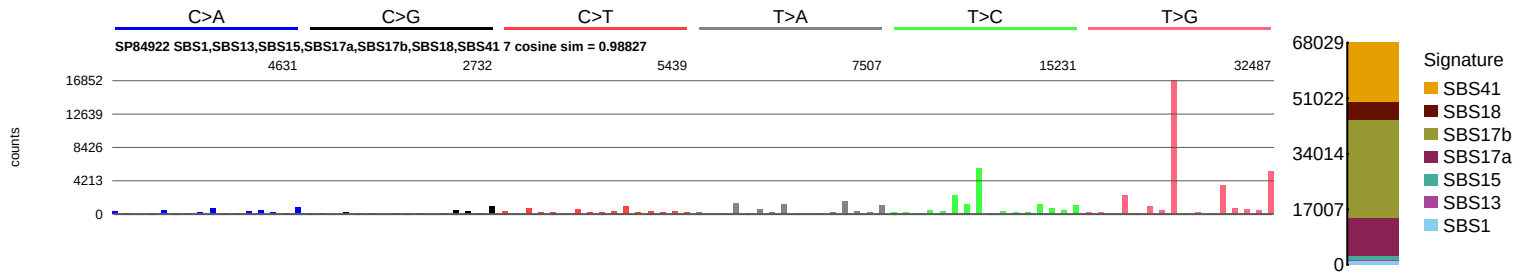
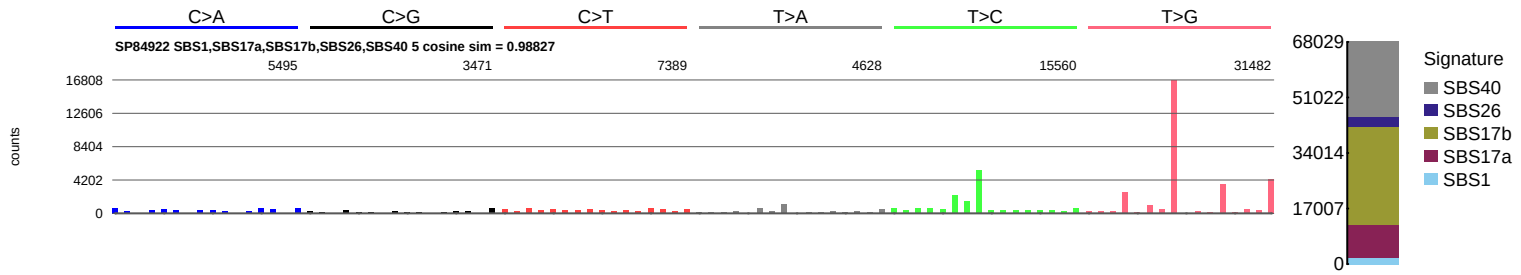
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98828
5476
3565
7352
4606
15564
31467
counts
16808
12606
8404
4202
0
68029
51022
34014
17007
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98827
4631
2732
5439
7507
15231
32487
counts
16852
12639
8426
4213
0
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS26,SBS40 5 cosine sim = 0.98827
5495
3471
7389
4628
15560
31482
counts
16808
12606
8404
4202
0
68029
51022
34014
17007
0
Signature
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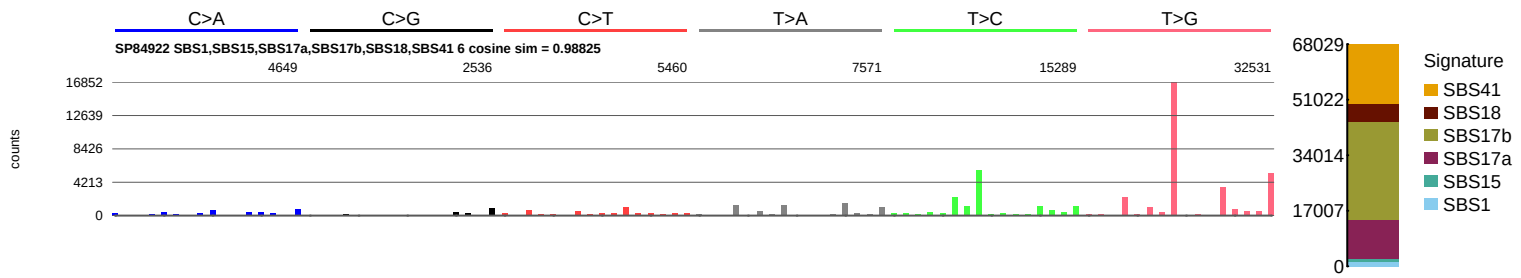
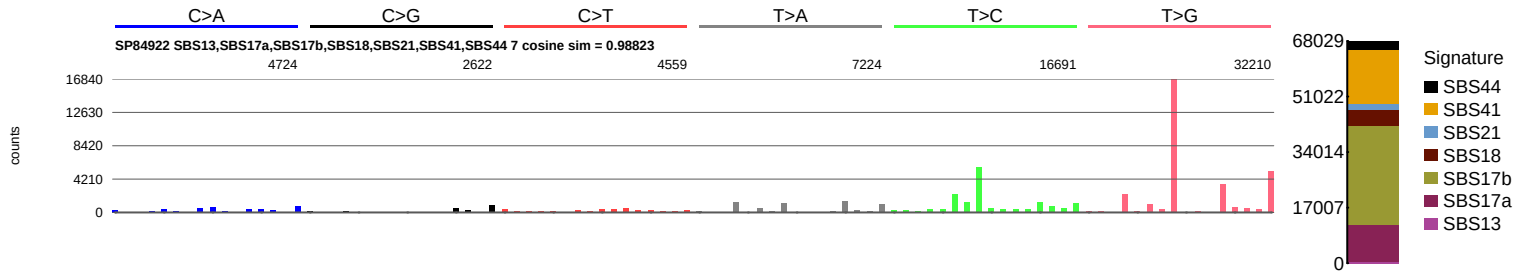
SBS26
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98825
4649
2536
5460
7571
15289
32531
counts
16852
12639
8426
4213
0
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98823
4724
2622
4559
7224
16691
32210
16840
12630
8420
4210
0
counts
68029
51022
34014
17007
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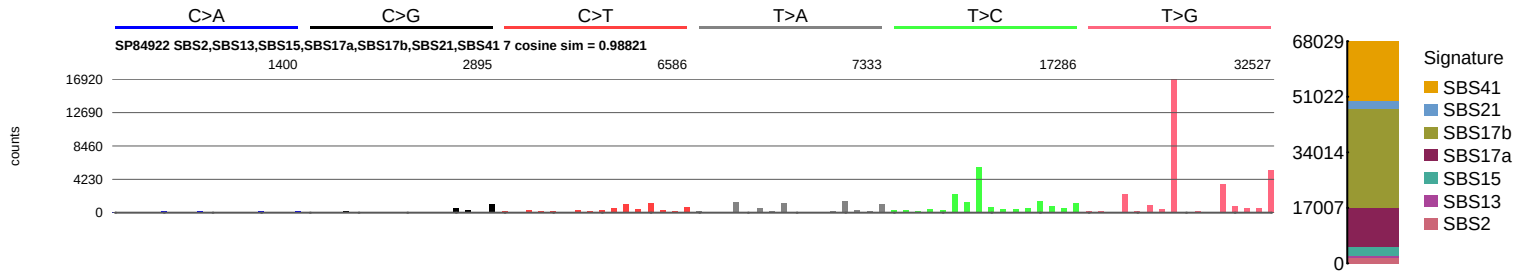
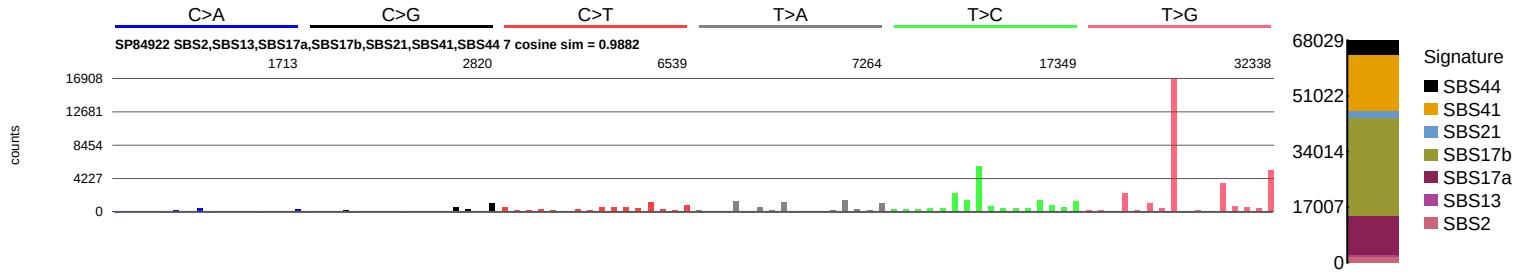
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98821
1400
2895
6586
7333
17286
32527
16920
12690
8460
4230
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.9882
1713
2820
6539
7264
17349
32338
16908
12681
8454
4227
0
counts
68029
51022
34014
17007
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Signature
SBS44
SBS41
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SBS17b
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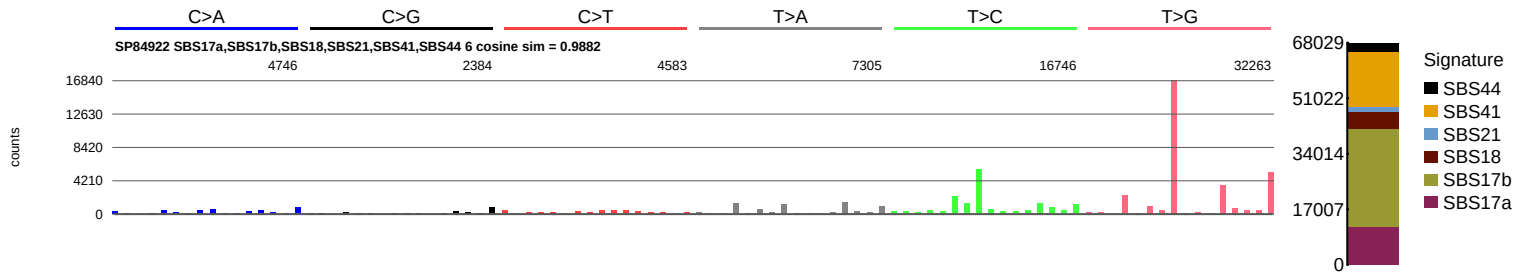
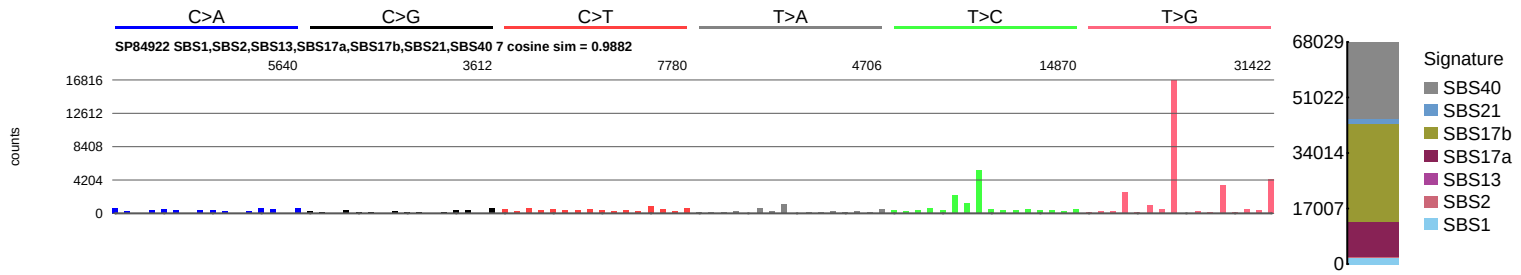
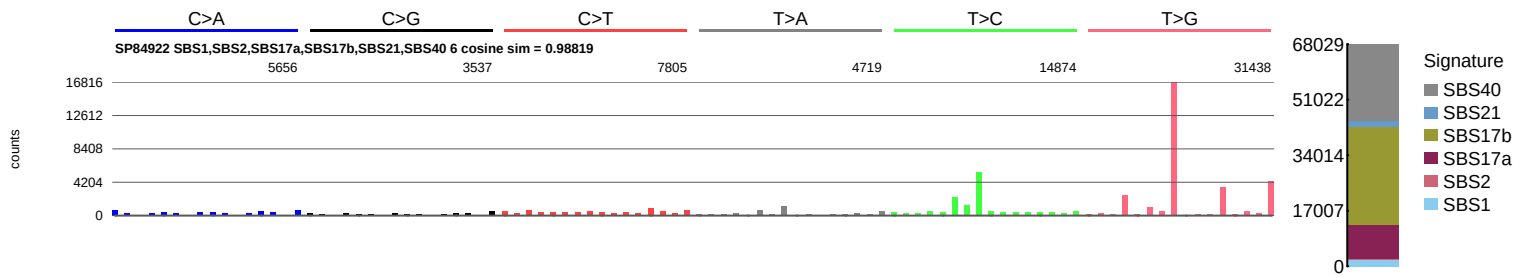
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 6 cosine sim = 0.9882
4746
2384
4583
7305
16746
32263
16840
12630
8420
4210
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.9882
5640
3612
7780
4706
14870
31422
16816
12612
8408
4204
0
counts
68029
51022
34014
17007
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98819
5656
3537
7805
4719
14874
31438
16816
12612
8408
4204
0
counts
68029
51022
34014
17007
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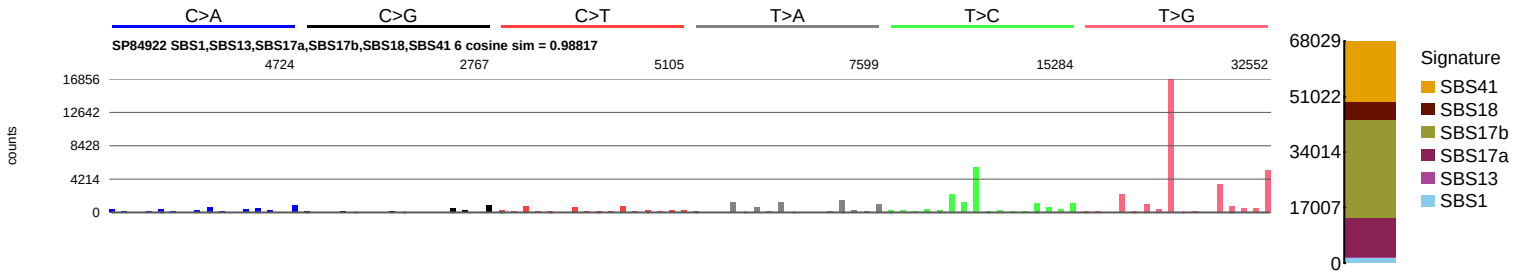
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98817
4724
2767
5105
7599
15284
32552
counts
16856
12642
8428
4214
0
68029
51022
34014
17007
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SBS18
SBS17b
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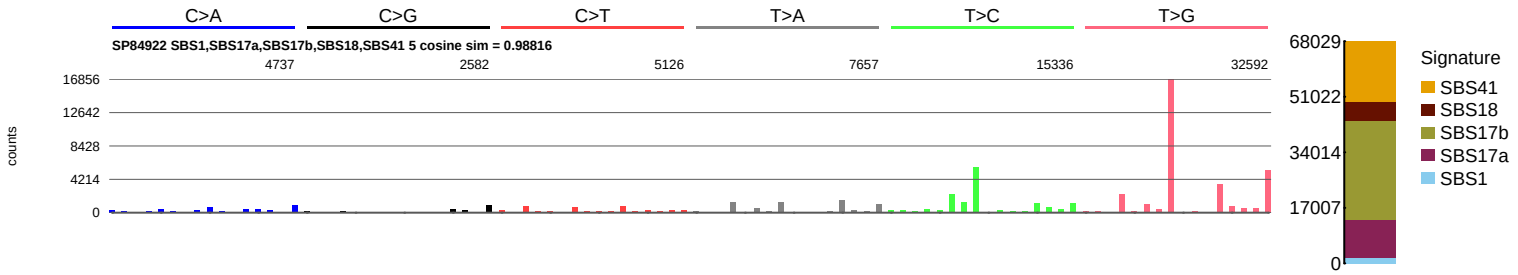
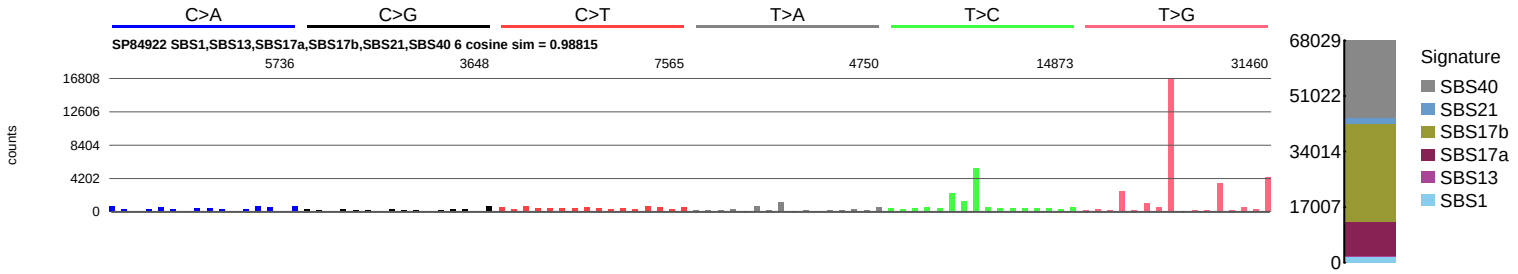
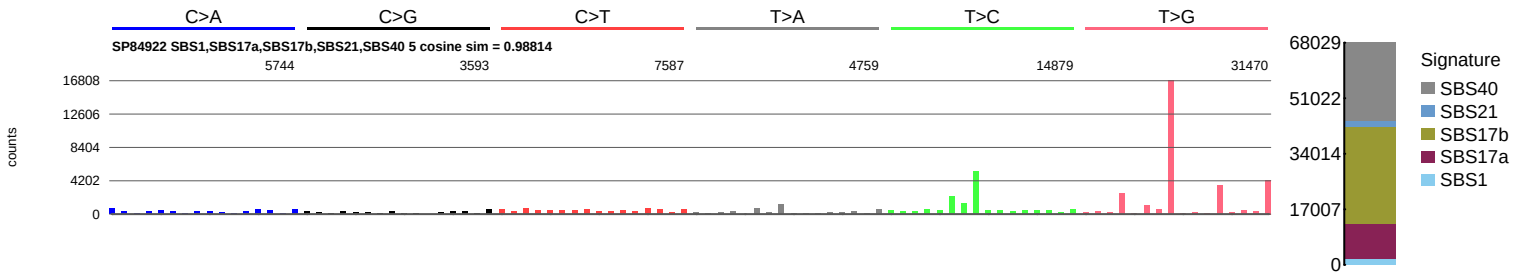
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98816
4737
2582
5126
7657
15336
32592
counts
16856
12642
8428
4214
0
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98815
5736
3648
7565
4750
14873
31460
counts
16808
12606
8404
4202
0
68029
51022
34014
17007
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS1


C>A
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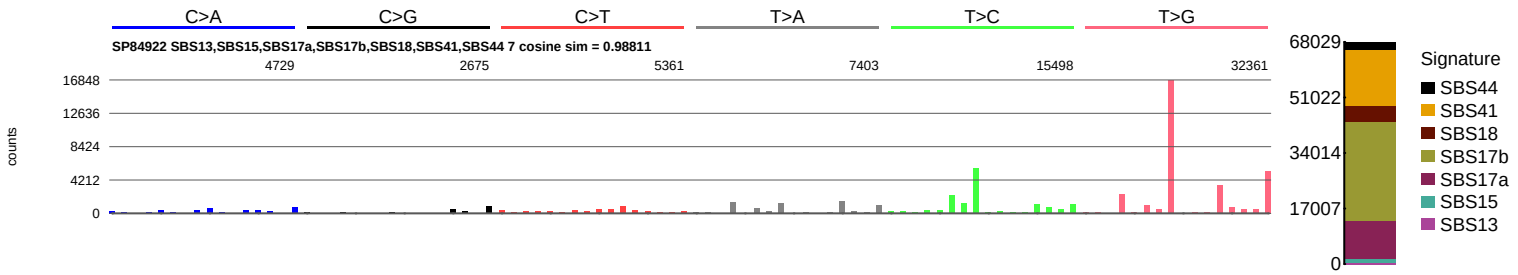
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS21,SBS40 5 cosine sim = 0.98814
5744
3593
7587
4759
14879
31470
counts
16808
12606
8404
4202
0
68029
51022
34014
17007
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS1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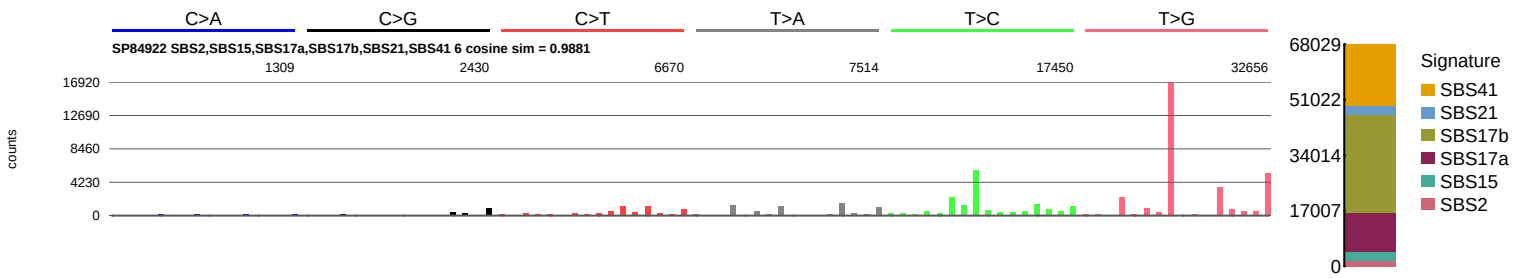
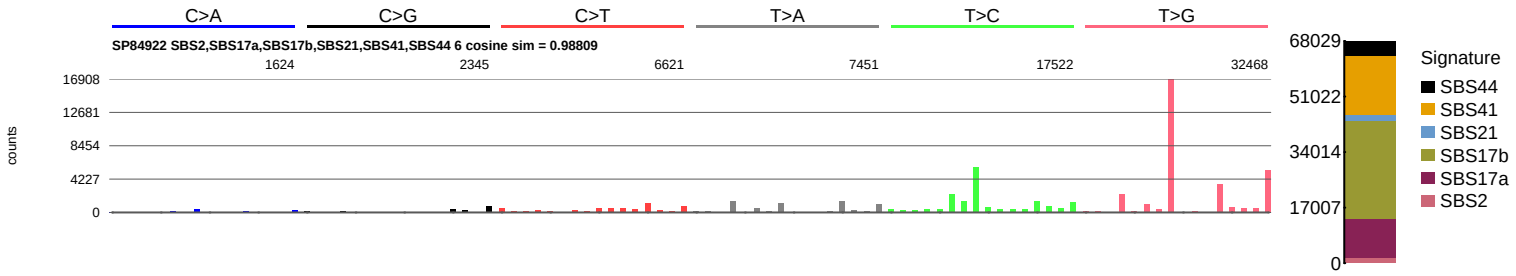
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98811
4729
2675
5361
7403
15498
32361
counts
16848
12636
8424
4212
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.9881
1309
2430
6670
7514
17450
32656
counts
16920
12690
8460
4230
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS2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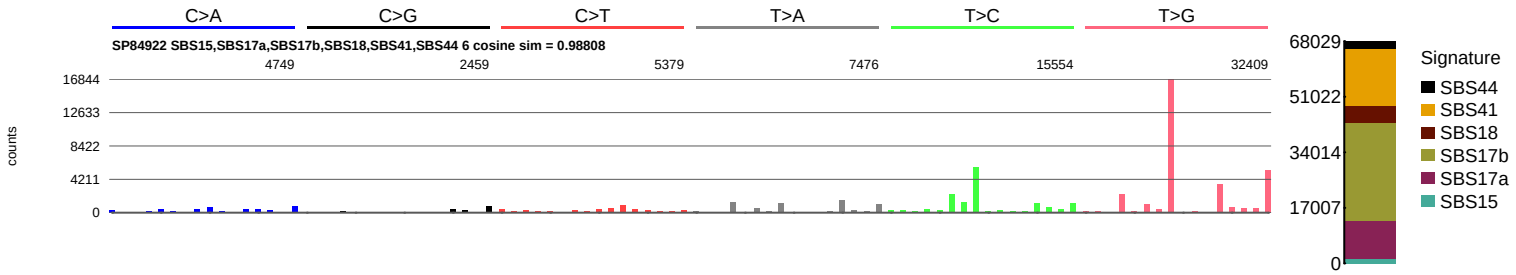
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS21,SBS41,SBS44 6 cosine sim = 0.98809
1624
2345
6621
7451
17522
32468
counts
16908
12681
8454
4227
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS2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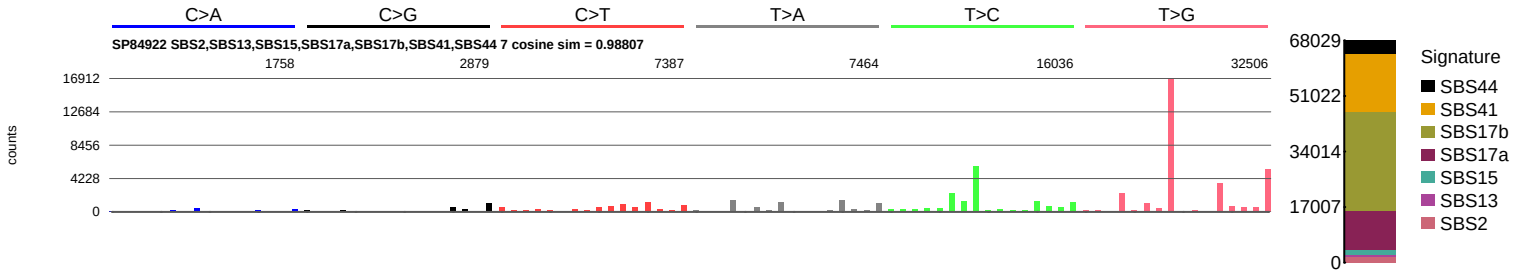
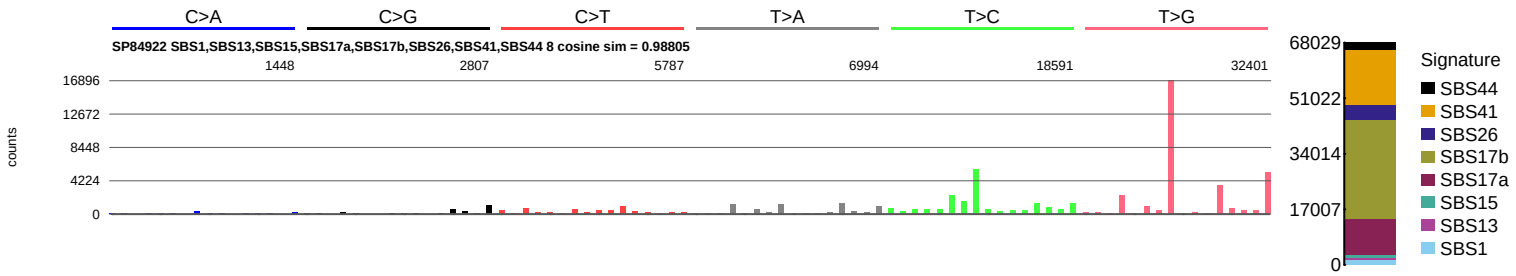
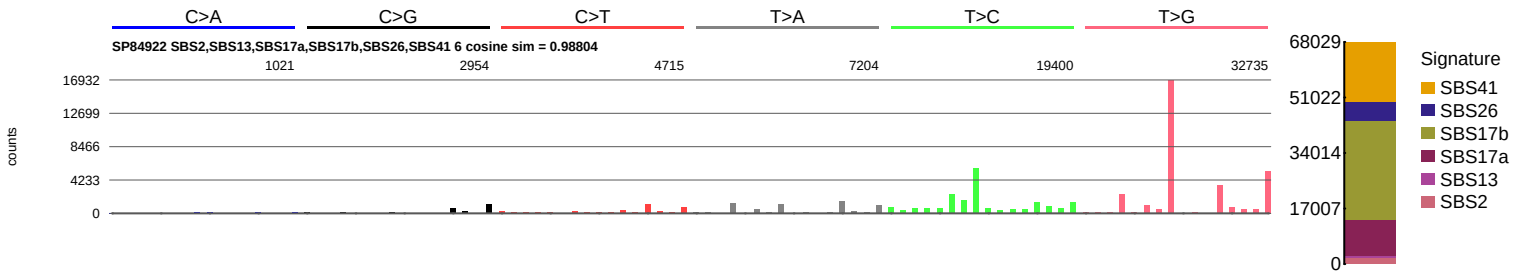
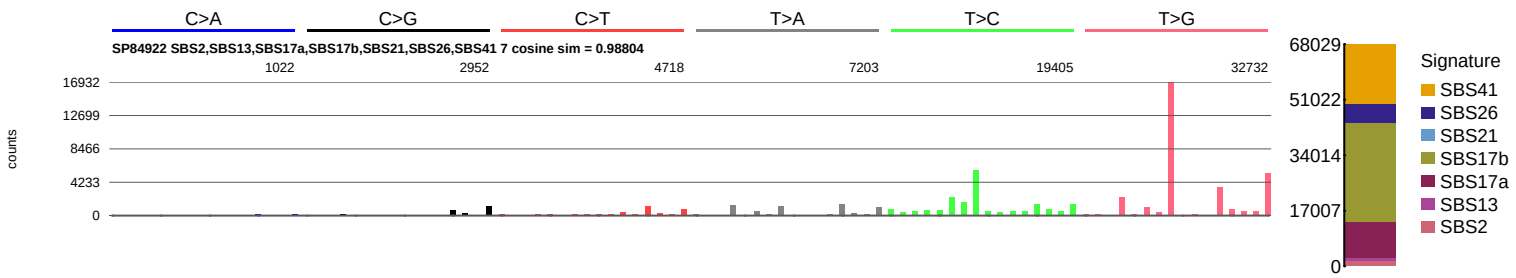
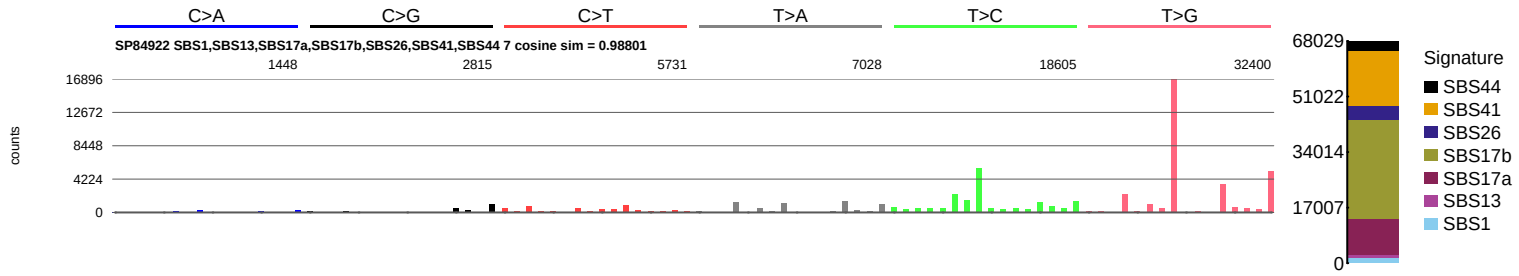
C>G
C>T
T>A
T>C
T>G
SP84922 SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98808
4749
2459
5379
7476
15554
32409
counts
16844
12633
8422
4211
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15

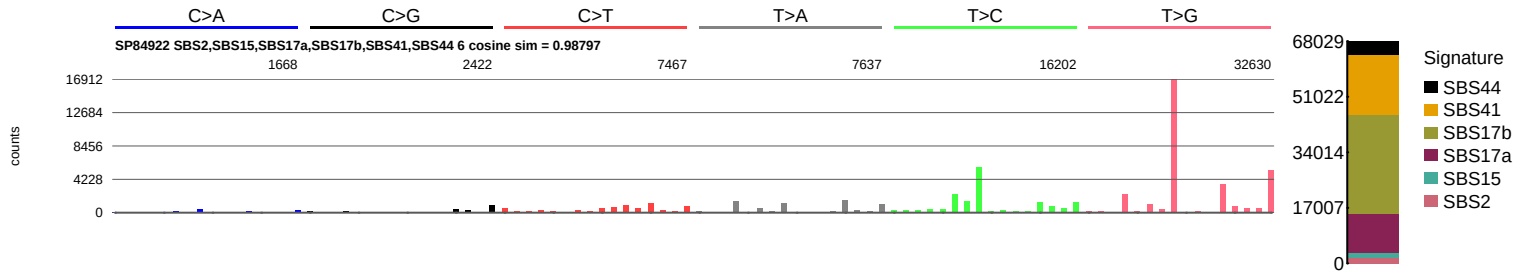

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98807
1758
2879
7387
7464
16036
32506
counts
16912
12684
8456
4228
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS2

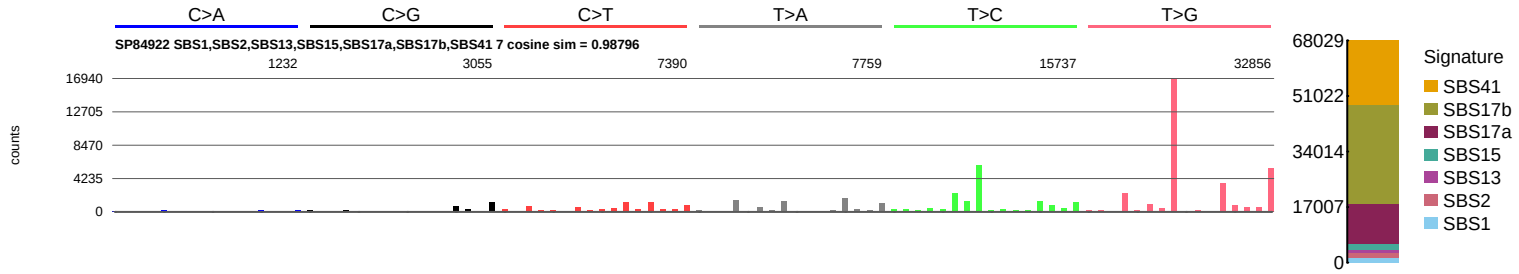

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98805
1448
2807
5787
6994
18591
32401
counts
16896
12672
8448
4224
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS1

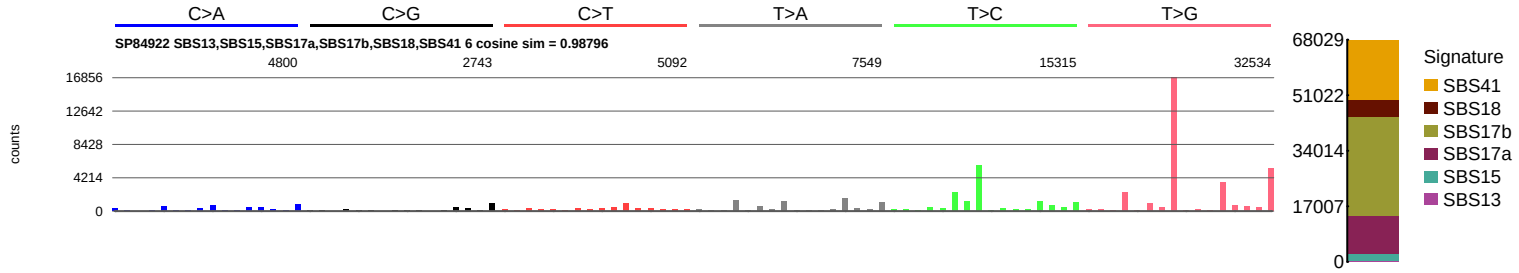

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98804
1021
2954
4715
7204
19400
32735
counts
16932
12699
8466
4233
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS2

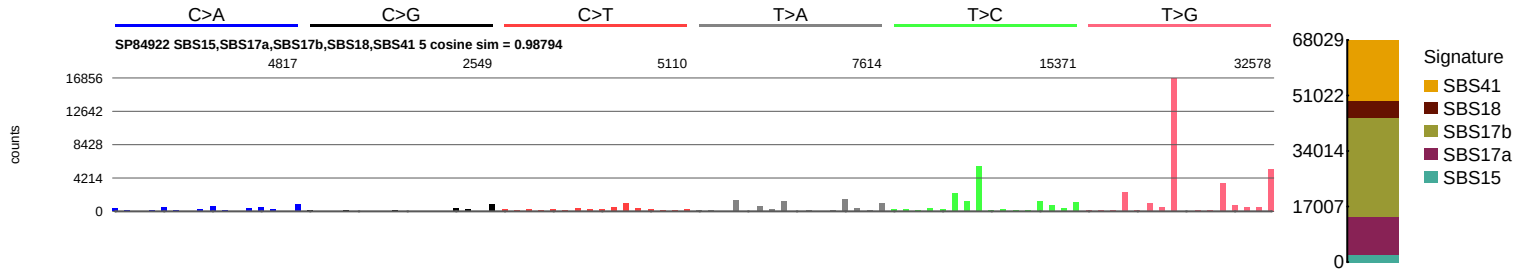

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41 7 cosine sim = 0.98804
1022
2952
4718
7203
19405
32732
counts
16932
12699
8466
4233
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2

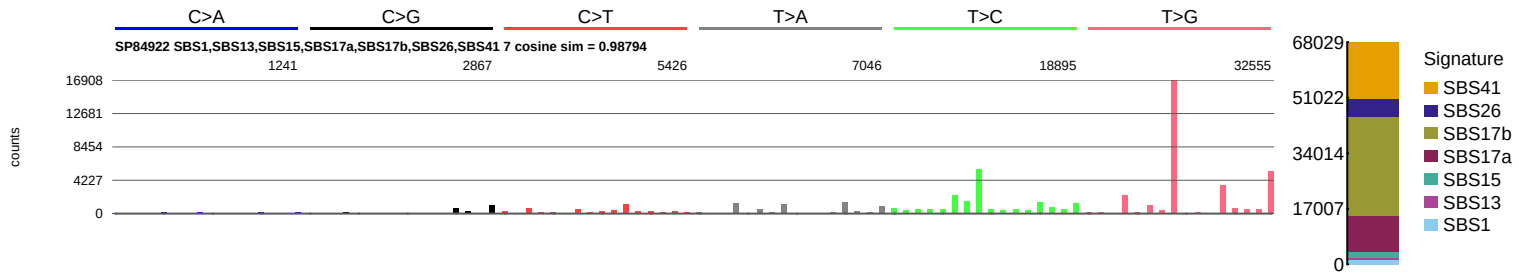

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98801
1448
2815
5731
7028
18605
32400
16896
12672
8448
4224
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS1

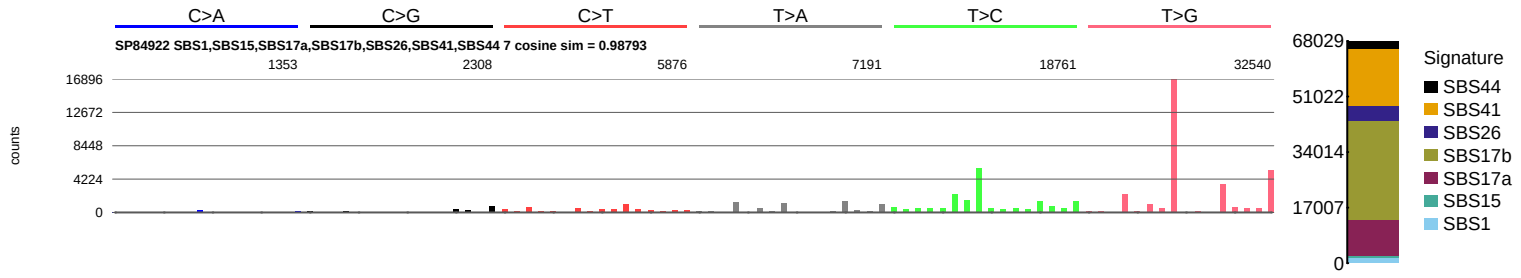

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS15,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98797
1668
2422
7467
7637
16202
32630
16912
12684
8456
4228
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS2

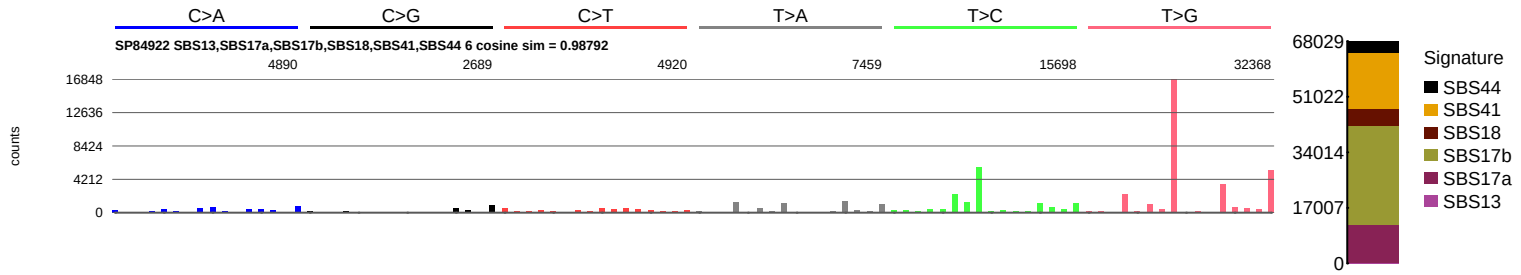

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98796
1232
3055
7390
7759
15737
32856
16940
12705
8470
4235
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

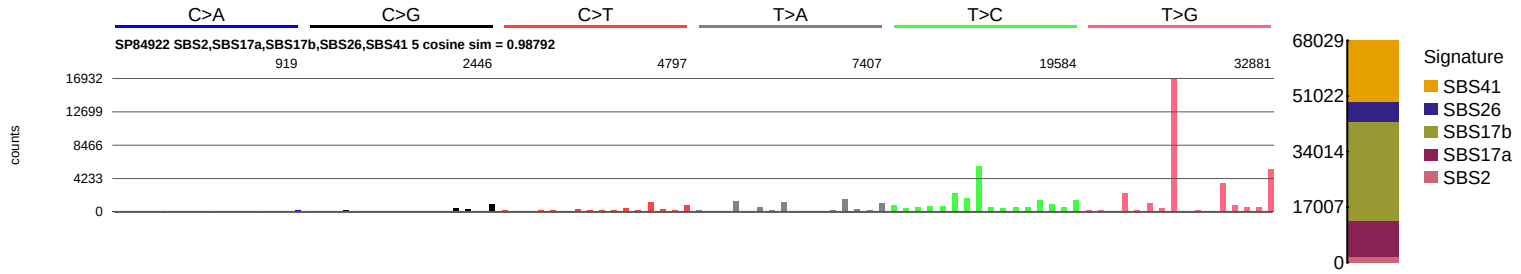

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98796
4800
2743
5092
7549
15315
32534
16856
12642
8428
4214
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13

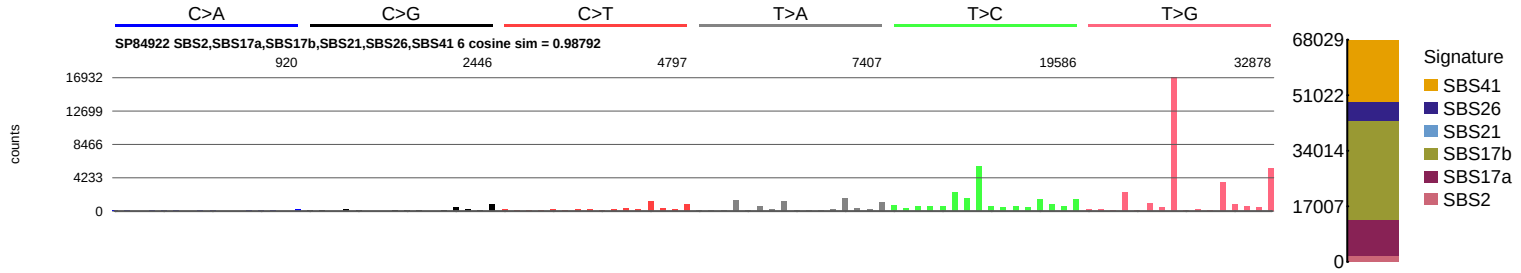

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS15,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98794
4817
2549
5110
7614
15371
32578
16856
12642
8428
4214
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15

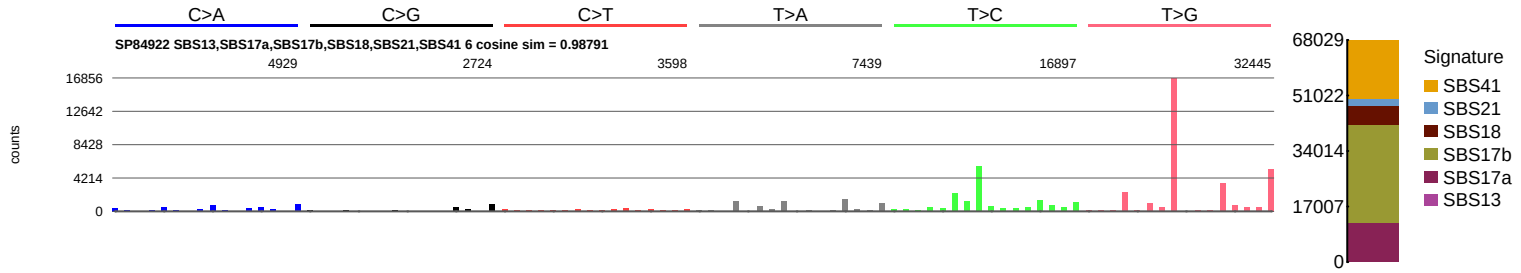

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98794
1241
2867
5426
7046
18895
32555
16908
12681
8454
4227
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS1

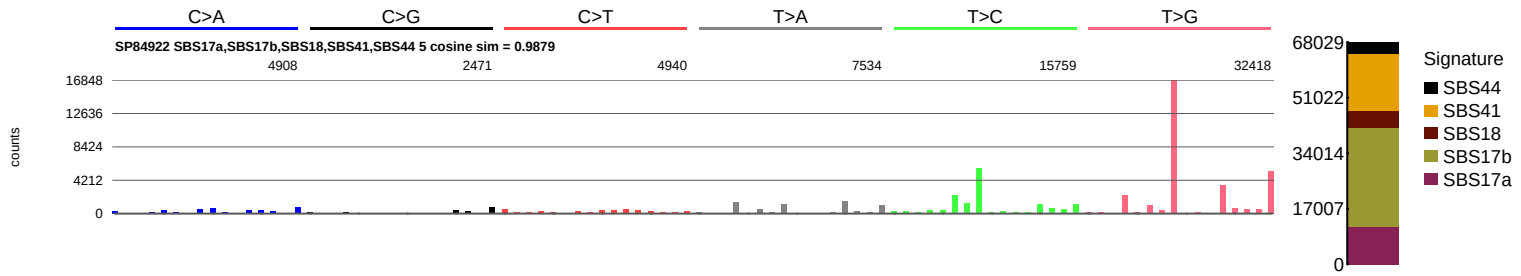

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98793
1353
2308
5876
7191
18761
32540
16896
12672
8448
4224
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS1

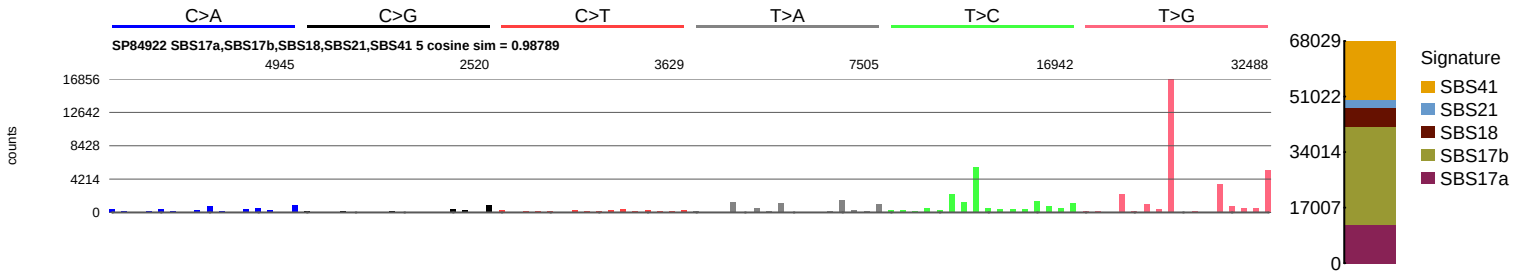

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98792
4890
2689
4920
7459
15698
32368
16848
12636
8424
4212
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13

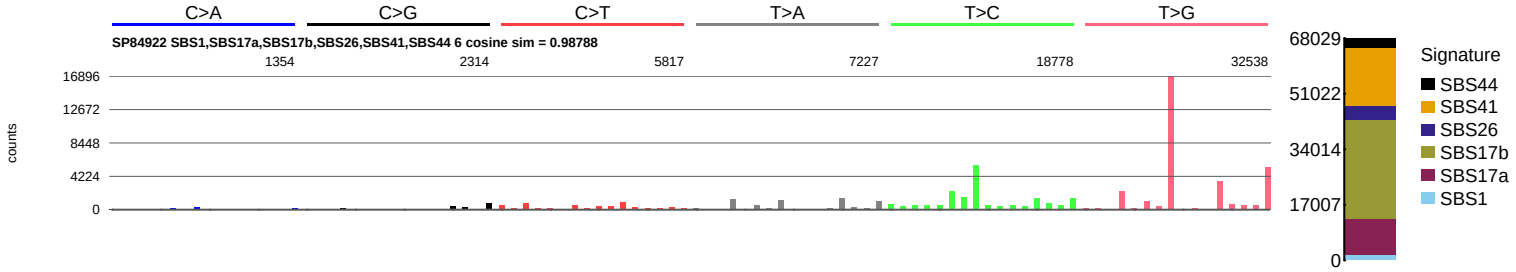

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS26,SBS41 5 cosine sim = 0.98792
919
2446
4797
7407
19584
32881
16932
12699
8466
4233
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS2

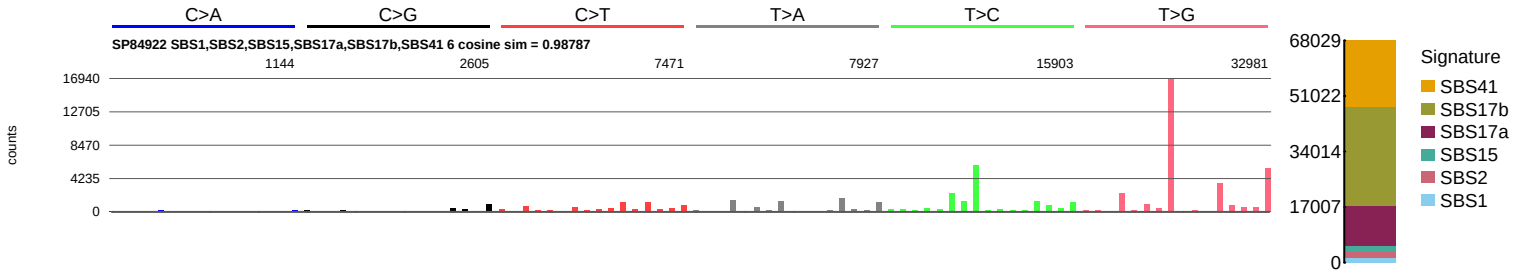

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS41 6 cosine sim = 0.98792
920
2446
4797
7407
19586
32878
16932
12699
8466
4233
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS2

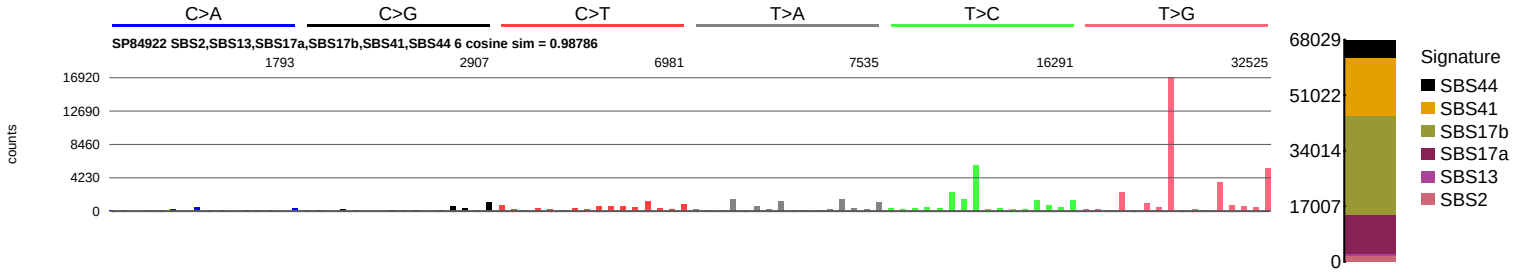

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98791
4929
2724
3598
7439
16897
32445
16856
12642
8428
4214
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13

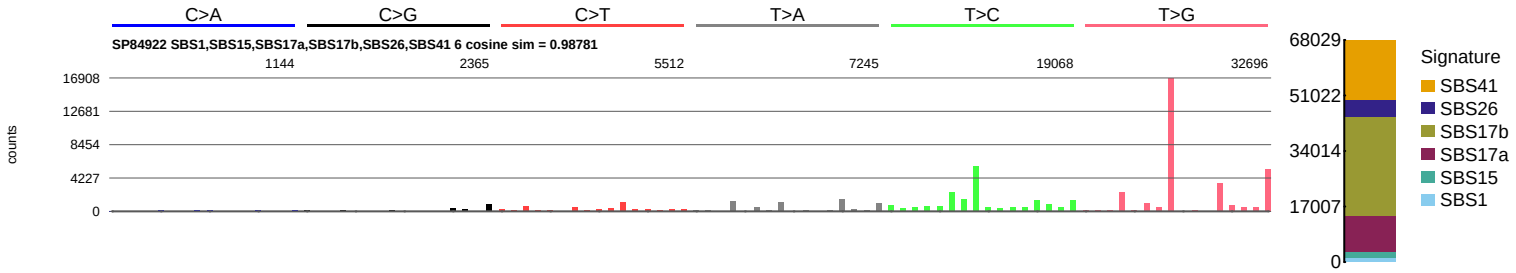

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS18,SBS41,SBS44 5 cosine sim = 0.9879
4908
2471
4940
7534
15759
32418
16848
12636
8424
4212
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS18,SBS21,SBS41 5 cosine sim = 0.98789
4945
2520
3629
7505
16942
32488
counts
16856
12642
8428
4214
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS26,SBS41,SBS44 6 cosine sim = 0.98788
1354
2314
5817
7227
18778
32538
counts
16896
12672
8448
4224
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98787
1144
2605
7471
7927
15903
32981
counts
16940
12705
8470
4235
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98786
1793
2907
6981
7535
16291
32525
counts
16920
12690
8460
4230
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98781
1144
2365
5512
7245
19068
32696
counts
16908
12681
8454
4227
0
68029
51022
34014
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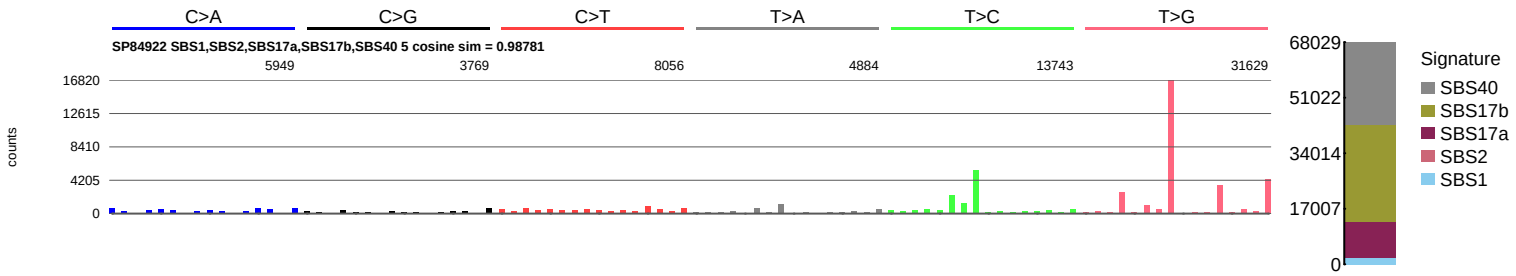
SBS1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C>G
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T>C
T>G
SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98781
5949
3769
8056
4884
13743
31629
counts
16820
12615
8410
4205
0
68029
51022
34014
17007
0
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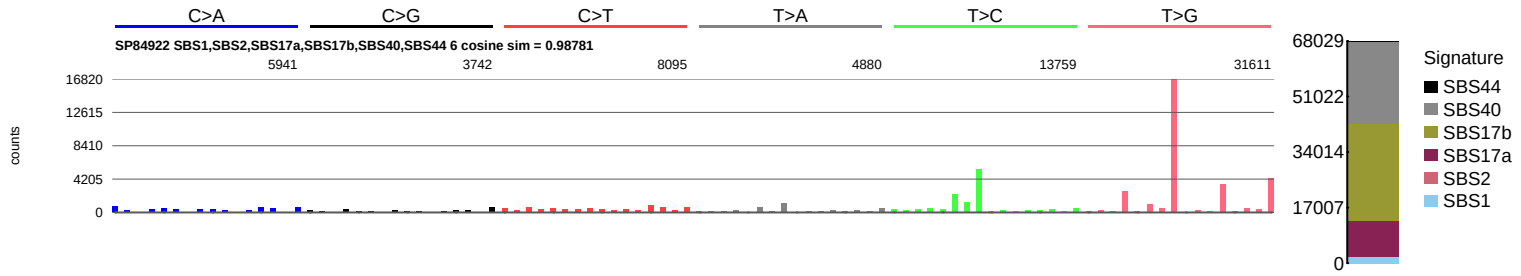
SBS17b
SBS17a
SBS2
SBS1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C>T
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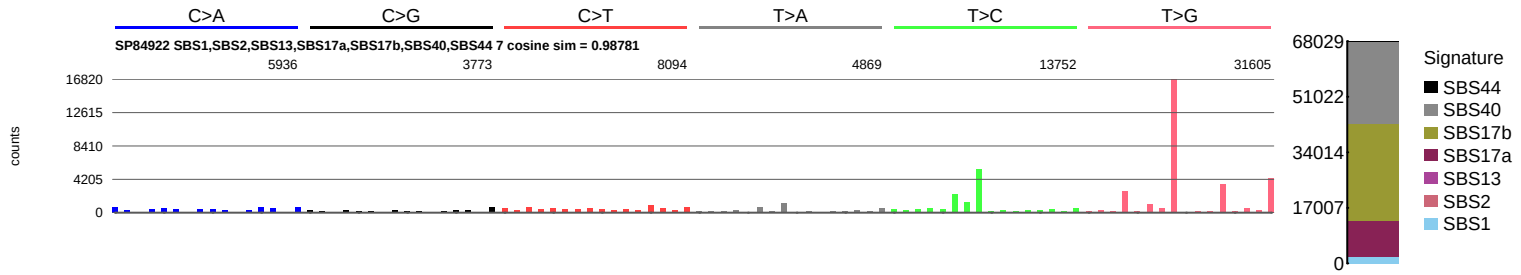
SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98781
5941
3742
8095
4880
13759
31611
16820
12615
8410
4205
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
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T>G
SP84922 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98781
5936
3773
8094
4869
13752
31605
16820
12615
8410
4205
0
counts
68029
51022
34014
17007
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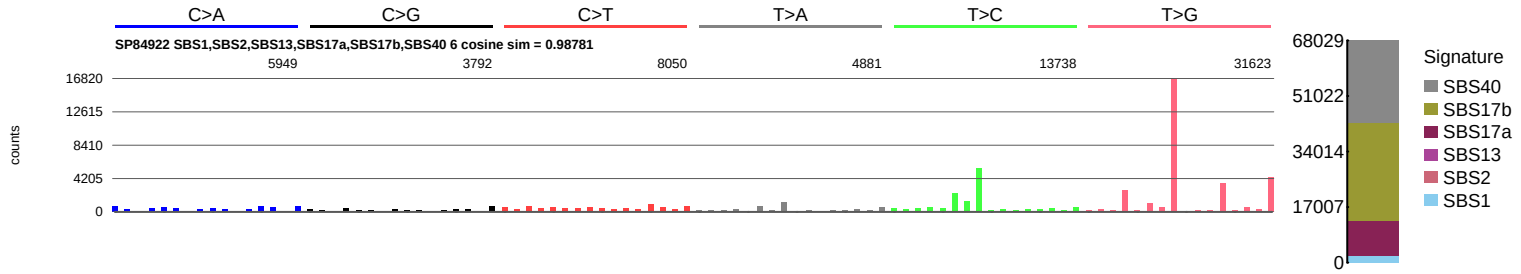
SBS17a
SBS13
SBS2
SBS1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C>G
C>T
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T>G
SP84922 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98781
5949
3792
8050
4881
13738
31623
16820
12615
8410
4205
0
counts
68029
51022
34014
17007
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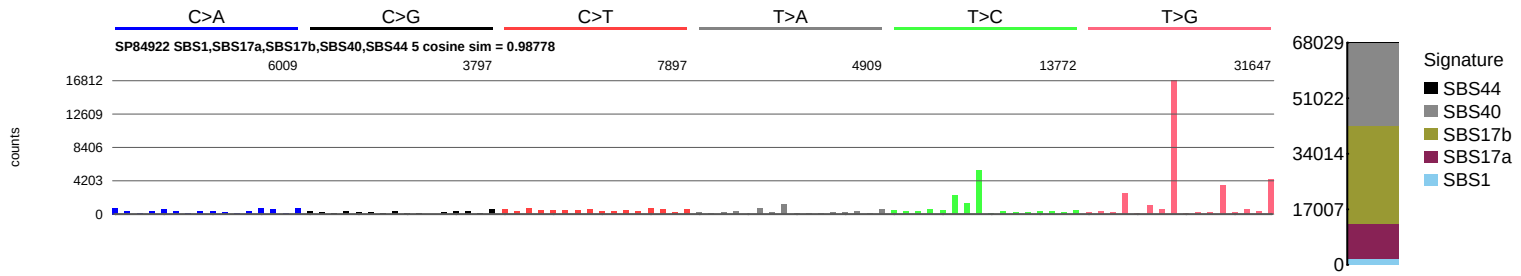
SBS40
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>G
C>T
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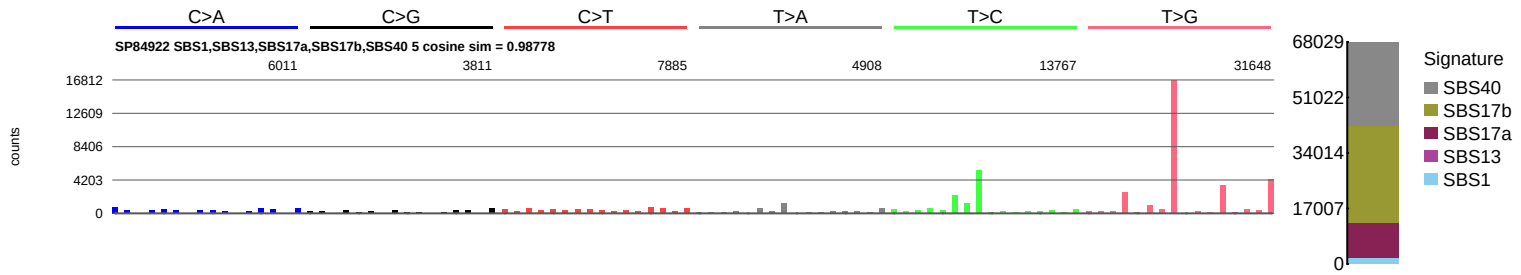
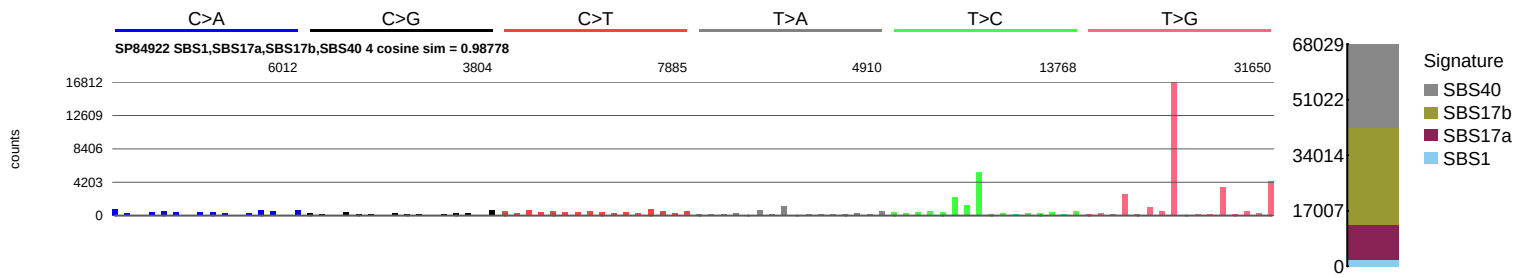
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS40,SBS44 5 cosine sim = 0.98778
6009
3797
7897
4909
13772
31647
16812
12609
8406
4203
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98778
6011
3811
7885
4908
13767
31648
16812
12609
8406
4203
0
counts
68029
51022
34014
17007
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS1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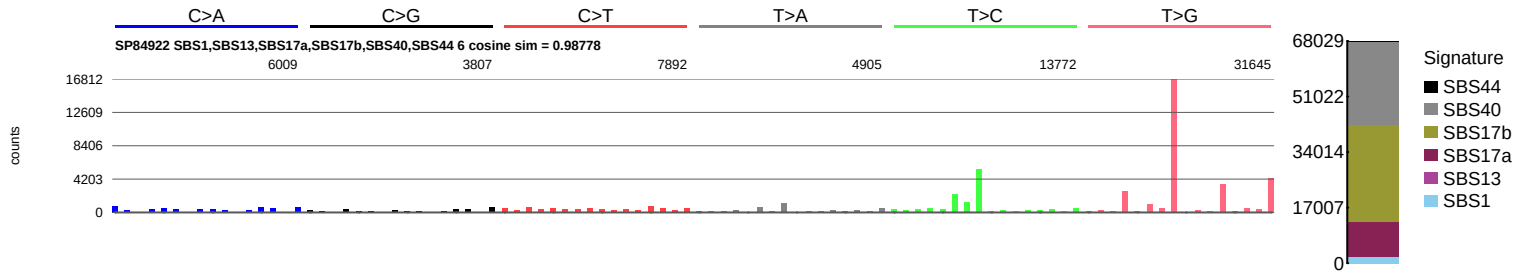
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS40 4 cosine sim = 0.98778
6012
3804
7885
4910
13768
31650
16812
12609
8406
4203
0
counts
68029
51022
34014
17007
0
Signature
SBS40
SBS17b
SBS17a
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98778
6009
3807
7892
4905
13772
31645
counts
16812
12609
8406
4203
0
68029
51022
34014
17007
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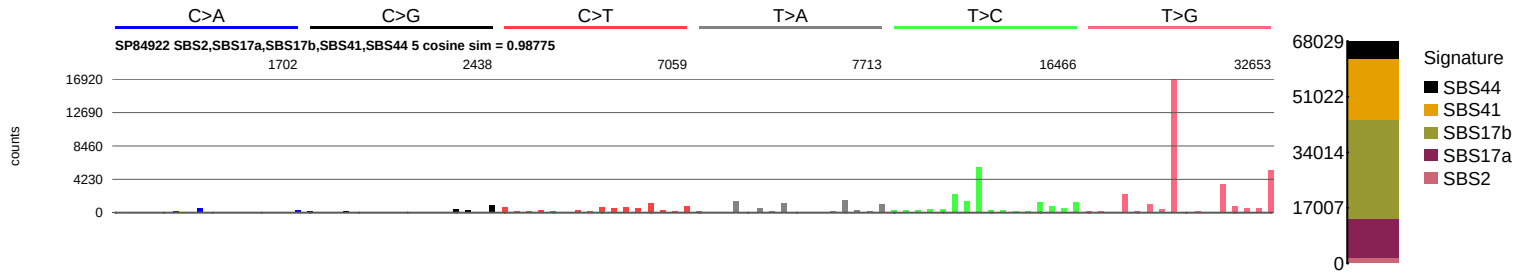
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS1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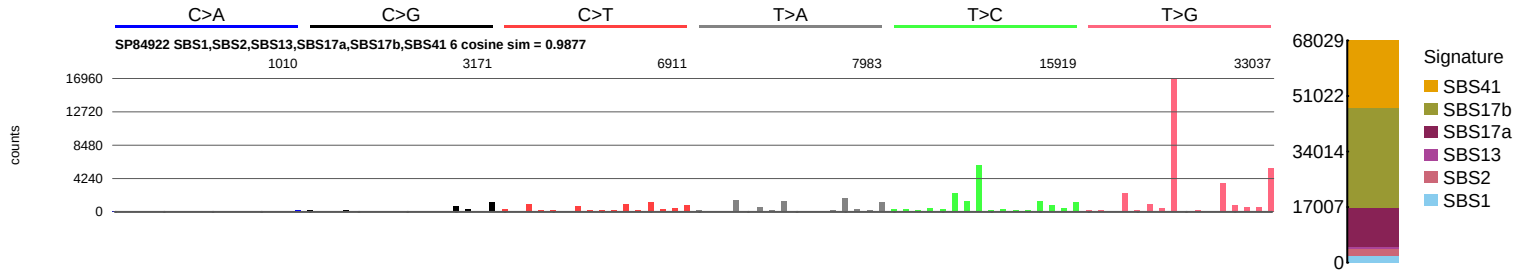
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS41,SBS44 5 cosine sim = 0.98775
1702
2438
7059
7713
16466
32653
counts
16920
12690
8460
4230
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS2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T>C
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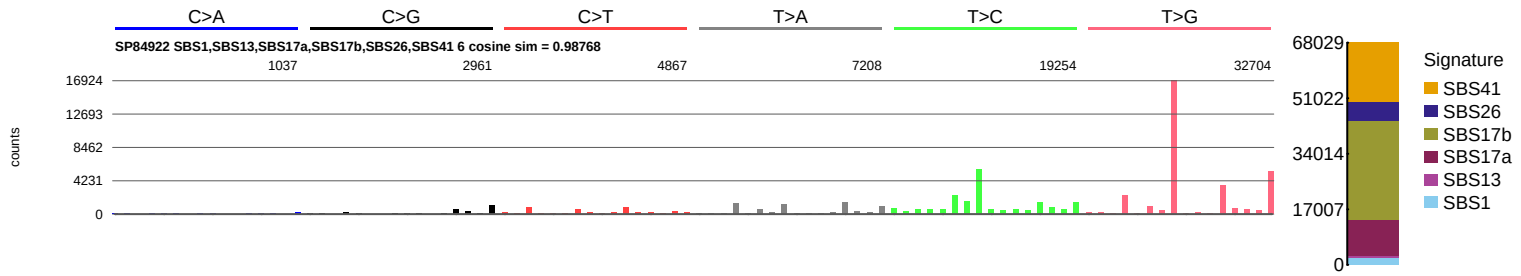
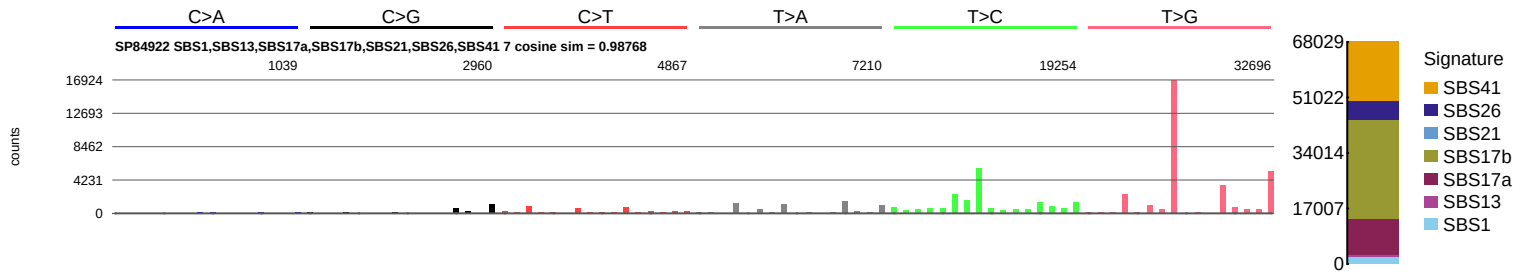
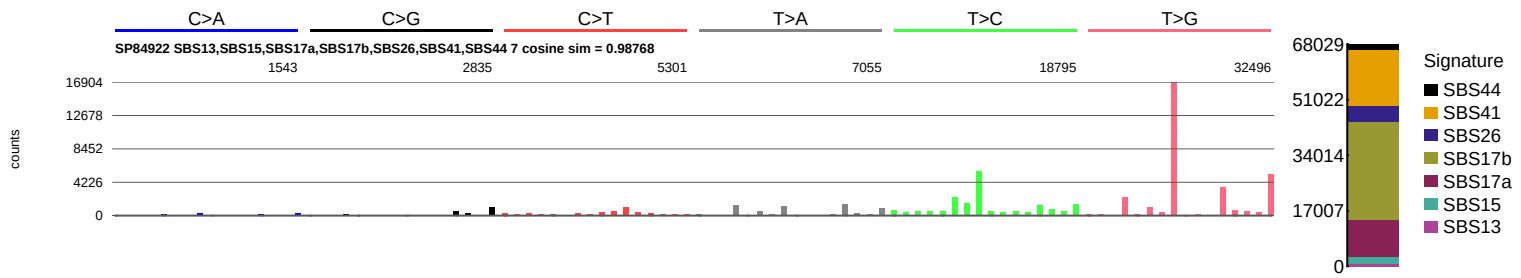
SP84922 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS41 6 cosine sim = 0.9877
1010
3171
6911
7983
15919
33037
counts
16960
12720
8480
4240
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98768
1037
2961
4867
7208
19254
32704
counts
16924
12693
8462
4231
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41 7 cosine sim = 0.98768
1039
2960
4867
7210
19254
32696
counts
16924
12693
8462
4231
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98768
1543
2835
5301
7055
18795
32496
counts
16904
12678
8452
4226
0
68029
51022
34014
17007
0
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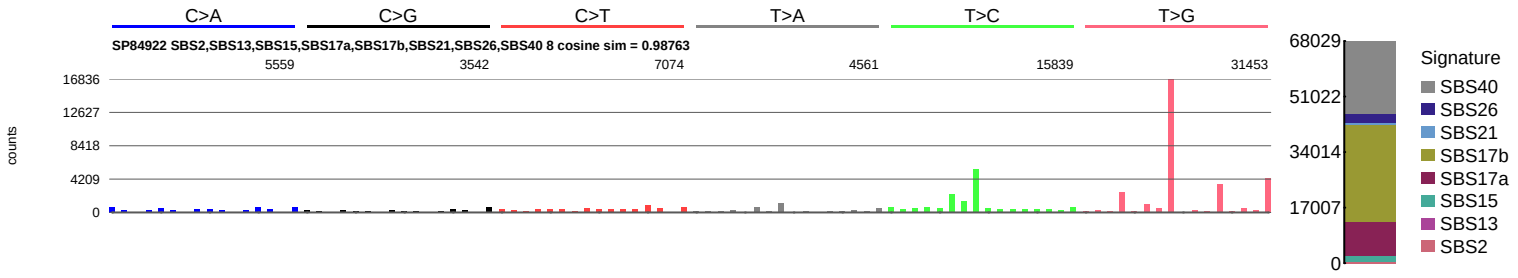
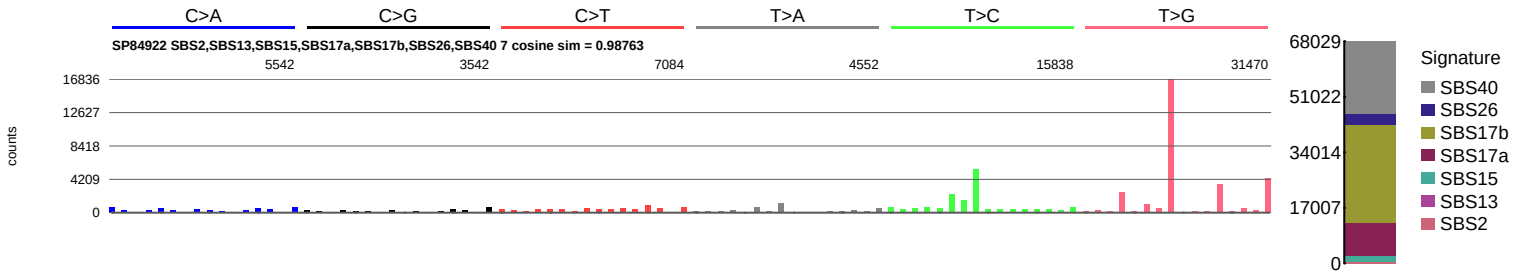
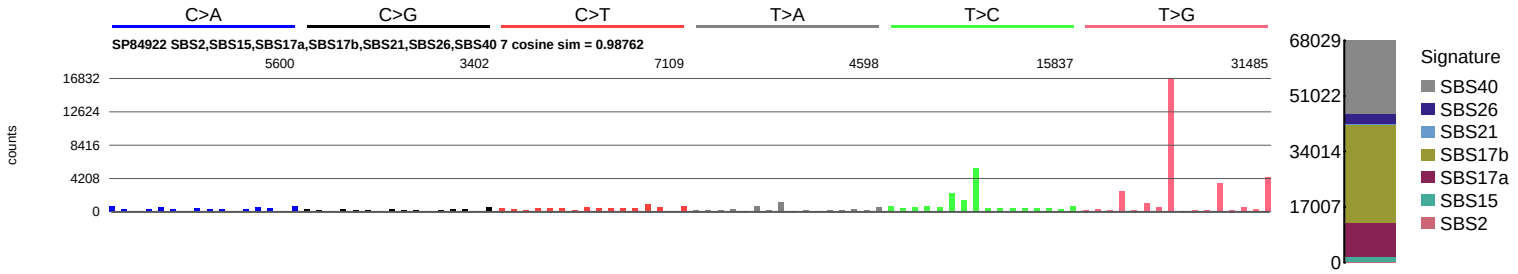
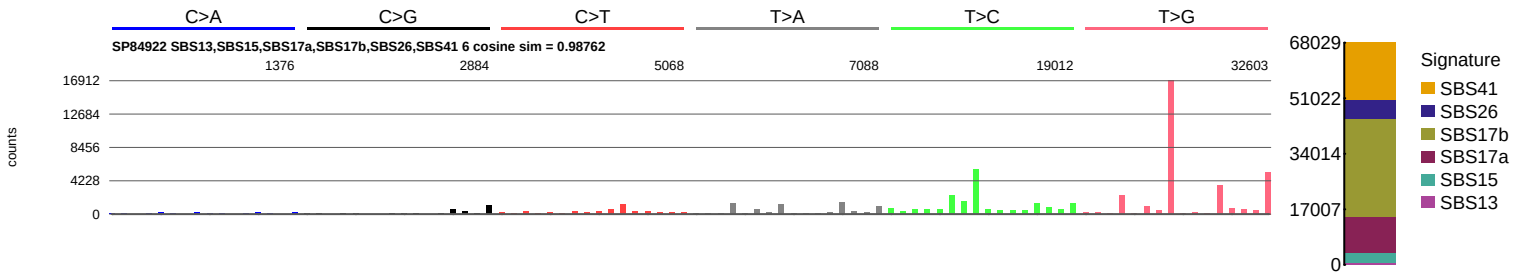
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 8 cosine sim = 0.98763
5559
3542
7074
4561
15839
31453
16836
12627
8418
4209
0
counts
68029
51022
34014
17007
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98763
5542
3542
7084
4552
15838
31470
16836
12627
8418
4209
0
counts
68029
51022
34014
17007
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.98762
5600
3402
7109
4598
15837
31485
16832
12624
8416
4208
0
counts
68029
51022
34014
17007
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98762
1376
2884
5068
7088
19012
32603
16912
12684
8456
4228
0
counts
68029
51022
34014
17007
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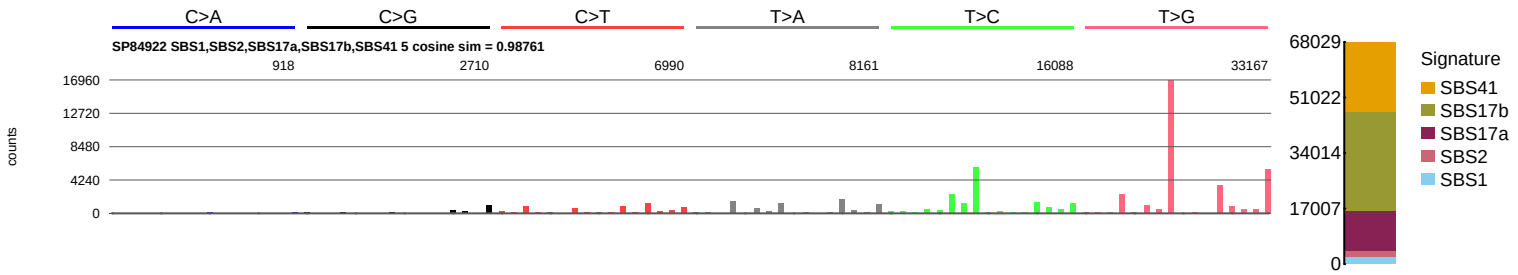
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98761
918
2710
6990
8161
16088
33167
16960
12720
8480
4240
0
counts
68029
51022
34014
17007
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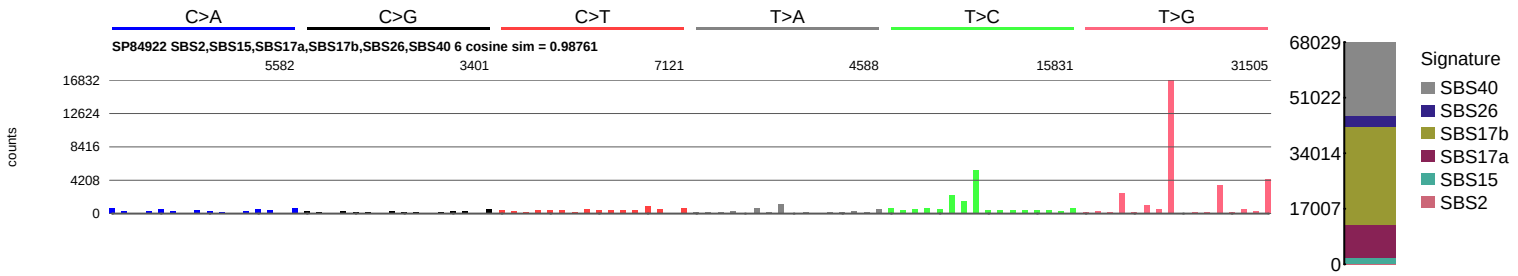
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98761
5582
3401
7121
4588
15831
31505
16832
12624
8416
4208
0
counts
68029
51022
34014
17007
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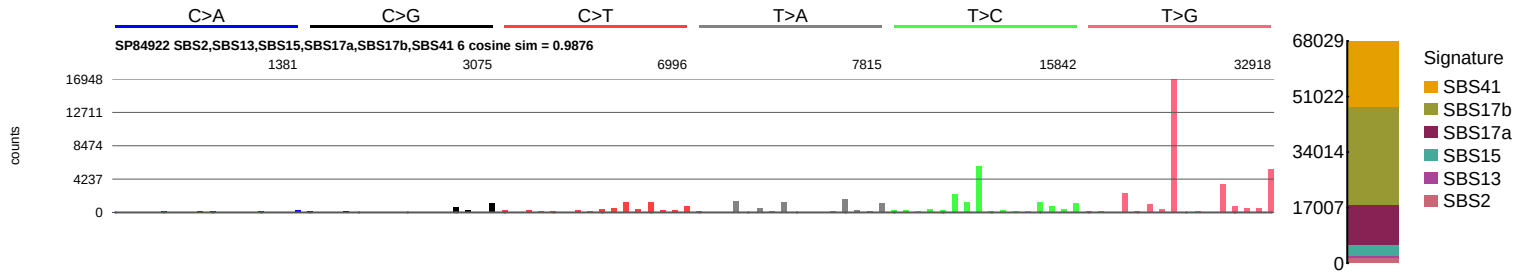
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2


C>A
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T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.9876
1381
3075
6996
7815
15842
32918
counts
16948
12711
8474
4237
0
68029
51022
34014
17007
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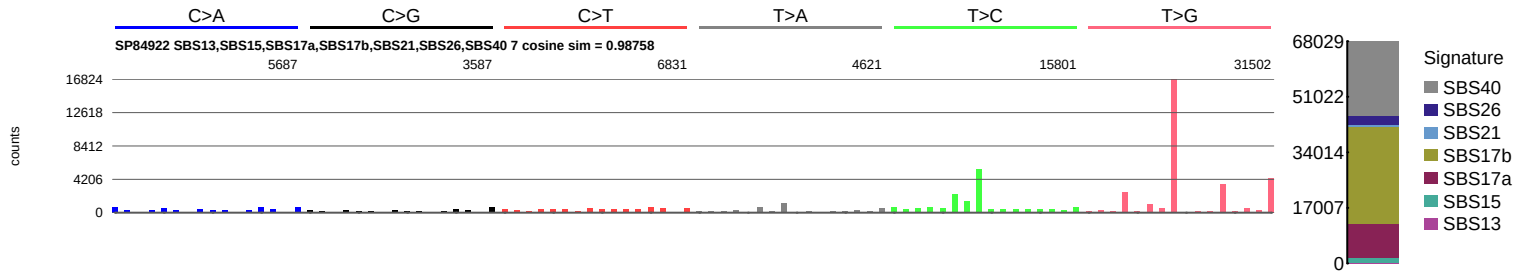
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.98758
5687
3587
6831
4621
15801
31502
counts
16824
12618
8412
4206
0
68029
51022
34014
17007
0
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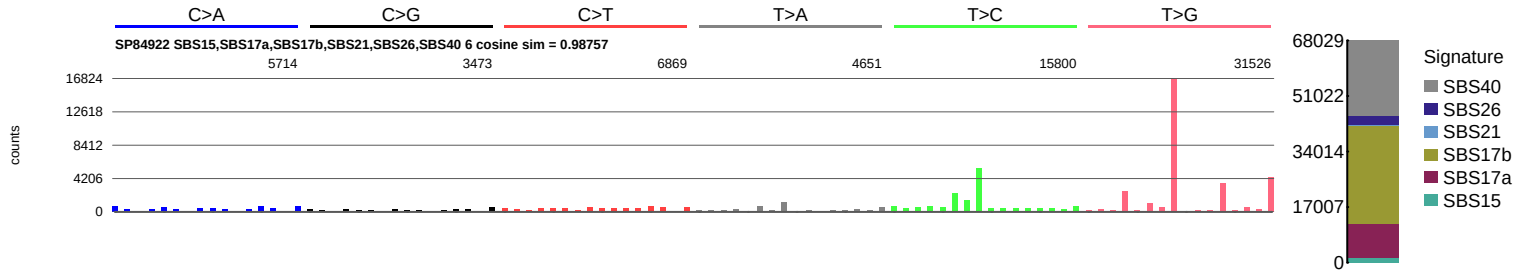
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 6 cosine sim = 0.98757
5714
3473
6869
4651
15800
31526
counts
16824
12618
8412
4206
0
68029
51022
34014
17007
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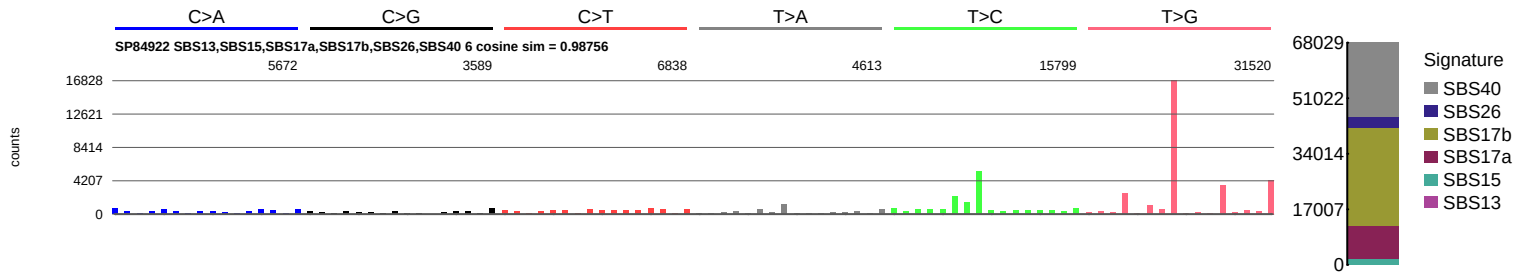
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98756
5672
3589
6838
4613
15799
31520
counts
16828
12621
8414
4207
0
68029
51022
34014
17007
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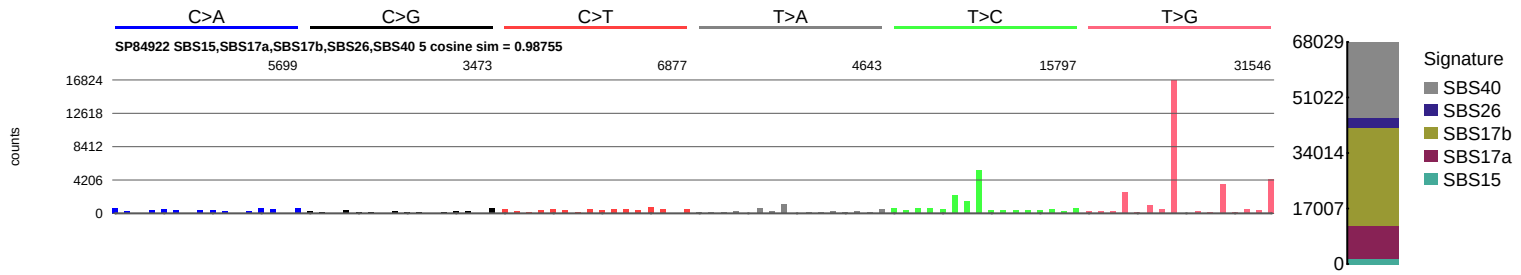
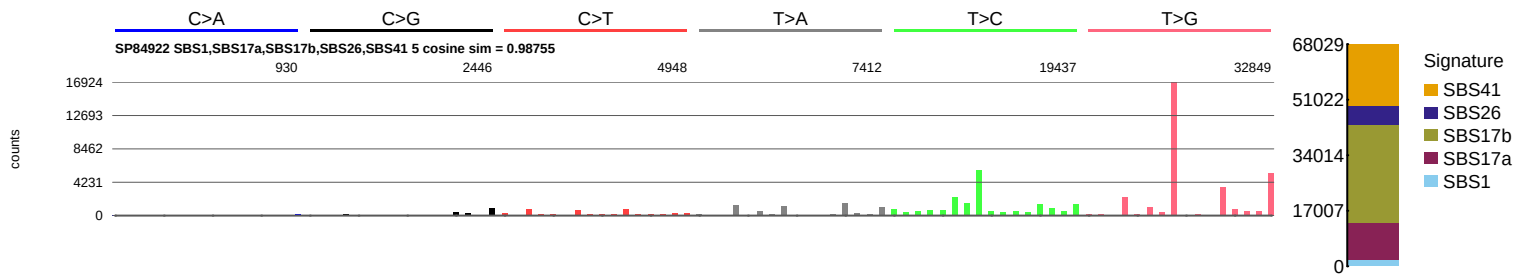
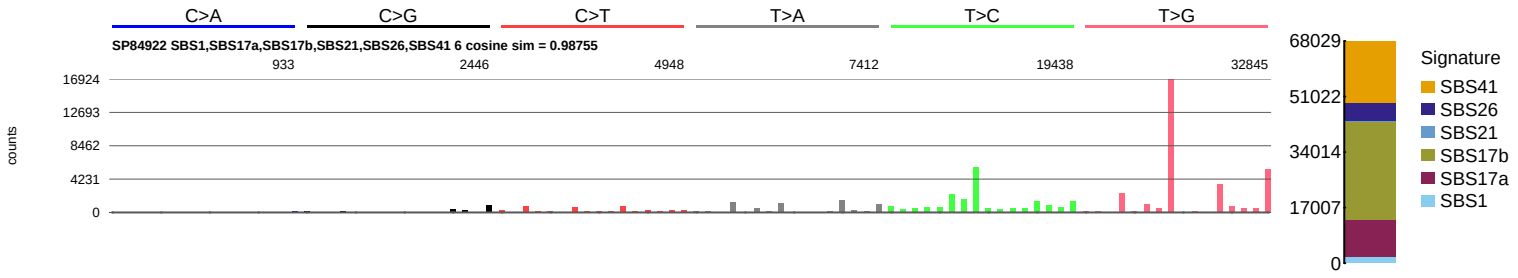
SBS26
SBS17b
SBS17a
SBS15
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS15,SBS17a,SBS17b,SBS26,SBS40 5 cosine sim = 0.98755
5699
3473
6877
4643
15797
31546
counts
16824
12618
8412
4206
0
68029
51022
34014
17007
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS26,SBS41 5 cosine sim = 0.98755
930
2446
4948
7412
19437
32849
counts
16924
12693
8462
4231
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS21,SBS26,SBS41 6 cosine sim = 0.98755
933
2446
4948
7412
19438
32845
16924
12693
8462
4231
0
counts
68029
51022
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SBS41
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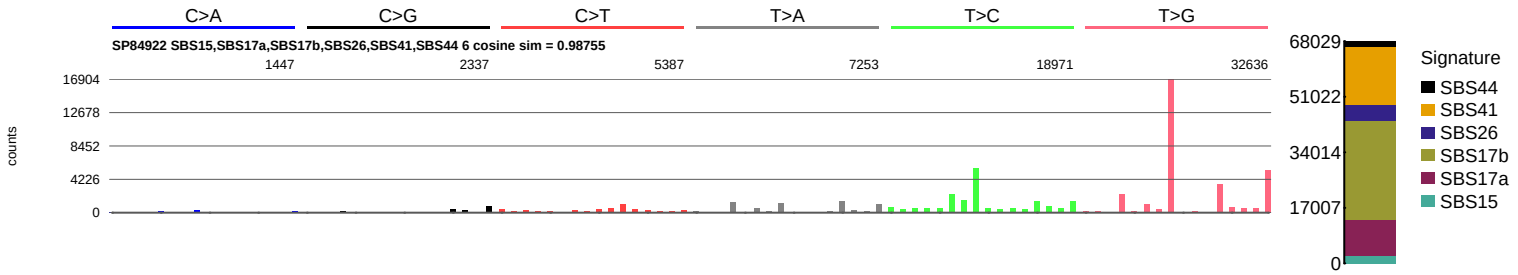
SBS1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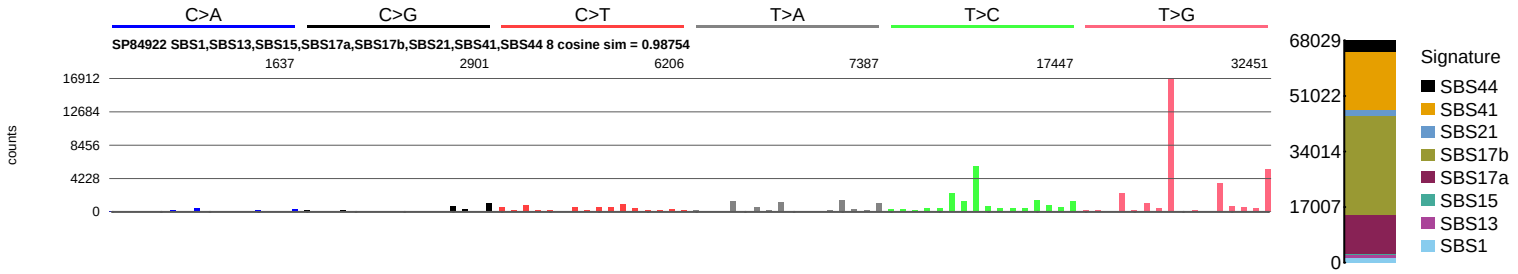
T>C
T>G
SP84922 SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 6 cosine sim = 0.98755
1447
2337
5387
7253
18971
32636
16904
12678
8452
4226
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15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C>T
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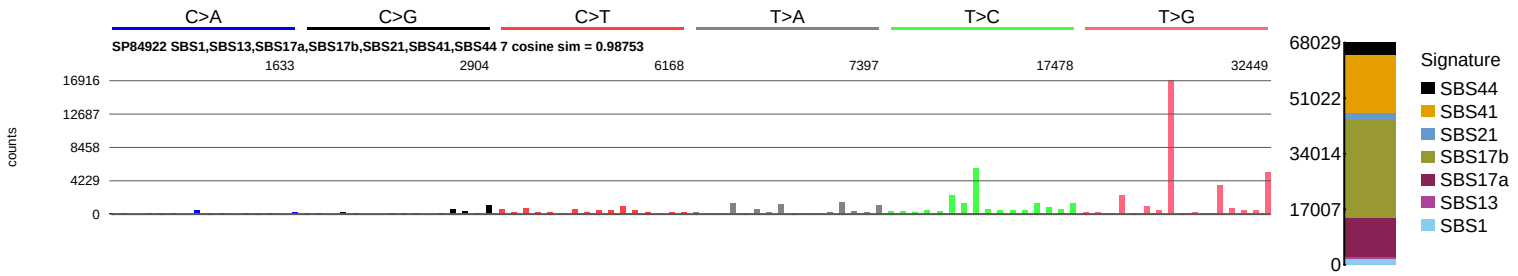
T>C
T>G
SP84922 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98754
1637
2901
6206
7387
17447
32451
16912
12684
8456
4228
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1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C>G
C>T
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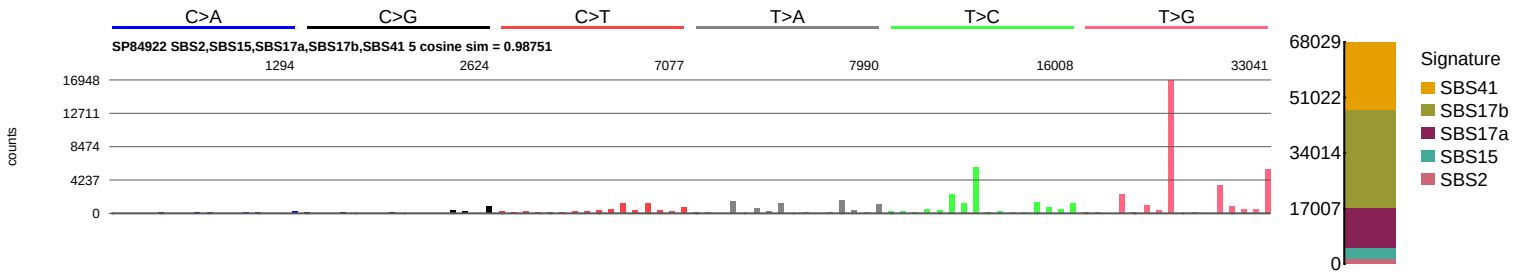
T>G
SP84922 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98753
1633
2904
6168
7397
17478
32449
16916
12687
8458
4229
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS1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C>G
C>T
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T>G
SP84922 SBS2,SBS15,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98751
1294
2624
7077
7990
16008
33041
16948
12711
8474
4237
0
counts
68029
51022
34014
17007
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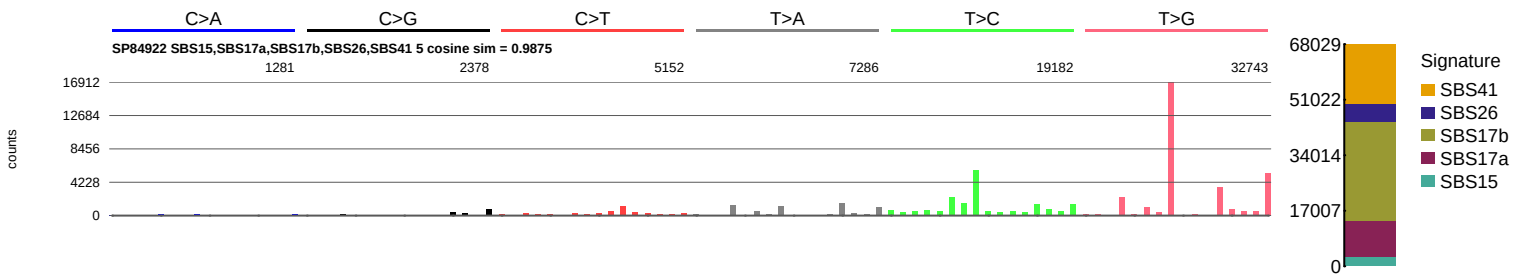
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS15,SBS17a,SBS17b,SBS26,SBS41 5 cosine sim = 0.9875
1281
2378
5152
7286
19182
32743
16912
12684
8456
4228
0
counts
68029
51022
34014
17007
0
Signature
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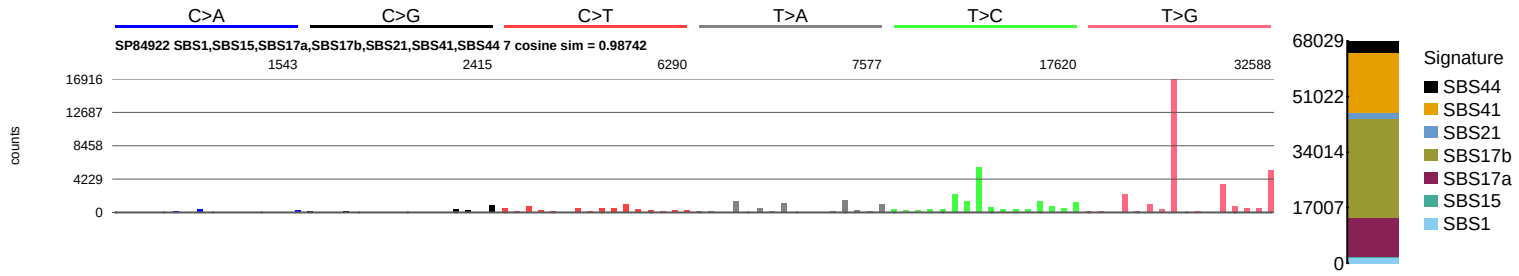
SBS26
SBS17b
SBS17a
SBS15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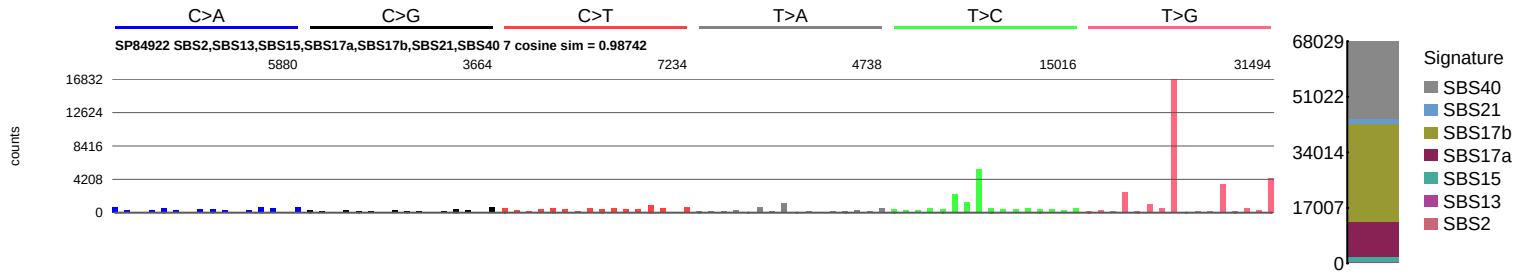
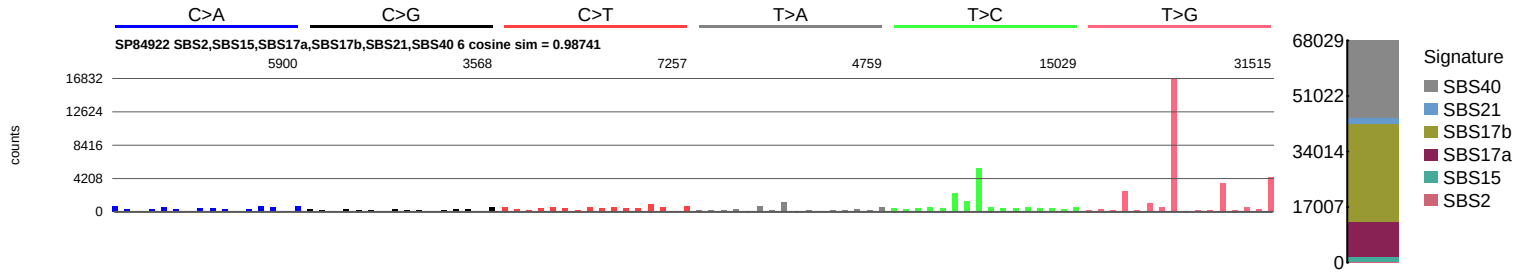
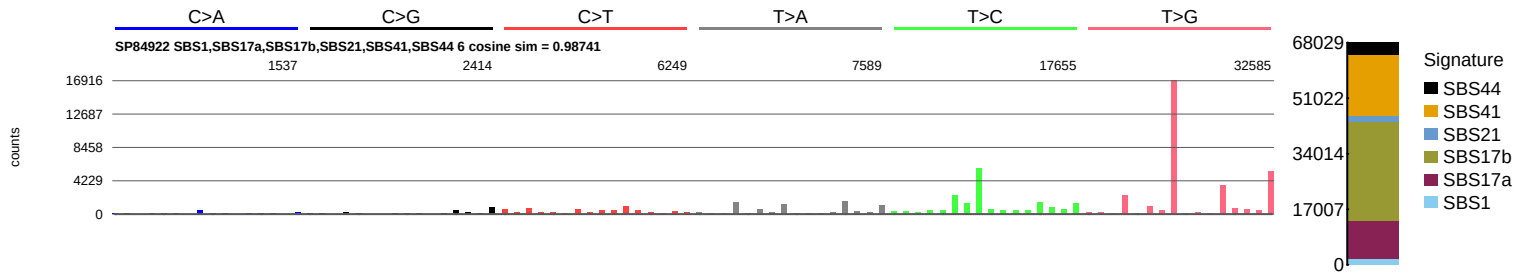
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98742
1543
2415
6290
7577
17620
32588
16916
12687
8458
4229
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98742
5880
3664
7234
4738
15016
31494
16832
12624
8416
4208
0
counts
68029
51022
34014
17007
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98741
5900
3568
7257
4759
15029
31515
16832
12624
8416
4208
0
counts
68029
51022
34014
17007
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS21,SBS41,SBS44 6 cosine sim = 0.98741
1537
2414
6249
7589
17655
32585
16916
12687
8458
4229
0
counts
68029
51022
34014
17007
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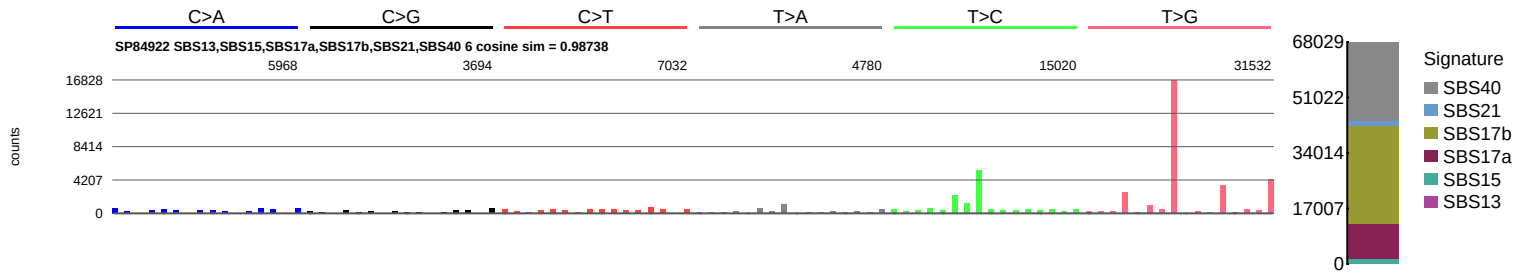
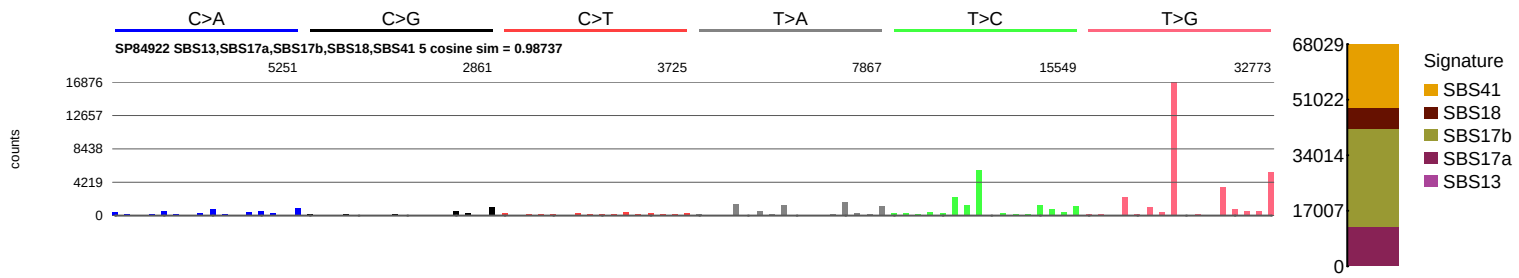
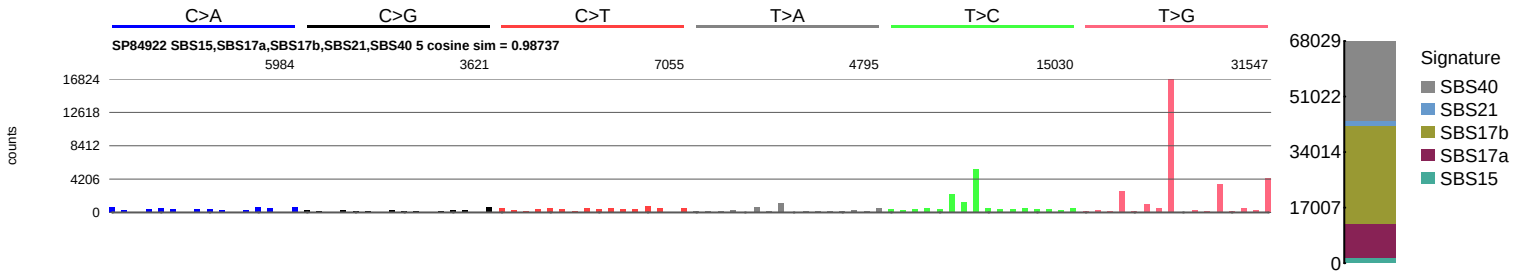
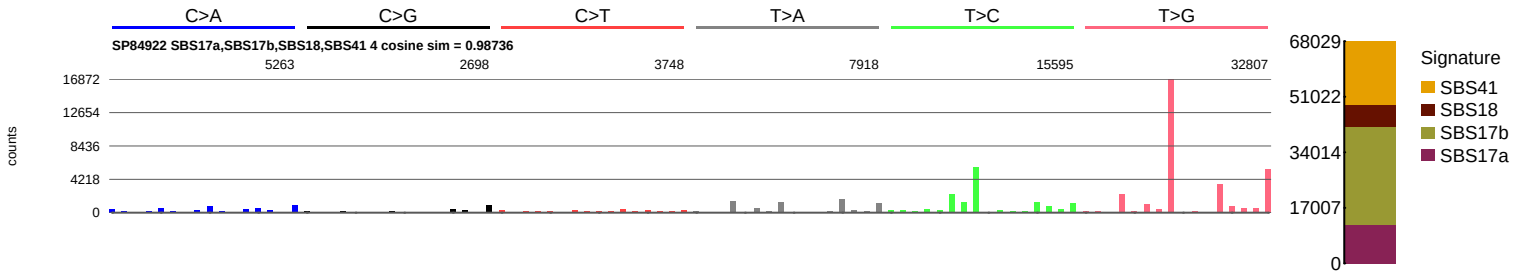
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98738
5968
3694
7032
4780
15020
31532
16828
12621
8414
4207
0
counts
68029
51022
34014
17007
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98737
5251
2861
3725
7867
15549
32773
16876
12657
8438
4219
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS15,SBS17a,SBS17b,SBS21,SBS40 5 cosine sim = 0.98737
5984
3621
7055
4795
15030
31547
16824
12618
8412
4206
0
counts
68029
51022
34014
17007
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS18,SBS41 4 cosine sim = 0.98736
5263
2698
3748
7918
15595
32807
16872
12654
8436
4218
0
counts
68029
51022
34014
17007
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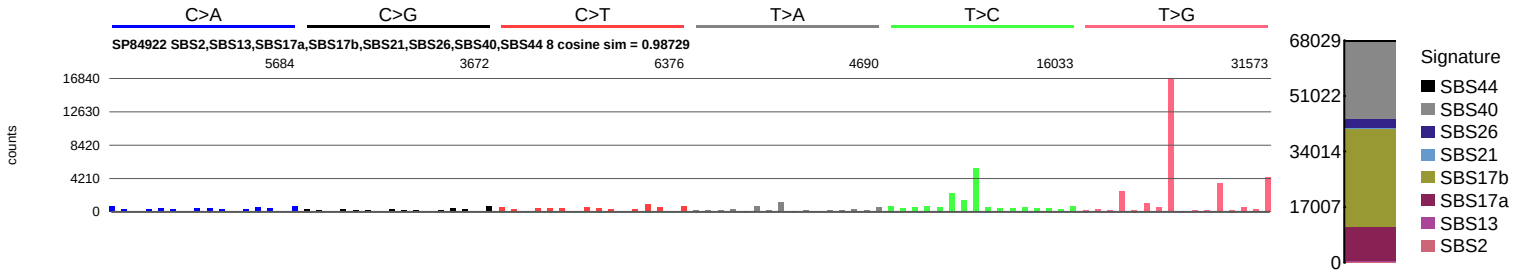
Signature
SBS41
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98729
5684
3672
6376
4690
16033
31573
16840
12630
8420
4210
0
counts
68029
51022
34014
17007
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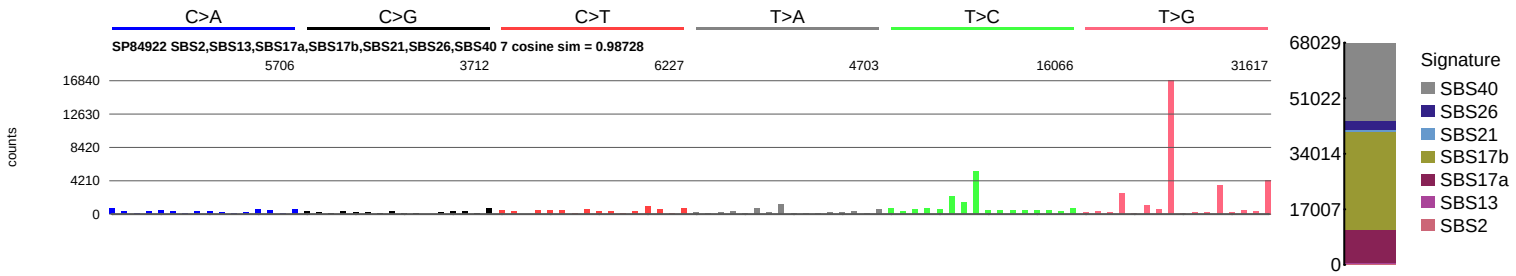
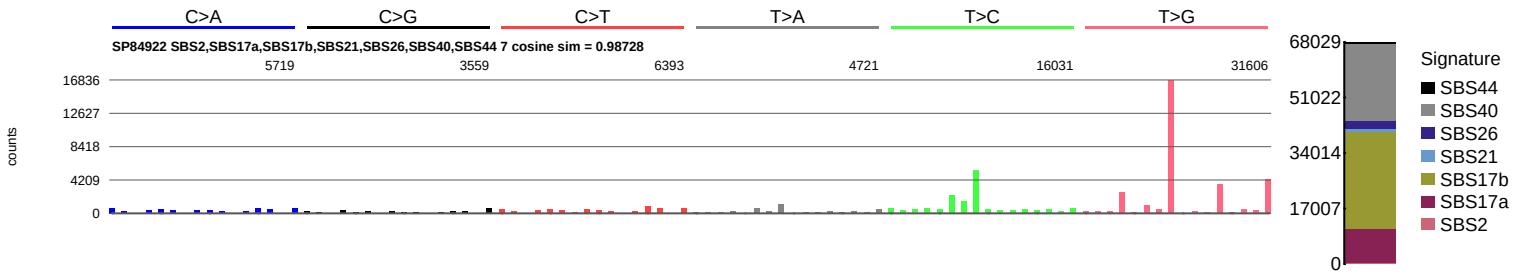
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.98728
5706
3712
6227
4703
16066
31617
16840
12630
8420
4210
0
counts
68029
51022
34014
17007
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 7 cosine sim = 0.98728
5719
3559
6393
4721
16031
31606
16836
12627
8418
4209
0
counts
68029
51022
34014
17007
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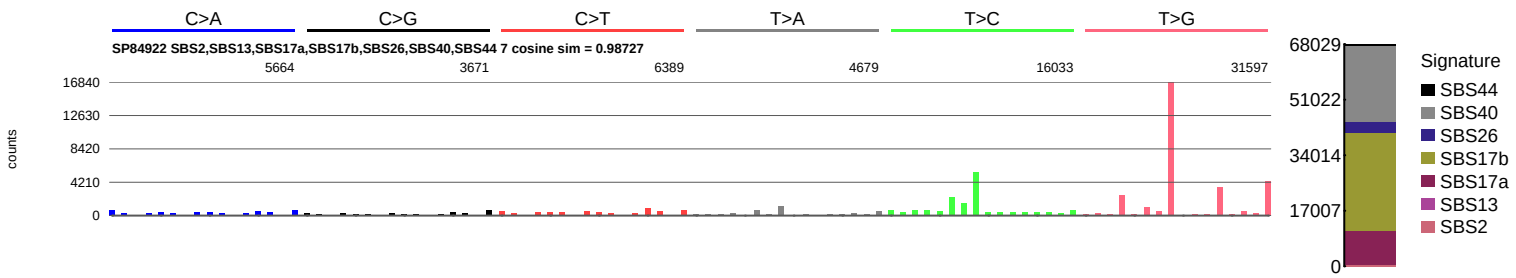
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98727
5664
3671
6389
4679
16033
31597
16840
12630
8420
4210
0
counts
68029
51022
34014
17007
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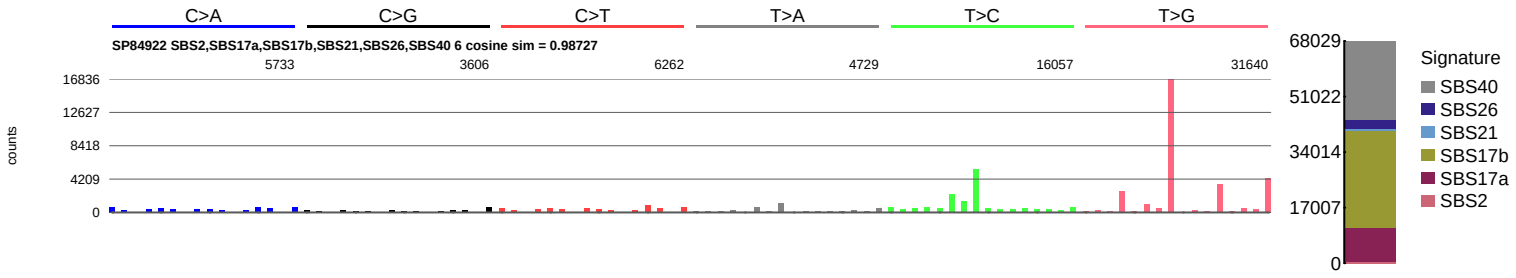
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40 6 cosine sim = 0.98727
5733
3606
6262
4729
16057
31640
counts
16836
12627
8418
4209
0
68029
51022
34014
17007
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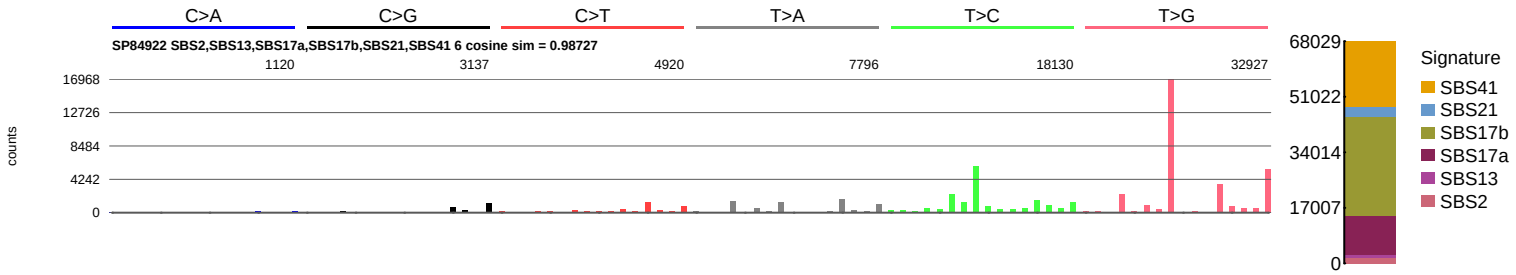
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98727
1120
3137
4920
7796
18130
32927
counts
16968
12726
8484
4242
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68029
51022
34014
17007
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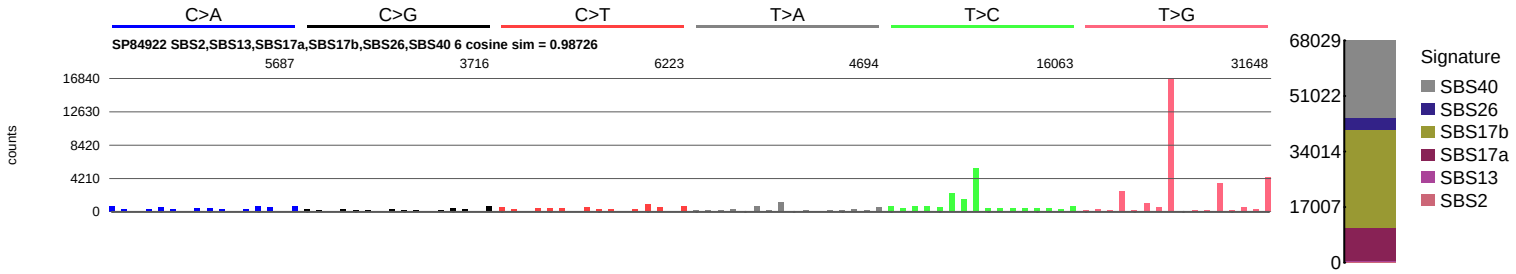
SBS21
SBS17b
SBS17a
SBS13
SBS2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C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98726
5687
3716
6223
4694
16063
31648
counts
16840
12630
8420
4210
0
68029
51022
34014
17007
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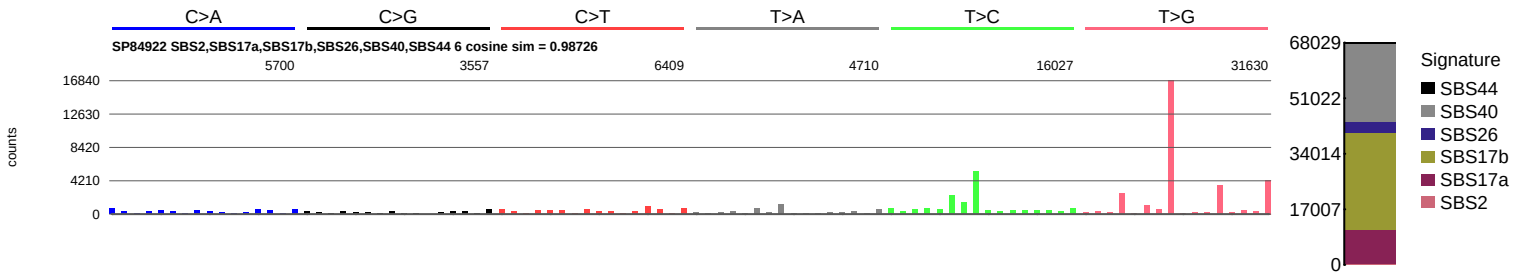
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS44 6 cosine sim = 0.98726
5700
3557
6409
4710
16027
31630
counts
16840
12630
8420
4210
0
68029
51022
34014
17007
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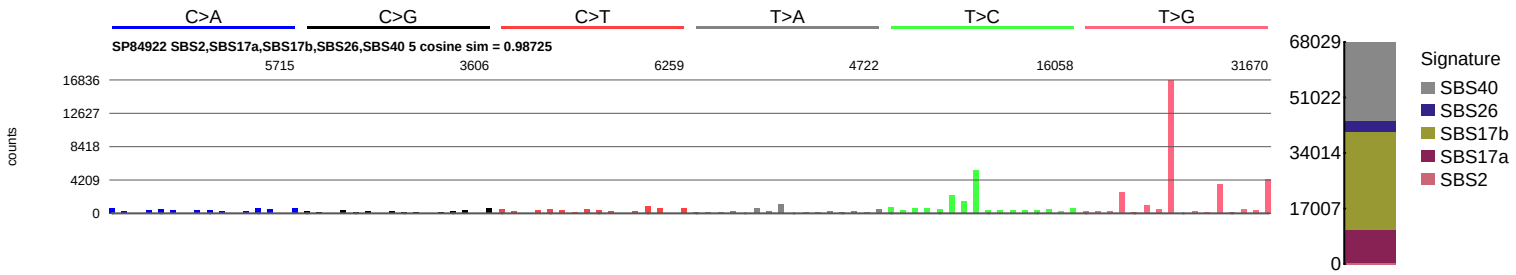
SBS40
SBS26
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS26,SBS40 5 cosine sim = 0.98725
5715
3606
6259
4722
16058
31670
counts
16836
12627
8418
4209
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68029
51022
34014
17007
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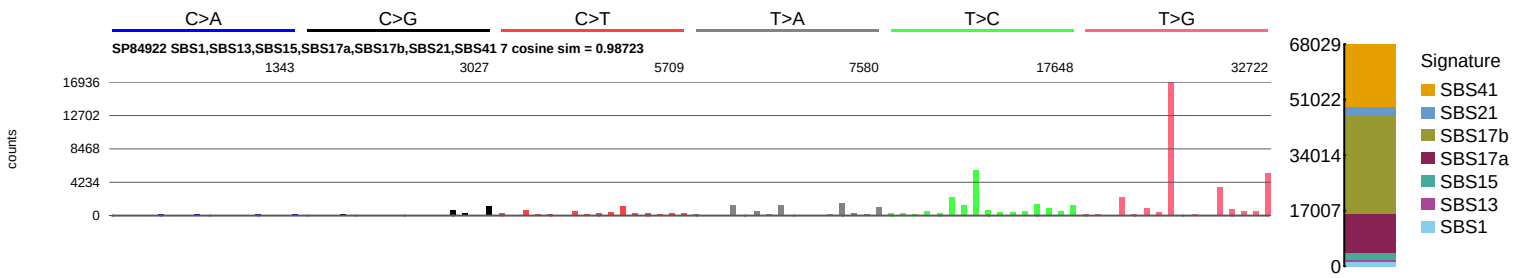
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98723
1343
3027
5709
7580
17648
32722
counts
16936
12702
8468
4234
0
68029
51022
34014
17007
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SBS21
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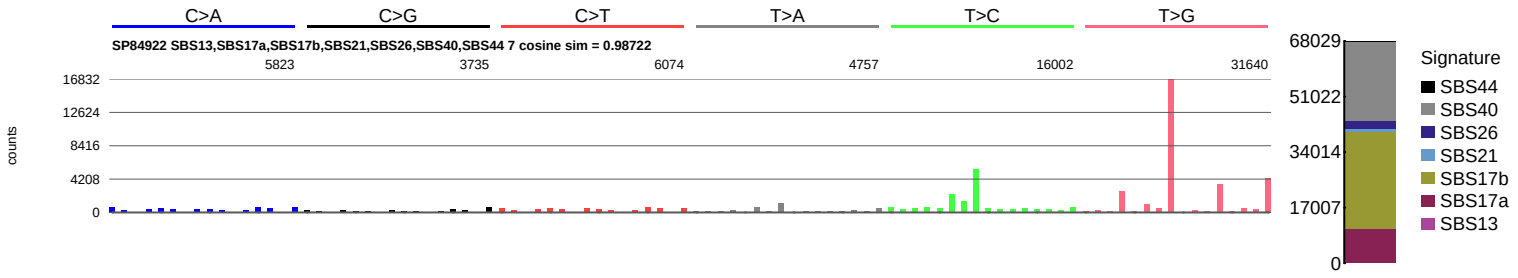
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 7 cosine sim = 0.98722
5823
3735
6074
4757
16002
31640
counts
16832
12624
8416
4208
0
68029
51022
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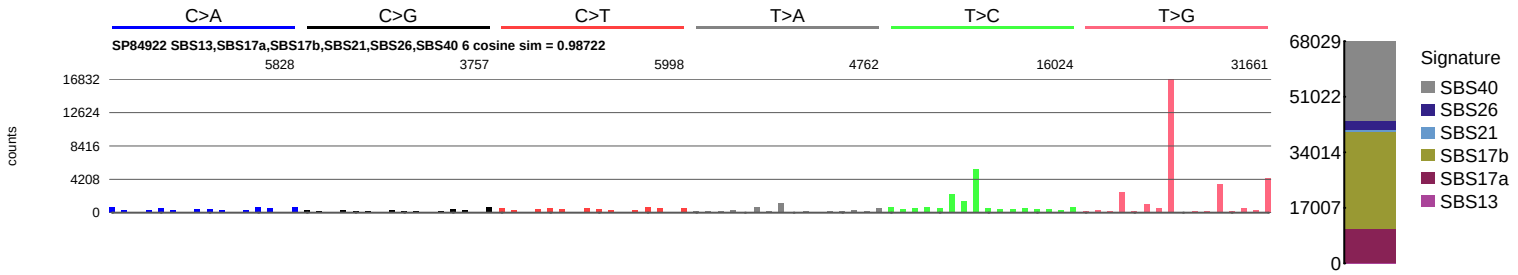
SBS21
SBS17b
SBS17a
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40 6 cosine sim = 0.98722
5828
3757
5998
4762
16024
31661
counts
16832
12624
8416
4208
0
68029
51022
34014
17007
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SBS21
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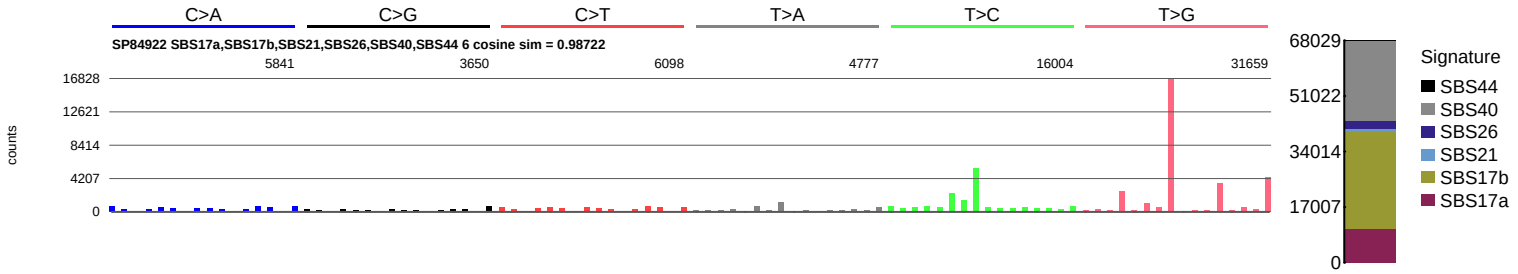
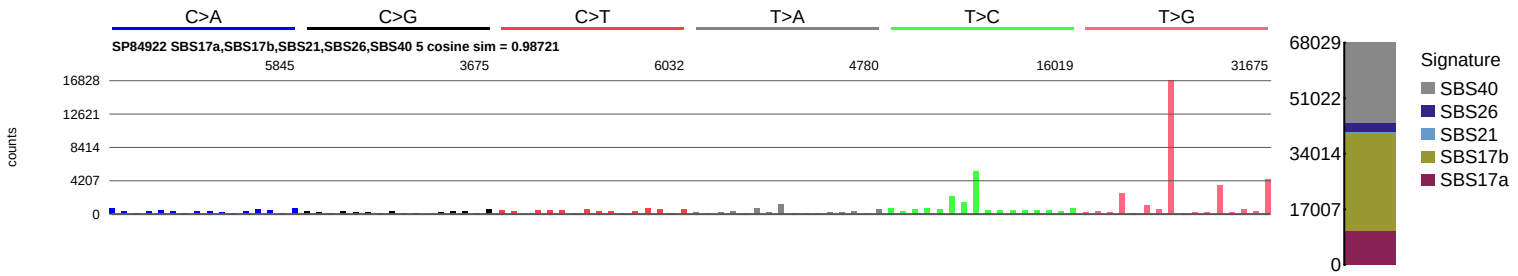
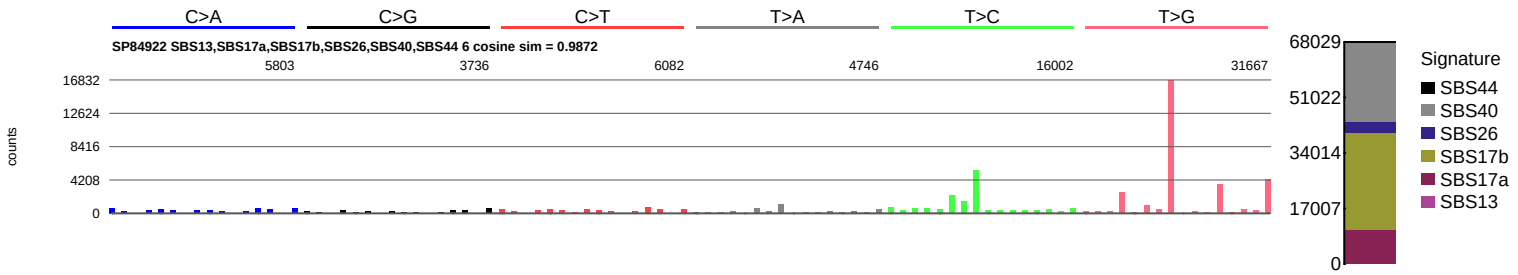
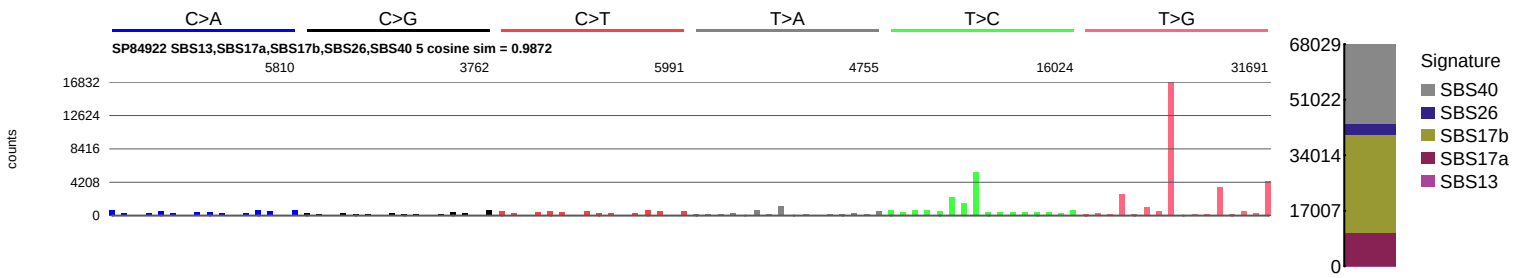
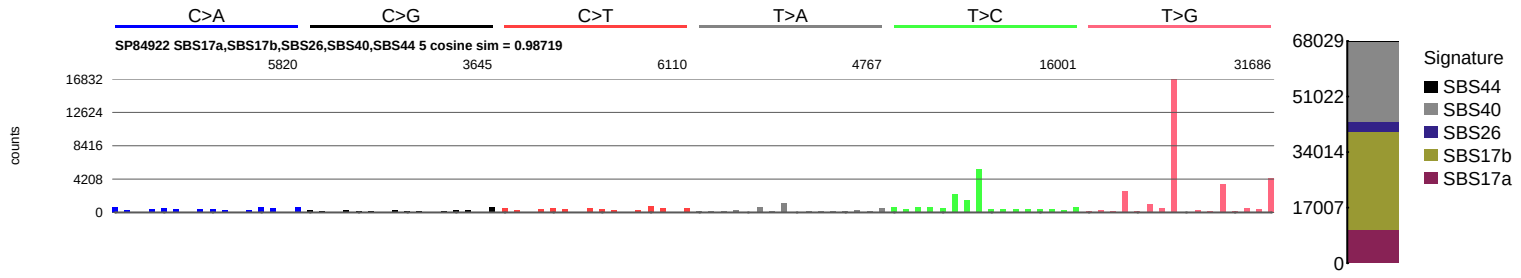
SBS17a
SBS13

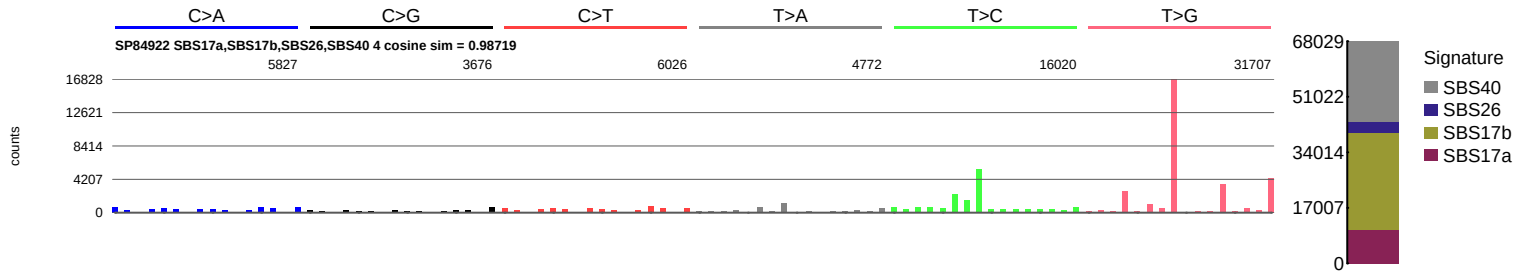

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 6 cosine sim = 0.98722
5841
3650
6098
4777
16004
31659
counts
16828
12621
8414
4207
0
68029
51022
34014
17007
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a

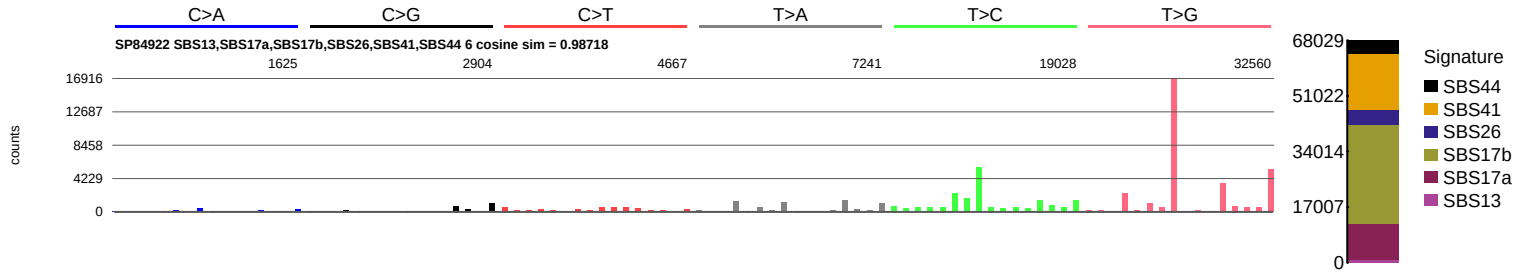

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS21,SBS26,SBS40 5 cosine sim = 0.98721
5845
3675
6032
4780
16019
31675
counts
16828
12621
8414
4207
0
68029
51022
34014
17007
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a

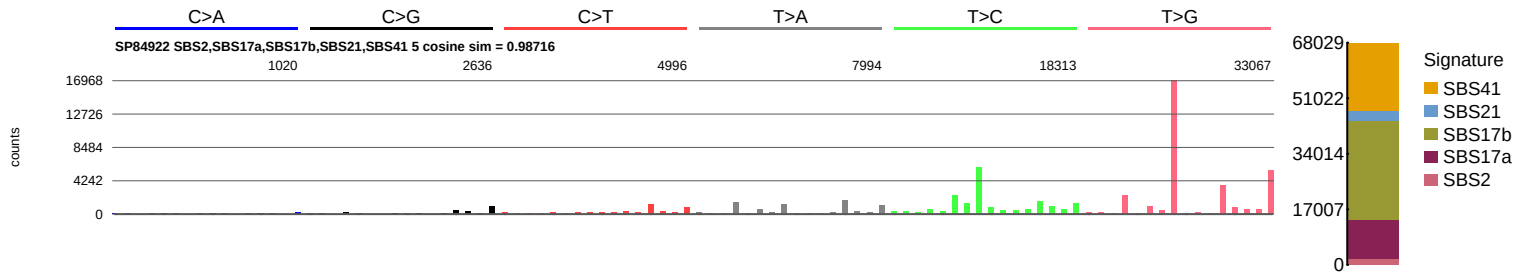

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 6 cosine sim = 0.9872
5803
3736
6082
4746
16002
31667
counts
16832
12624
8416
4208
0
68029
51022
34014
17007
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13

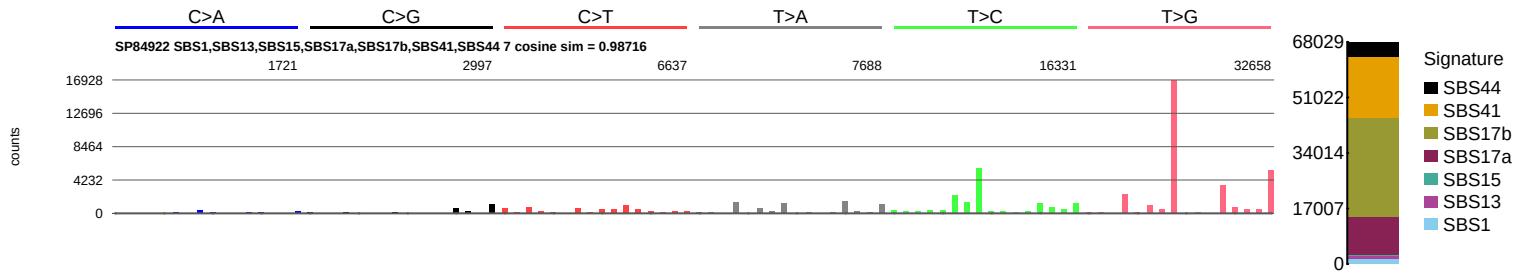

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS17a,SBS17b,SBS26,SBS40 5 cosine sim = 0.9872
5810
3762
5991
4755
16024
31691
counts
16832
12624
8416
4208
0
68029
51022
34014
17007
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13

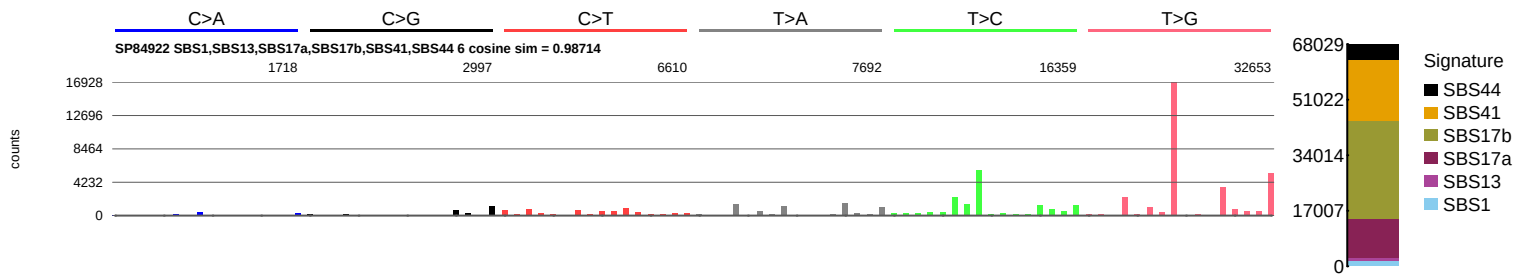

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS26,SBS40,SBS44 5 cosine sim = 0.98719
5820
3645
6110
4767
16001
31686
16832
12624
8416
4208
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a

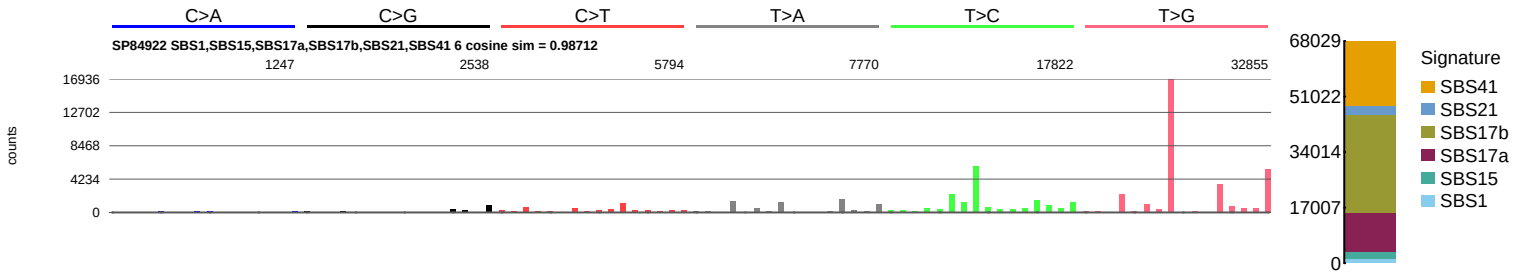

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS26,SBS40 4 cosine sim = 0.98719
5827
3676
6026
4772
16020
31707
16828
12621
8414
4207
0
counts
68029
51022
34014
17007
0
Signature
SBS40
SBS26
SBS17b
SBS17a

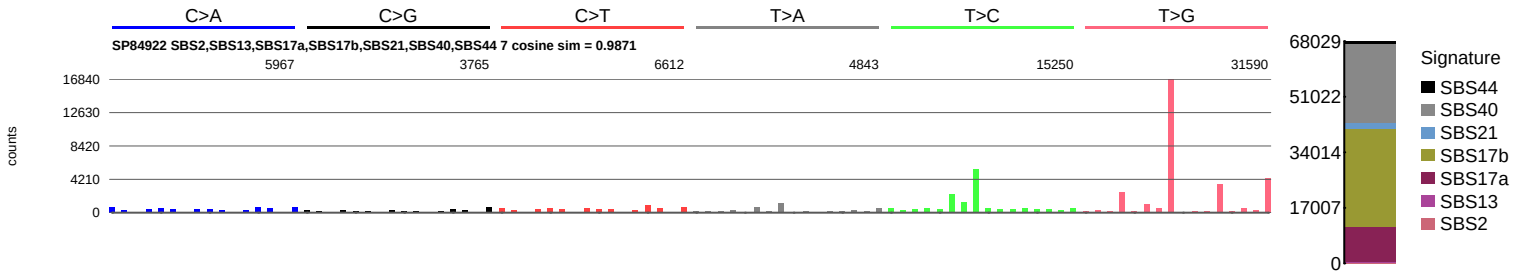

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 6 cosine sim = 0.98718
1625
2904
4667
7241
19028
32560
16916
12687
8458
4229
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13

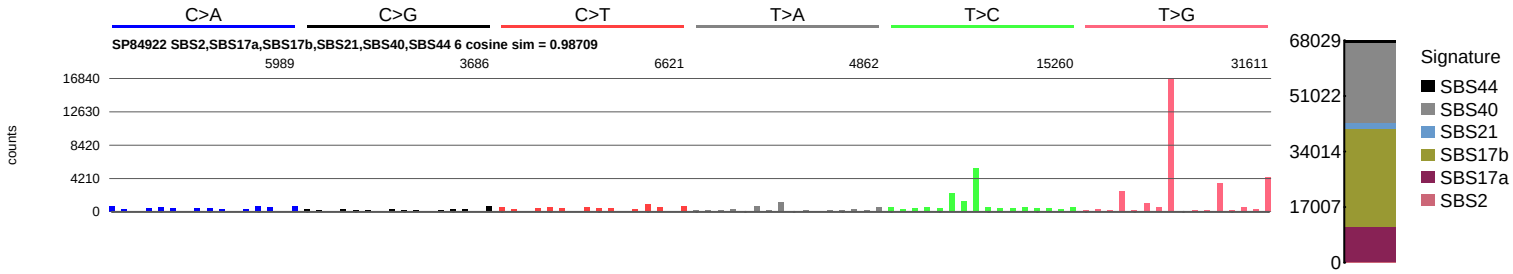

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS21,SBS41 5 cosine sim = 0.98716
1020
2636
4996
7994
18313
33067
16968
12726
8484
4242
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS2

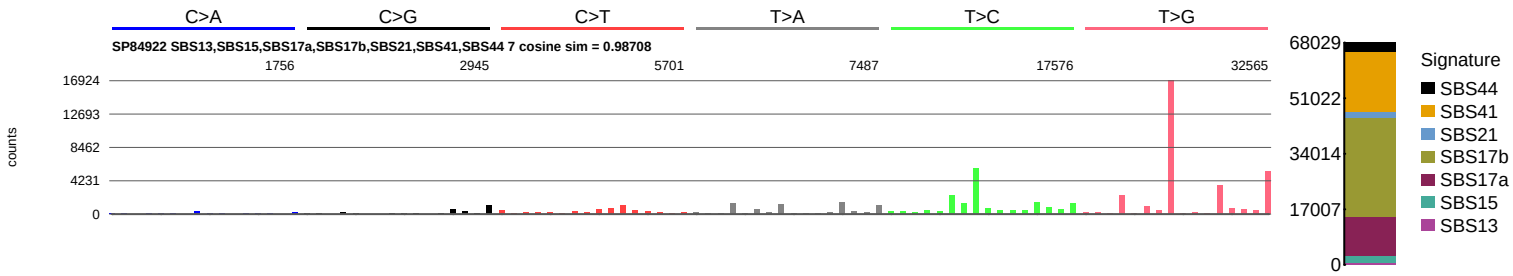

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98716
1721
2997
6637
7688
16331
32658
16928
12696
8464
4232
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS1

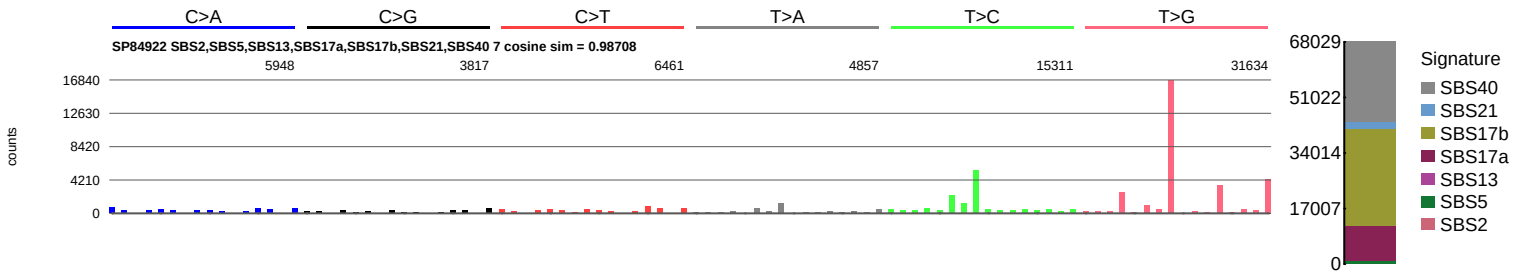

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98714
1718
2997
6610
7692
16359
32653
16928
12696
8464
4232
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS1

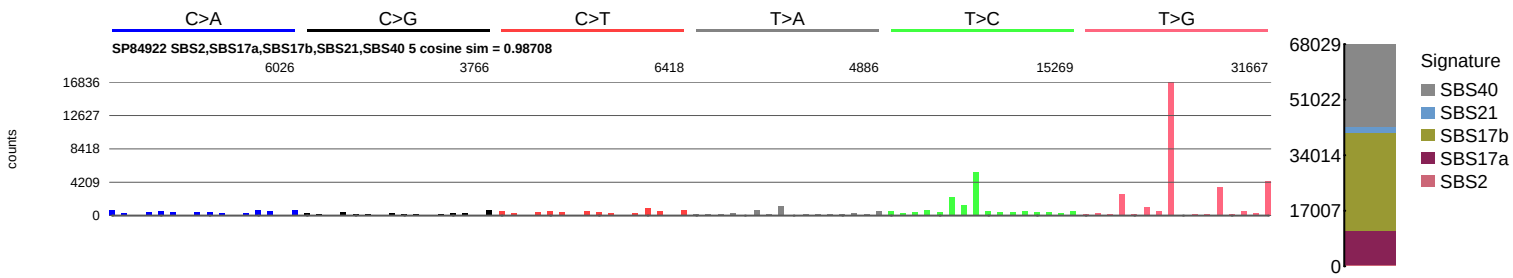

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98712
1247
2538
5794
7770
17822
32855
16936
12702
8468
4234
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS1

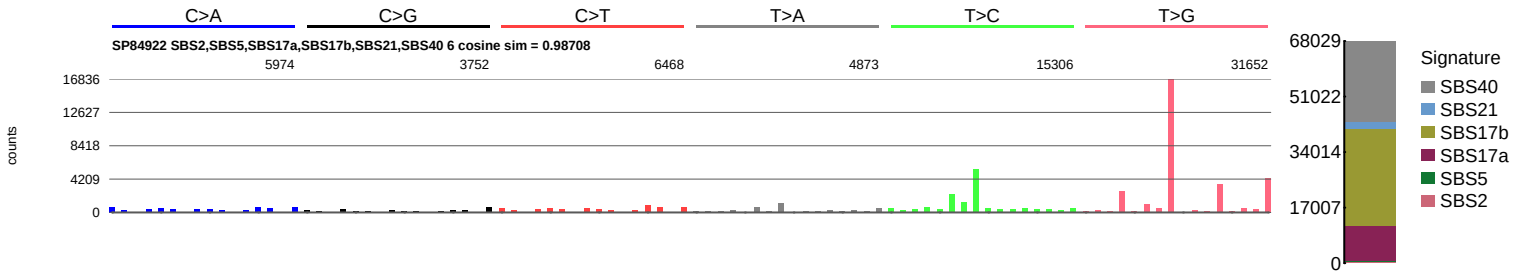

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.9871
5967
3765
6612
4843
15250
31590
16840
12630
8420
4210
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS2

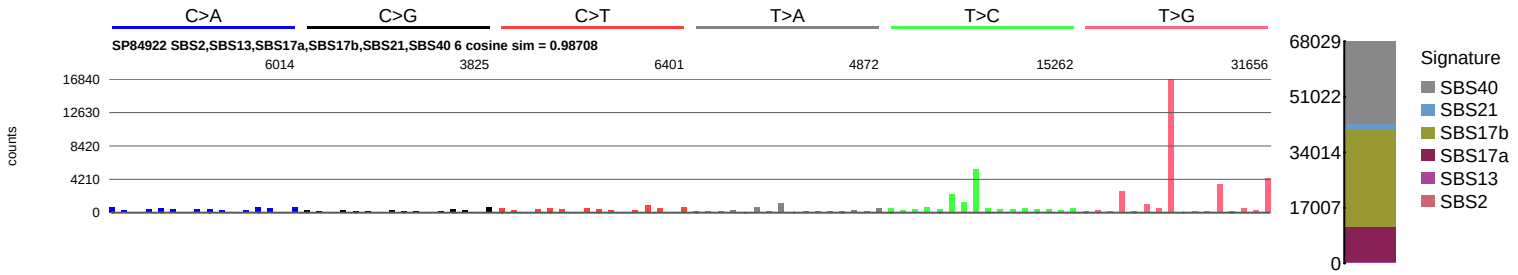

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS44 6 cosine sim = 0.98709
5989
3686
6621
4862
15260
31611
16840
12630
8420
4210
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS2

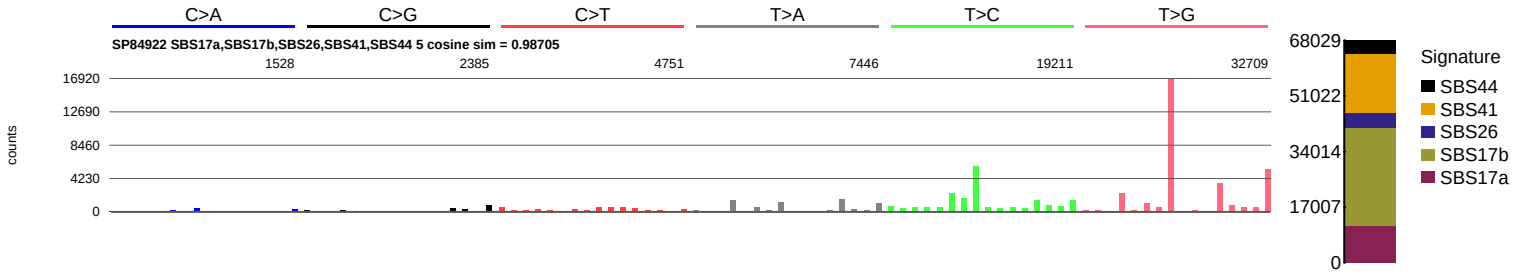

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98708
1756
2945
5701
7487
17576
32565
16924
12693
8462
4231
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13

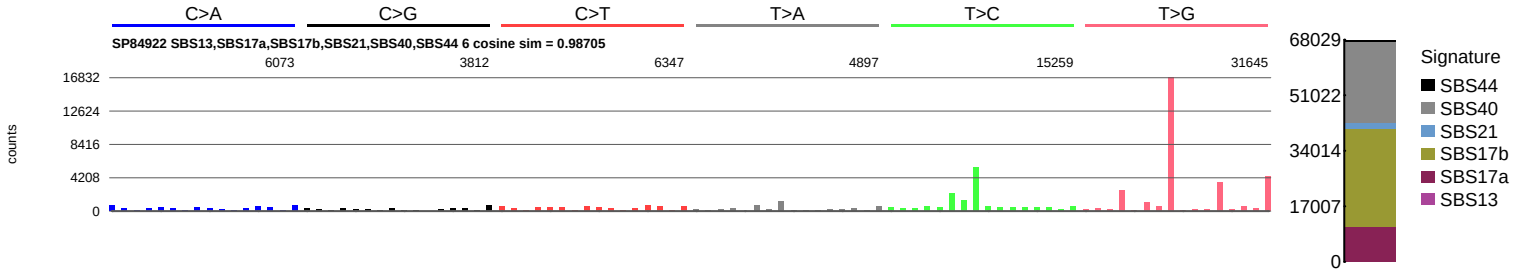

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98708
5948
3817
6461
4857
15311
31634
16840
12630
8420
4210
0
counts
68029
51022
34014
17007
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2

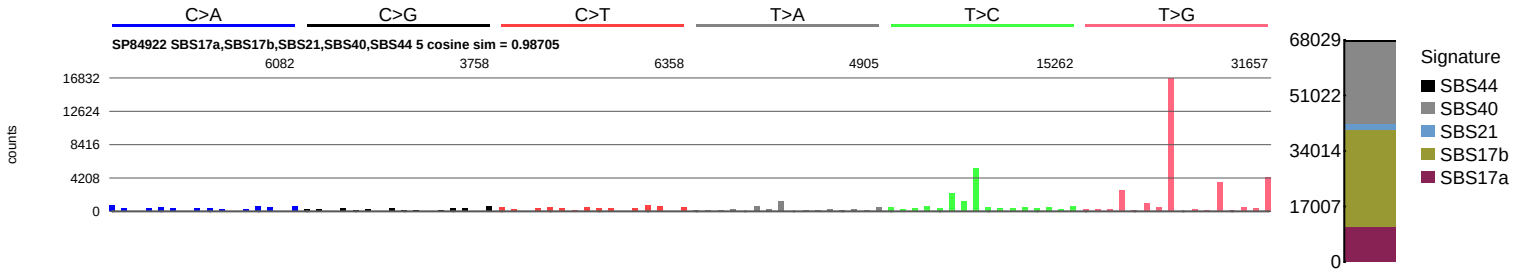

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS21,SBS40 5 cosine sim = 0.98708
6026
3766
6418
4886
15269
31667
16836
12627
8418
4209
0
counts
68029
51022
34014
17007
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS2

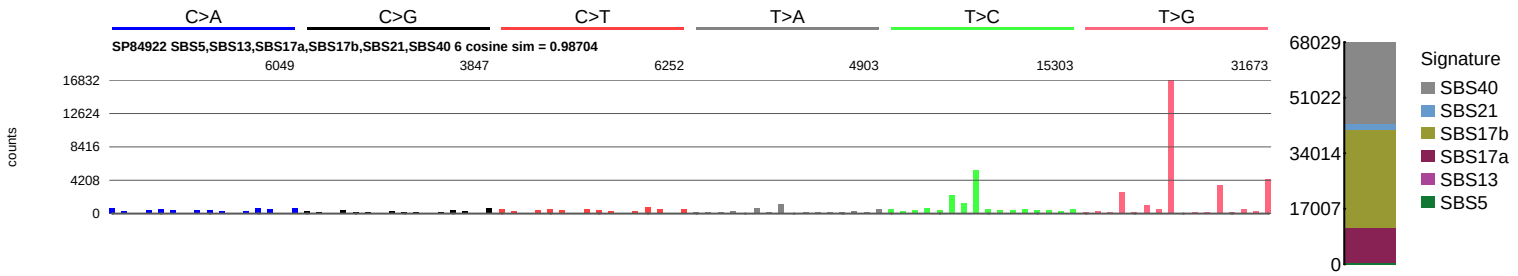

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98708
5974
3752
6468
4873
15306
31652
counts
16836
12627
8418
4209
0
68029
51022
34014
17007
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2

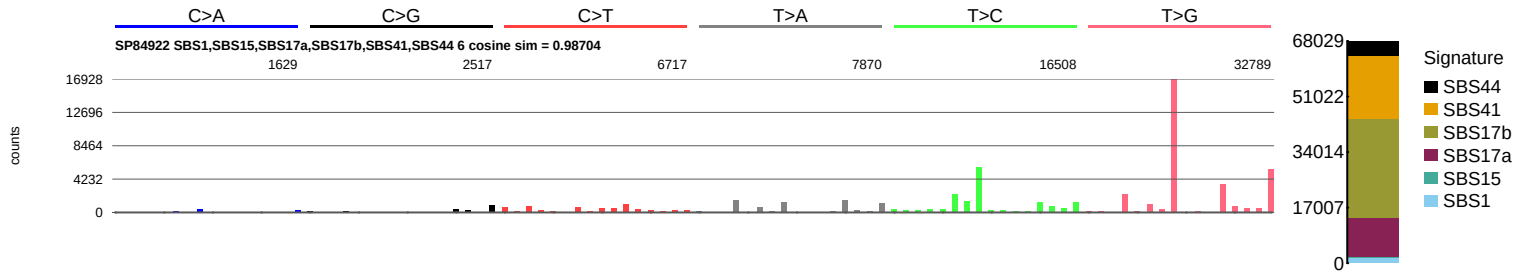

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98708
6014
3825
6401
4872
15262
31656
counts
16840
12630
8420
4210
0
68029
51022
34014
17007
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS2

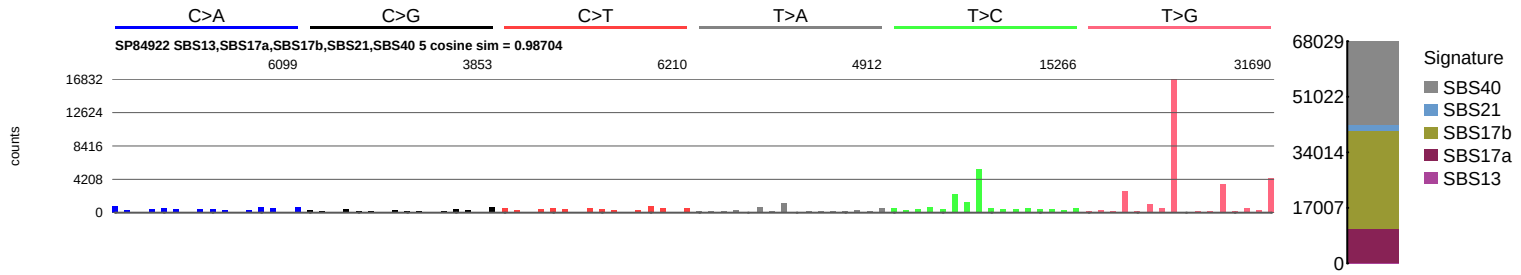

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS26,SBS41,SBS44 5 cosine sim = 0.98705
1528
2385
4751
7446
19211
32709
counts
16920
12690
8460
4230
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a

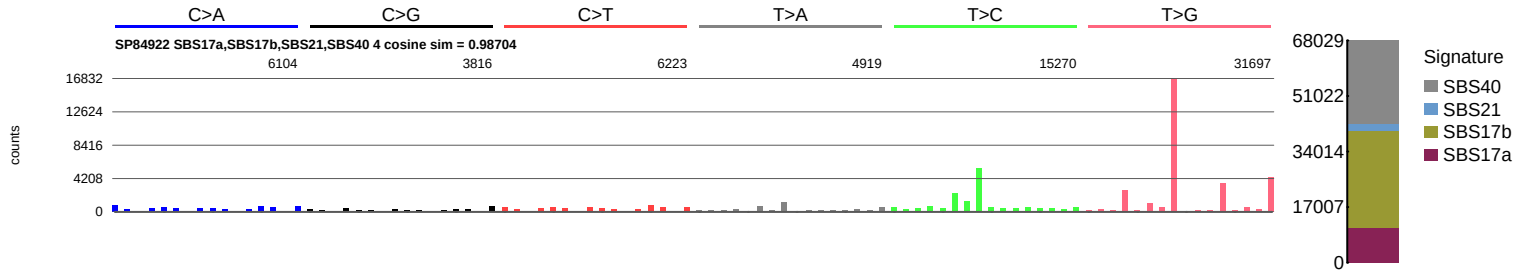

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 6 cosine sim = 0.98705
6073
3812
6347
4897
15259
31645
counts
16832
12624
8416
4208
0
68029
51022
34014
17007
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13

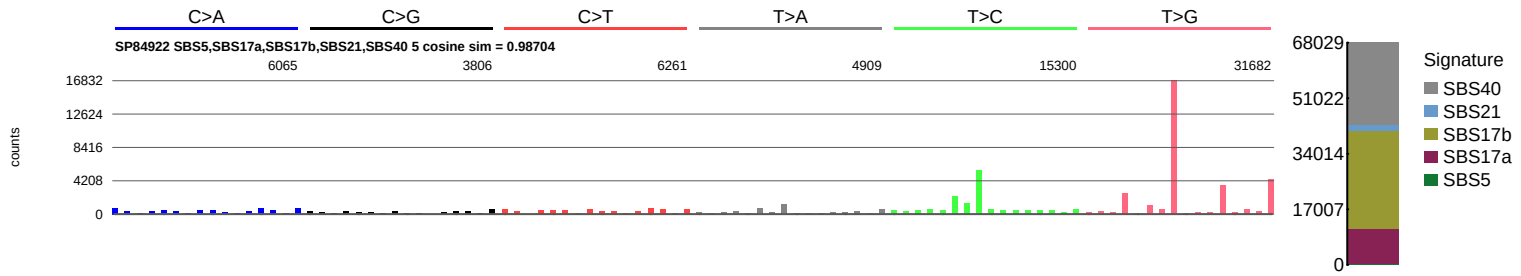

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS21,SBS40,SBS44 5 cosine sim = 0.98705
6082
3758
6358
4905
15262
31657
counts
16832
12624
8416
4208
0
68029
51022
34014
17007
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a

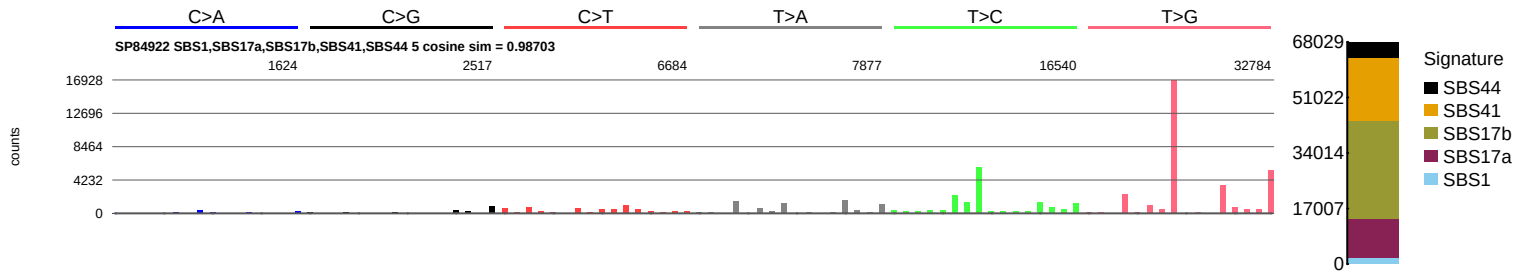

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98704
6049
3847
6252
4903
15303
31673
counts
16832
12624
8416
4208
0
68029
51022
34014
17007
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS15,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98704
1629
2517
6717
7870
16508
32789
counts
16928
12696
8464
4232
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS17a,SBS17b,SBS21,SBS40 5 cosine sim = 0.98704
6099
3853
6210
4912
15266
31690
counts
16832
12624
8416
4208
0
68029
51022
34014
17007
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS21,SBS40 4 cosine sim = 0.98704
6104
3816
6223
4919
15270
31697
counts
16832
12624
8416
4208
0
68029
51022
34014
17007
0
Signature
SBS40
SBS21
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS17a,SBS17b,SBS21,SBS40 5 cosine sim = 0.98704
6065
3806
6261
4909
15300
31682
counts
16832
12624
8416
4208
0
68029
51022
34014
17007
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS41,SBS44 5 cosine sim = 0.98703
1624
2517
6684
7877
16540
32784
counts
16928
12696
8464
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SBS44
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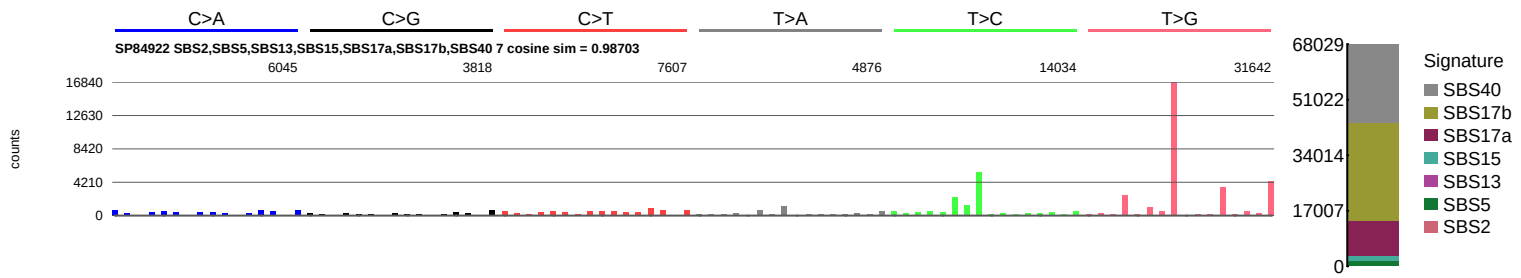
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98703
6045
3818
7607
4876
14034
31642
counts
16840
12630
8420
4210
0
68029
51022
34014
17007
0
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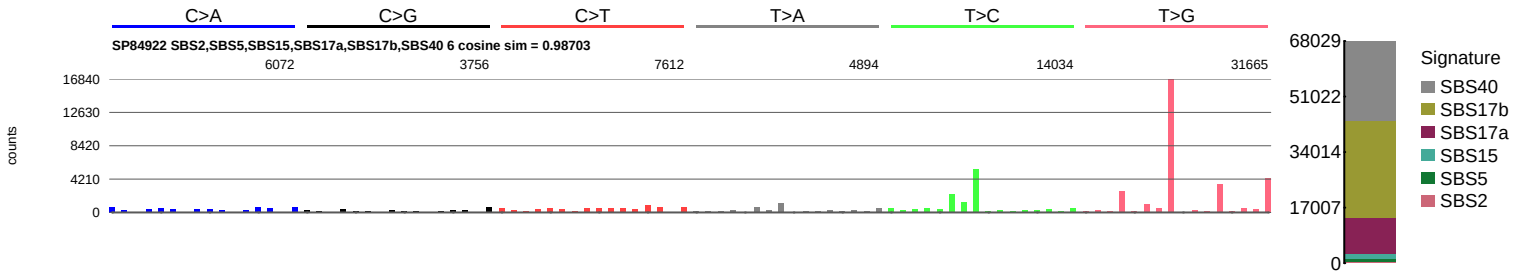
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98703
6072
3756
7612
4894
14034
31665
counts
16840
12630
8420
4210
0
68029
51022
34014
17007
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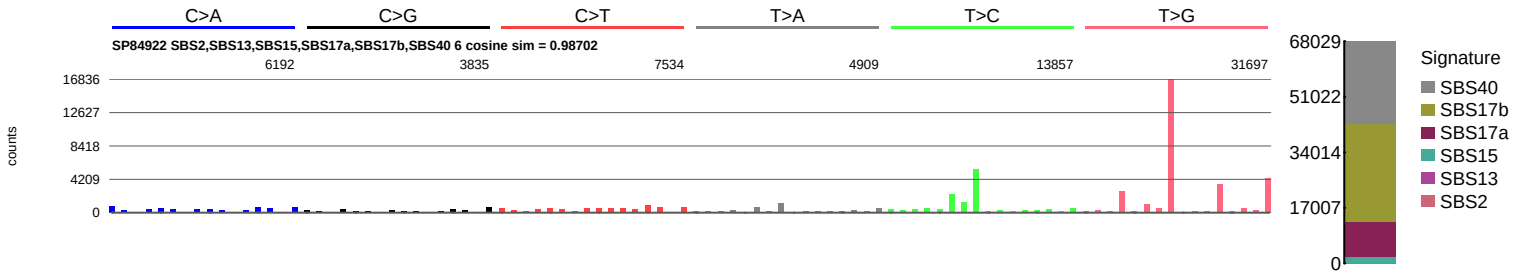
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2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C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98702
6192
3835
7534
4909
13857
31697
counts
16836
12627
8418
4209
0
68029
51022
34014
17007
0
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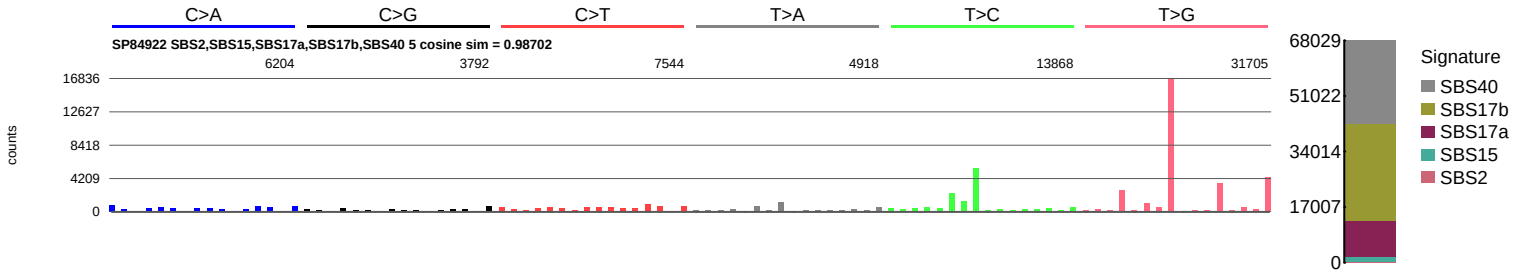
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS15,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98702
6204
3792
7544
4918
13868
31705
counts
16836
12627
8418
4209
0
68029
51022
34014
17007
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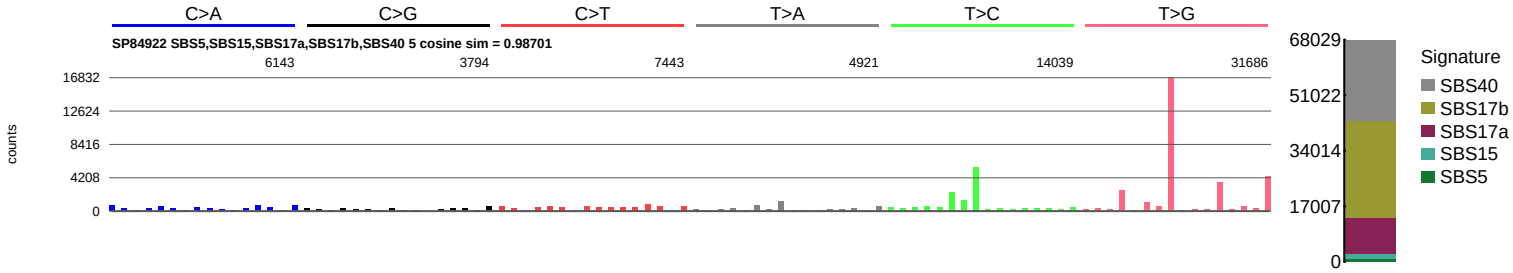
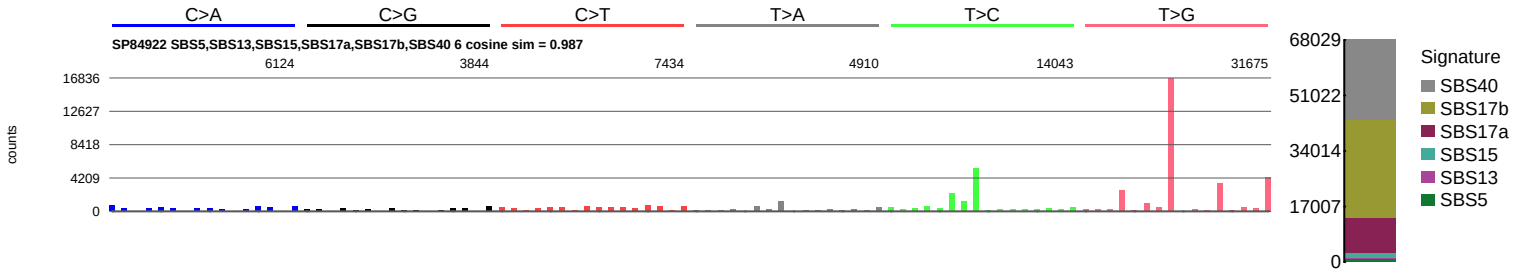
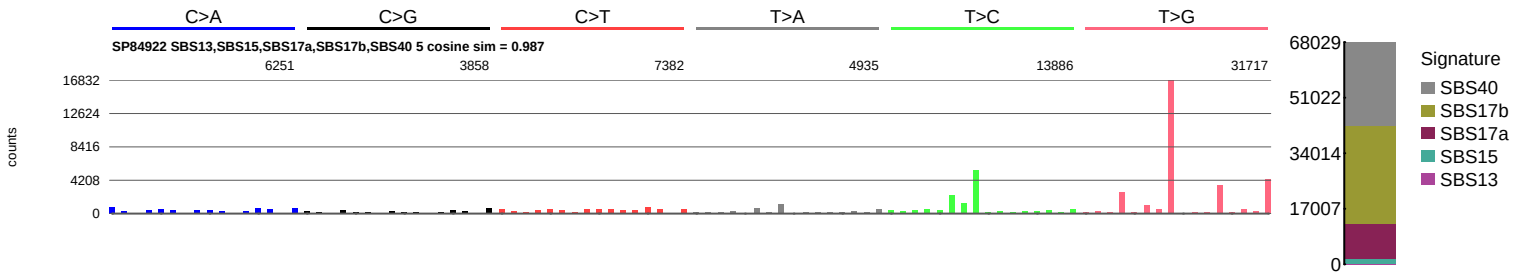
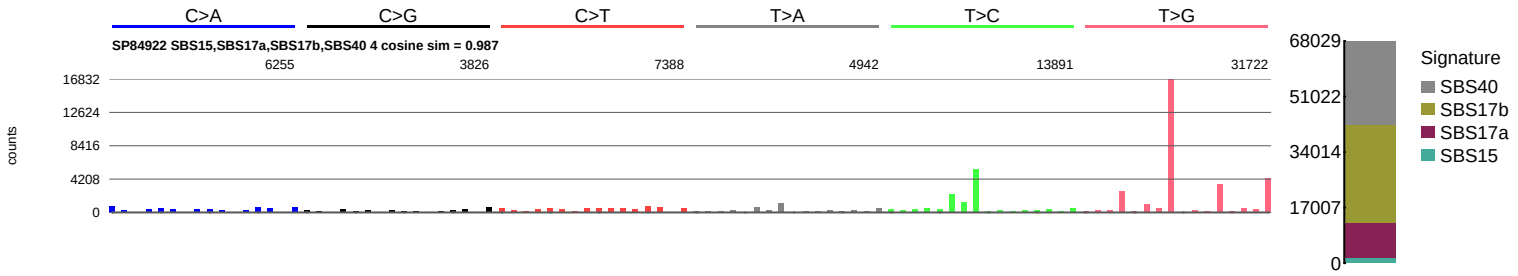
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS15,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98701
6143
3794
7443
4921
14039
31686
counts
16832
12624
8416
4208
0
68029
51022
34014
17007
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.987
6124
3844
7434
4910
14043
31675
counts
16836
12627
8418
4209
0
68029
51022
34014
17007
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS15,SBS17a,SBS17b,SBS40 5 cosine sim = 0.987
6251
3858
7382
4935
13886
31717
counts
16832
12624
8416
4208
0
68029
51022
34014
17007
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS15,SBS17a,SBS17b,SBS40 4 cosine sim = 0.987
6255
3826
7388
4942
13891
31722
16832
12624
8416
4208
0
counts
68029
51022
34014
17007
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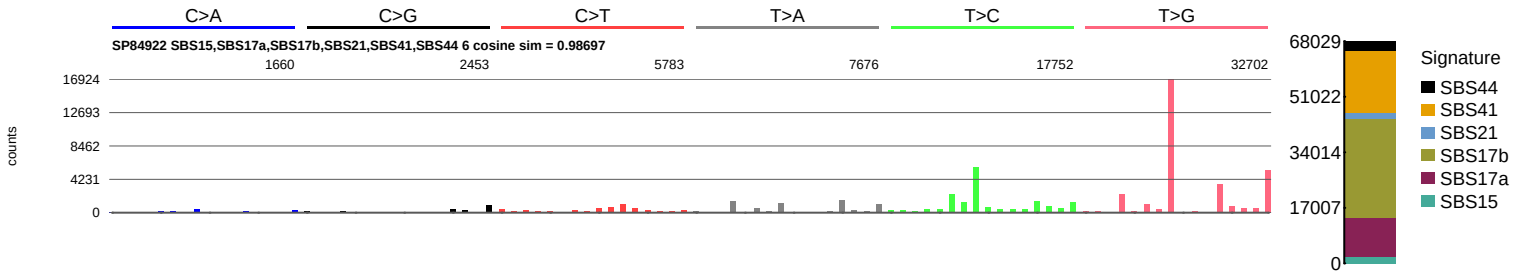
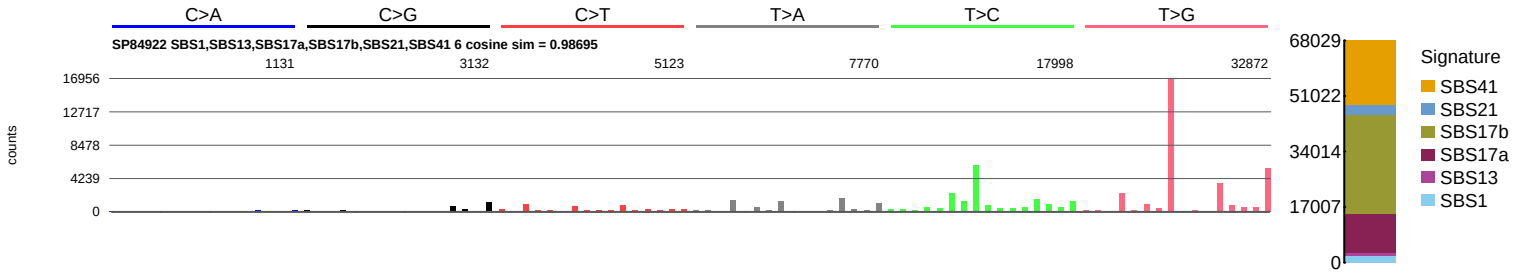
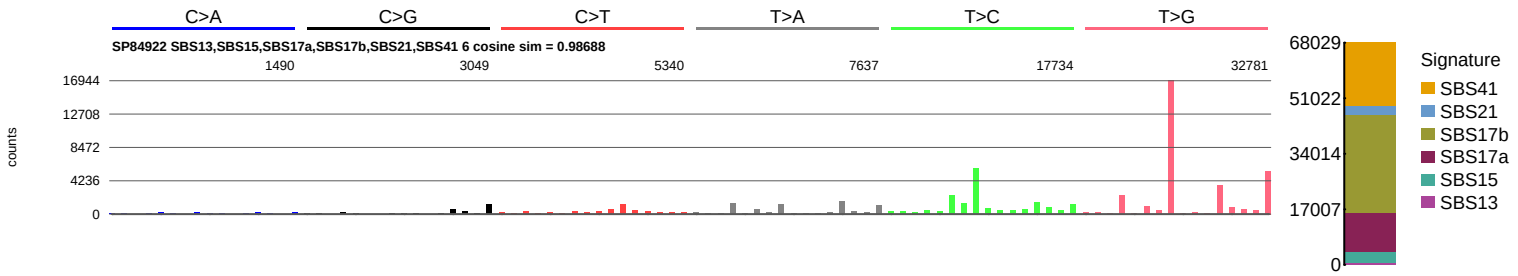
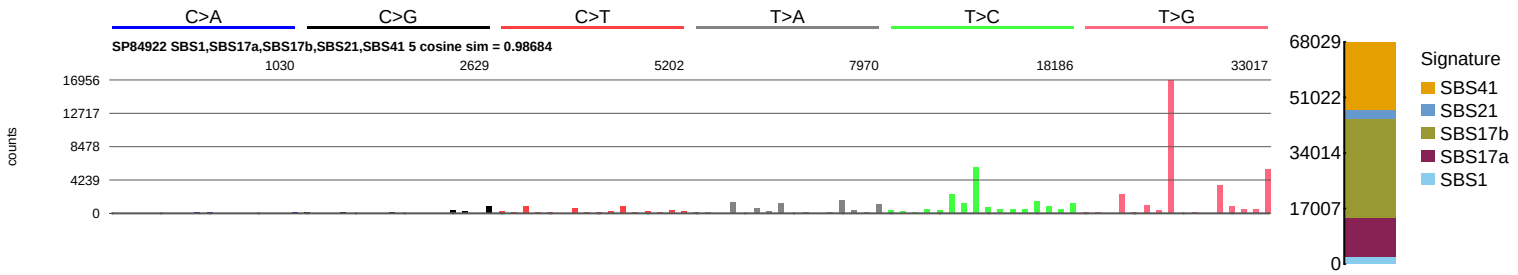
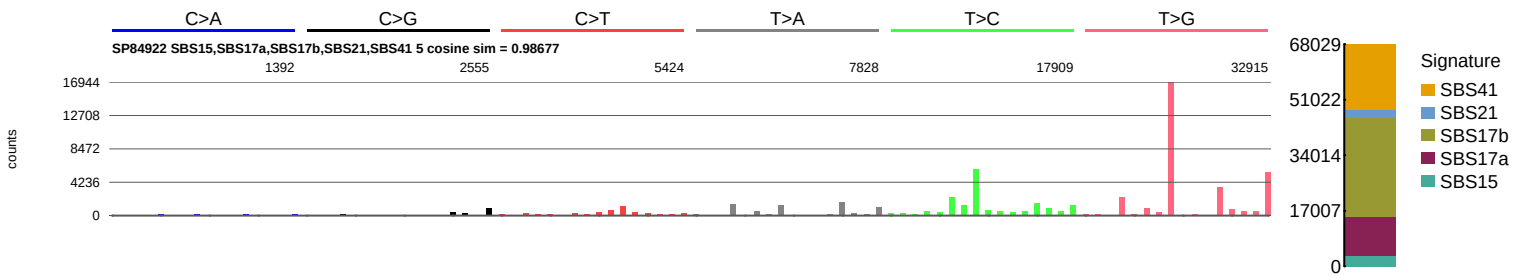
Signature
SBS40
SBS17b
SBS17a
SBS15

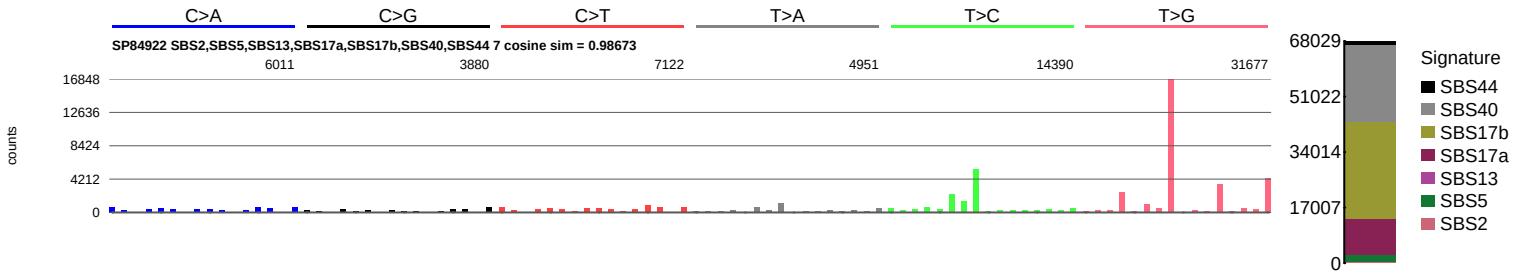

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 6 cosine sim = 0.98697
1660
2453
5783
7676
17752
32702
16924
12693
8462
4231
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15

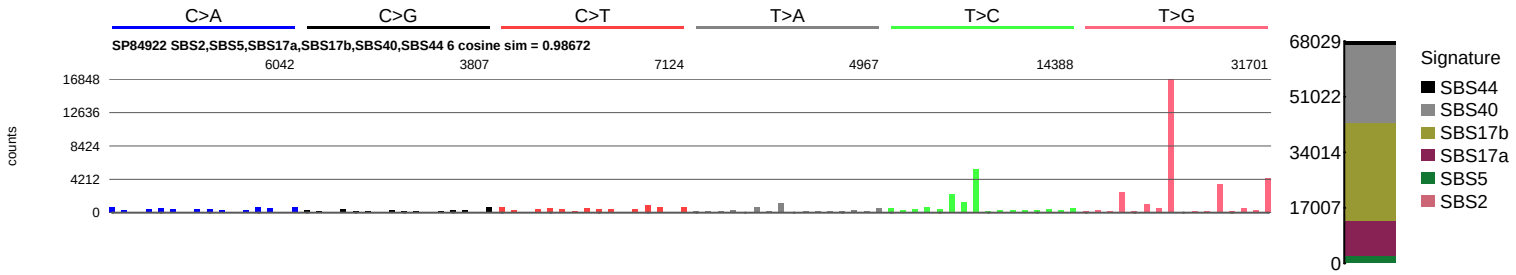

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98695
1131
3132
5123
7770
17998
32872
16956
12717
8478
4239
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS1

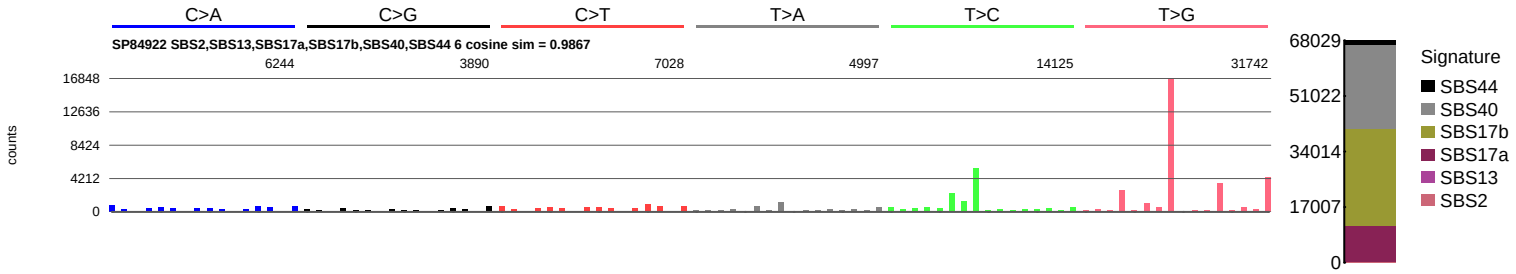

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98688
1490
3049
5340
7637
17734
32781
16944
12708
8472
4236
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13

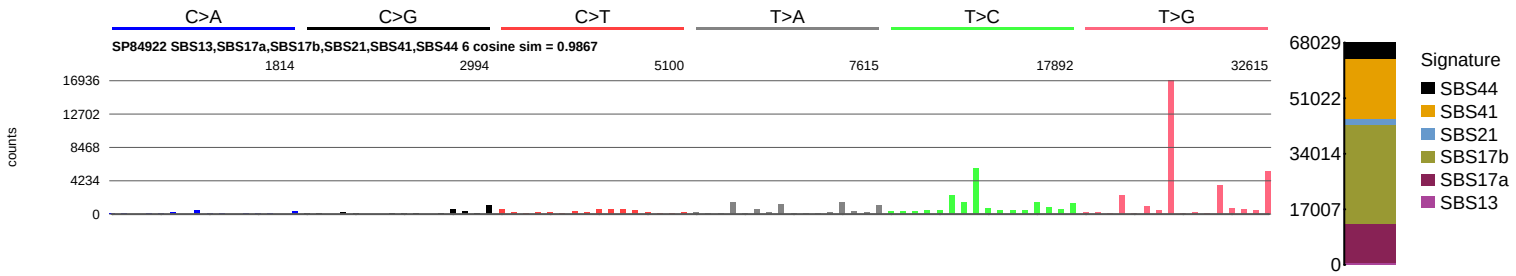

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS21,SBS41 5 cosine sim = 0.98684
1030
2629
5202
7970
18186
33017
16956
12717
8478
4239
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS1

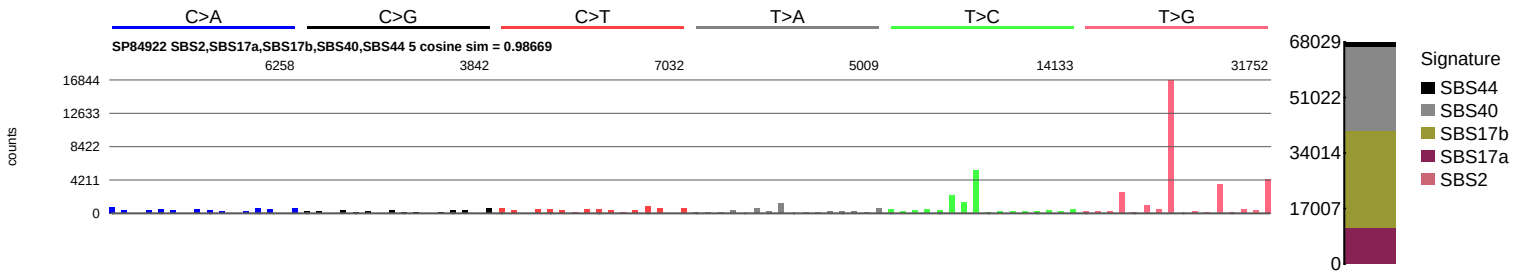

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS15,SBS17a,SBS17b,SBS21,SBS41 5 cosine sim = 0.98677
1392
2555
5424
7828
17909
32915
16944
12708
8472
4236
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15

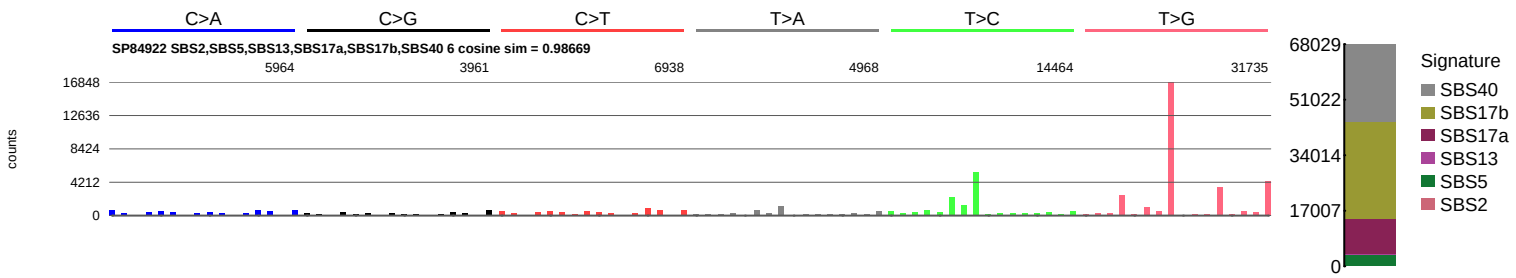

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98673
6011
3880
7122
4951
14390
31677
counts
16848
12636
8424
4212
0
68029
51022
34014
17007
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS2

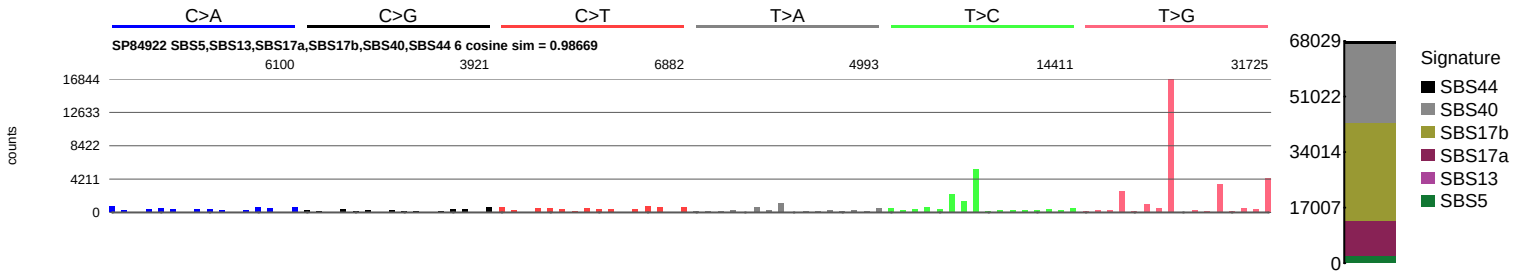

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98672
6042
3807
7124
4967
14388
31701
counts
16848
12636
8424
4212
0
68029
51022
34014
17007
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.9867
6244
3890
7028
4997
14125
31742
counts
16848
12636
8424
4212
0
68029
51022
34014
17007
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 6 cosine sim = 0.9867
1814
2994
5100
7615
17892
32615
counts
16936
12702
8468
4234
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS40,SBS44 5 cosine sim = 0.98669
6258
3842
7032
5009
14133
31752
counts
16844
12633
8422
4211
0
68029
51022
34014
17007
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98669
5964
3961
6938
4968
14464
31735
counts
16848
12636
8424
4212
0
68029
51022
34014
17007
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98669
6100
3921
6882
4993
14411
31725
counts
16844
12633
8422
4211
0
68029
51022
34014
17007
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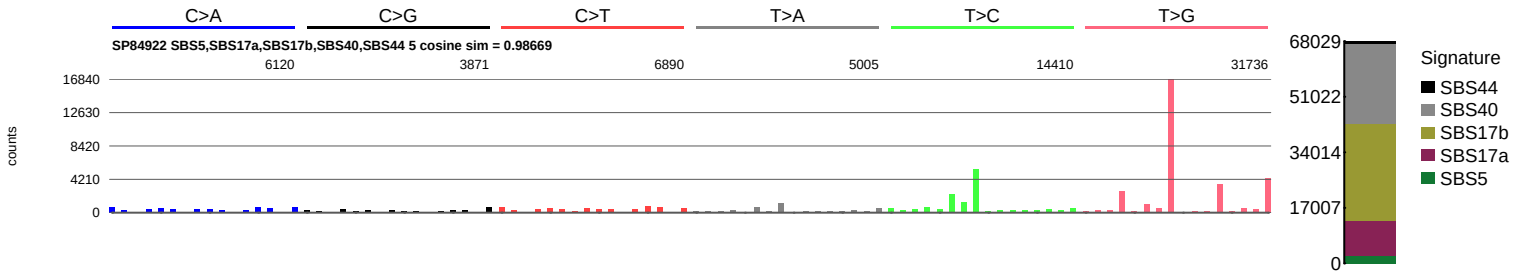
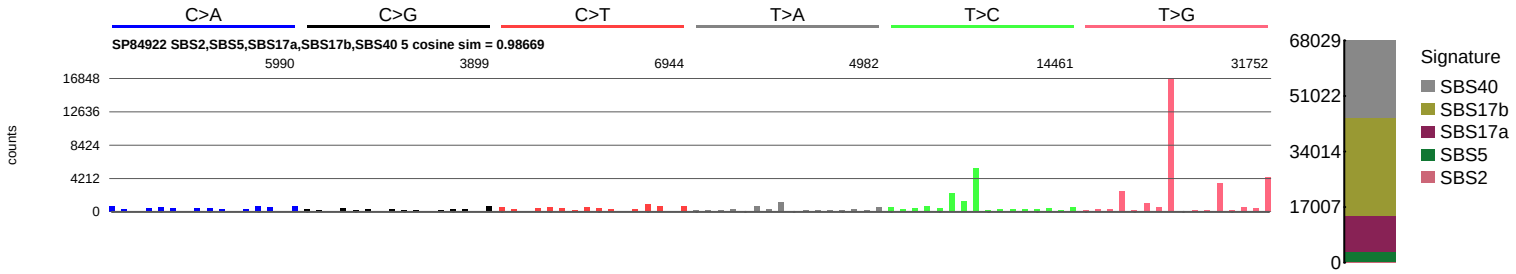
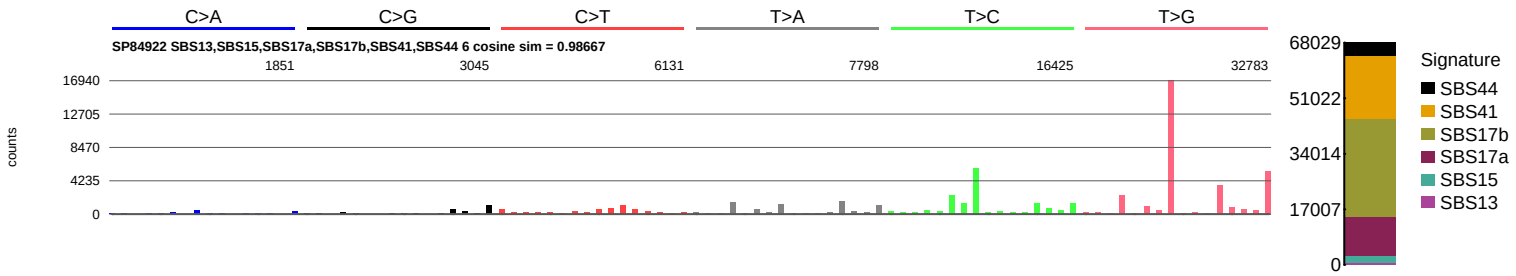
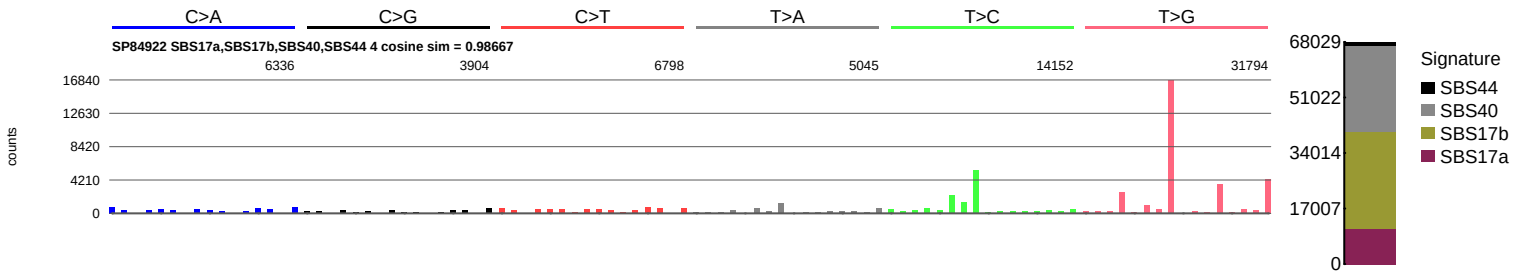
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS17a,SBS17b,SBS40,SBS44 5 cosine sim = 0.98669
6120
3871
6890
5005
14410
31736
counts
16840
12630
8420
4210
0
68029
51022
34014
17007
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98669
5990
3899
6944
4982
14461
31752
counts
16848
12636
8424
4212
0
68029
51022
34014
17007
0
Signature
SBS40
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS15,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98667
1851
3045
6131
7798
16425
32783
counts
16940
12705
8470
4235
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS40,SBS44 4 cosine sim = 0.98667
6336
3904
6798
5045
14152
31794
counts
16840
12630
8420
4210
0
68029
51022
34014
17007
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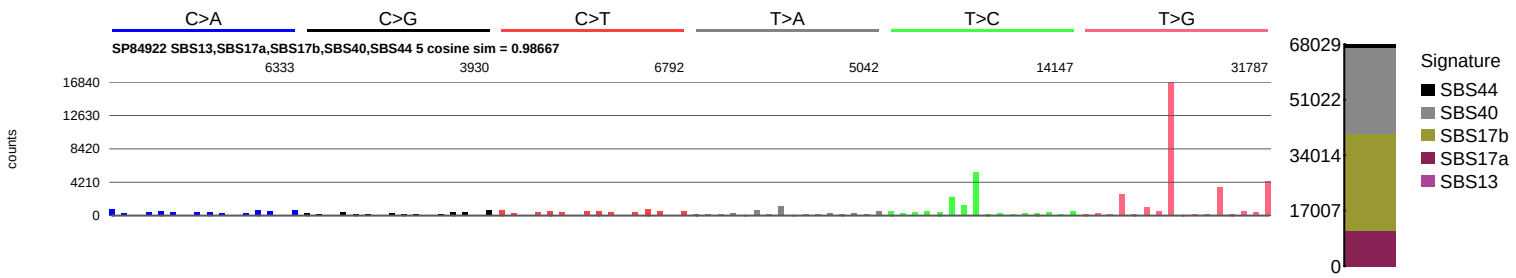
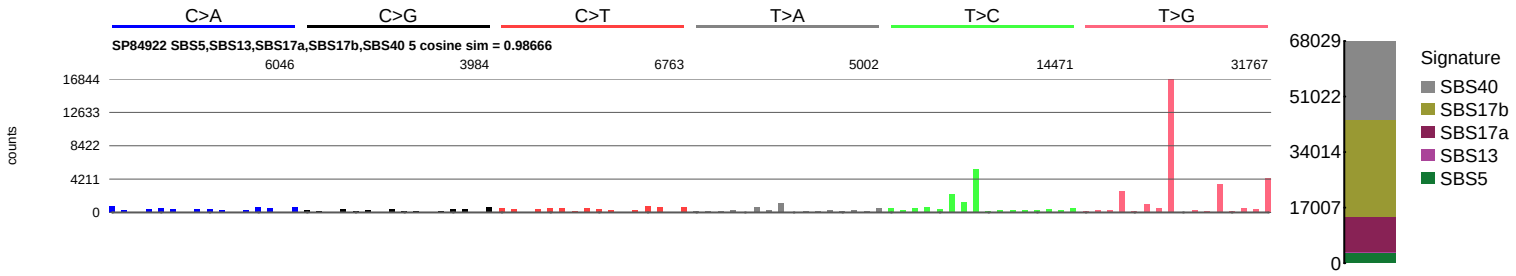
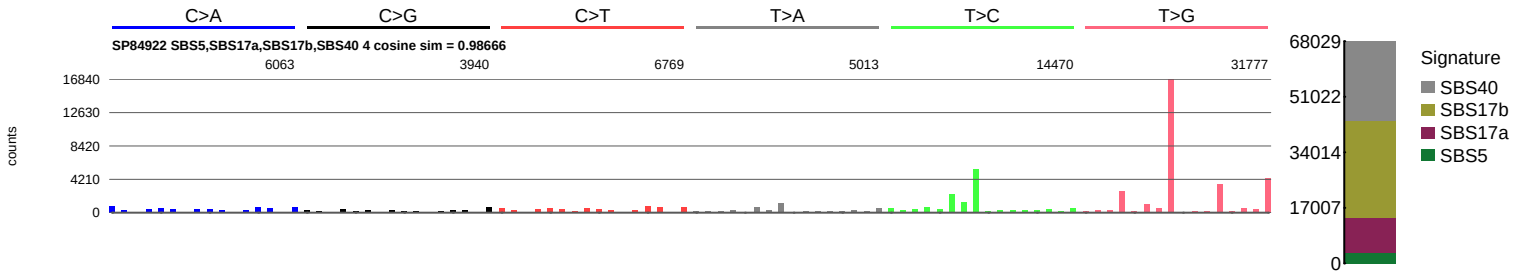
Signature
SBS44
SBS40
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS17a,SBS17b,SBS40,SBS44 5 cosine sim = 0.98667
6333
3930
6792
5042
14147
31787
counts
16840
12630
8420
4210
0
68029
51022
34014
17007
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98666
6046
3984
6763
5002
14471
31767
16844
12633
8422
4211
0
counts
68029
51022
34014
17007
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS5


C>A
C>G
C>T
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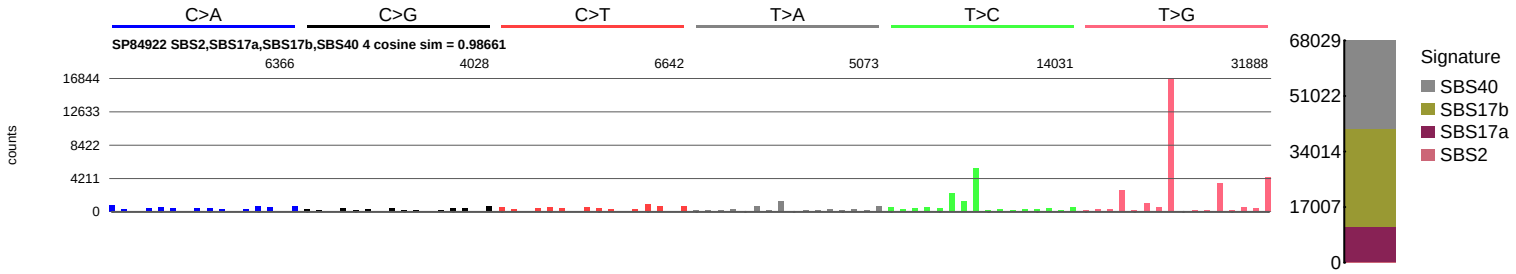
T>C
T>G
SP84922 SBS5,SBS17a,SBS17b,SBS40 4 cosine sim = 0.98666
6063
3940
6769
5013
14470
31777
16840
12630
8420
4210
0
counts
68029
51022
34014
17007
0
Signature
SBS40
SBS17b
SBS17a
SBS5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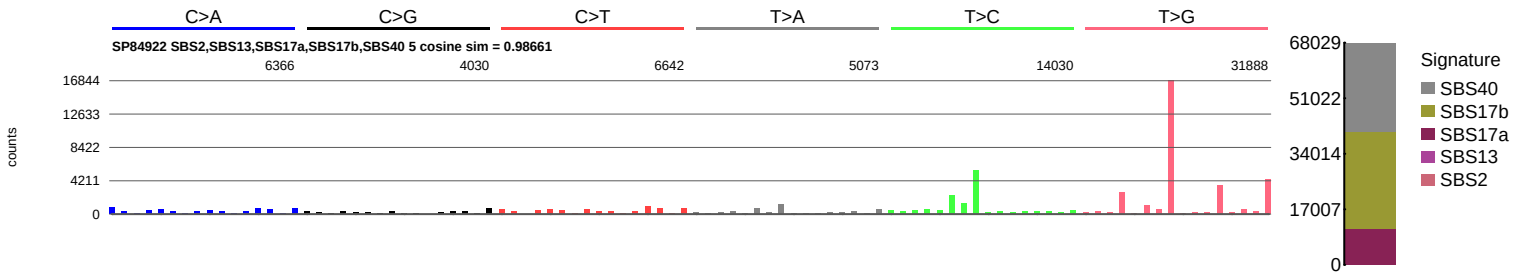
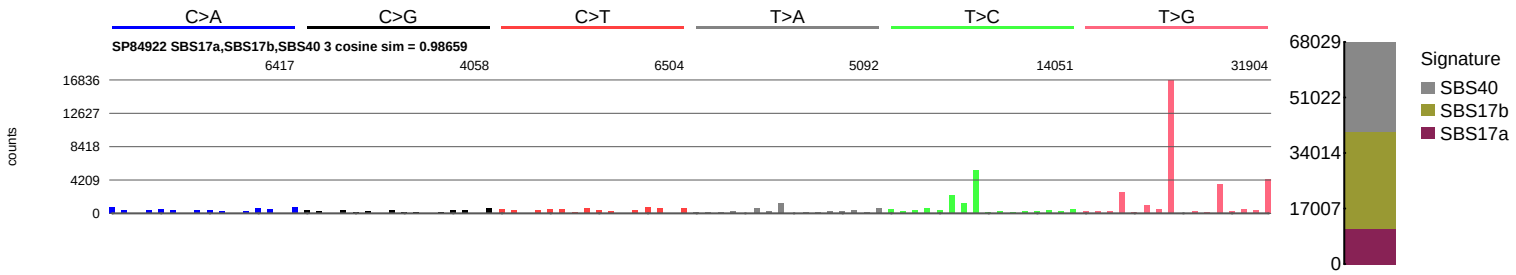
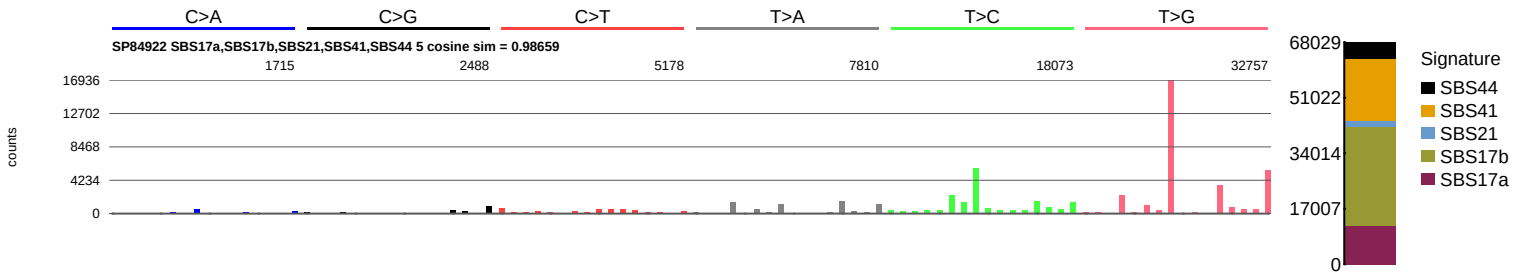
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS40 4 cosine sim = 0.98661
6366
4028
6642
5073
14031
31888
16844
12633
8422
4211
0
counts
68029
51022
34014
17007
0
Signature
SBS40
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98661
6366
4030
6642
5073
14030
31888
16844
12633
8422
4211
0
counts
68029
51022
34014
17007
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS40 3 cosine sim = 0.98659
6417
4058
6504
5092
14051
31904
16836
12627
8418
4209
0
counts
68029
51022
34014
17007
0
Signature
SBS40
SBS17b
SBS17a


C>A
C>G
C>T
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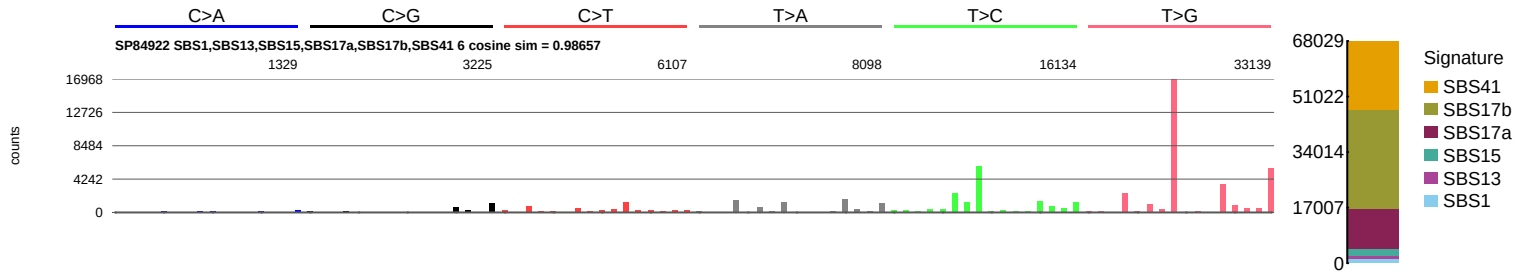
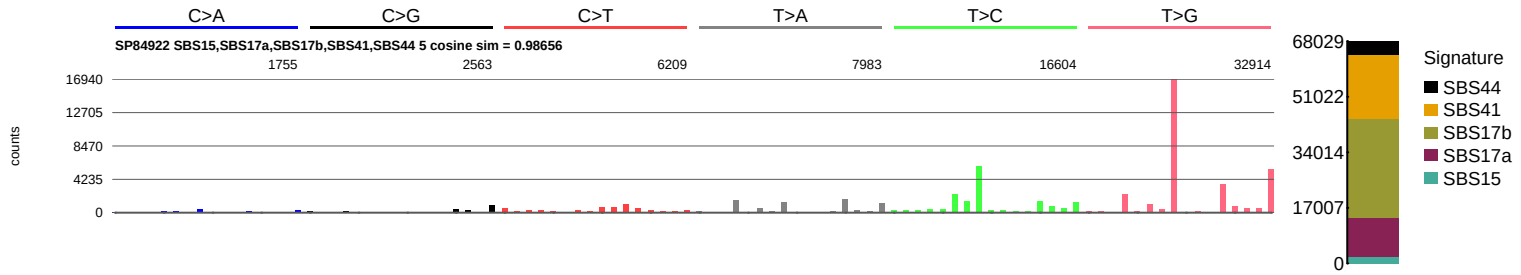
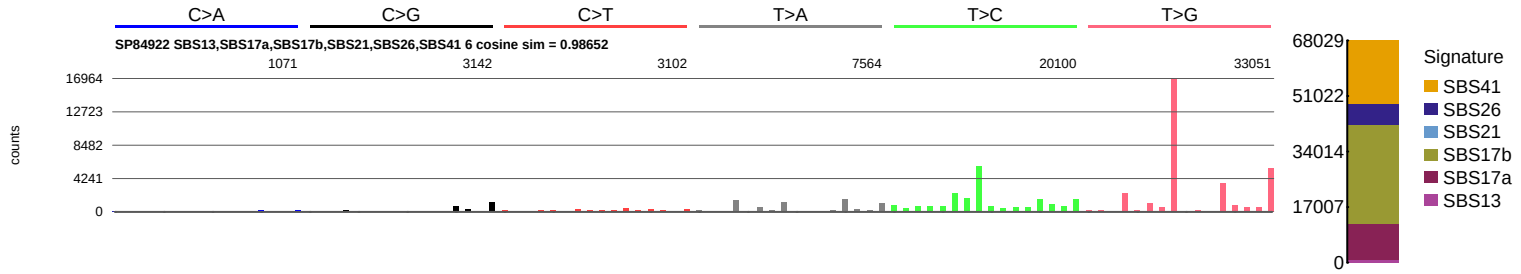
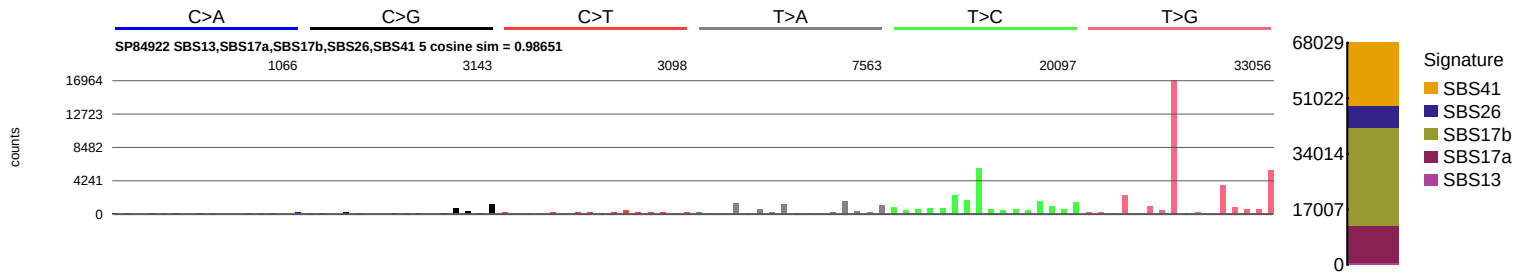
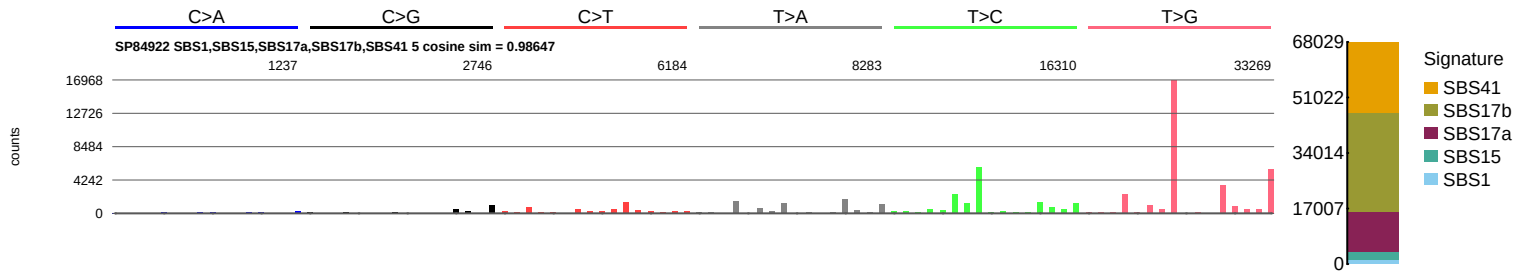
T>C
T>G
SP84922 SBS17a,SBS17b,SBS21,SBS41,SBS44 5 cosine sim = 0.98659
1715
2488
5178
7810
18073
32757
16936
12702
8468
4234
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a

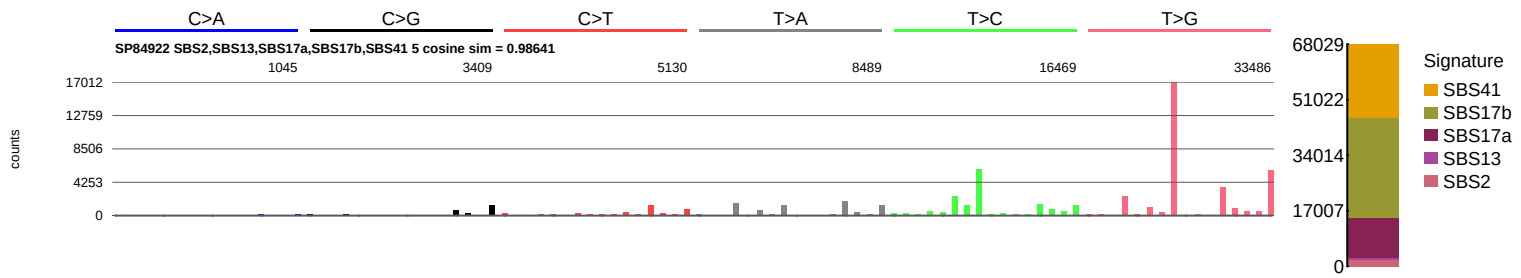

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98657
1329
3225
6107
8098
16134
33139
16968
12726
8484
4242
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS1

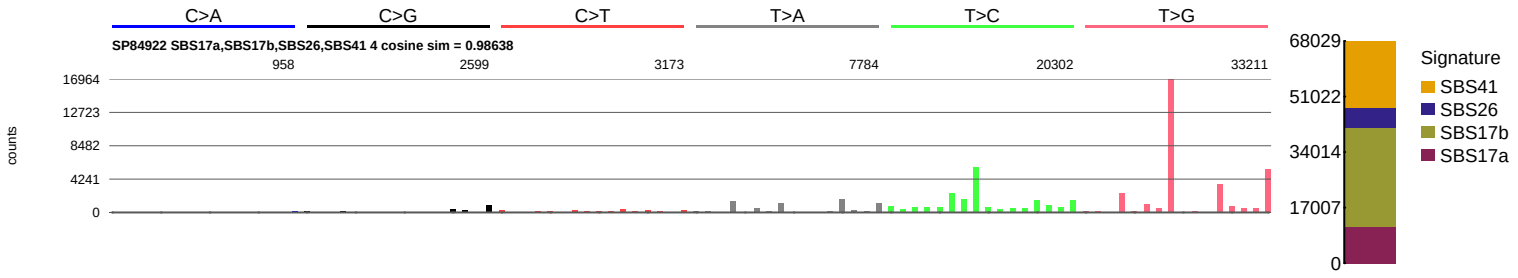

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS15,SBS17a,SBS17b,SBS41,SBS44 5 cosine sim = 0.98656
1755
2563
6209
7983
16604
32914
16940
12705
8470
4235
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15

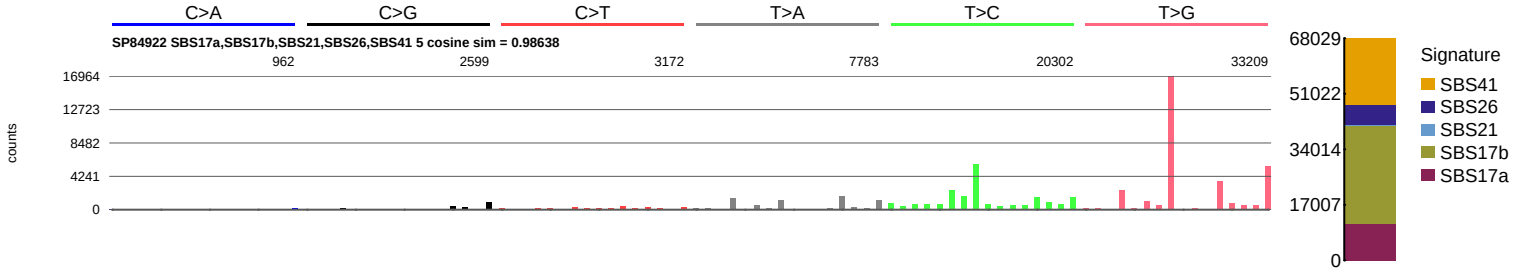

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41 6 cosine sim = 0.98652
1071
3142
3102
7564
20100
33051
16964
12723
8482
4241
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13

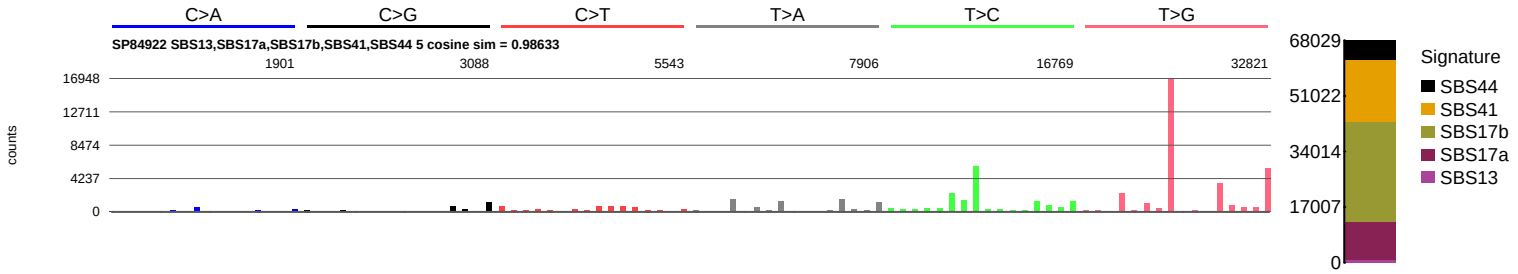

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS17a,SBS17b,SBS26,SBS41 5 cosine sim = 0.98651
1066
3143
3098
7563
20097
33056
16964
12723
8482
4241
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS13

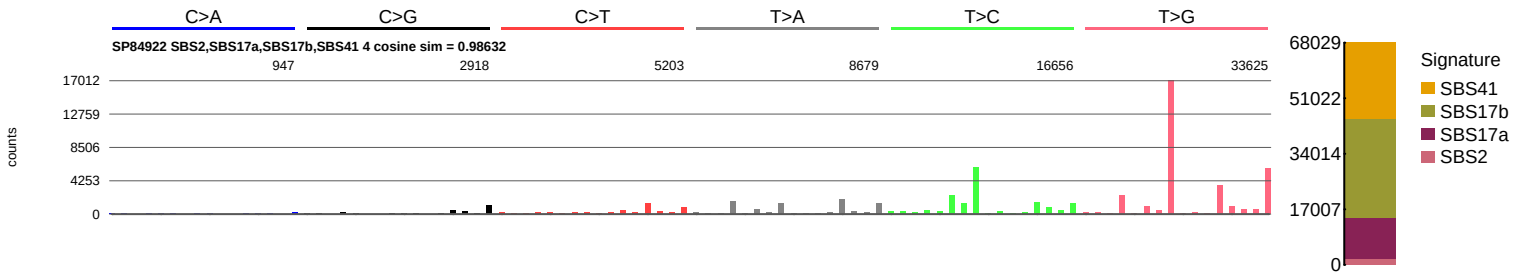

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS15,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98647
1237
2746
6184
8283
16310
33269
16968
12726
8484
4242
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS1

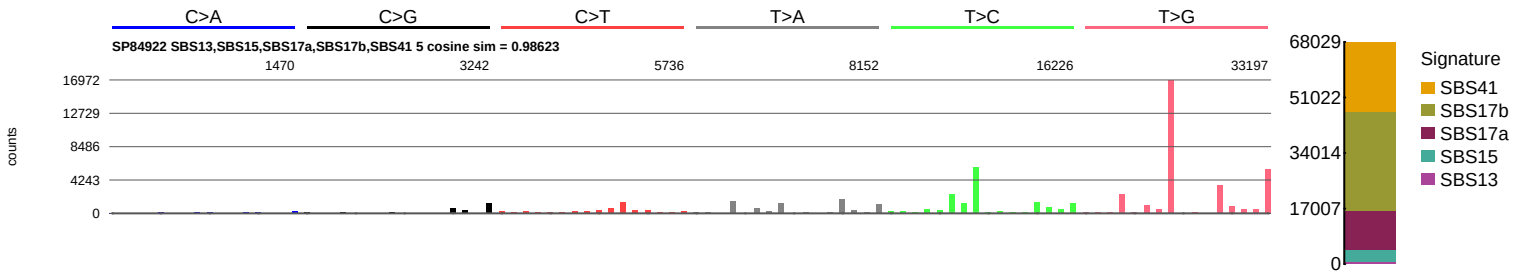

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98641
1045
3409
5130
8489
16469
33486
17012
12759
8506
4253
0
counts
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS2

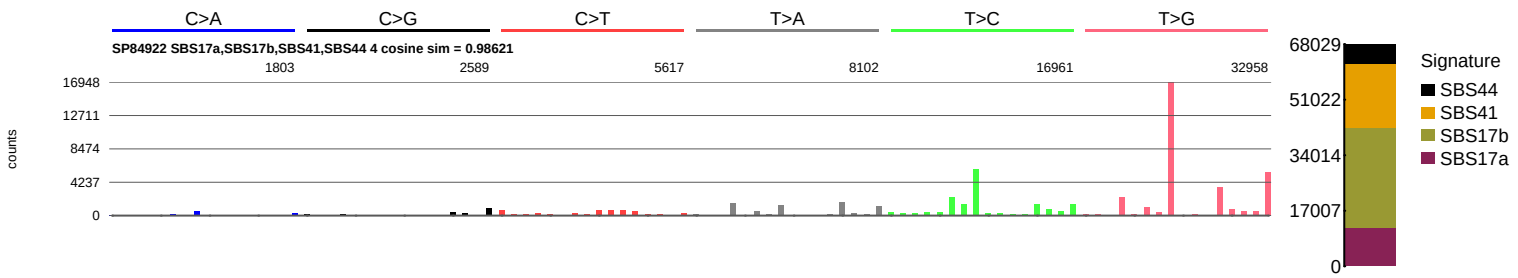

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS26,SBS41 4 cosine sim = 0.98638
958
2599
3173
7784
20302
33211
counts
16964
12723
8482
4241
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS17b
SBS17a

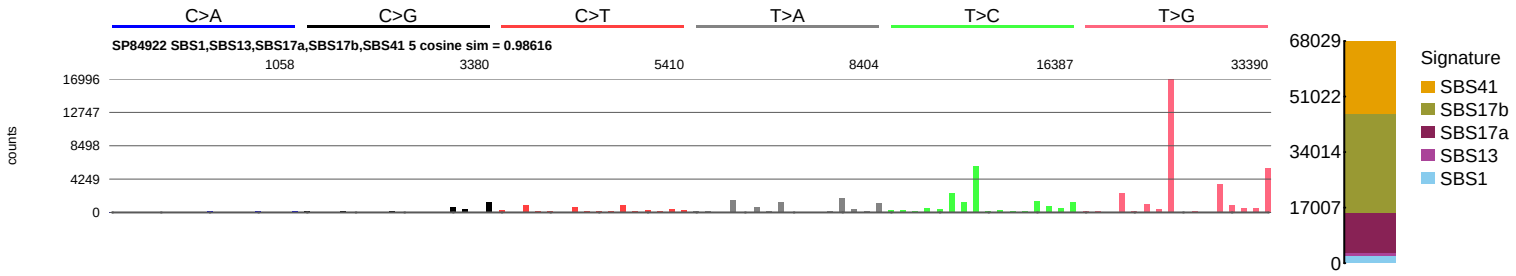

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS21,SBS26,SBS41 5 cosine sim = 0.98638
962
2599
3172
7783
20302
33209
counts
16964
12723
8482
4241
0
68029
51022
34014
17007
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a

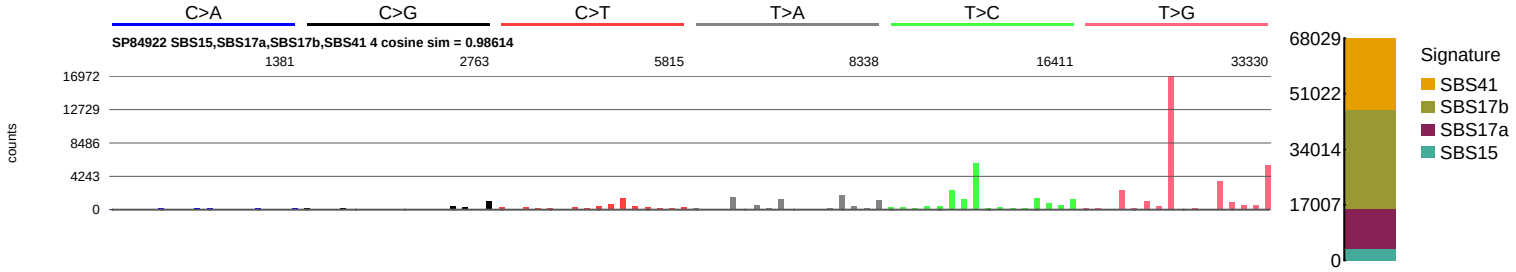

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS17a,SBS17b,SBS41,SBS44 5 cosine sim = 0.98633
1901
3088
5543
7906
16769
32821
counts
16948
12711
8474
4237
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13

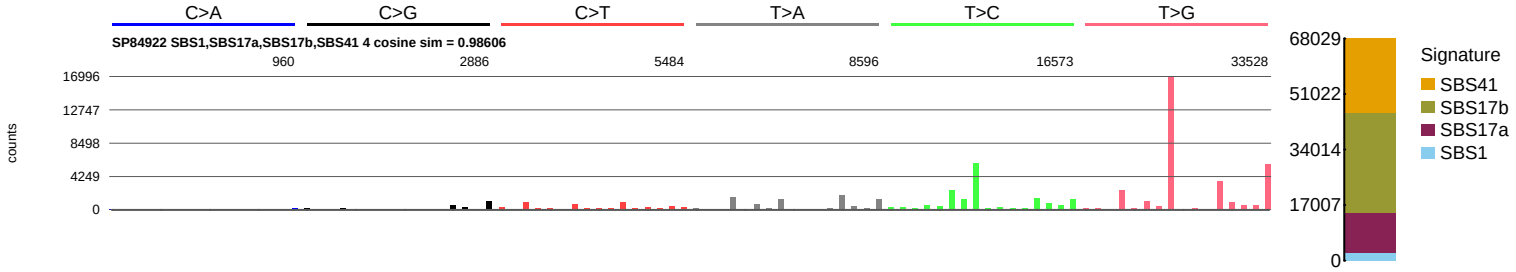

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS17a,SBS17b,SBS41 4 cosine sim = 0.98632
947
2918
5203
8679
16656
33625
counts
17012
12759
8506
4253
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS2

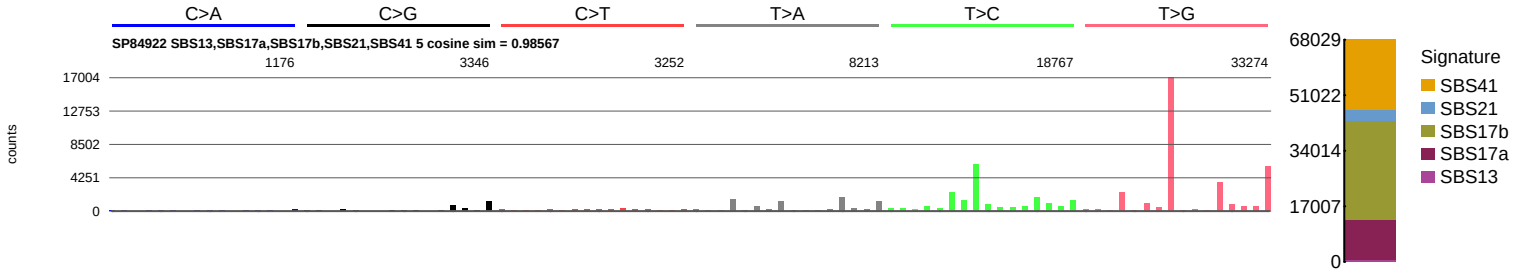

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS15,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98623
1470
3242
5736
8152
16226
33197
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13

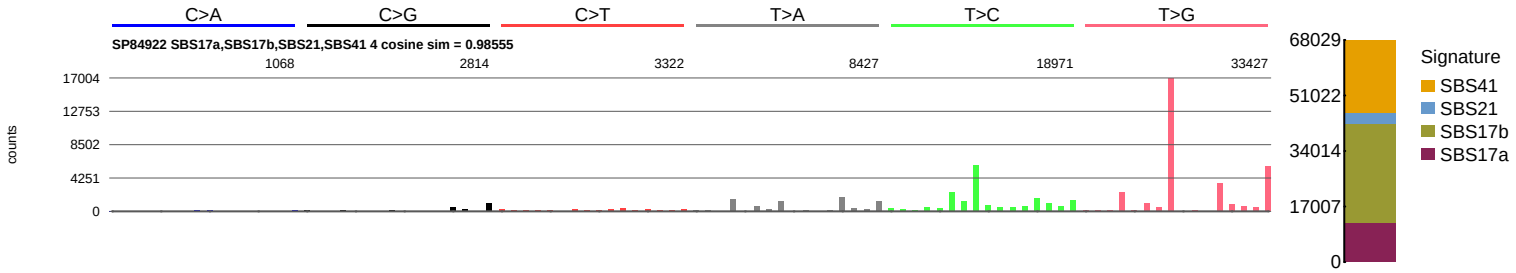

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS41,SBS44 4 cosine sim = 0.98621
1803
2589
5617
8102
16961
32958
counts
16948
12711
8474
4237
0
68029
51022
34014
17007
0
Signature
SBS44
SBS41
SBS17b
SBS17a

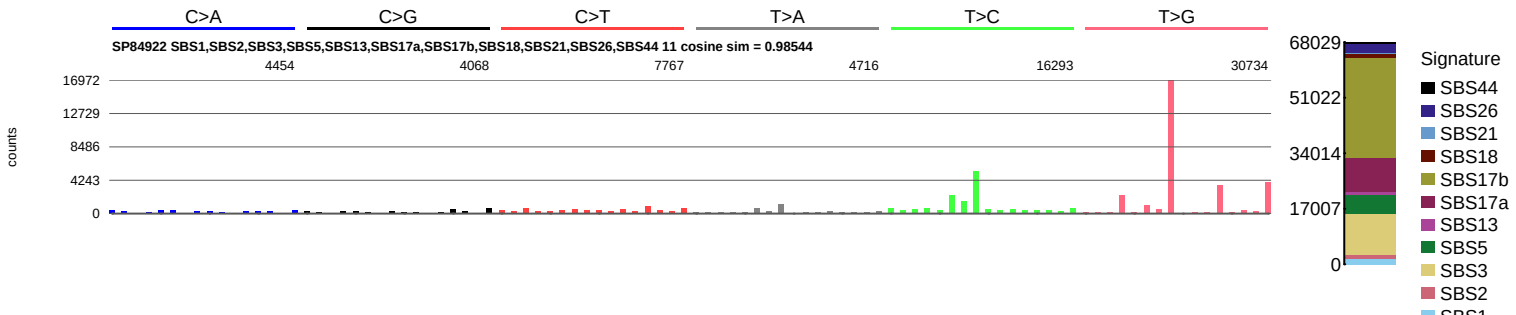

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS13,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98616
1058
3380
5410
8404
16387
33390
counts
16996
12747
8498
4249
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS1

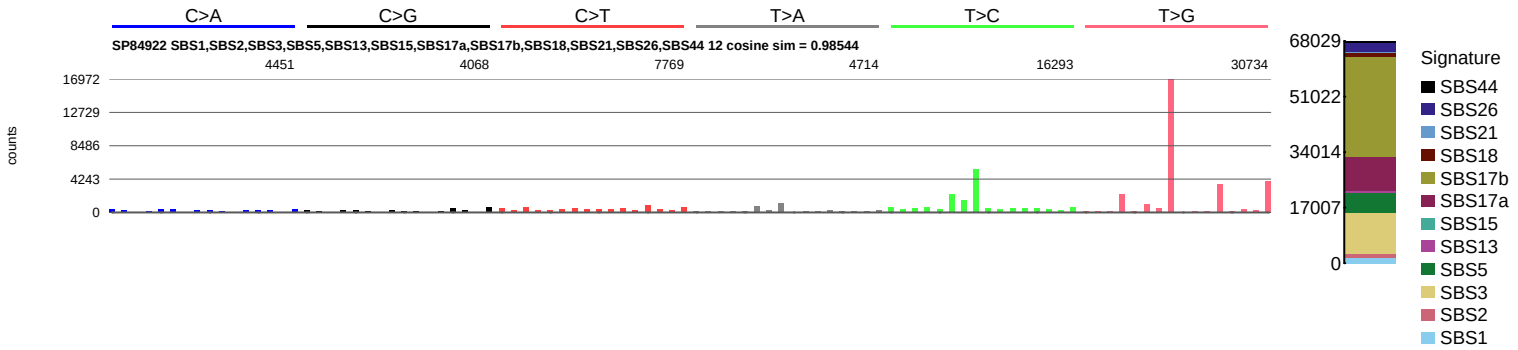

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS15,SBS17a,SBS17b,SBS41 4 cosine sim = 0.98614
1381
2763
5815
8338
16411
33330
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS15

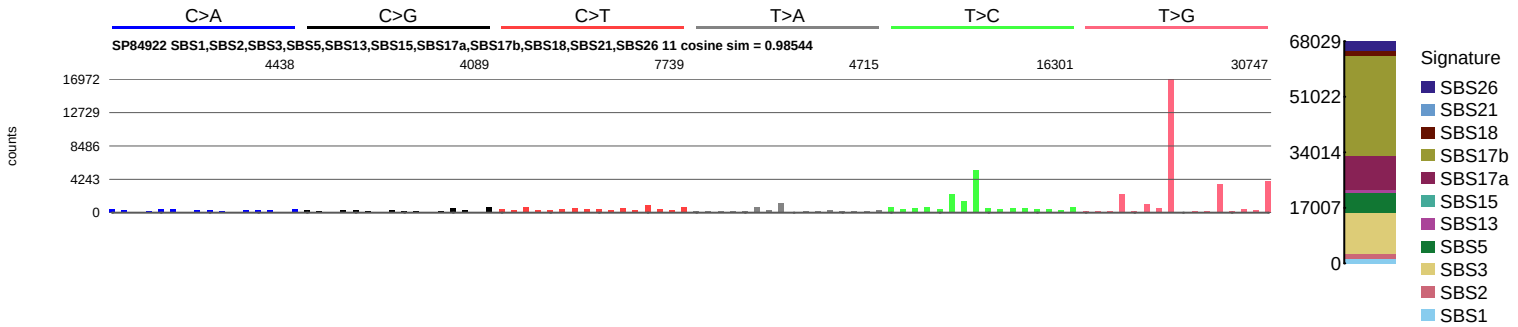

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS17a,SBS17b,SBS41 4 cosine sim = 0.98606
960
2886
5484
8596
16573
33528
counts
16996
12747
8498
4249
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS17a,SBS17b,SBS21,SBS41 5 cosine sim = 0.98567
1176
3346
3252
8213
18767
33274
counts
17004
12753
8502
4251
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS21,SBS41 4 cosine sim = 0.98555
1068
2814
3322
8427
18971
33427
counts
17004
12753
8502
4251
0
68029
51022
34014
17007
0
Signature
SBS41
SBS21
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 11 cosine sim = 0.98544
4454
4068
7767
4716
16293
30734
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 12 cosine sim = 0.98544
4451
4068
7769
4714
16293
30734
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
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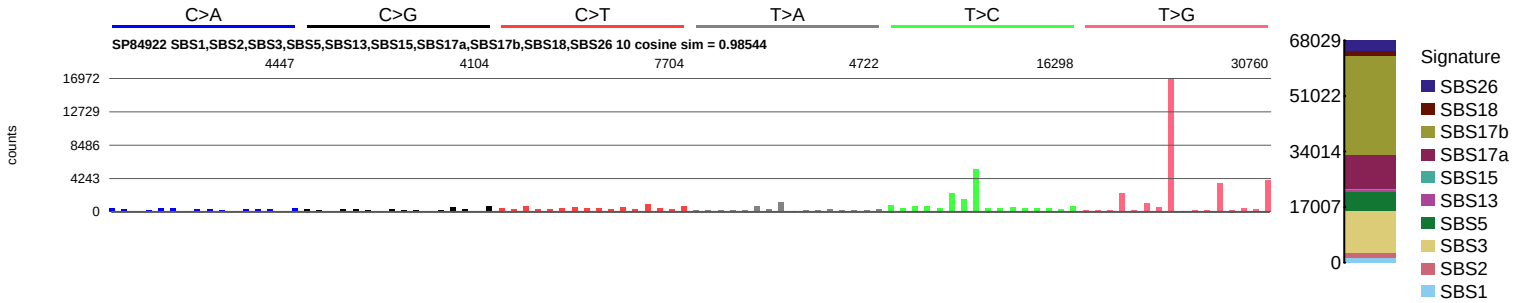
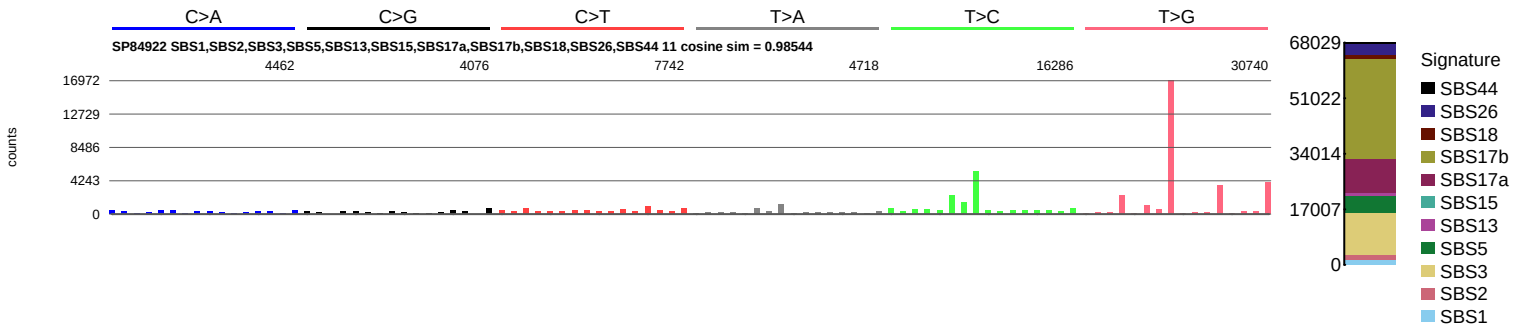
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 11 cosine sim = 0.98544
4438
4089
7739
4715
16301
30747
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 10 cosine sim = 0.98544
4447
4104
7704
4722
16298
30760
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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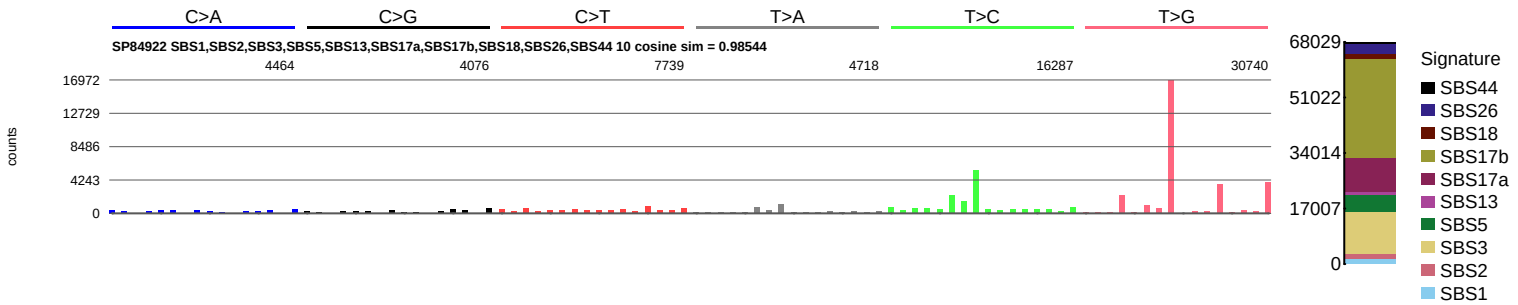
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 11 cosine sim = 0.98544
4462
4076
7742
4718
16286
30740
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98544
4464
4076
7739
4718
16287
30740
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
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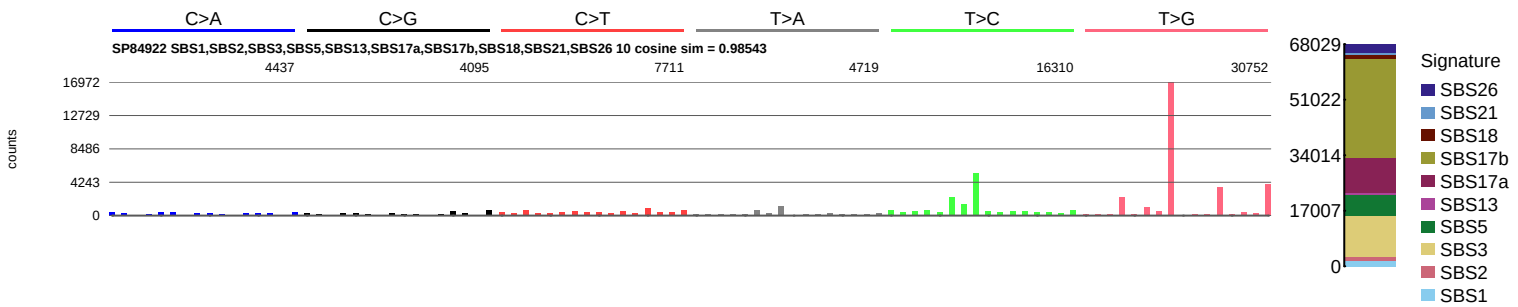
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26 10 cosine sim = 0.98543
4437
4095
7711
4719
16310
30752
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
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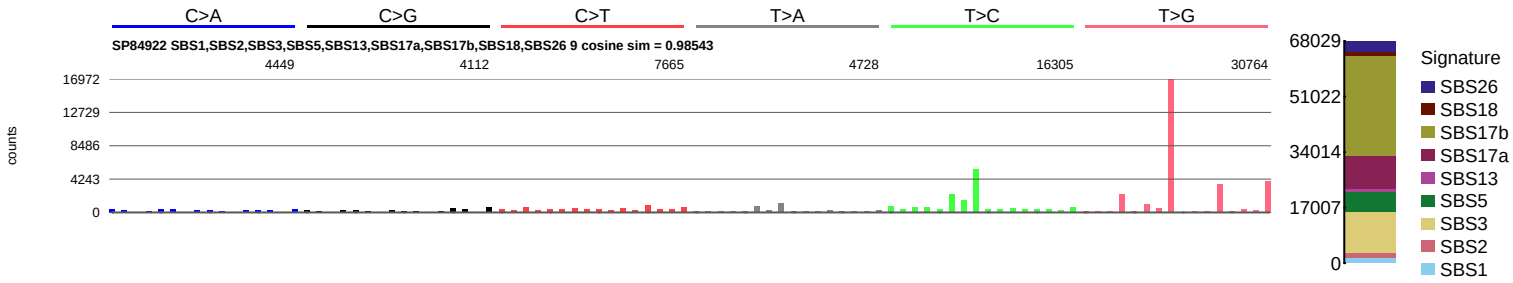
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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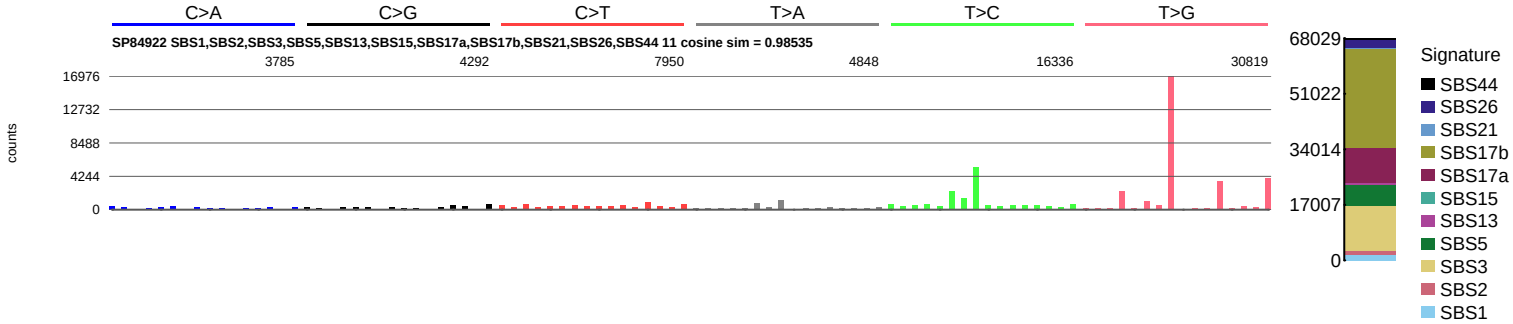
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98543
4449
4112
7665
4728
16305
30764
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS44 11 cosine sim = 0.98535
3785
4292
7950
4848
16336
30819
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS44
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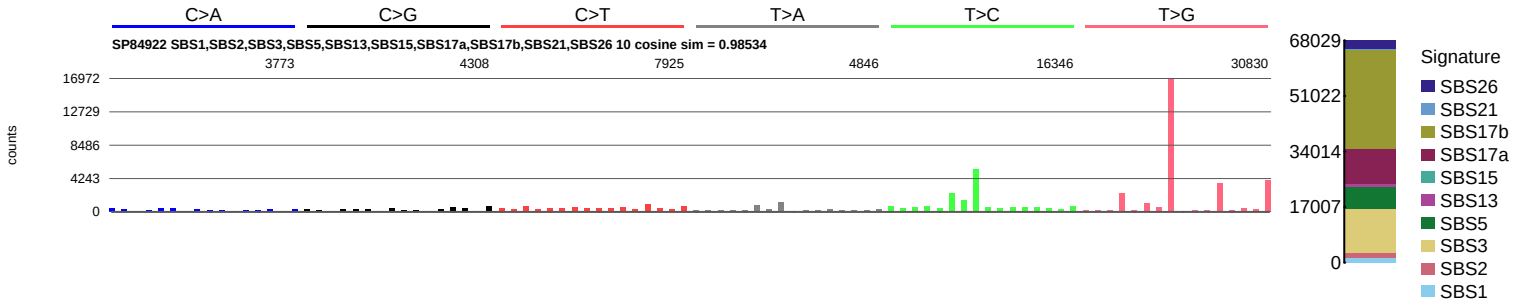
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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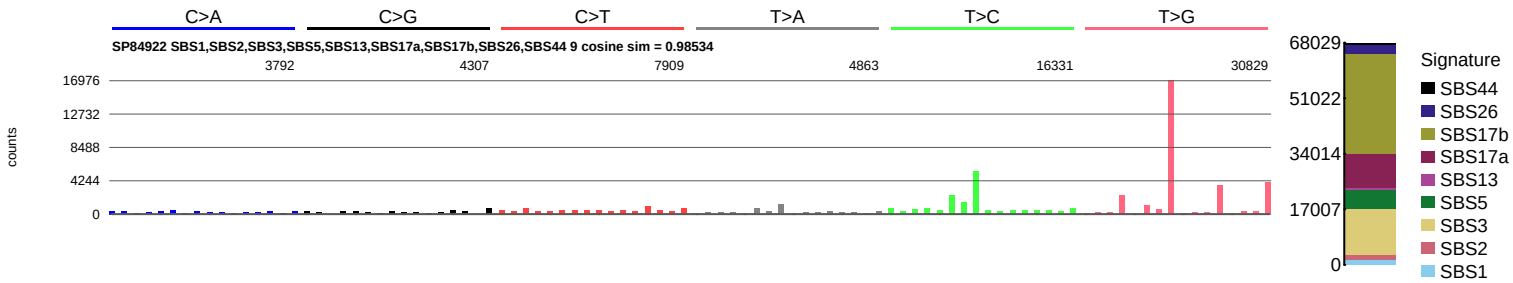
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26 10 cosine sim = 0.98534
3773
4308
7925
4846
16346
30830
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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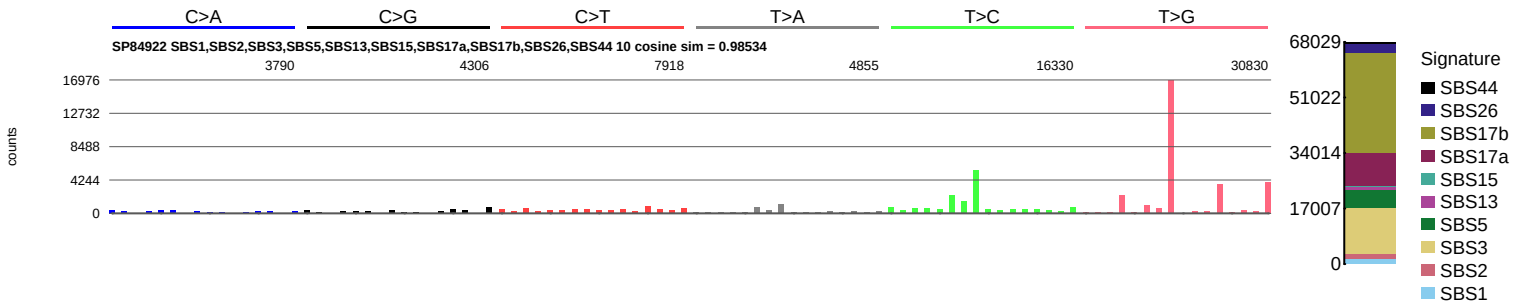
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS44 9 cosine sim = 0.98534
3792
4307
7909
4863
16331
30829
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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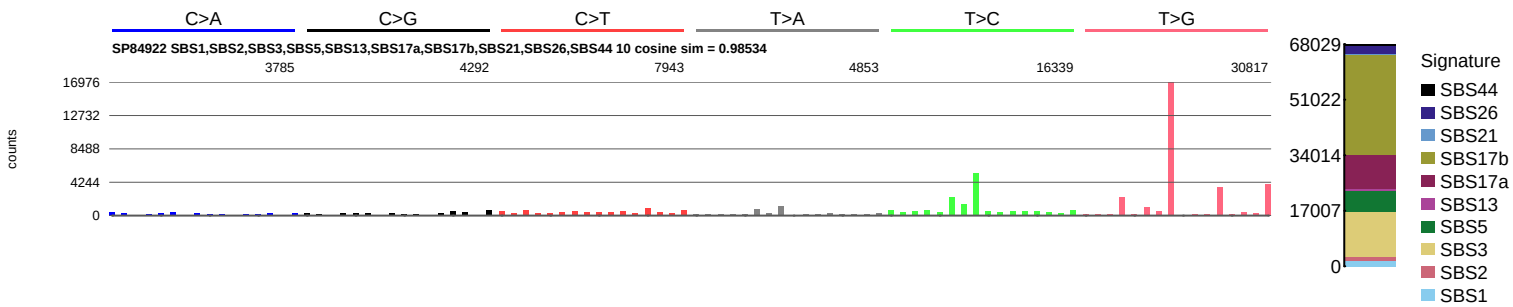
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS44 10 cosine sim = 0.98534
3790
4306
7918
4855
16330
30830
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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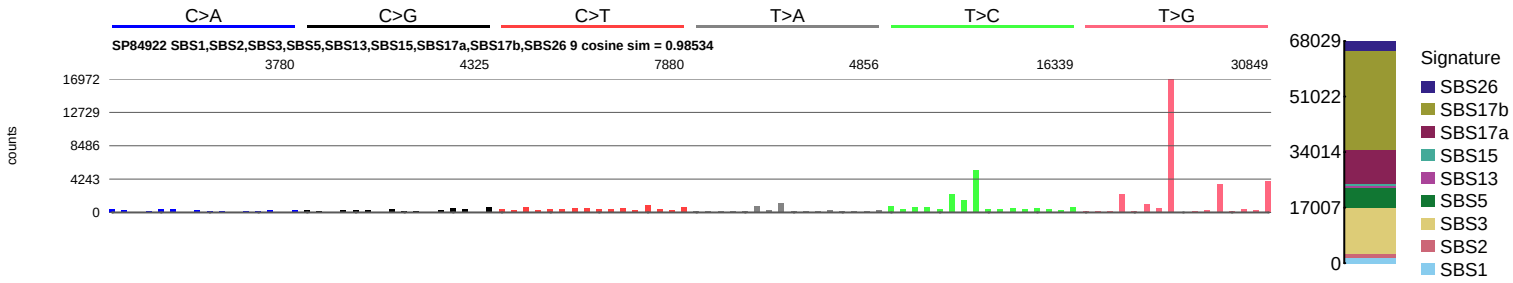
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS44 10 cosine sim = 0.98534
3785
4292
7943
4853
16339
30817
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26 9 cosine sim = 0.98534
3780
4325
7880
4856
16339
30849
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
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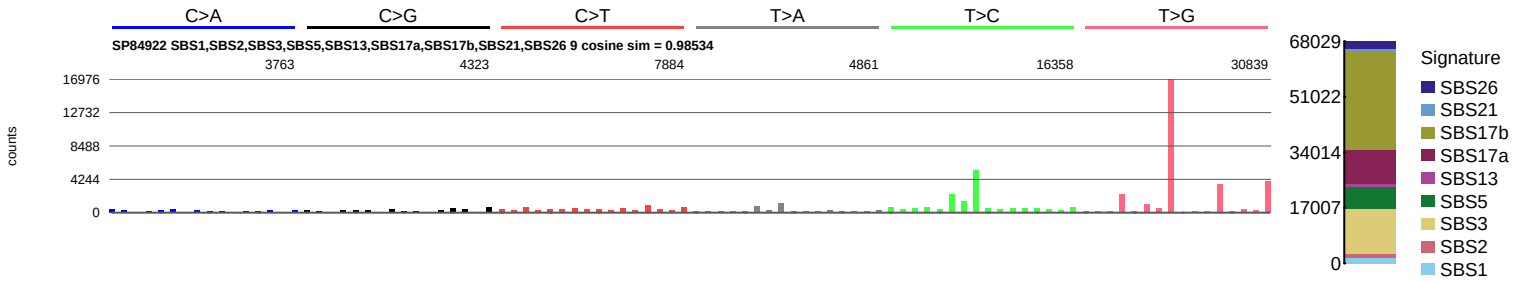
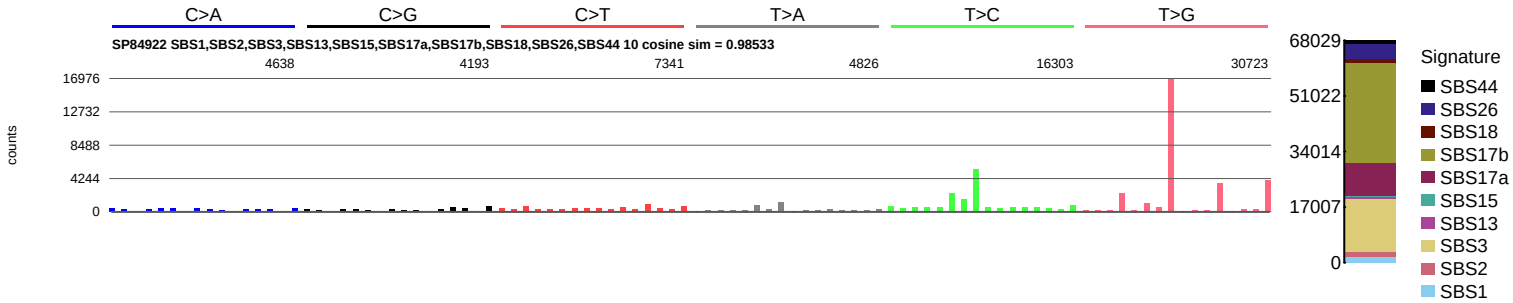
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS26 9 cosine sim = 0.98534
3763
4323
7884
4861
16358
30839
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98533
4638
4193
7341
4826
16303
30723
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
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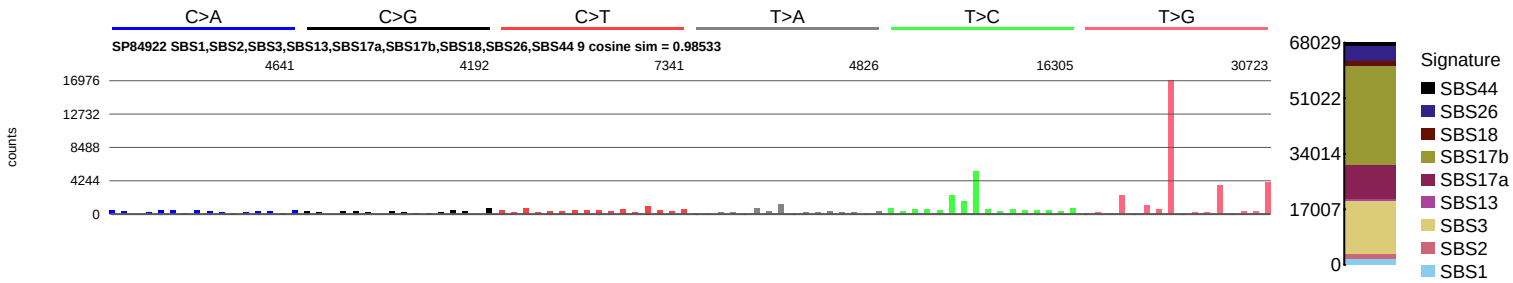
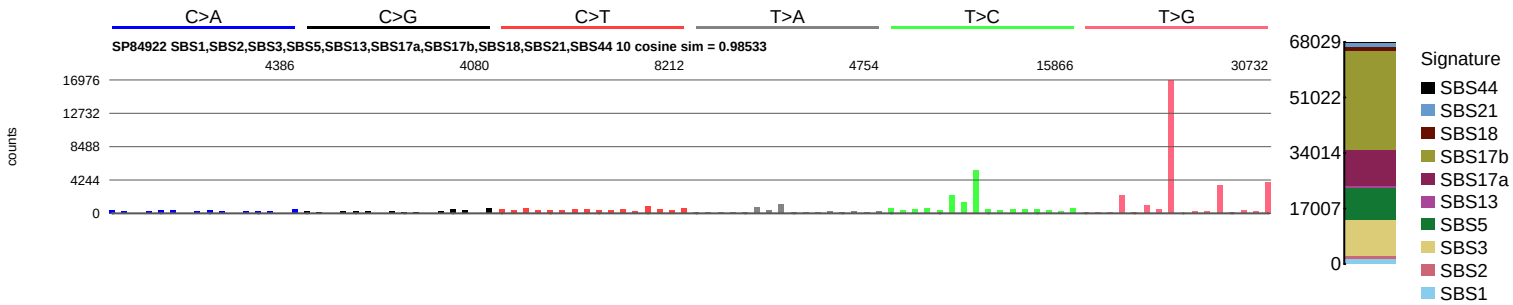
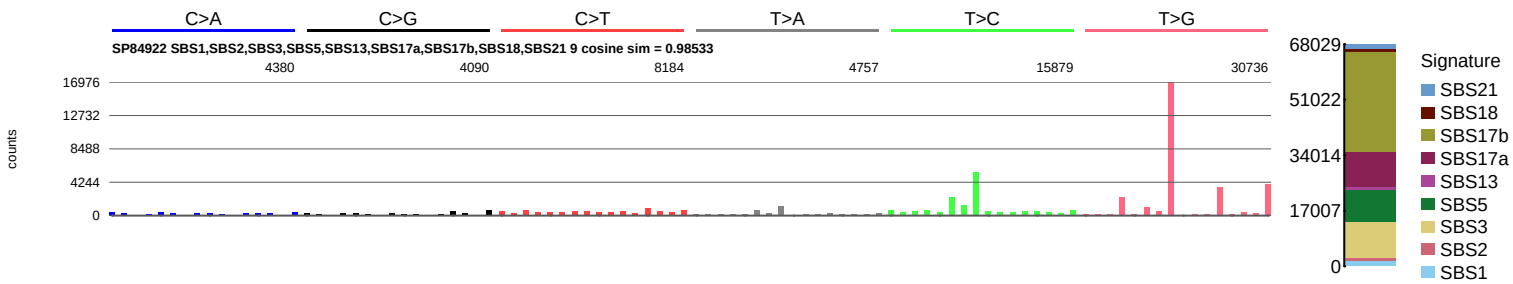
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98533
4641
4192
7341
4826
16305
30723
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98533
4386
4080
8212
4754
15866
30732
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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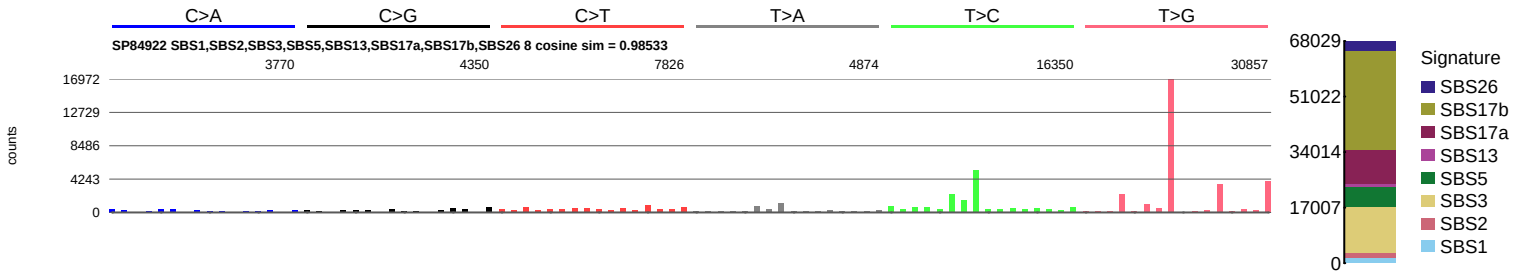
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98533
4380
4090
8184
4757
15879
30736
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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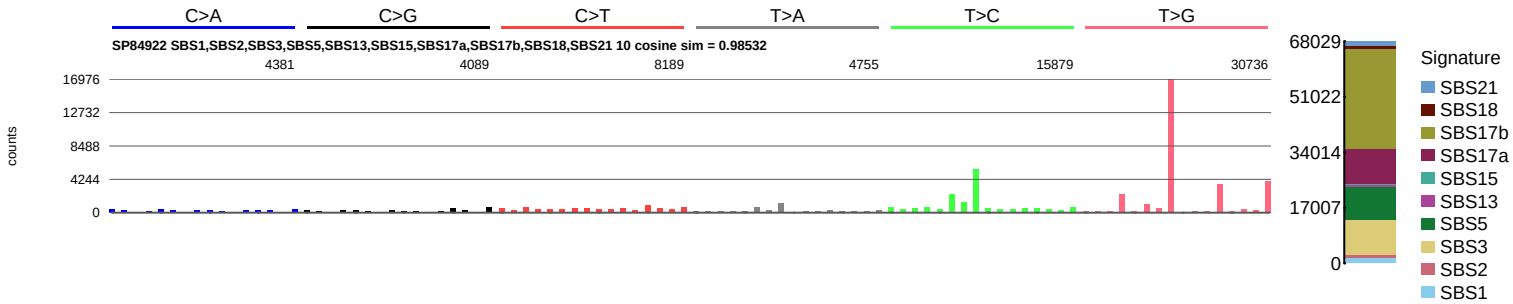
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26 8 cosine sim = 0.98533
3770
4350
7826
4874
16350
30857
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.98532
4381
4089
8189
4755
15879
30736
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
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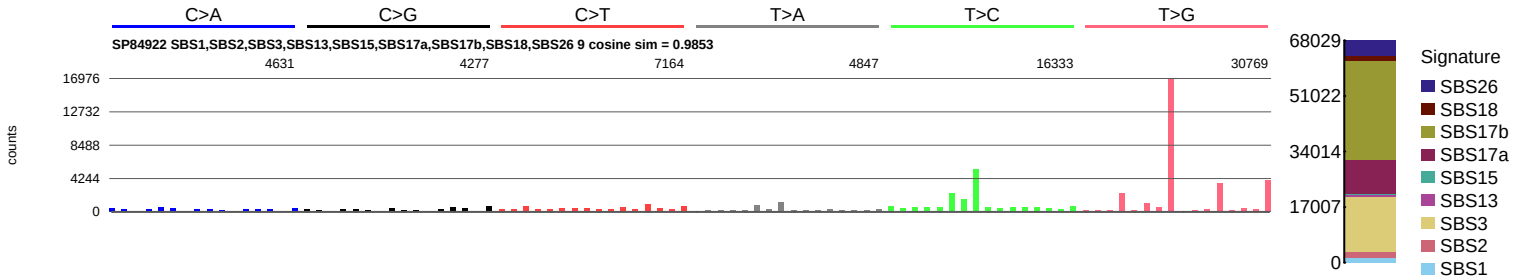
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.9853
4631
4277
7164
4847
16333
30769
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
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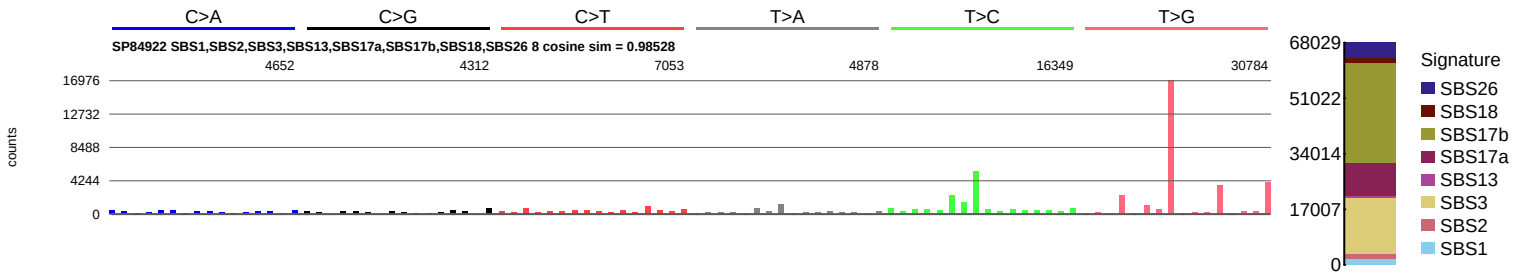
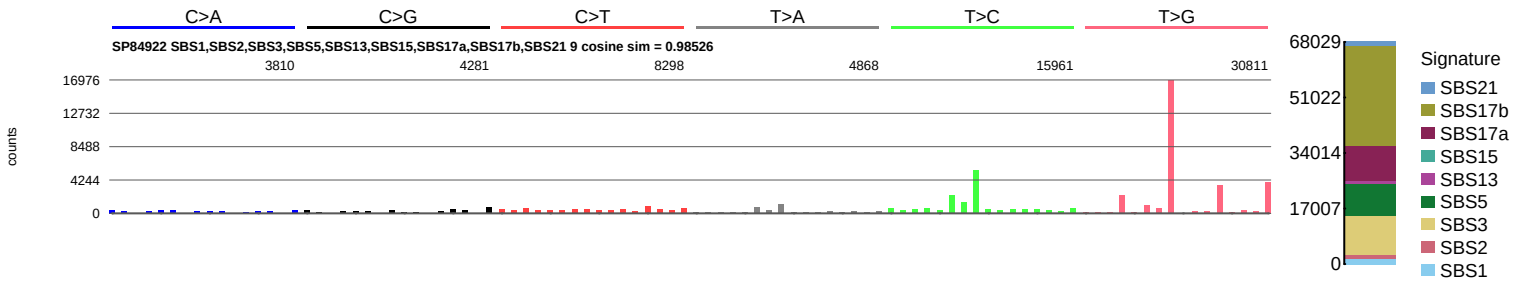
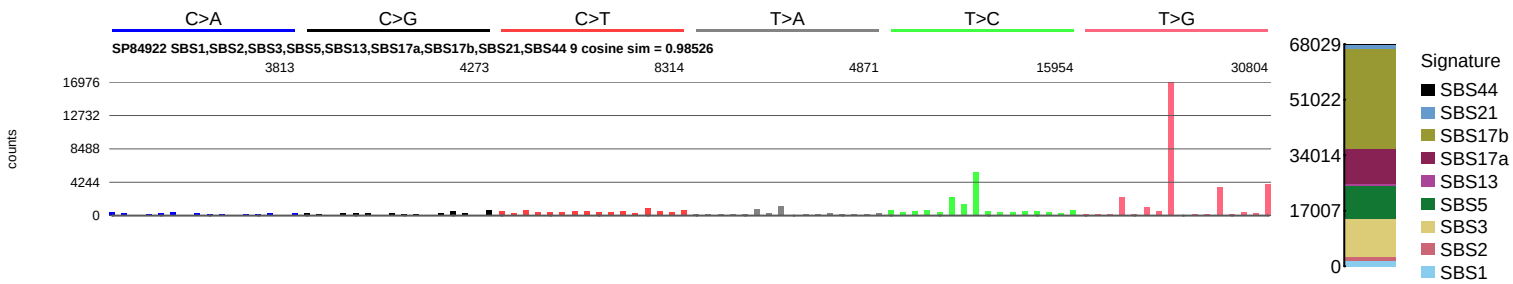
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98528
4652
4312
7053
4878
16349
30784
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 9 cosine sim = 0.98526
3810
4281
8298
4868
15961
30811
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 9 cosine sim = 0.98526
3813
4273
8314
4871
15954
30804
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
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Signature
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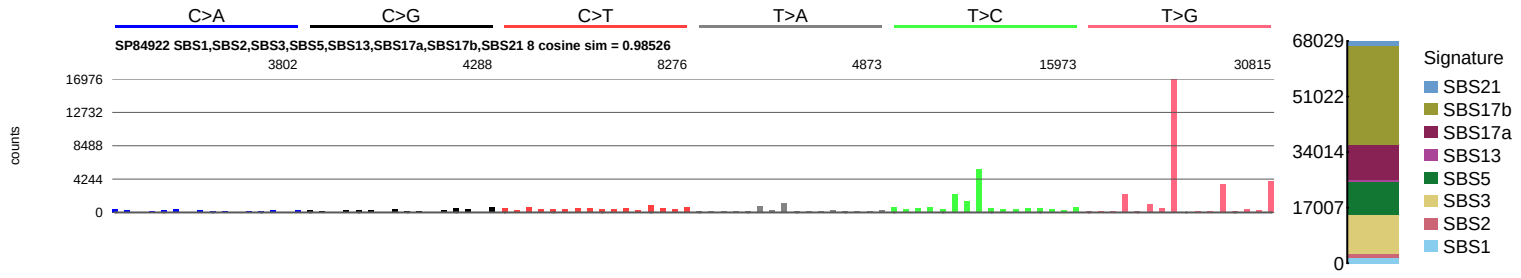
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98526
3802
4288
8276
4873
15973
30815
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
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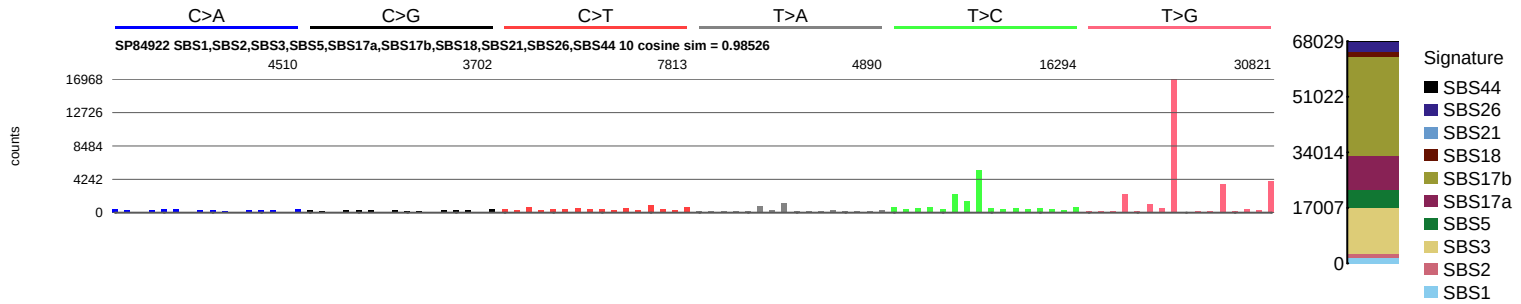
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.98526
4510
3702
7813
4890
16294
30821
counts
16968
12726
8484
4242
0
68029
51022
34014
17007
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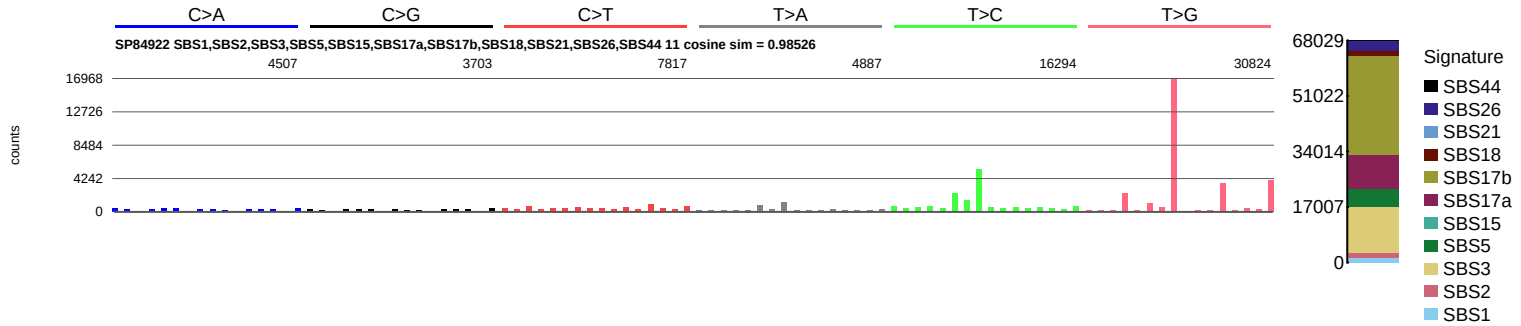
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 11 cosine sim = 0.98526
4507
3703
7817
4887
16294
30824
counts
16968
12726
8484
4242
0
68029
51022
34014
17007
0
Signature
SBS44
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SBS18
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SBS15
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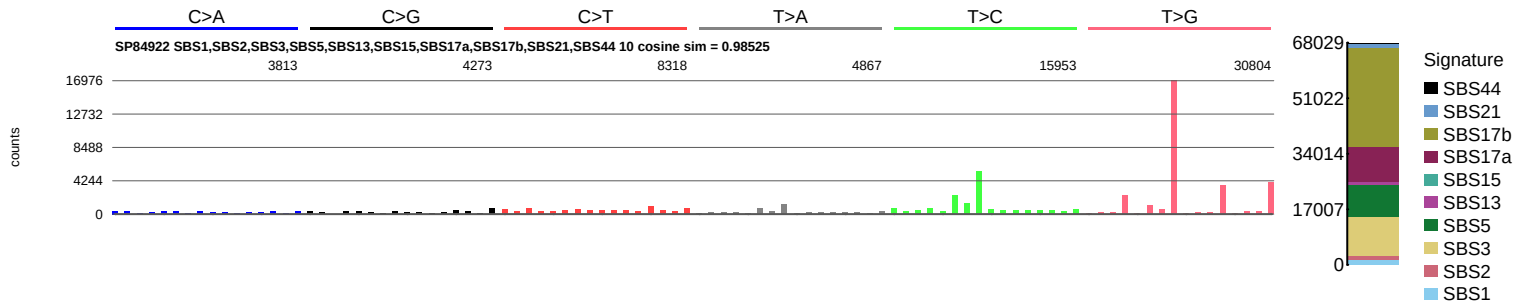
SBS1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T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS44 10 cosine sim = 0.98525
3813
4273
8318
4867
15953
30804
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
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SBS17b
SBS17a
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SBS13
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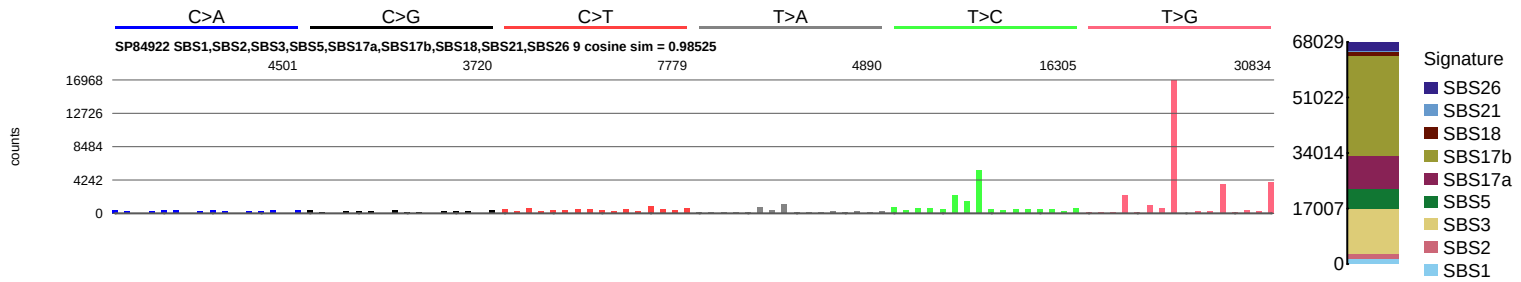
SBS1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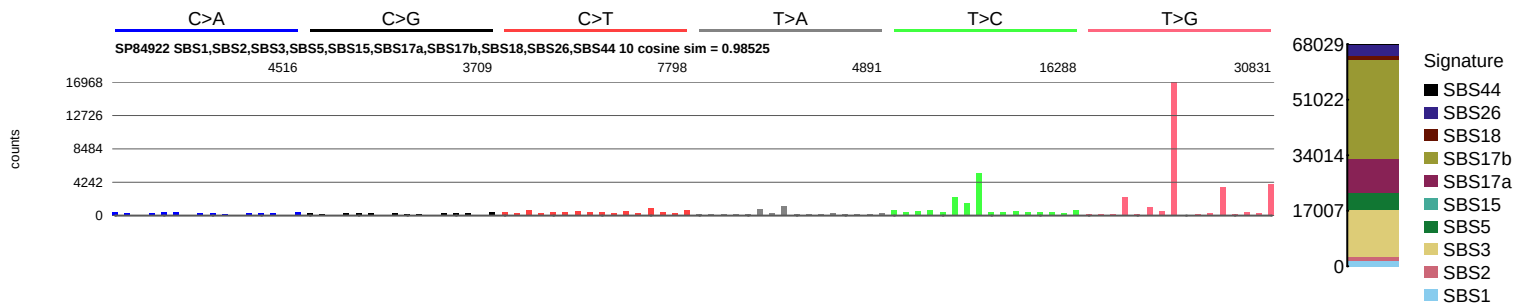
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98525
4501
3720
7779
4890
16305
30834
counts
16968
12726
8484
4242
0
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
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T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98525
4516
3709
7798
4891
16288
30831
counts
16968
12726
8484
4242
0
68029
51022
34014
17007
0
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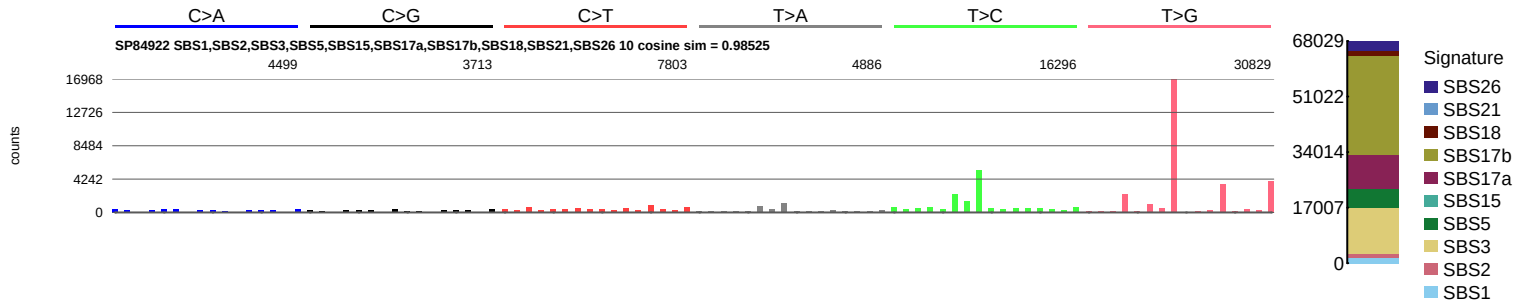
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 10 cosine sim = 0.98525
4499
3713
7803
4886
16296
30829
counts
16968
12726
8484
4242
0
68029
51022
34014
17007
0
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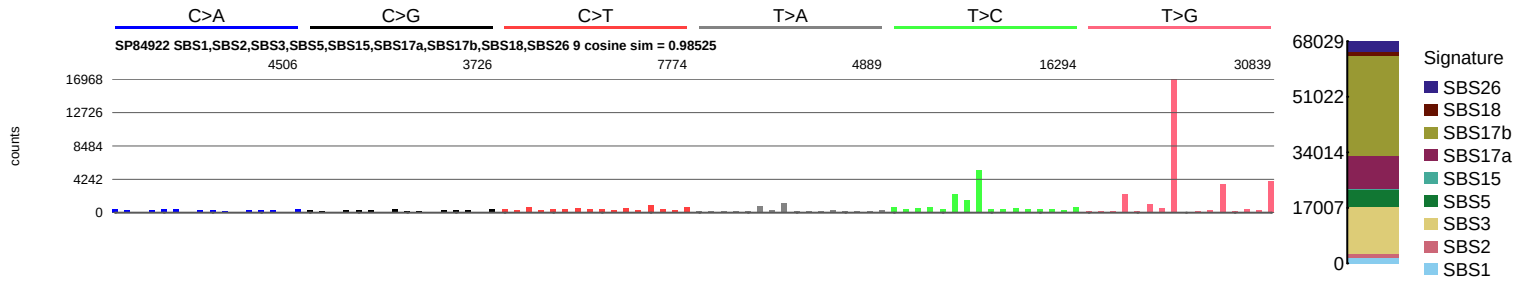
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98525
4506
3726
7774
4889
16294
30839
counts
16968
12726
8484
4242
0
68029
51022
34014
17007
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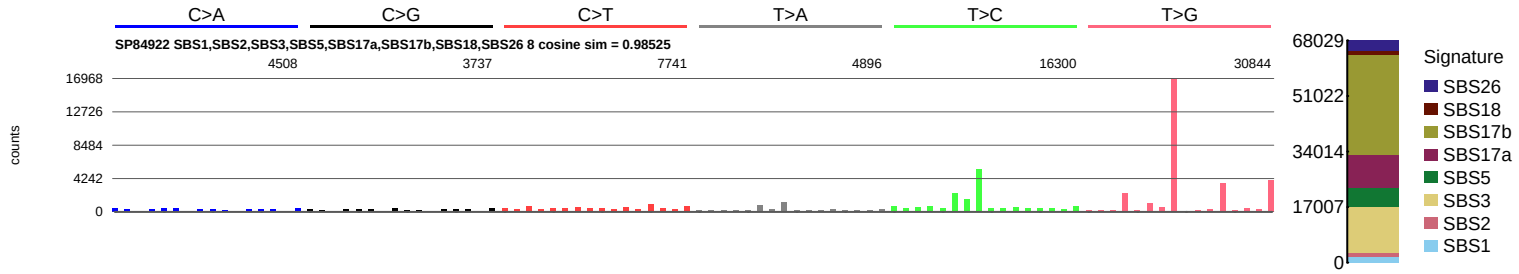
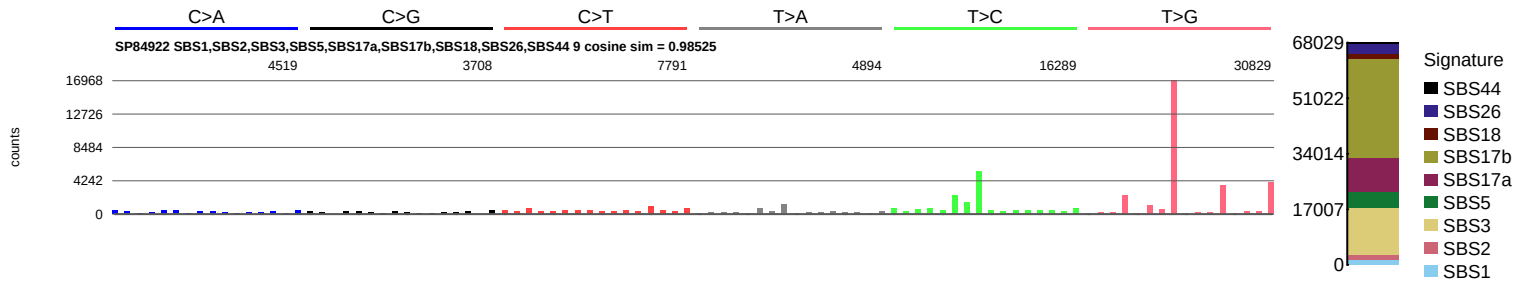
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98525
4508
3737
7741
4896
16300
30844
counts
16968
12726
8484
4242
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98525
4519
3708
7791
4894
16289
30829
counts
16968
12726
8484
4242
0
68029
51022
34014
17007
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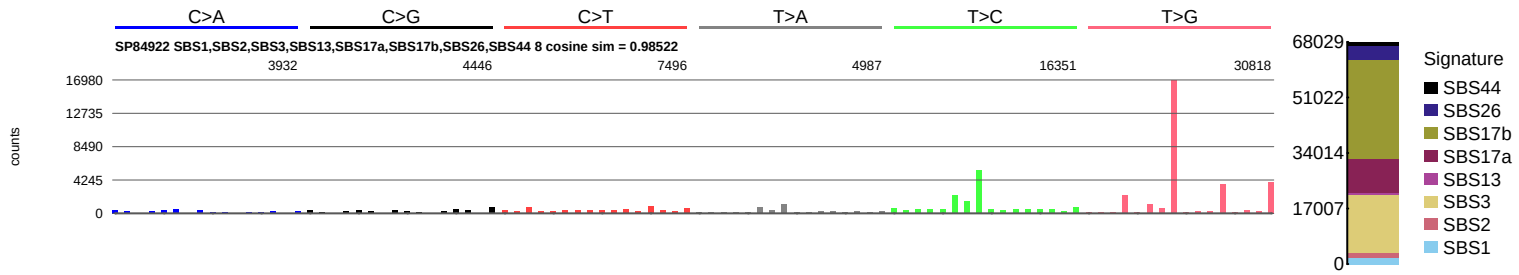
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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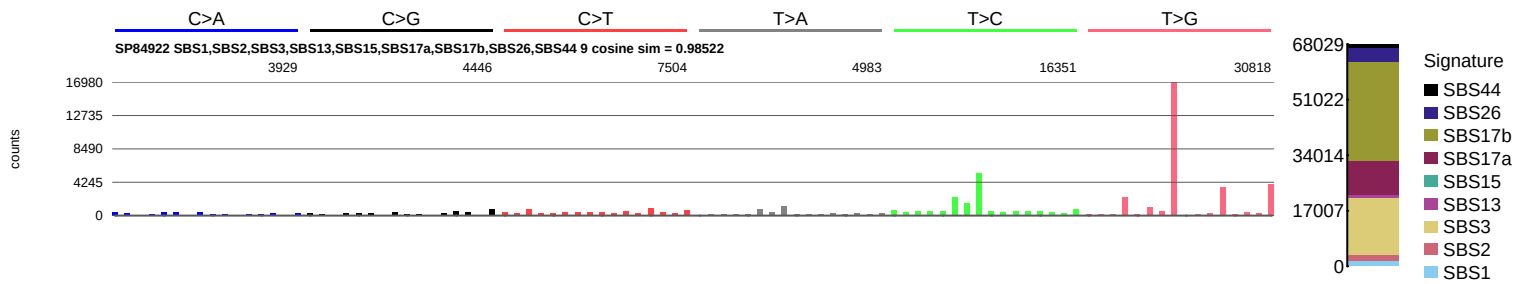
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98522
3932
4446
7496
4987
16351
30818
counts
16980
12735
8490
4245
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS44 9 cosine sim = 0.98522
3929
4446
7504
4983
16351
30818
counts
16980
12735
8490
4245
0
68029
51022
34014
17007
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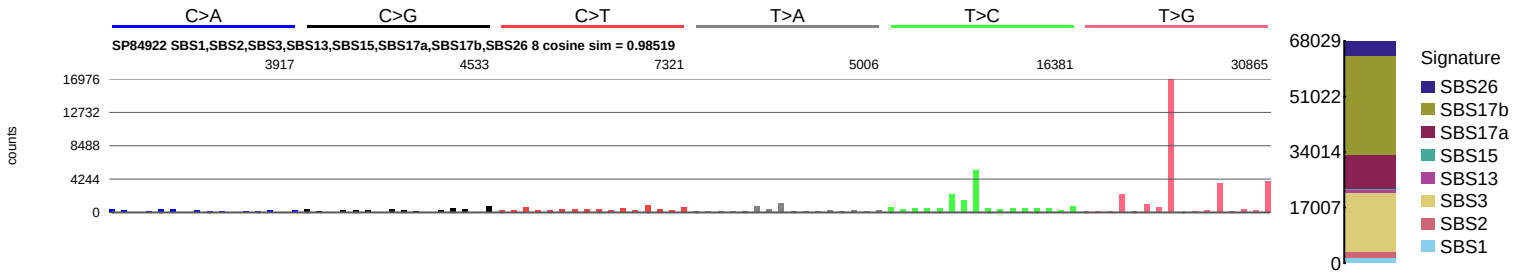
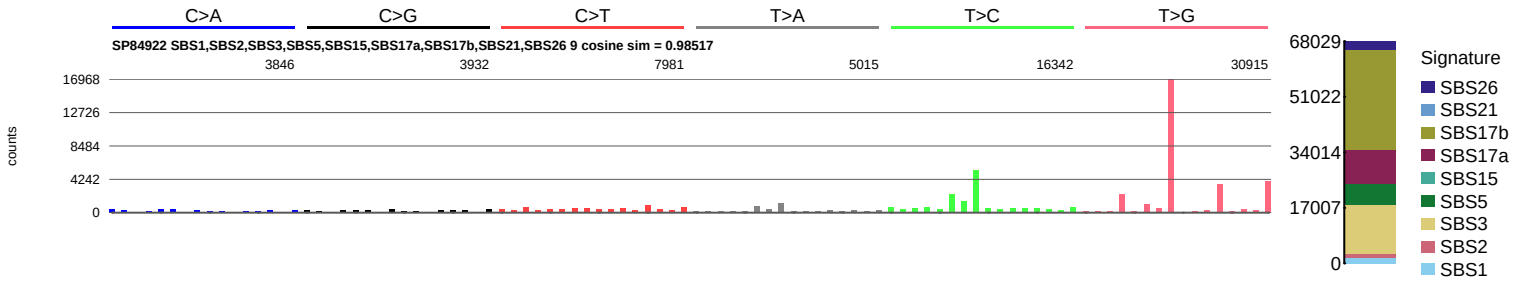
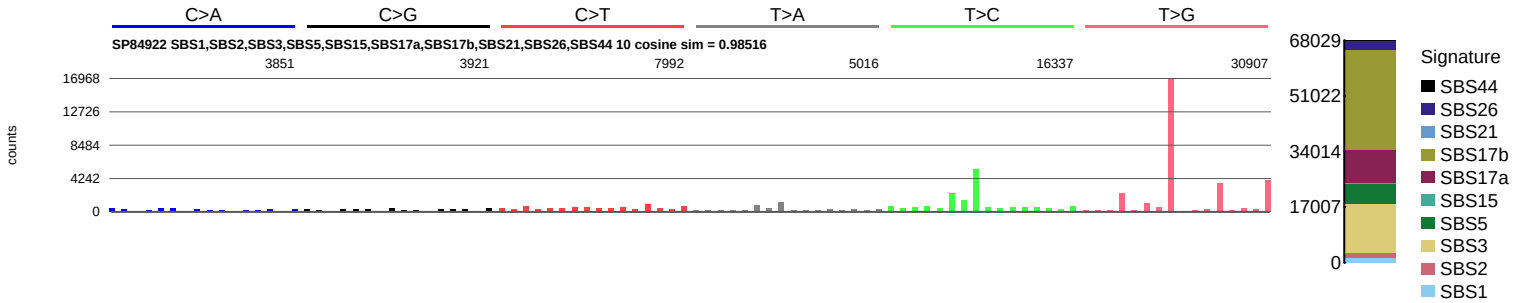
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26 8 cosine sim = 0.98519
3917
4533
7321
5006
16381
30865
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS26 9 cosine sim = 0.98517
3846
3932
7981
5015
16342
30915
counts
16968
12726
8484
4242
0
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS44 10 cosine sim = 0.98516
3851
3921
7992
5016
16337
30907
counts
16968
12726
8484
4242
0
68029
51022
34014
17007
0
Signature
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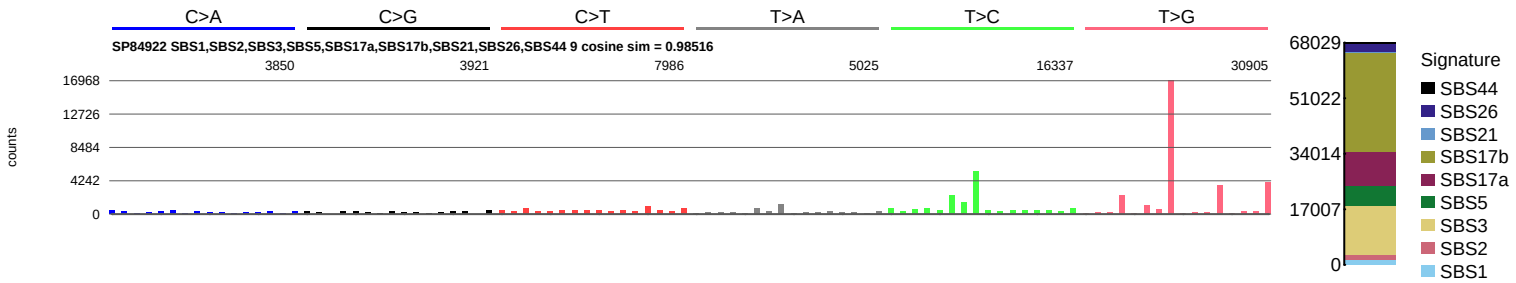
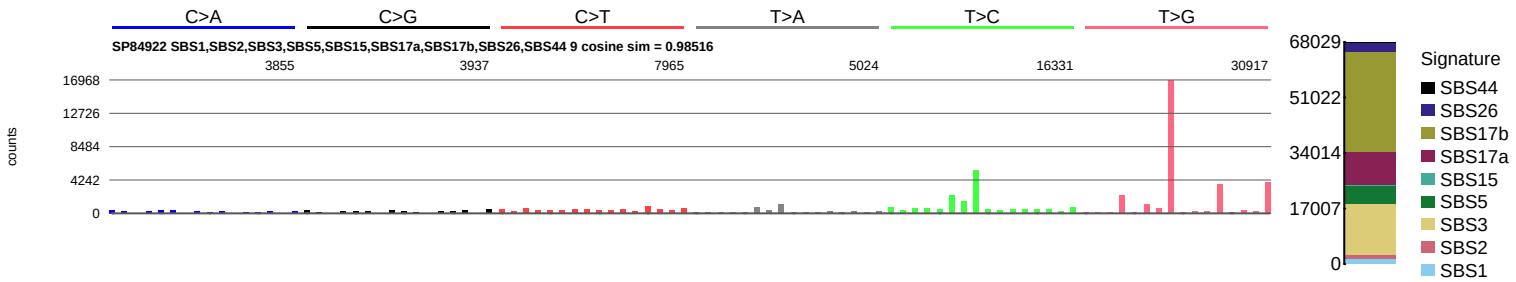
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS26,SBS44 9 cosine sim = 0.98516
3850
3921
7986
5025
16337
30905
counts
16968
12726
8484
4242
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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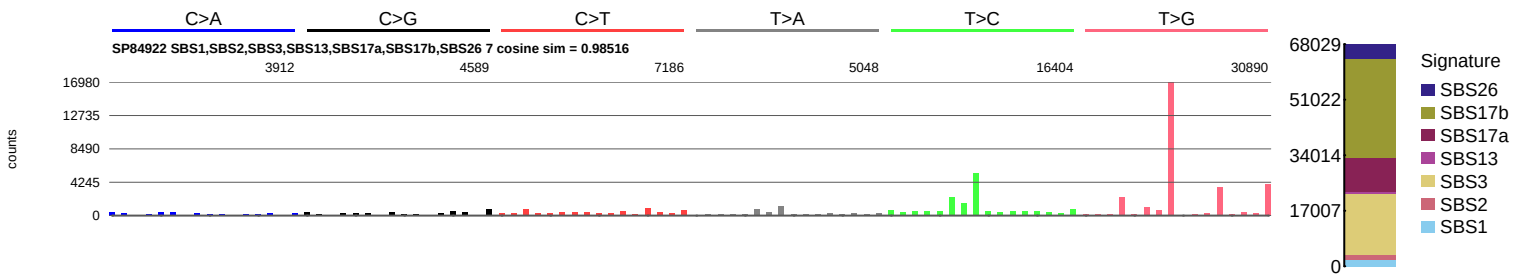
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS44 9 cosine sim = 0.98516
3855
3937
7965
5024
16331
30917
counts
16968
12726
8484
4242
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98516
3912
4589
7186
5048
16404
30890
counts
16980
12735
8490
4245
0
68029
51022
34014
17007
0
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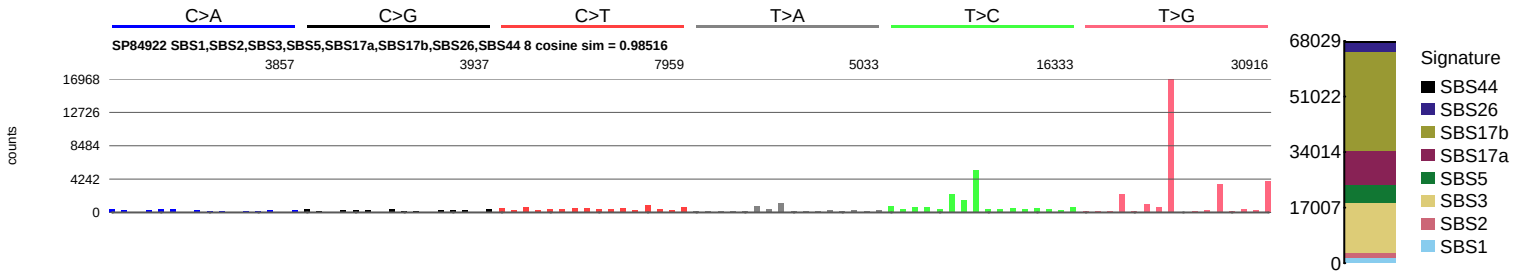
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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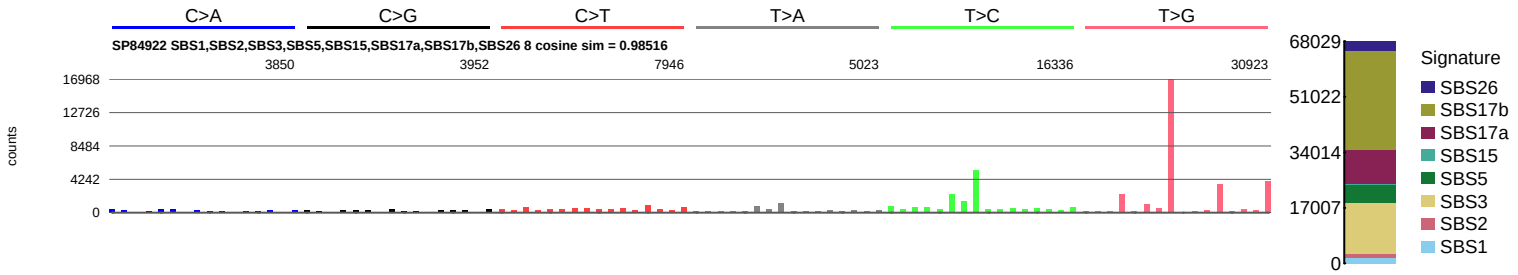
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98516
3857
3937
7959
5033
16333
30916
counts
16968
12726
8484
4242
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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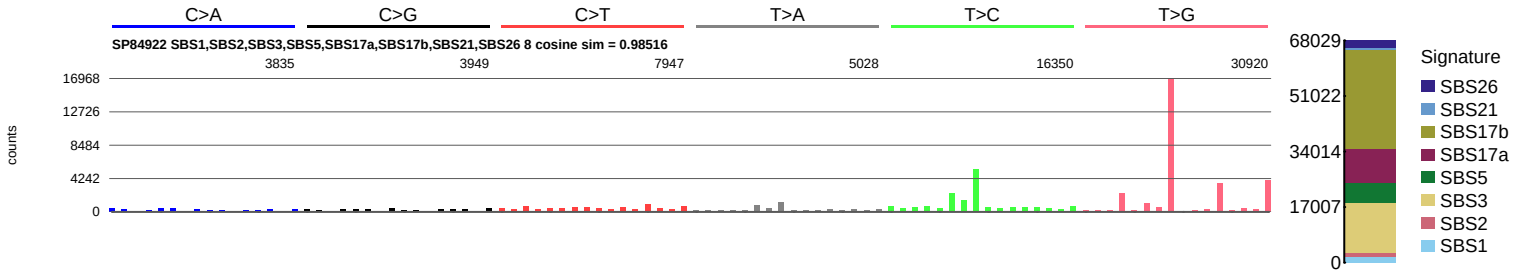
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26 8 cosine sim = 0.98516
3850
3952
7946
5023
16336
30923
counts
16968
12726
8484
4242
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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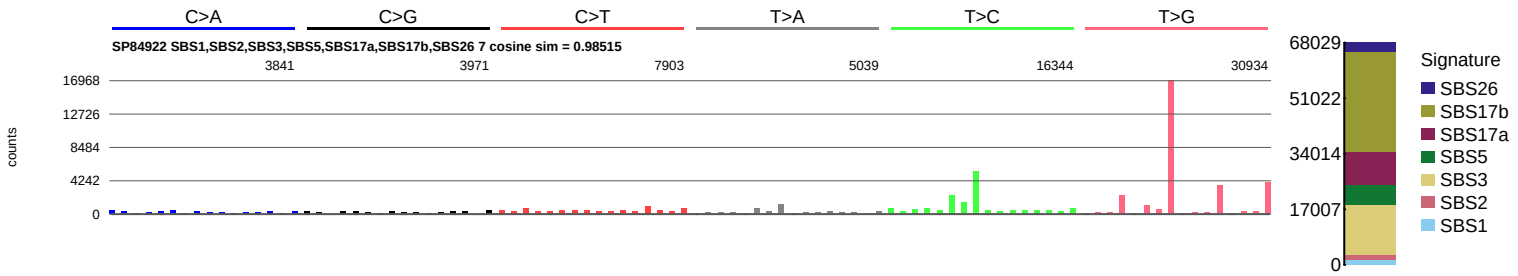
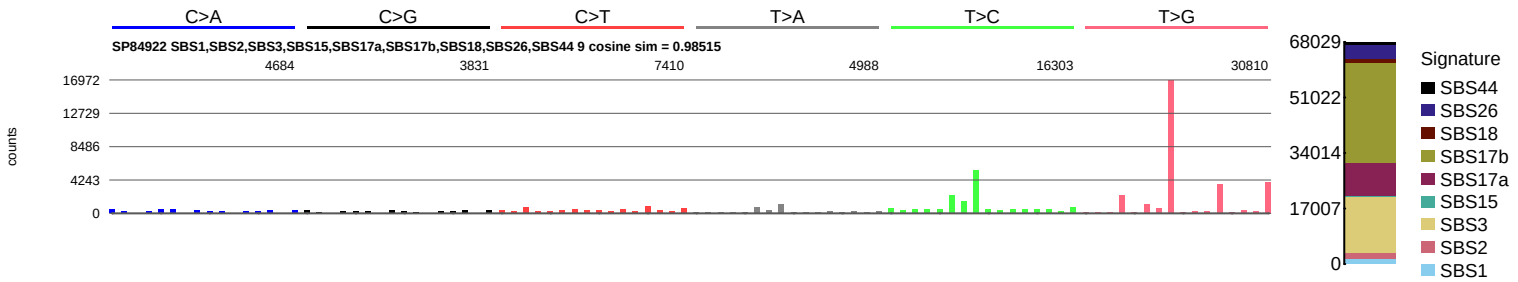
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS26 8 cosine sim = 0.98516
3835
3949
7947
5028
16350
30920
counts
16968
12726
8484
4242
0
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98515
3841
3971
7903
5039
16344
30934
counts
16968
12726
8484
4242
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98515
4684
3831
7410
4988
16303
30810
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
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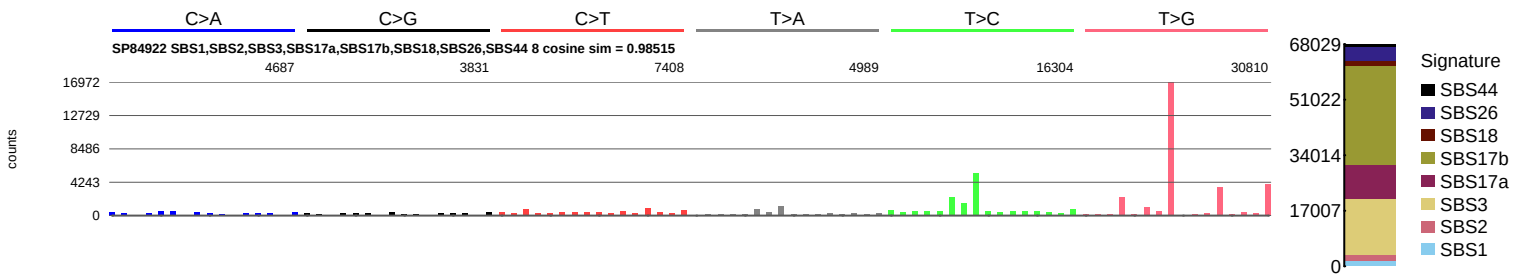
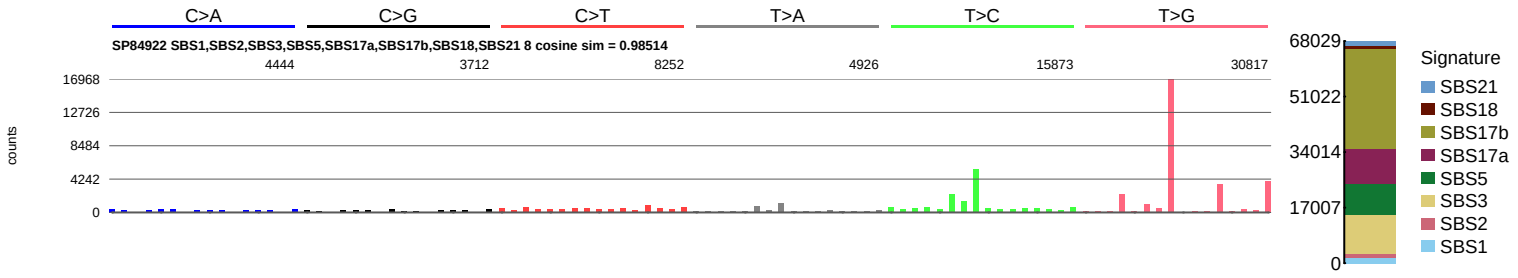
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98515
4687
3831
7408
4989
16304
30810
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98514
4444
3712
8252
4926
15873
30817
counts
16968
12726
8484
4242
0
68029
51022
34014
17007
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Signature
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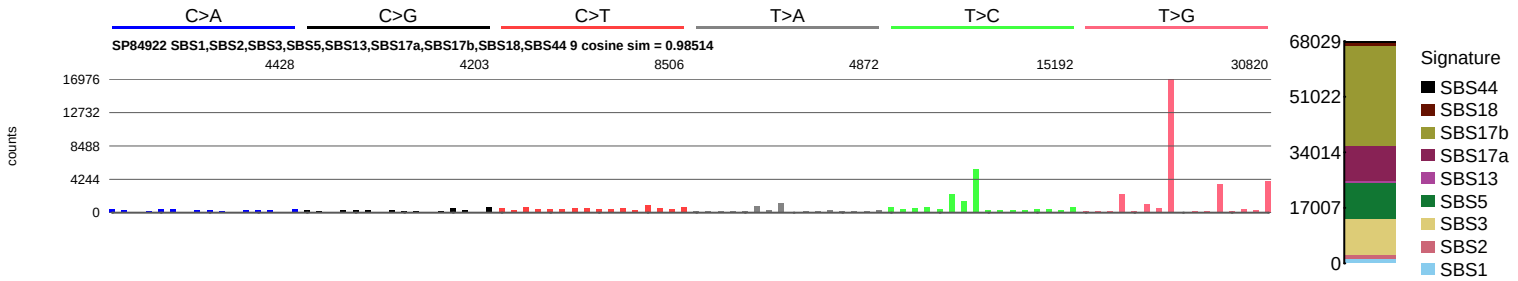
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98514
4428
4203
8506
4872
15192
30820
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
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Signature
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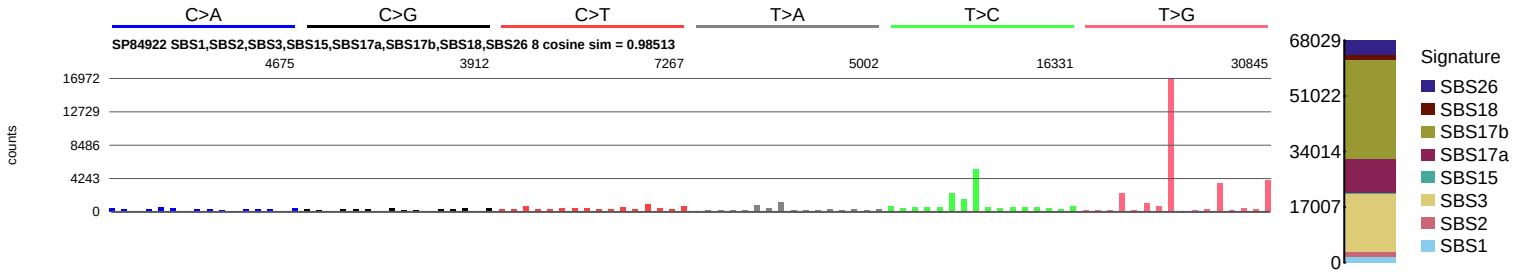
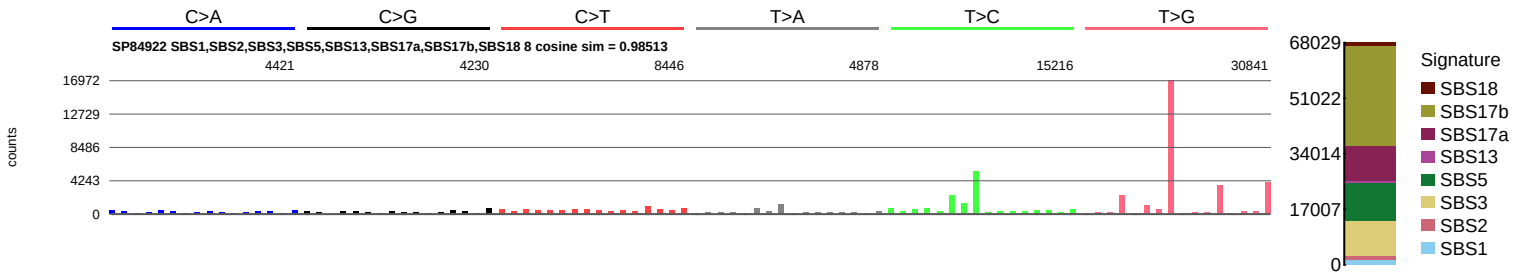
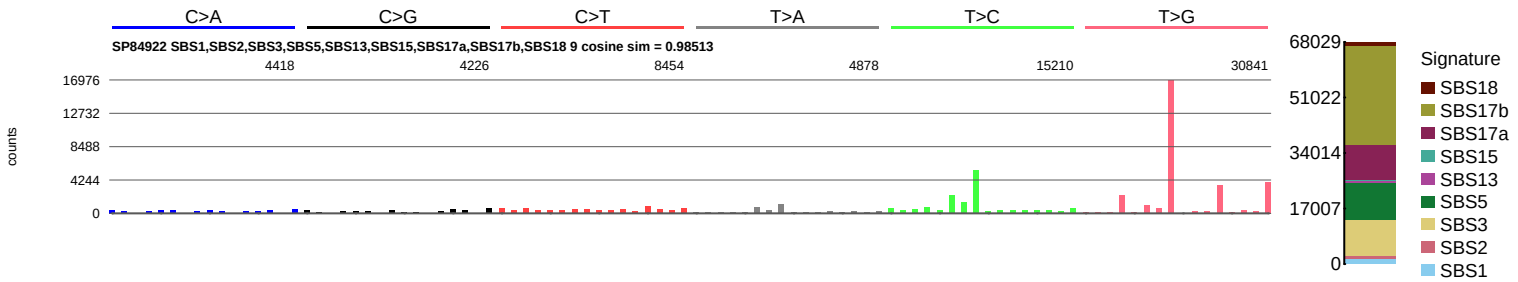
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98513
4675
3912
7267
5002
16331
30845
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98513
4421
4230
8446
4878
15216
30841
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 9 cosine sim = 0.98513
4418
4226
8454
4878
15210
30841
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
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SBS17b
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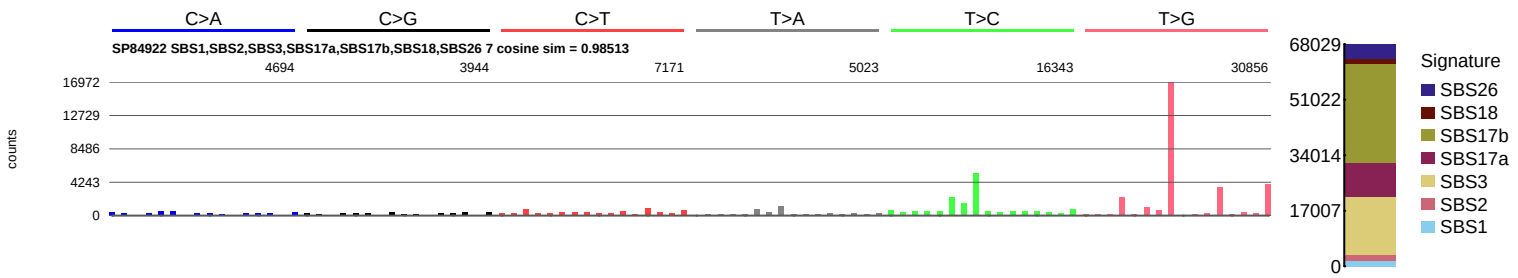
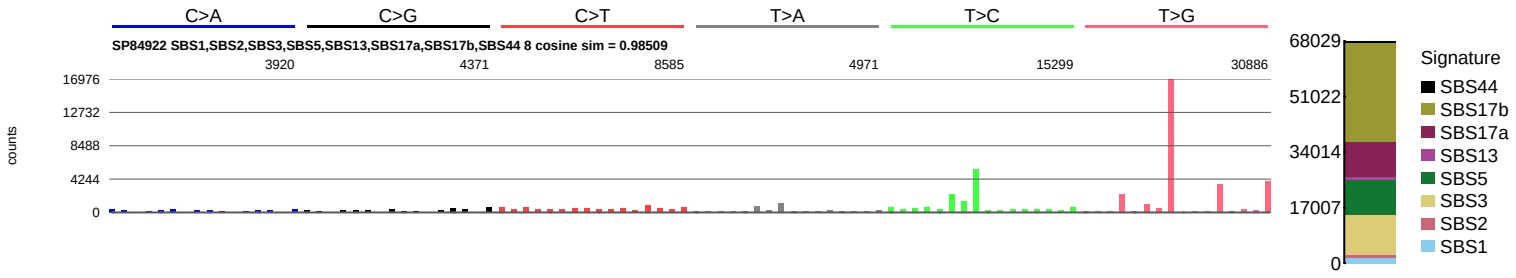
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98513
4694
3944
7171
5023
16343
30856
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS44 8 cosine sim = 0.98509
3920
4371
8585
4971
15299
30886
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
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Signature
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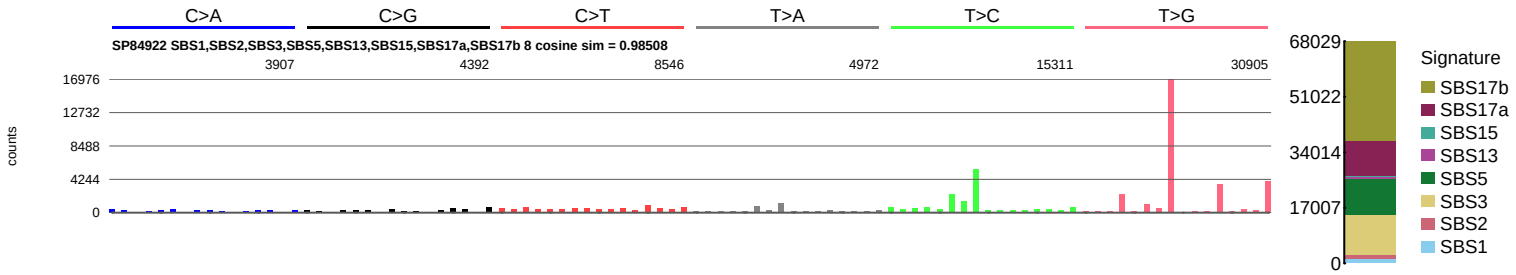
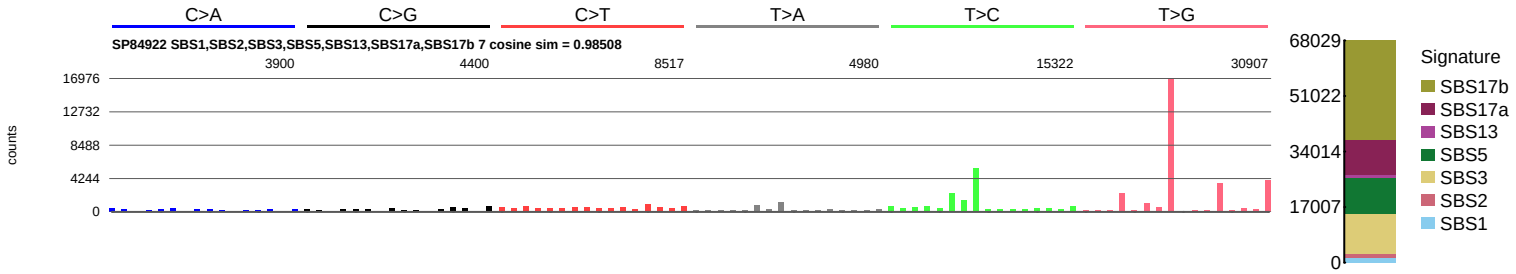
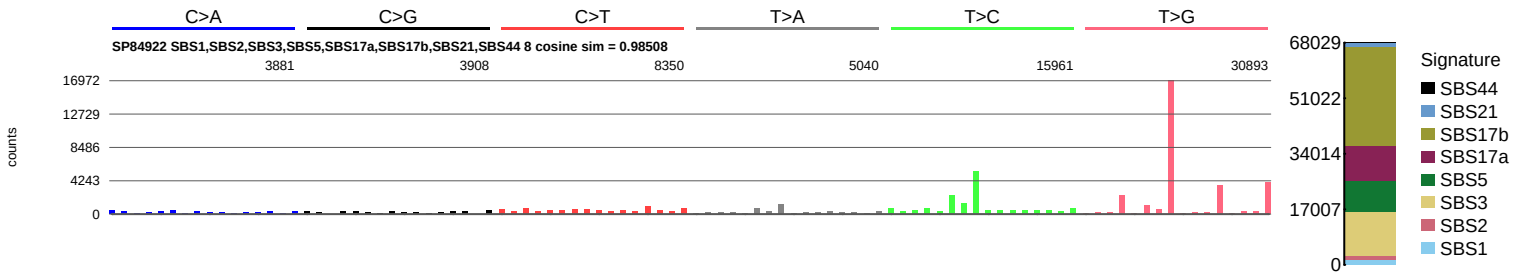
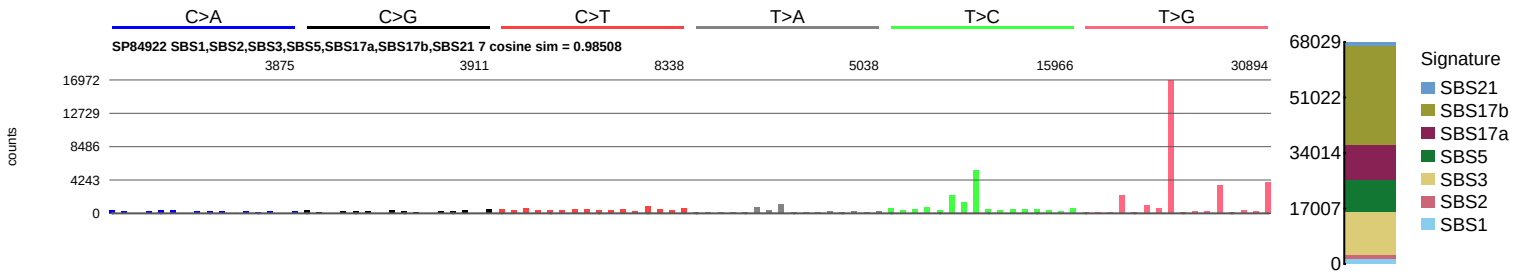
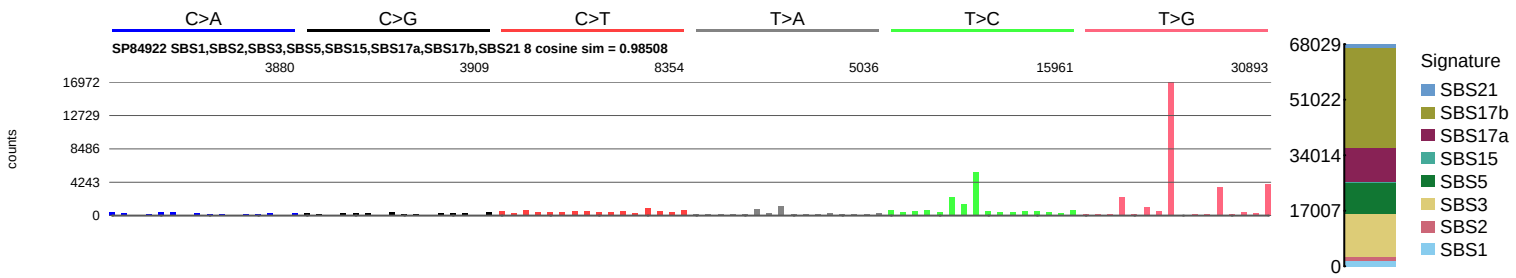
SBS3
SBS2
SBS1

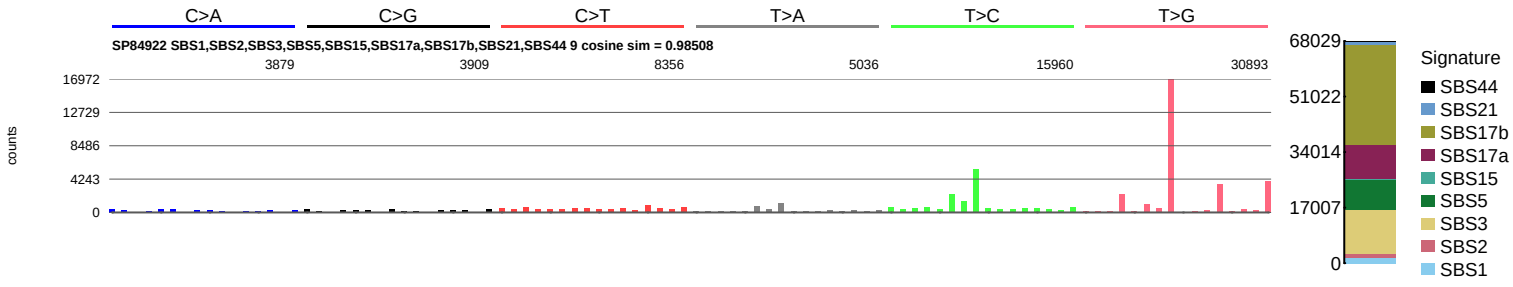

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b 8 cosine sim = 0.98508
3907
4392
8546
4972
15311
30905
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

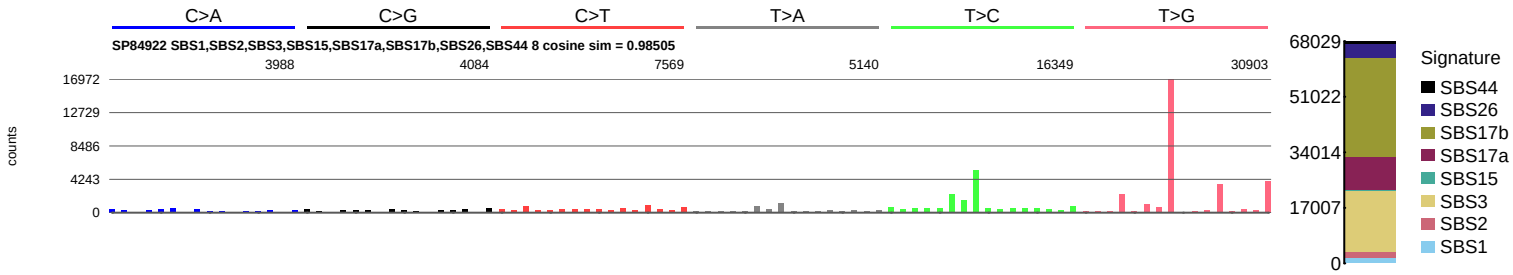

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b 7 cosine sim = 0.98508
3900
4400
8517
4980
15322
30907
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

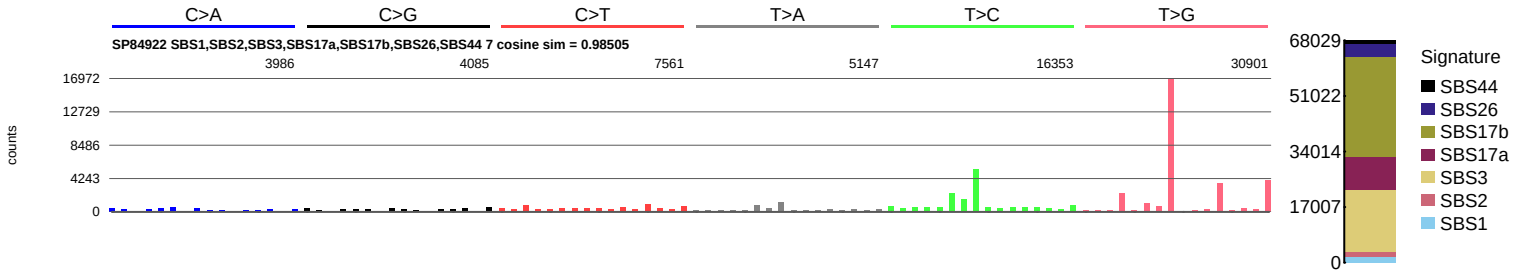

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98508
3881
3908
8350
5040
15961
30893
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

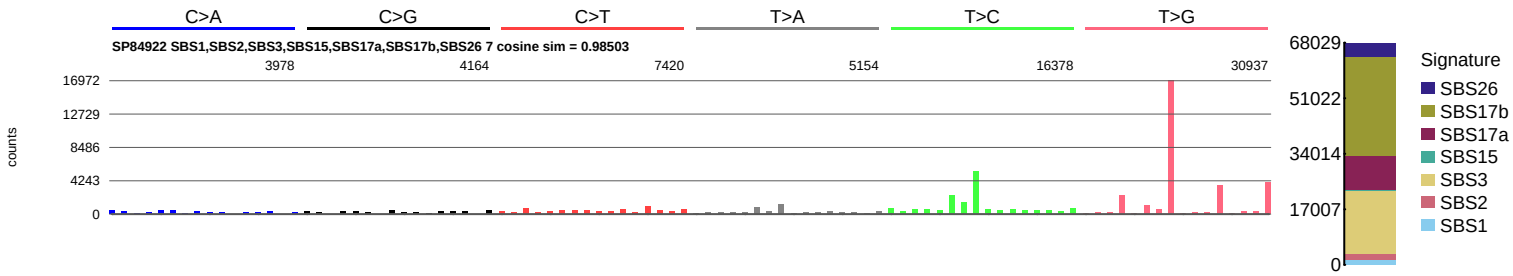

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98508
3875
3911
8338
5038
15966
30894
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

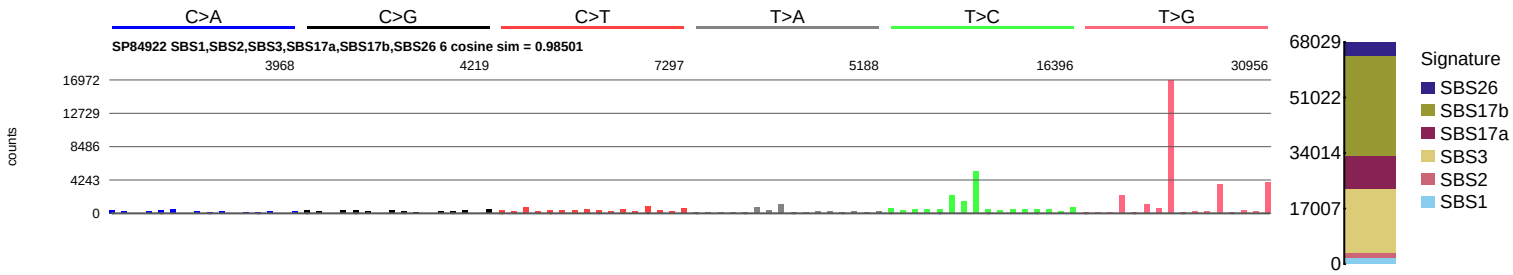

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98508
3880
3909
8354
5036
15961
30893
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

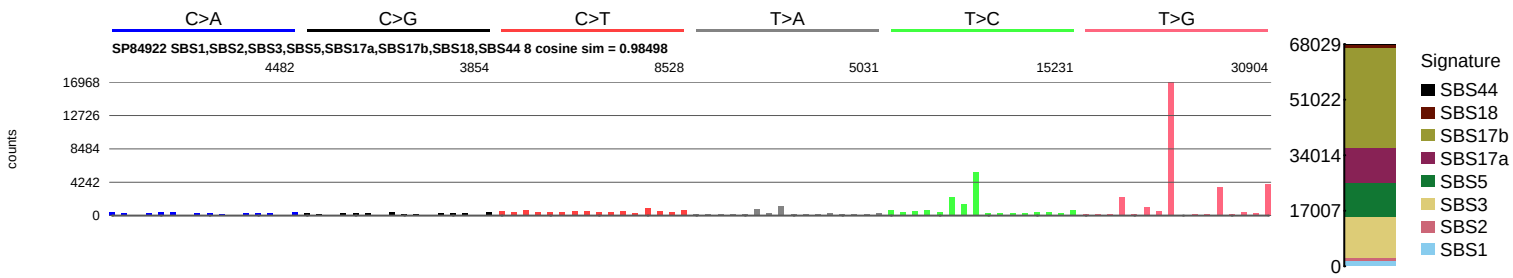

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS44 9 cosine sim = 0.98508
3879
3909
8356
5036
15960
30893
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

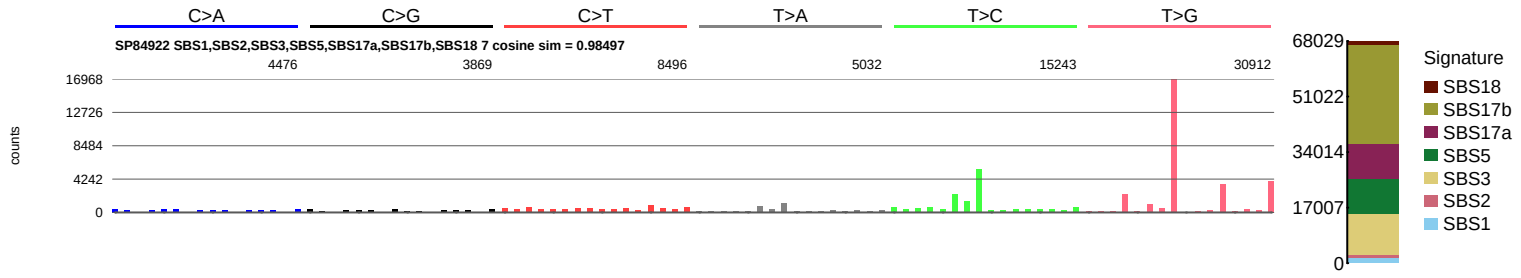

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98505
3988
4084
7569
5140
16349
30903
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

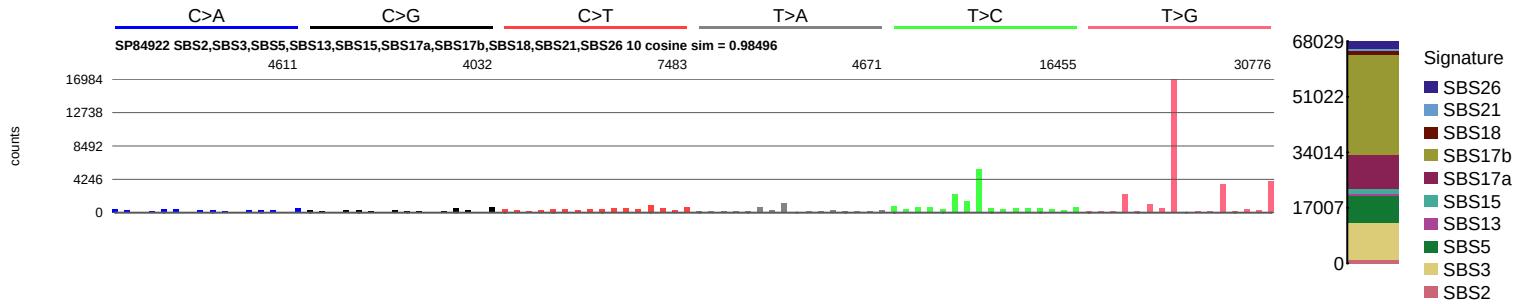

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98505
3986
4085
7561
5147
16353
30901
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1

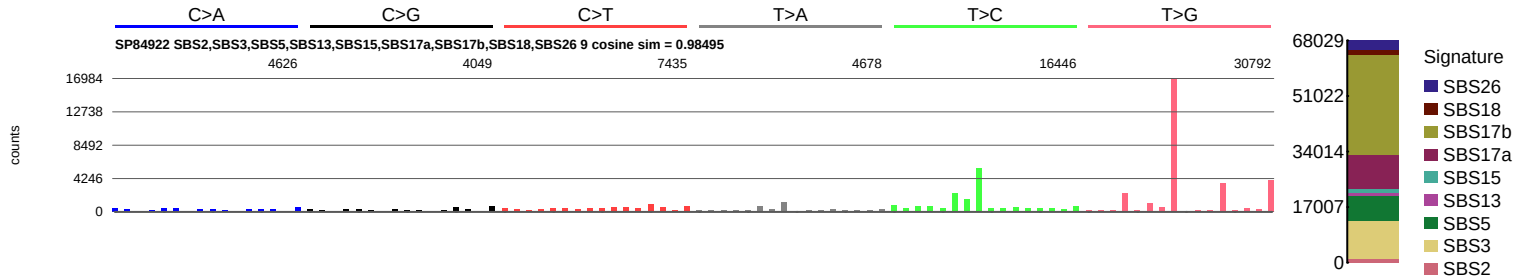

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98503
3978
4164
7420
5154
16378
30937
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

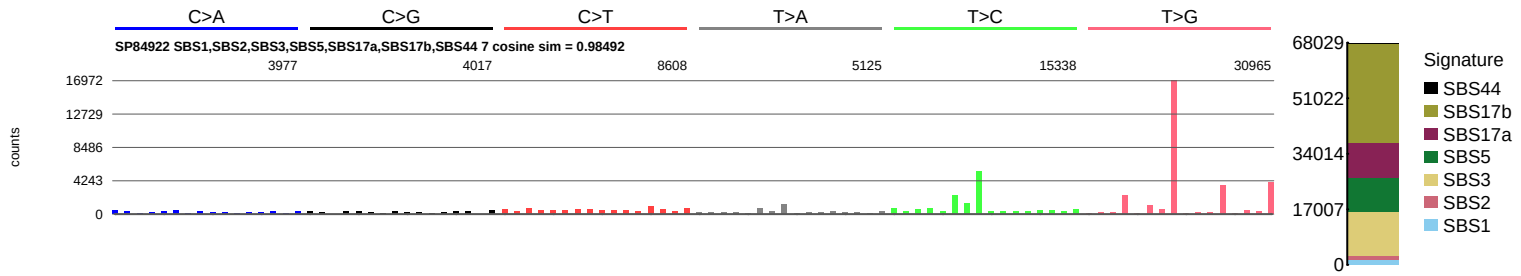

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98501
3968
4219
7297
5188
16396
30956
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1

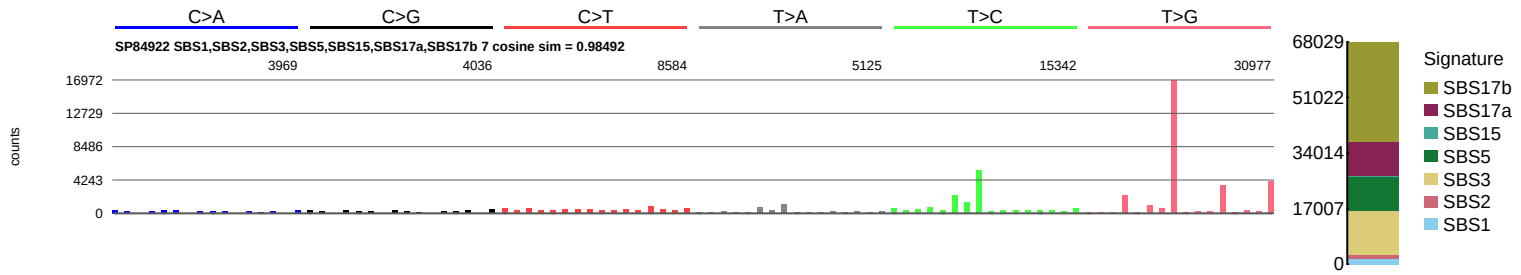

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98498
4482
3854
8528
5031
15231
30904
counts
16968
12726
8484
4242
0
68029
51022
34014
17007
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

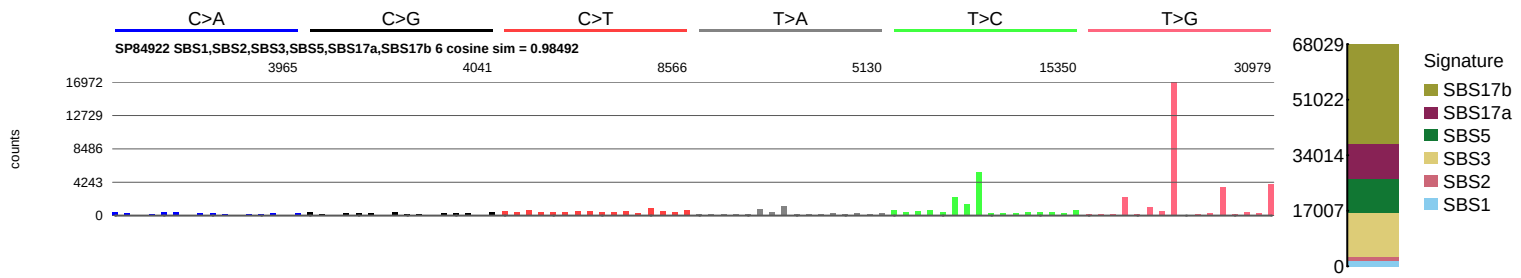

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98497
4476
3869
8496
5032
15243
30912
16968
12726
8484
4242
0
counts
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

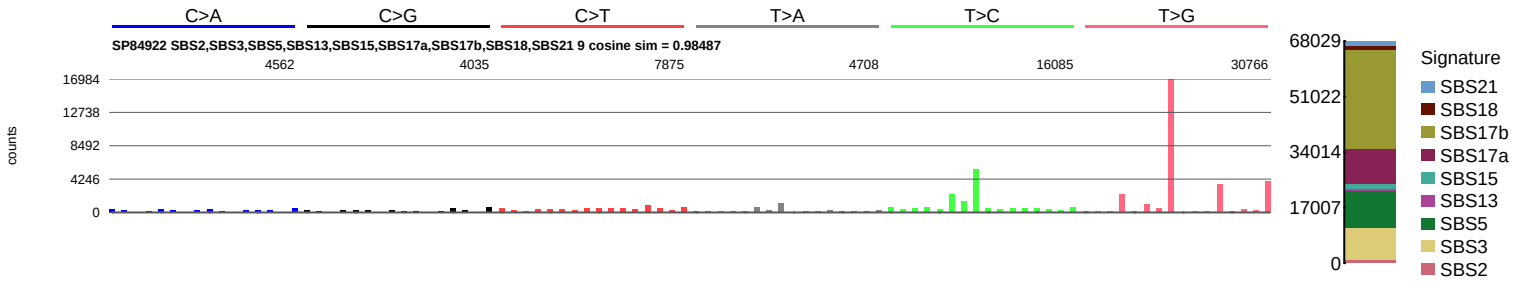

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 10 cosine sim = 0.98496
4611
4032
7483
4671
16455
30776
16984
12738
8492
4246
0
counts
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

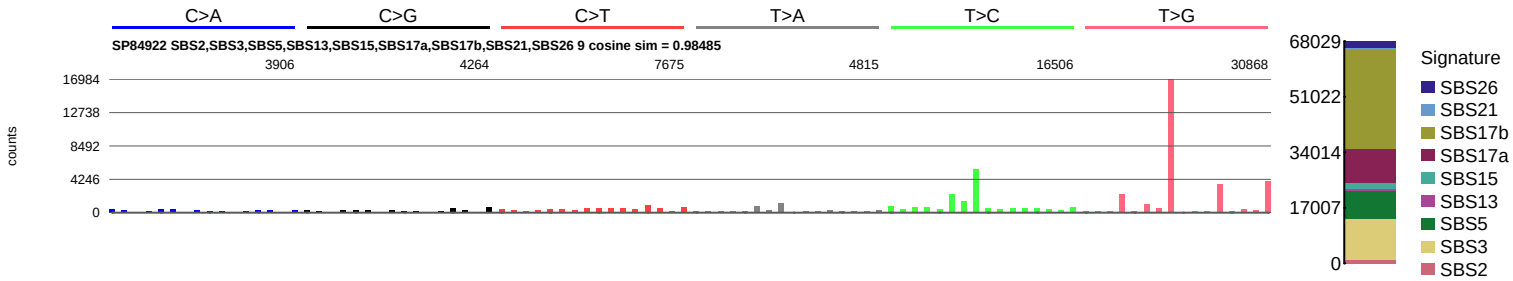

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98495
4626
4049
7435
4678
16446
30792
16984
12738
8492
4246
0
counts
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

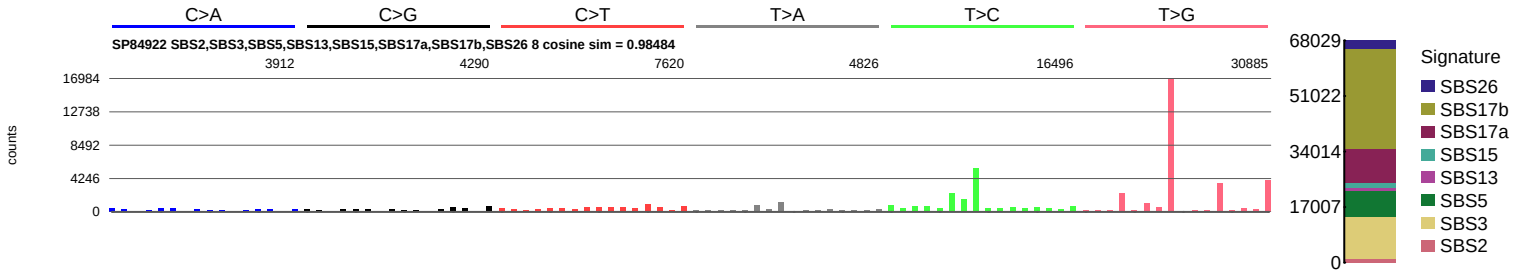

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98492
3977
4017
8608
5125
15338
30965
16972
12729
8486
4243
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

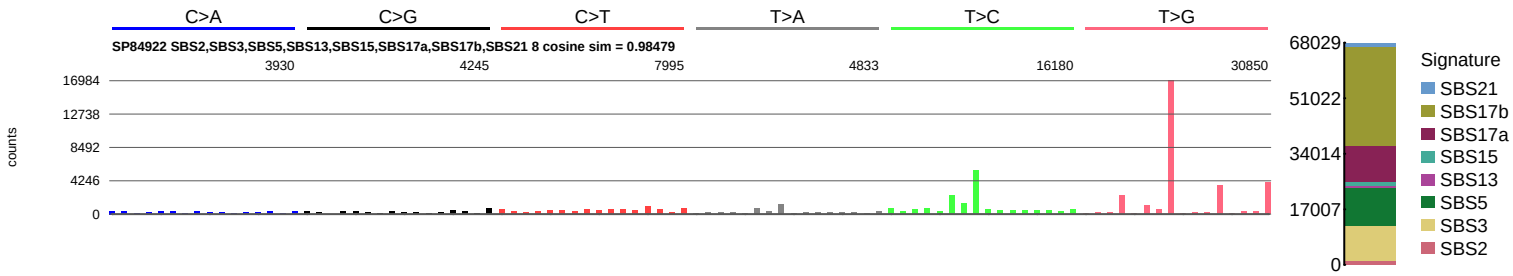

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98492
3969
4036
8584
5125
15342
30977
16972
12729
8486
4243
0
counts
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b 6 cosine sim = 0.98492
3965
4041
8566
5130
15350
30979
16972
12729
8486
4243
0
counts
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98487
4562
4035
7875
4708
16085
30766
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26 9 cosine sim = 0.98485
3906
4264
7675
4815
16506
30868
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26 8 cosine sim = 0.98484
3912
4290
7620
4826
16496
30885
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2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C>G
C>T
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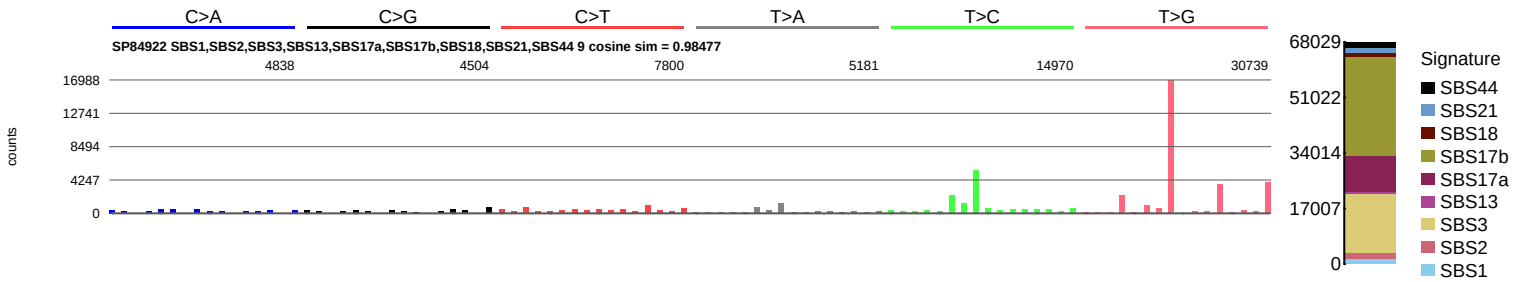
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98479
3930
4245
7995
4833
16180
30850
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2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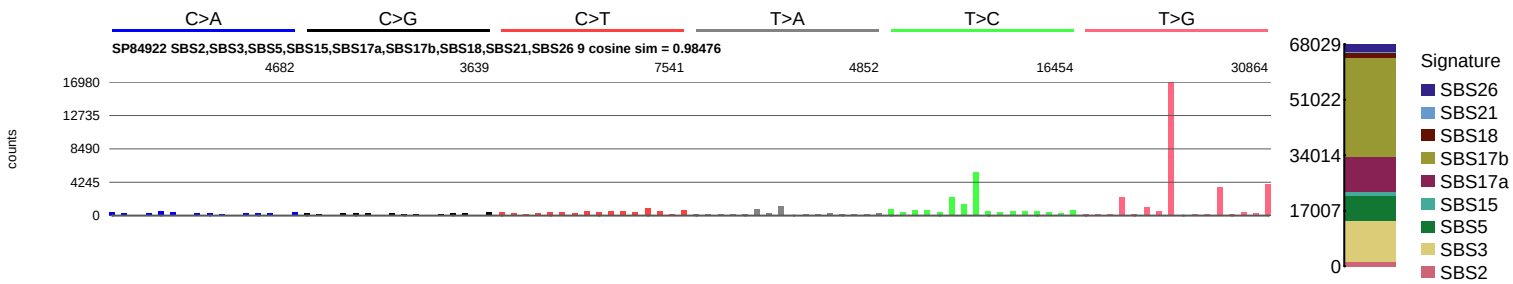
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98477
4838
4504
7800
5181
14970
30739
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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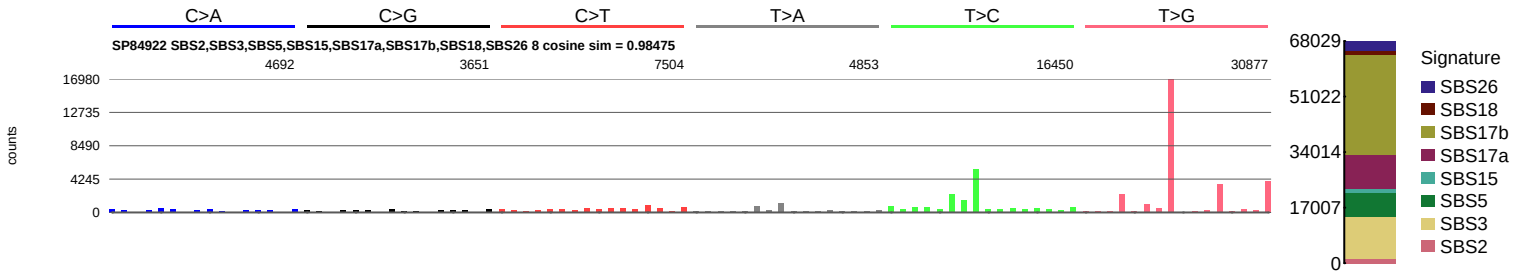
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98476
4682
3639
7541
4852
16454
30864
counts
16980
12735
8490
4245
0
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2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C>G
C>T
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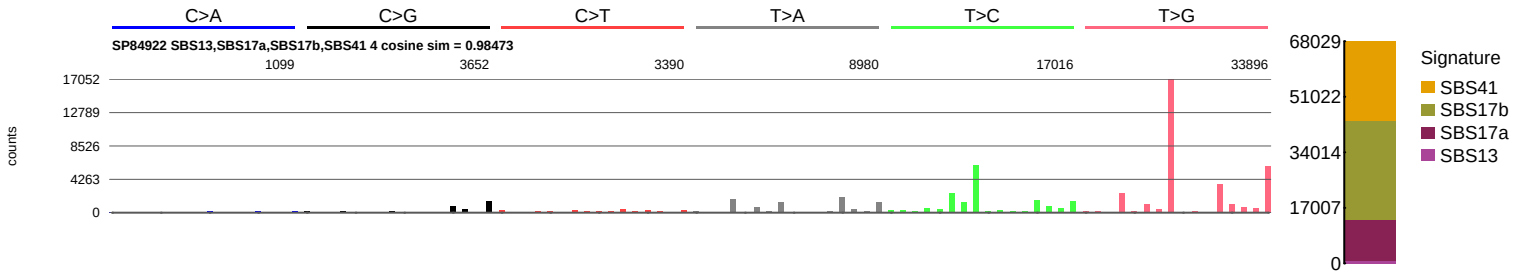
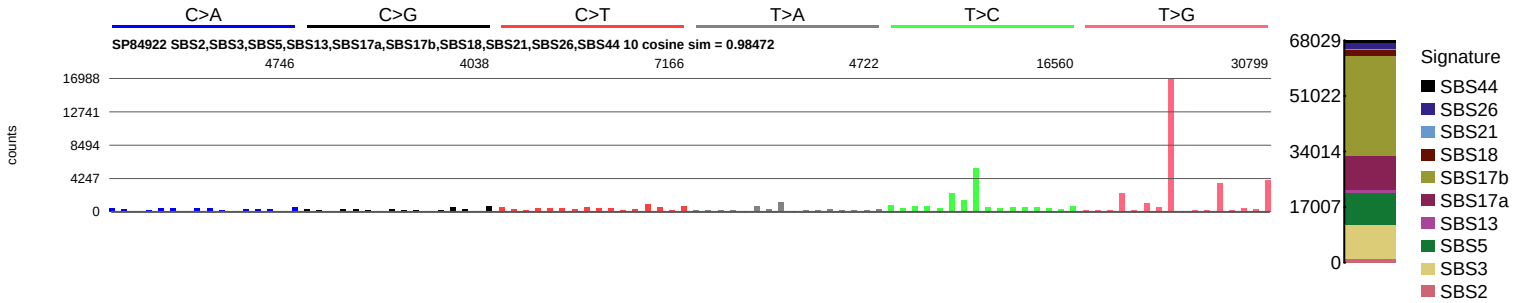
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98475
4692
3651
7504
4853
16450
30877
counts
16980
12735
8490
4245
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS13,SBS17a,SBS17b,SBS41 4 cosine sim = 0.98473
1099
3652
3390
8980
17016
33896
counts
17052
12789
8526
4263
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a
SBS13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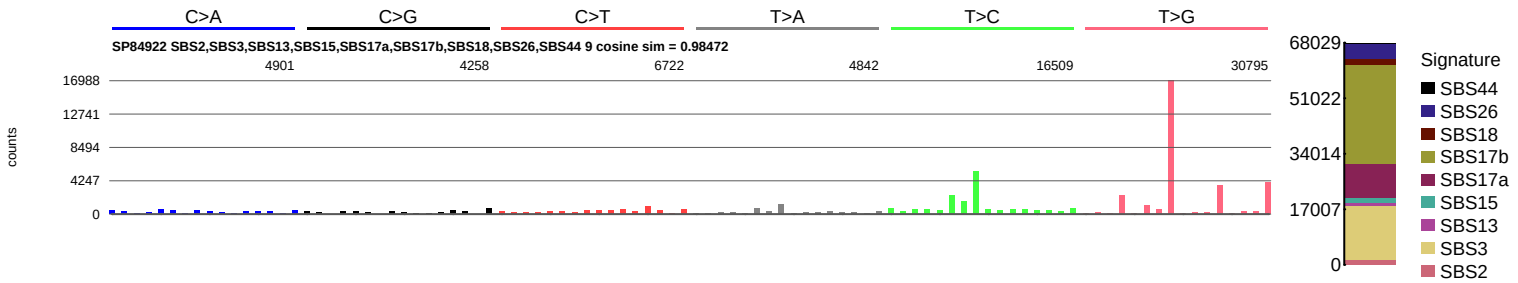
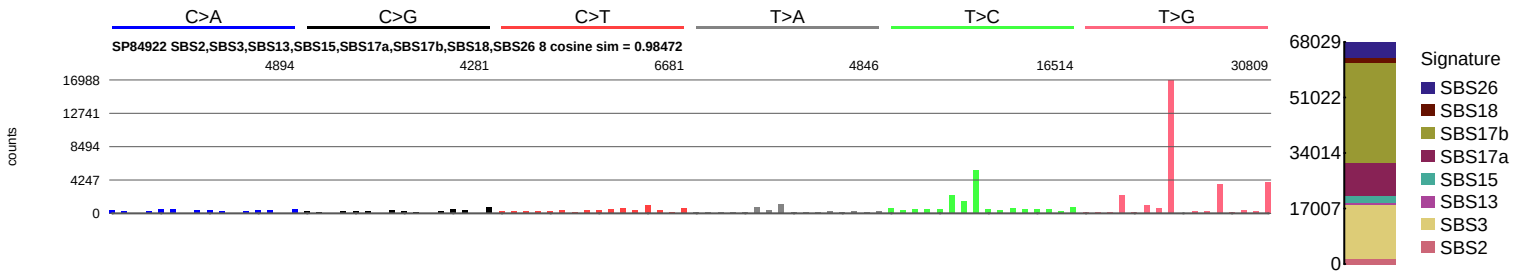
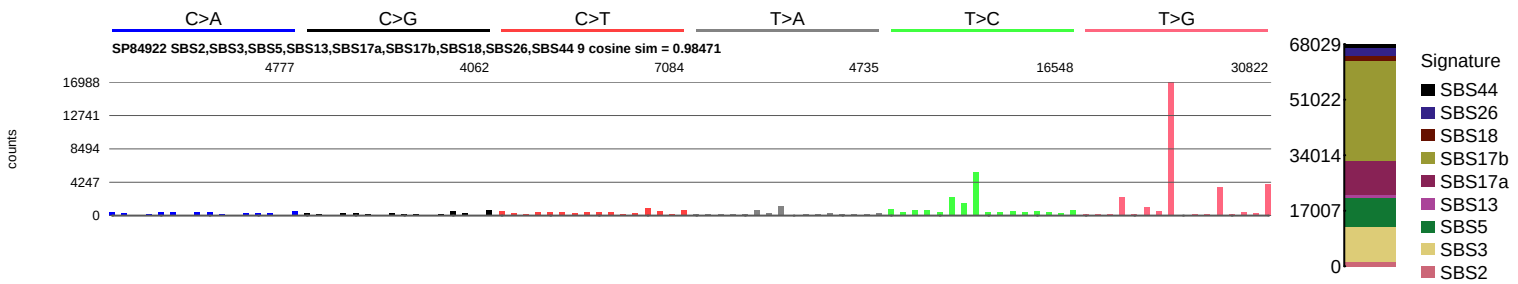
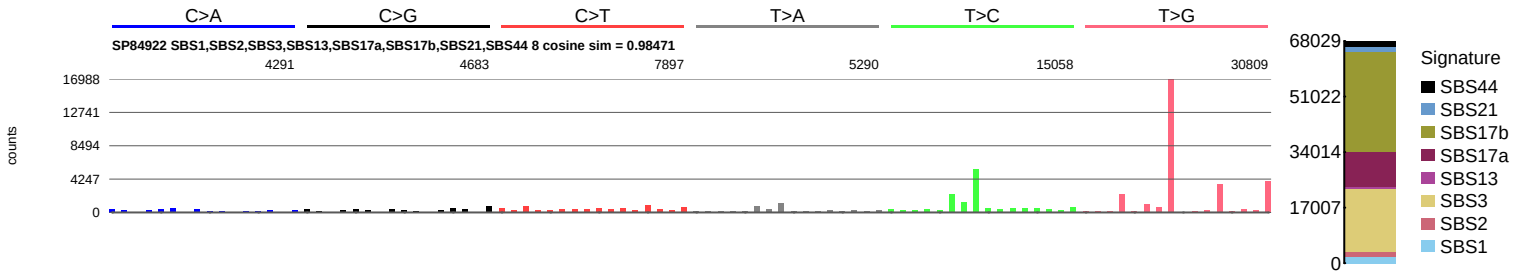
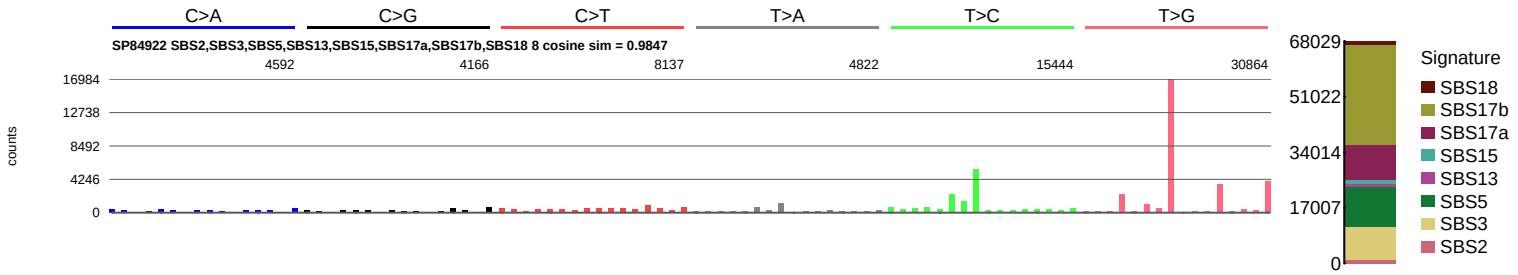
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.98472
4746
4038
7166
4722
16560
30799
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

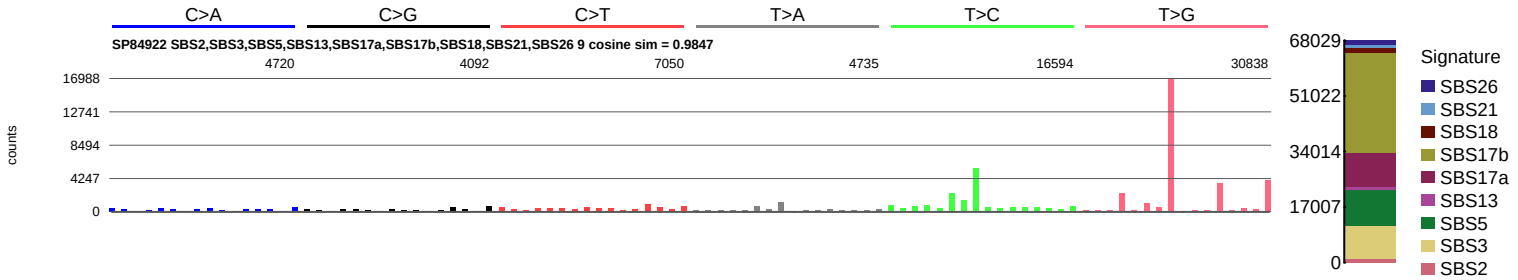

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98472
4901
4258
6722
4842
16509
30795
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

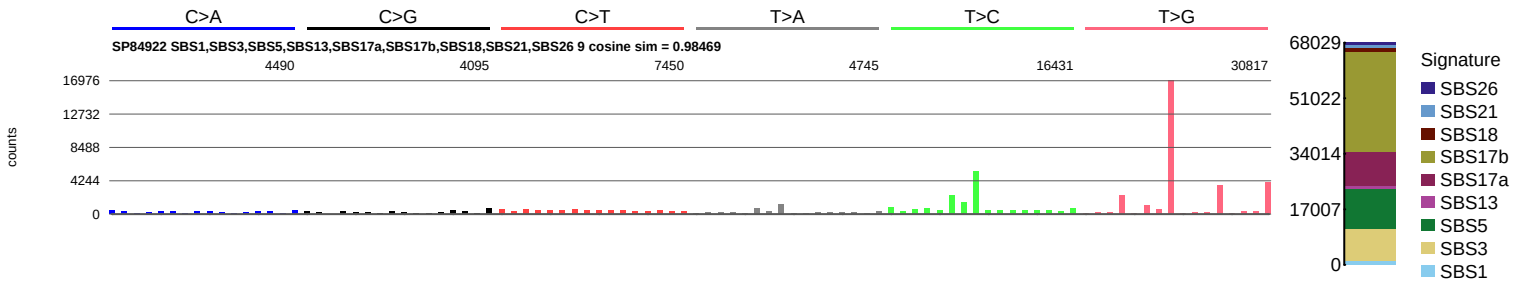

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98472
4894
4281
6681
4846
16514
30809
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

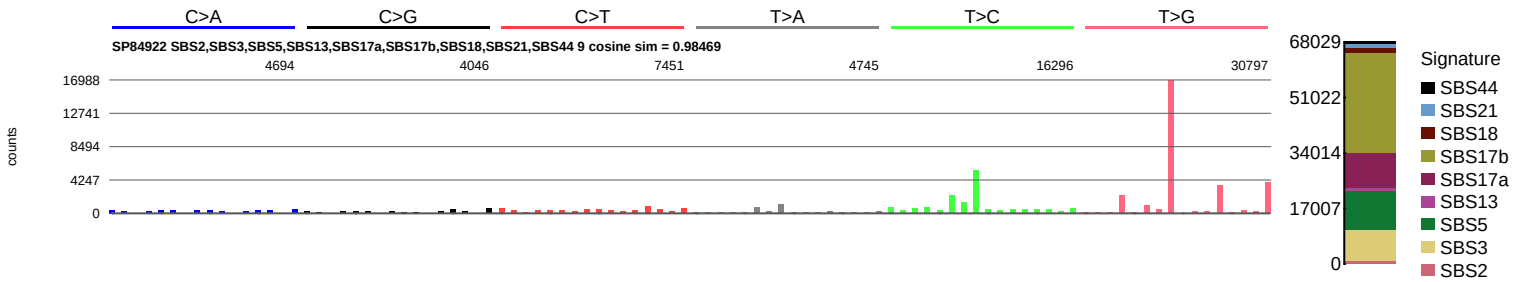

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98471
4777
4062
7084
4735
16548
30822
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

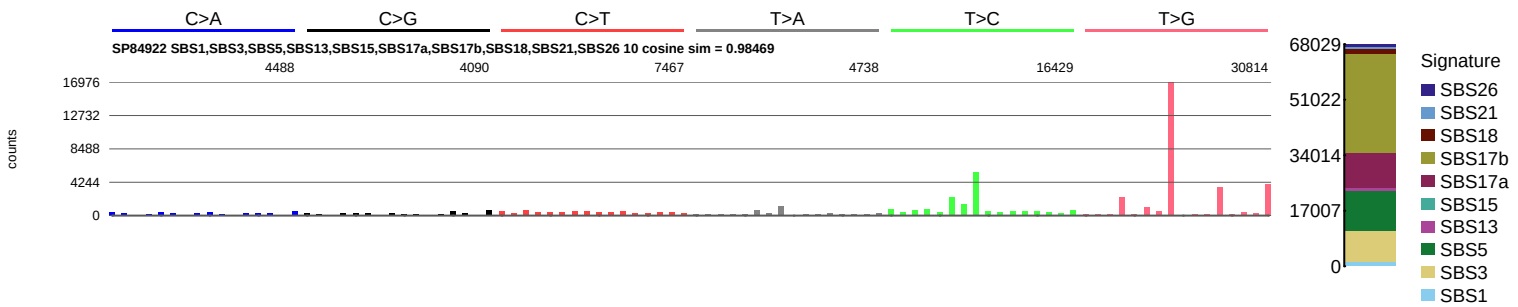

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98471
4291
4683
7897
5290
15058
30809
16988
12741
8494
4247
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

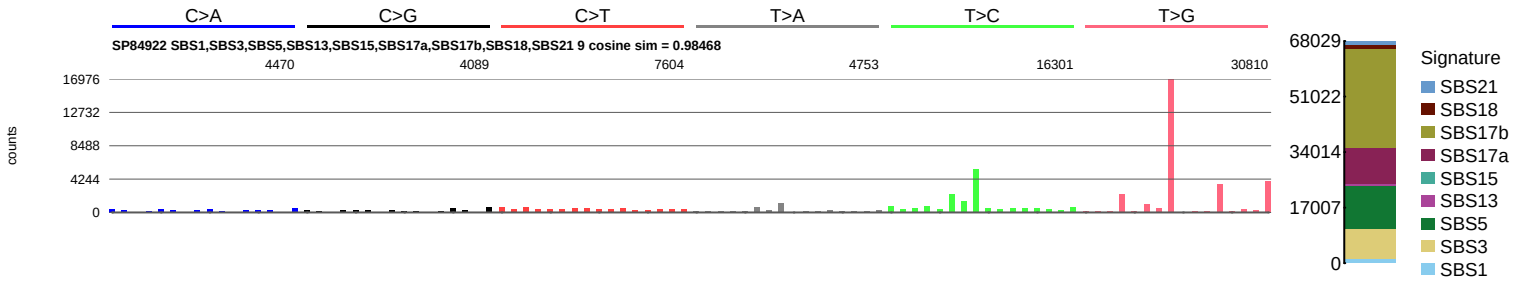

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.9847
4592
4166
8137
4822
15444
30864
16984
12738
8492
4246
0
counts
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

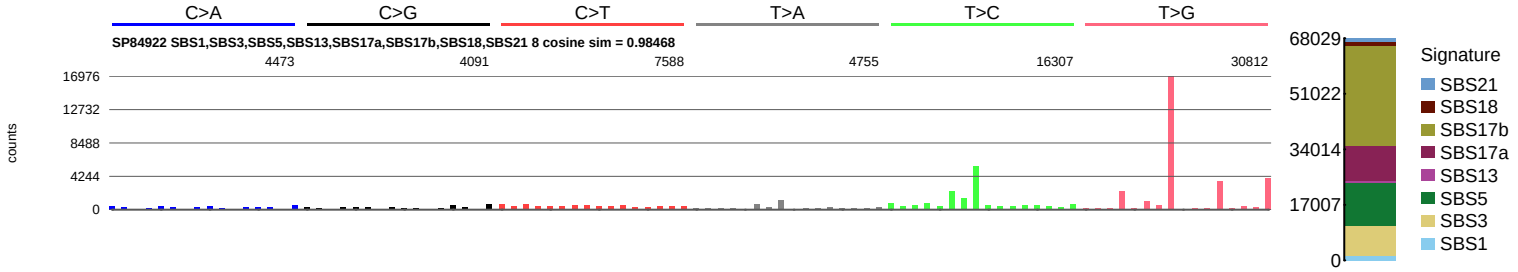

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.9847
4720
4092
7050
4735
16594
30838
16988
12741
8494
4247
0
counts
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

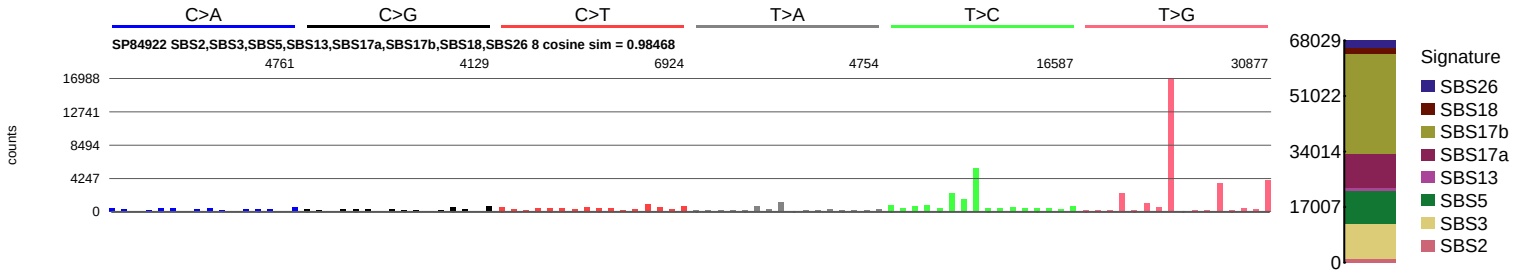

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98469
4490
4095
7450
4745
16431
30817
16976
12732
8488
4244
0
counts
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

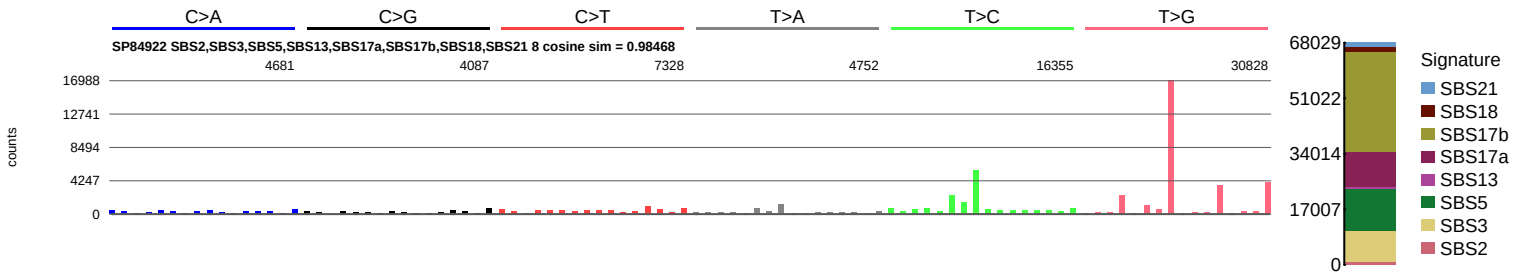

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98469
4694
4046
7451
4745
16296
30797
16988
12741
8494
4247
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 10 cosine sim = 0.98469
4488
4090
7467
4738
16429
30814
16976
12732
8488
4244
0
counts
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98468
4470
4089
7604
4753
16301
30810
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98468
4473
4091
7588
4755
16307
30812
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98468
4761
4129
6924
4754
16587
30877
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98468
4681
4087
7328
4752
16355
30828
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
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Signature
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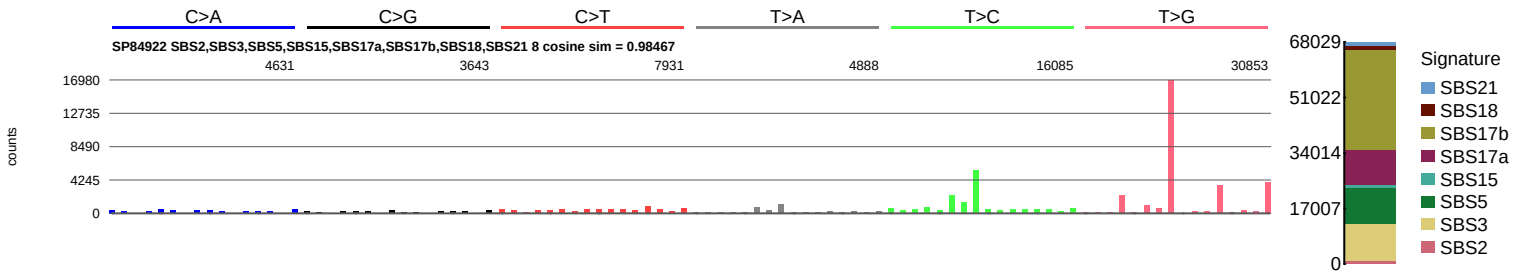
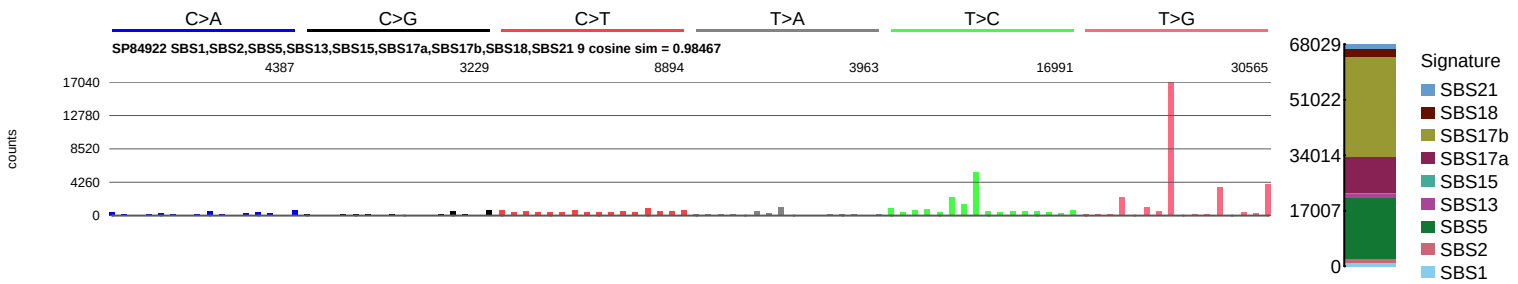
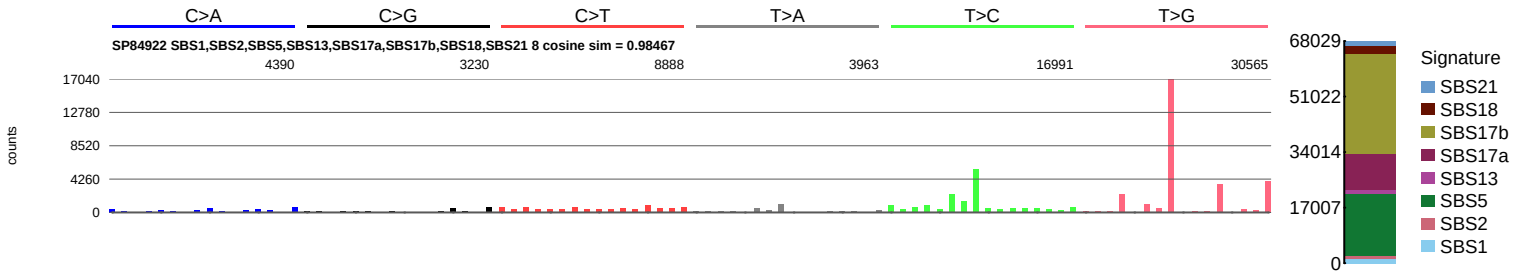
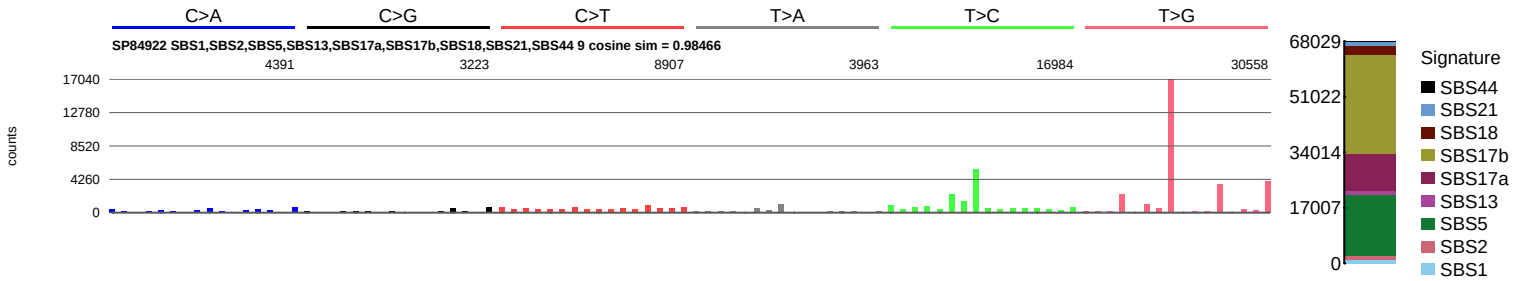
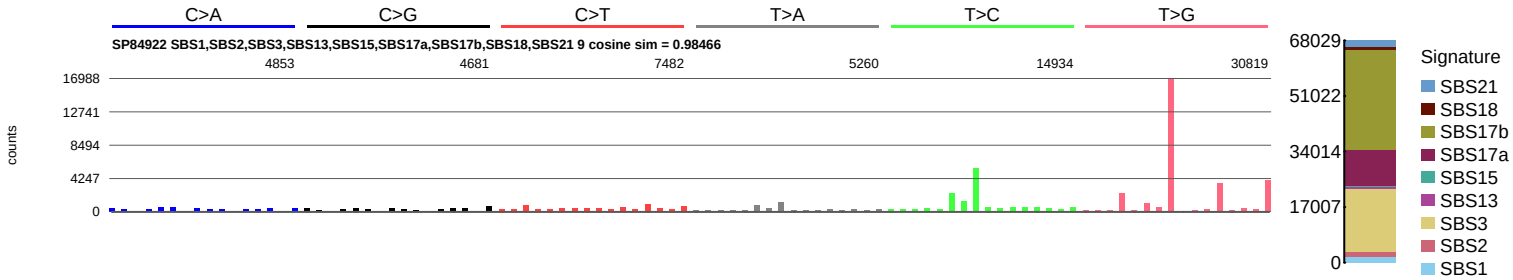
SBS3
SBS2

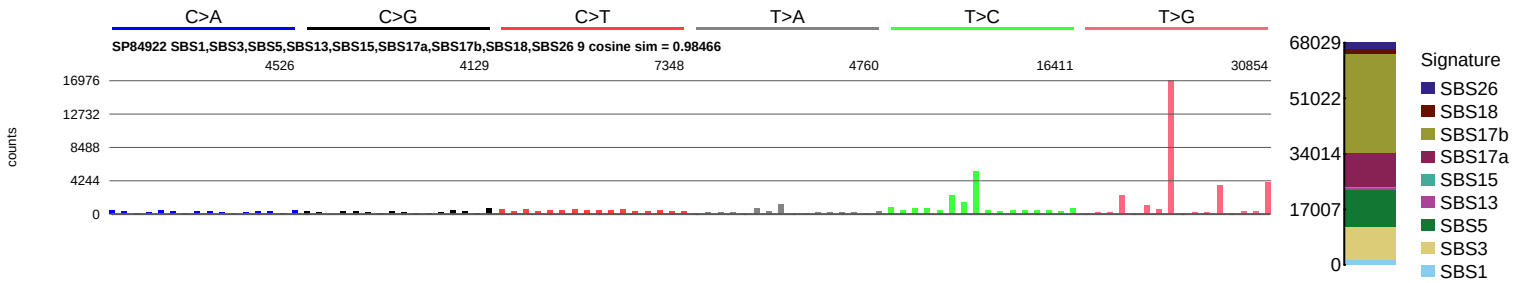

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98467
4631
3643
7931
4888
16085
30853
counts
16980
12735
8490
4245
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98467
4387
3229
8894
3963
16991
30565
counts
17040
12780
8520
4260
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98467
4390
3230
8888
3963
16991
30565
counts
17040
12780
8520
4260
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98466
4391
3223
8907
3963
16984
30558
counts
17040
12780
8520
4260
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98466
4853
4681
7482
5260
14934
30819
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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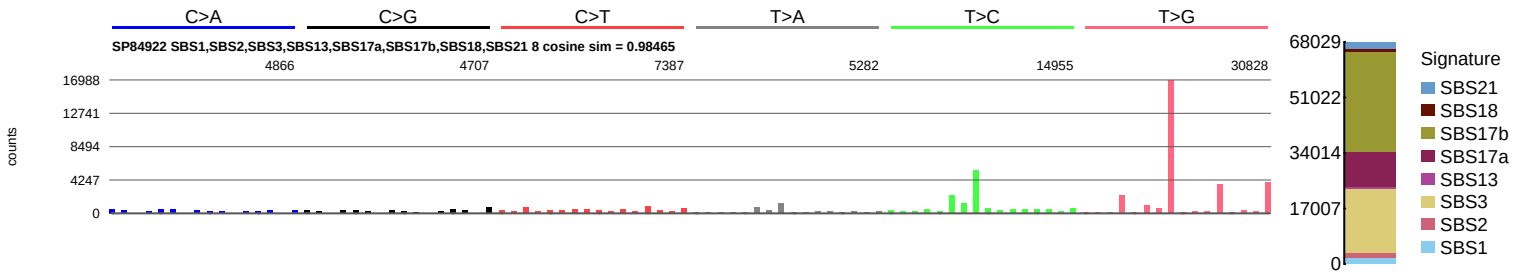
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98466
4526
4129
7348
4760
16411
30854
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1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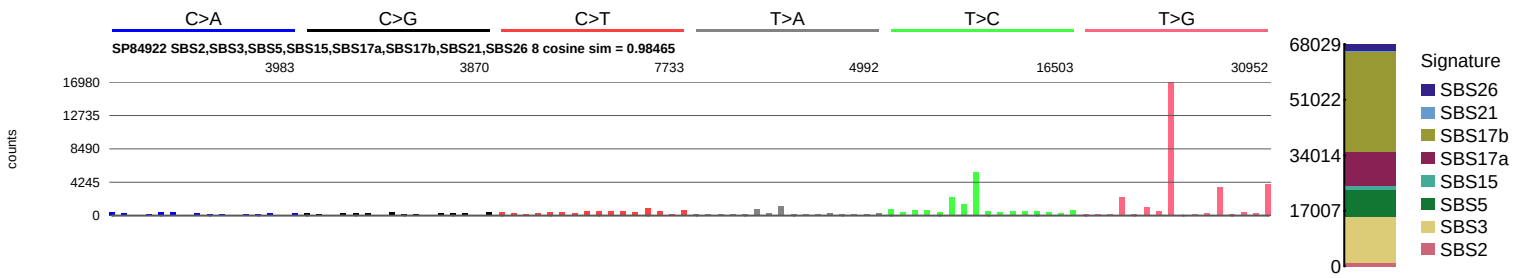
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98465
4866
4707
7387
5282
14955
30828
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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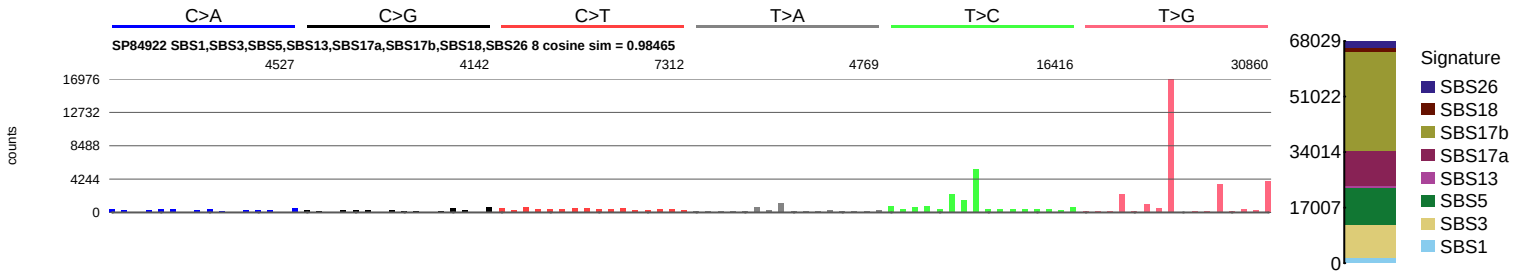
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS26 8 cosine sim = 0.98465
3983
3870
7733
4992
16503
30952
counts
16980
12735
8490
4245
0
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2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C>G
C>T
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T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98465
4527
4142
7312
4769
16416
30860
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
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Signature
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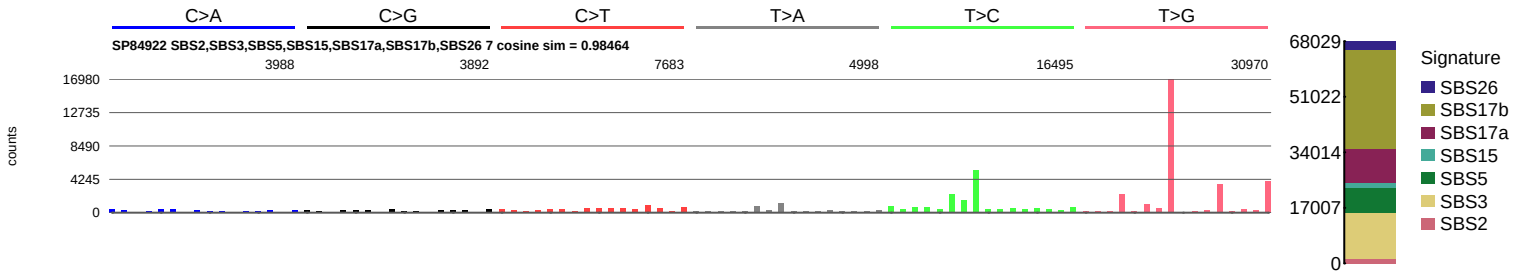
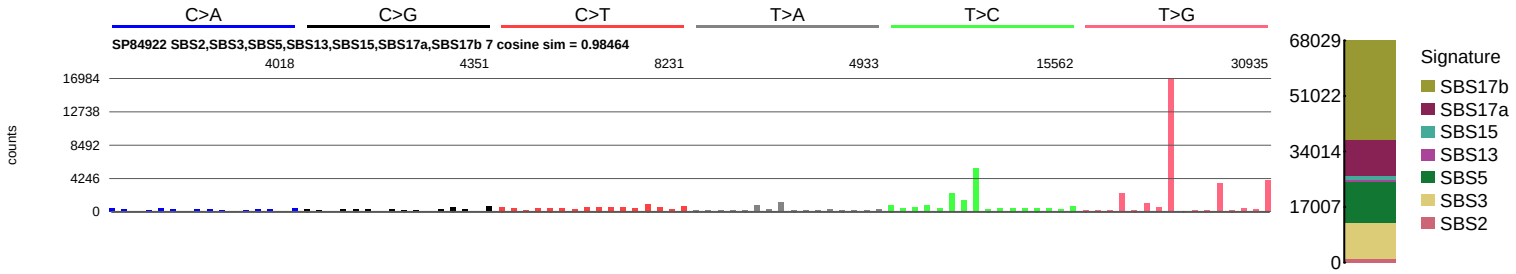
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98464
3988
3892
7683
4998
16495
30970
counts
16980
12735
8490
4245
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2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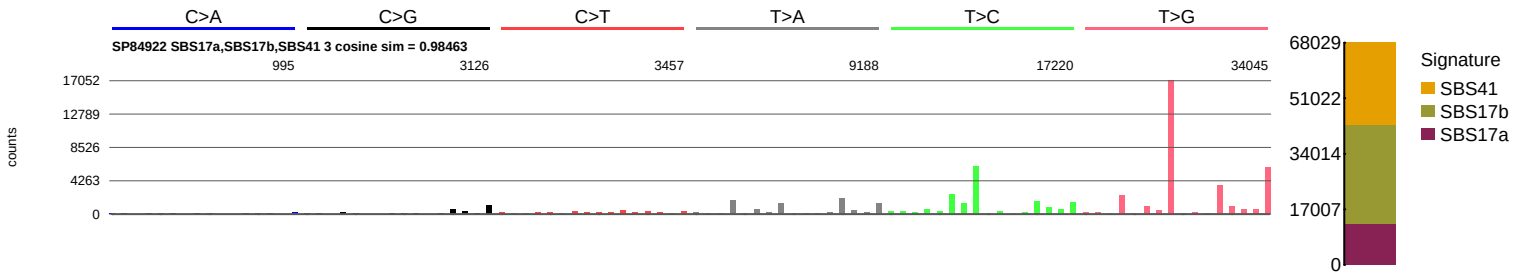
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98464
4018
4351
8231
4933
15562
30935
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2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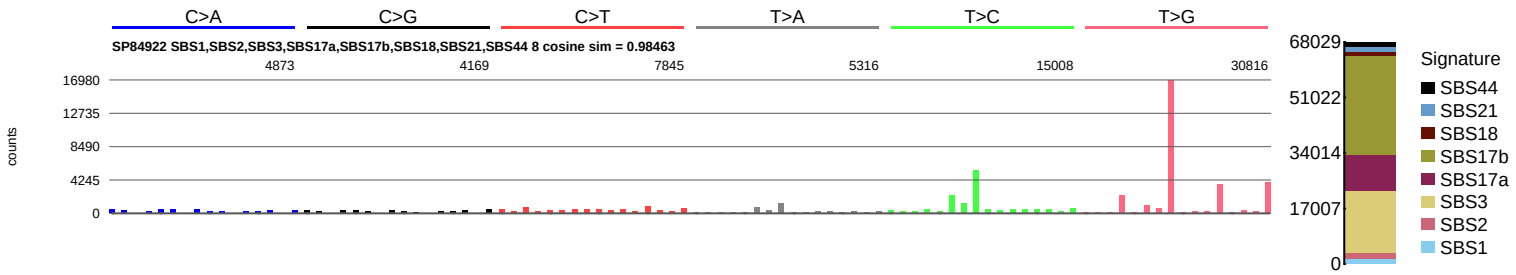
C>G
C>T
T>A
T>C
T>G
SP84922 SBS17a,SBS17b,SBS41 3 cosine sim = 0.98463
995
3126
3457
9188
17220
34045
counts
17052
12789
8526
4263
0
68029
51022
34014
17007
0
Signature
SBS41
SBS17b
SBS17a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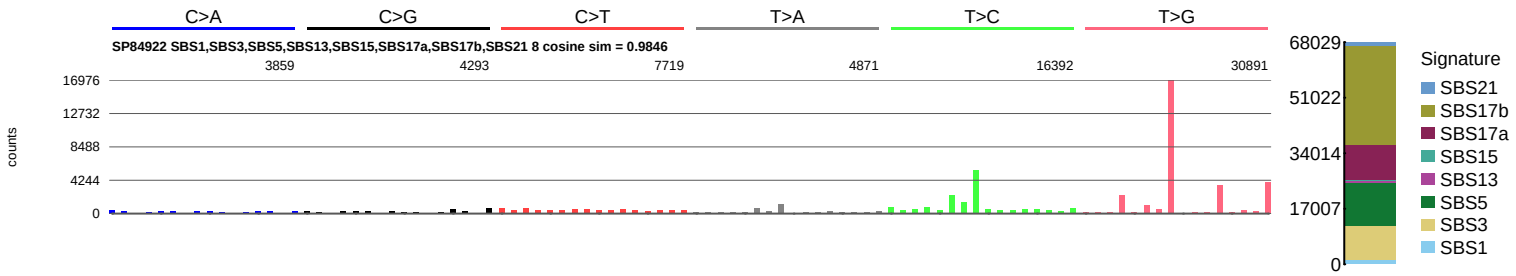
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98463
4873
4169
7845
5316
15008
30816
counts
16980
12735
8490
4245
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.9846
3859
4293
7719
4871
16392
30891
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
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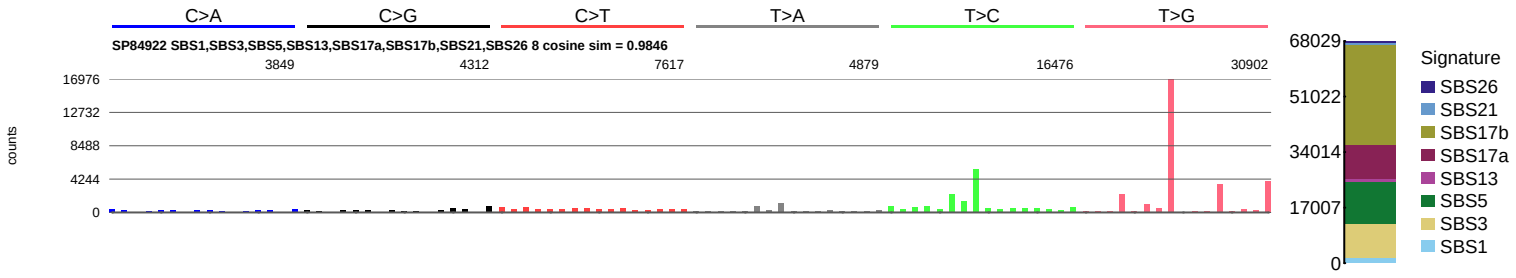
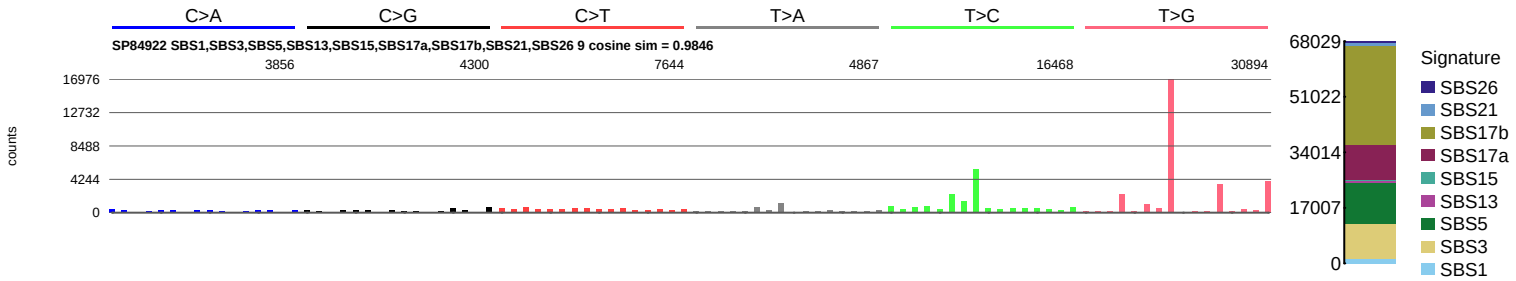
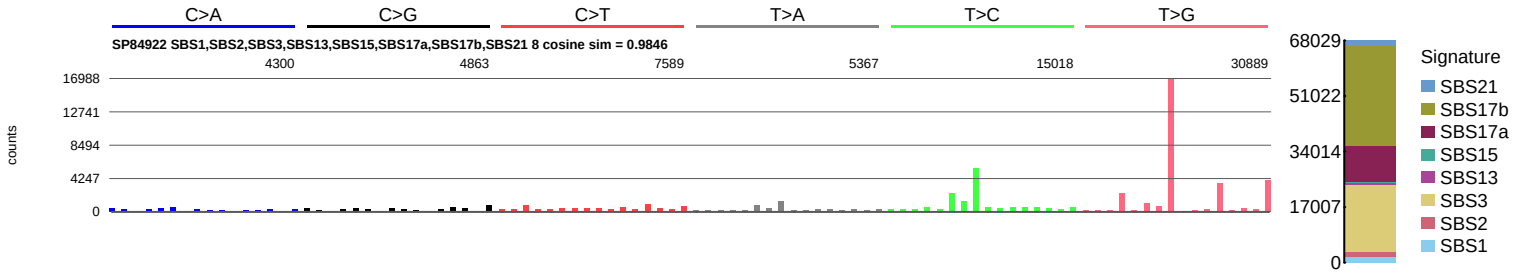
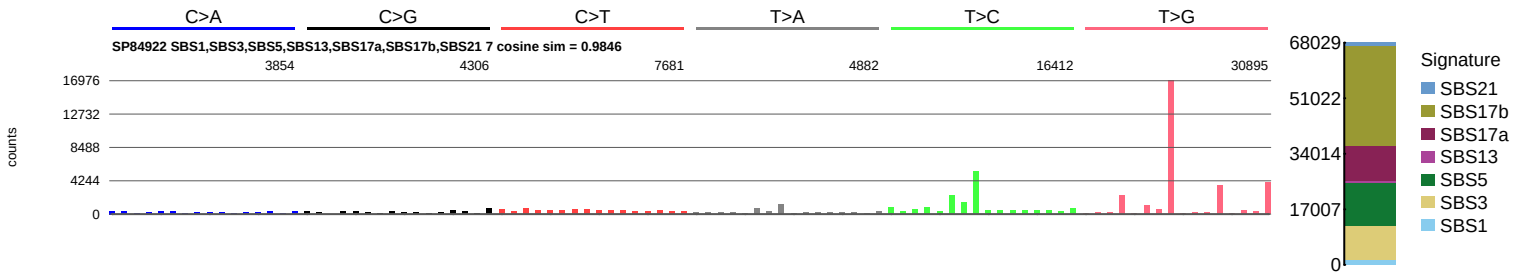
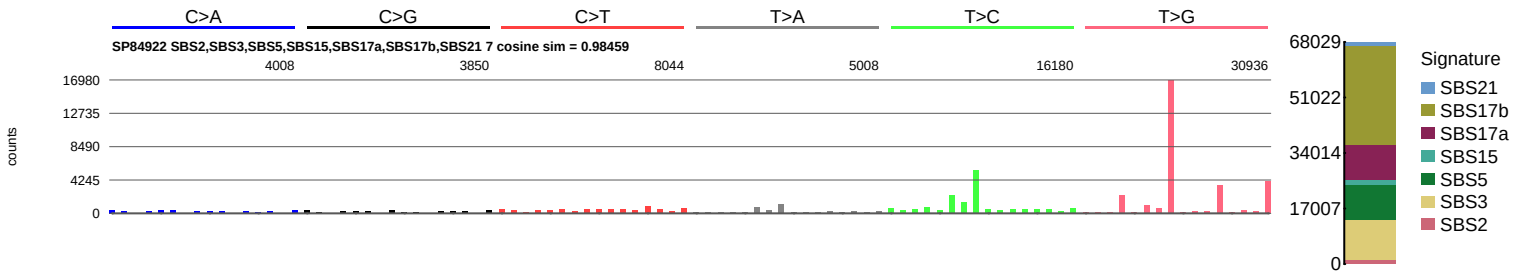
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

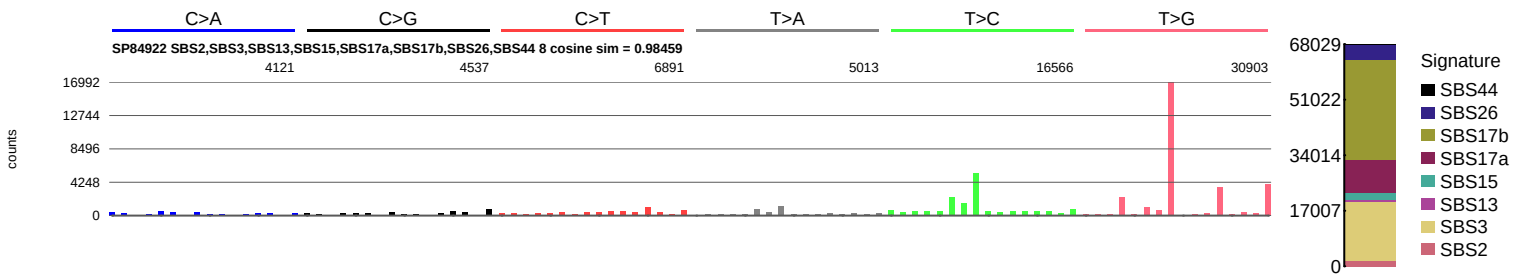

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS26 8 cosine sim = 0.9846
3849
4312
7617
4879
16476
30902
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

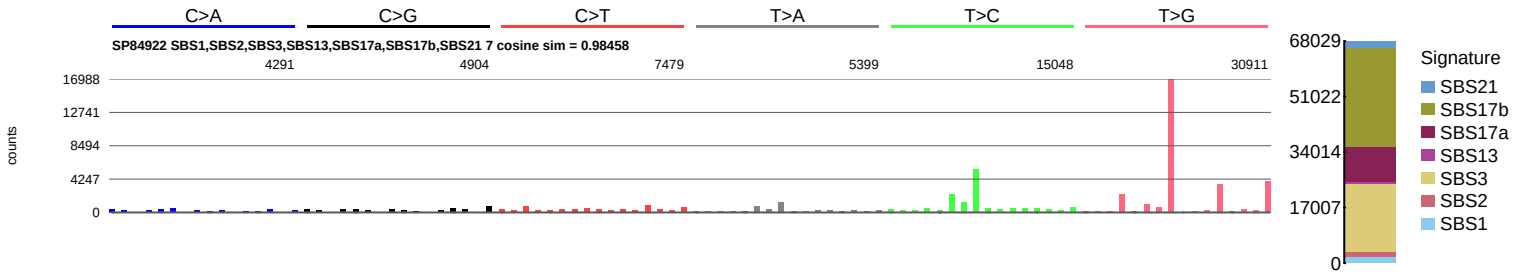

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26 9 cosine sim = 0.9846
3856
4300
7644
4867
16468
30894
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

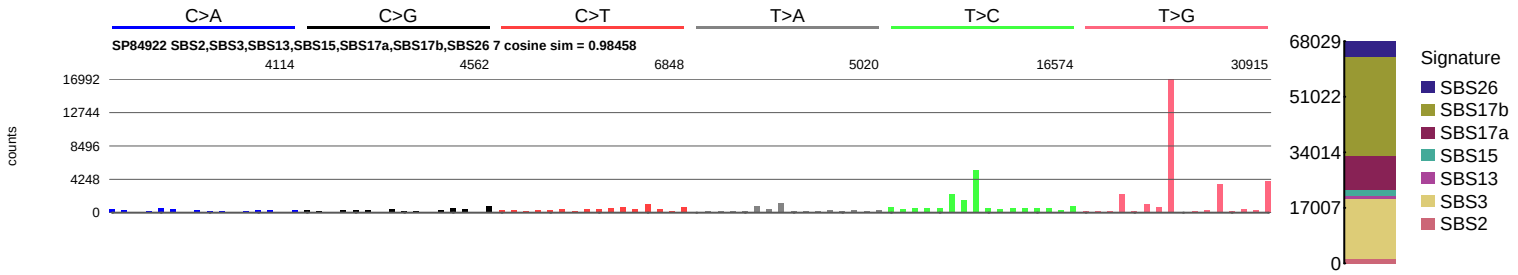

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.9846
4300
4863
7589
5367
15018
30889
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

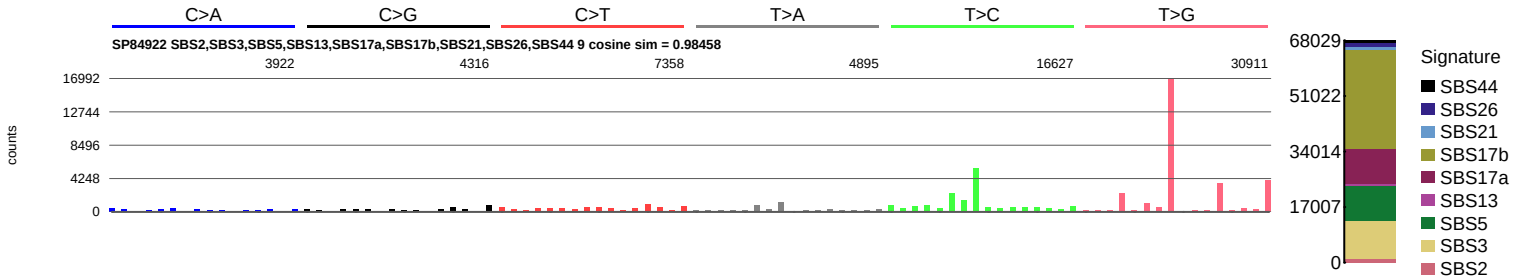

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21 7 cosine sim = 0.9846
3854
4306
7681
4882
16412
30895
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

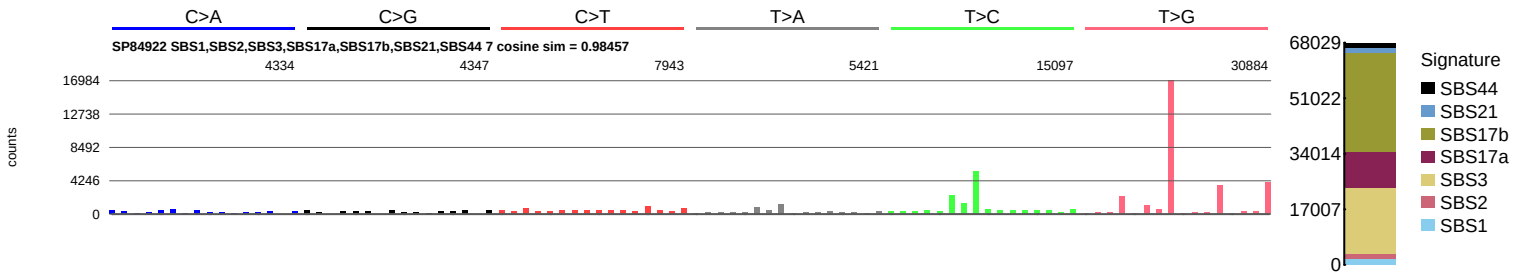

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98459
4008
3850
8044
5008
16180
30936
counts
16980
12735
8490
4245
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

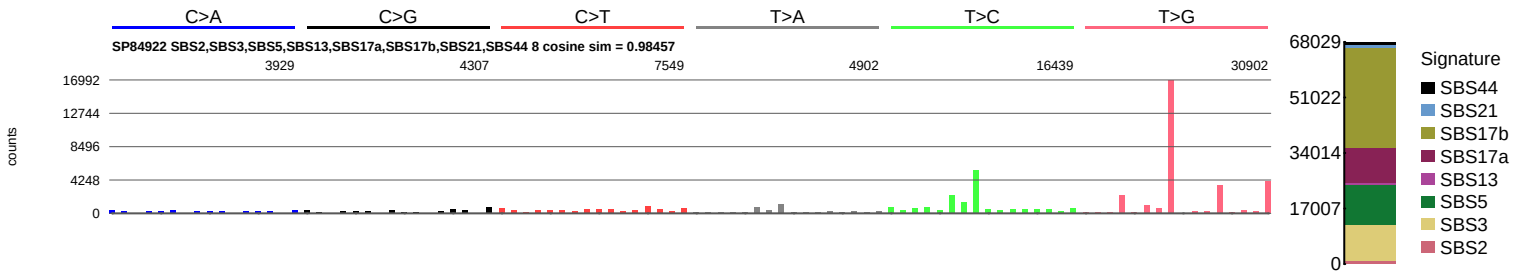

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98459
4121
4537
6891
5013
16566
30903
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

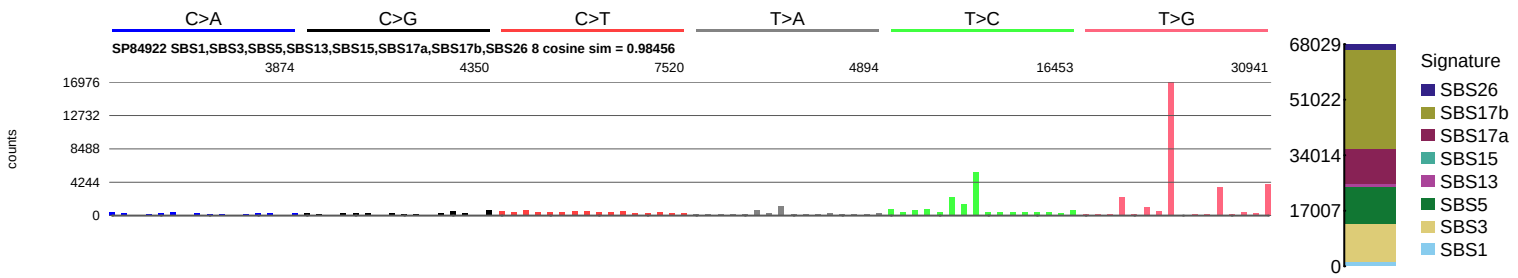

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98458
4291
4904
7479
5399
15048
30911
16988
12741
8494
4247
0
counts
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

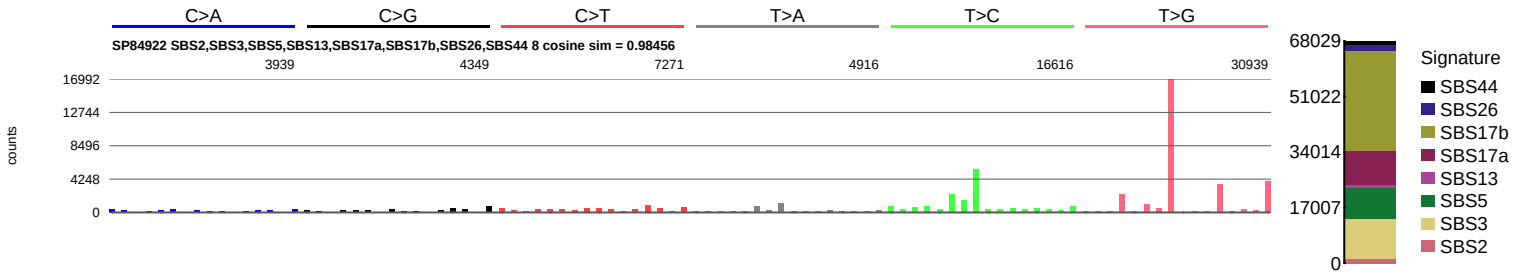

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98458
4114
4562
6848
5020
16574
30915
16992
12744
8496
4248
0
counts
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

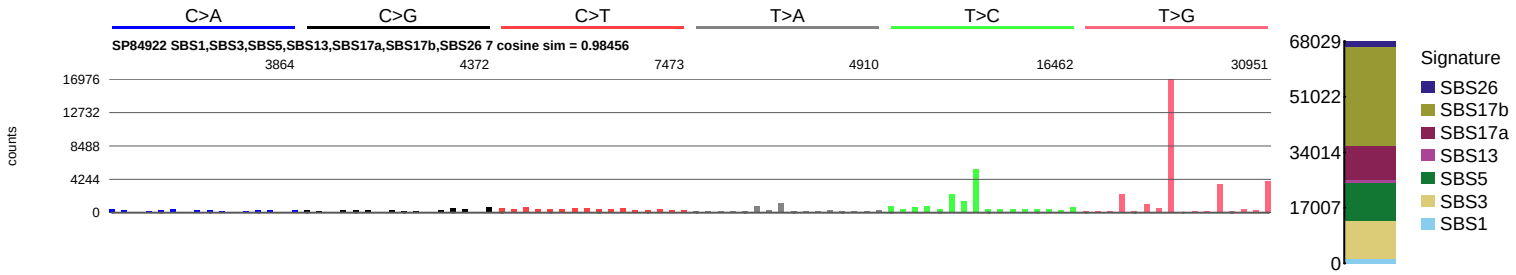

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS44 9 cosine sim = 0.98458
3922
4316
7358
4895
16627
30911
16992
12744
8496
4248
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

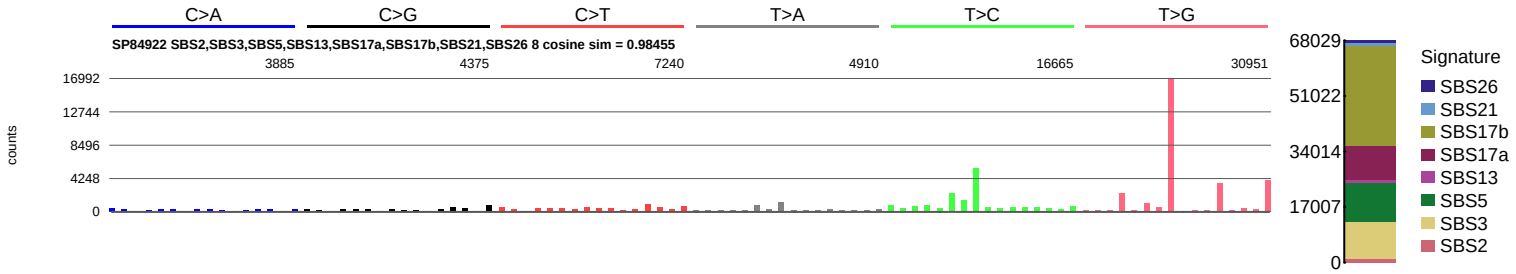

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98457
4334
4347
7943
5421
15097
30884
16984
12738
8492
4246
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98457
3929
4307
7549
4902
16439
30902
16992
12744
8496
4248
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26 8 cosine sim = 0.98456
3874
4350
7520
4894
16453
30941
16976
12732
8488
4244
0
counts
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98456
3939
4349
7271
4916
16616
30939
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98456
3864
4372
7473
4910
16462
30951
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS26 8 cosine sim = 0.98455
3885
4375
7240
4910
16665
30951
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS17b
SBS17a
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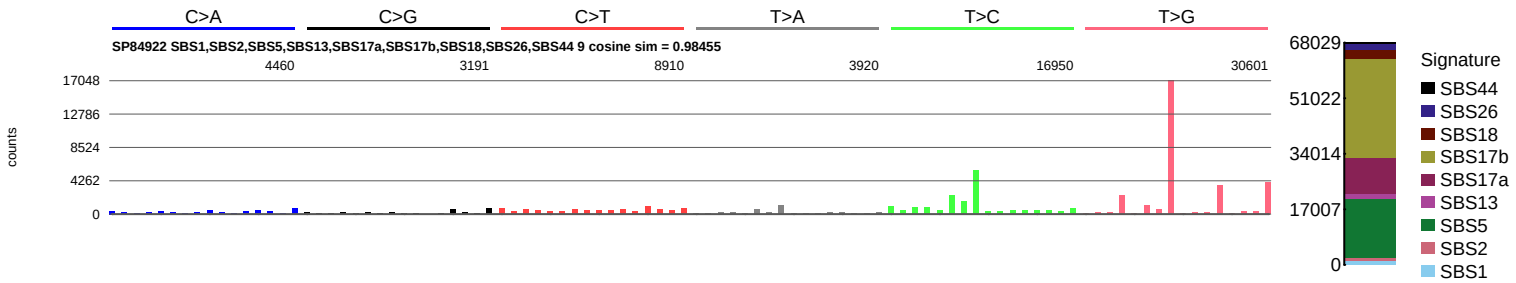
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98455
4460
3191
8910
3920
16950
30601
counts
17048
12786
8524
4262
0
68029
51022
34014
17007
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Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
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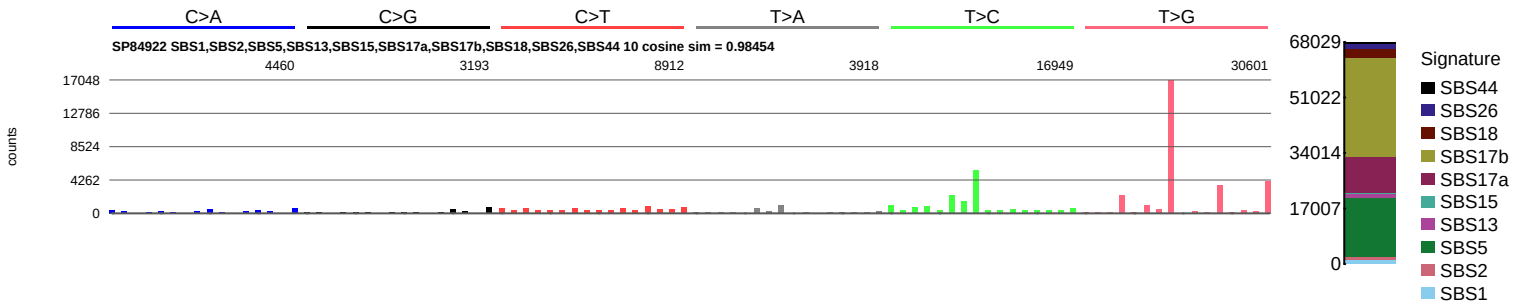
SBS2
SBS1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T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98454
4460
3193
8912
3918
16949
30601
counts
17048
12786
8524
4262
0
68029
51022
34014
17007
0
Signature
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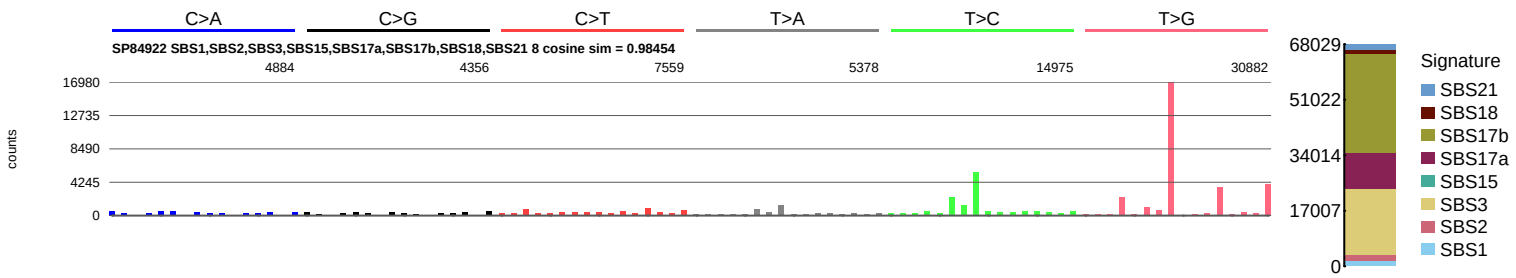
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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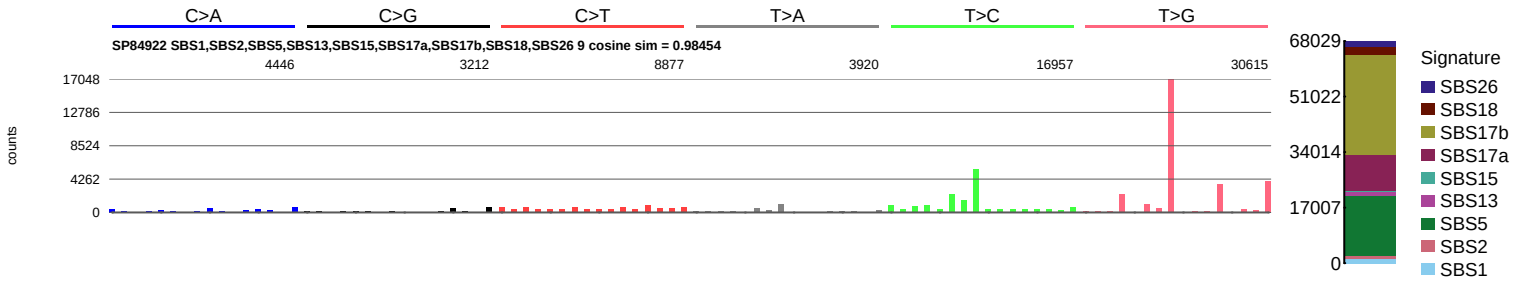
SP84922 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98454
4884
4356
7559
5378
14975
30882
counts
16980
12735
8490
4245
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98454
4446
3212
8877
3920
16957
30615
counts
17048
12786
8524
4262
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
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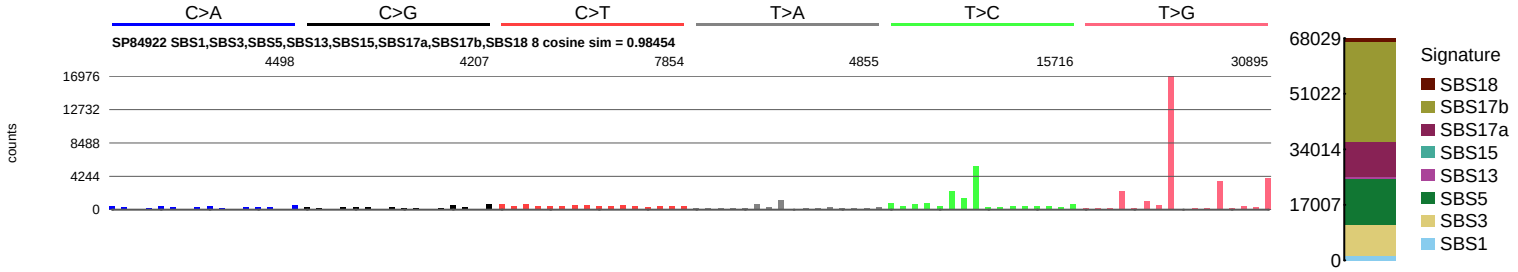
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98454
4498
4207
7854
4855
15716
30895
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS18
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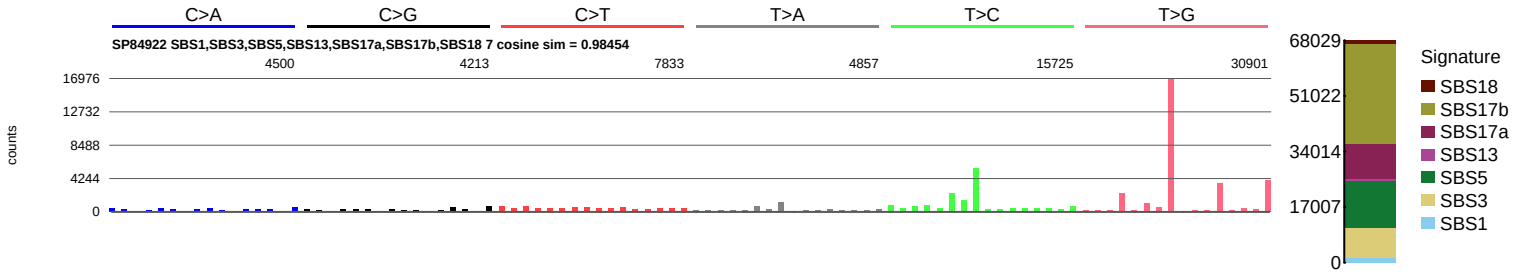
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98454
4500
4213
7833
4857
15725
30901
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
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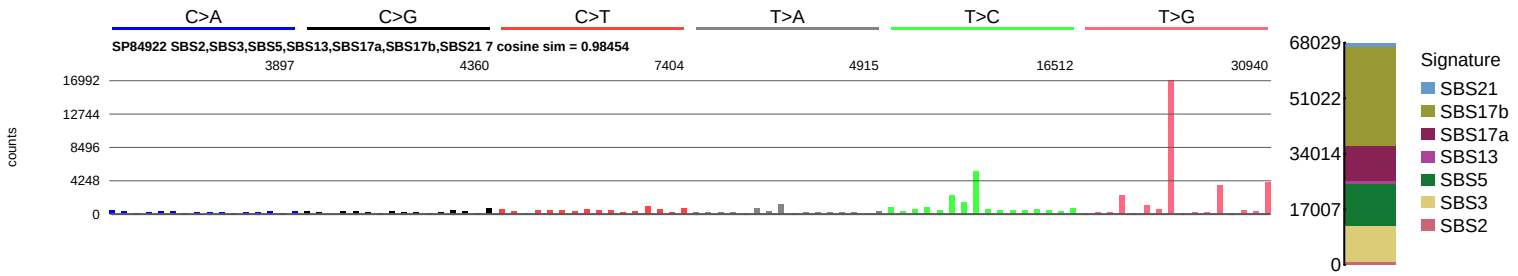
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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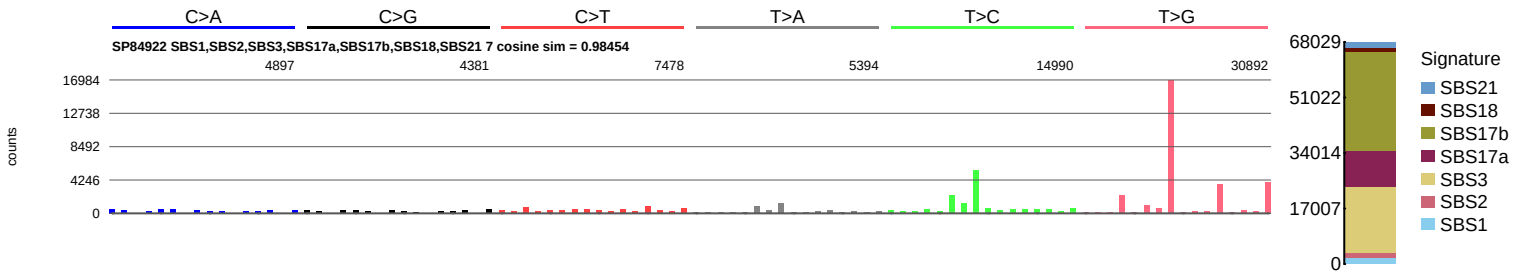
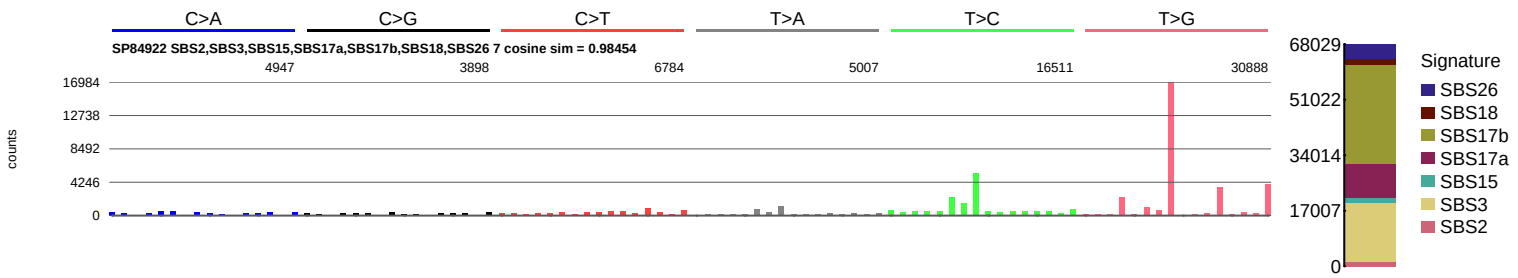
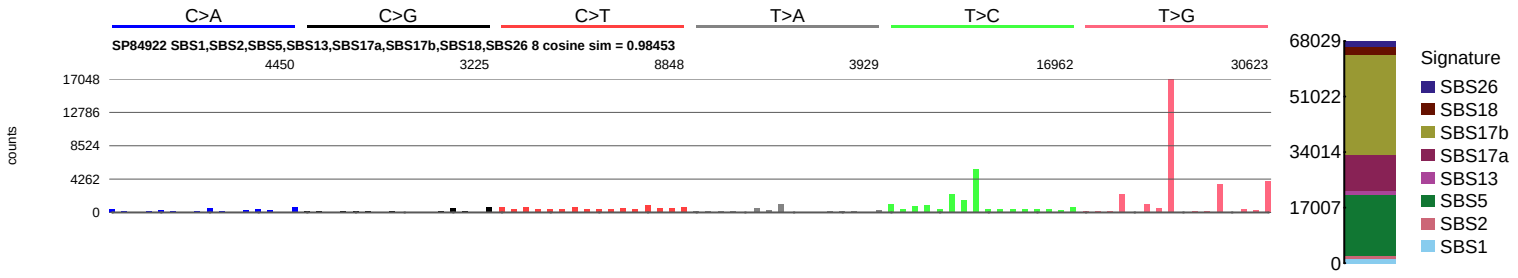
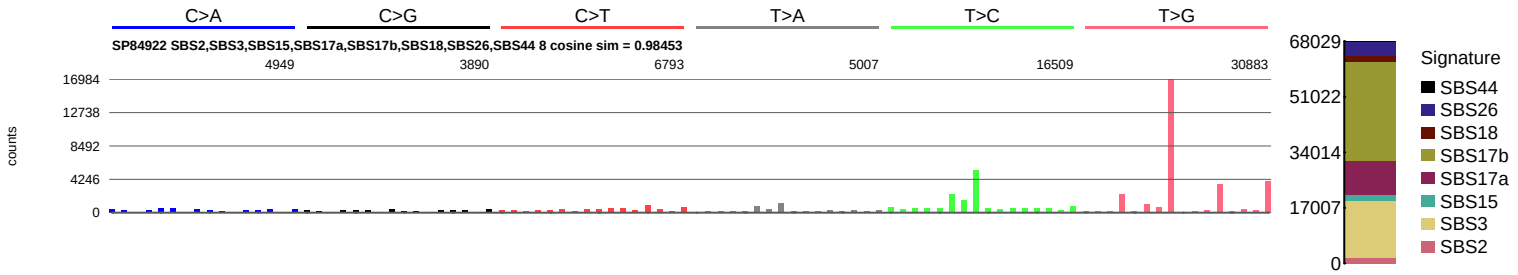
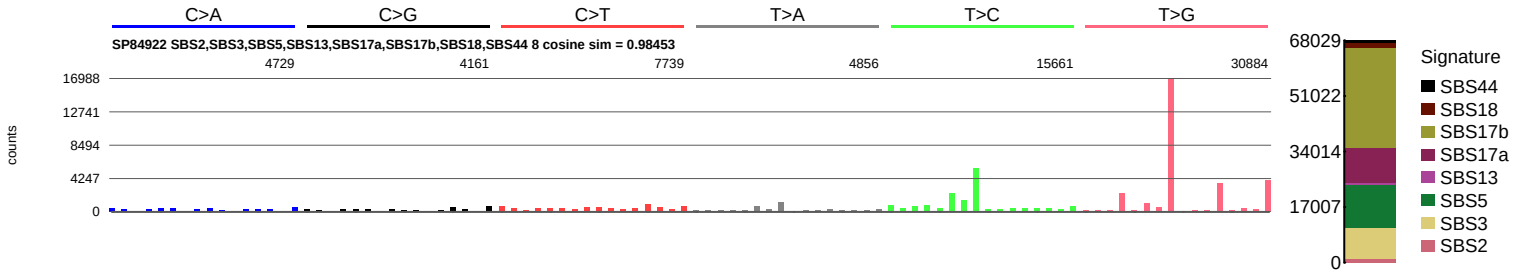
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98454
3897
4360
7404
4915
16512
30940
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

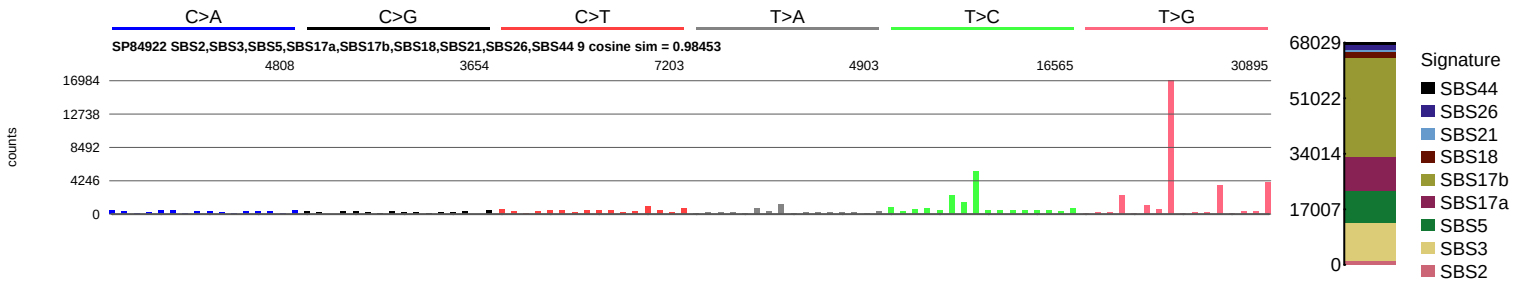

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98454
4897
4381
7478
5394
14990
30892
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

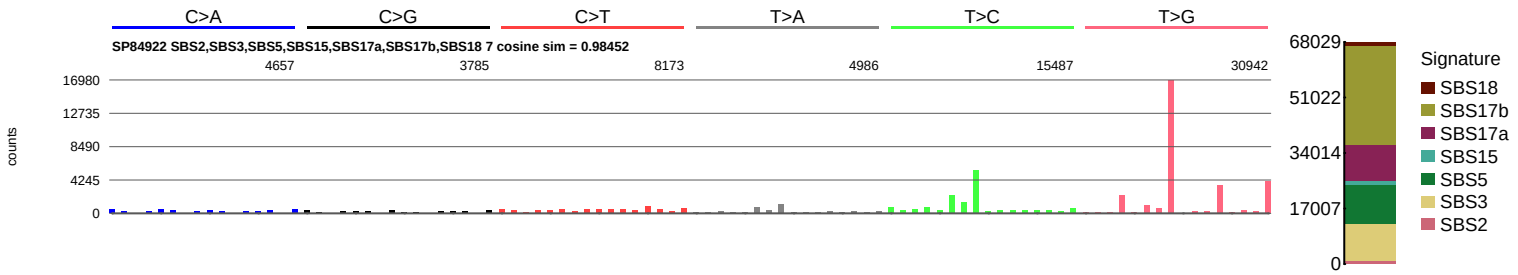

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98454
4947
3898
6784
5007
16511
30888
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

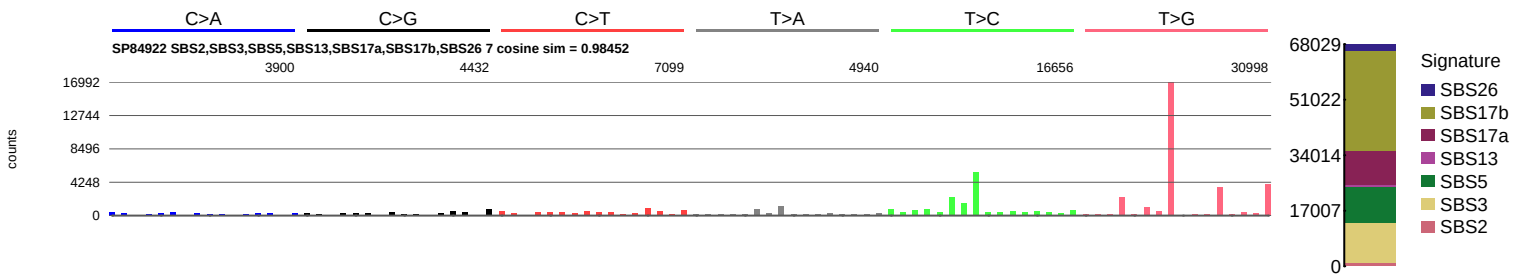

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98453
4450
3225
8848
3929
16962
30623
counts
17048
12786
8524
4262
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

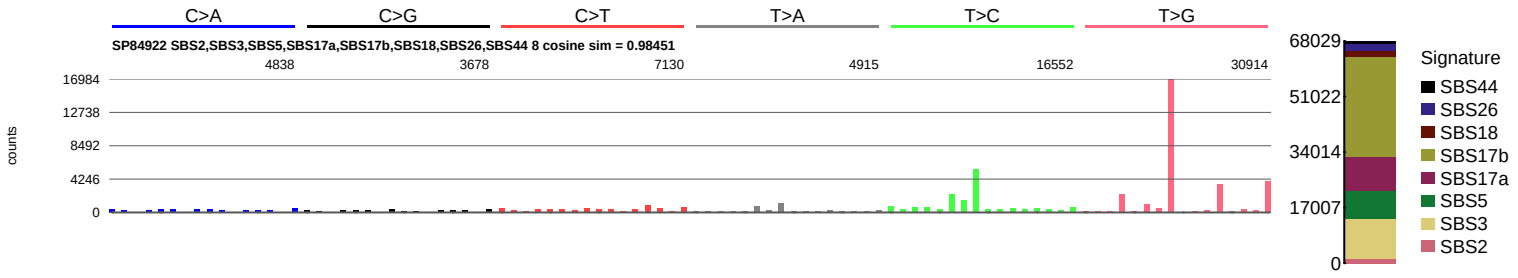

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98453
4949
3890
6793
5007
16509
30883
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98453
4729
4161
7739
4856
15661
30884
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98453
4808
3654
7203
4903
16565
30895
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98452
4657
3785
8173
4986
15487
30942
counts
16980
12735
8490
4245
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98452
3900
4432
7099
4940
16656
30998
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2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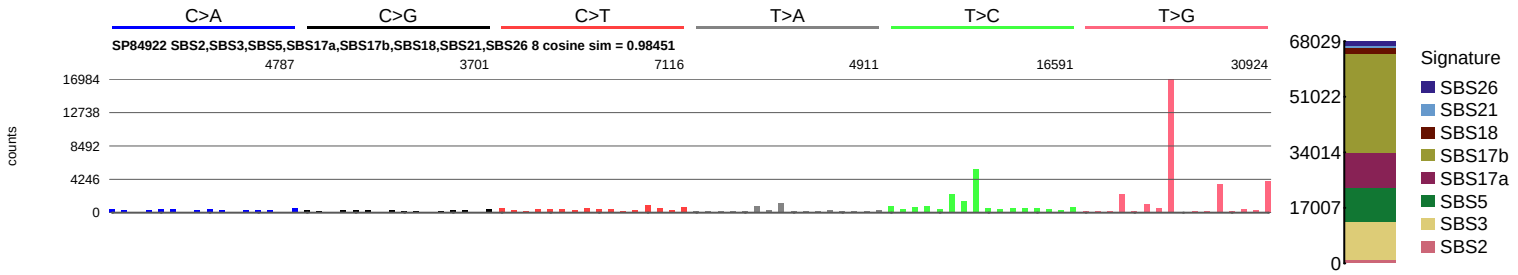
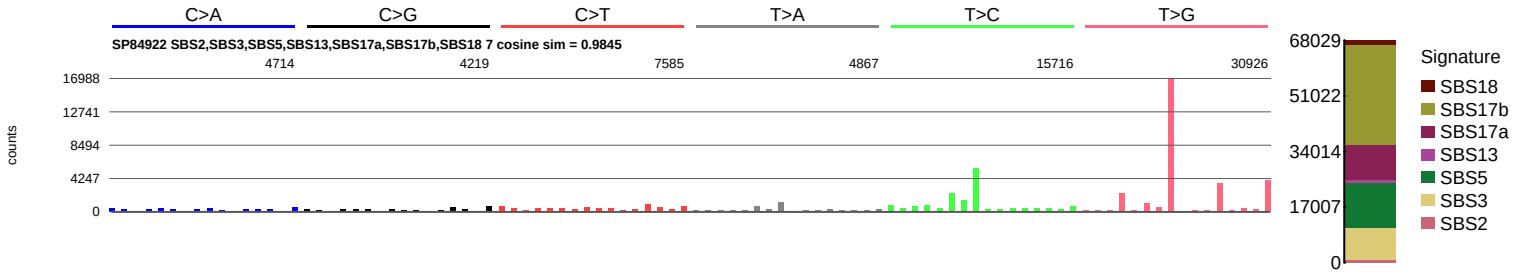
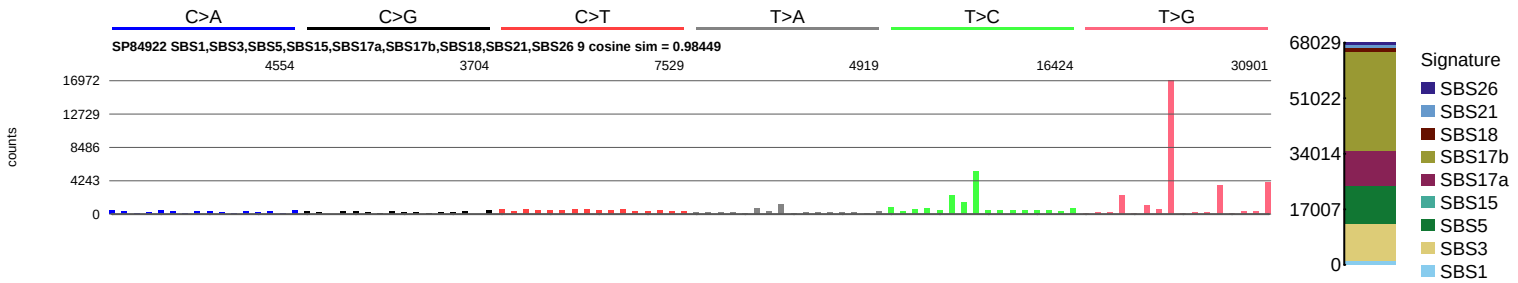
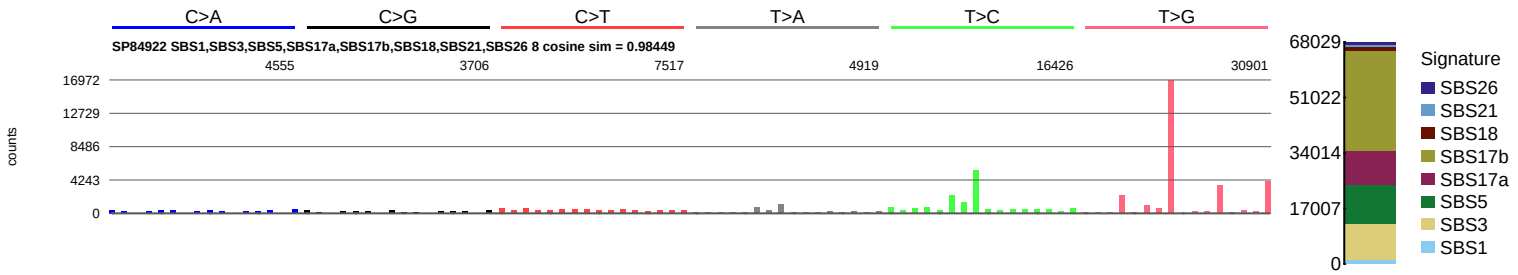
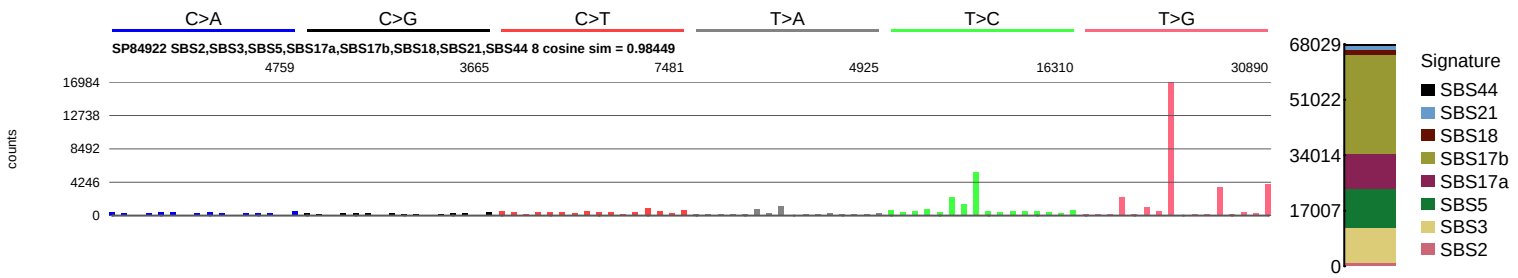
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98451
4838
3678
7130
4915
16552
30914
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2

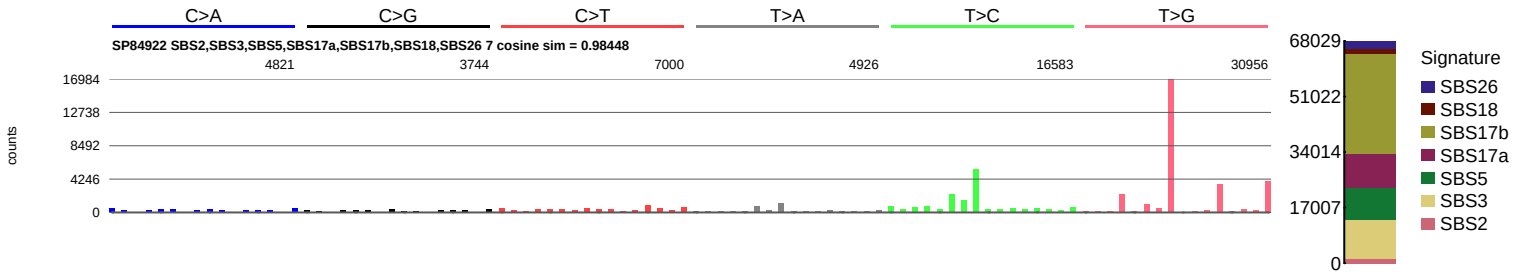

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98451
4787
3701
7116
4911
16591
30924
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2

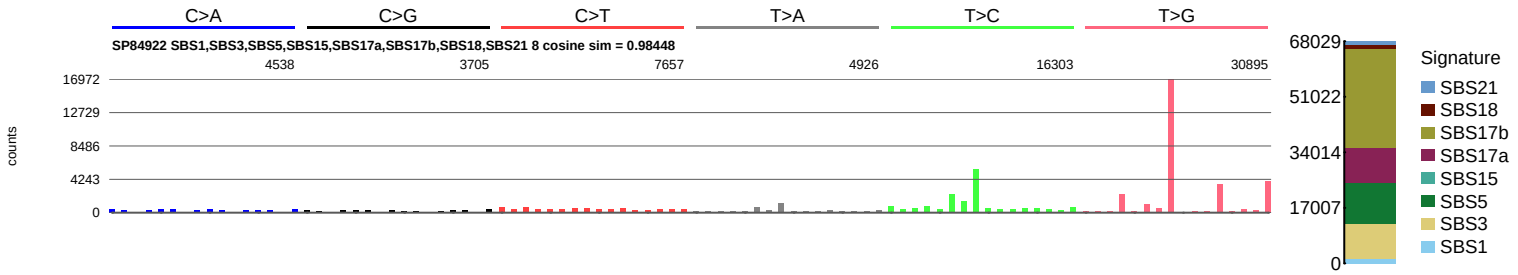

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.9845
4714
4219
7585
4867
15716
30926
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

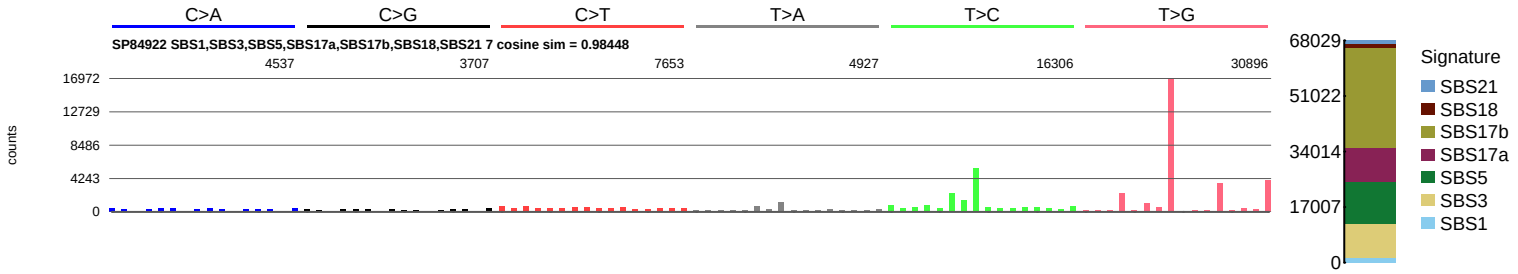

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98449
4554
3704
7529
4919
16424
30901
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98449
4555
3706
7517
4919
16426
30901
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98449
4759
3665
7481
4925
16310
30890
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98448
4821
3744
7000
4926
16583
30956
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98448
4538
3705
7657
4926
16303
30895
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98448
4537
3707
7653
4927
16306
30896
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
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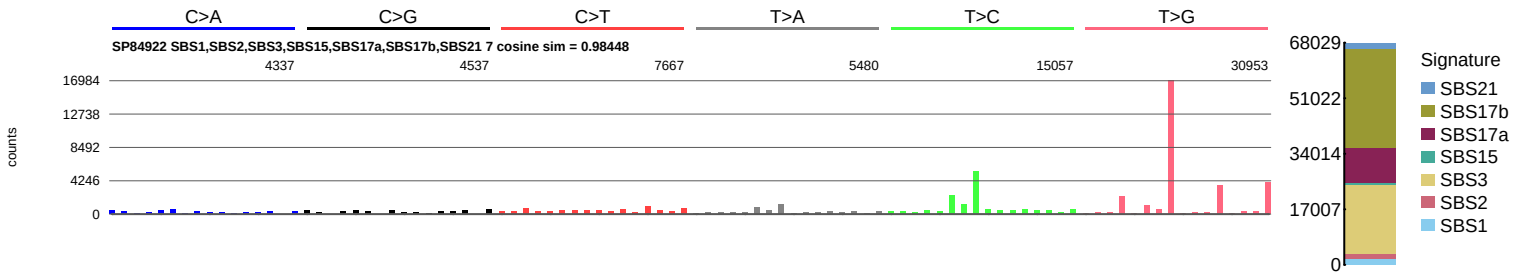
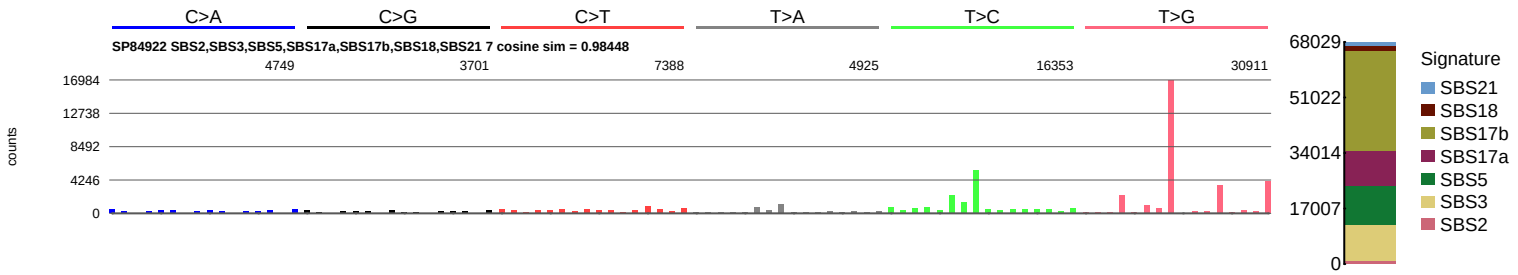
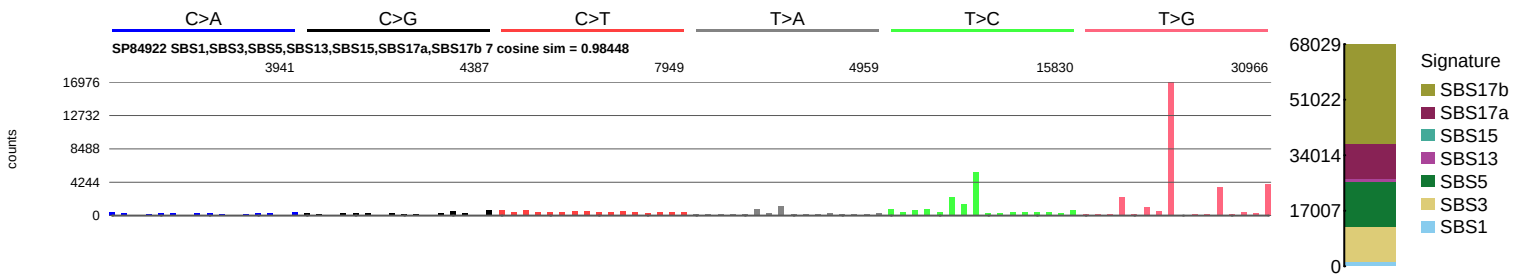
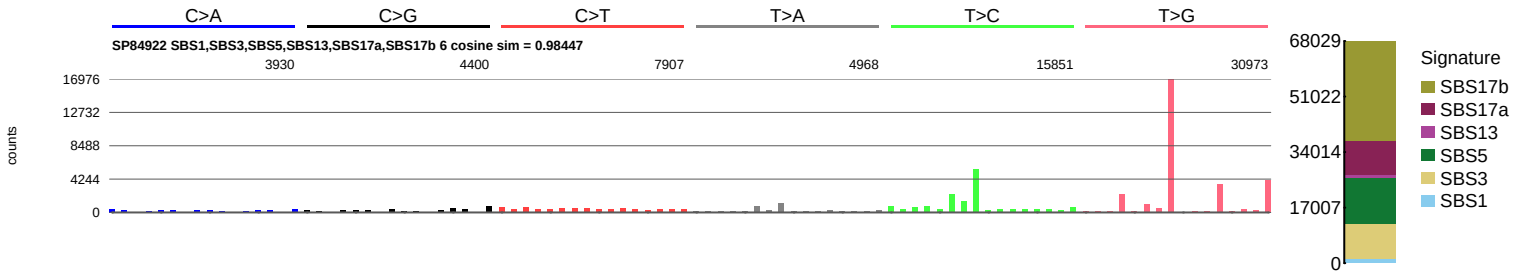
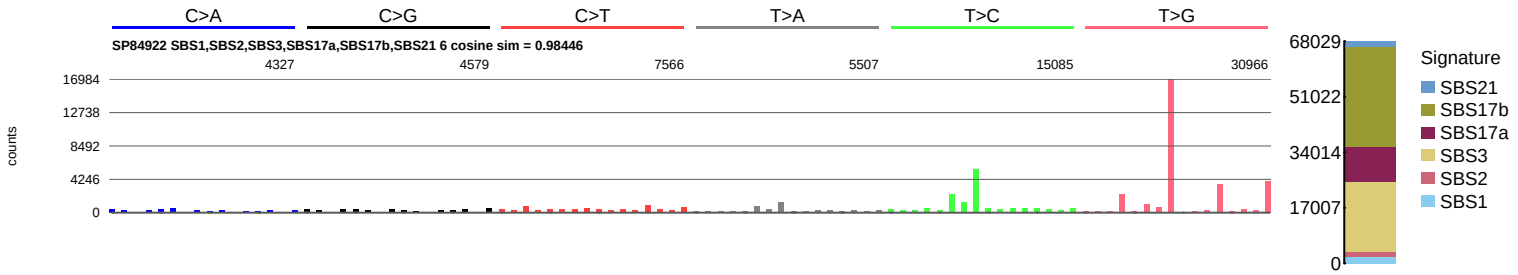
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

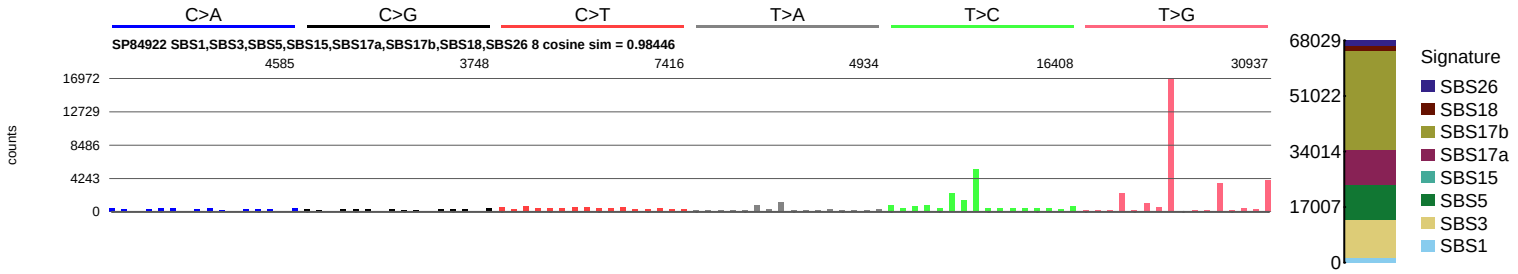

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98448
4337
4537
7667
5480
15057
30953
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

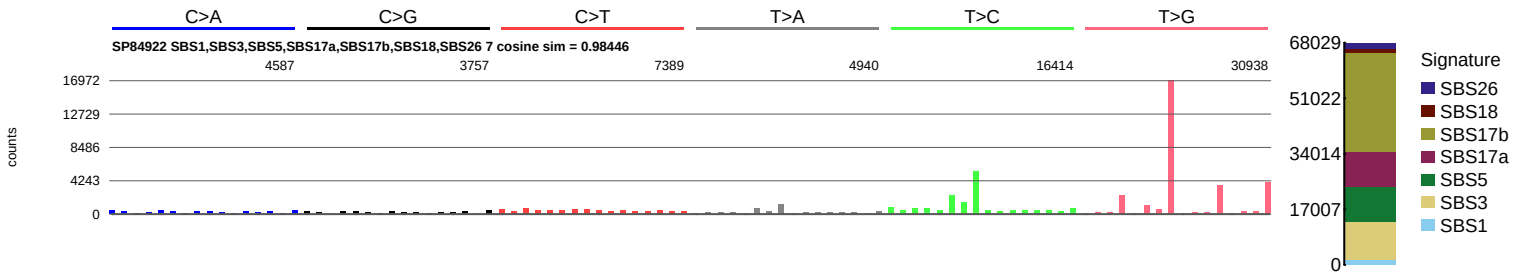

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98448
4749
3701
7388
4925
16353
30911
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2

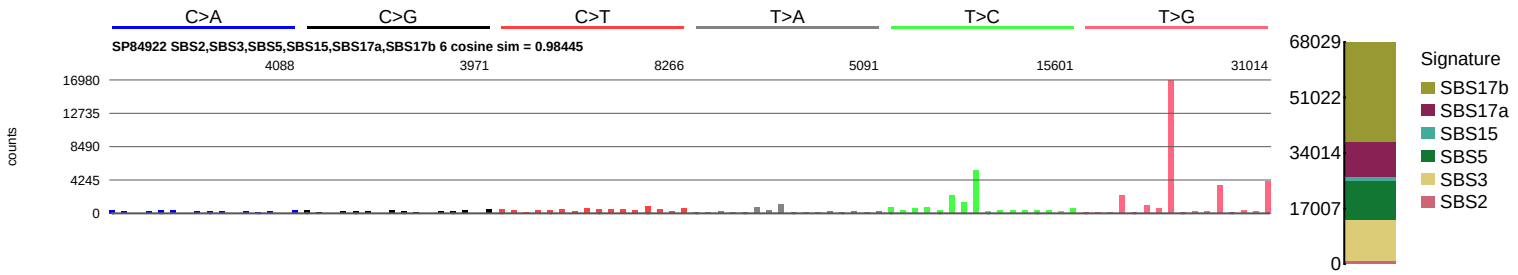

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98448
3941
4387
7949
4959
15830
30966
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

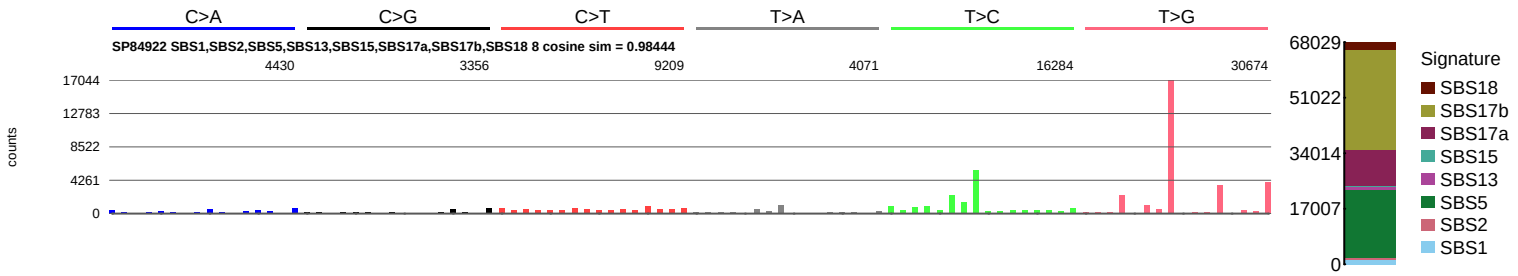

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b 6 cosine sim = 0.98447
3930
4400
7907
4968
15851
30973
counts
16976
12732
8488
4244
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

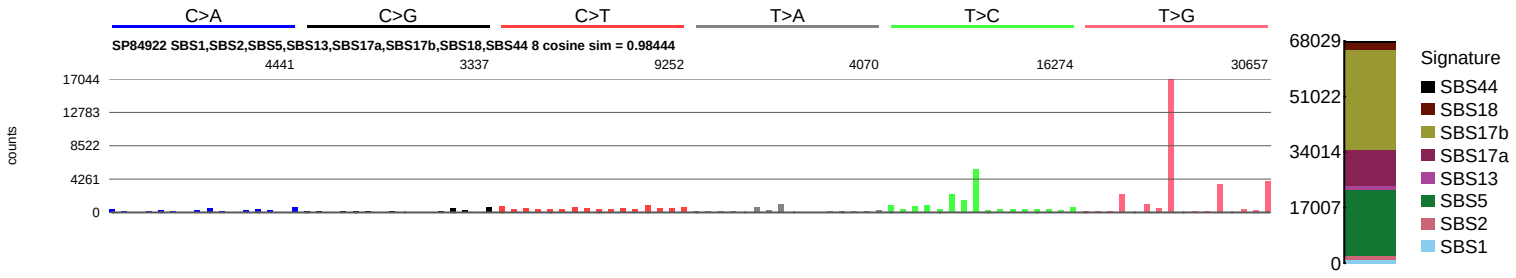

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98446
4327
4579
7566
5507
15085
30966
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

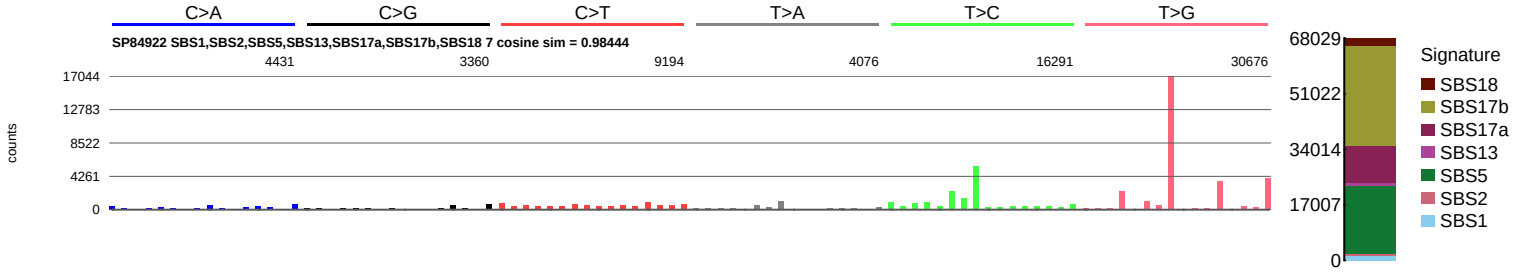

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98446
4585
3748
7416
4934
16408
30937
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

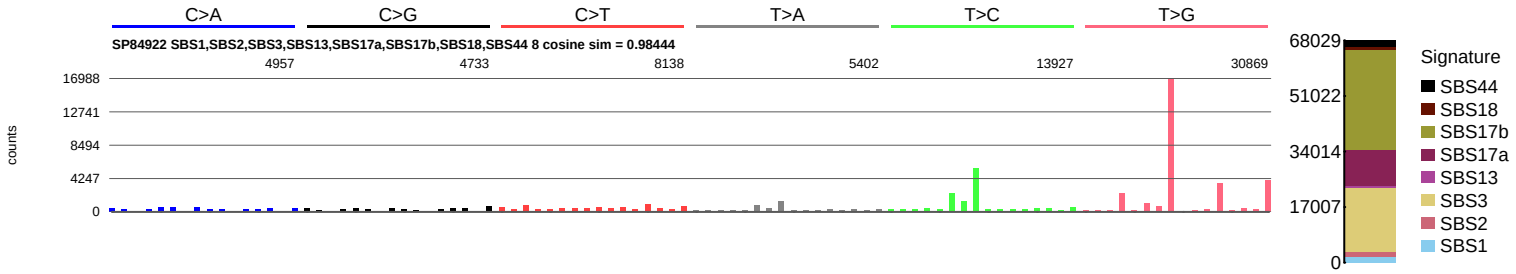

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98446
4587
3757
7389
4940
16414
30938
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

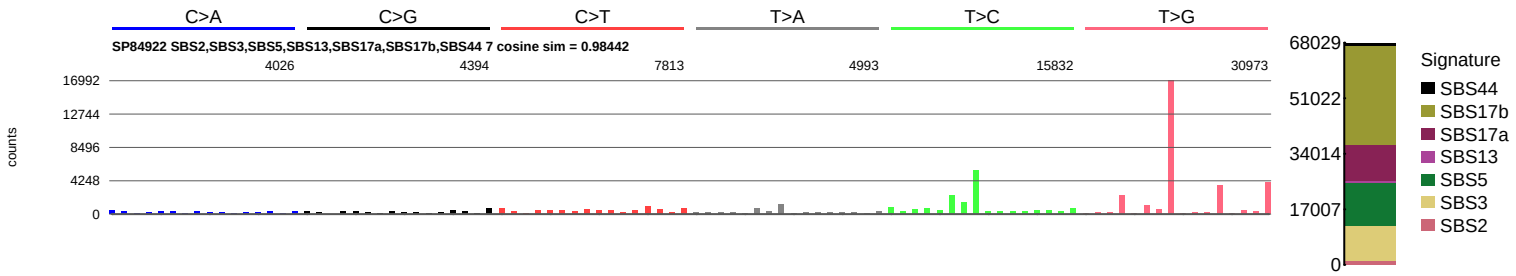

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98445
4088
3971
8266
5091
15601
31014
counts
16980
12735
8490
4245
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

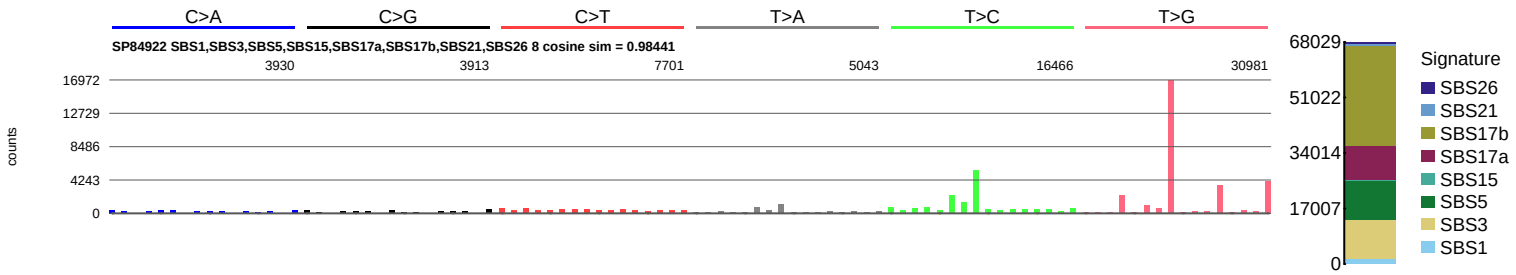

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98444
4430
3356
9209
4071
16284
30674
counts
17044
12783
8522
4261
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

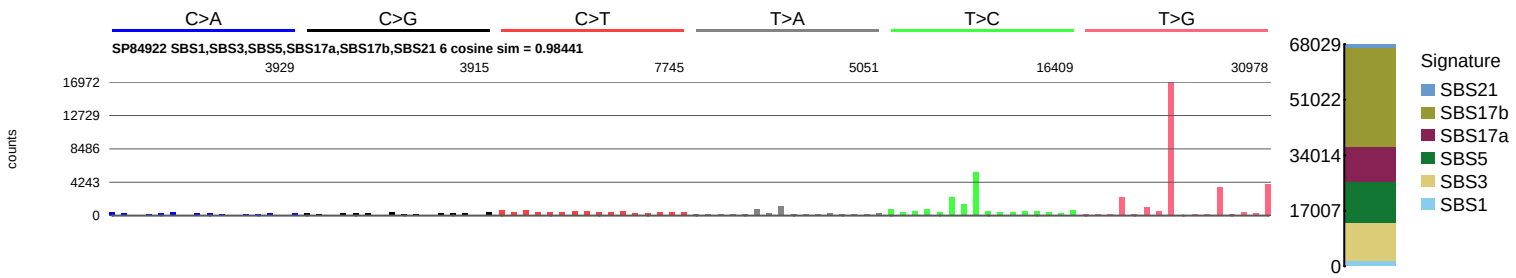

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98444
4441
3337
9252
4070
16274
30657
counts
17044
12783
8522
4261
0
68029
51022
34014
17007
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

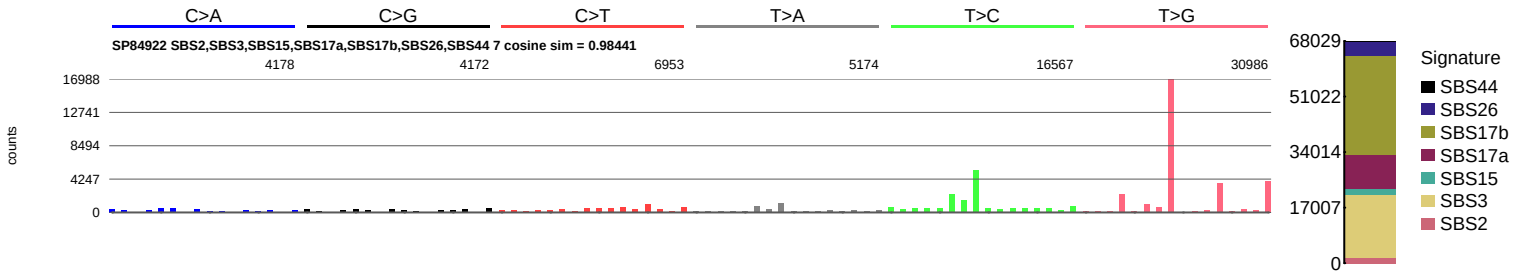

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98444
4431
3360
9194
4076
16291
30676
counts
17044
12783
8522
4261
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

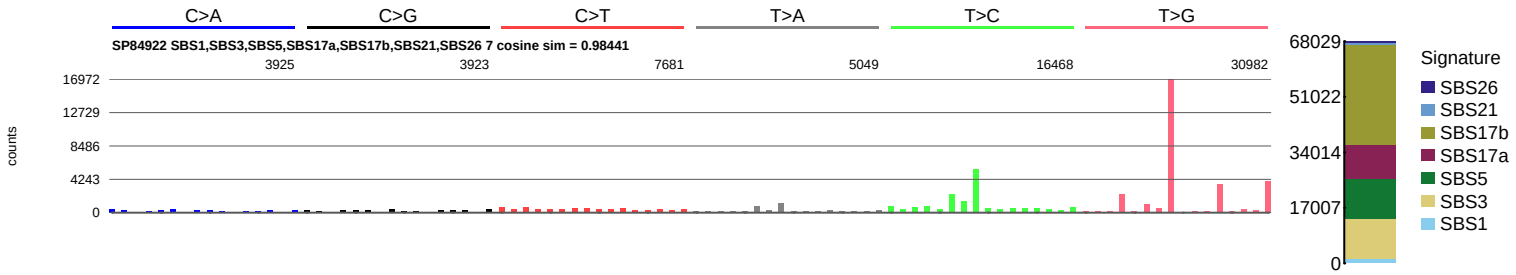

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98444
4957
4733
8138
5402
13927
30869
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

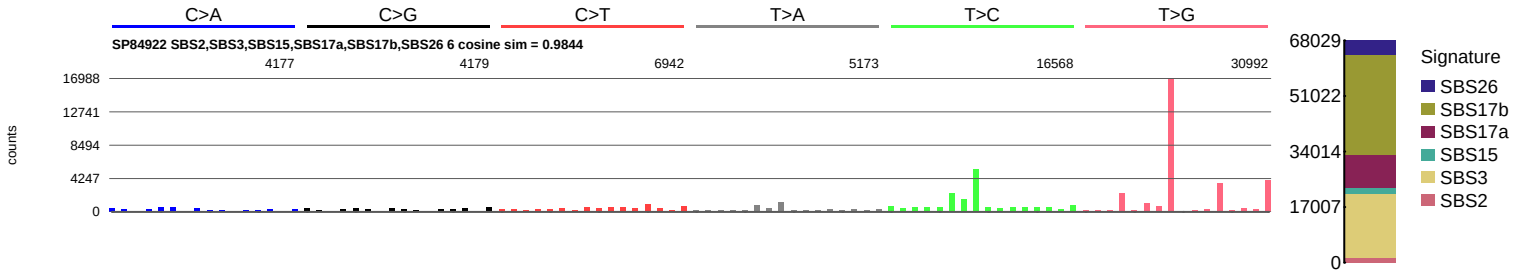

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98442
4026
4394
7813
4993
15832
30973
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

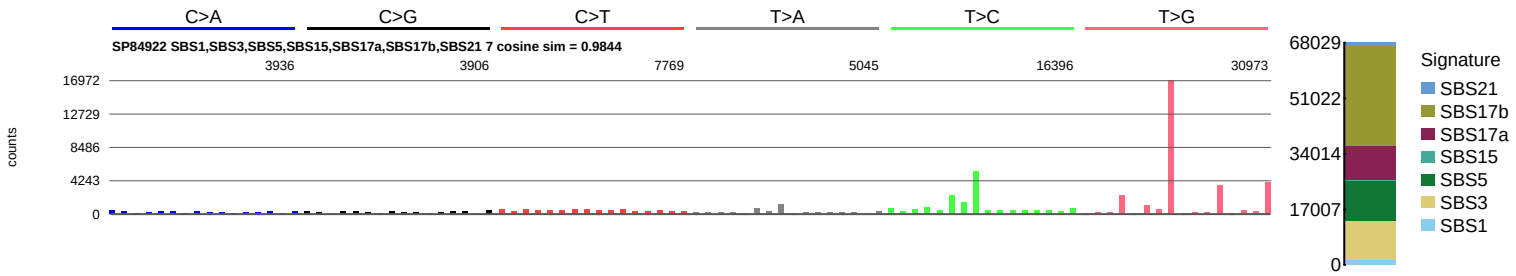

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS26 8 cosine sim = 0.98441
3930
3913
7701
5043
16466
30981
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

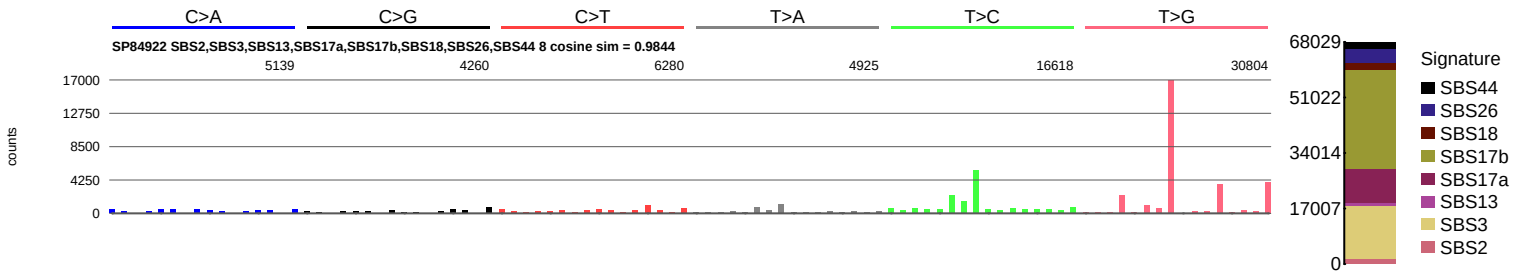

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98441
3929
3915
7745
5051
16409
30978
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS1

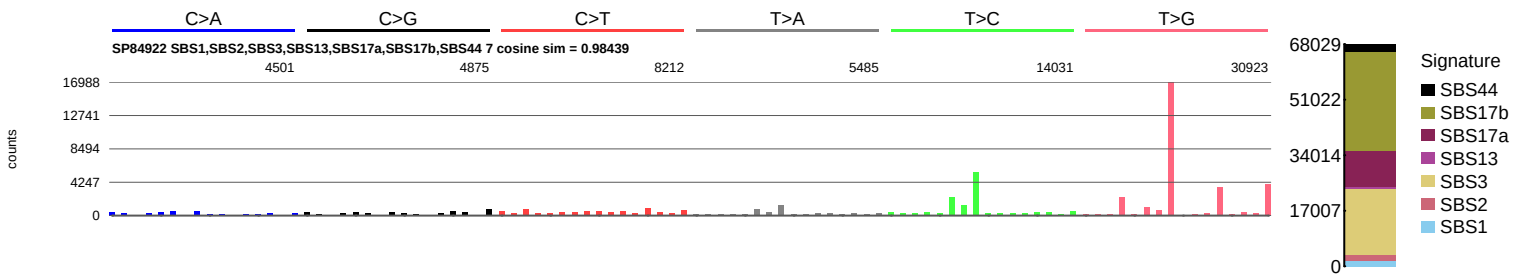

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98441
4178
4172
6953
5174
16567
30986
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2

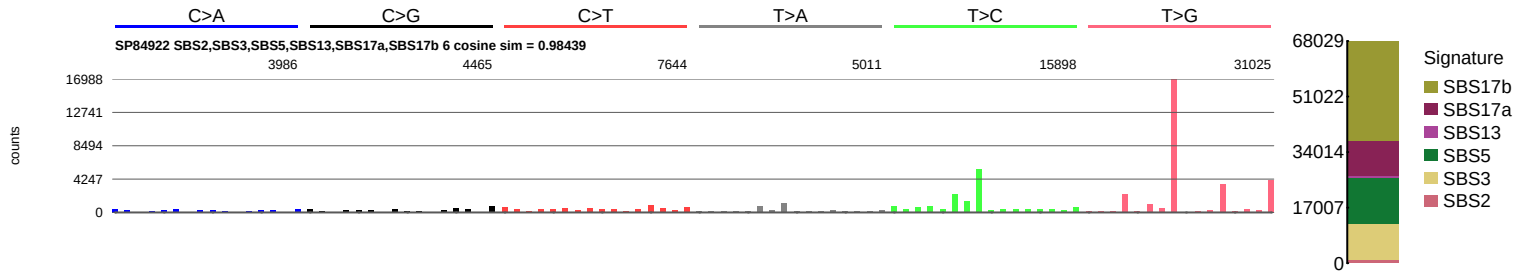

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS26 7 cosine sim = 0.98441
3925
3923
7681
5049
16468
30982
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS1

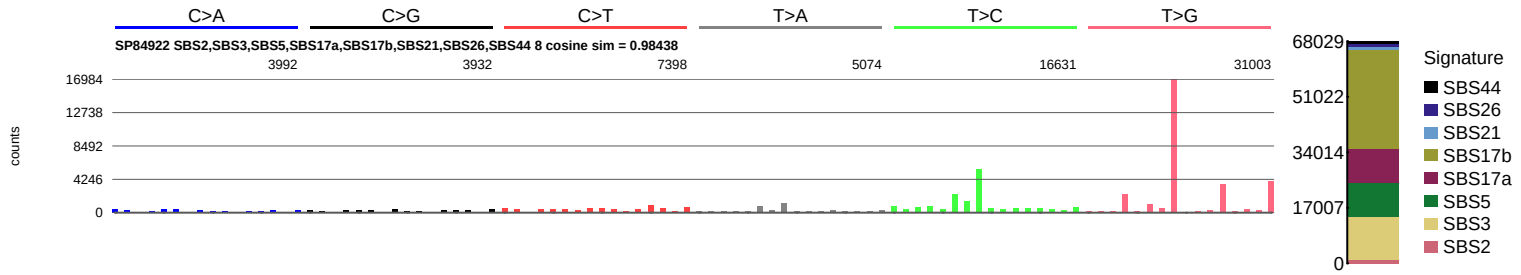

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26 6 cosine sim = 0.9844
4177
4179
6942
5173
16568
30992
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2

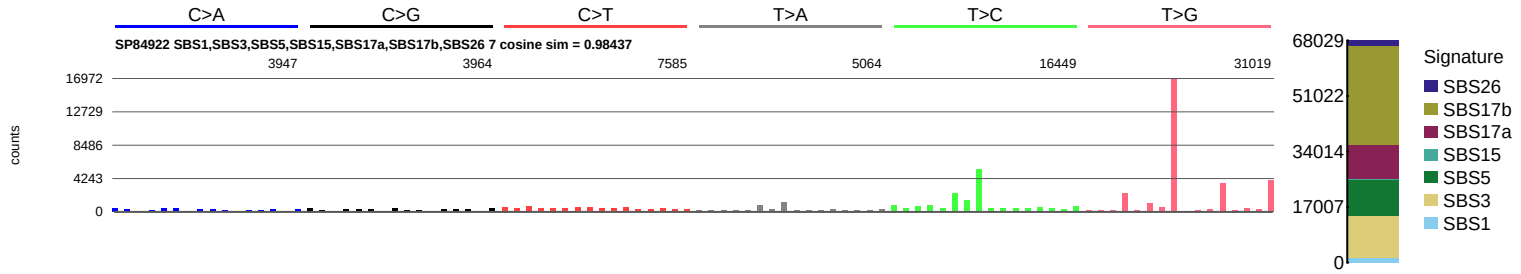

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.9844
3936
3906
7769
5045
16396
30973
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

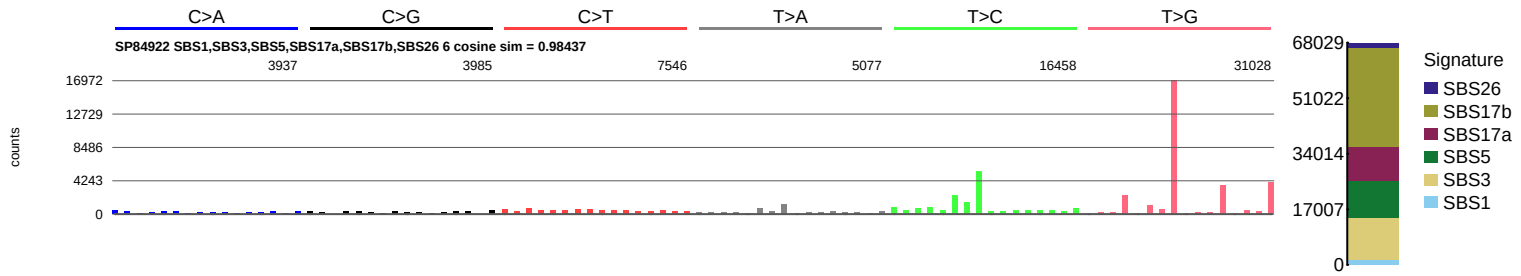

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.9844
5139
4260
6280
4925
16618
30804
counts
17000
12750
8500
4250
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2

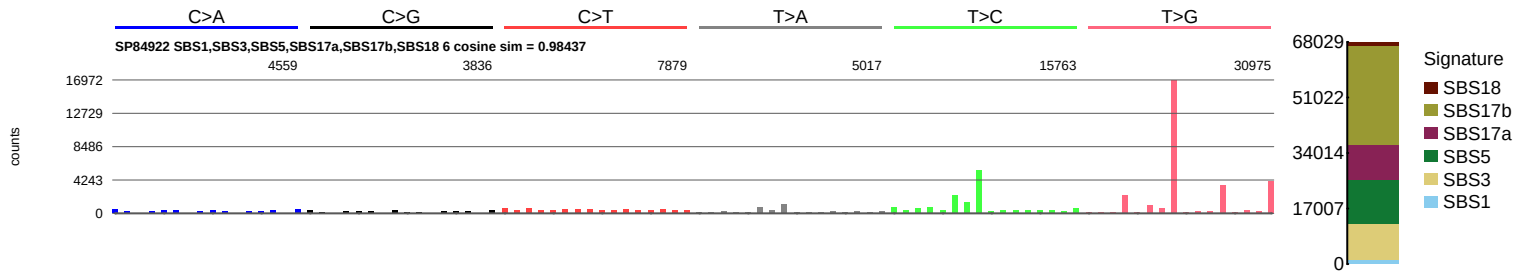

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98439
4501
4875
8212
5485
14031
30923
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

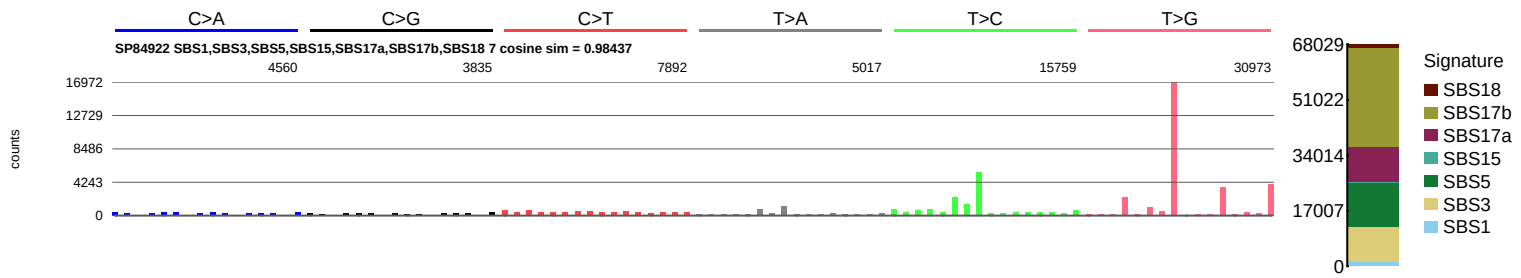

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b 6 cosine sim = 0.98439
3986
4465
7644
5011
15898
31025
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS26,SBS44 8 cosine sim = 0.98438
3992
3932
7398
5074
16631
31003
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98437
3947
3964
7585
5064
16449
31019
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98437
3937
3985
7546
5077
16458
31028
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98437
4559
3836
7879
5017
15763
30975
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98437
4560
3835
7892
5017
15759
30973
counts
16972
12729
8486
4243
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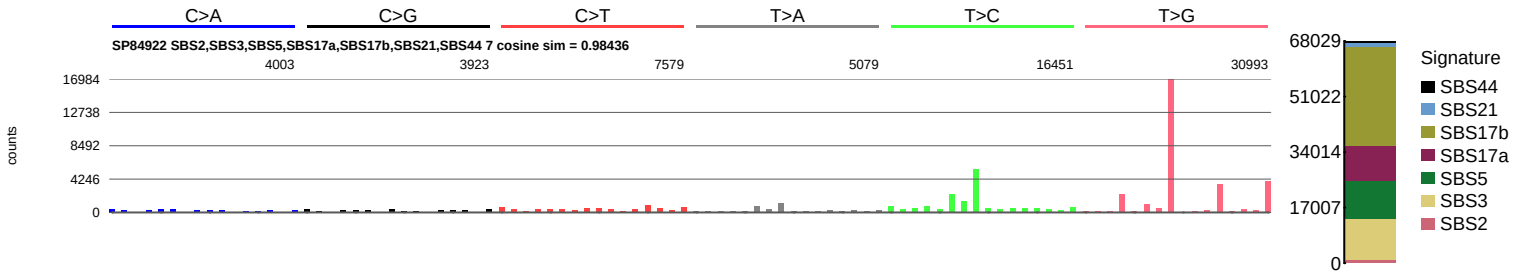
SBS1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SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98436
4003
3923
7579
5079
16451
30993
counts
16984
12738
8492
4246
0
68029
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17007
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SBS17b
SBS17a
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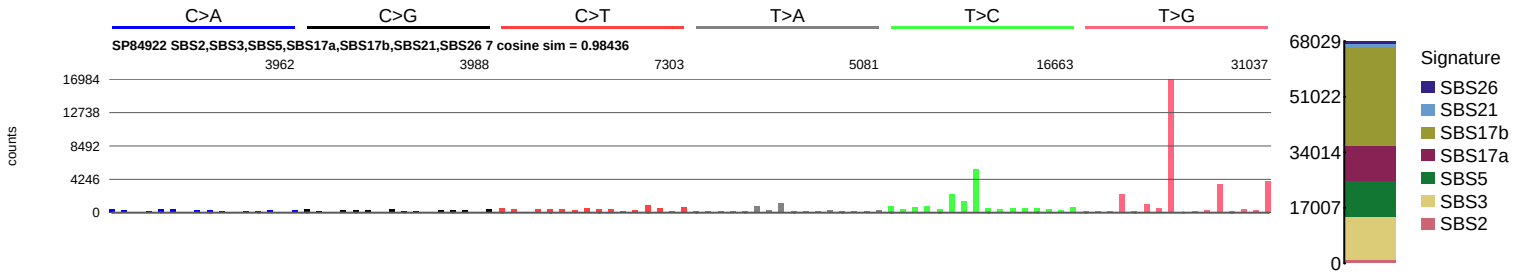
SBS2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SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS26 7 cosine sim = 0.98436
3962
3988
7303
5081
16663
31037
counts
16984
12738
8492
4246
0
68029
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34014
17007
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Signature
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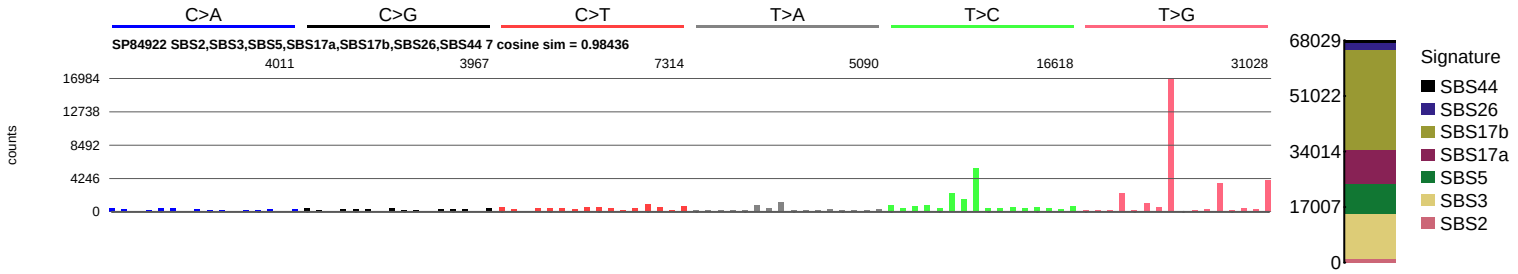
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2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SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98436
4011
3967
7314
5090
16618
31028
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
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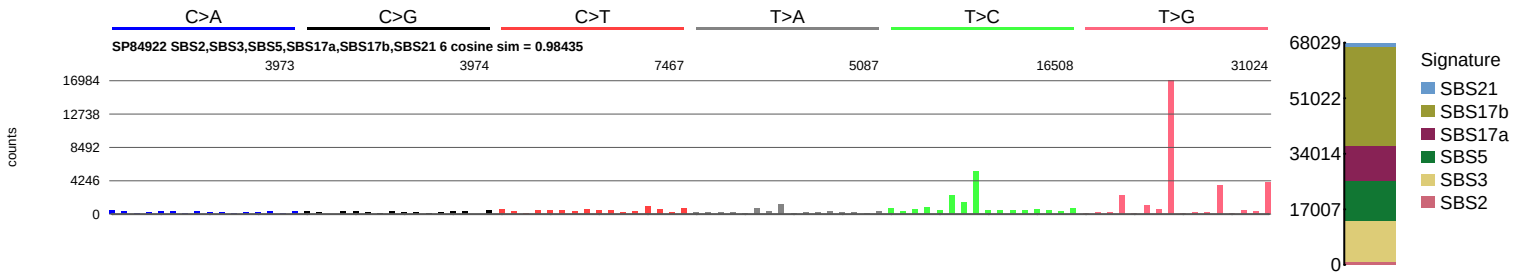
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2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SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98435
3973
3974
7467
5087
16508
31024
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
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SBS21
SBS17b
SBS17a
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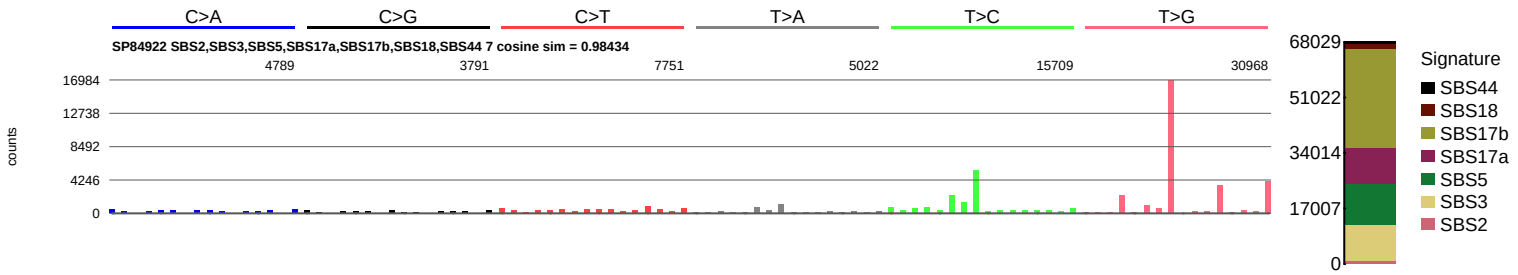
SBS3
SBS2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SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98434
4789
3791
7751
5022
15709
30968
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
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Signature
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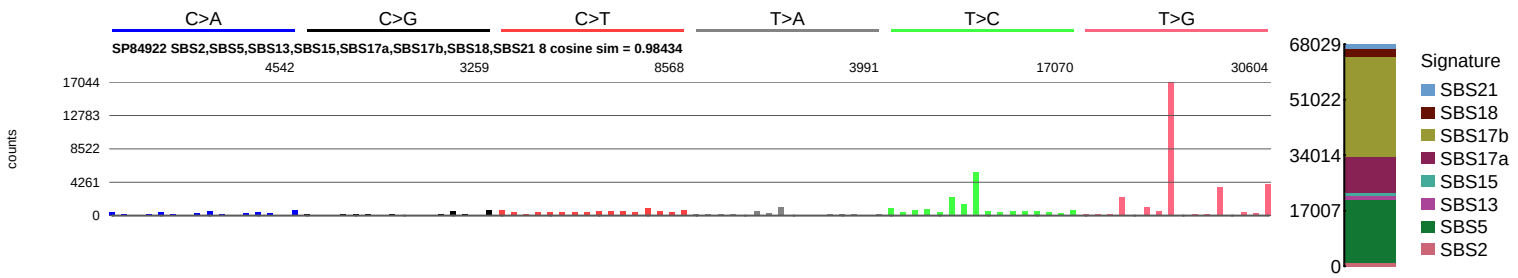
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2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SP84922 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98434
4542
3259
8568
3991
17070
30604
counts
17044
12783
8522
4261
0
68029
51022
34014
17007
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Signature
SBS21
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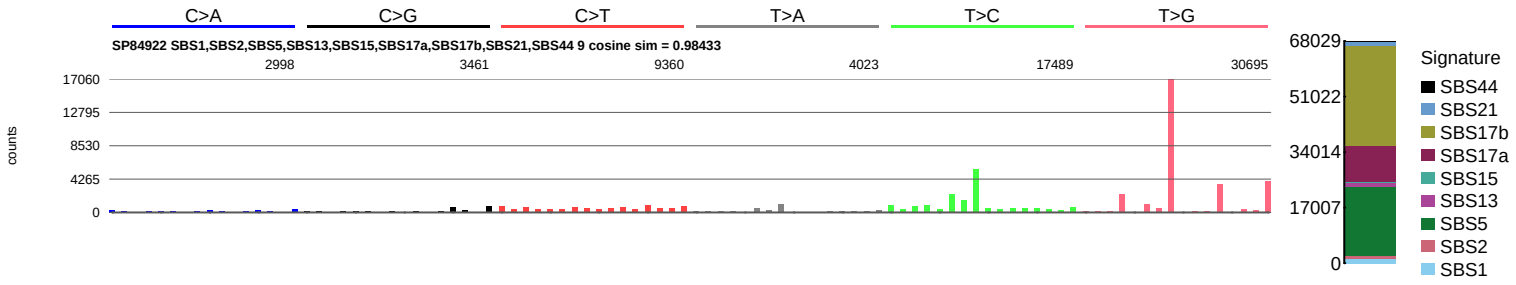
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2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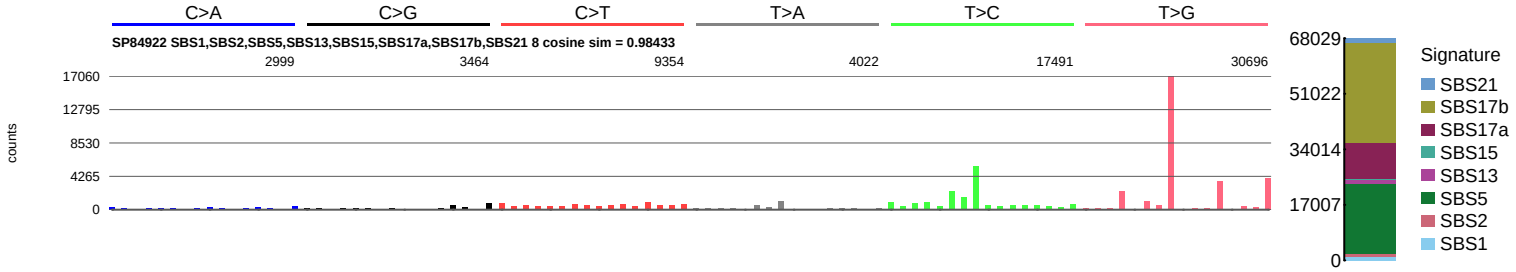
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS44 9 cosine sim = 0.98433
2998
3461
9360
4023
17489
30695
counts
17060
12795
8530
4265
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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SP84922 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98433
2999
3464
9354
4022
17491
30696
counts
17060
12795
8530
4265
0
68029
51022
34014
17007
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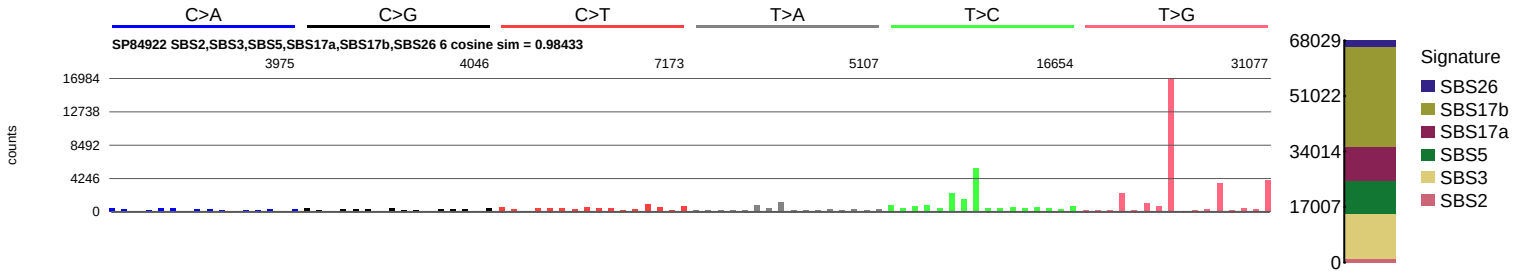
SBS2
SBS1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C>G
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T>G
SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98433
3975
4046
7173
5107
16654
31077
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
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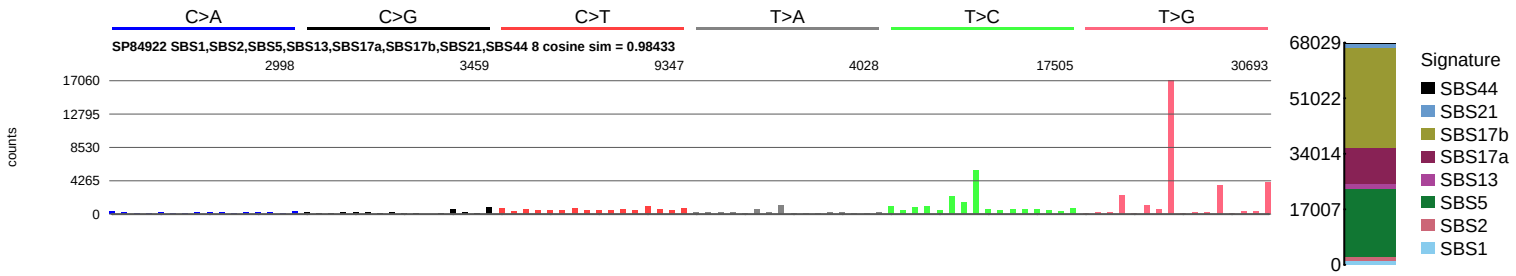
Signature
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98433
2998
3459
9347
4028
17505
30693
counts
17060
12795
8530
4265
0
68029
51022
34014
17007
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Signature
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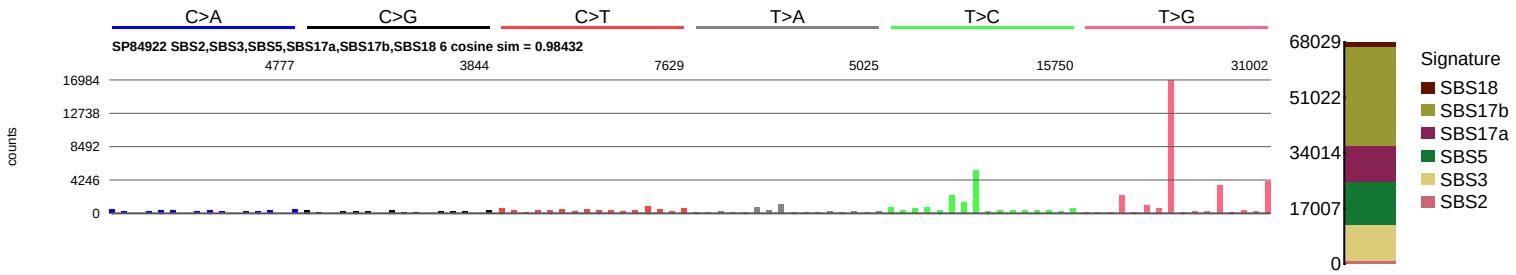
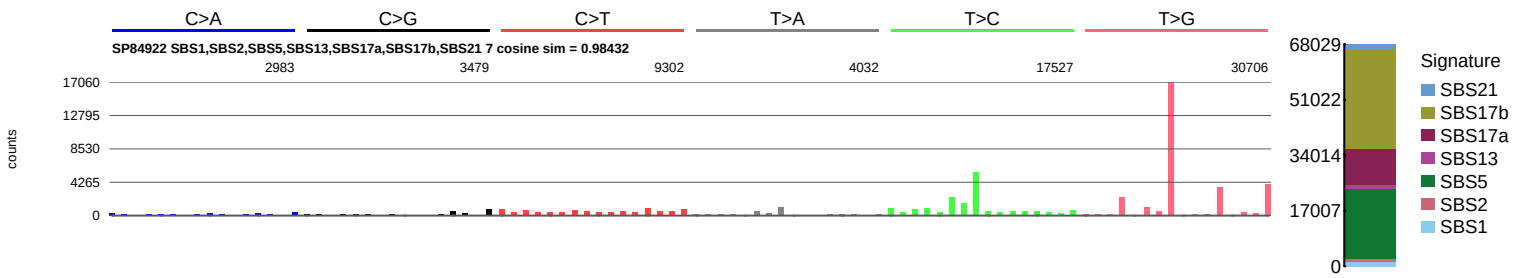
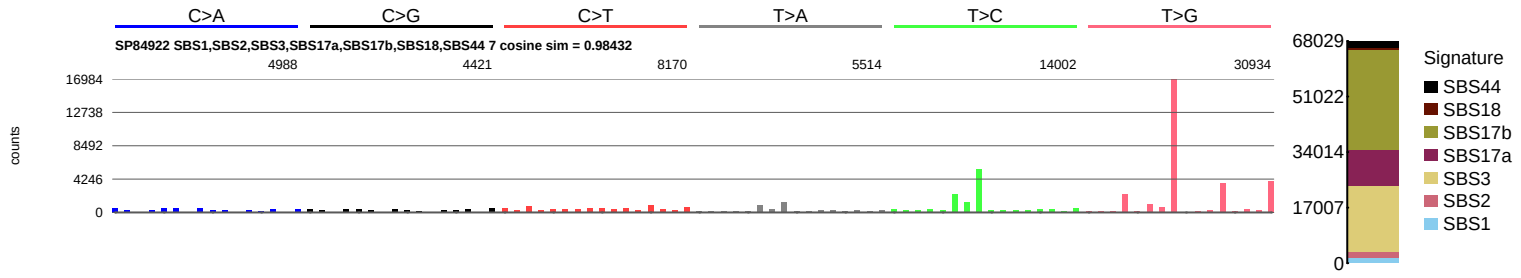
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98432
4777
3844
7629
5025
15750
31002
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98432
2983
3479
9302
4032
17527
30706
counts
17060
12795
8530
4265
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98432
4988
4421
8170
5514
14002
30934
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
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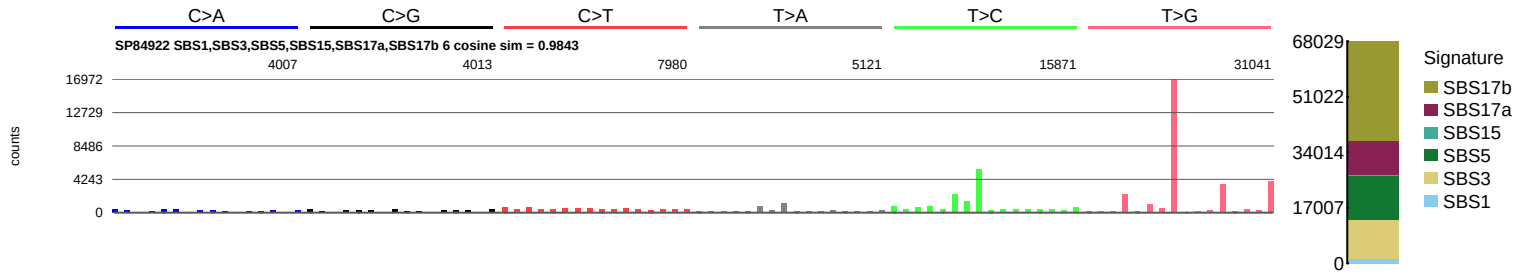
SBS44
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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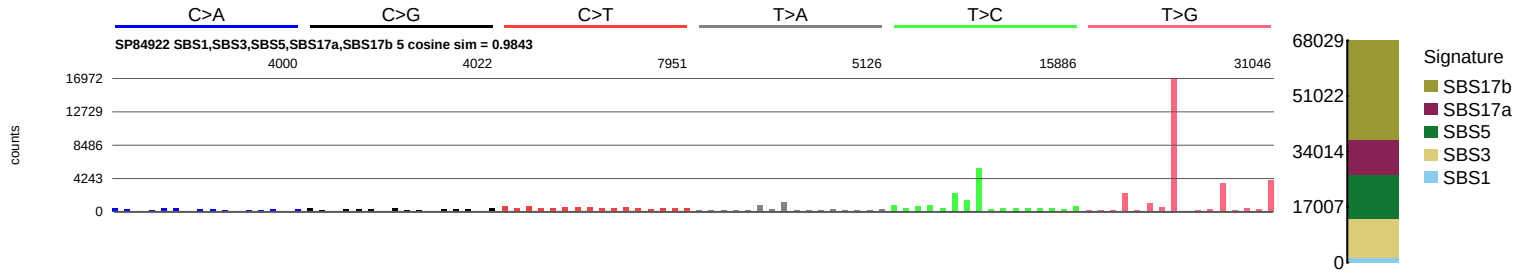
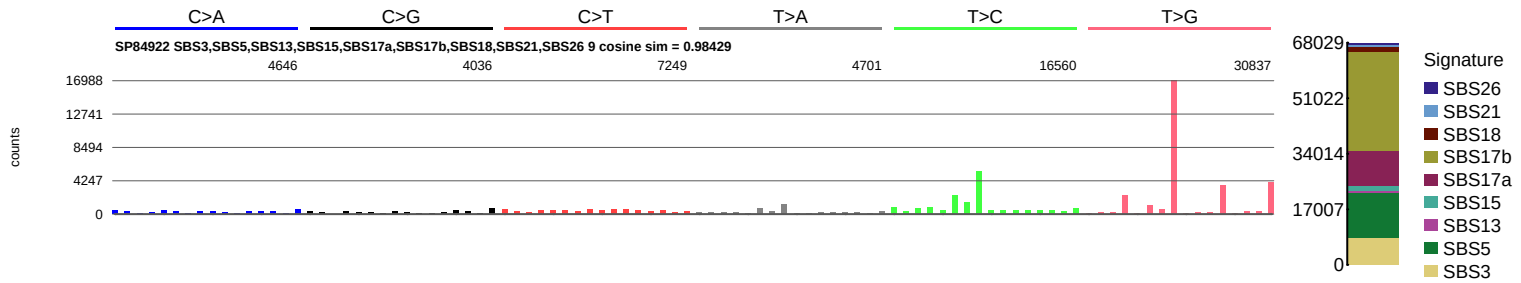
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b 6 cosine sim = 0.9843
4007
4013
7980
5121
15871
31041
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS5,SBS17a,SBS17b 5 cosine sim = 0.9843
4000
4022
7951
5126
15886
31046
counts
16972
12729
8486
4243
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS5
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98429
4646
4036
7249
4701
16560
30837
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
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Signature
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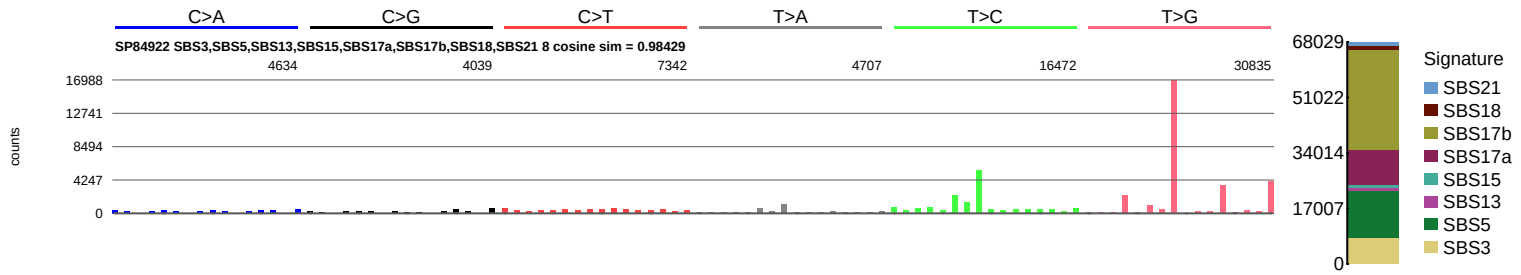
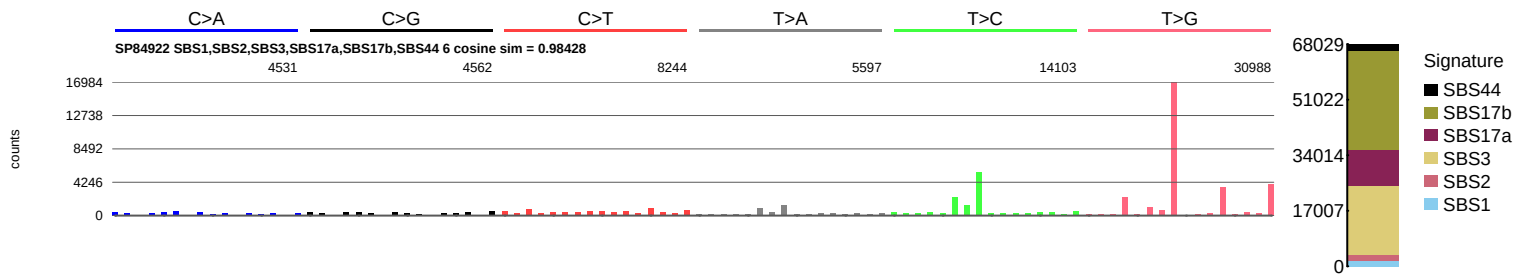
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98429
4634
4039
7342
4707
16472
30835
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98428
4531
4562
8244
5597
14103
30988
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
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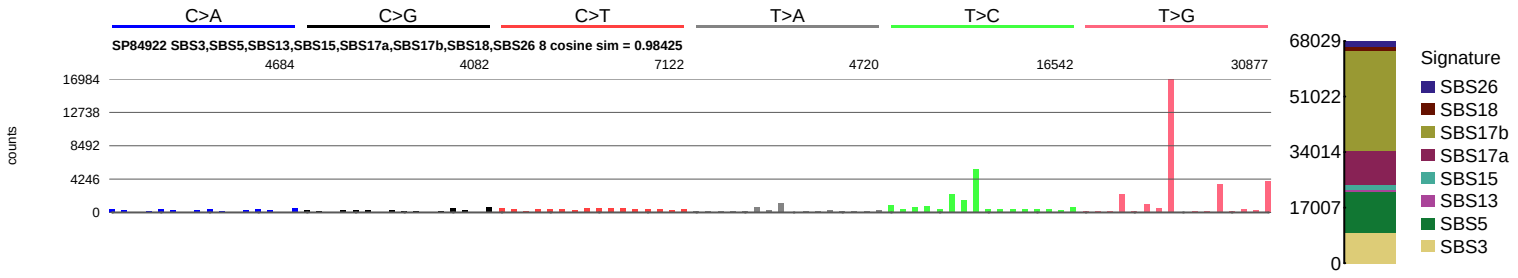
SBS44
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98425
4684
4082
7122
4720
16542
30877
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
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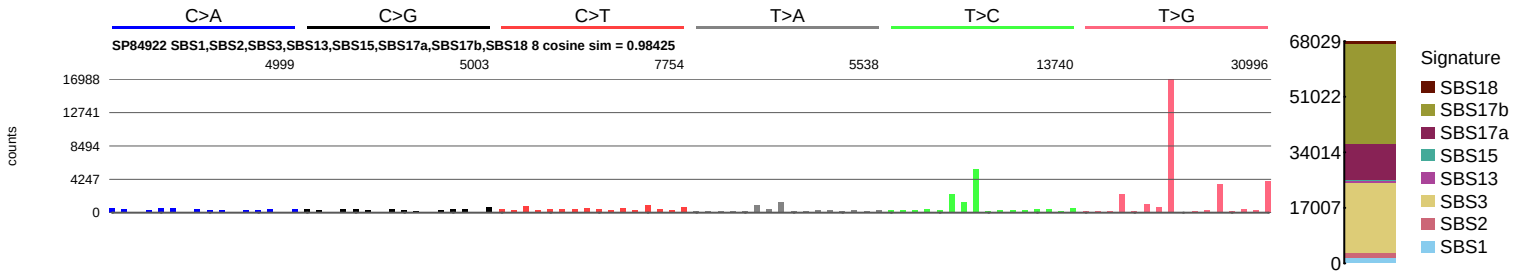
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98425
4999
5003
7754
5538
13740
30996
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
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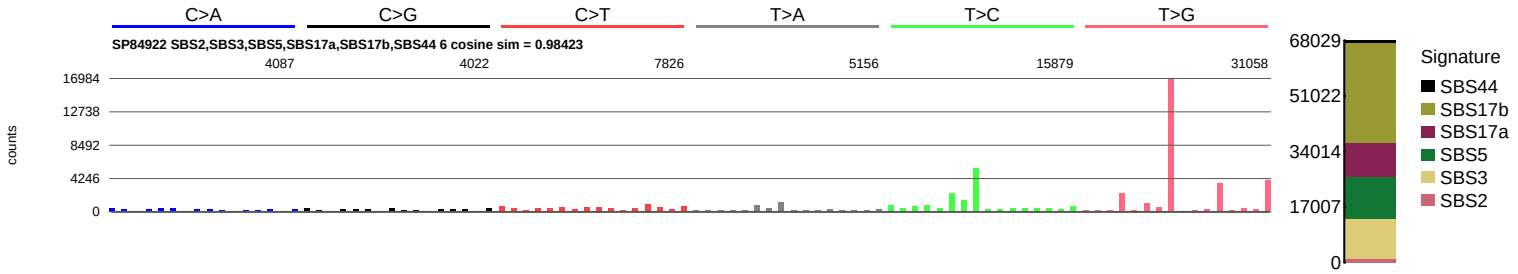
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98423
4087
4022
7826
5156
15879
31058
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
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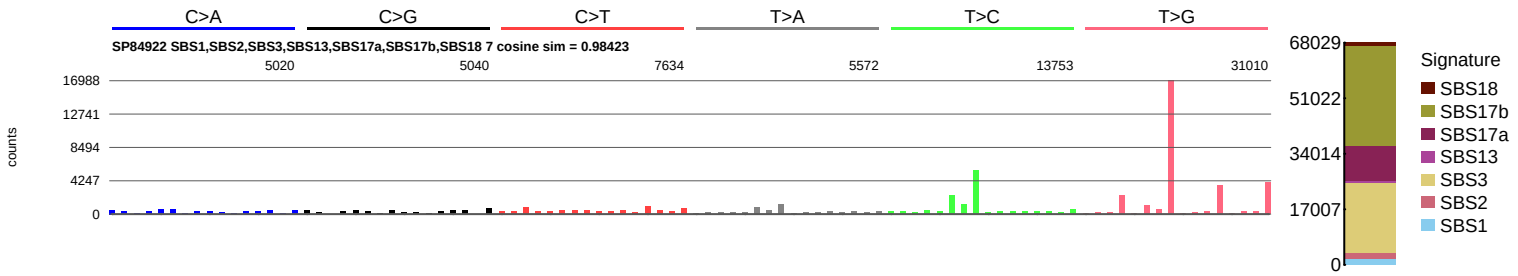
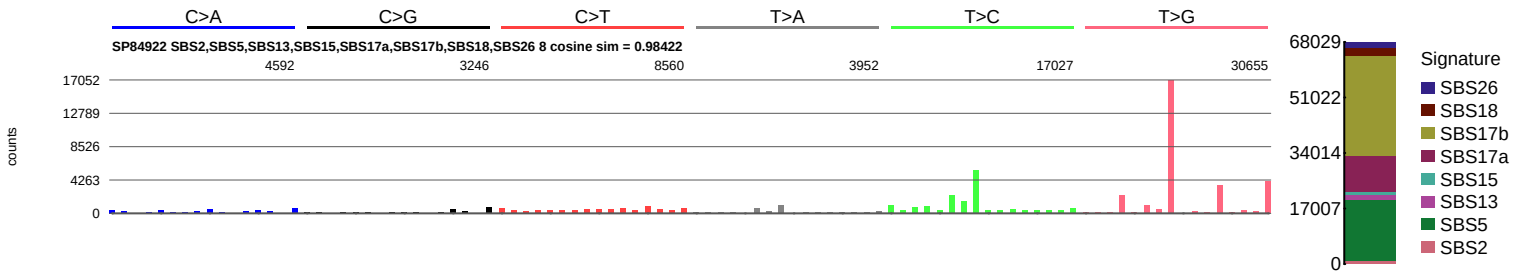
SBS44
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98423
5020
5040
7634
5572
13753
31010
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98422
4592
3246
8560
3952
17027
30655
counts
17052
12789
8526
4263
0
68029
51022
34014
17007
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Signature
SBS26
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SBS15
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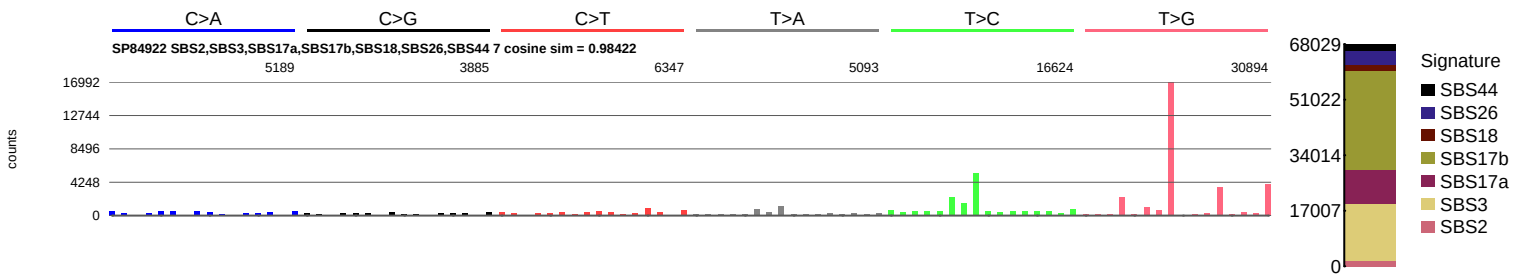
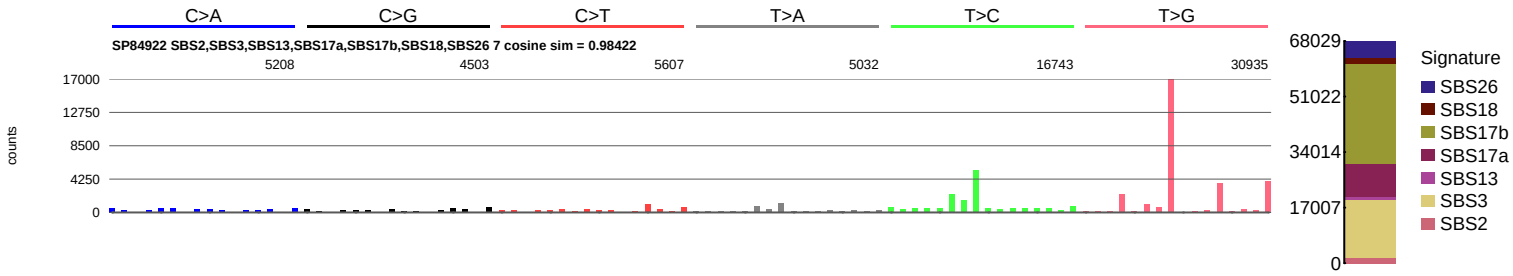
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98422
5189
3885
6347
5093
16624
30894
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98422
5208
4503
5607
5032
16743
30935
counts
17000
12750
8500
4250
0
68029
51022
34014
17007
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Signature
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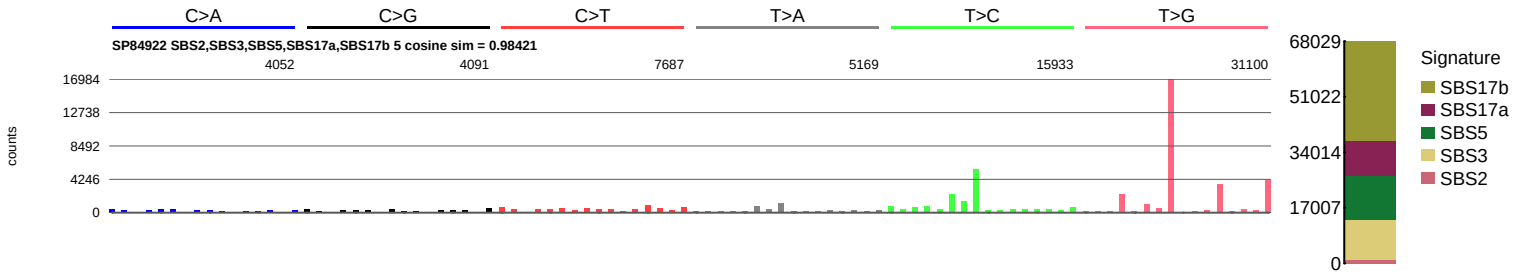
SBS3
SBS2


C>A
C>G
C>T
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SP84922 SBS2,SBS3,SBS5,SBS17a,SBS17b 5 cosine sim = 0.98421
4052
4091
7687
5169
15933
31100
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
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SBS17a
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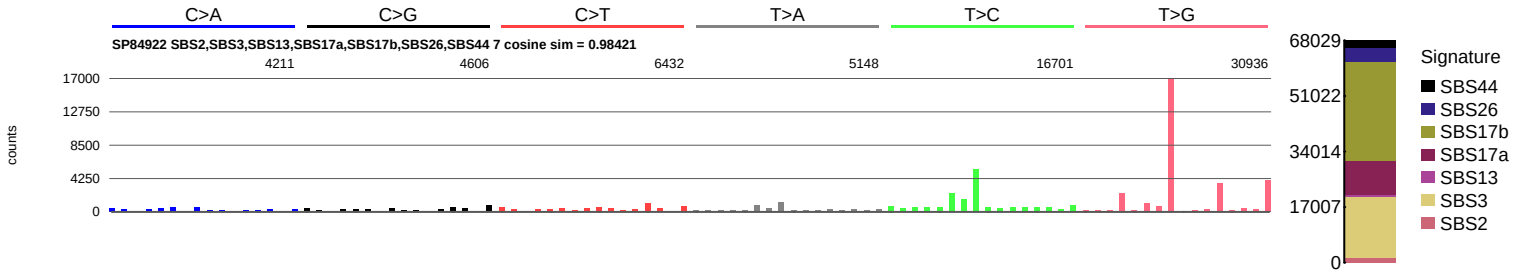
SBS3
SBS2


C>A
C>G
C>T
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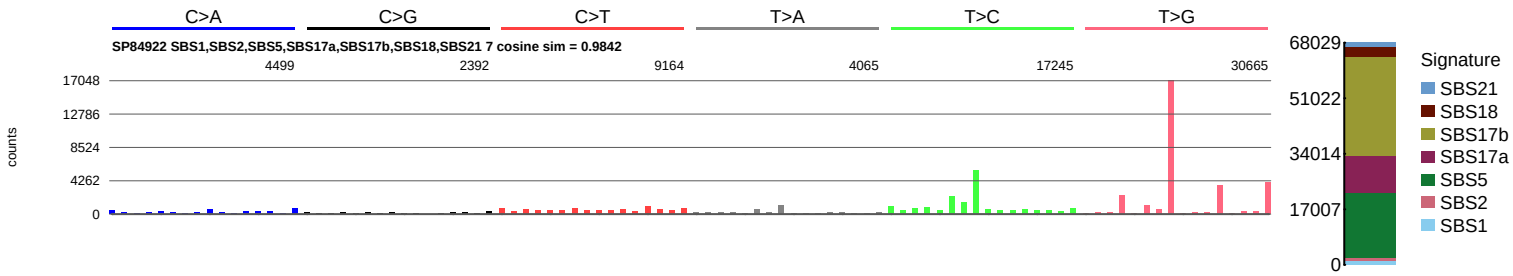
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98421
4211
4606
6432
5148
16701
30936
counts
17000
12750
8500
4250
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2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C>G
C>T
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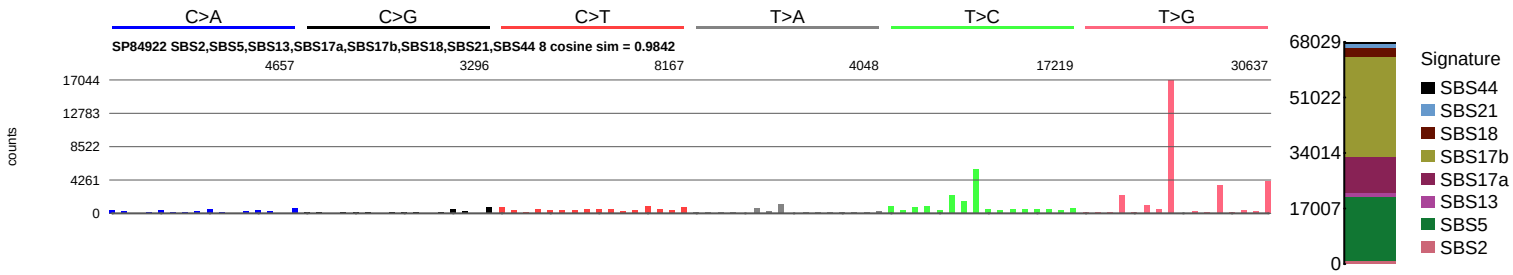
T>G
SP84922 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.9842
4499
2392
9164
4065
17245
30665
counts
17048
12786
8524
4262
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>T
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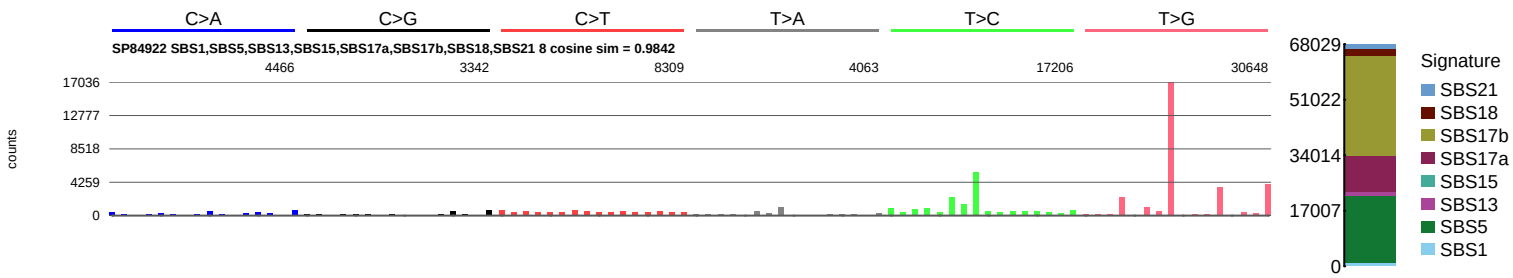
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.9842
4657
3296
8167
4048
17219
30637
counts
17044
12783
8522
4261
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2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C>G
C>T
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T>C
T>G
SP84922 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.9842
4466
3342
8309
4063
17206
30648
counts
17036
12777
8518
4259
0
68029
51022
34014
17007
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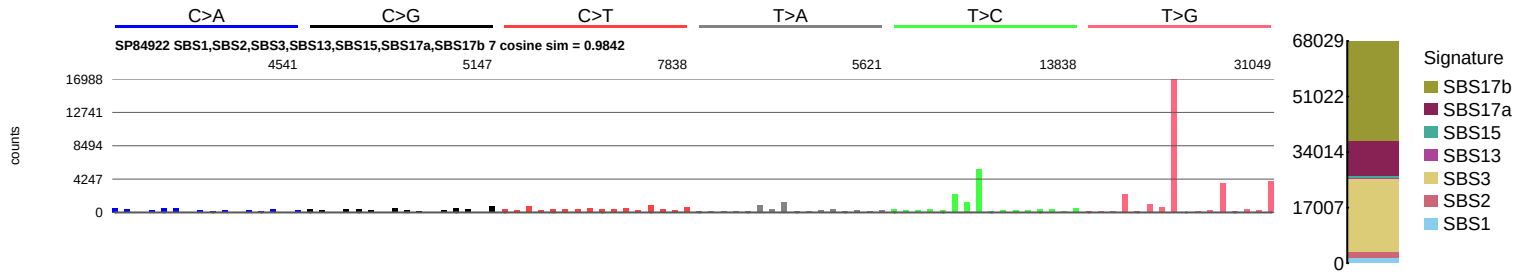
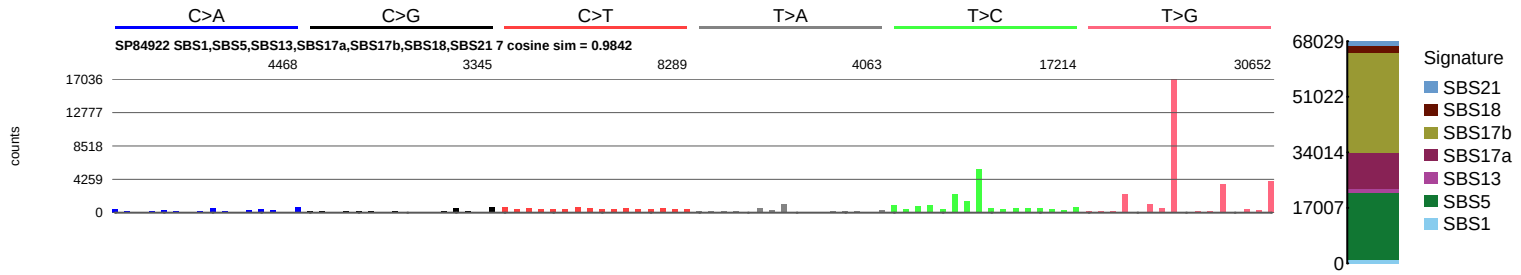
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b 7 cosine sim = 0.9842
4541
5147
7838
5621
13838
31049
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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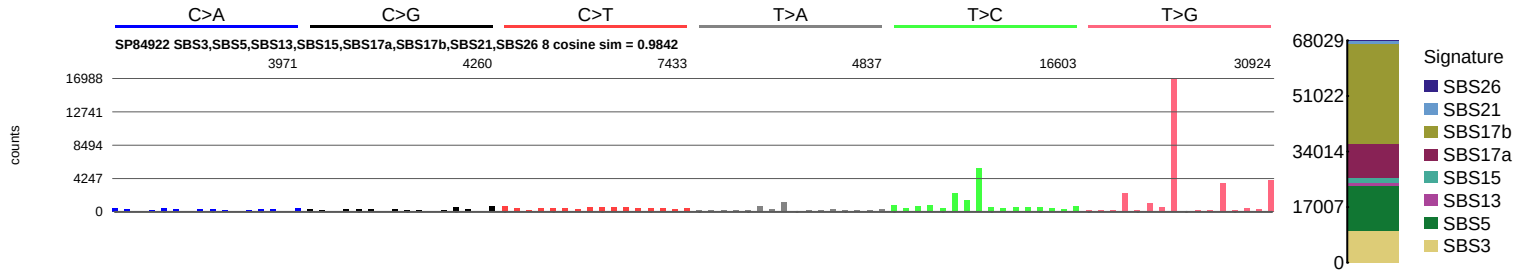
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.9842
4468
3345
8289
4063
17214
30652
counts
17036
12777
8518
4259
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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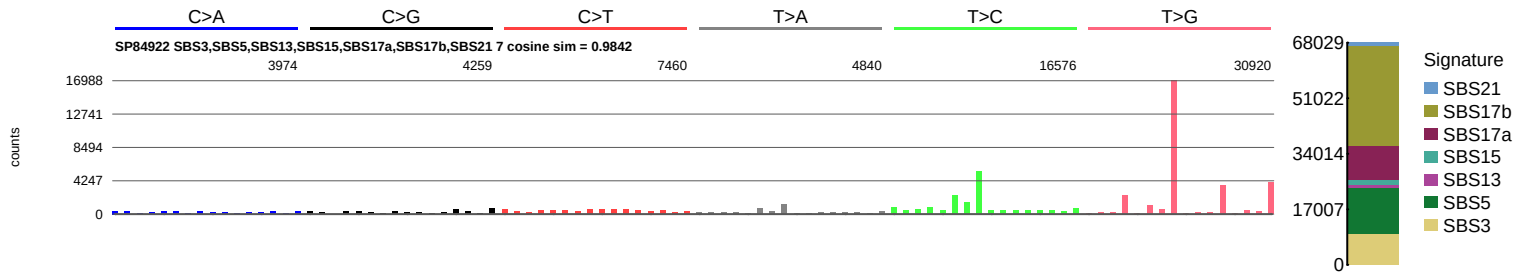
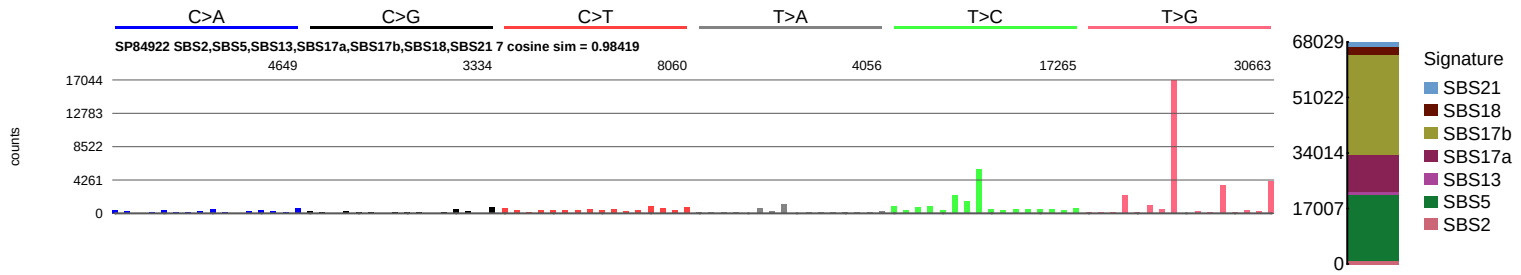
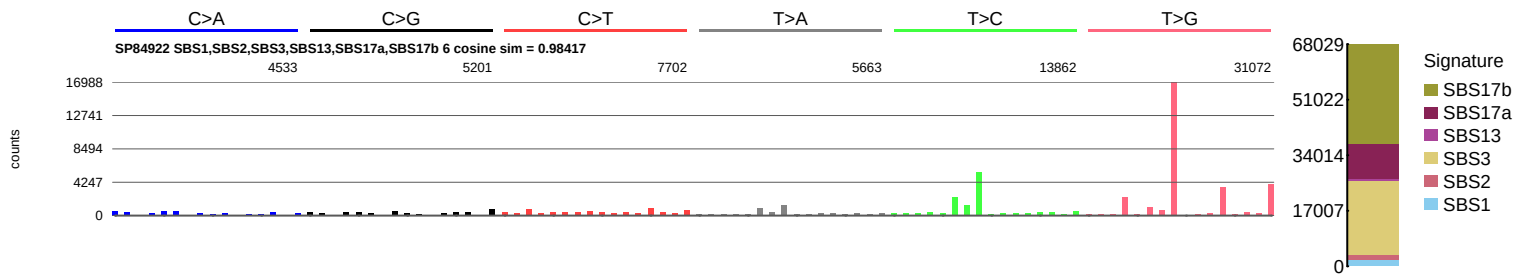
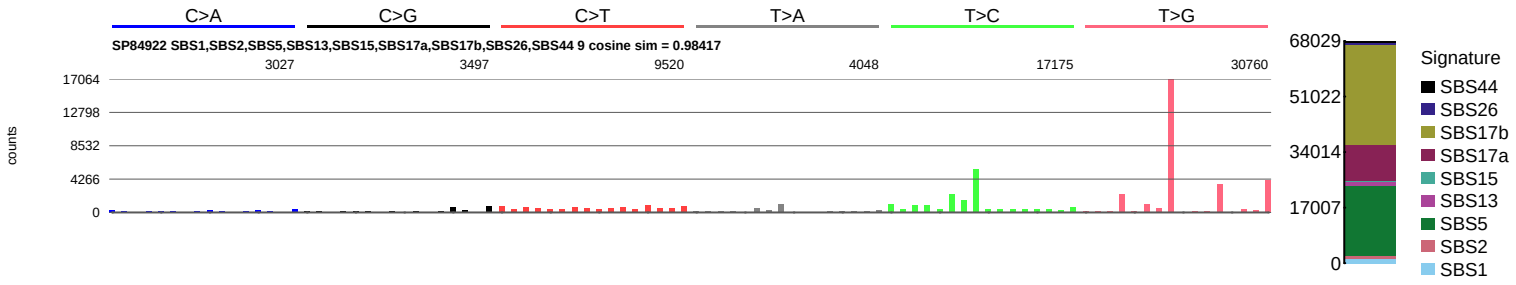
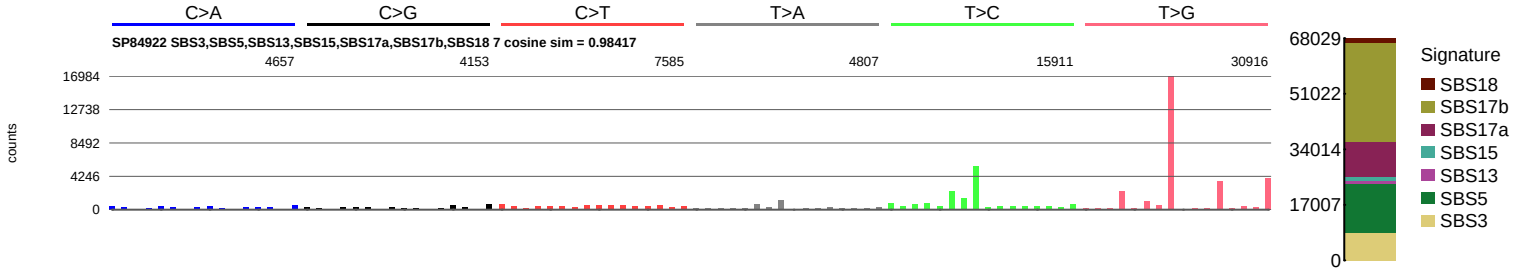
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26 8 cosine sim = 0.9842
3971
4260
7433
4837
16603
30924
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

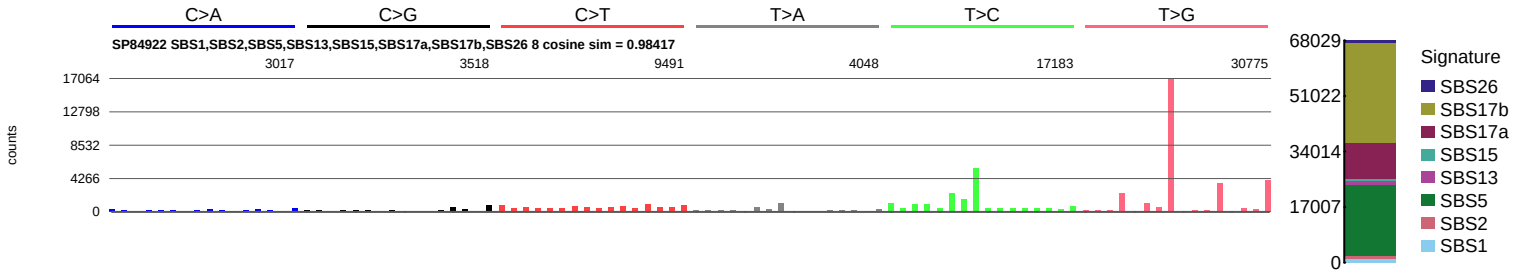

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.9842
3974
4259
7460
4840
16576
30920
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98419
4649
3334
8060
4056
17265
30663
counts
17044
12783
8522
4261
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b 6 cosine sim = 0.98417
4533
5201
7702
5663
13862
31072
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS44 9 cosine sim = 0.98417
3027
3497
9520
4048
17175
30760
counts
17064
12798
8532
4266
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98417
4657
4153
7585
4807
15911
30916
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26 8 cosine sim = 0.98417
3017
3518
9491
4048
17183
30775
counts
17064
12798
8532
4266
0
68029
51022
34014
17007
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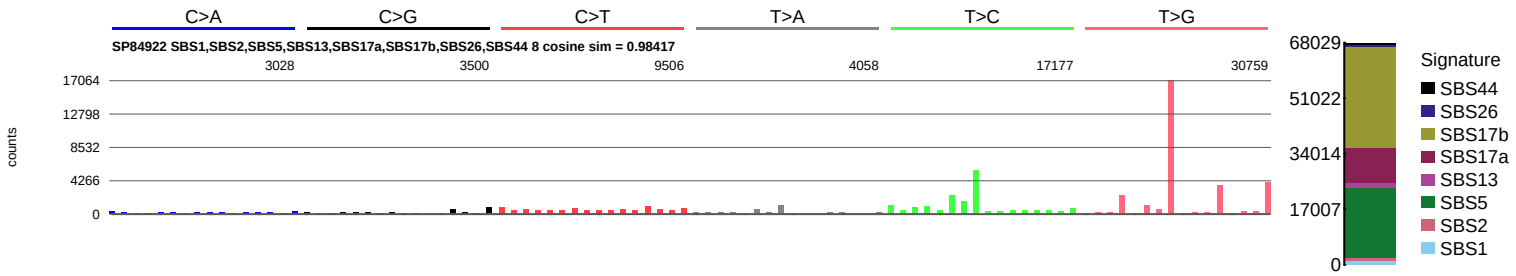
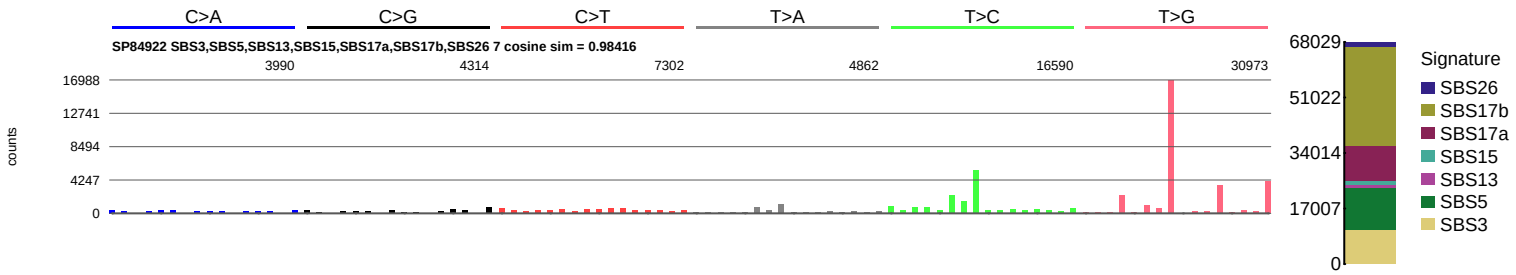
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98417
3028
3500
9506
4058
17177
30759
counts
17064
12798
8532
4266
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98416
3990
4314
7302
4862
16590
30973
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
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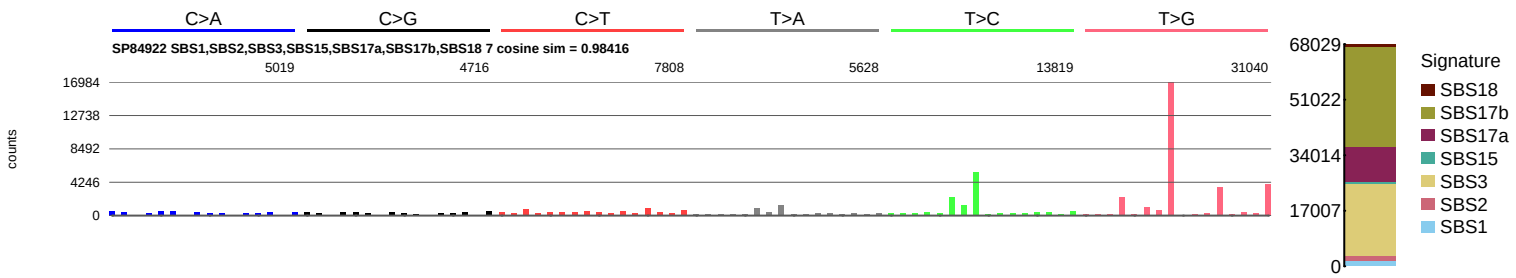
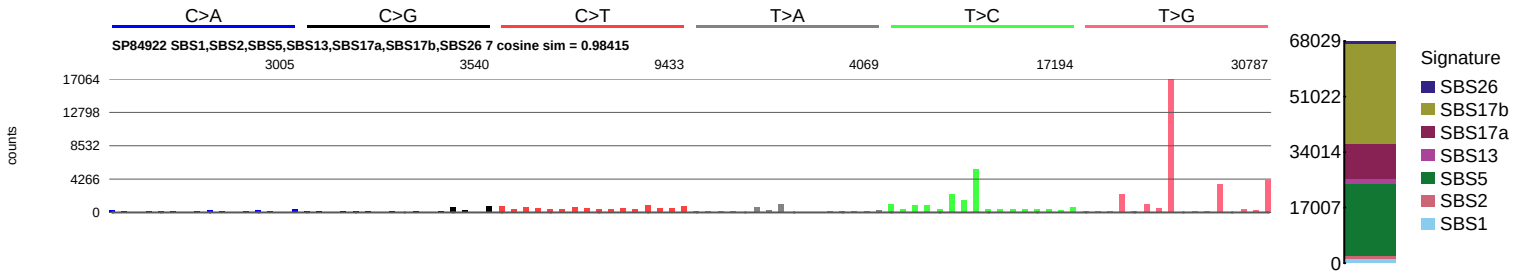
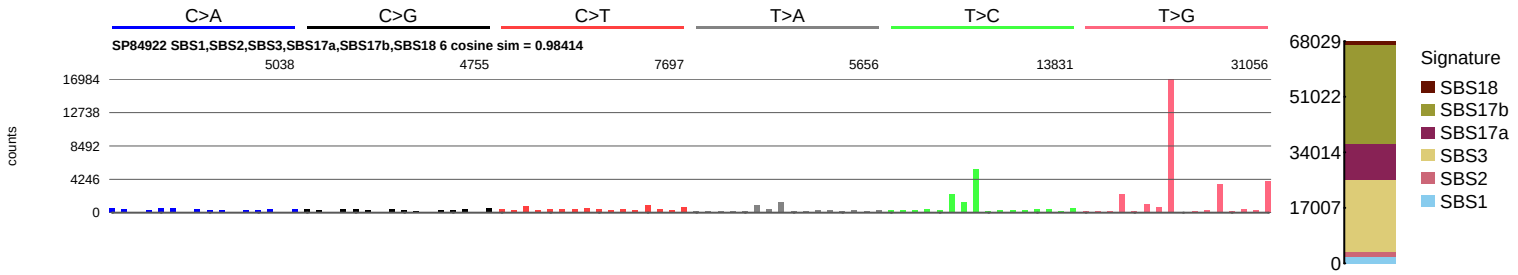
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98416
5019
4716
7808
5628
13819
31040
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98415
3005
3540
9433
4069
17194
30787
counts
17064
12798
8532
4266
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98414
5038
4755
7697
5656
13831
31056
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
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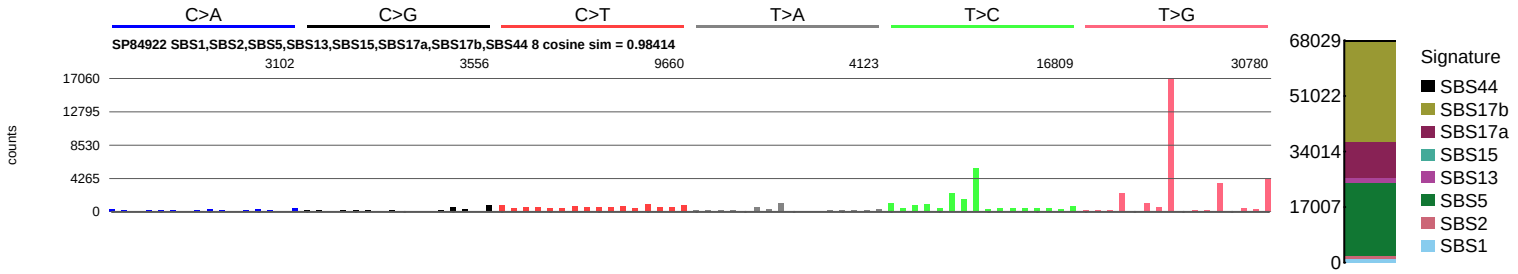
Signature
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS44 8 cosine sim = 0.98414
3102
3556
9660
4123
16809
30780
counts
17060
12795
8530
4265
0
68029
51022
34014
17007
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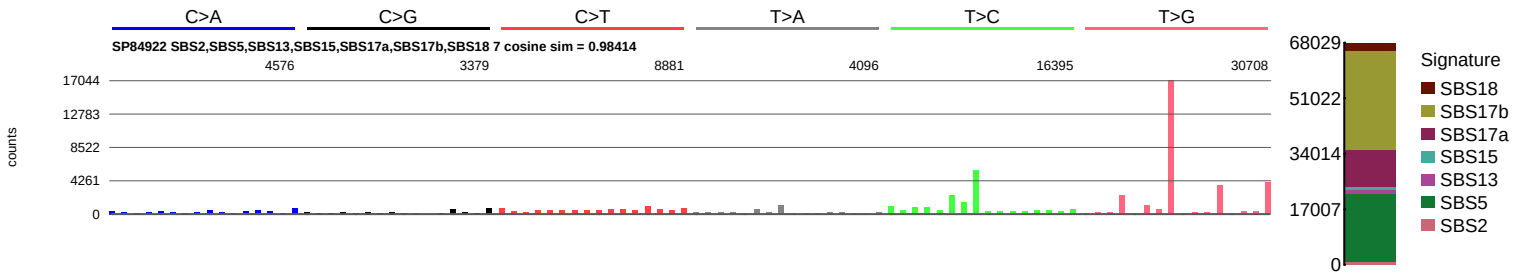
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98414
4576
3379
8881
4096
16395
30708
counts
17044
12783
8522
4261
0
68029
51022
34014
17007
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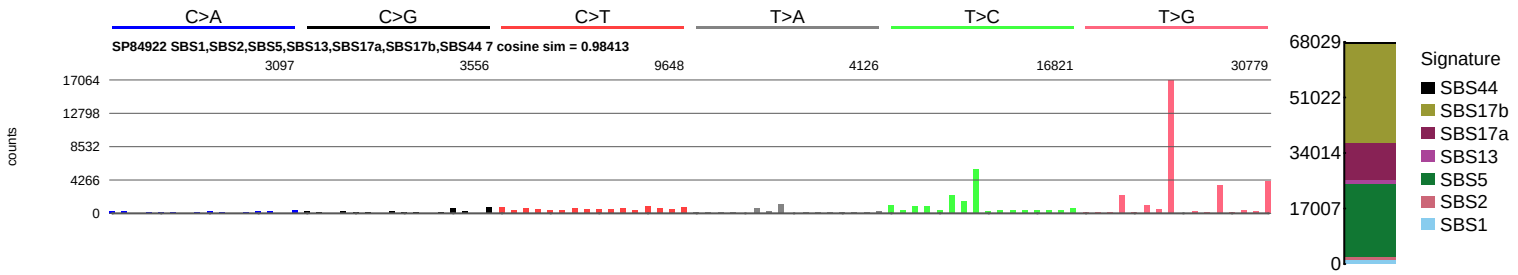
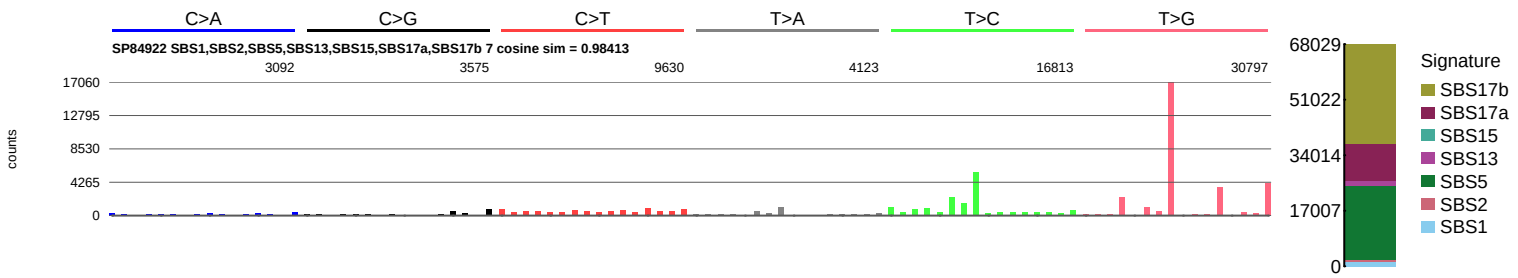
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98413
3097
3556
9648
4126
16821
30779
counts
17064
12798
8532
4266
0
68029
51022
34014
17007
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98413
3092
3575
9630
4123
16813
30797
counts
17060
12795
8530
4265
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
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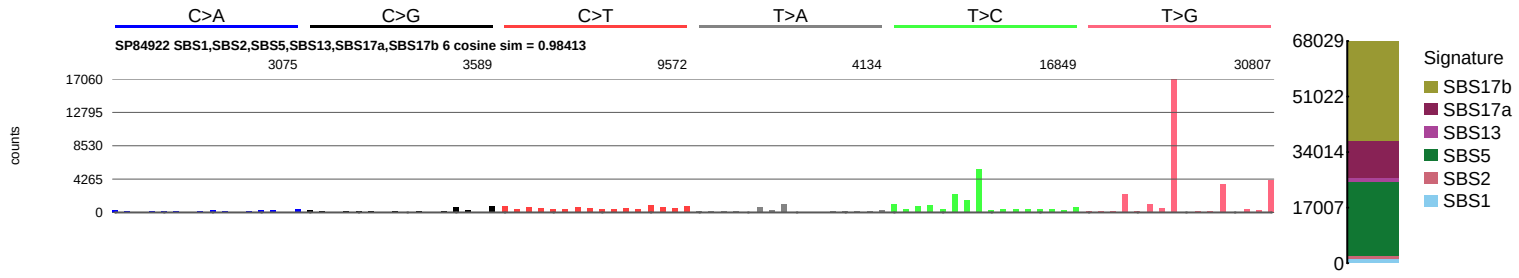
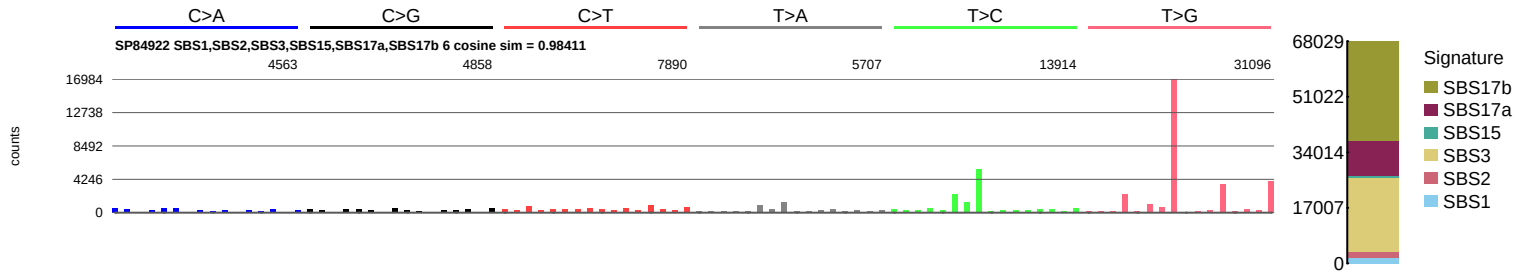
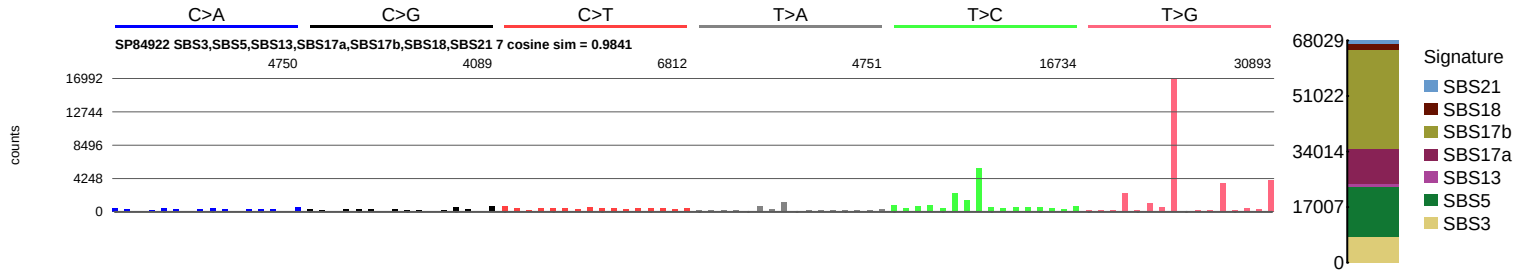
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b 6 cosine sim = 0.98413
3075
3589
9572
4134
16849
30807
counts
17060
12795
8530
4265
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98411
4563
4858
7890
5707
13914
31096
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
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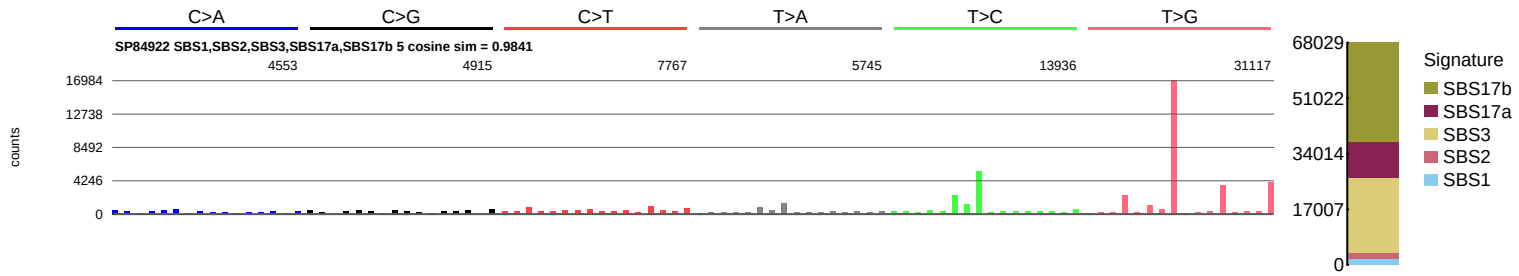
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.9841
4750
4089
6812
4751
16734
30893
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3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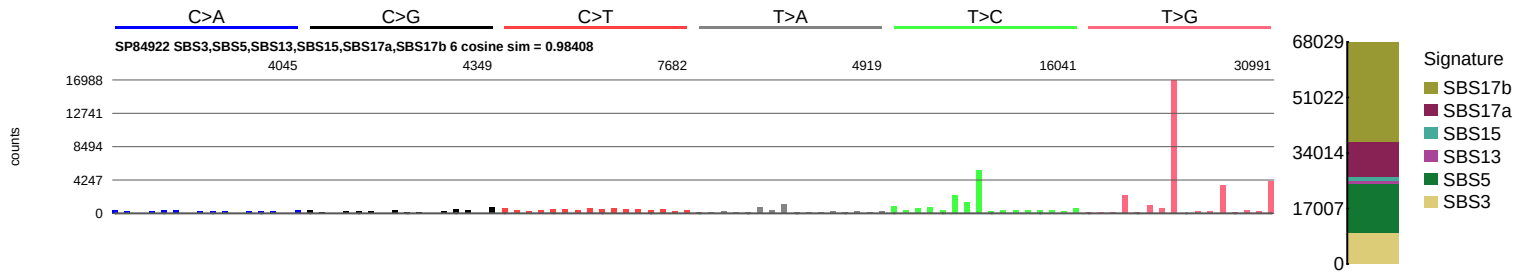
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS3,SBS17a,SBS17b 5 cosine sim = 0.9841
4553
4915
7767
5745
13936
31117
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS3
SBS2
SBS1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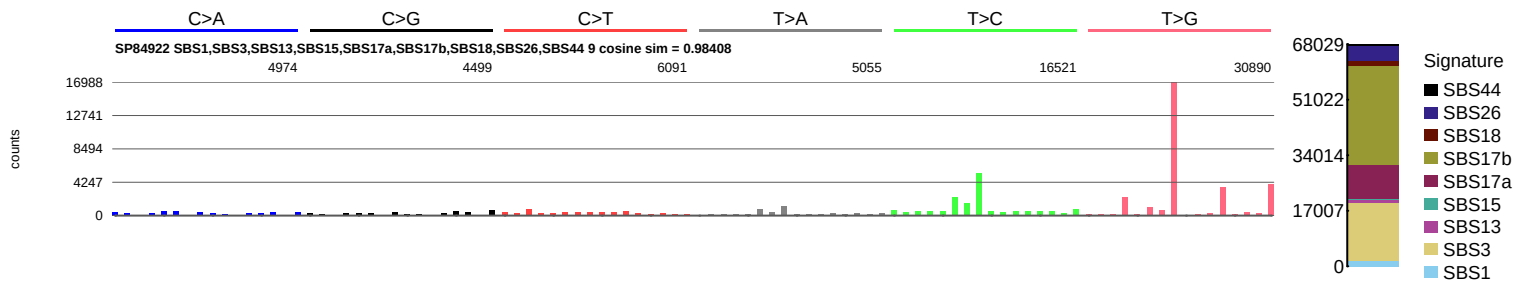
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98408
4045
4349
7682
4919
16041
30991
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3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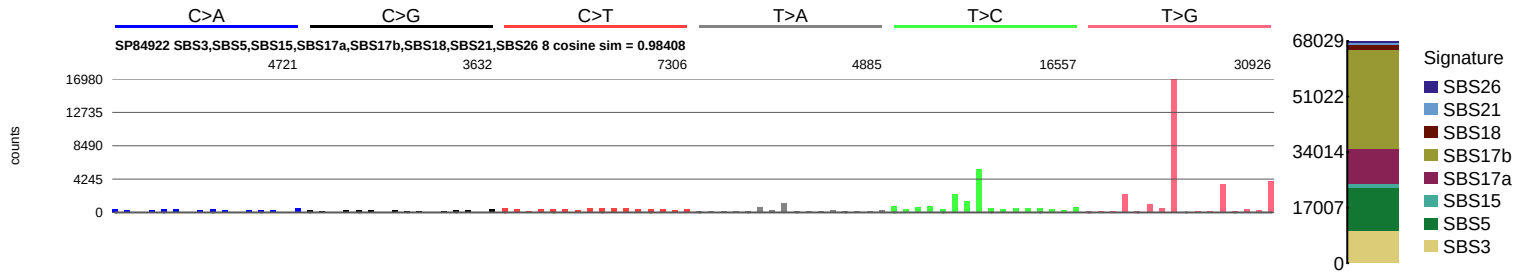
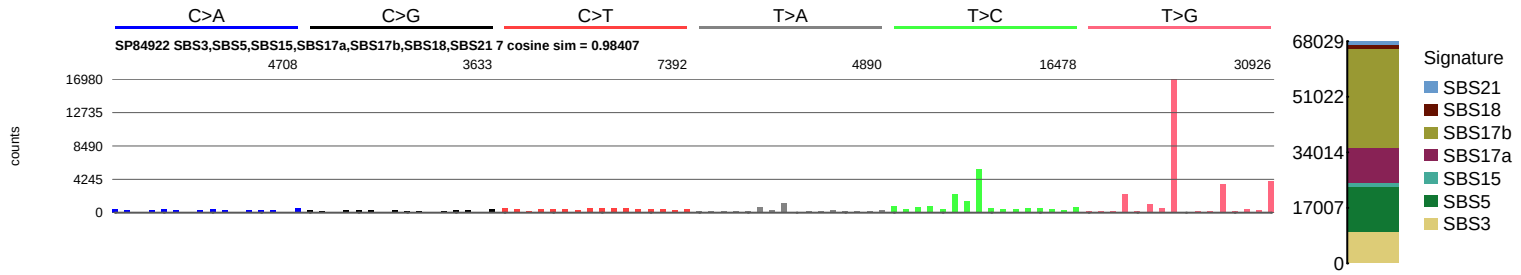
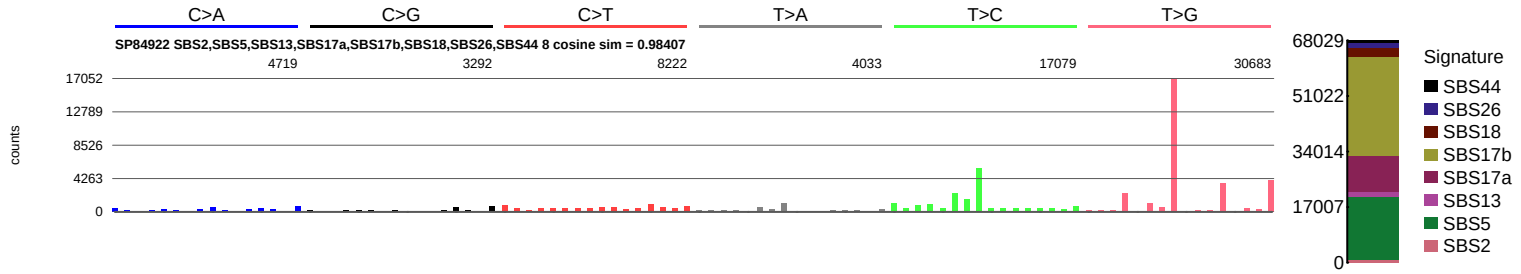
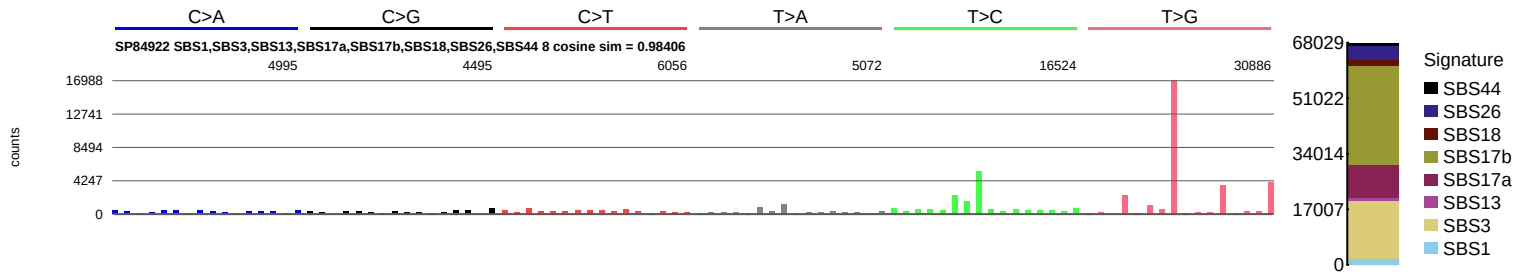
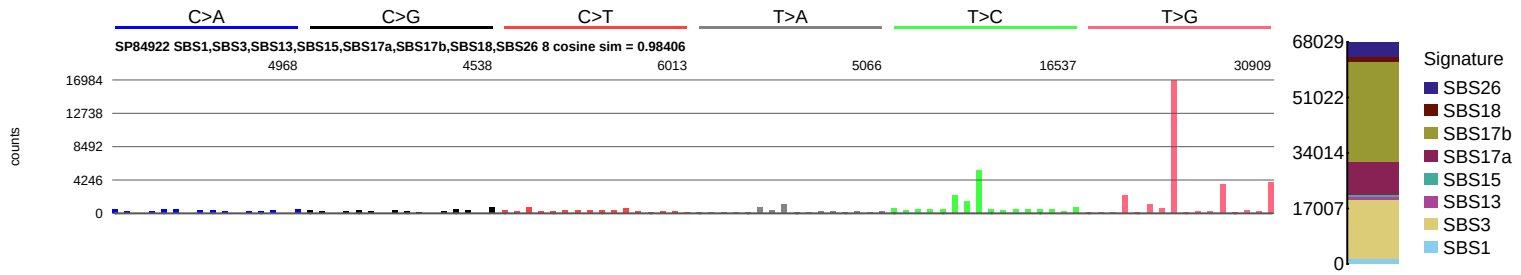
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98408
4974
4499
6091
5055
16521
30890
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

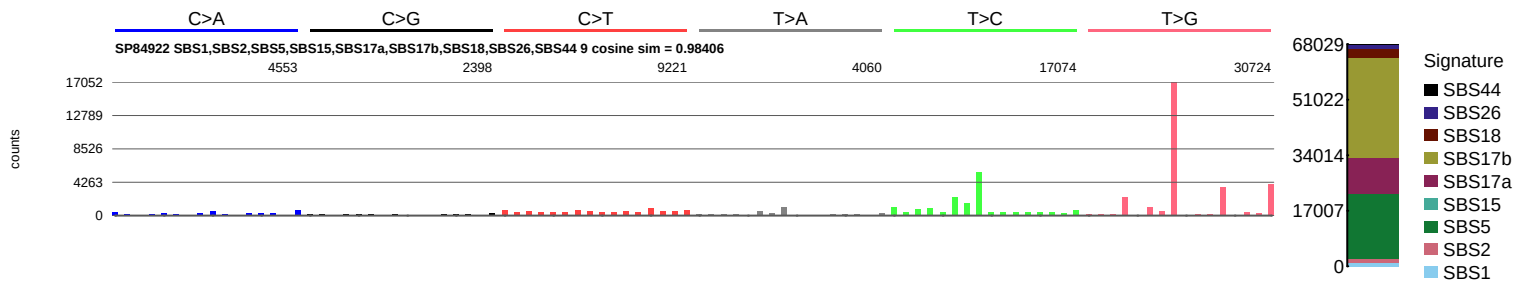

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98408
4721
3632
7306
4885
16557
30926
counts
16980
12735
8490
4245
0
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3

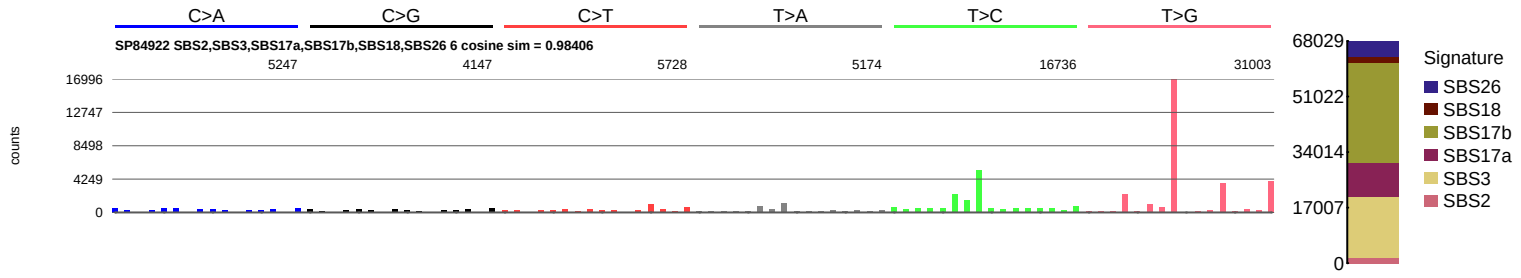

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98407
4708
3633
7392
4890
16478
30926
counts
16980
12735
8490
4245
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3

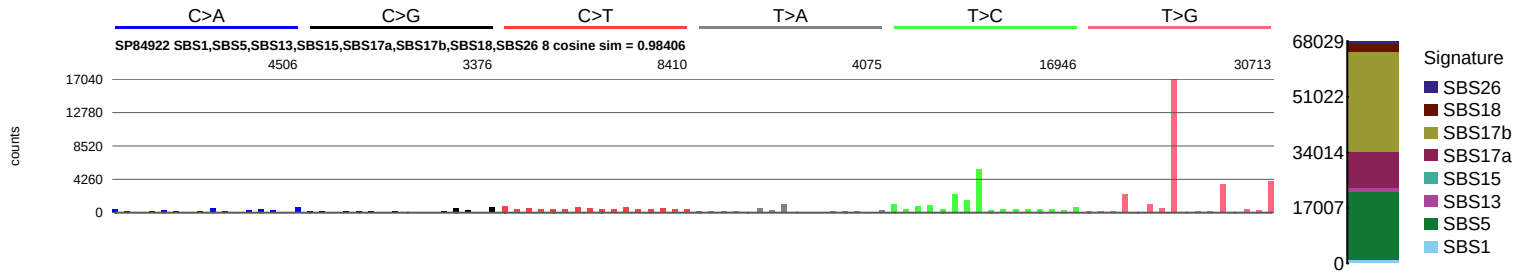

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98407
4719
3292
8222
4033
17079
30683
counts
17052
12789
8526
4263
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

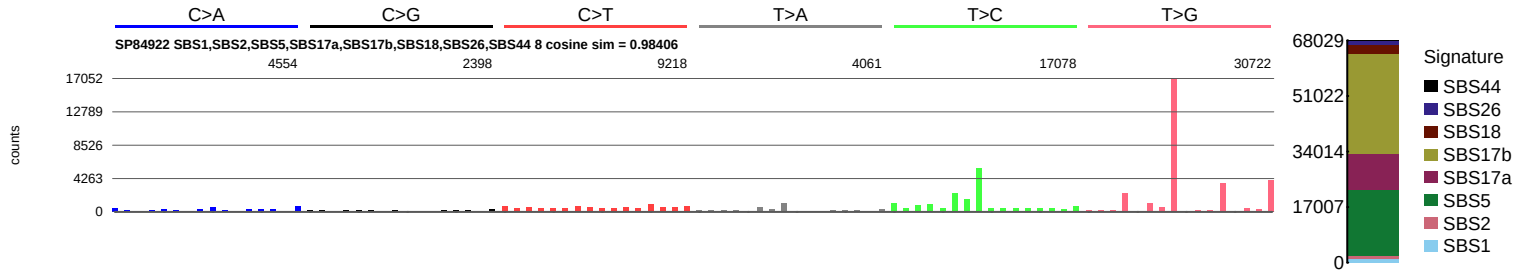

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98406
4995
4495
6056
5072
16524
30886
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

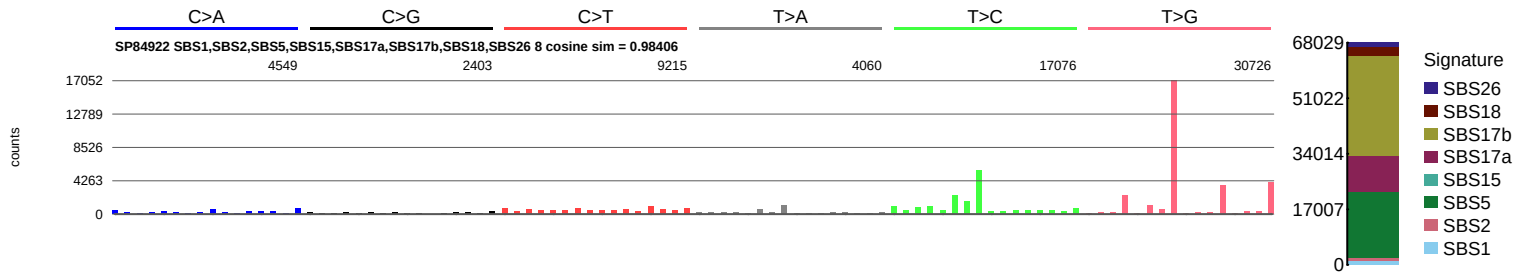

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98406
4968
4538
6013
5066
16537
30909
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

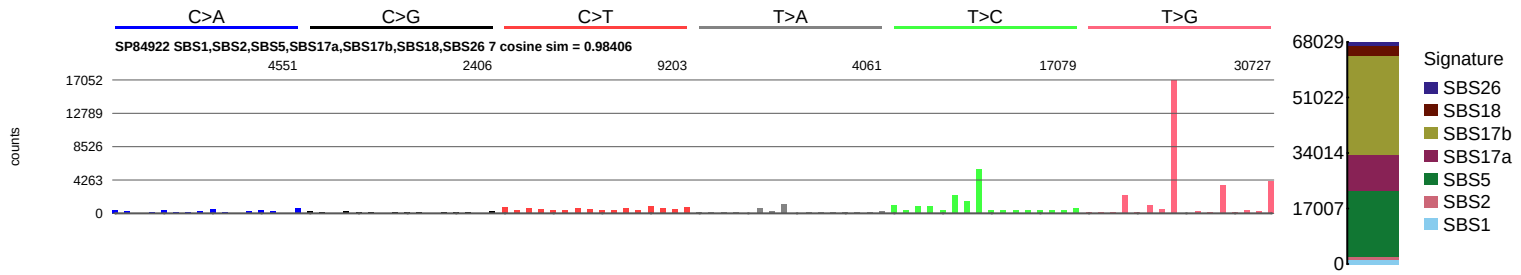

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98406
4553
2398
9221
4060
17074
30724
counts
17052
12789
8526
4263
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

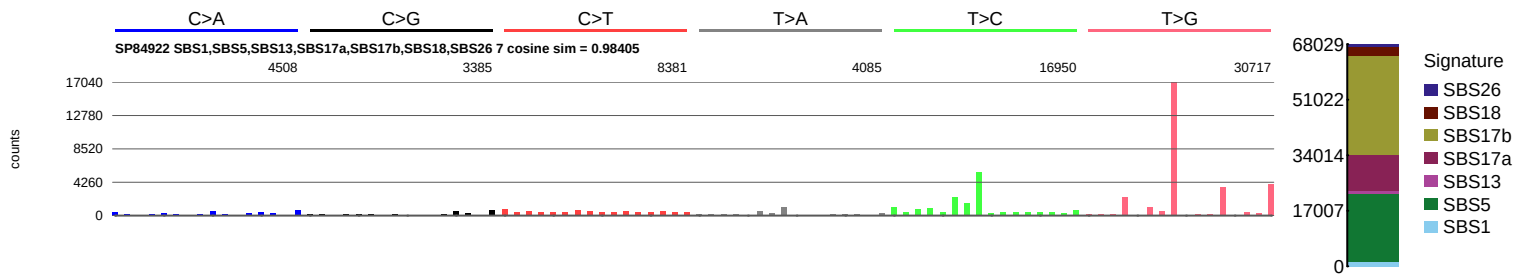

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98406
5247
4147
5728
5174
16736
31003
counts
16996
12747
8498
4249
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2

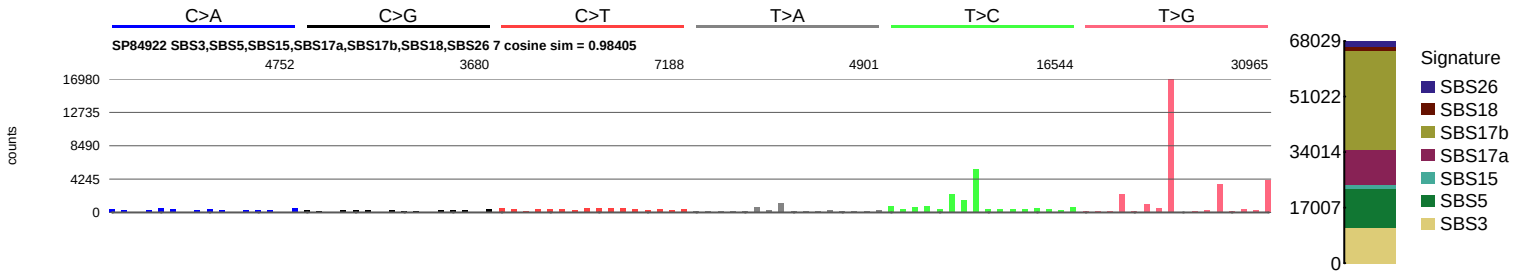

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98406
4506
3376
8410
4075
16946
30713
counts
17040
12780
8520
4260
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

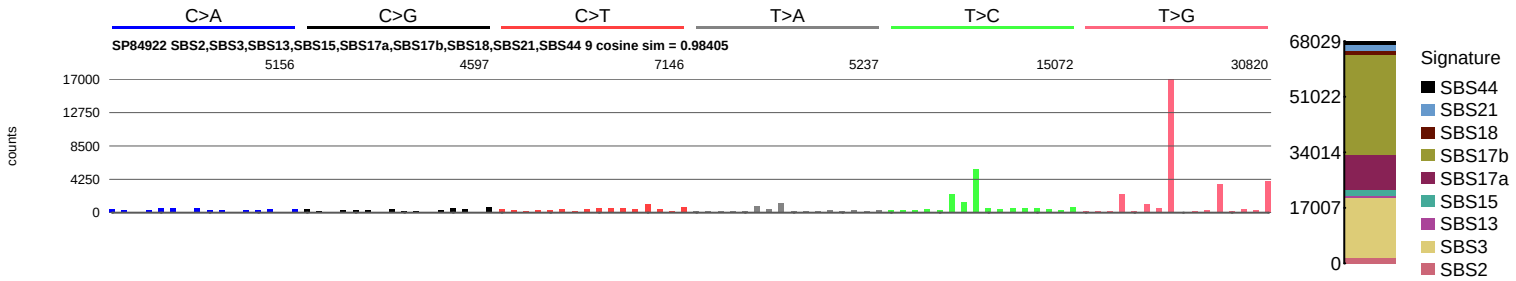

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98406
4554
2398
9218
4061
17078
30722
counts
17052
12789
8526
4263
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

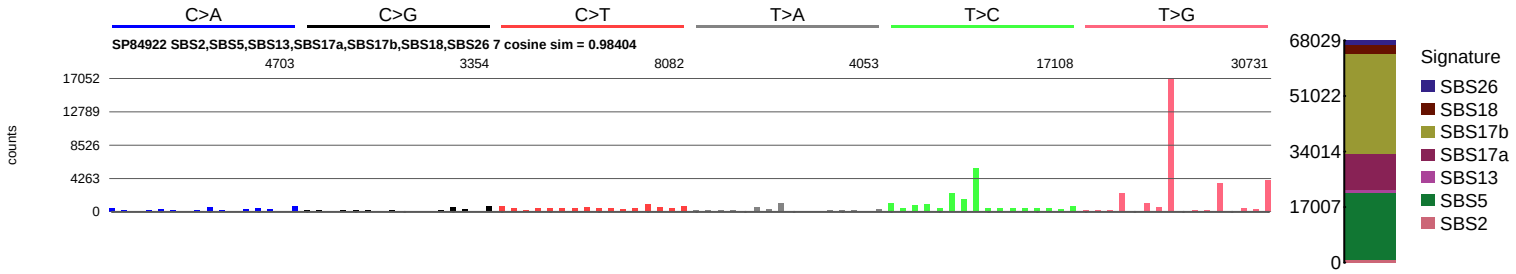

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98406
4549
2403
9215
4060
17076
30726
counts
17052
12789
8526
4263
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

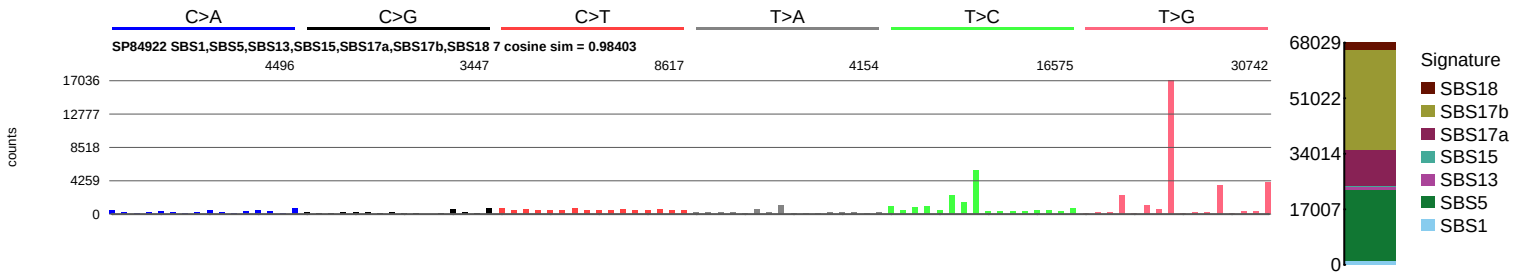

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98406
4551
2406
9203
4061
17079
30727
counts
17052
12789
8526
4263
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

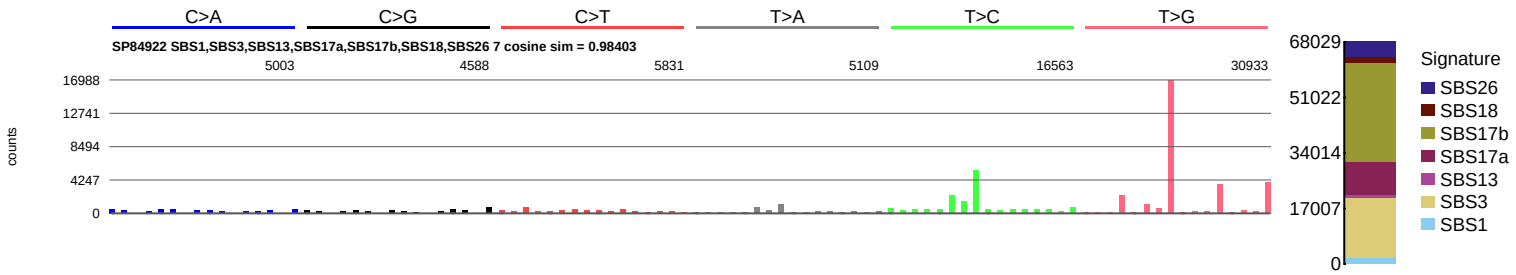

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98405
4508
3385
8381
4085
16950
30717
counts
17040
12780
8520
4260
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

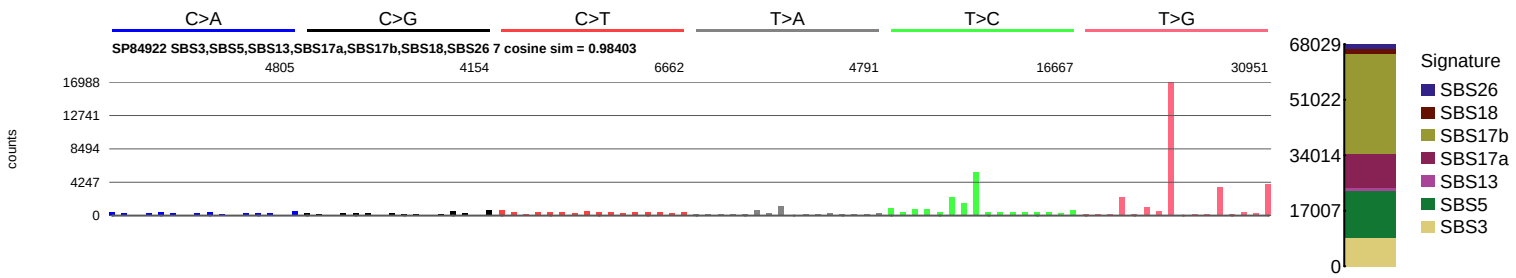

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98405
4752
3680
7188
4901
16544
30965
counts
16980
12735
8490
4245
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3

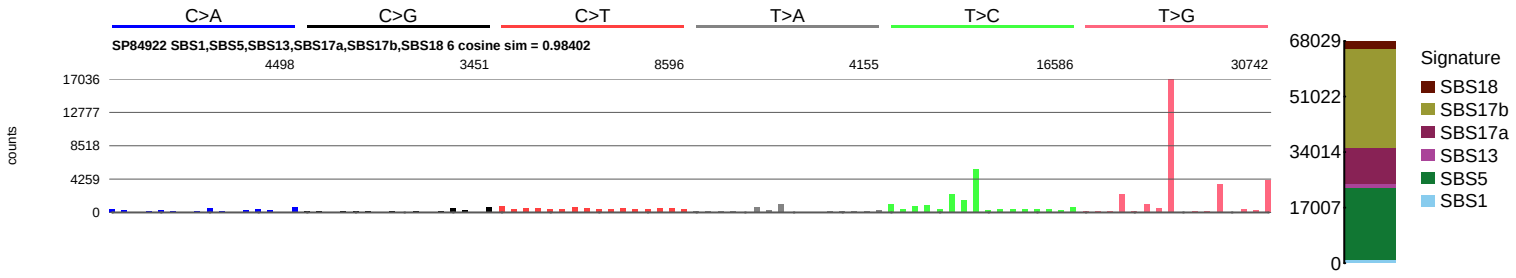

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98405
5156
4597
7146
5237
15072
30820
counts
17000
12750
8500
4250
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

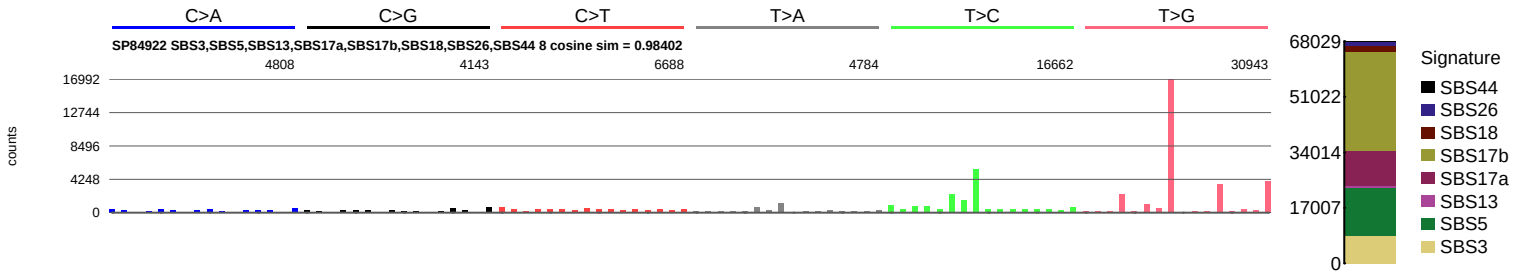

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98404
4703
3354
8082
4053
17108
30731
counts
17052
12789
8526
4263
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

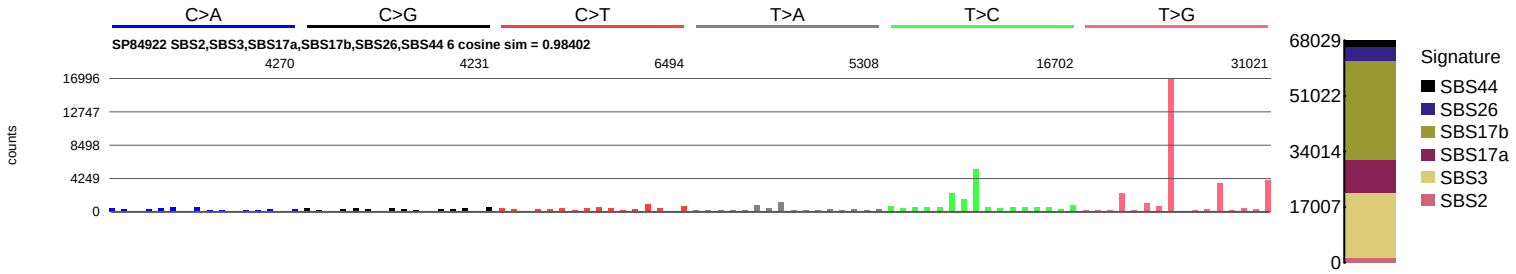

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98403
4496
3447
8617
4154
16575
30742
counts
17036
12777
8518
4259
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

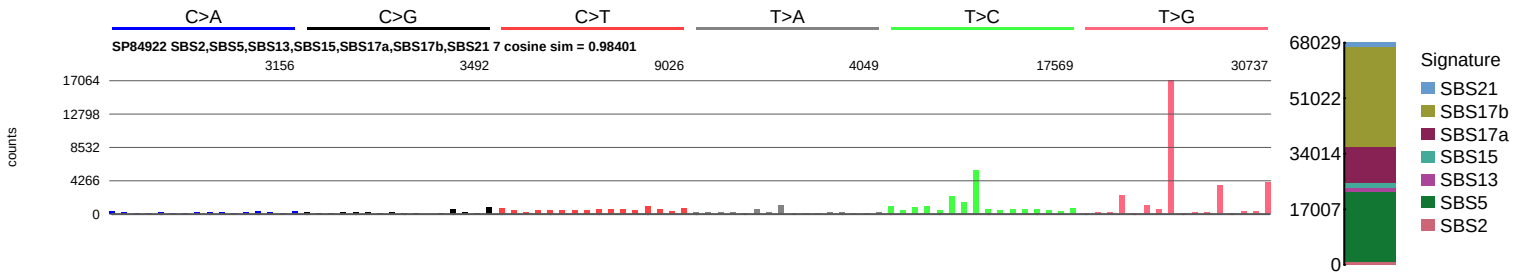

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98403
5003
4588
5831
5109
16563
30933
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98403
4805
4154
6662
4791
16667
30951
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98402
4498
3451
8596
4155
16586
30742
counts
17036
12777
8518
4259
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98402
4808
4143
6688
4784
16662
30943
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS44 6 cosine sim = 0.98402
4270
4231
6494
5308
16702
31021
counts
16996
12747
8498
4249
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS3
SBS2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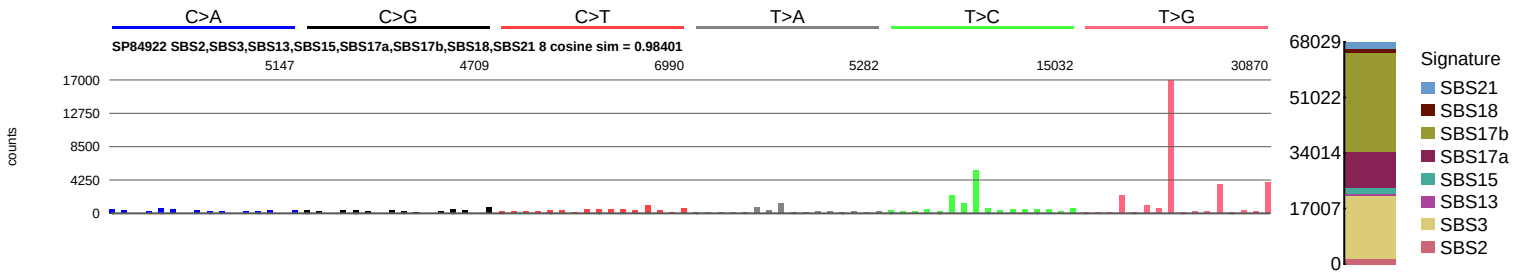
T>G
SP84922 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98401
3156
3492
9026
4049
17569
30737
counts
17064
12798
8532
4266
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2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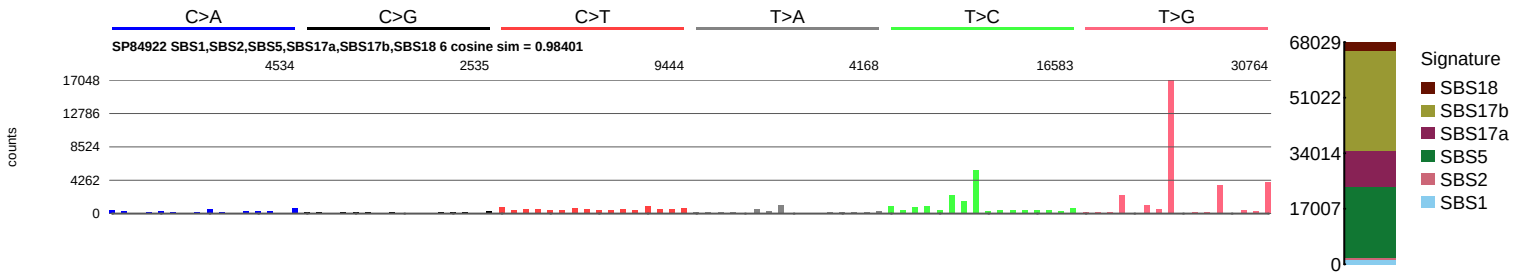
T>G
SP84922 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98401
5147
4709
6990
5282
15032
30870
counts
17000
12750
8500
4250
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2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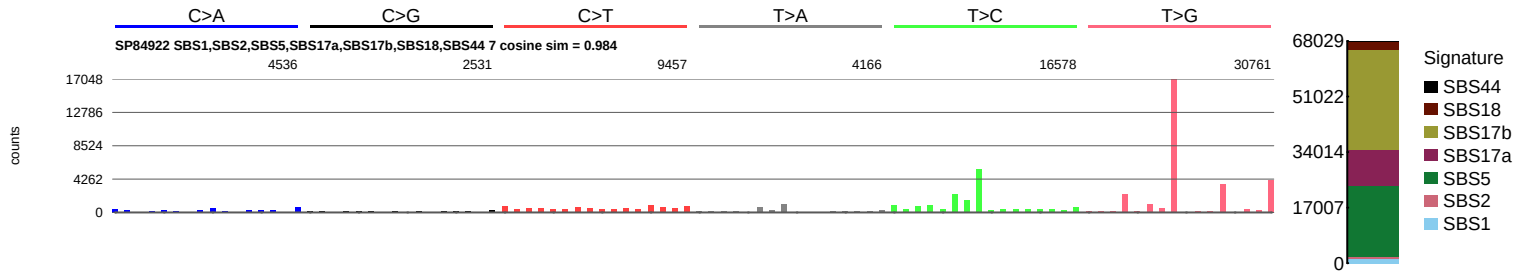
T>G
SP84922 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98401
4534
2535
9444
4168
16583
30764
counts
17048
12786
8524
4262
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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SP84922 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.984
4536
2531
9457
4166
16578
30761
counts
17048
12786
8524
4262
0
68029
51022
34014
17007
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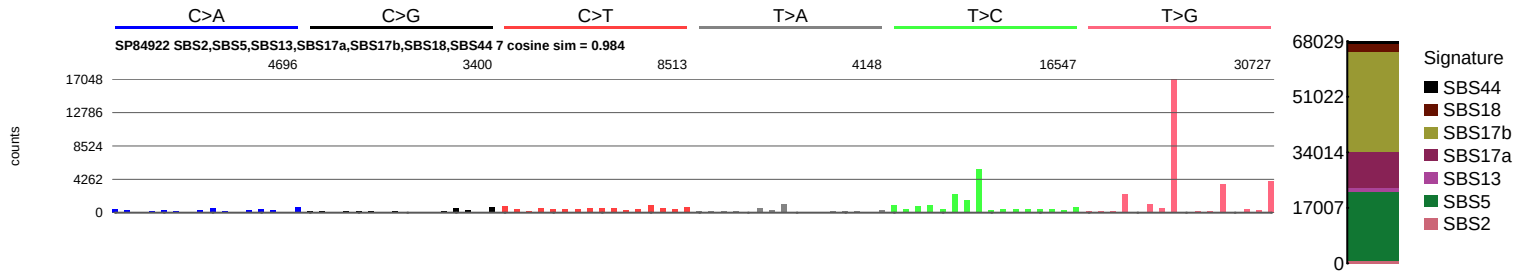
SBS2
SBS1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C>G
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T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.984
4696
3400
8513
4148
16547
30727
counts
17048
12786
8524
4262
0
68029
51022
34014
17007
0
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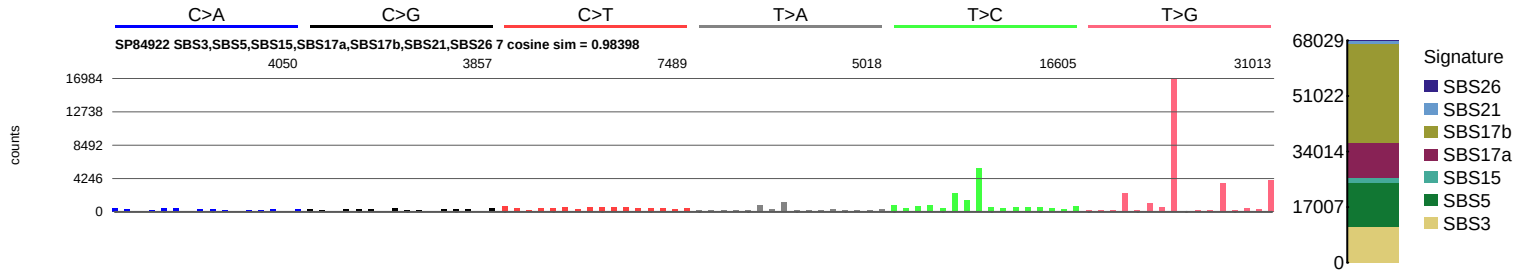
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2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C>T
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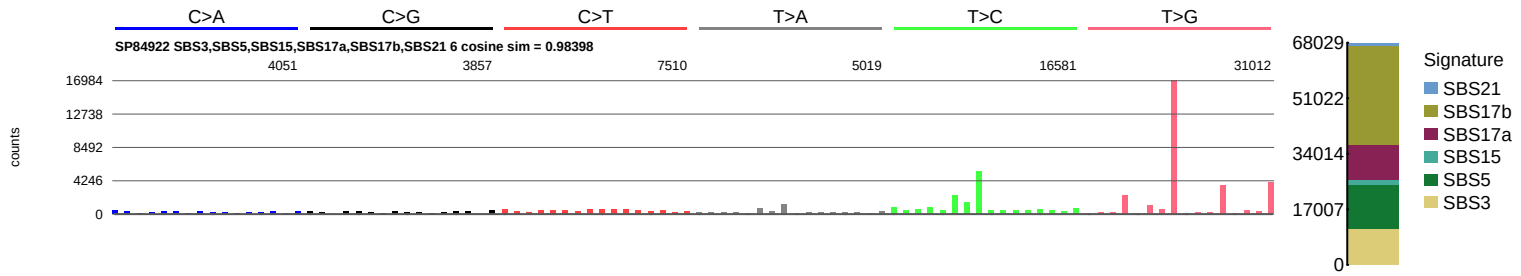
T>C
T>G
SP84922 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS26 7 cosine sim = 0.98398
4050
3857
7489
5018
16605
31013
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3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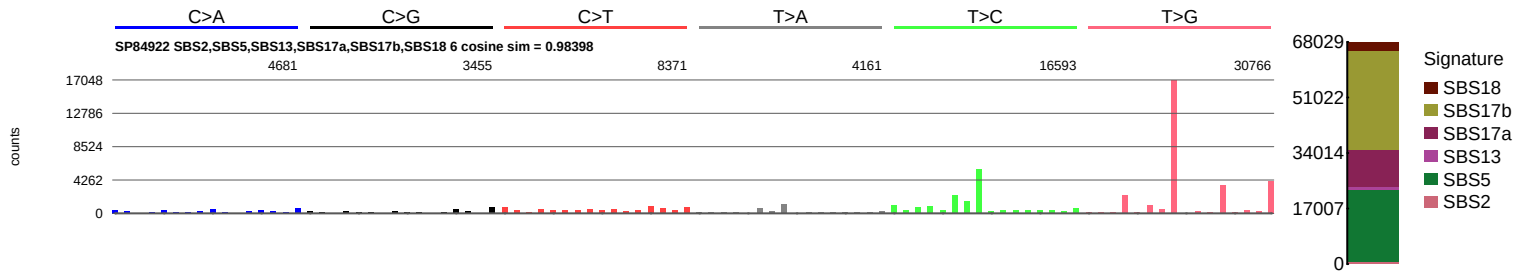
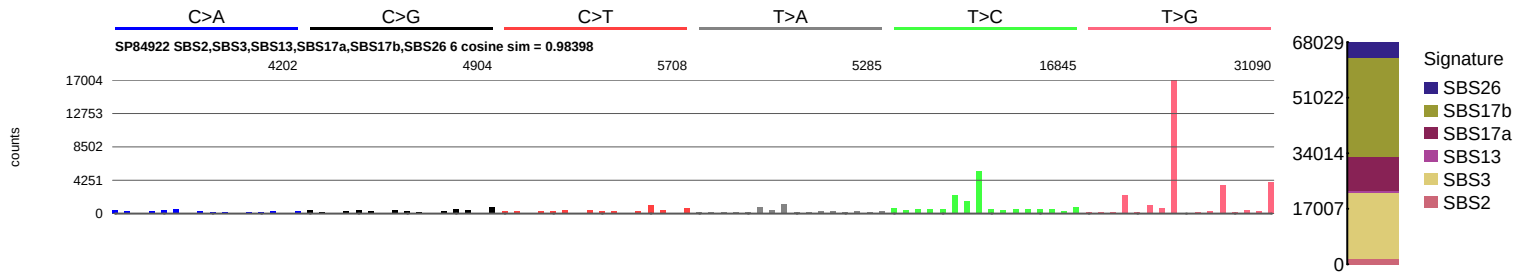
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98398
4051
3857
7510
5019
16581
31012
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98398
4681
3455
8371
4161
16593
30766
counts
17048
12786
8524
4262
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2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C>G
C>T
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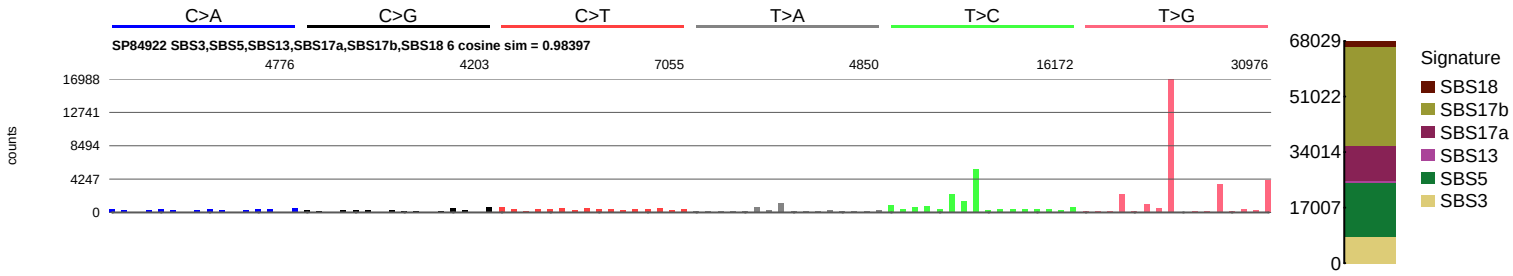
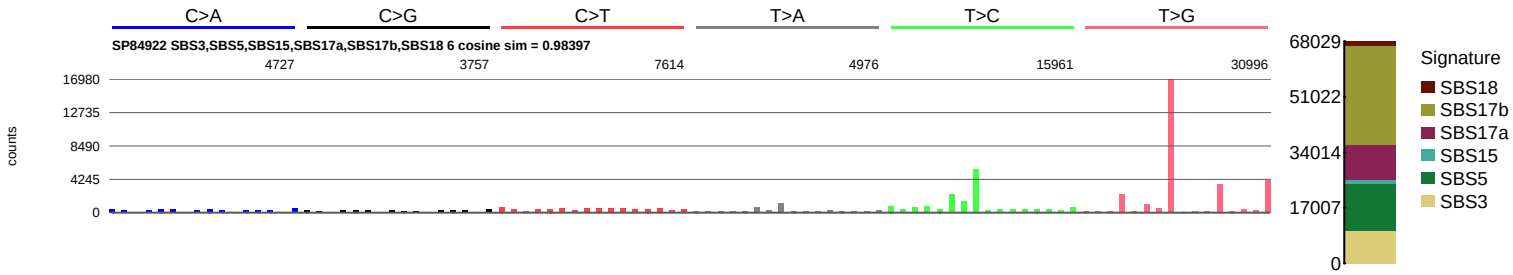
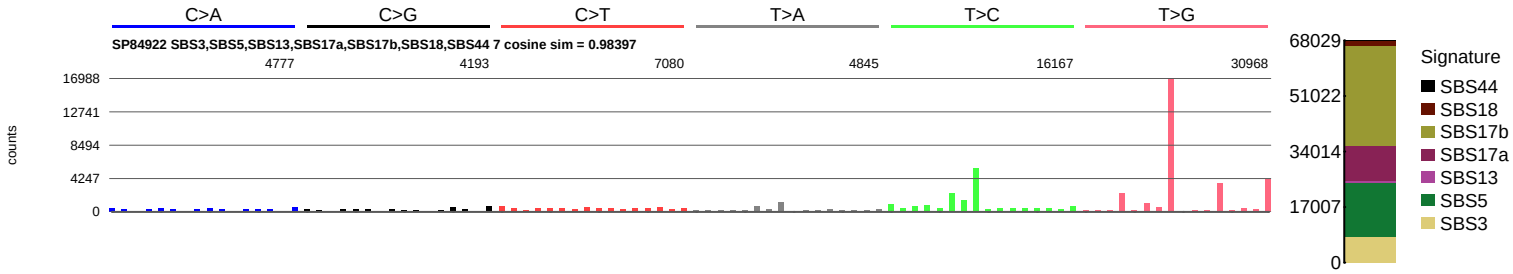
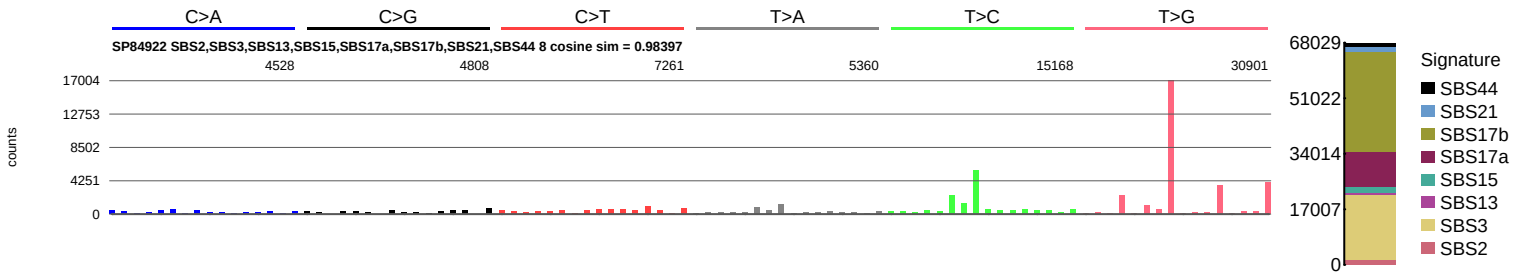
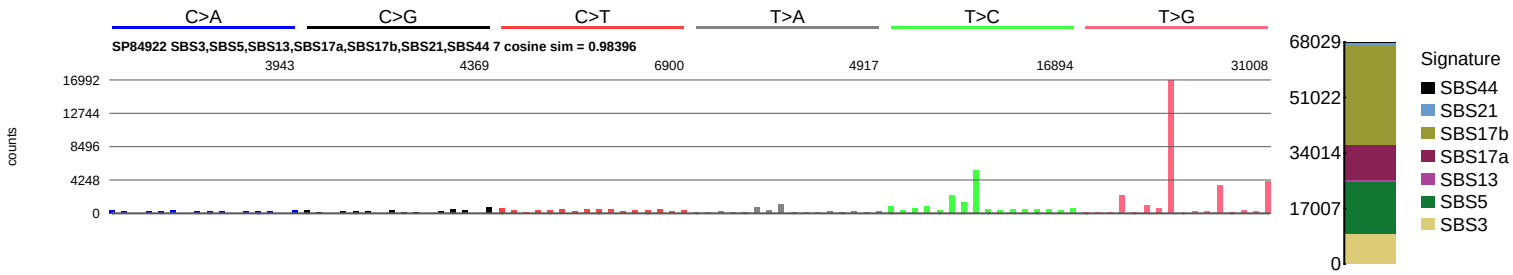
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98398
4202
4904
5708
5285
16845
31090
counts
17004
12753
8502
4251
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98397
4776
4203
7055
4850
16172
30976
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98397
4727
3757
7614
4976
15961
30996
counts
16980
12735
8490
4245
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98397
4777
4193
7080
4845
16167
30968
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98397
4528
4808
7261
5360
15168
30901
counts
17004
12753
8502
4251
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2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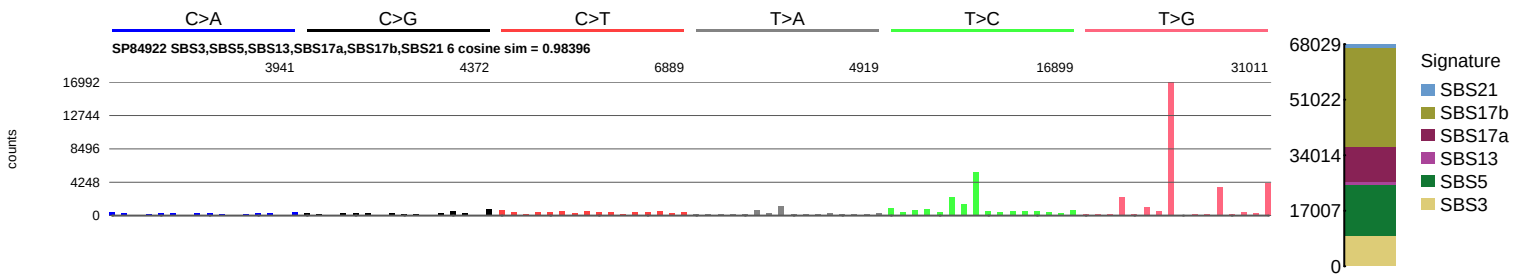
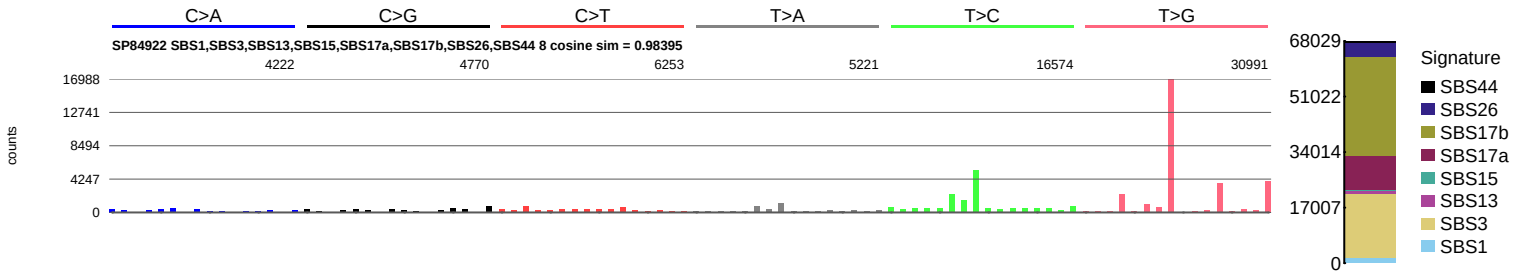
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98396
3943
4369
6900
4917
16894
31008
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98396
3941
4372
6889
4919
16899
31011
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3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C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98395
4222
4770
6253
5221
16574
30991
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
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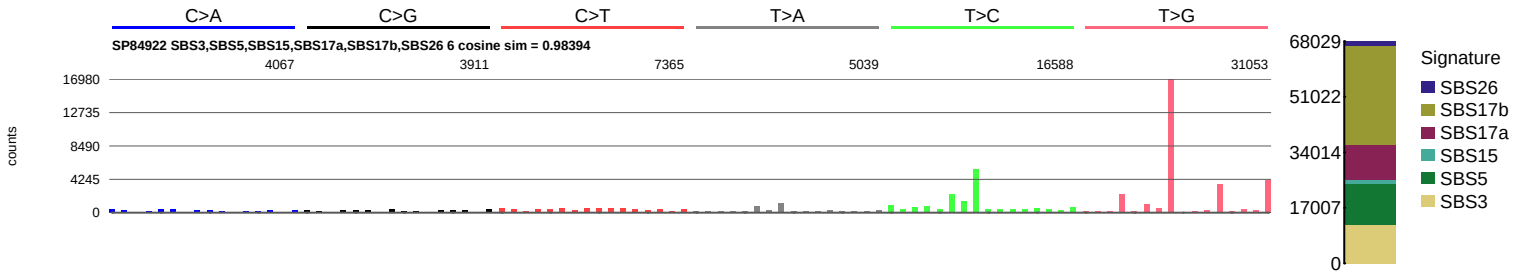
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1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C>G
C>T
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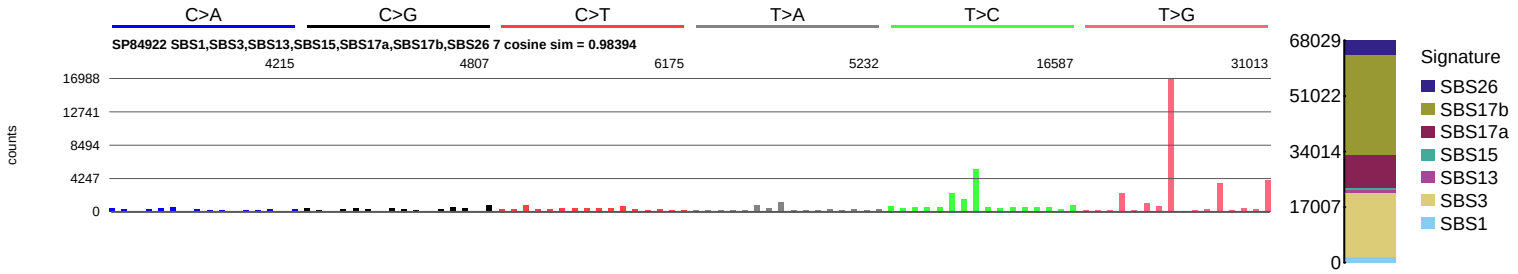
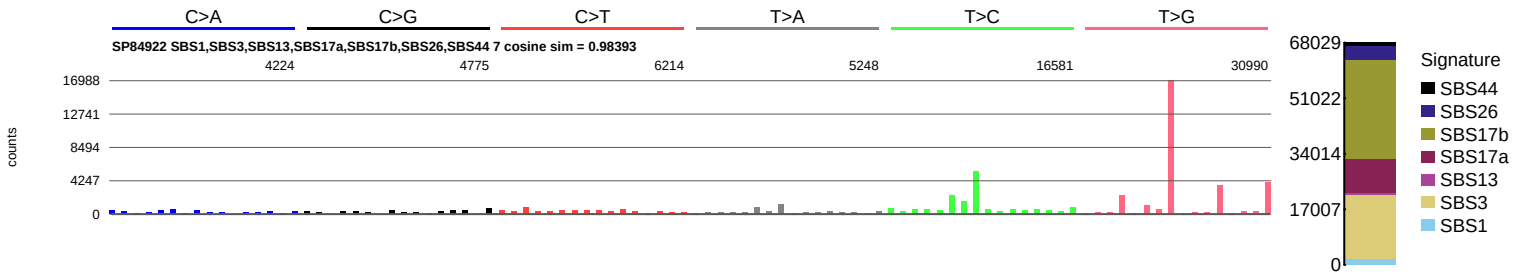
T>C
T>G
SP84922 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98394
4067
3911
7365
5039
16588
31053
counts
16980
12735
8490
4245
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98394
4215
4807
6175
5232
16587
31013
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98393
4224
4775
6214
5248
16581
30990
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
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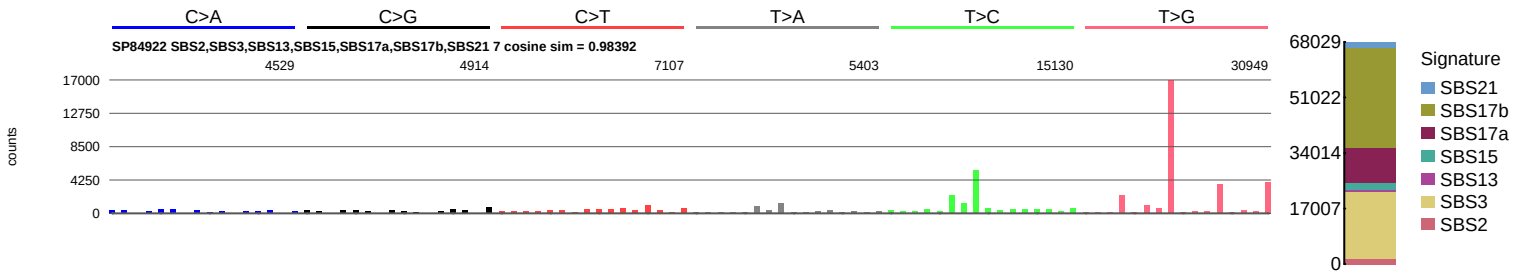
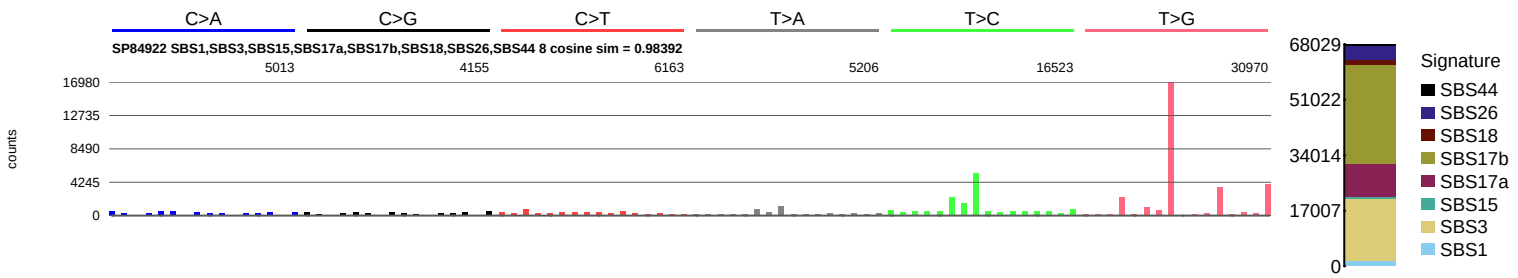
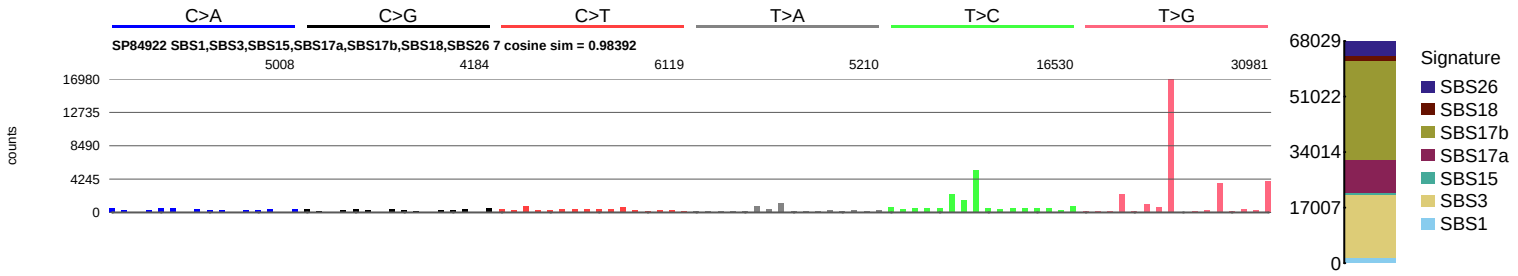
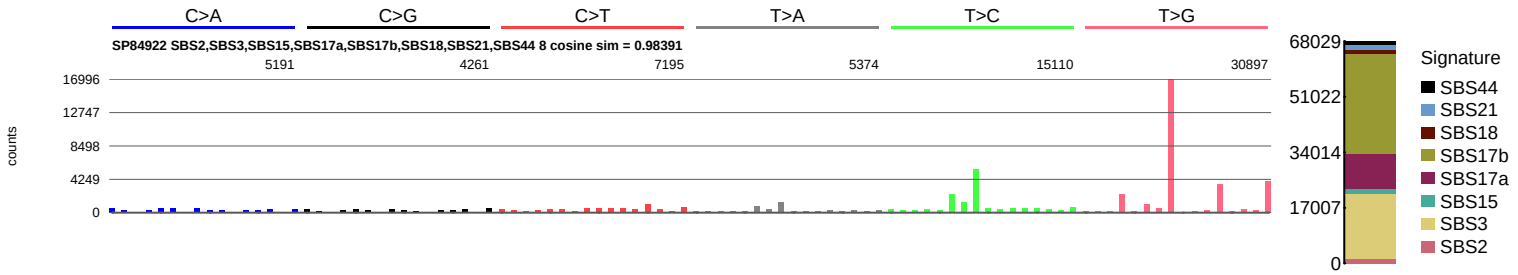
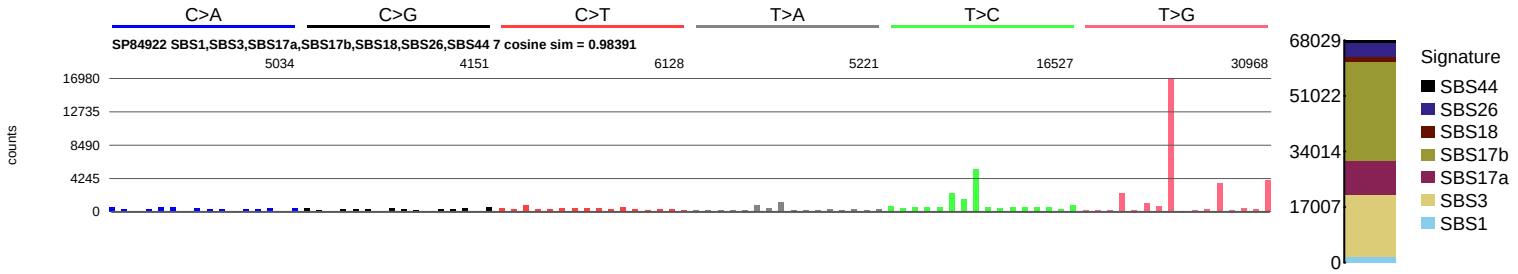
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS1

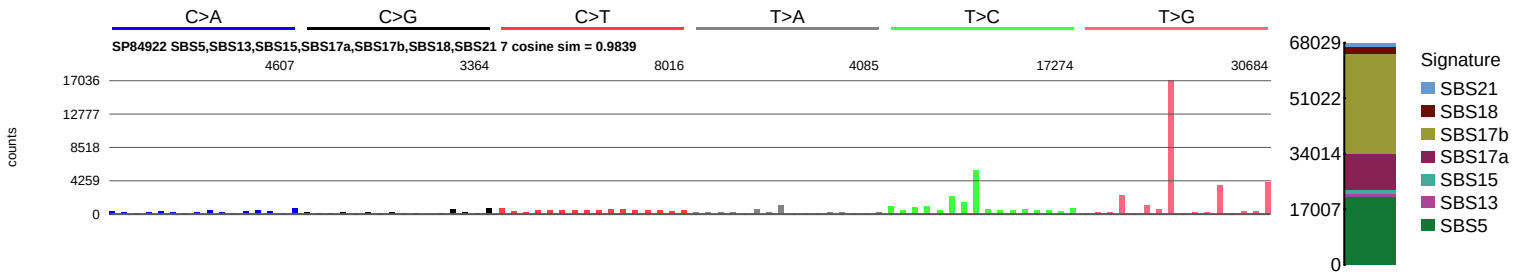

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98392
4529
4914
7107
5403
15130
30949
counts
17000
12750
8500
4250
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

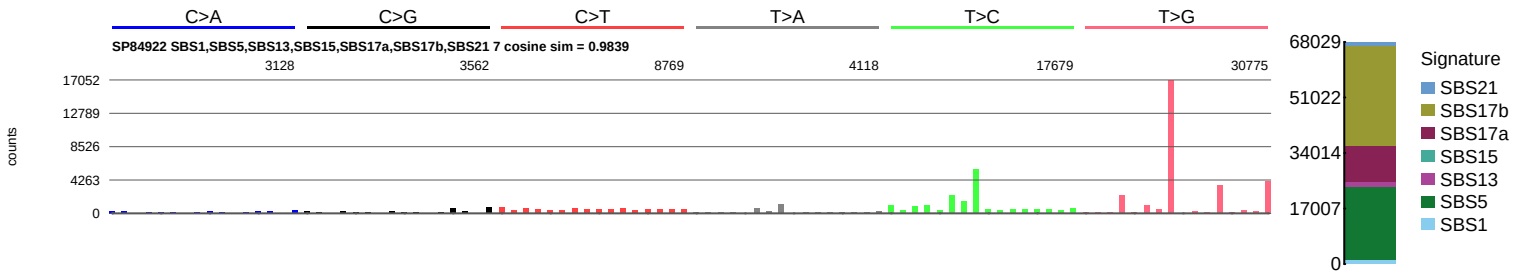

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98392
5013
4155
6163
5206
16523
30970
counts
16980
12735
8490
4245
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98392
5008
4184
6119
5210
16530
30981
counts
16980
12735
8490
4245
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98391
5191
4261
7195
5374
15110
30897
counts
16996
12747
8498
4249
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98391
5034
4151
6128
5221
16527
30968
counts
16980
12735
8490
4245
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.9839
4607
3364
8016
4085
17274
30684
counts
17036
12777
8518
4259
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5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C>G
C>T
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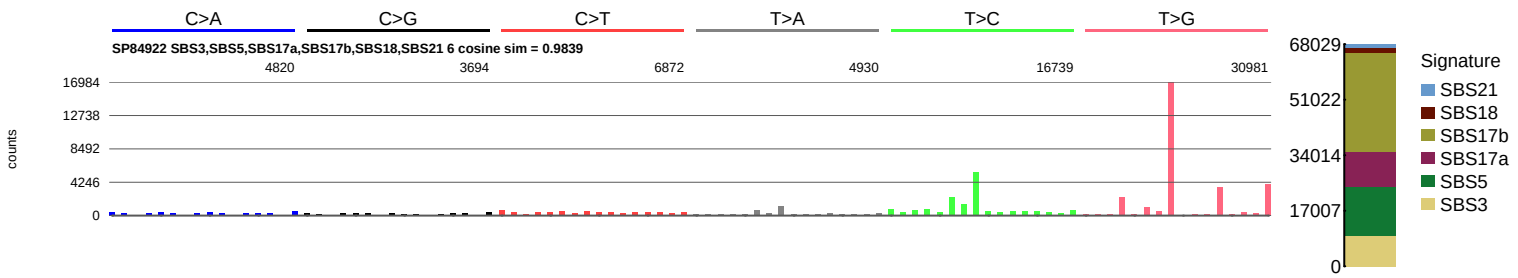
T>C
T>G
SP84922 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.9839
3128
3562
8769
4118
17679
30775
counts
17052
12789
8526
4263
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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C>G
C>T
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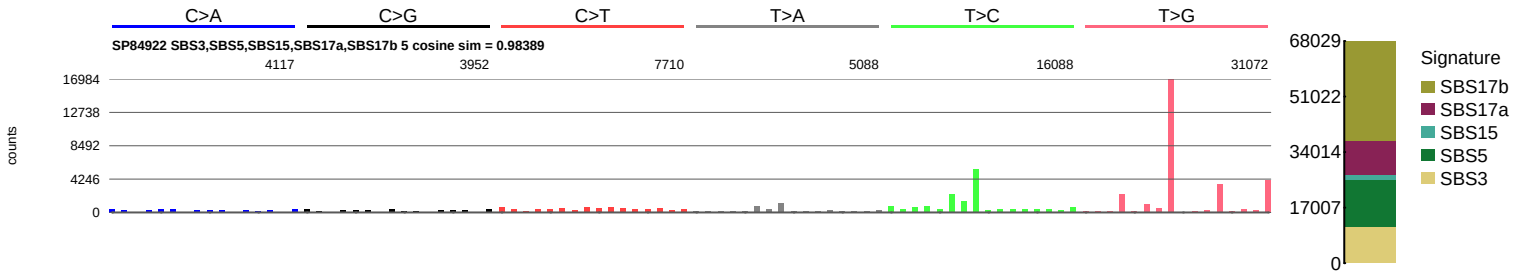
T>C
T>G
SP84922 SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.9839
4820
3694
6872
4930
16739
30981
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3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C>G
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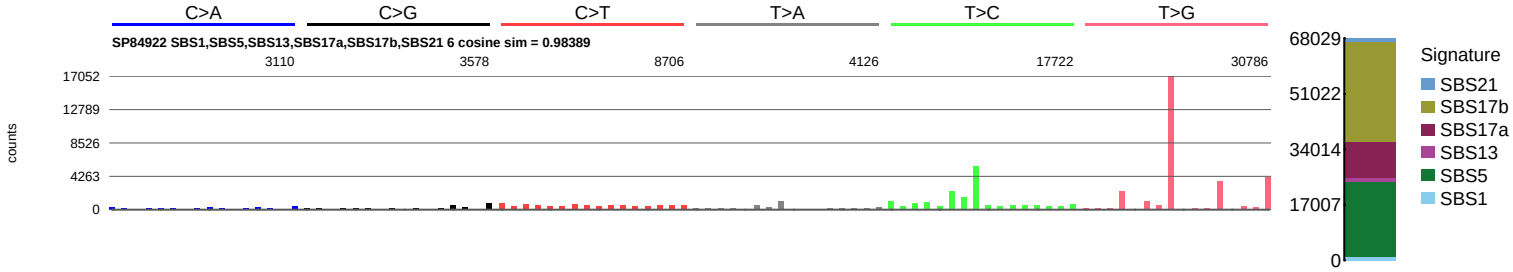
SP84922 SBS3,SBS5,SBS15,SBS17a,SBS17b 5 cosine sim = 0.98389
4117
3952
7710
5088
16088
31072
16984
12738
8492
4246
0
counts
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS3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C>G
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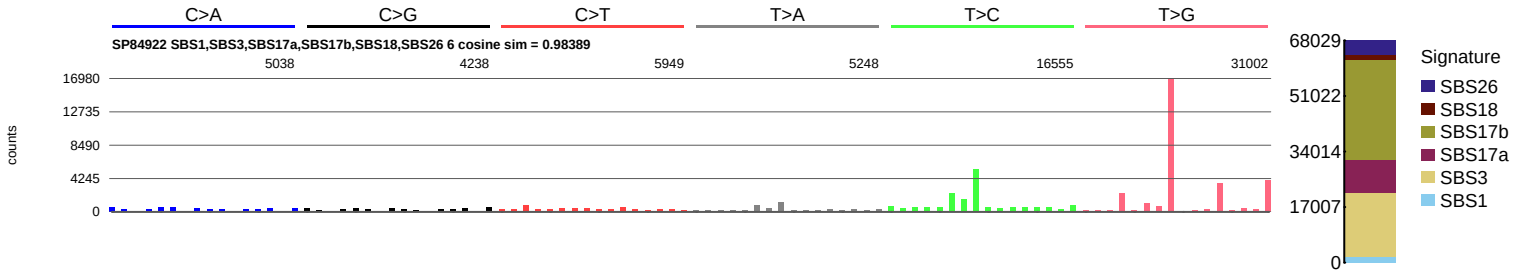
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98389
3110
3578
8706
4126
17722
30786
17052
12789
8526
4263
0
counts
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1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C>T
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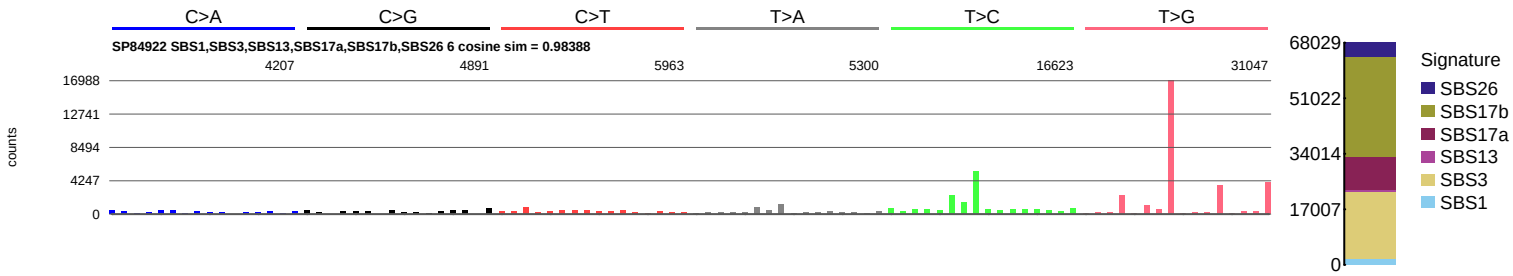
SP84922 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98389
5038
4238
5949
5248
16555
31002
16980
12735
8490
4245
0
counts
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1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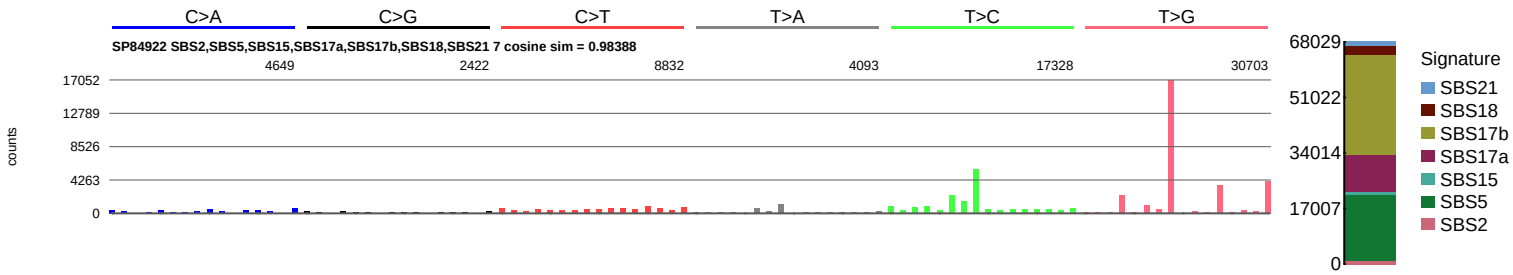
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98388
4207
4891
5963
5300
16623
31047
16988
12741
8494
4247
0
counts
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS1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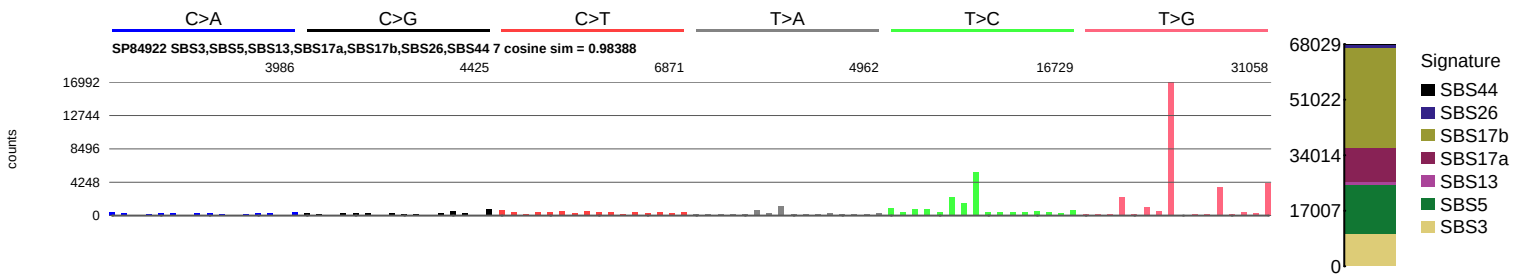
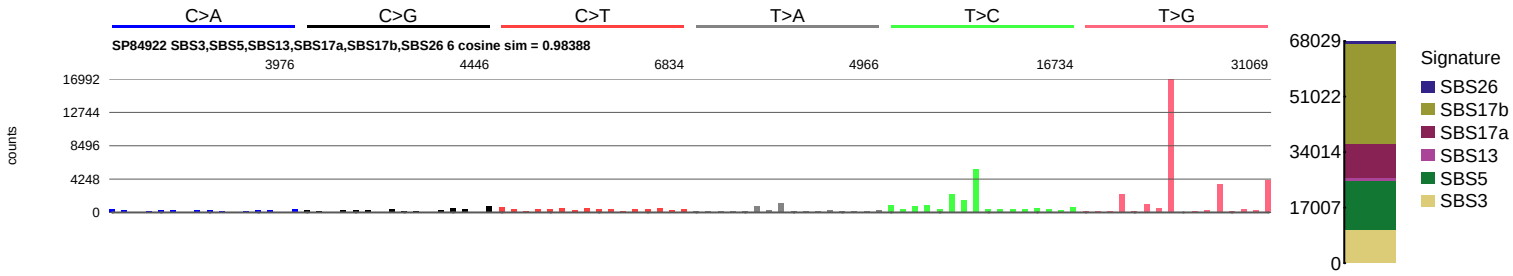
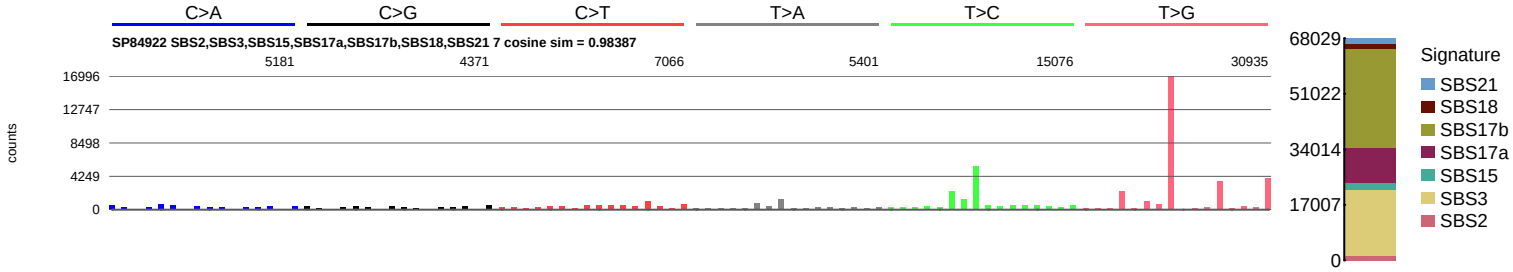
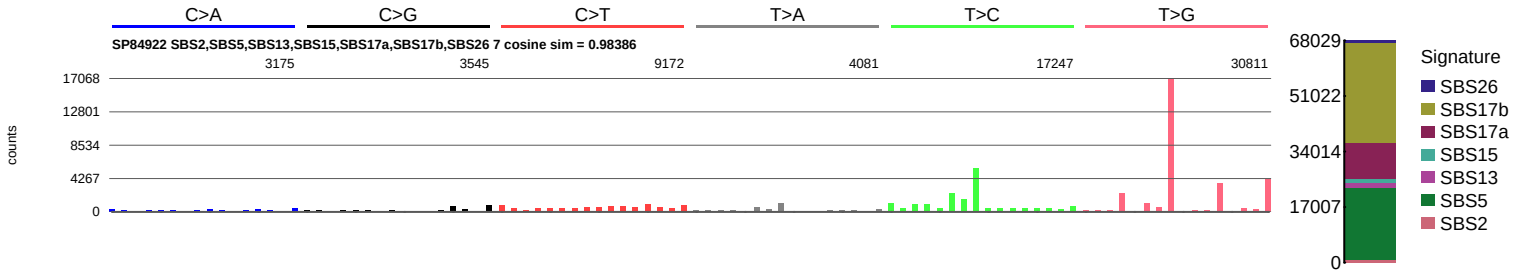
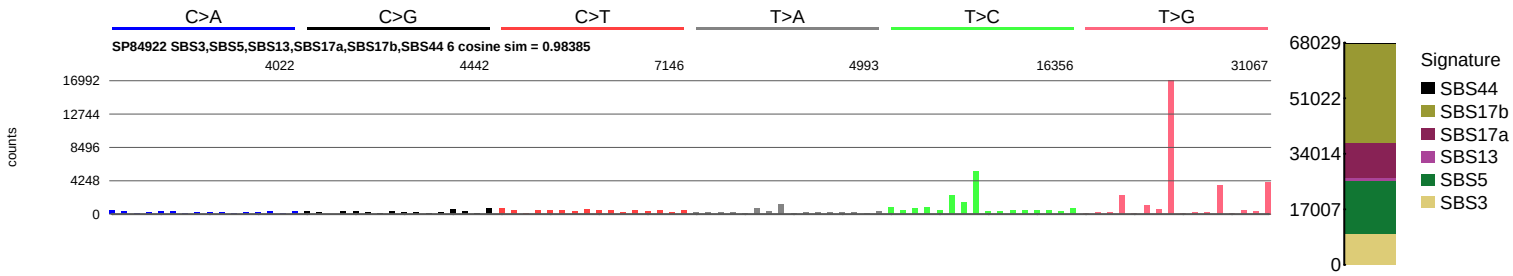
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98388
4649
2422
8832
4093
17328
30703
17052
12789
8526
4263
0
counts
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2

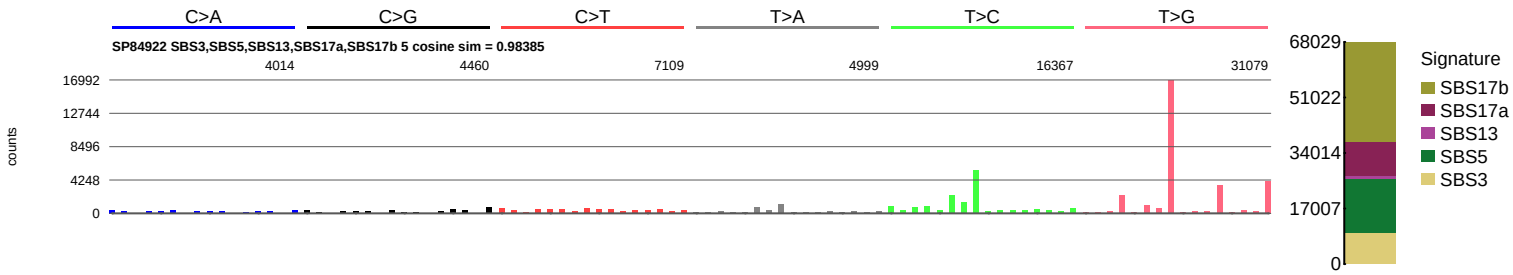

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98388
3986
4425
6871
4962
16729
31058
16992
12744
8496
4248
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3

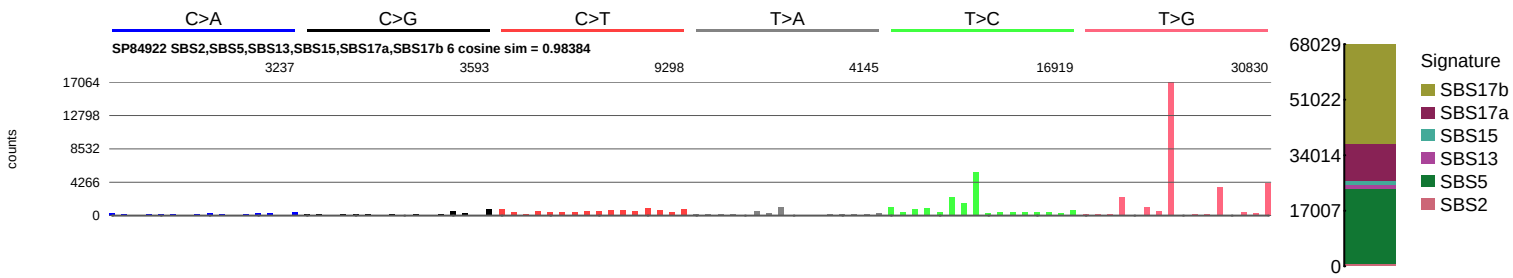

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98388
3976
4446
6834
4966
16734
31069
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3

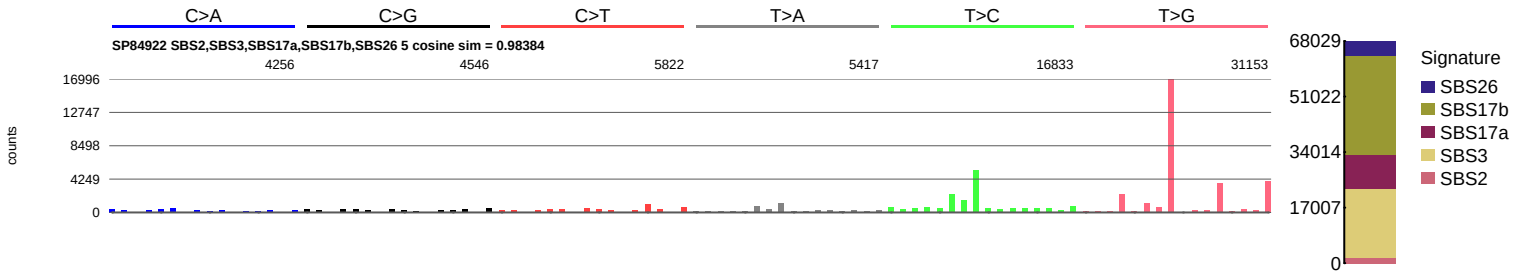

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98387
5181
4371
7066
5401
15076
30935
counts
16996
12747
8498
4249
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

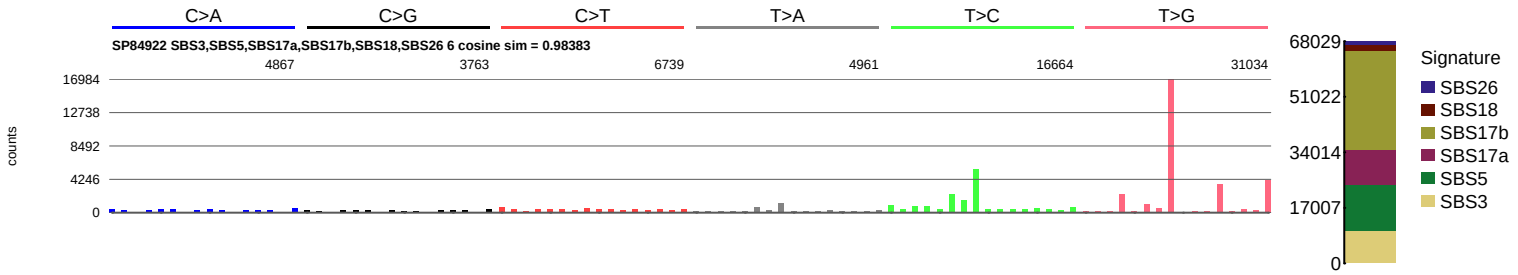

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98386
3175
3545
9172
4081
17247
30811
counts
17068
12801
8534
4267
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98385
4022
4442
7146
4993
16356
31067
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS13,SBS17a,SBS17b 5 cosine sim = 0.98385
4014
4460
7109
4999
16367
31079
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98384
3237
3593
9298
4145
16919
30830
counts
17064
12798
8532
4266
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS26 5 cosine sim = 0.98384
4256
4546
5822
5417
16833
31153
counts
16996
12747
8498
4249
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS3
SBS2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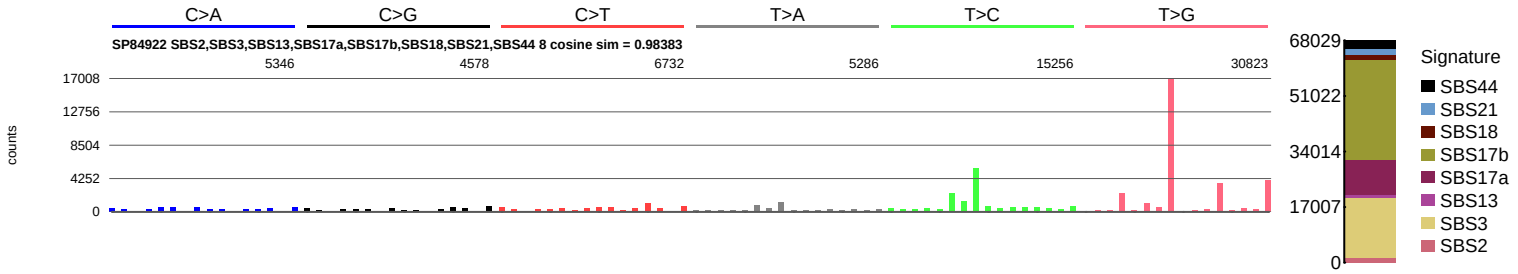
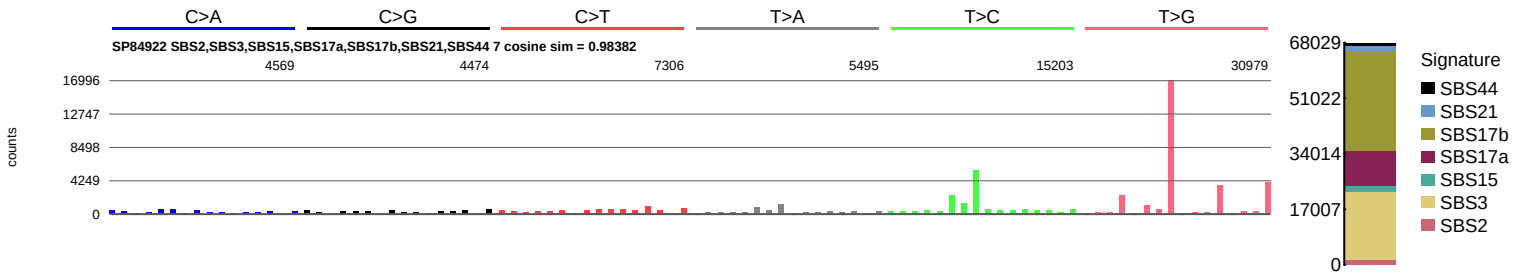
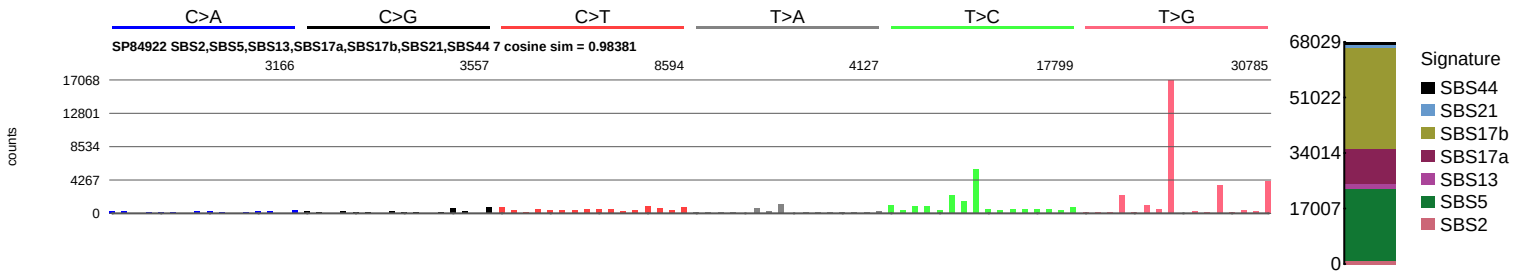
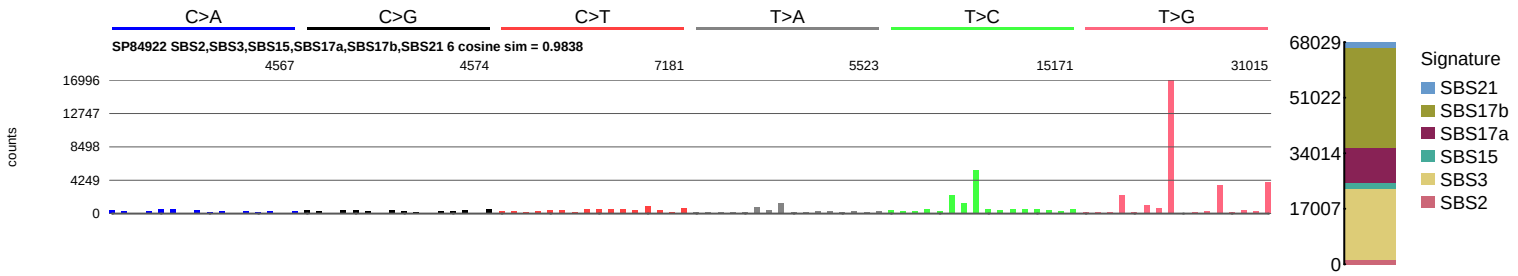
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98383
4867
3763
6739
4961
16664
31034
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98383
5346
4578
6732
5286
15256
30823
counts
17008
12756
8504
4252
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98382
4569
4474
7306
5495
15203
30979
counts
16996
12747
8498
4249
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98381
3166
3557
8594
4127
17799
30785
counts
17068
12801
8534
4267
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21 6 cosine sim = 0.9838
4567
4574
7181
5523
15171
31015
counts
16996
12747
8498
4249
0
68029
51022
34014
17007
0
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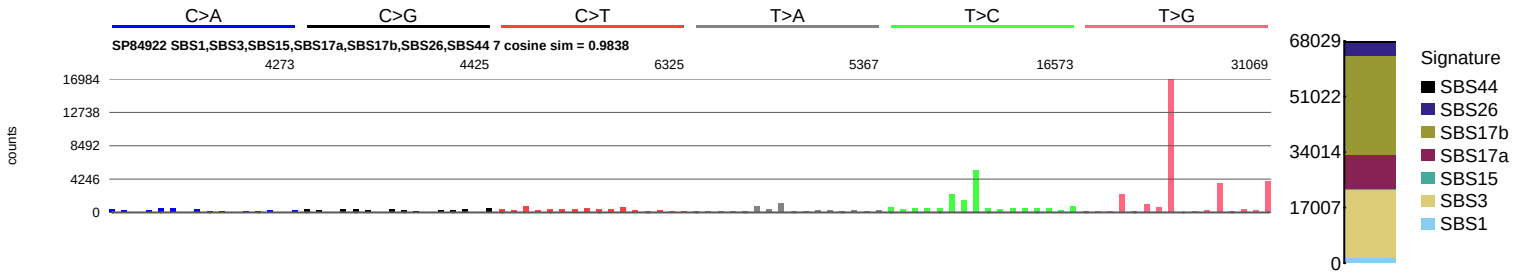
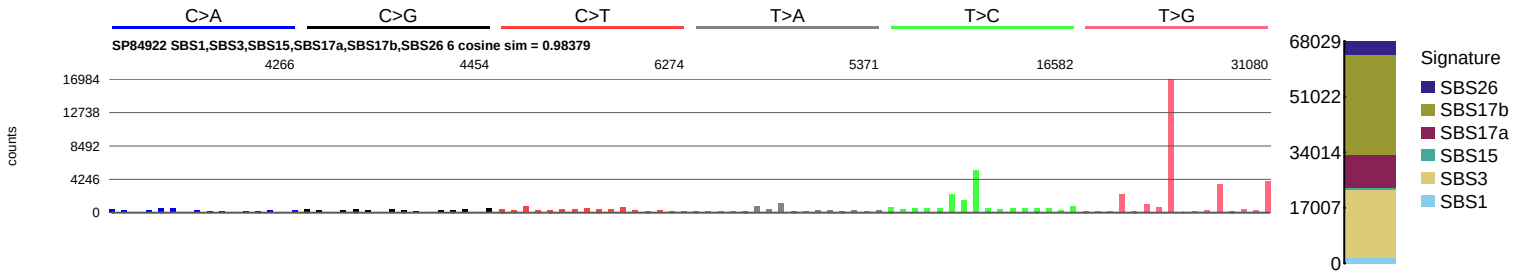
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.9838
4273
4425
6325
5367
16573
31069
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98379
4266
4454
6274
5371
16582
31080
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
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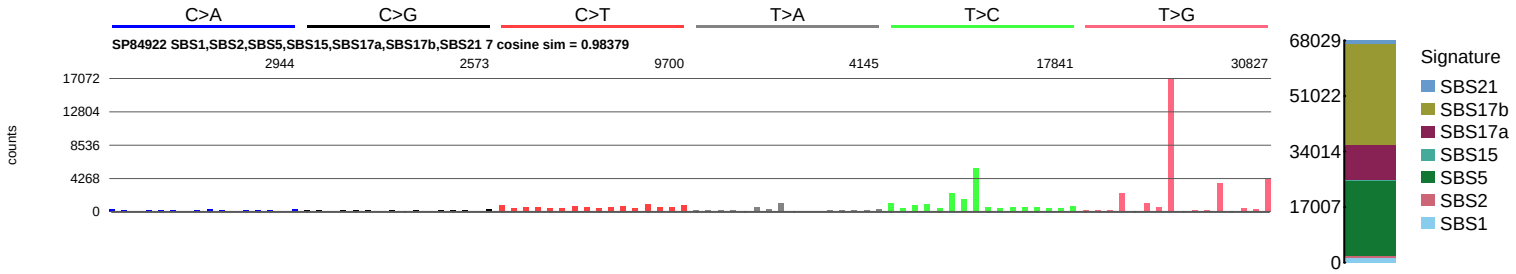
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98379
2944
2573
9700
4145
17841
30827
counts
17072
12804
8536
4268
0
68029
51022
34014
17007
0
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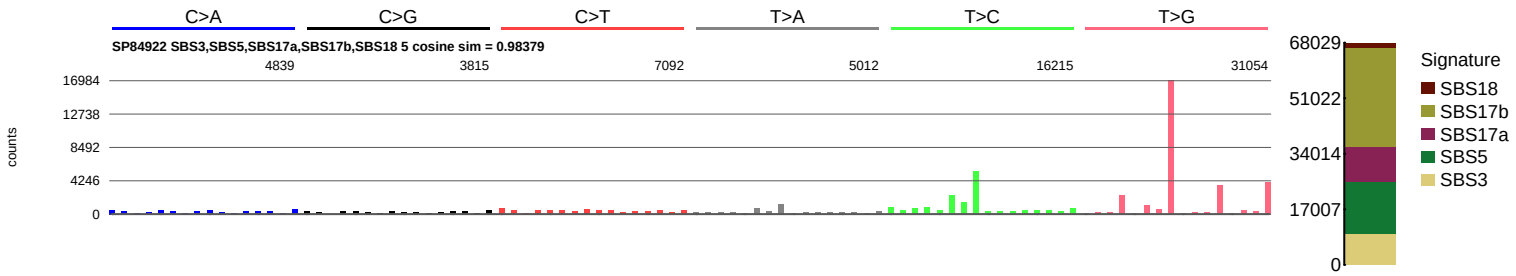
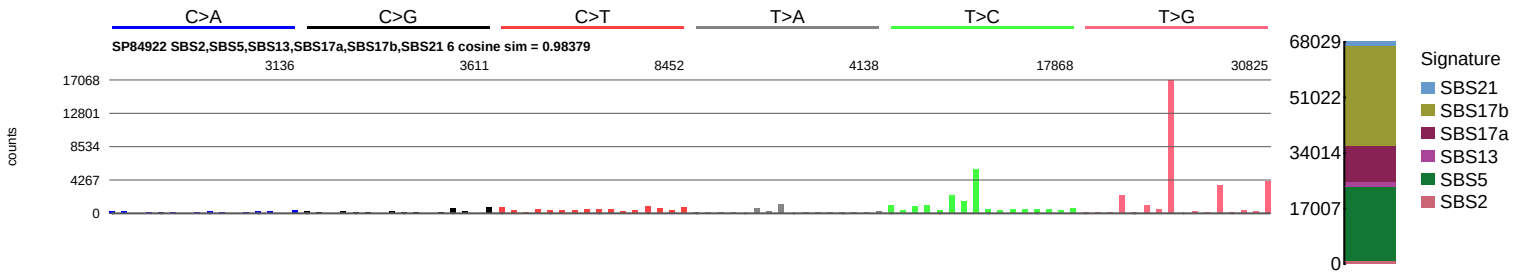
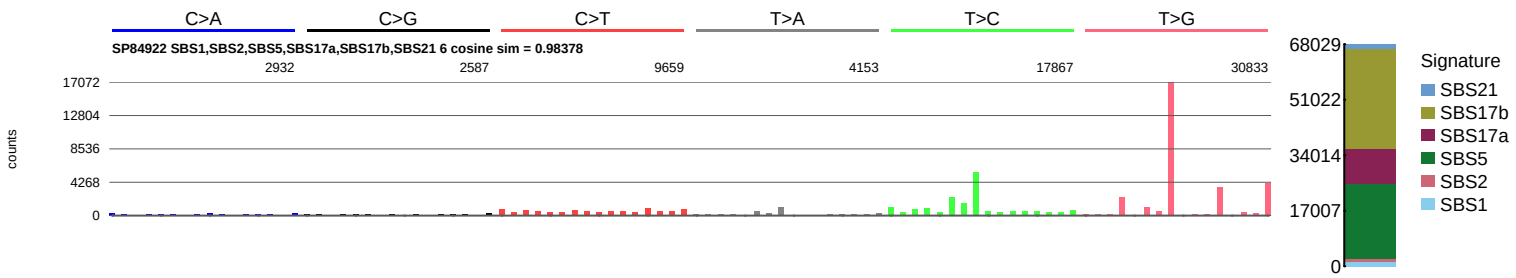
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98379
4839
3815
7092
5012
16215
31054
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98379
3136
3611
8452
4138
17868
30825
counts
17068
12801
8534
4267
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98378
2932
2587
9659
4153
17867
30833
counts
17072
12804
8536
4268
0
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51022
34014
17007
0
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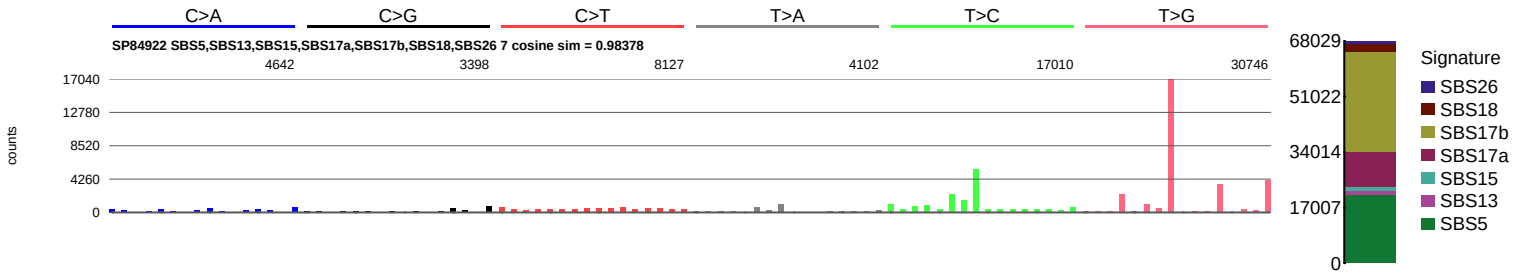
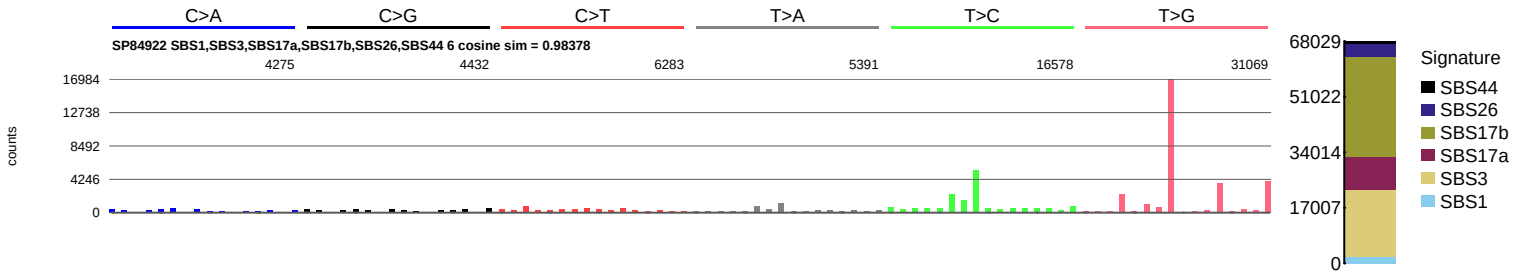
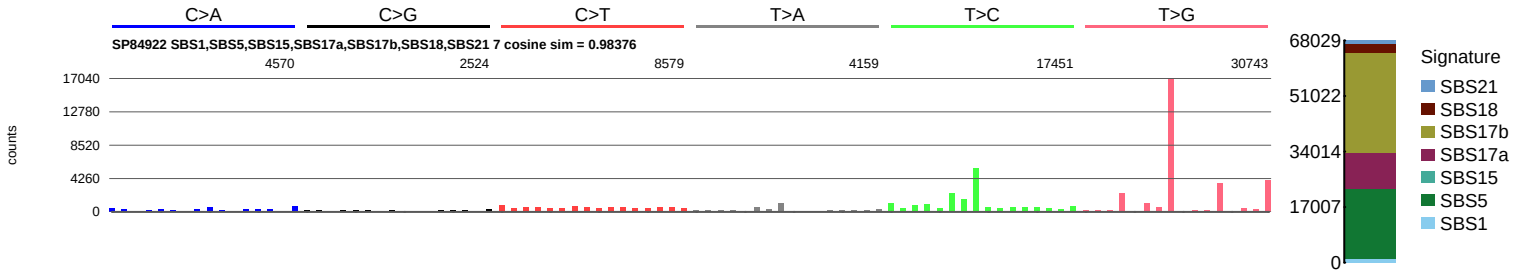
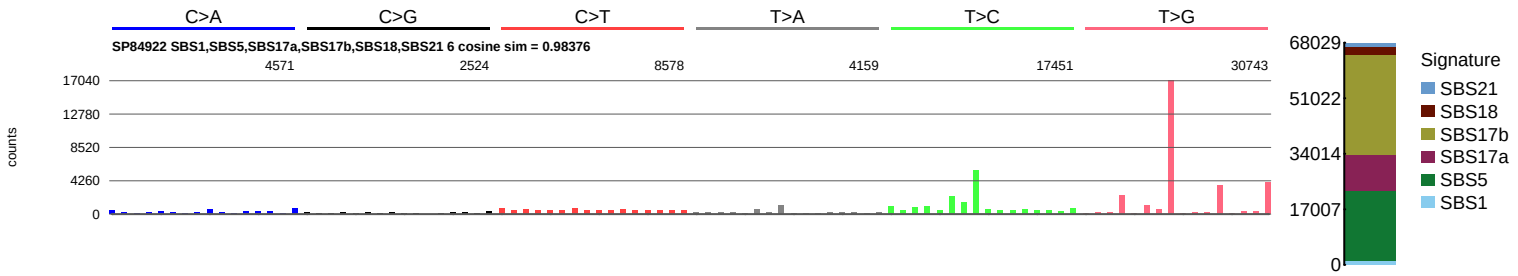
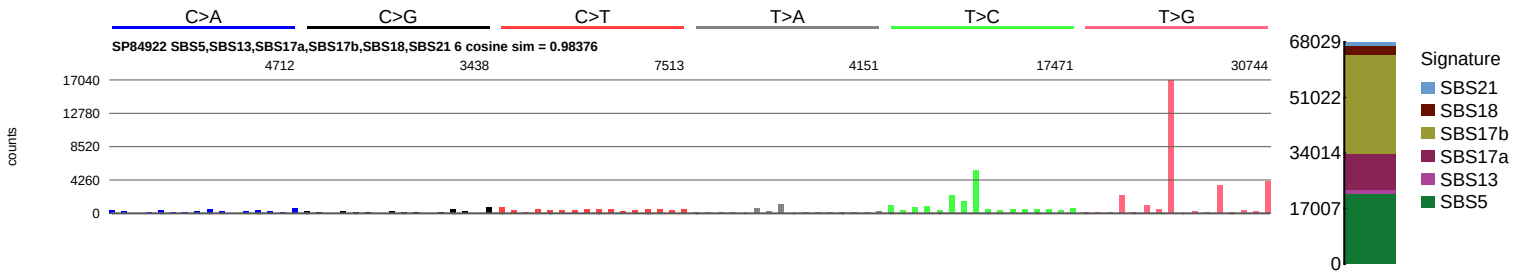
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1

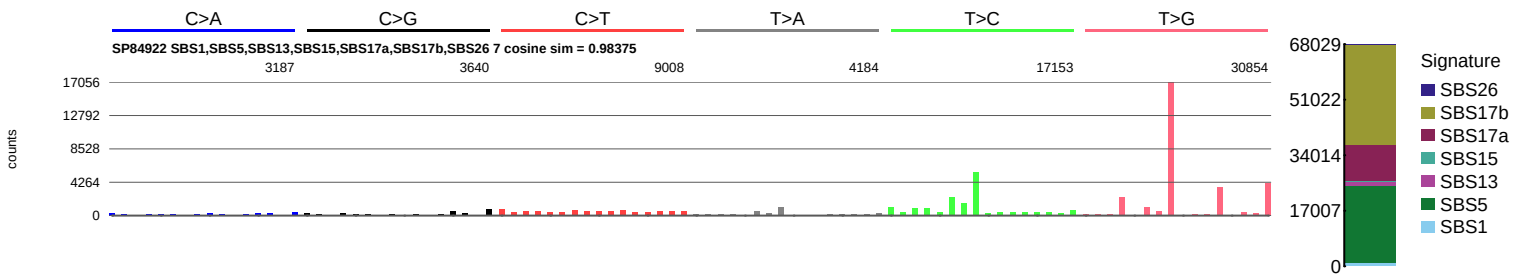

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98378
4642
3398
8127
4102
17010
30746
counts
17040
12780
8520
4260
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5

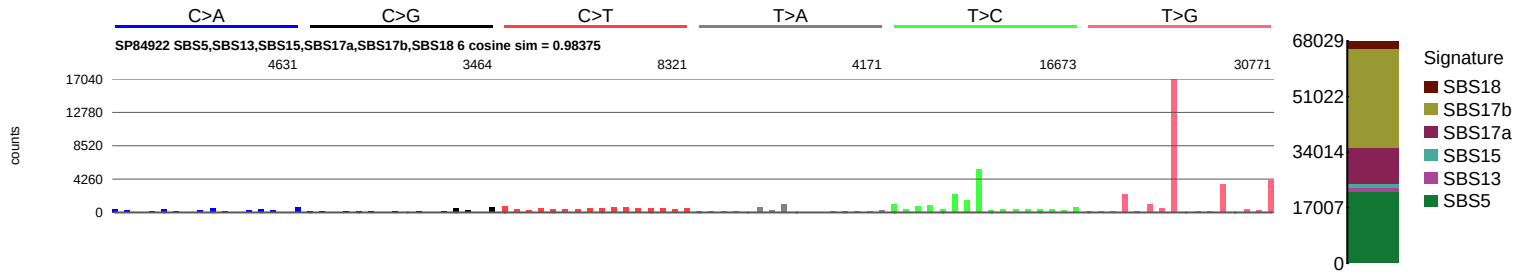

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS44 6 cosine sim = 0.98378
4275
4432
6283
5391
16578
31069
counts
16984
12738
8492
4246
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS3
SBS1

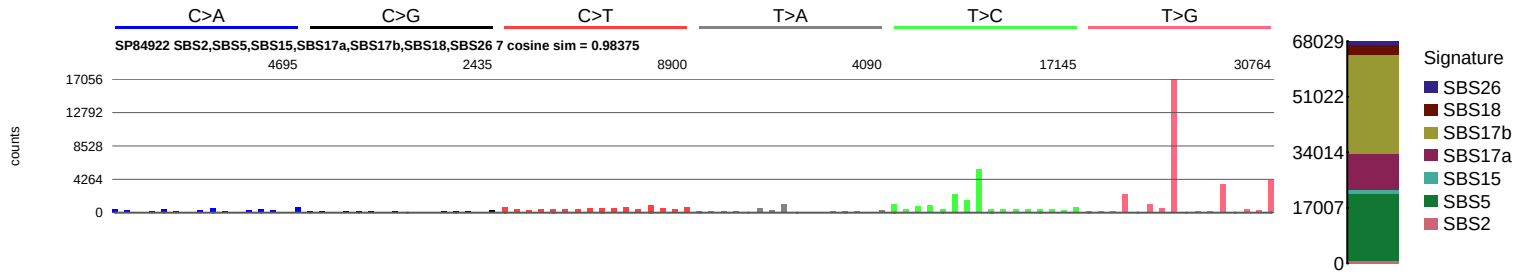

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98376
4570
2524
8579
4159
17451
30743
counts
17040
12780
8520
4260
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

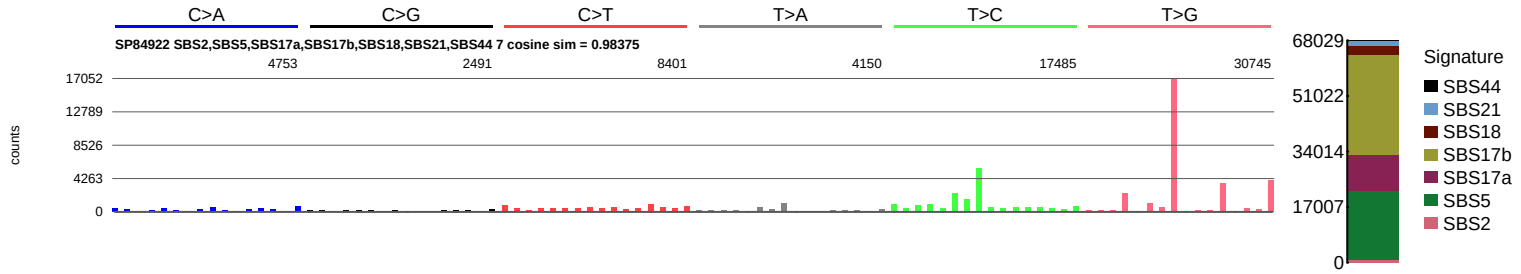

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98376
4571
2524
8578
4159
17451
30743
counts
17040
12780
8520
4260
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

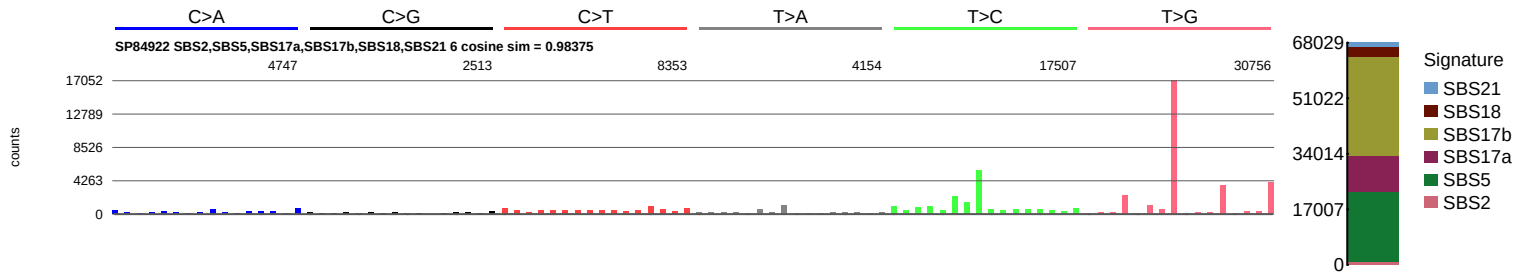

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98376
4712
3438
7513
4151
17471
30744
counts
17040
12780
8520
4260
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5

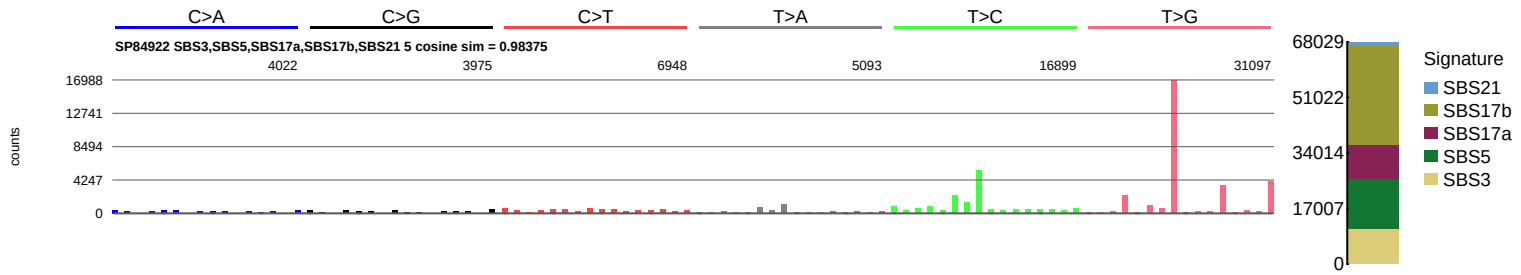

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98375
3187
3640
9008
4184
17153
30854
counts
17056
12792
8528
4264
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

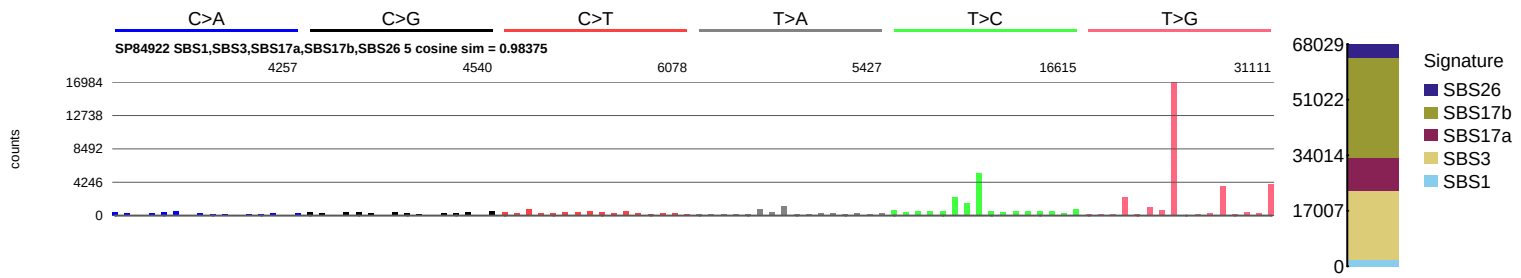

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98375
4631
3464
8321
4171
16673
30771
17040
12780
8520
4260
0
counts
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5

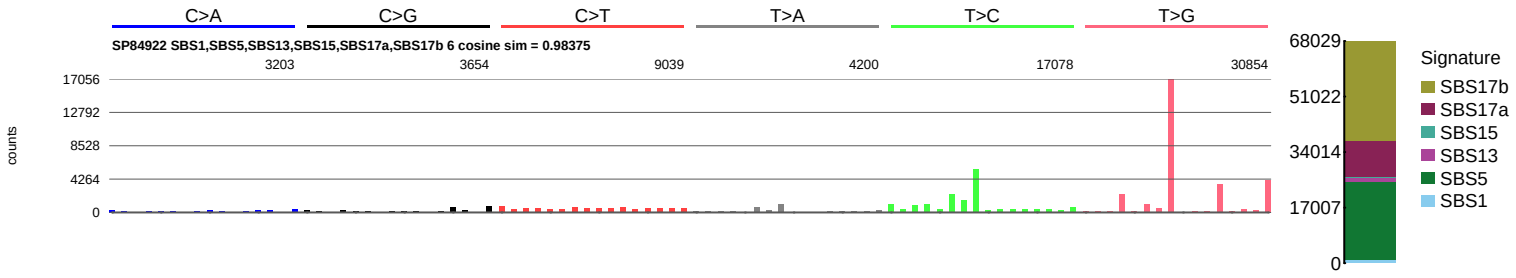

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98375
4695
2435
8900
4090
17145
30764
17056
12792
8528
4264
0
counts
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2

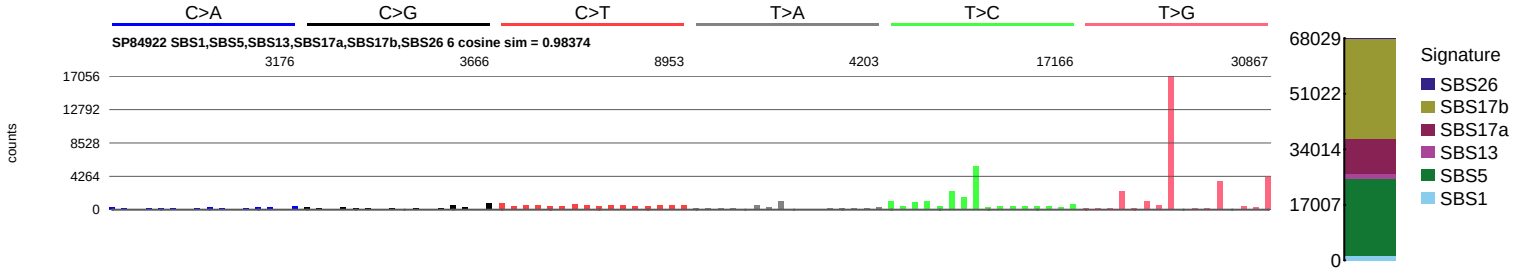

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98375
4753
2491
8401
4150
17485
30745
17052
12789
8526
4263
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2

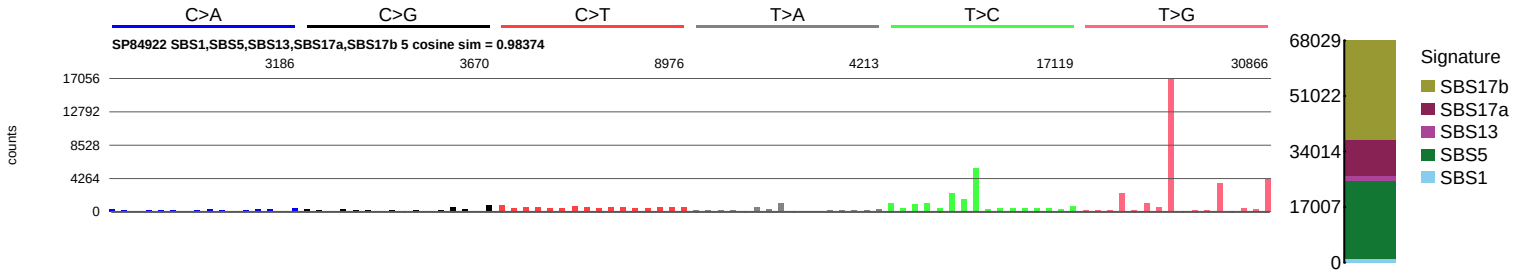

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98375
4747
2513
8353
4154
17507
30756
17052
12789
8526
4263
0
counts
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2

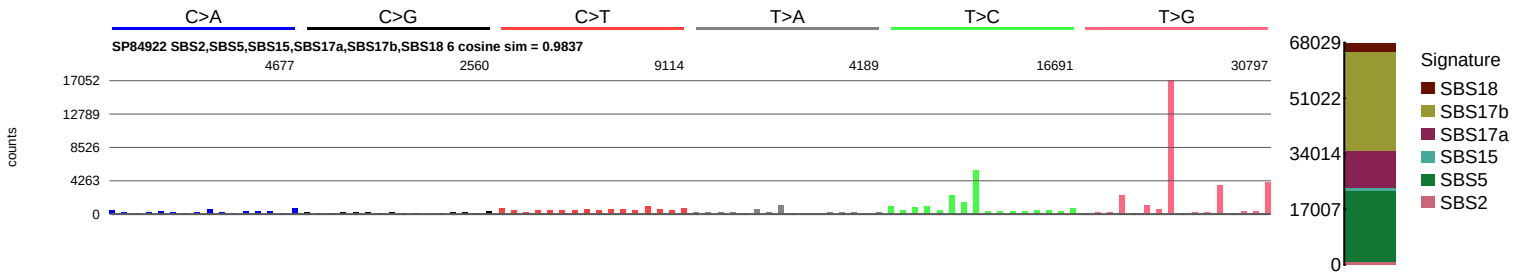

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS17a,SBS17b,SBS21 5 cosine sim = 0.98375
4022
3975
6948
5093
16899
31097
16988
12741
8494
4247
0
counts
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS3

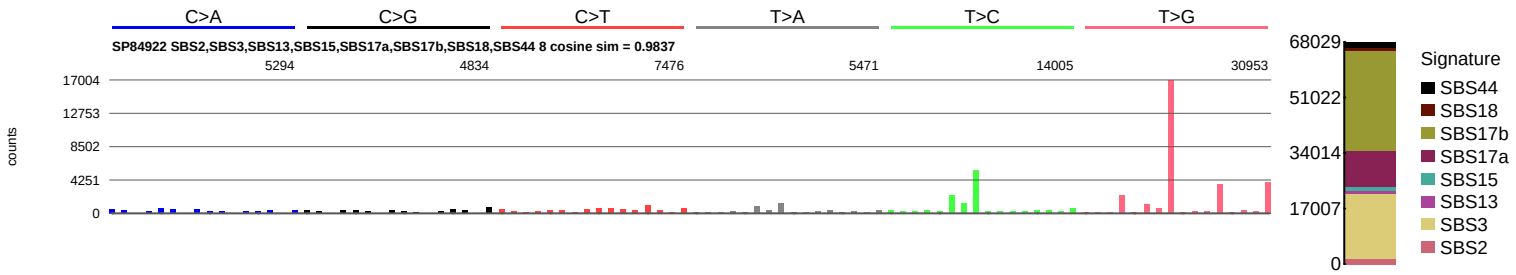

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS17a,SBS17b,SBS26 5 cosine sim = 0.98375
4257
4540
6078
5427
16615
31111
16984
12738
8492
4246
0
counts
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS3
SBS1

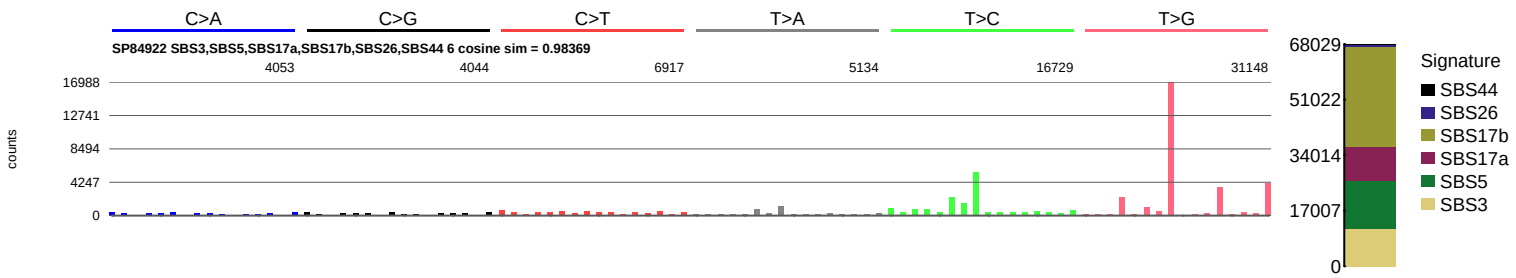

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98375
3203
3654
9039
4200
17078
30854
counts
17056
12792
8528
4264
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

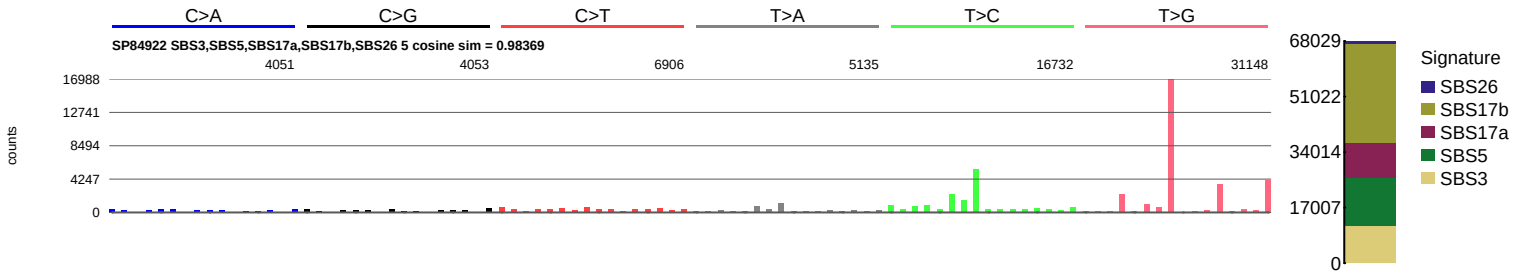

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98374
3176
3666
8953
4203
17166
30867
counts
17056
12792
8528
4264
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS1

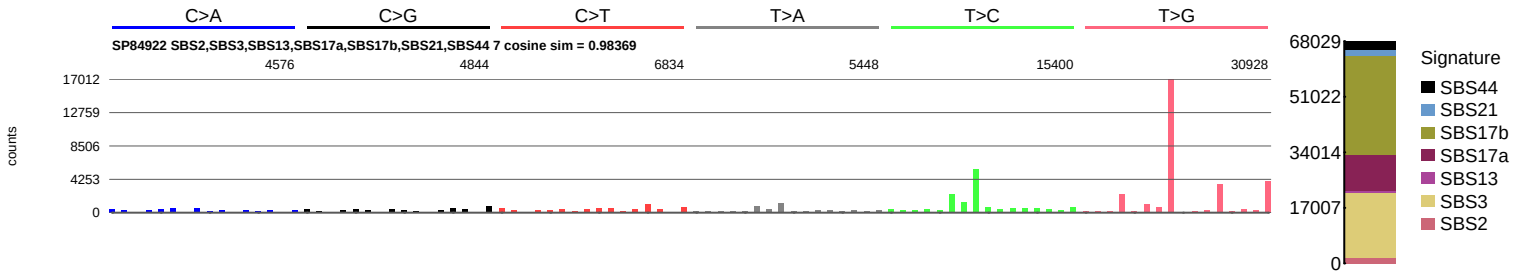

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS13,SBS17a,SBS17b 5 cosine sim = 0.98374
3186
3670
8976
4213
17119
30866
counts
17056
12792
8528
4264
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS1

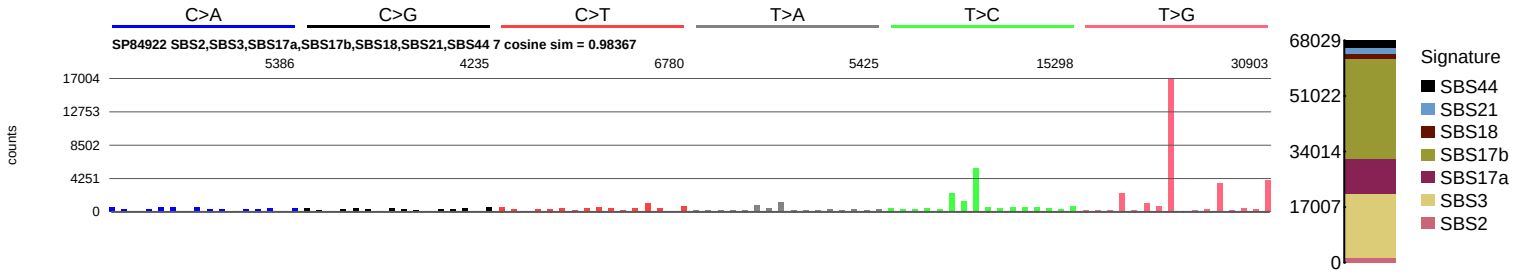

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.9837
4677
2560
9114
4189
16691
30797
counts
17052
12789
8526
4263
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2

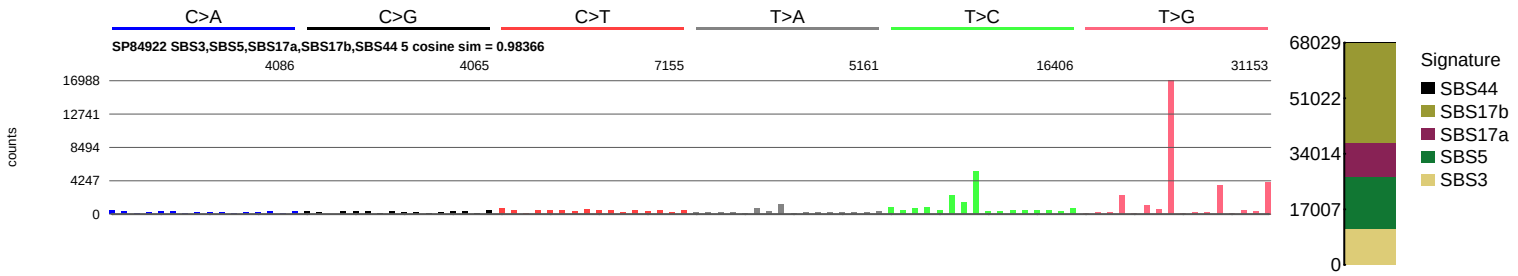

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.9837
5294
4834
7476
5471
14005
30953
counts
17004
12753
8502
4251
0
68029
51022
34014
17007
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

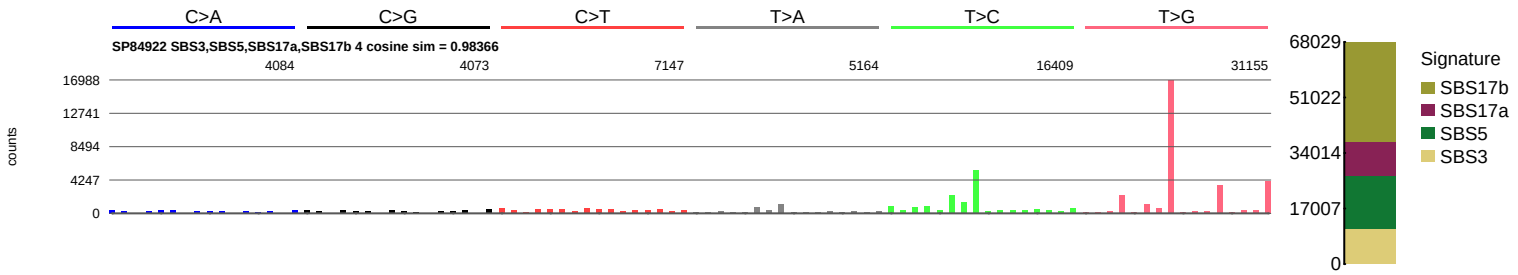

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS44 6 cosine sim = 0.98369
4053
4044
6917
5134
16729
31148
counts
16988
12741
8494
4247
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS5
SBS3

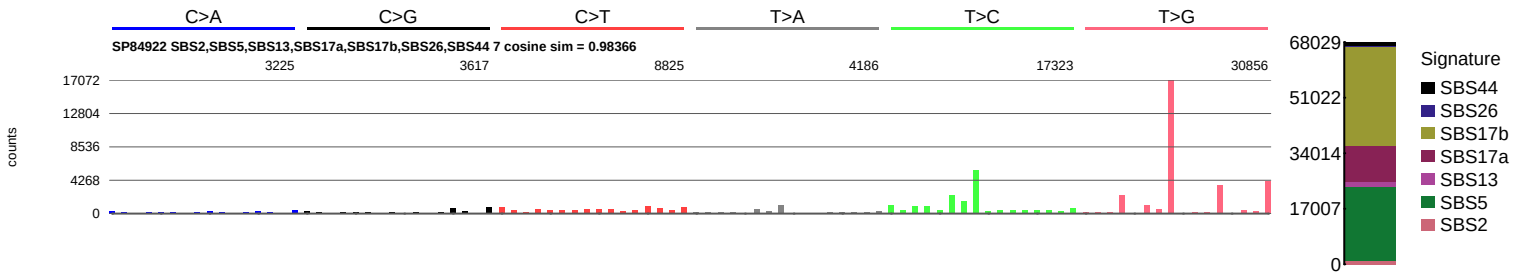

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS17a,SBS17b,SBS26 5 cosine sim = 0.98369
4051
4053
6906
5135
16732
31148
16988
12741
8494
4247
0
counts
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS5
SBS3

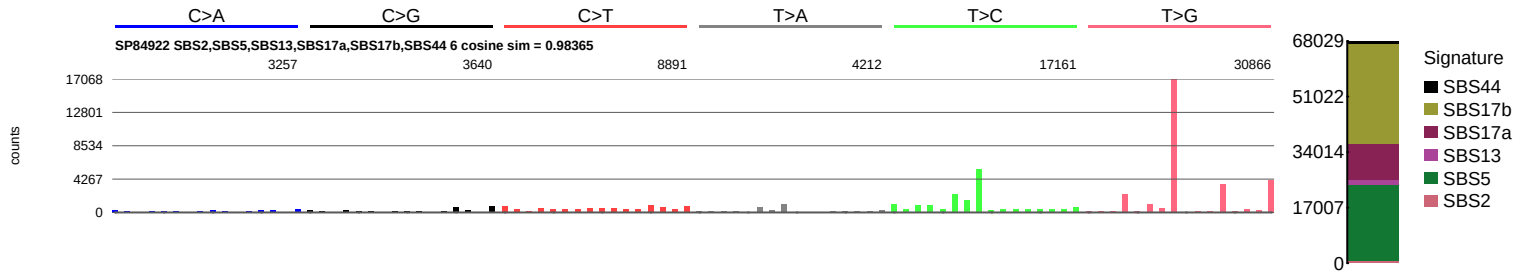

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98369
4576
4844
6834
5448
15400
30928
17012
12759
8506
4253
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2

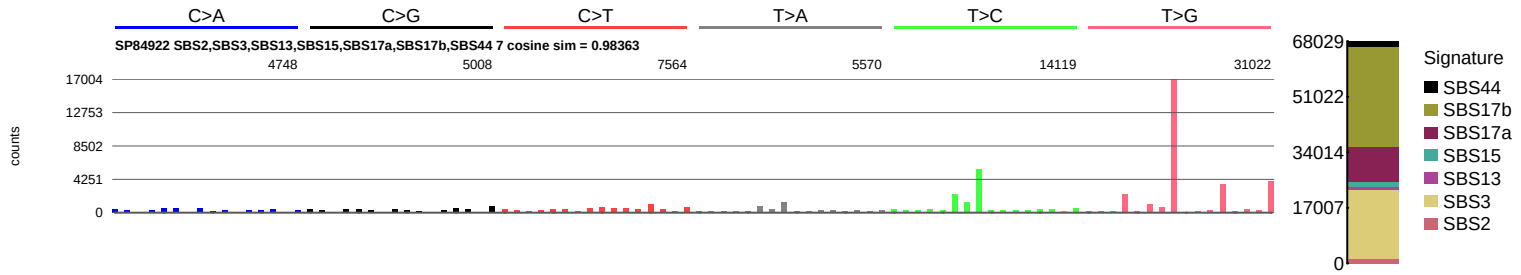

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98367
5386
4235
6780
5425
15298
30903
17004
12753
8502
4251
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2

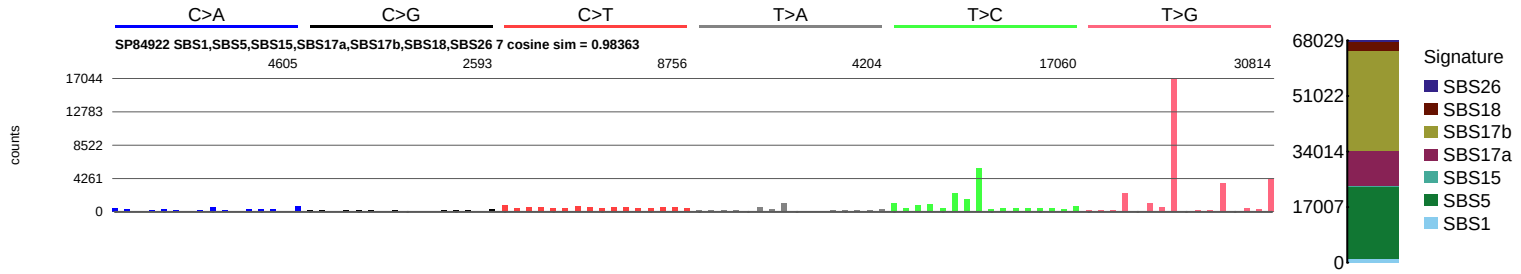

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS17a,SBS17b,SBS44 5 cosine sim = 0.98366
4086
4065
7155
5161
16406
31153
16988
12741
8494
4247
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS5,SBS17a,SBS17b 4 cosine sim = 0.98366
4084
4073
7147
5164
16409
31155
16988
12741
8494
4247
0
counts
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98366
3225
3617
8825
4186
17323
30856
17072
12804
8536
4268
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98365
3257
3640
8891
4212
17161
30866
counts
17068
12801
8534
4267
0
68029
51022
34014
17007
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98363
4748
5008
7564
5570
14119
31022
counts
17004
12753
8502
4251
0
68029
51022
34014
17007
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98363
4605
2593
8756
4204
17060
30814
counts
17044
12783
8522
4261
0
68029
51022
34014
17007
0
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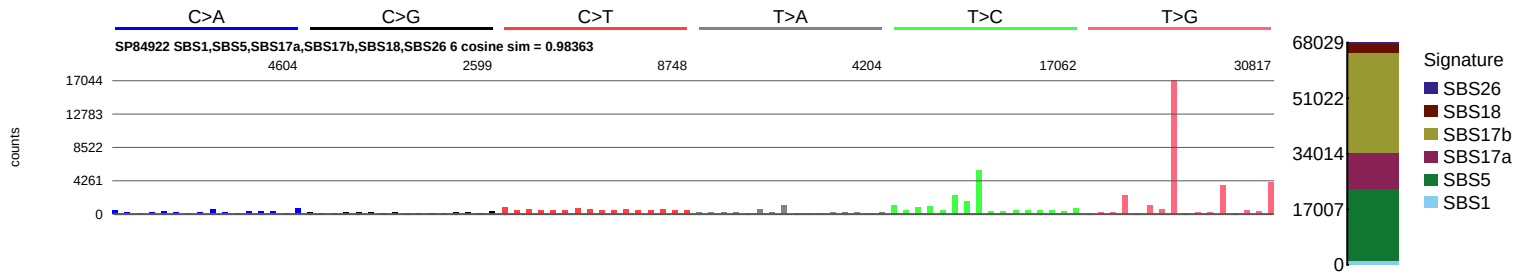
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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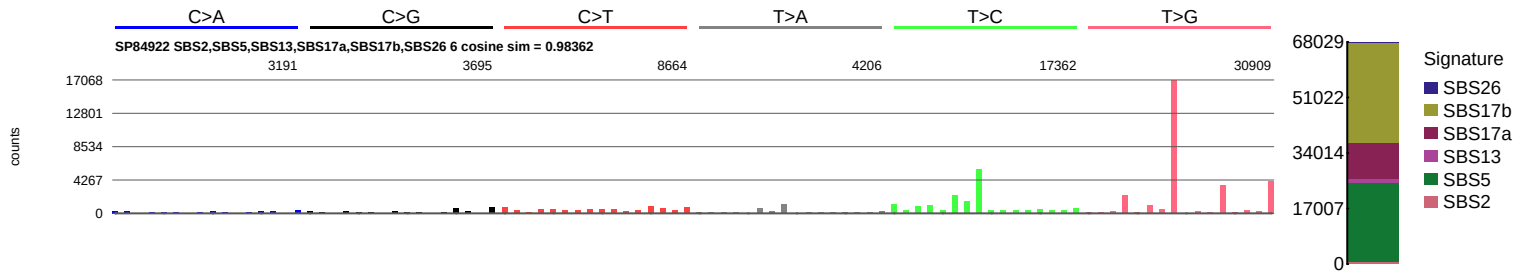
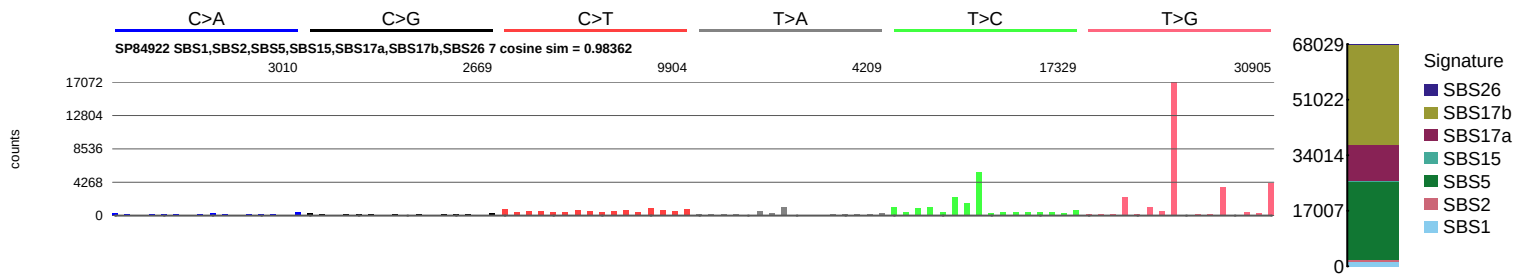
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98363
4604
2599
8748
4204
17062
30817
counts
17044
12783
8522
4261
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98362
3191
3695
8664
4206
17362
30909
counts
17068
12801
8534
4267
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS2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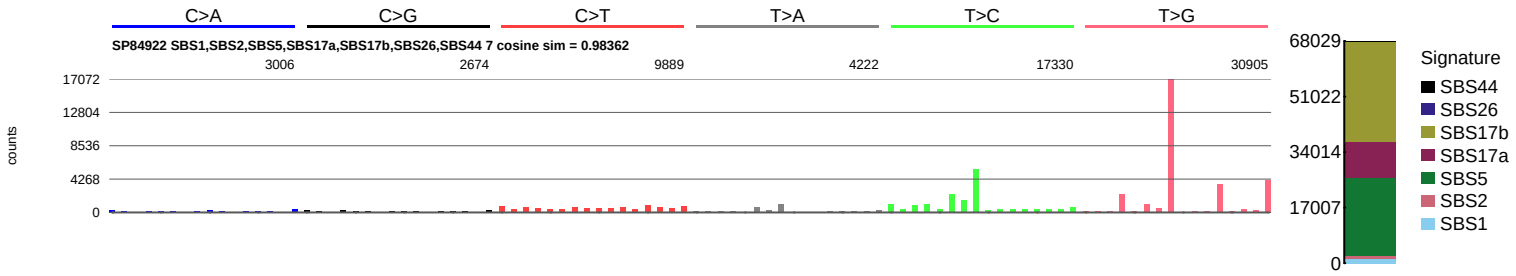
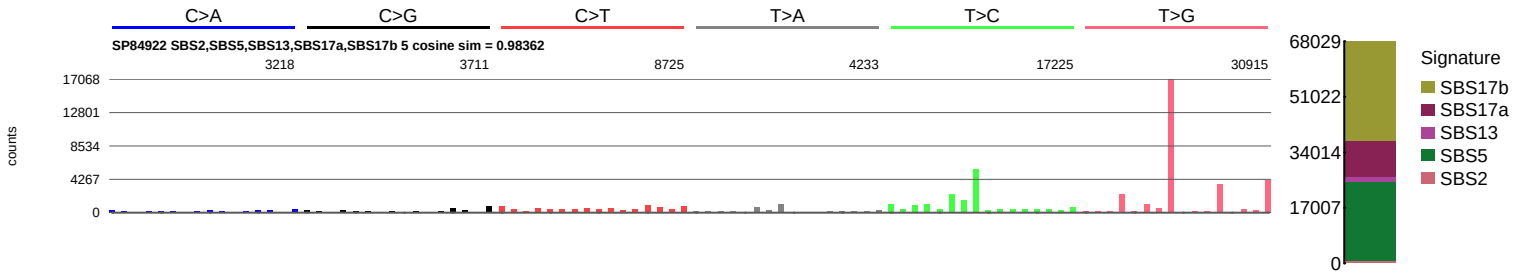
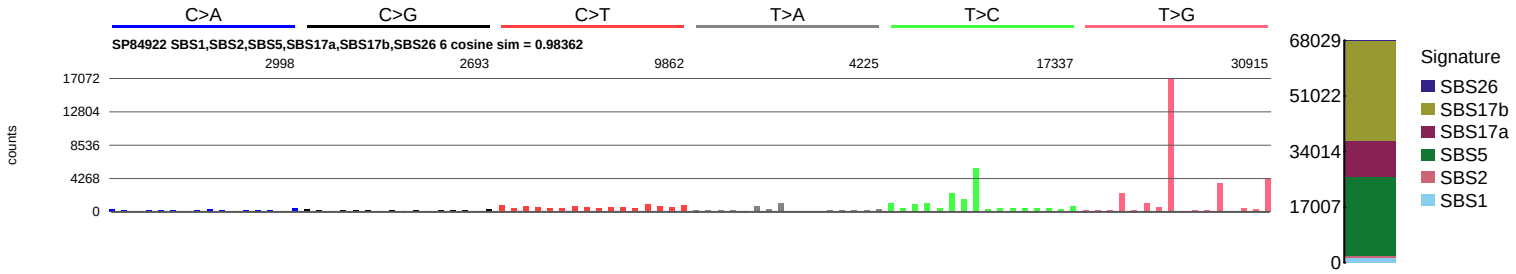
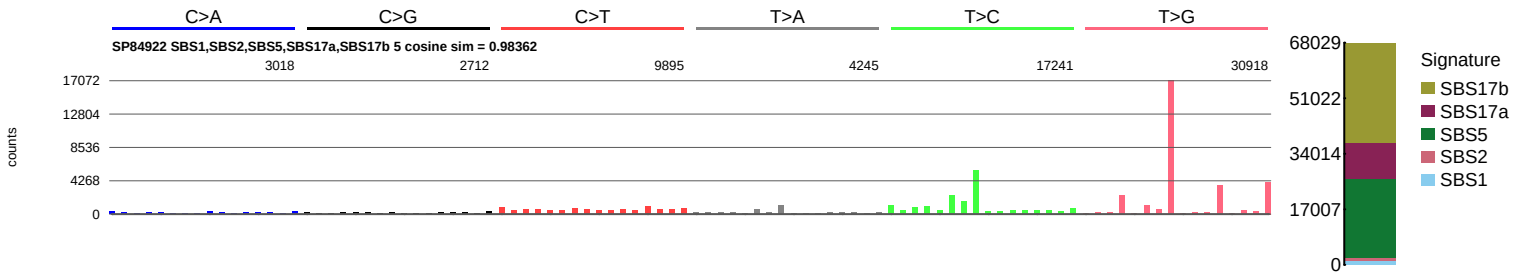
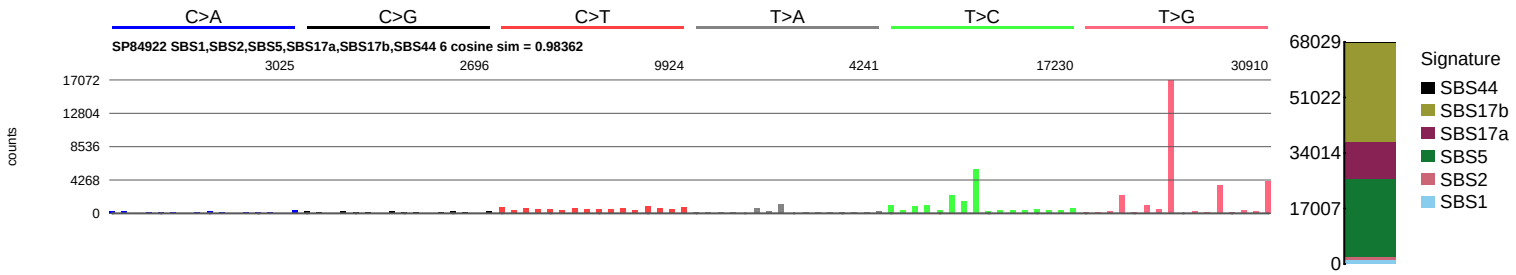
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98362
3010
2669
9904
4209
17329
30905
counts
17072
12804
8536
4268
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

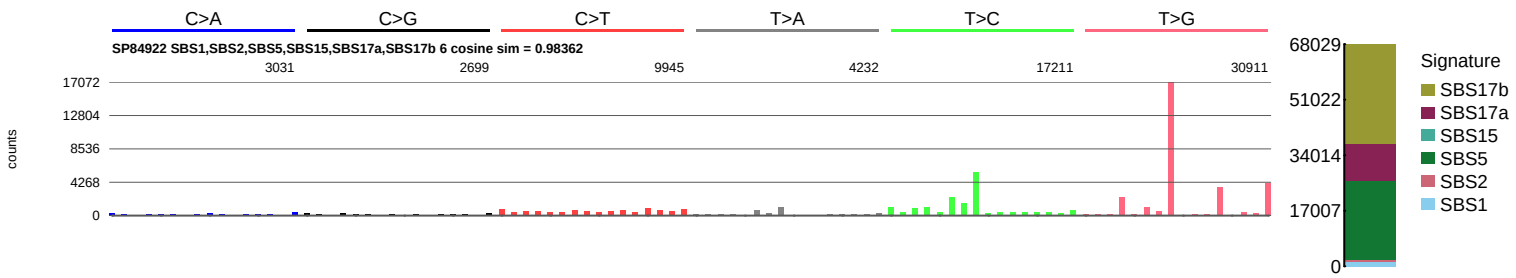

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98362
3006
2674
9889
4222
17330
30905
counts
17072
12804
8536
4268
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS5
SBS2
SBS1

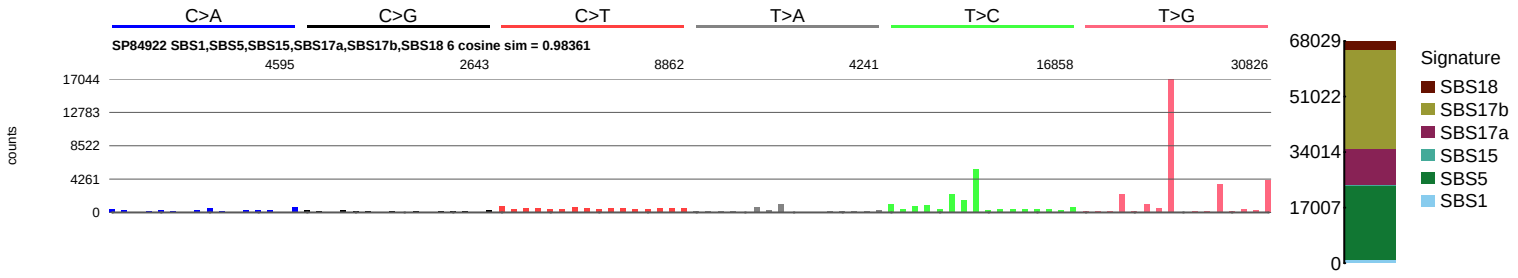

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS13,SBS17a,SBS17b 5 cosine sim = 0.98362
3218
3711
8725
4233
17225
30915
counts
17068
12801
8534
4267
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS2

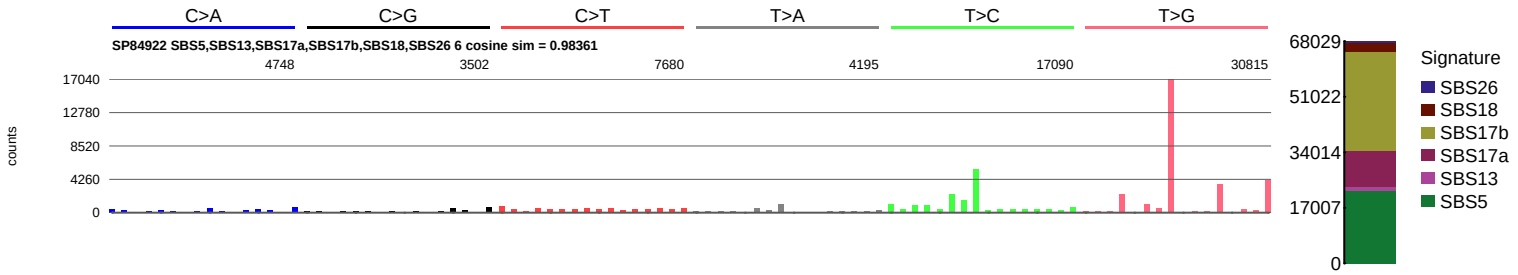

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98362
2998
2693
9862
4225
17337
30915
counts
17072
12804
8536
4268
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS5
SBS2
SBS1

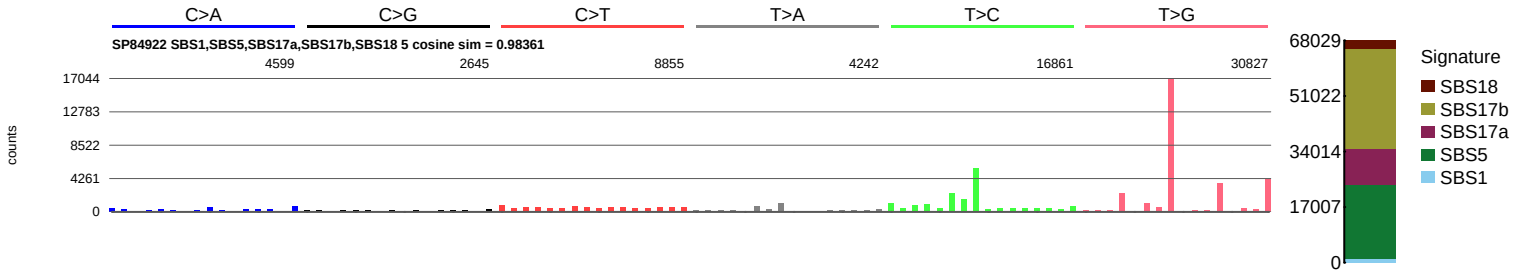

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS17a,SBS17b 5 cosine sim = 0.98362
3018
2712
9895
4245
17241
30918
counts
17072
12804
8536
4268
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98362
3025
2696
9924
4241
17230
30910
counts
17072
12804
8536
4268
0
68029
51022
34014
17007
0
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98362
3031
2699
9945
4232
17211
30911
counts
17072
12804
8536
4268
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98361
4595
2643
8862
4241
16858
30826
counts
17044
12783
8522
4261
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98361
4748
3502
7680
4195
17090
30815
counts
17040
12780
8520
4260
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98361
4599
2645
8855
4242
16861
30827
counts
17044
12783
8522
4261
0
68029
51022
34014
17007
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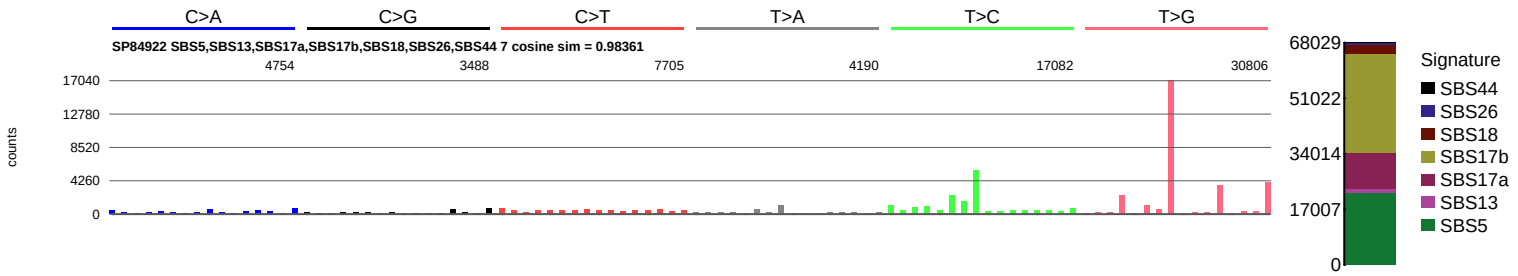
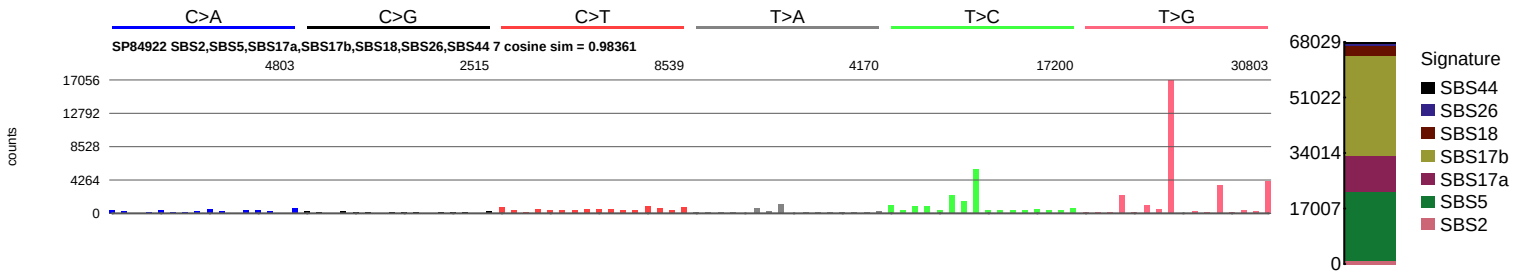
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98361
4754
3488
7705
4190
17082
30806
counts
17040
12780
8520
4260
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5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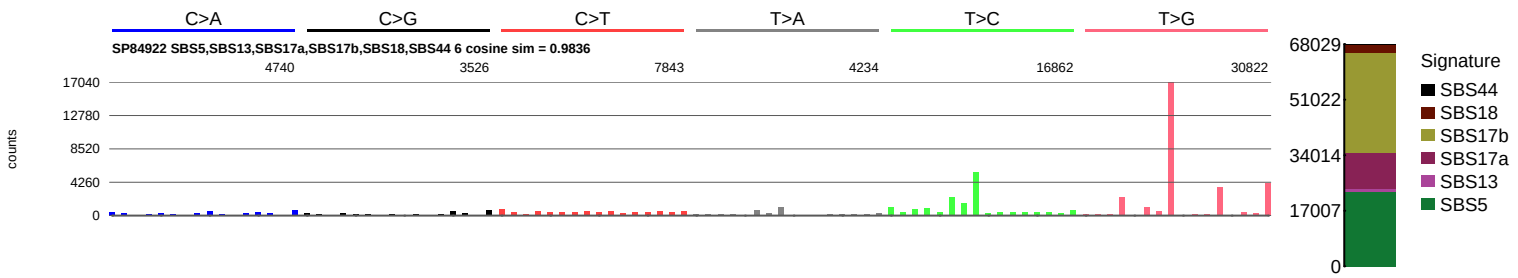
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98361
4803
2515
8539
4170
17200
30803
counts
17056
12792
8528
4264
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2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C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.9836
4740
3526
7843
4234
16862
30822
counts
17040
12780
8520
4260
0
68029
51022
34014
17007
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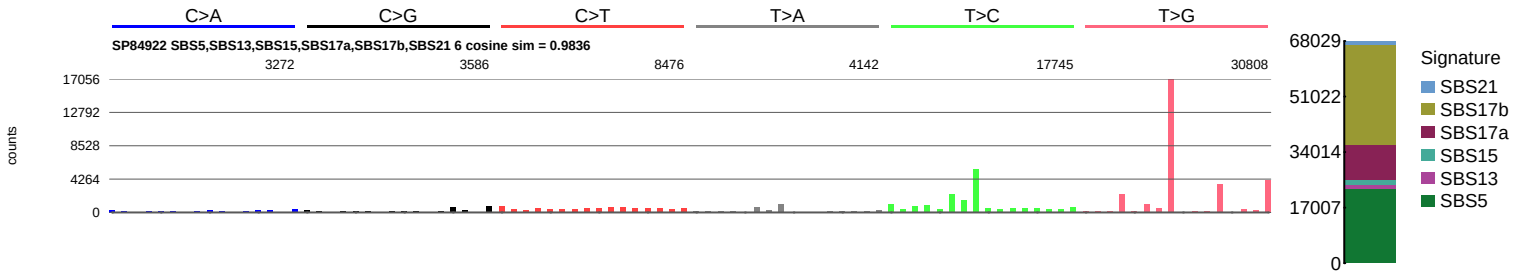
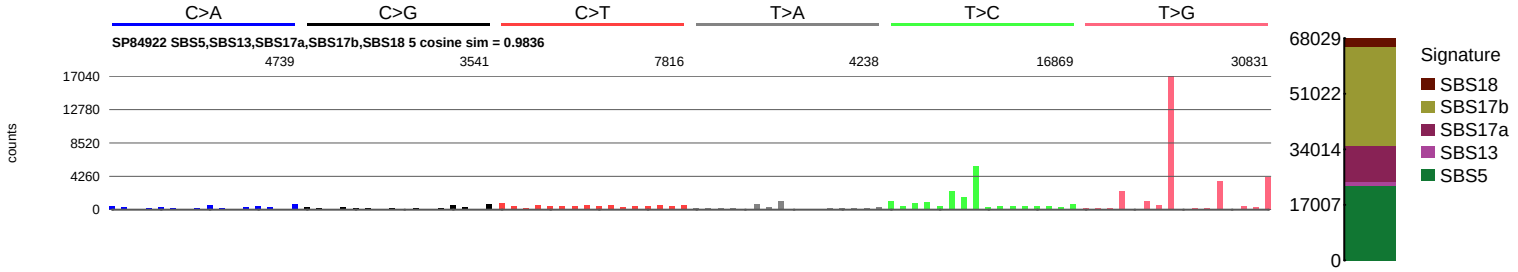
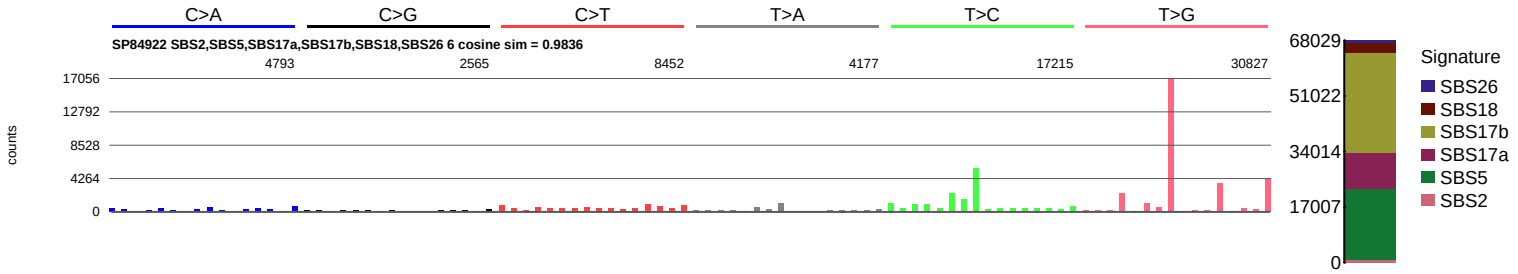
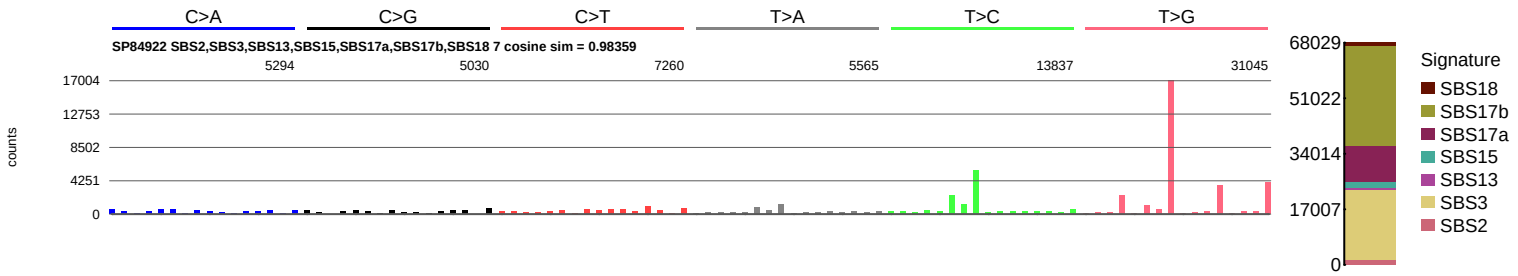
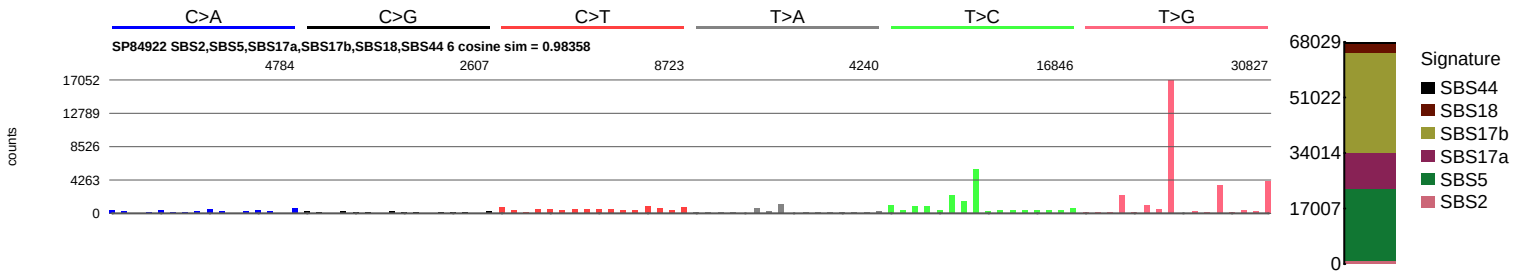
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5

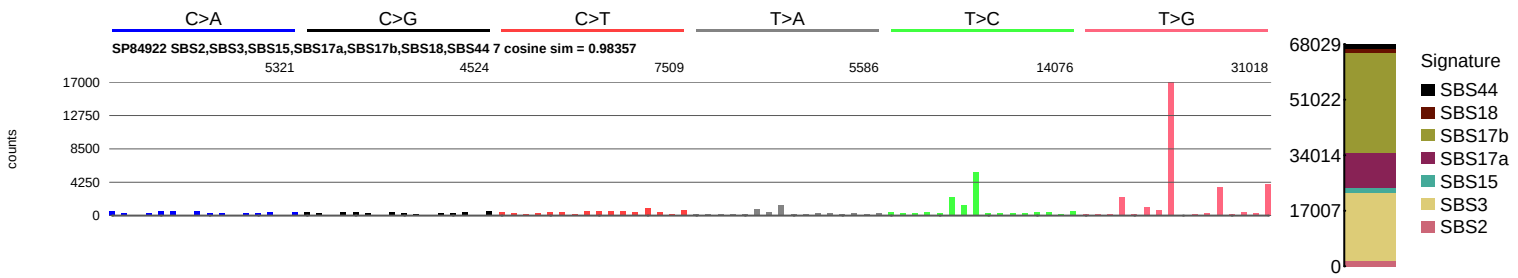

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 6 cosine sim = 0.9836
3272
3586
8476
4142
17745
30808
counts
17056
12792
8528
4264
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5

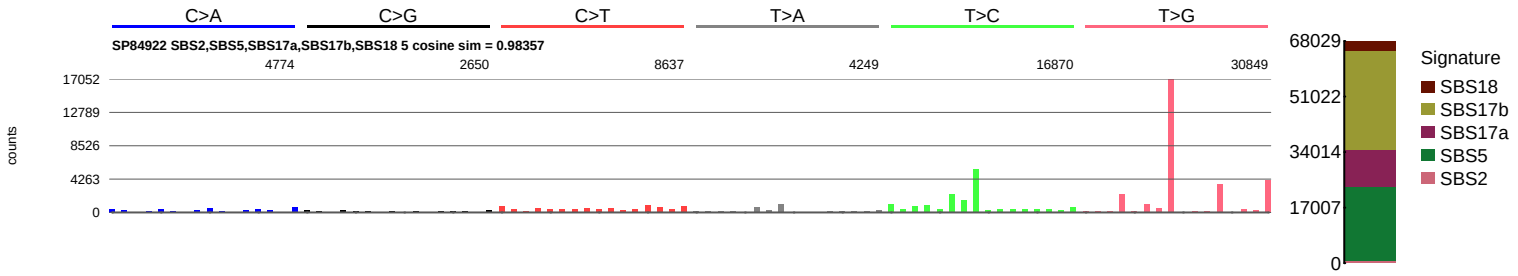

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS17a,SBS17b,SBS18 5 cosine sim = 0.9836
4739
3541
7816
4238
16869
30831
counts
17040
12780
8520
4260
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5

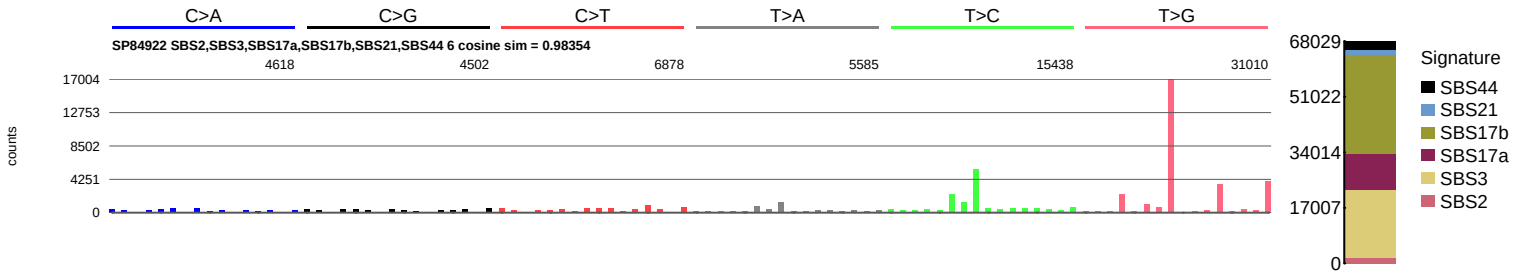

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.9836
4793
2565
8452
4177
17215
30827
counts
17056
12792
8528
4264
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2

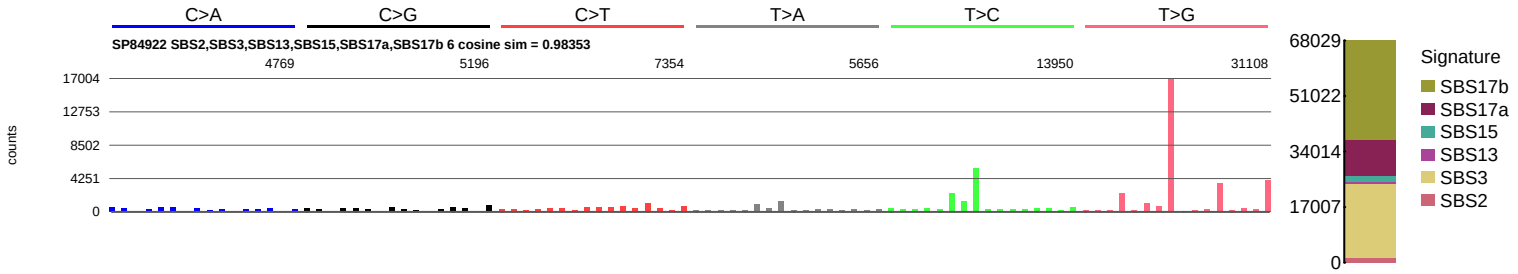

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98359
5294
5030
7260
5565
13837
31045
counts
17004
12753
8502
4251
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

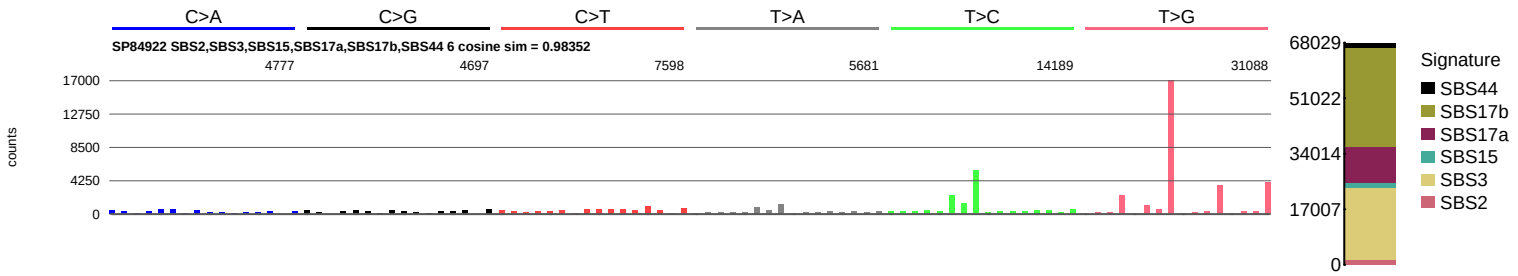

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98358
4784
2607
8723
4240
16846
30827
counts
17052
12789
8526
4263
0
68029
51022
34014
17007
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98357
5321
4524
7509
5586
14076
31018
counts
17000
12750
8500
4250
0
68029
51022
34014
17007
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98357
4774
2650
8637
4249
16870
30849
counts
17052
12789
8526
4263
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS44 6 cosine sim = 0.98354
4618
4502
6878
5585
15438
31010
counts
17004
12753
8502
4251
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98353
4769
5196
7354
5656
13950
31108
counts
17004
12753
8502
4251
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2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C>G
C>T
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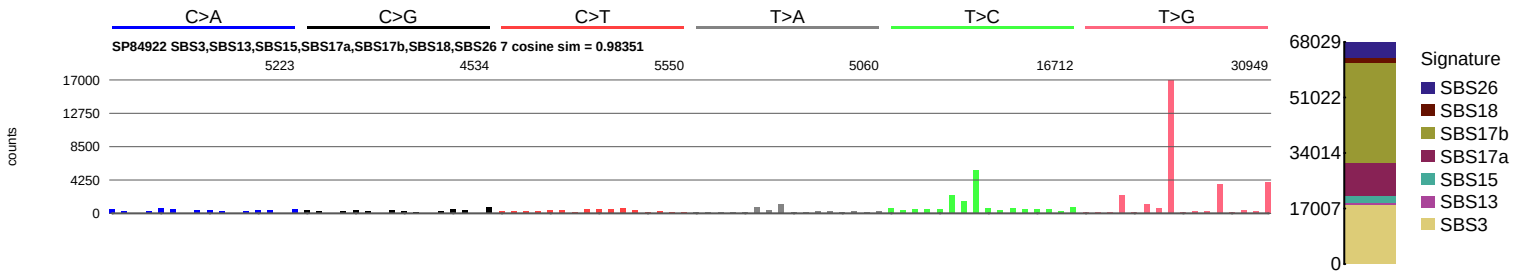
T>C
T>G
SP84922 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98352
4777
4697
7598
5681
14189
31088
counts
17000
12750
8500
4250
0
68029
51022
34014
17007
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS3
SBS2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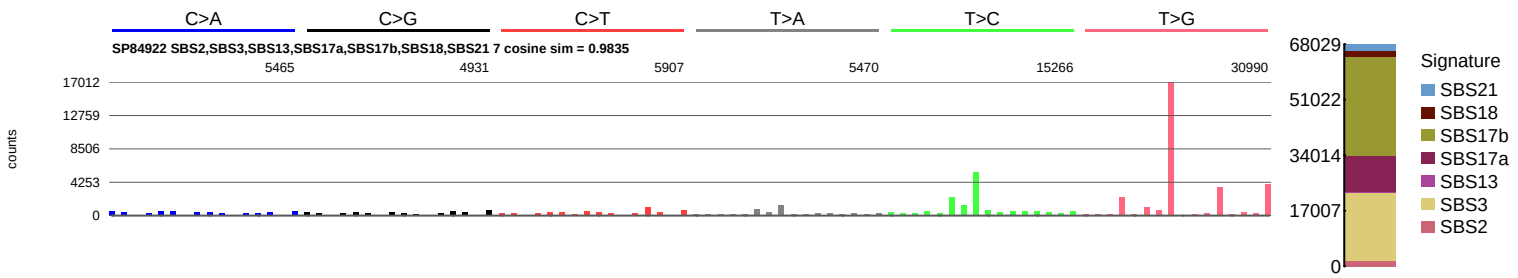
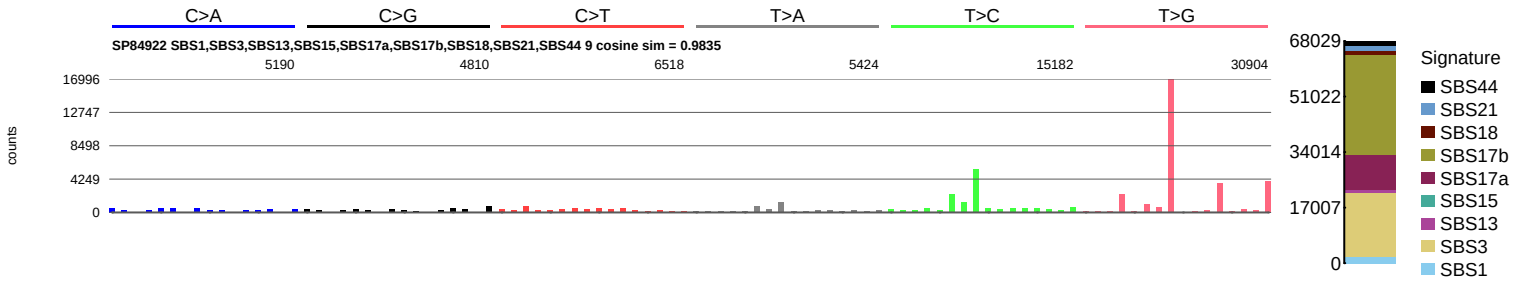
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98351
5223
4534
5550
5060
16712
30949
counts
17000
12750
8500
4250
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.9835
5465
4931
5907
5470
15266
30990
counts
17012
12759
8506
4253
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2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C>G
C>T
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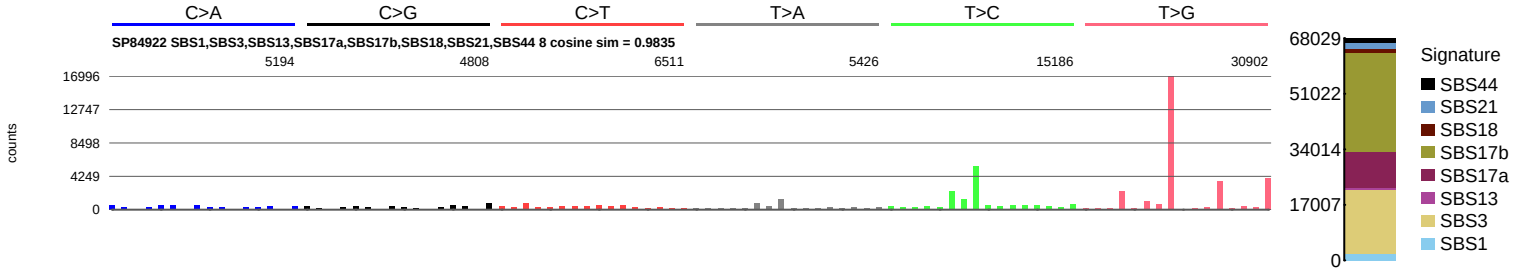
T>C
T>G
SP84922 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.9835
5190
4810
6518
5424
15182
30904
16996
12747
8498
4249
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1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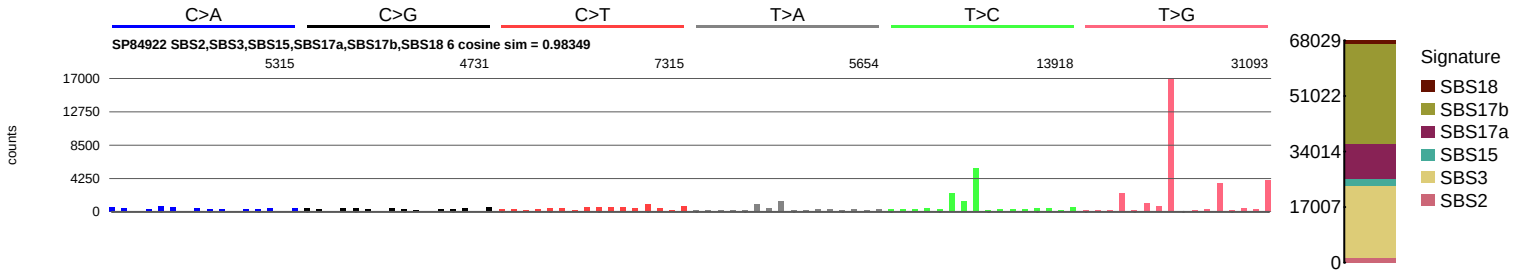
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.9835
5194
4808
6511
5426
15186
30902
16996
12747
8498
4249
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1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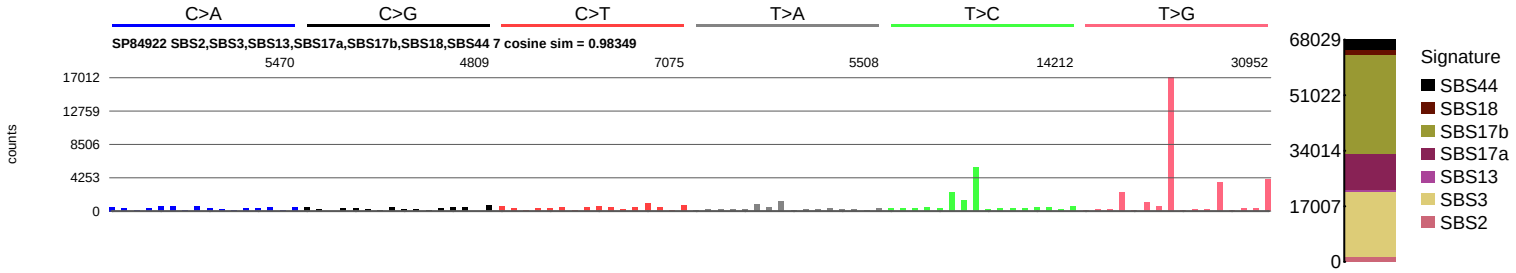
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98349
5315
4731
7315
5654
13918
31093
17000
12750
8500
4250
0
counts
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2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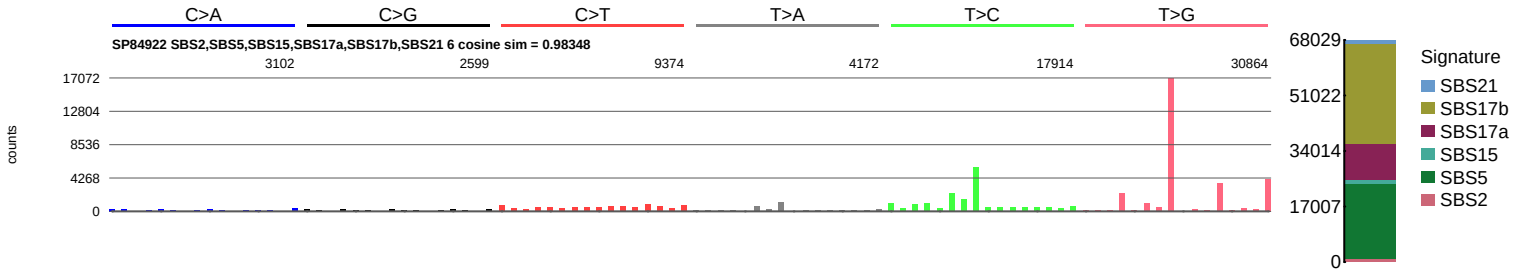
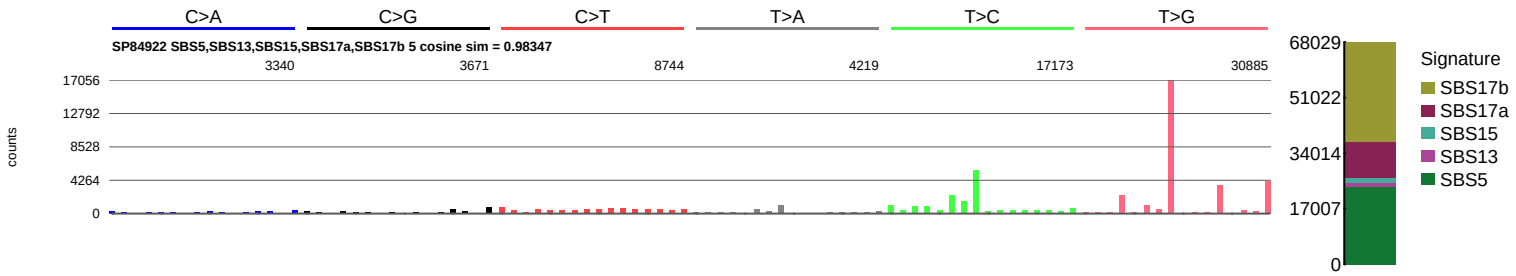
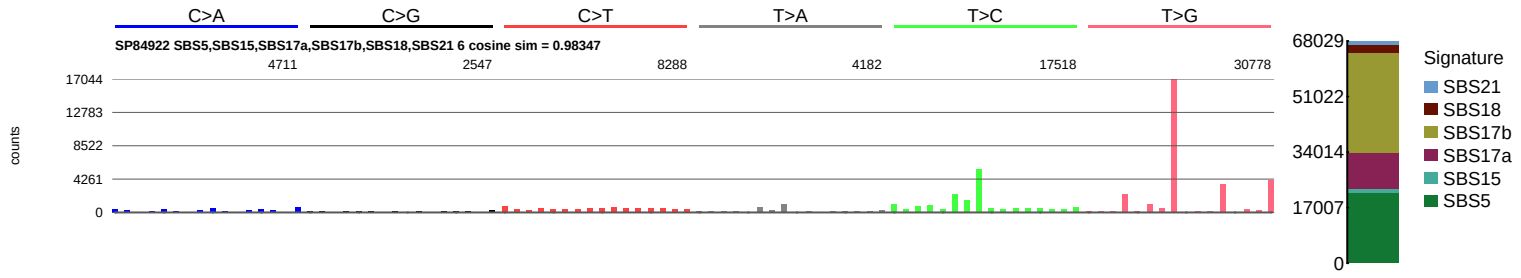
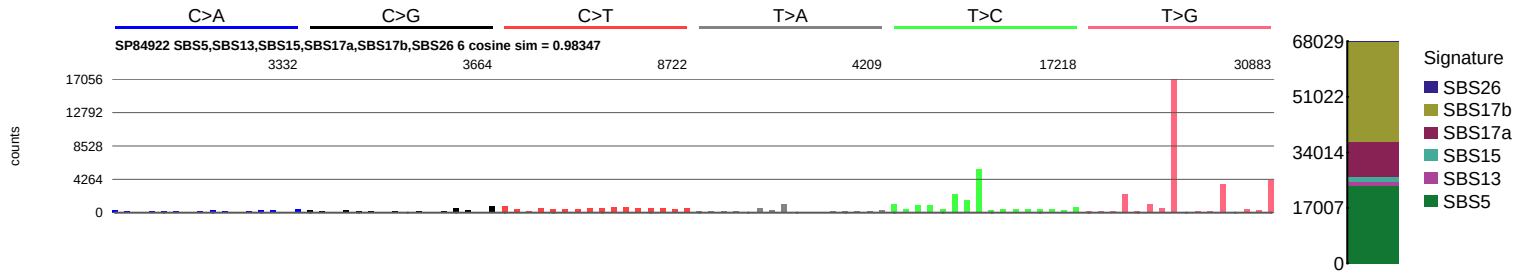
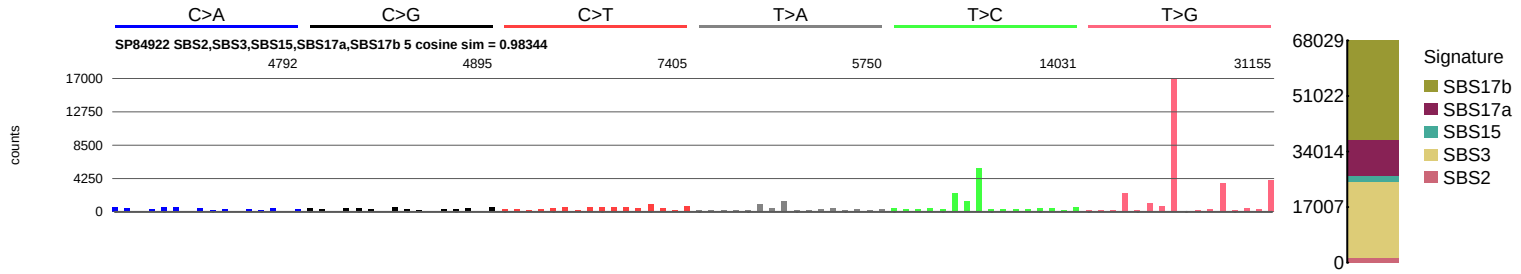
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98349
5470
4809
7075
5508
14212
30952
17012
12759
8506
4253
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2

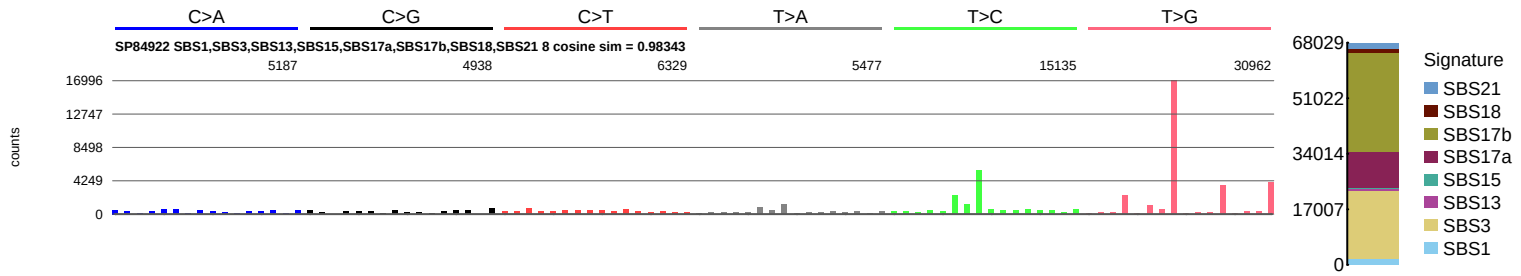

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98348
3102
2599
9374
4172
17914
30864
17072
12804
8536
4268
0
counts
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2

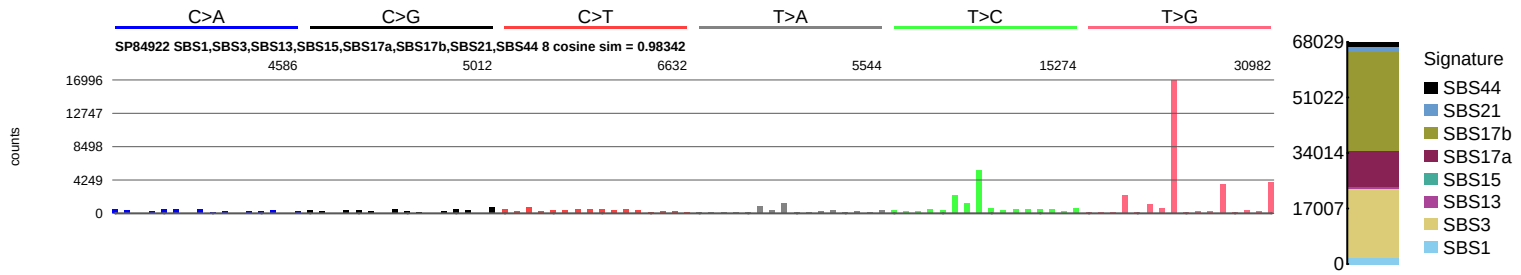

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS15,SBS17a,SBS17b 5 cosine sim = 0.98347
3340
3671
8744
4219
17173
30885
17056
12792
8528
4264
0
counts
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98347
4711
2547
8288
4182
17518
30778
counts
17044
12783
8522
4261
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98347
3332
3664
8722
4209
17218
30883
counts
17056
12792
8528
4264
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS15,SBS17a,SBS17b 5 cosine sim = 0.98344
4792
4895
7405
5750
14031
31155
counts
17000
12750
8500
4250
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98343
5187
4938
6329
5477
15135
30962
counts
16996
12747
8498
4249
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98342
4586
5012
6632
5544
15274
30982
counts
16996
12747
8498
4249
0
68029
51022
34014
17007
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Signature
SBS44
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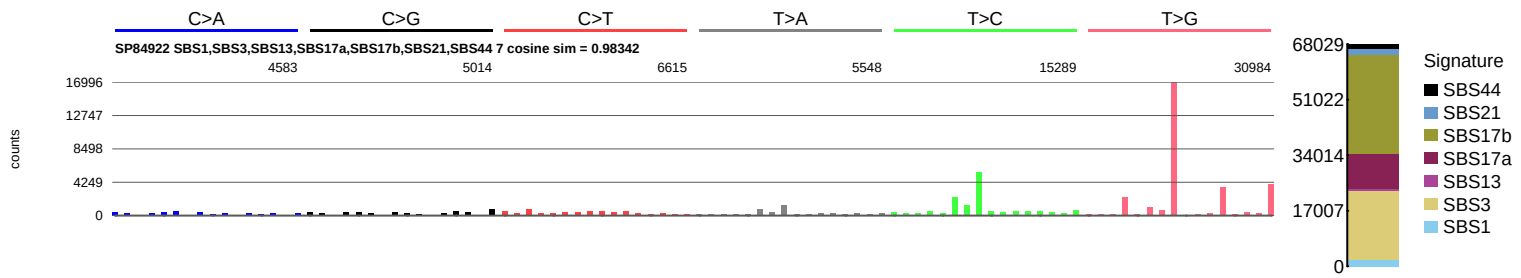
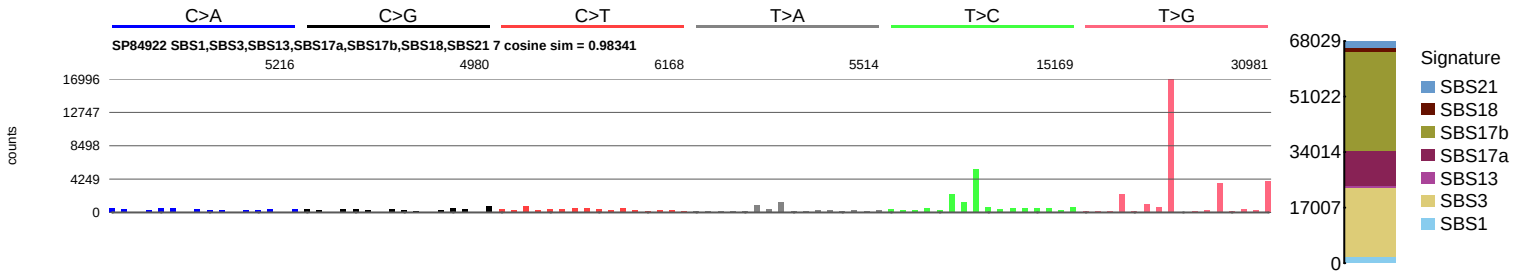
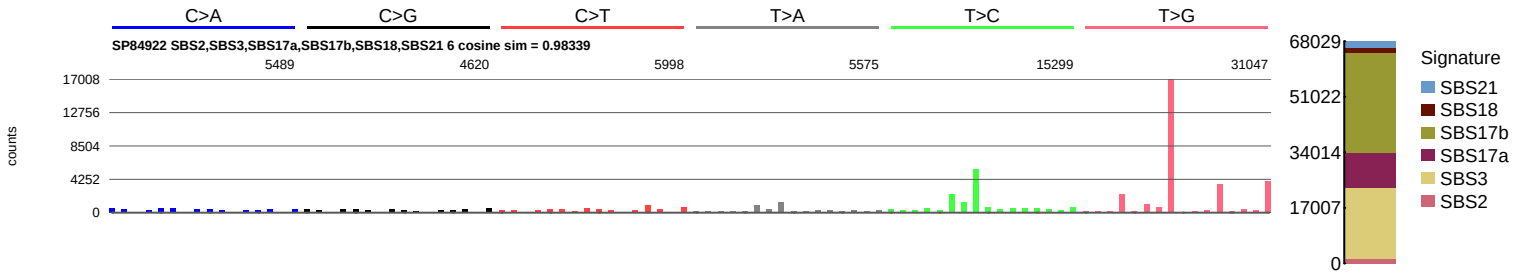
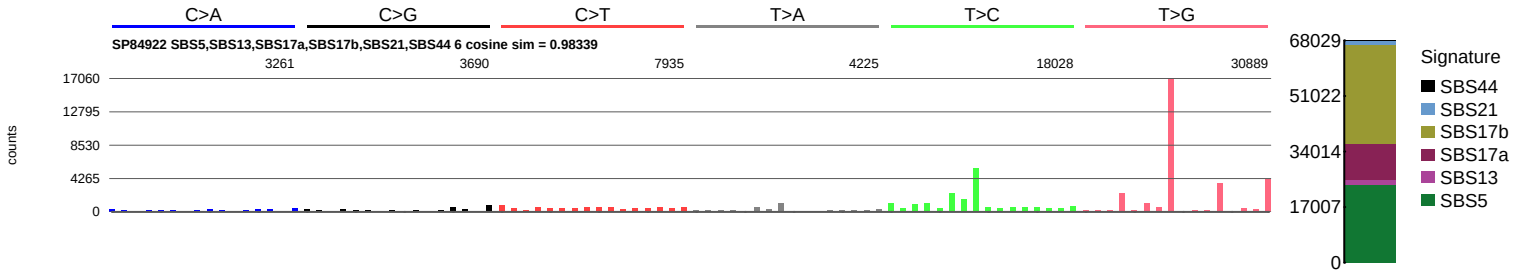
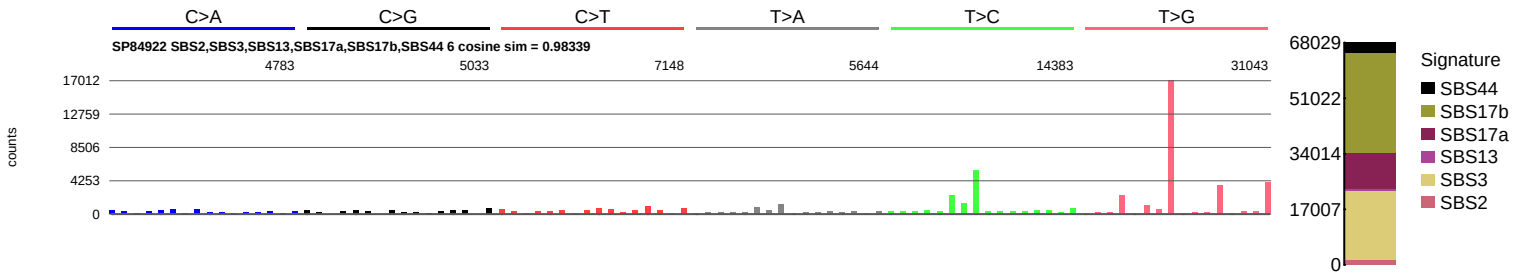
SBS13
SBS3
SBS1

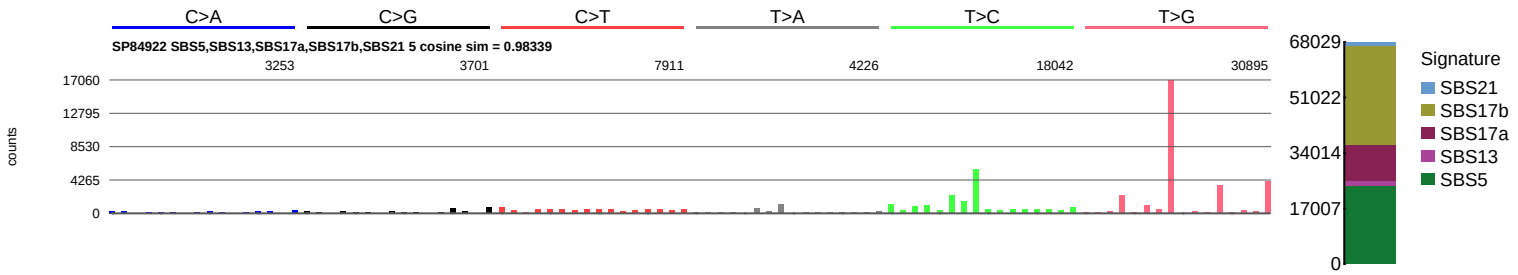

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98342
4583
5014
6615
5548
15289
30984
counts
16996
12747
8498
4249
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS1

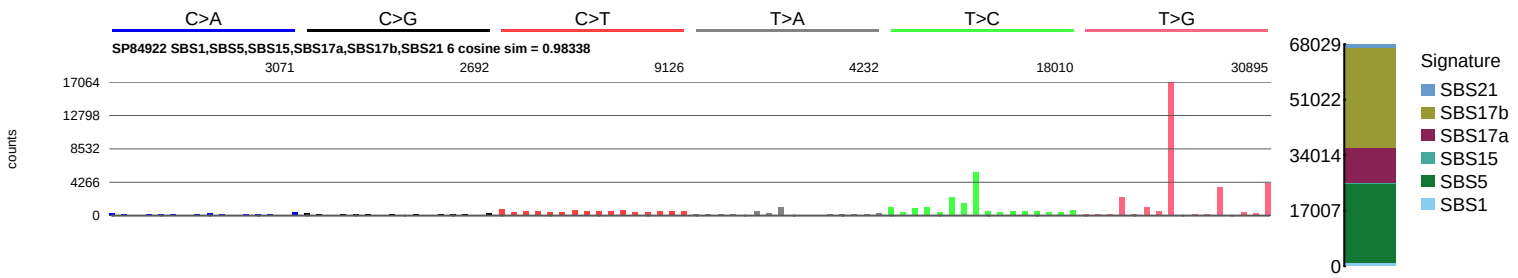

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98341
5216
4980
6168
5514
15169
30981
16996
12747
8498
4249
0
counts
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

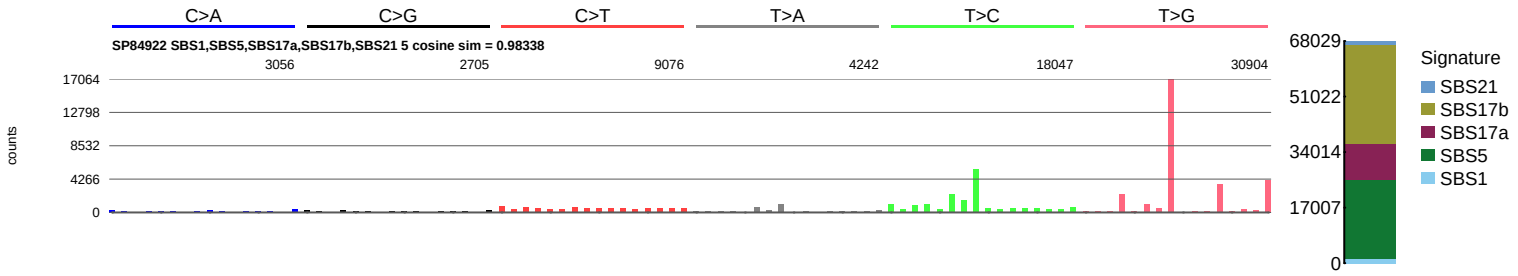

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98339
5489
4620
5998
5575
15299
31047
17008
12756
8504
4252
0
counts
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2

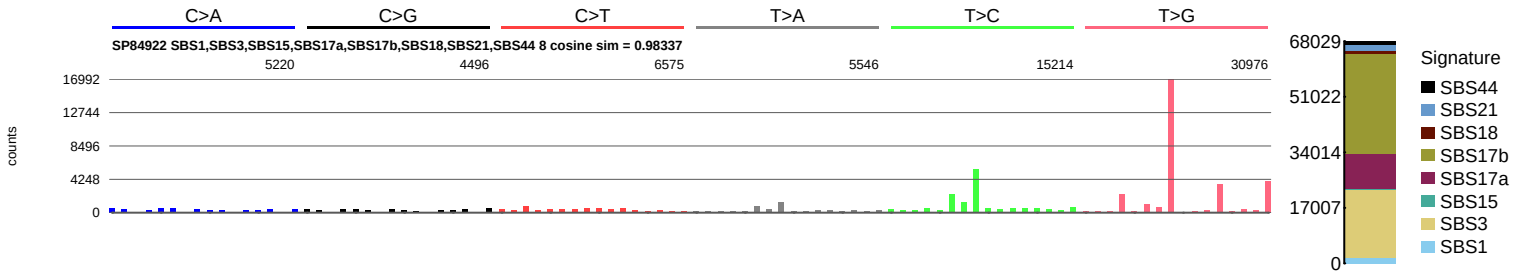

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 6 cosine sim = 0.98339
3261
3690
7935
4225
18028
30889
17060
12795
8530
4265
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS5

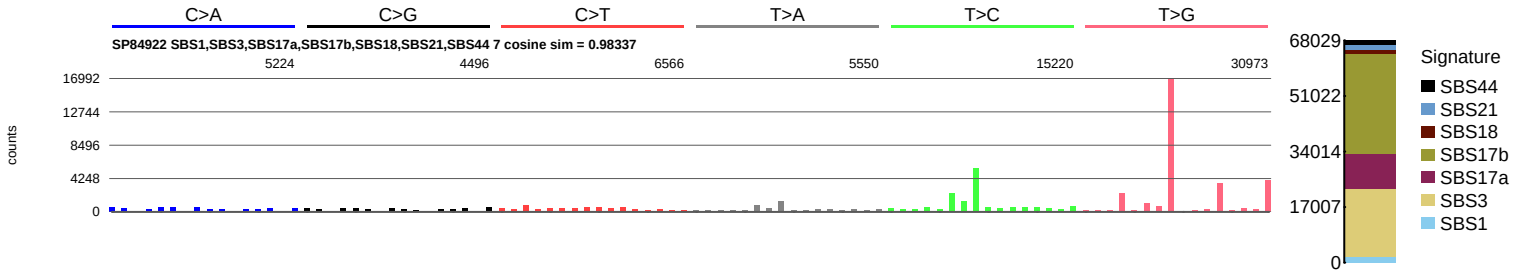

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98339
4783
5033
7148
5644
14383
31043
17012
12759
8506
4253
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS3
SBS2

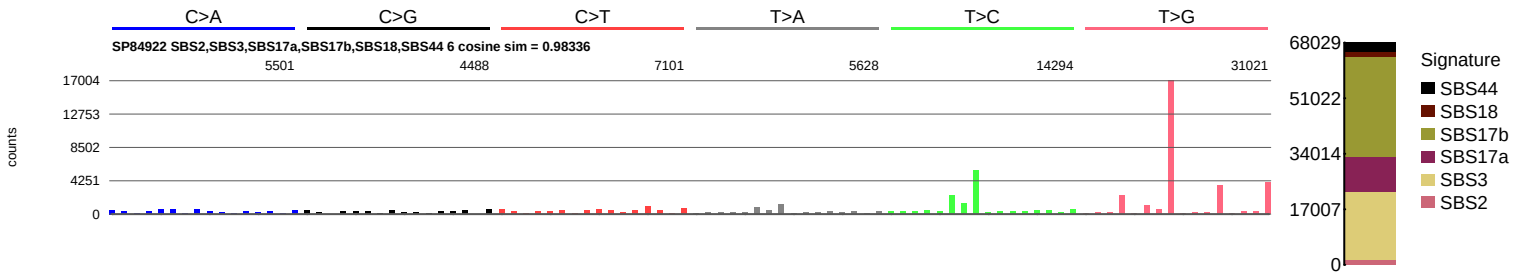

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS17a,SBS17b,SBS21 5 cosine sim = 0.98339
3253
3701
7911
4226
18042
30895
17060
12795
8530
4265
0
counts
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5

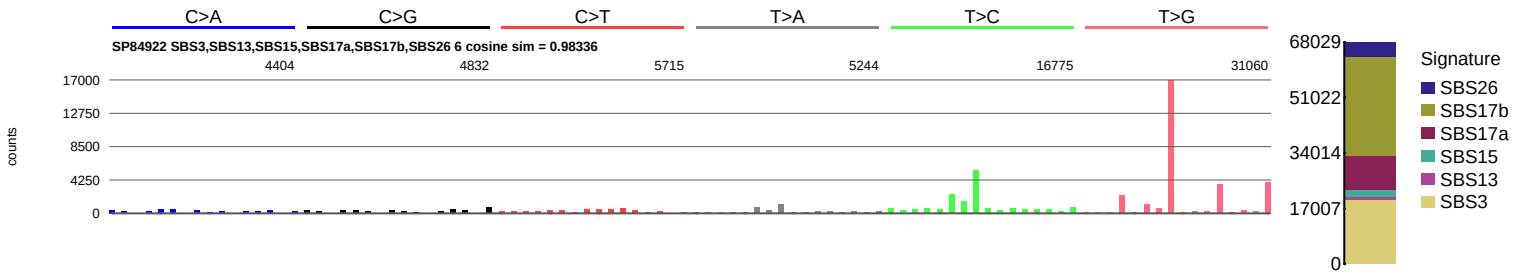

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98338
3071
2692
9126
4232
18010
30895
17064
12798
8532
4266
0
counts
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS17a,SBS17b,SBS21 5 cosine sim = 0.98338
3056
2705
9076
4242
18047
30904
counts
17064
12798
8532
4266
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98337
5220
4496
6575
5546
15214
30976
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98337
5224
4496
6566
5550
15220
30973
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98336
5501
4488
7101
5628
14294
31021
counts
17004
12753
8502
4251
0
68029
51022
34014
17007
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98336
4404
4832
5715
5244
16775
31060
counts
17000
12750
8500
4250
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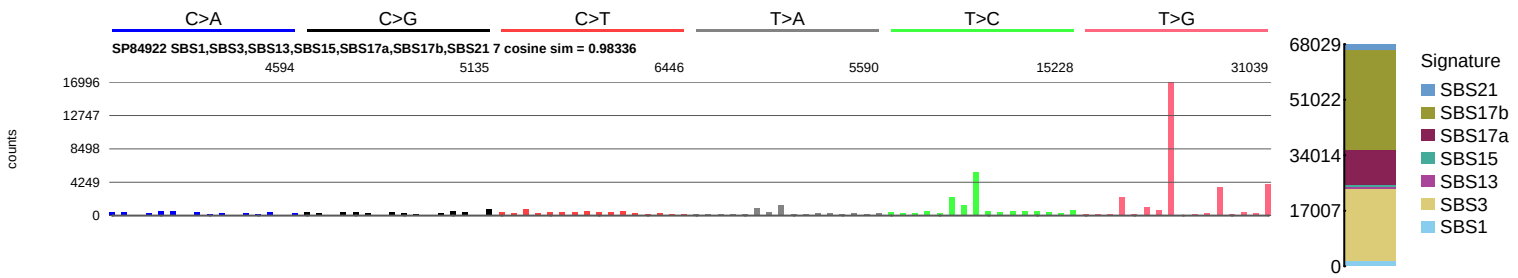
SBS13
SBS3


C>A
C>G
C>T
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T>G
SP84922 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98336
4594
5135
6446
5590
15228
31039
counts
16996
12747
8498
4249
0
68029
51022
34014
17007
0
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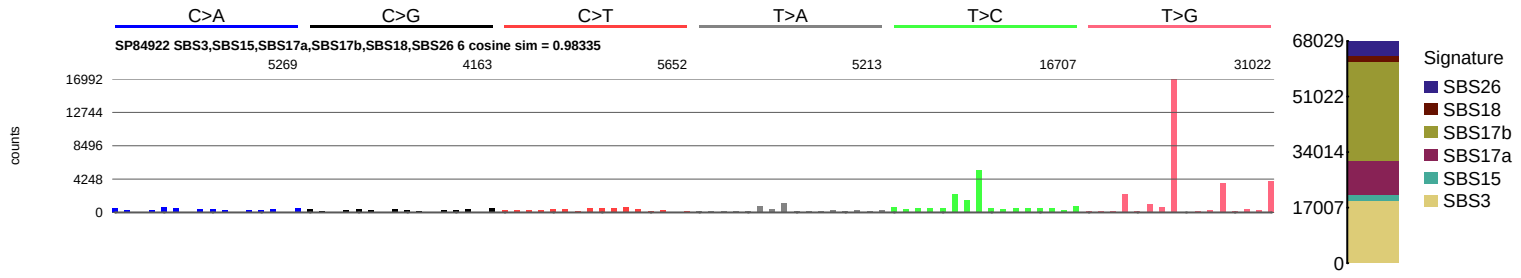
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1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C>G
C>T
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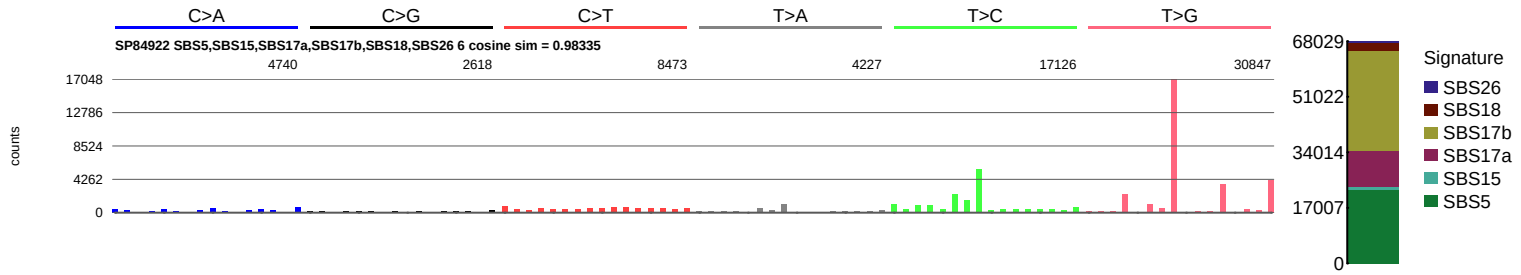
T>C
T>G
SP84922 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98335
5269
4163
5652
5213
16707
31022
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98335
4740
2618
8473
4227
17126
30847
counts
17048
12786
8524
4262
0
68029
51022
34014
17007
0
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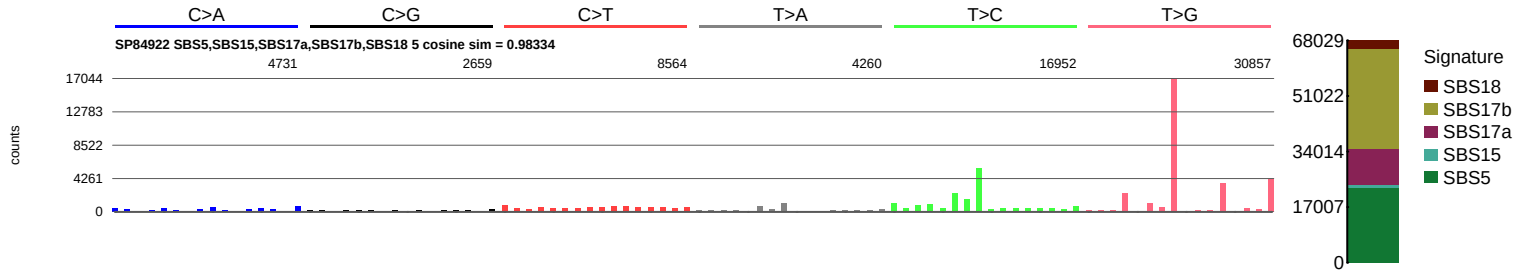
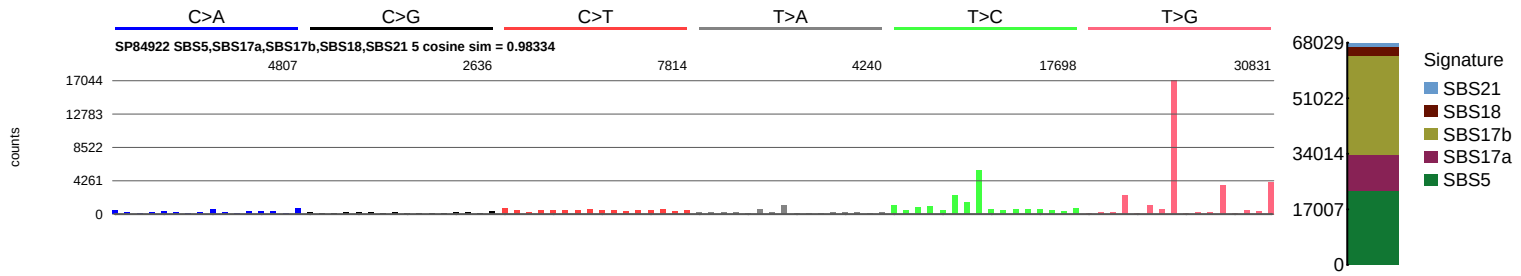
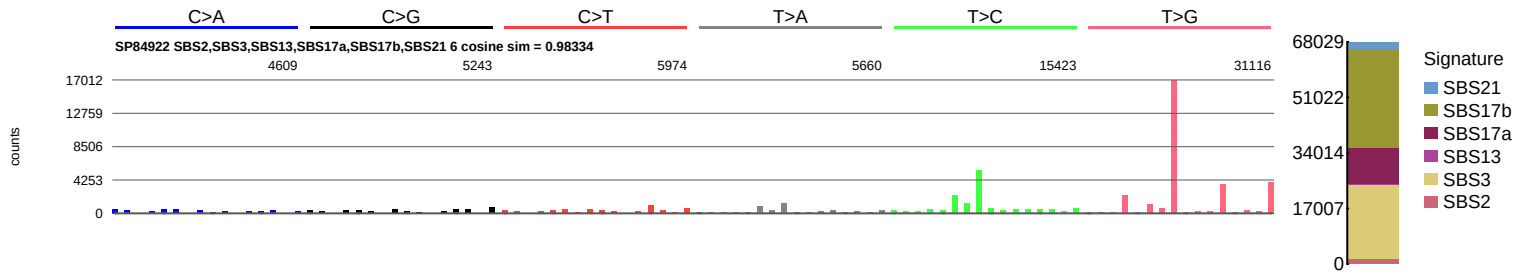
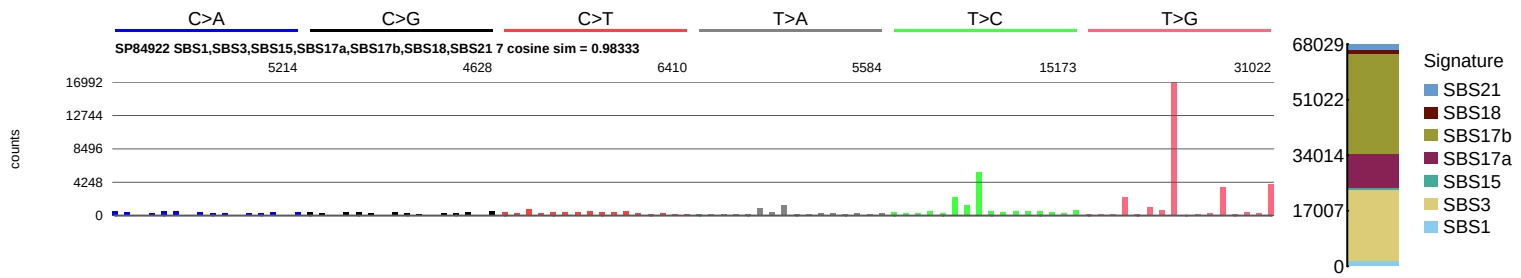
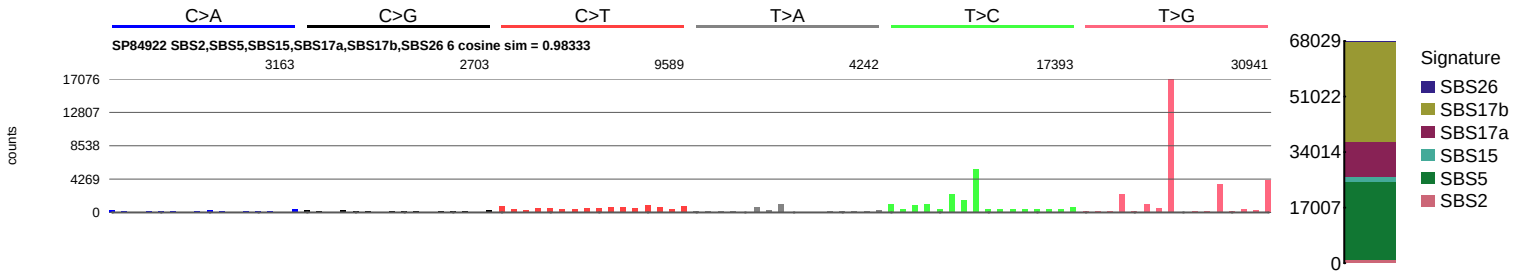
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5

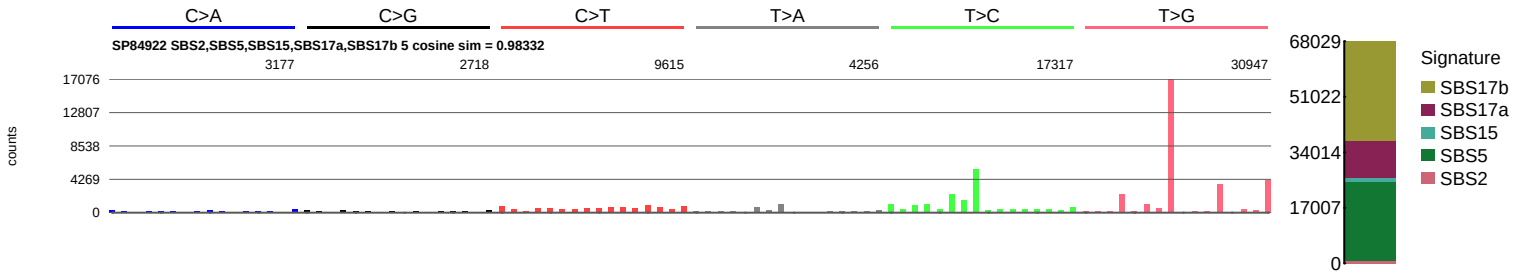

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS15,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98334
4731
2659
8564
4260
16952
30857
counts
17044
12783
8522
4261
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5

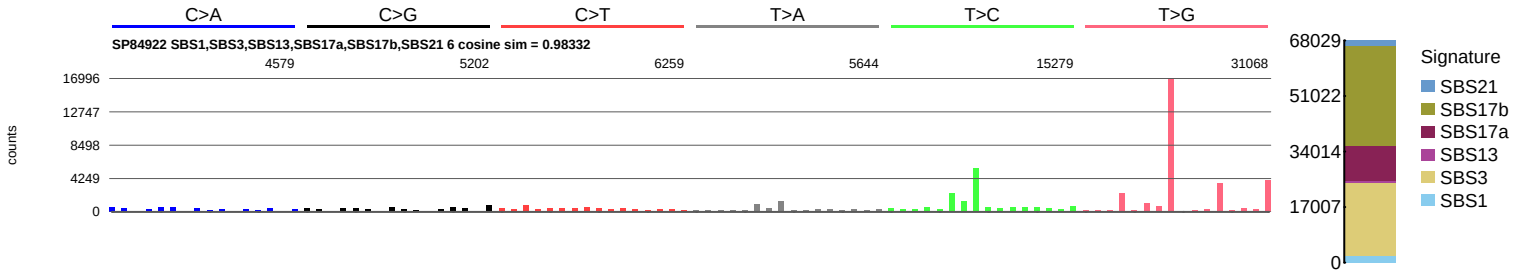

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS17a,SBS17b,SBS18,SBS21 5 cosine sim = 0.98334
4807
2636
7814
4240
17698
30831
counts
17044
12783
8522
4261
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5

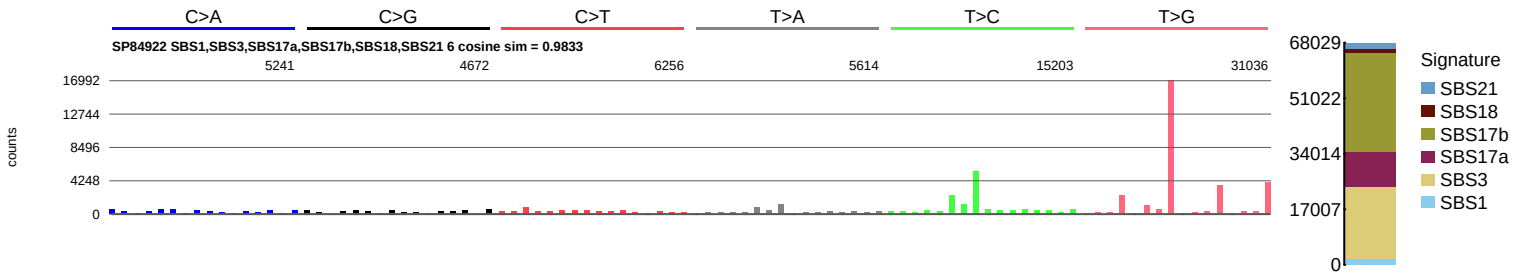

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98334
4609
5243
5974
5660
15423
31116
counts
17012
12759
8506
4253
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2

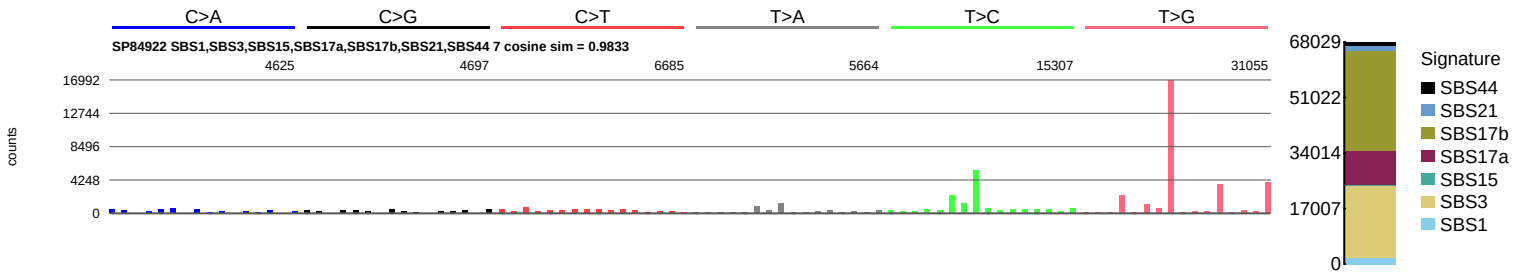

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98333
5214
4628
6410
5584
15173
31022
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

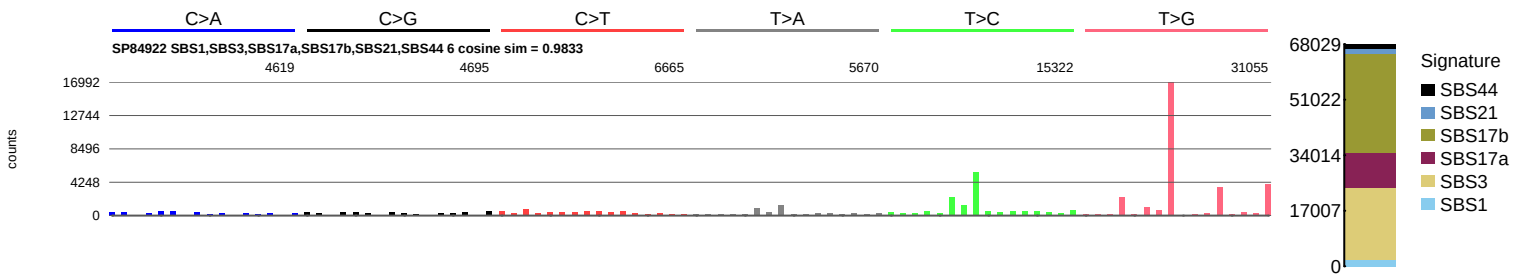

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98333
3163
2703
9589
4242
17393
30941
counts
17076
12807
8538
4269
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS2

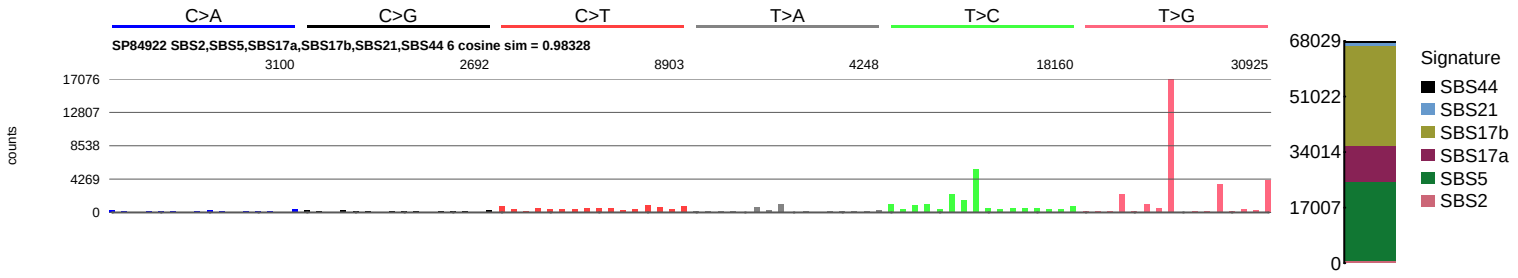

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS15,SBS17a,SBS17b 5 cosine sim = 0.98332
3177
2718
9615
4256
17317
30947
counts
17076
12807
8538
4269
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS2

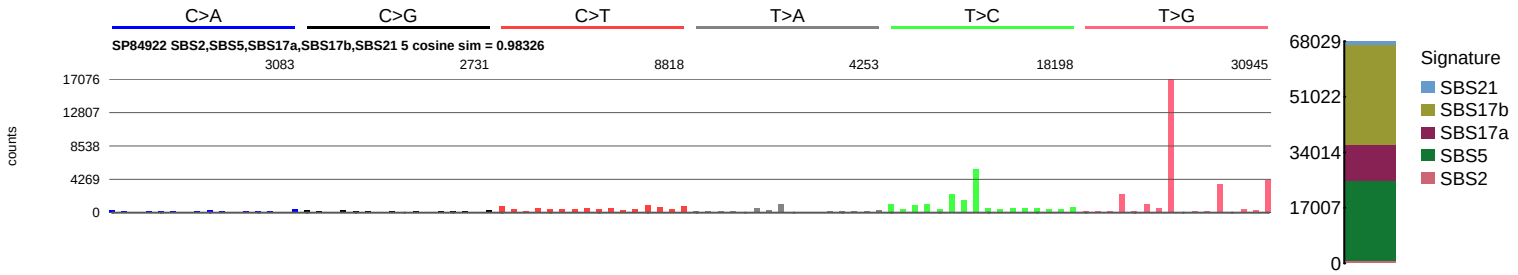

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98332
4579
5202
6259
5644
15279
31068
counts
16996
12747
8498
4249
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS1

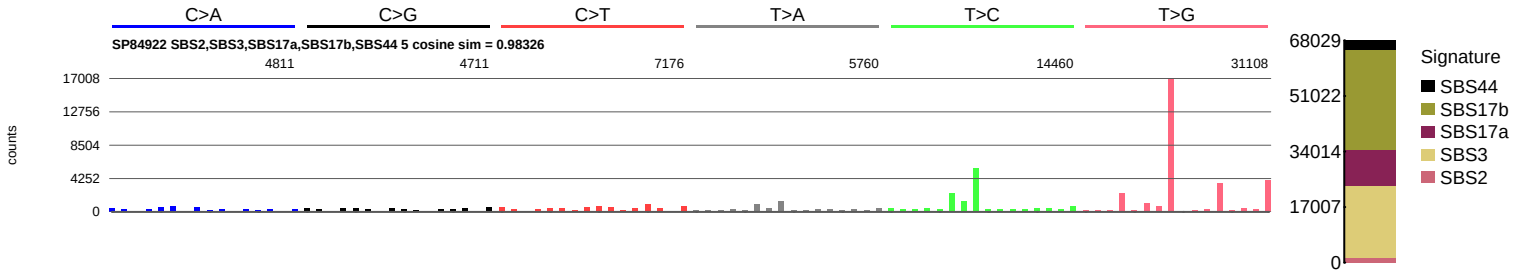

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.9833
5241
4672
6256
5614
15203
31036
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

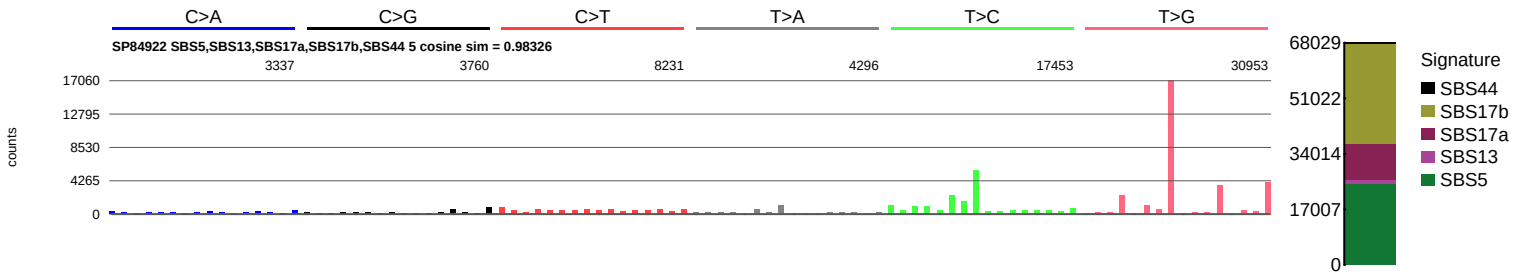

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.9833
4625
4697
6685
5664
15307
31055
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1

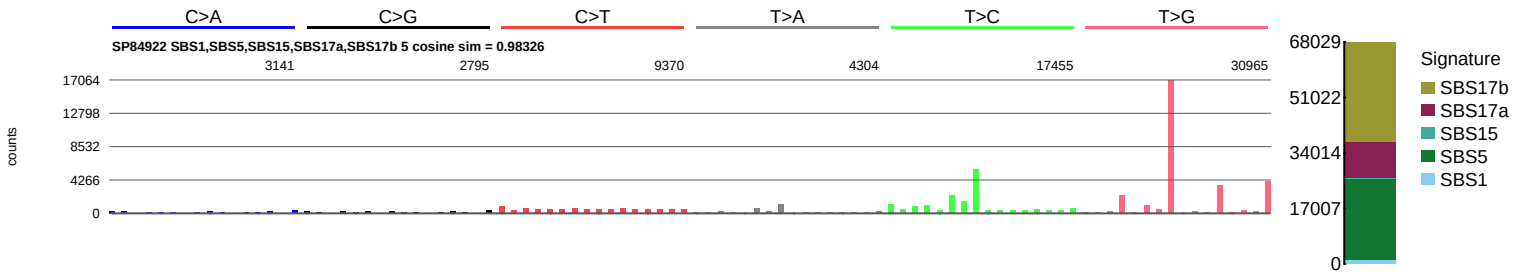

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS44 6 cosine sim = 0.9833
4619
4695
6665
5670
15322
31055
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS3
SBS1

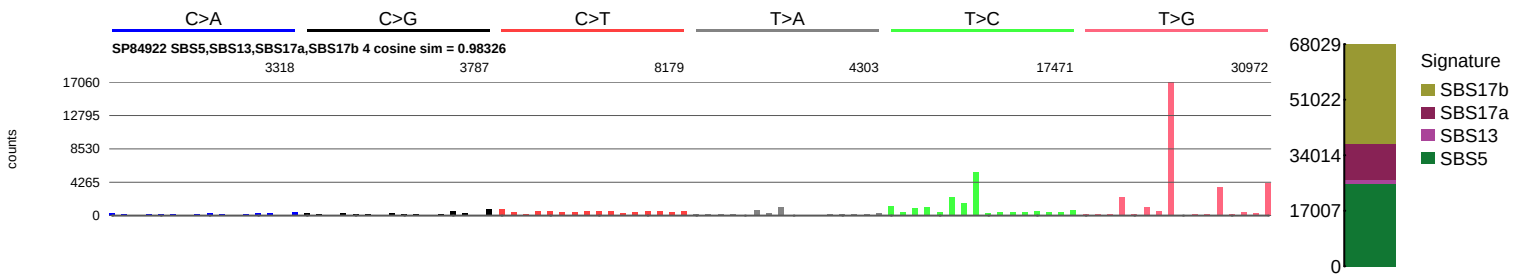

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS44 6 cosine sim = 0.98328
3100
2692
8903
4248
18160
30925
counts
17076
12807
8538
4269
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS2

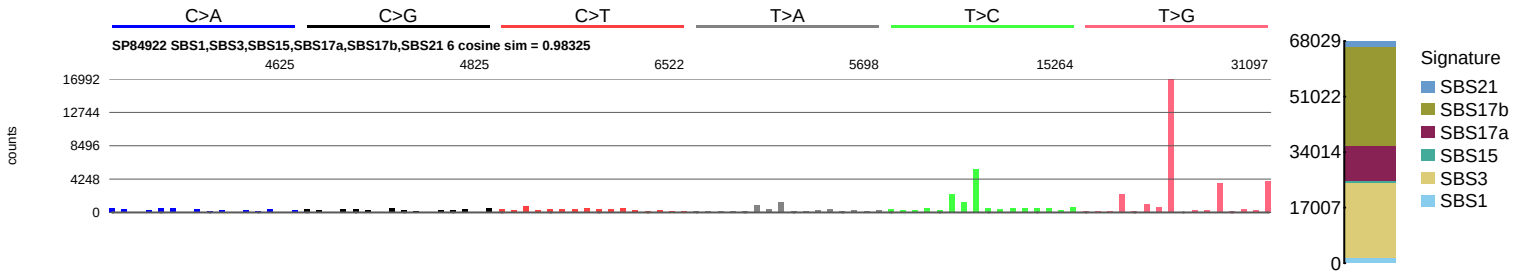

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS5,SBS17a,SBS17b,SBS21 5 cosine sim = 0.98326
3083
2731
8818
4253
18198
30945
counts
17076
12807
8538
4269
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS2

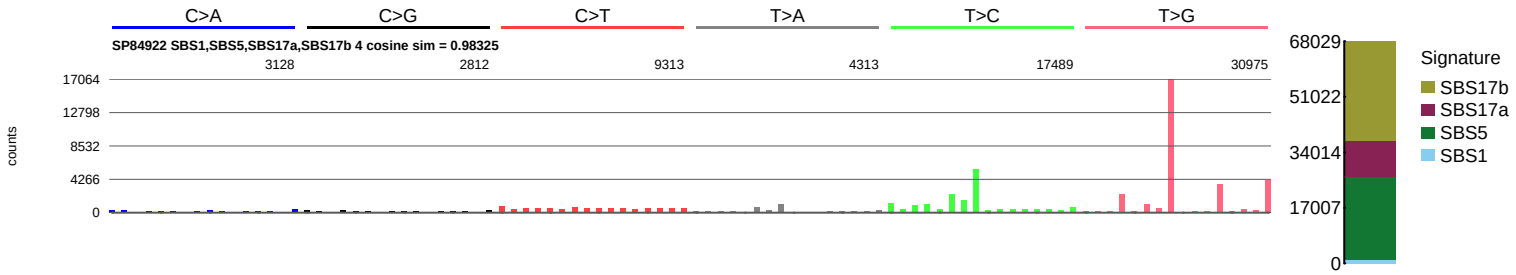

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS44 5 cosine sim = 0.98326
4811
4711
7176
5760
14460
31108
counts
17008
12756
8504
4252
0
68029
51022
34014
17007
0
Signature
SBS44
SBS17b
SBS17a
SBS3
SBS2

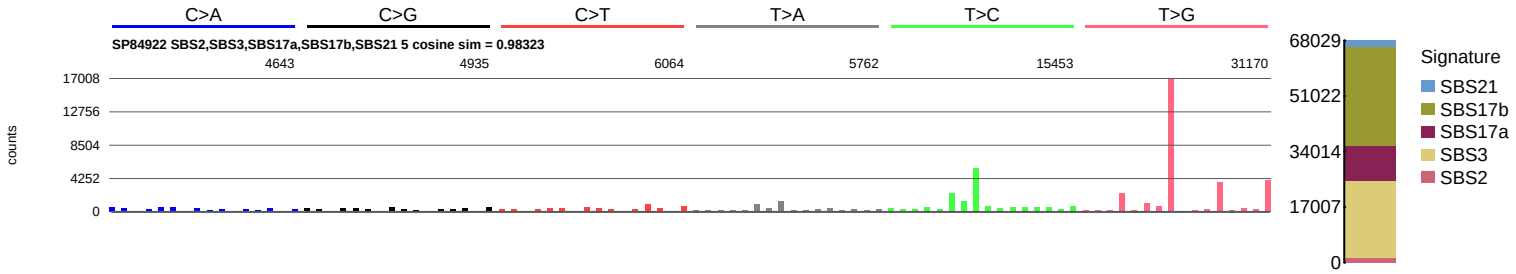

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS17a,SBS17b,SBS44 5 cosine sim = 0.98326
3337
3760
8231
4296
17453
30953
counts
17060
12795
8530
4265
0
68029
51022
34014
17007
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5

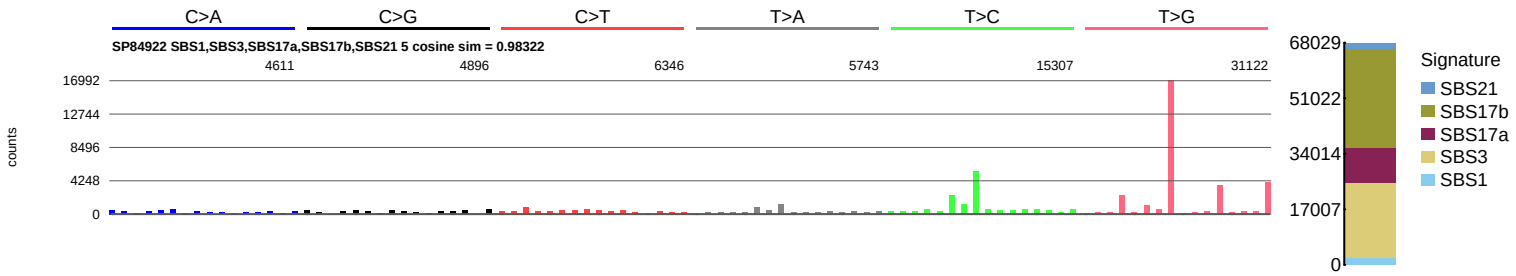

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS15,SBS17a,SBS17b 5 cosine sim = 0.98326
3141
2795
9370
4304
17455
30965
counts
17064
12798
8532
4266
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS1

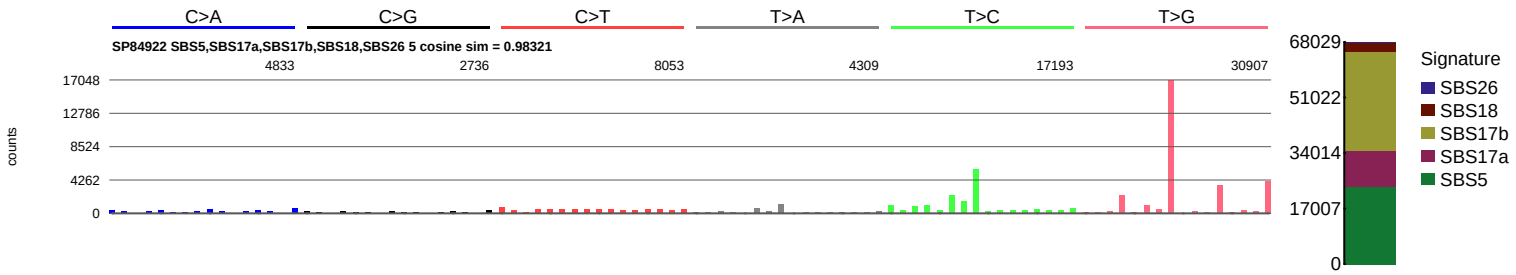

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS13,SBS17a,SBS17b 4 cosine sim = 0.98326
3318
3787
8179
4303
17471
30972
counts
17060
12795
8530
4265
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS13
SBS5

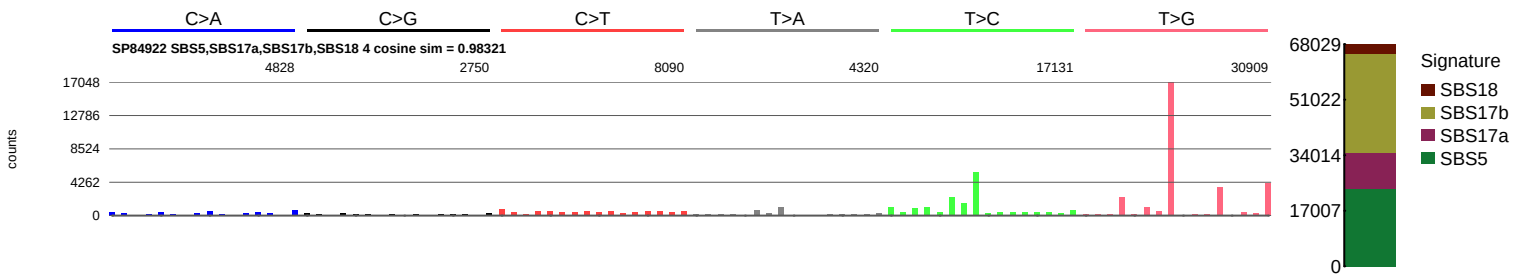

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98325
4625
4825
6522
5698
15264
31097
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1

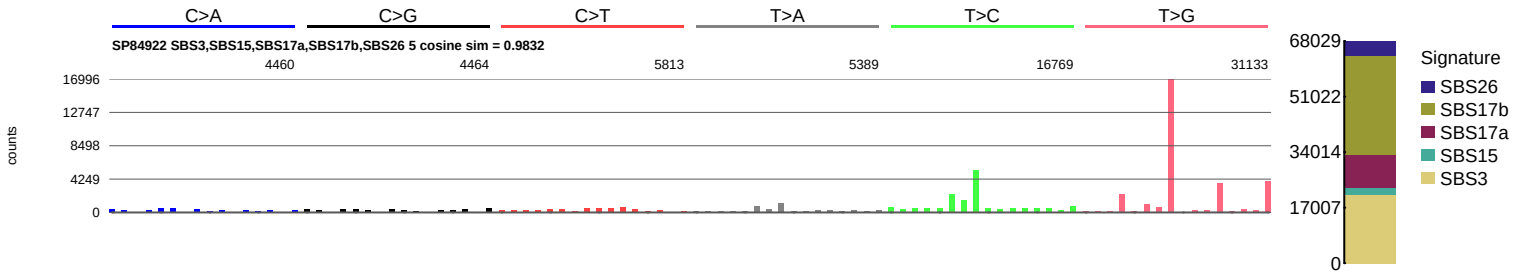

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS5,SBS17a,SBS17b 4 cosine sim = 0.98325
3128
2812
9313
4313
17489
30975
counts
17064
12798
8532
4266
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS5
SBS1

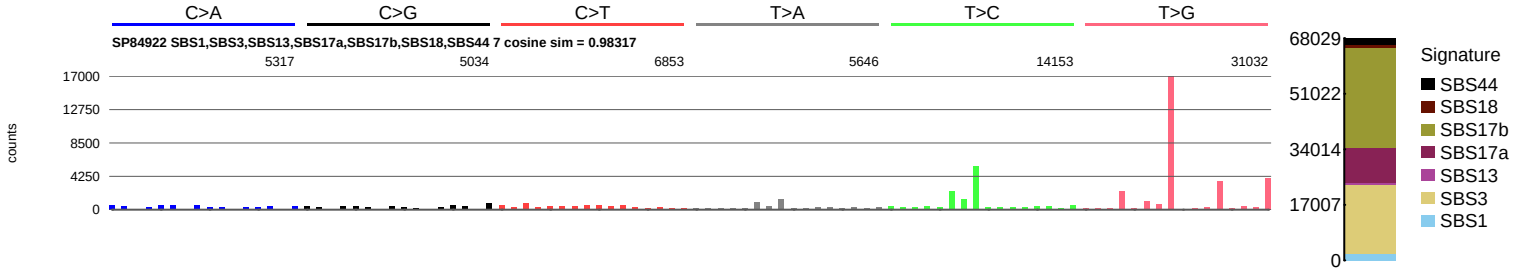

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS21 5 cosine sim = 0.98323
4643
4935
6064
5762
15453
31170
counts
17008
12756
8504
4252
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS17a,SBS17b,SBS21 5 cosine sim = 0.98322
4611
4896
6346
5743
15307
31122
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS17a,SBS17b,SBS18,SBS26 5 cosine sim = 0.98321
4833
2736
8053
4309
17193
30907
counts
17048
12786
8524
4262
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS17a,SBS17b,SBS18 4 cosine sim = 0.98321
4828
2750
8090
4320
17131
30909
counts
17048
12786
8524
4262
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS15,SBS17a,SBS17b,SBS26 5 cosine sim = 0.9832
4460
4464
5813
5389
16769
31133
16996
12747
8498
4249
0
counts
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS3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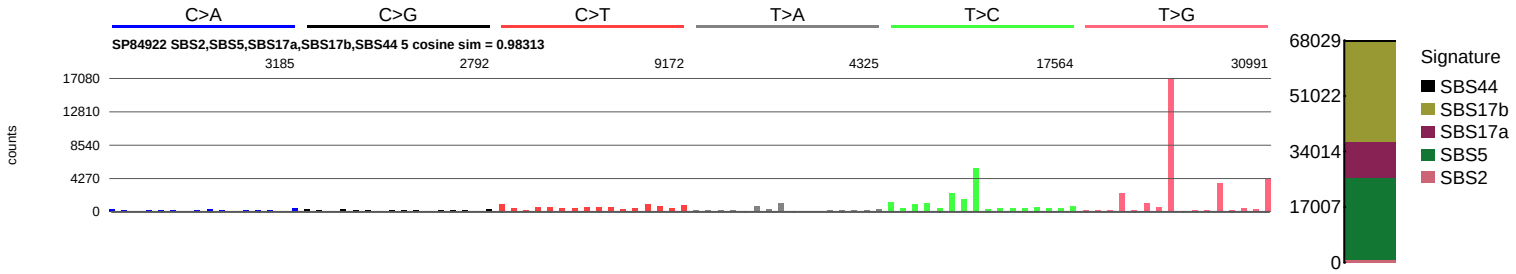
SP84922 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98317
5317
5034
6853
5646
14153
31032
17000
12750
8500
4250
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1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SP84922 SBS2,SBS5,SBS17a,SBS17b,SBS44 5 cosine sim = 0.98313
3185
2792
9172
4325
17564
30991
17080
12810
8540
4270
0
counts
68029
51022
34014
17007
0
Signature
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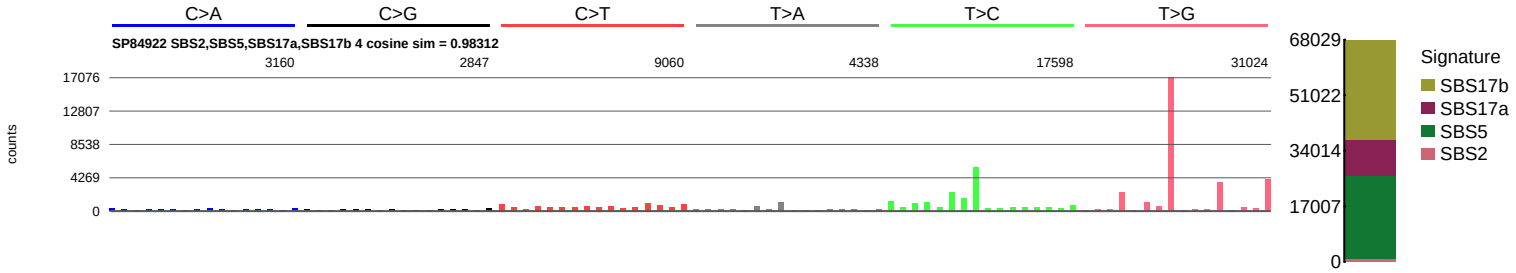
SBS17b
SBS17a
SBS5
SBS2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SP84922 SBS2,SBS5,SBS17a,SBS17b 4 cosine sim = 0.98312
3160
2847
9060
4338
17598
31024
17076
12807
8538
4269
0
counts
68029
51022
34014
17007
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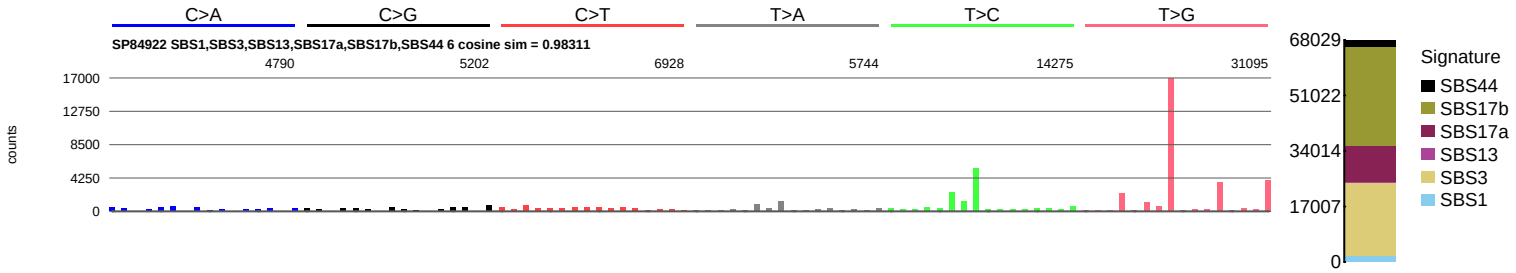
Signature
SBS17b
SBS17a
SBS5
SBS2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T>C
T>G
SP84922 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98311
4790
5202
6928
5744
14275
31095
17000
12750
8500
4250
0
counts
68029
51022
34014
17007
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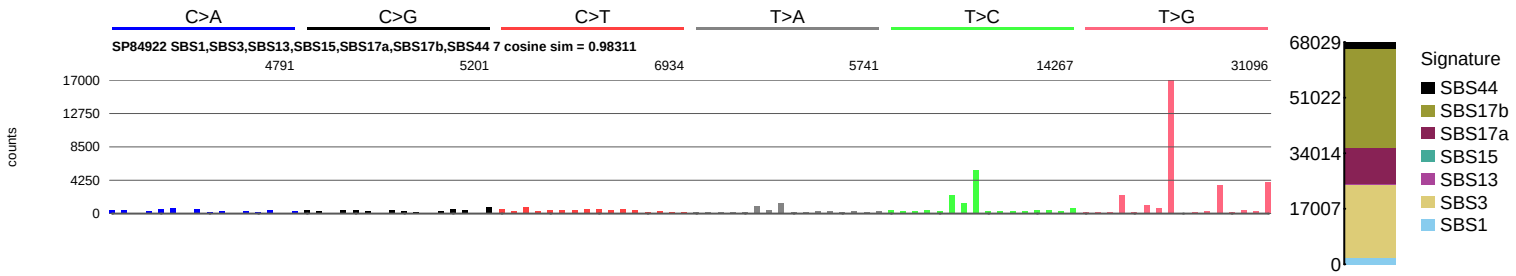
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS3
SBS1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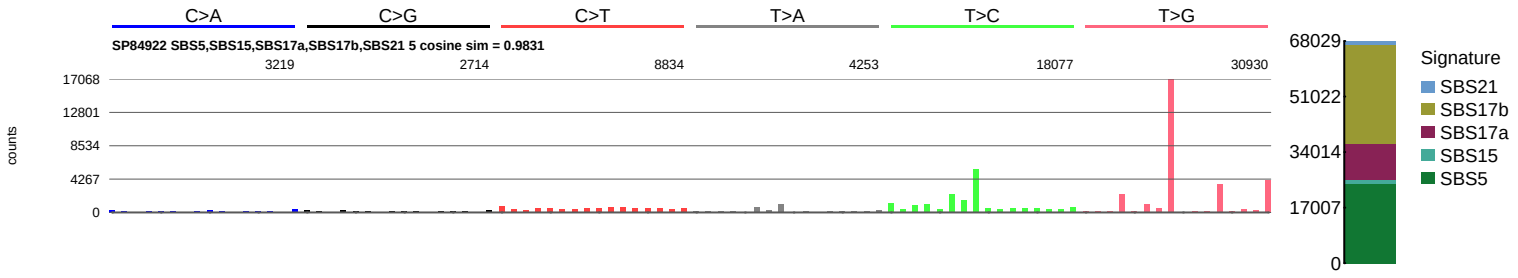
SP84922 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98311
4791
5201
6934
5741
14267
31096
17000
12750
8500
4250
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1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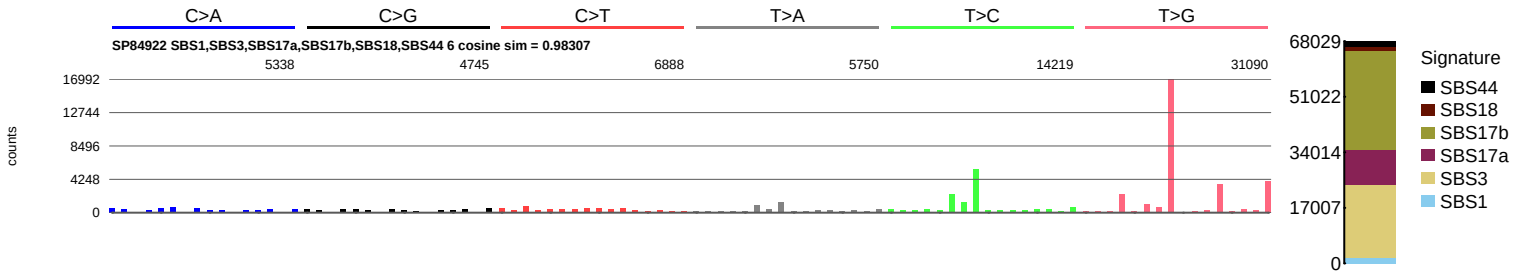
T>G
SP84922 SBS5,SBS15,SBS17a,SBS17b,SBS21 5 cosine sim = 0.9831
3219
2714
8834
4253
18077
30930
17068
12801
8534
4267
0
counts
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5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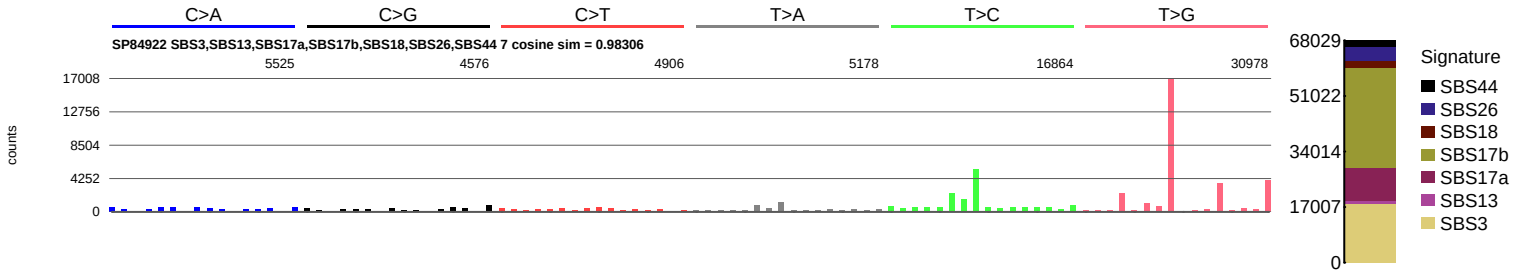
T>C
T>G
SP84922 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98307
5338
4745
6888
5750
14219
31090
16992
12744
8496
4248
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS3
SBS1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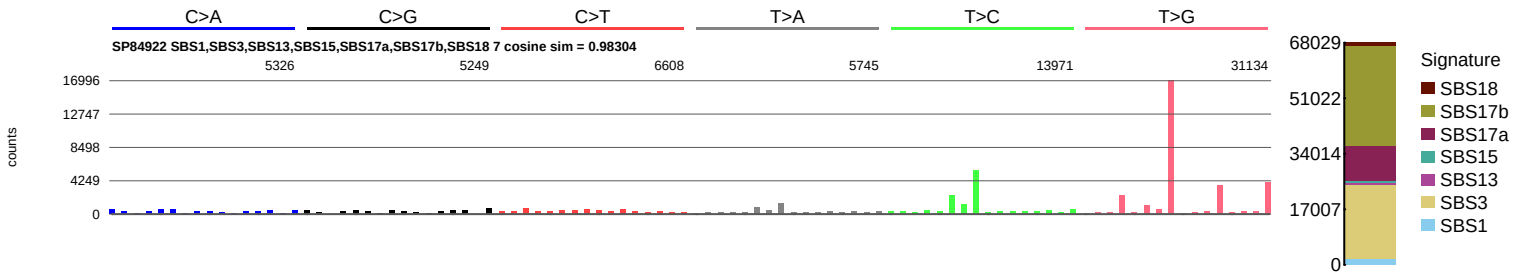
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98306
5525
4576
4906
5178
16864
30978
17008
12756
8504
4252
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3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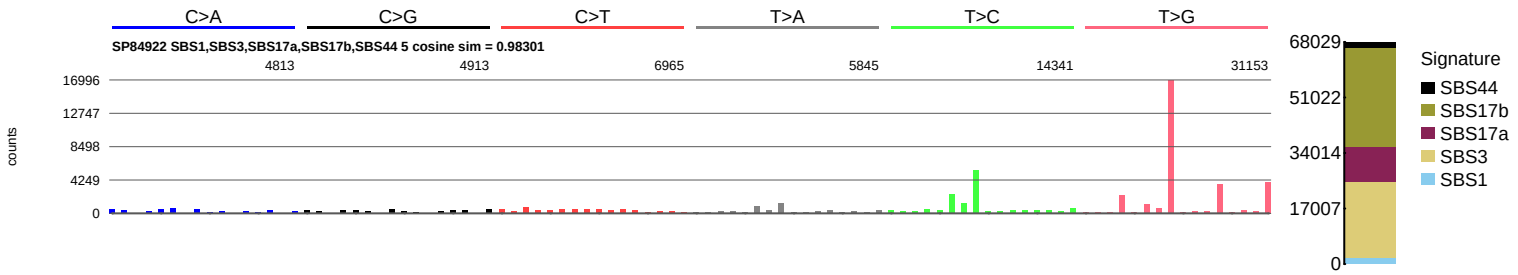
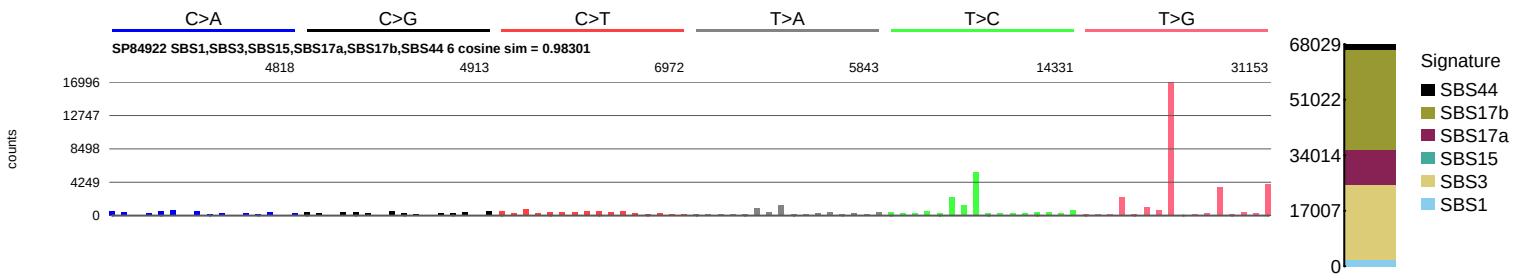
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98304
5326
5249
6608
5745
13971
31134
16996
12747
8498
4249
0
counts
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS17a,SBS17b,SBS44 5 cosine sim = 0.98301
4813
4913
6965
5845
14341
31153
16996
12747
8498
4249
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS17b
SBS17a
SBS3
SBS1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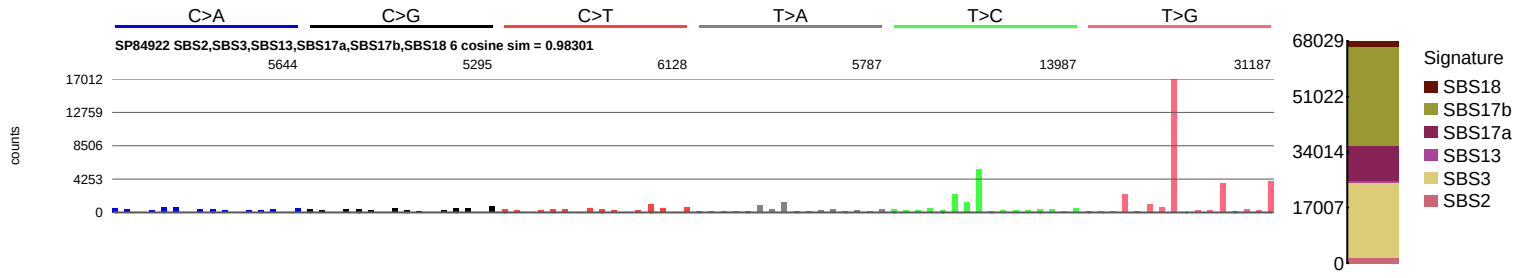
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98301
4818
4913
6972
5843
14331
31153
16996
12747
8498
4249
0
counts
68029
51022
34014
17007
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS3
SBS1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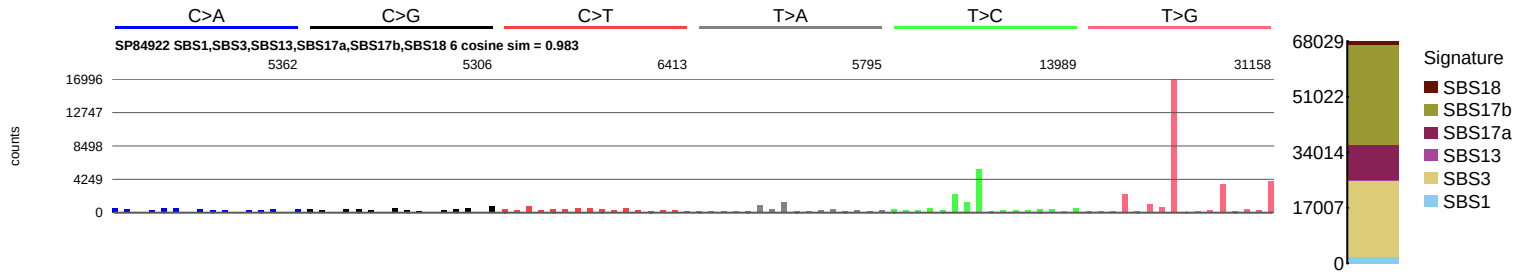
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98301
5644
5295
6128
5787
13987
31187
counts
17012
12759
8506
4253
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18 6 cosine sim = 0.983
5362
5306
6413
5795
13989
31158
counts
16996
12747
8498
4249
0
68029
51022
34014
17007
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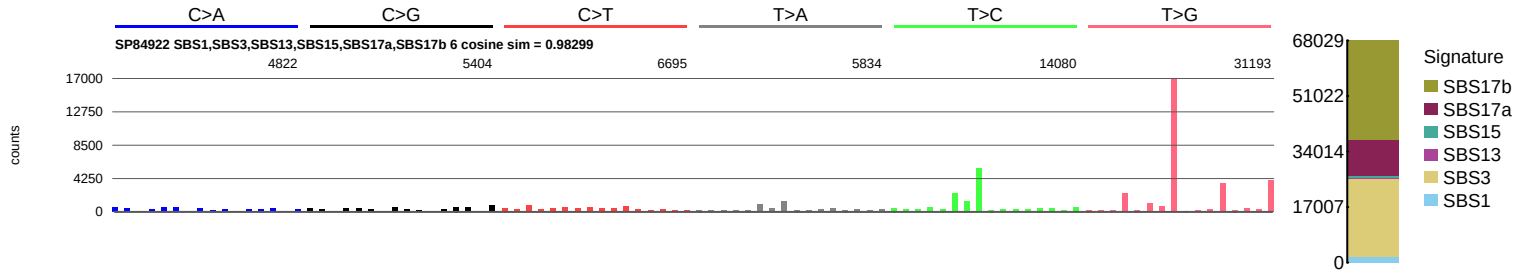
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1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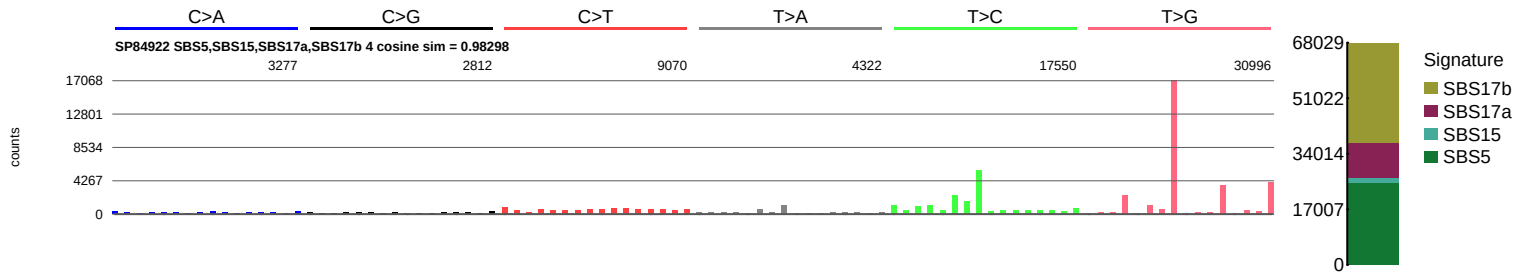
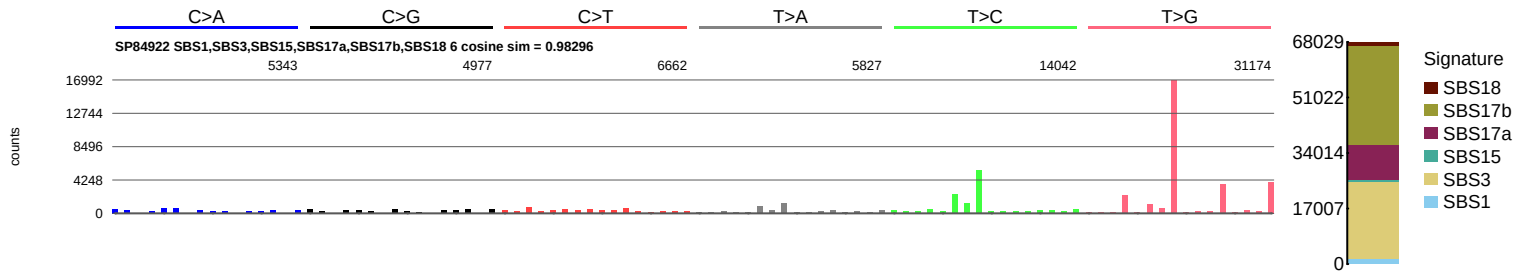
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98299
4822
5404
6695
5834
14080
31193
counts
17000
12750
8500
4250
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS15,SBS17a,SBS17b 4 cosine sim = 0.98298
3277
2812
9070
4322
17550
30996
counts
17068
12801
8534
4267
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
SBS5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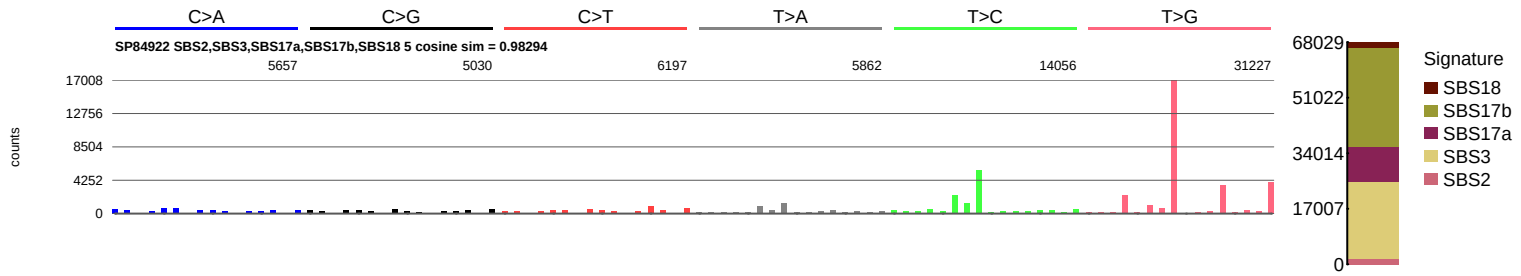
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98296
5343
4977
6662
5827
14042
31174
counts
16992
12744
8496
4248
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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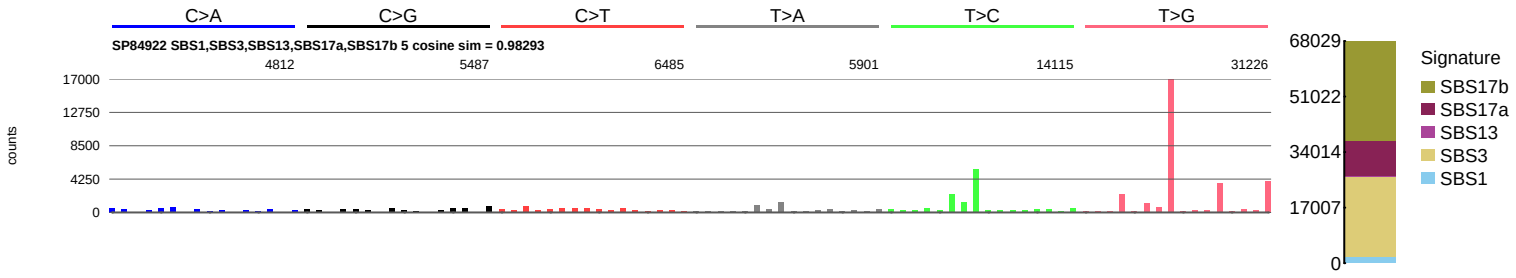
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98294
5657
5030
6197
5862
14056
31227
counts
17008
12756
8504
4252
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS3
SBS2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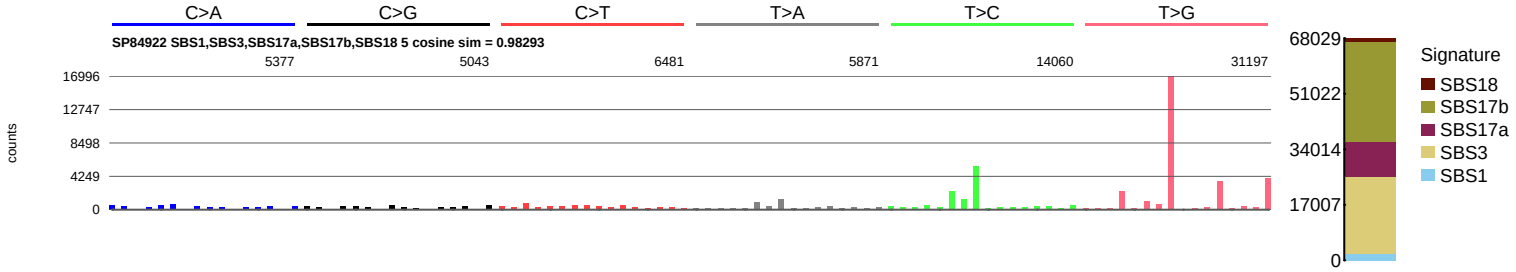
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS13,SBS17a,SBS17b 5 cosine sim = 0.98293
4812
5487
6485
5901
14115
31226
counts
17000
12750
8500
4250
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98293
5377
5043
6481
5871
14060
31197
counts
16996
12747
8498
4249
0
68029
51022
34014
17007
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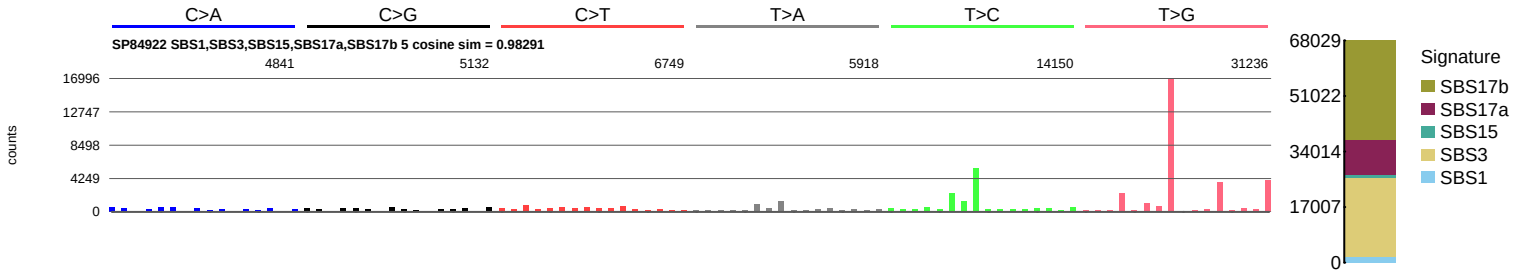
Signature
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS15,SBS17a,SBS17b 5 cosine sim = 0.98291
4841
5132
6749
5918
14150
31236
counts
16996
12747
8498
4249
0
68029
51022
34014
17007
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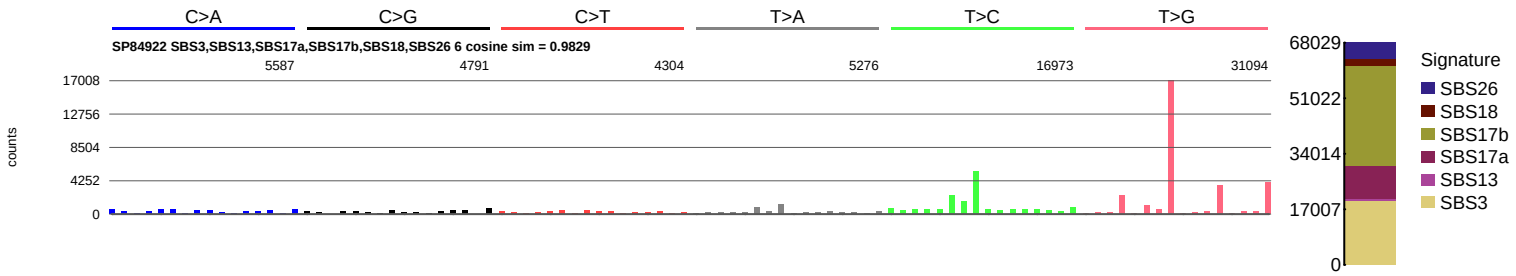
Signature
SBS17b
SBS17a
SBS15
SBS3
SBS1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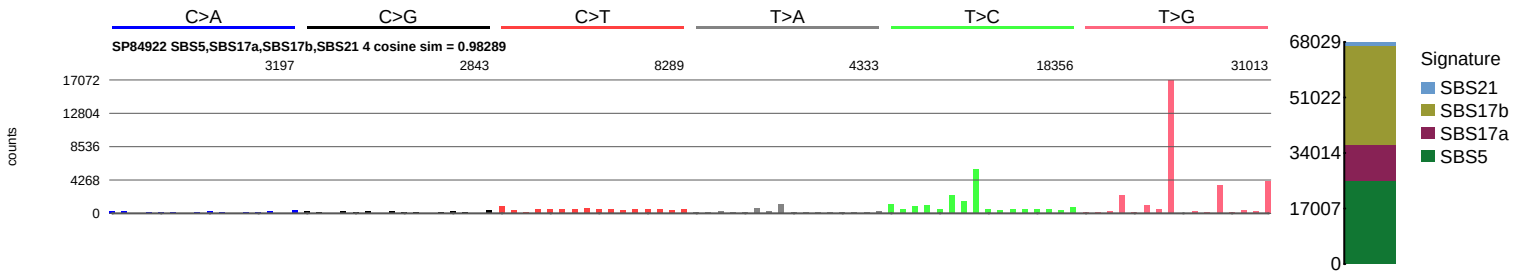
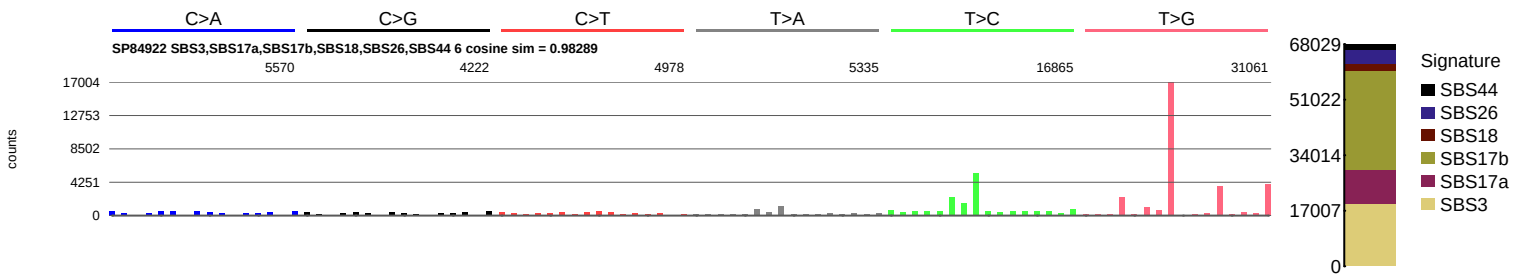
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.9829
5587
4791
4304
5276
16973
31094
counts
17008
12756
8504
4252
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS17a,SBS17b,SBS21 4 cosine sim = 0.98289
3197
2843
8289
4333
18356
31013
counts
17072
12804
8536
4268
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 6 cosine sim = 0.98289
5570
4222
4978
5335
16865
31061
counts
17004
12753
8502
4251
0
68029
51022
34014
17007
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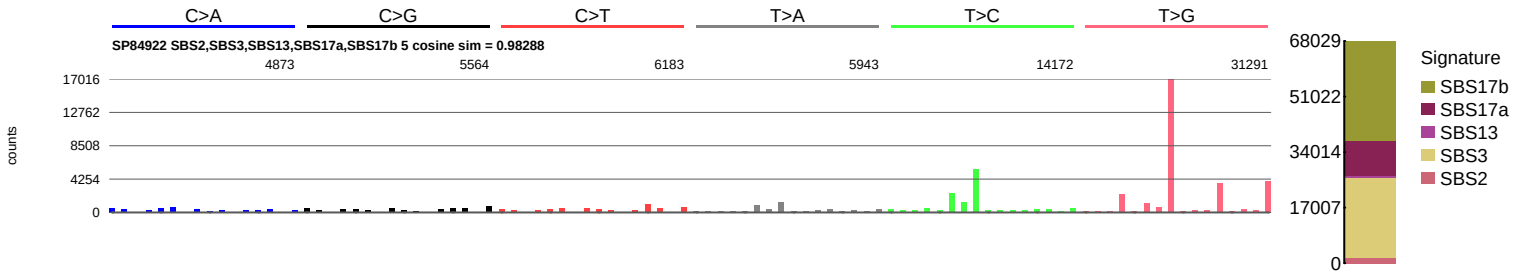
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3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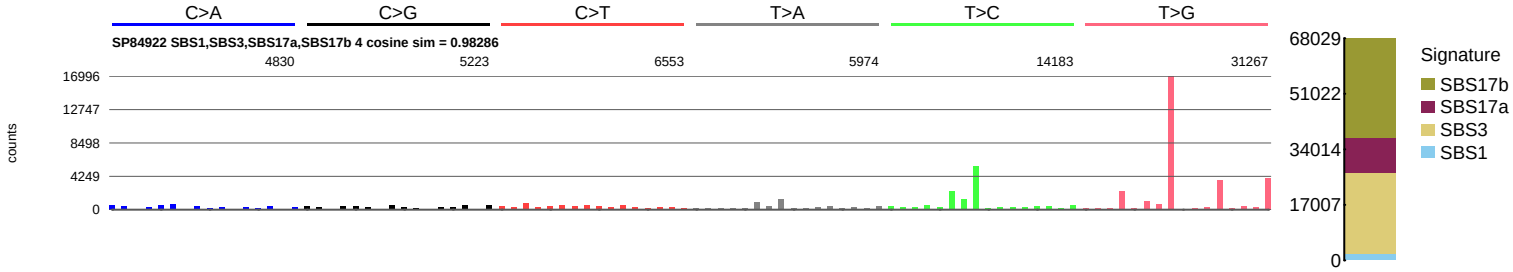
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS3,SBS13,SBS17a,SBS17b 5 cosine sim = 0.98288
4873
5564
6183
5943
14172
31291
counts
17016
12762
8508
4254
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS13
SBS3
SBS2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C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS3,SBS17a,SBS17b 4 cosine sim = 0.98286
4830
5223
6553
5974
14183
31267
counts
16996
12747
8498
4249
0
68029
51022
34014
17007
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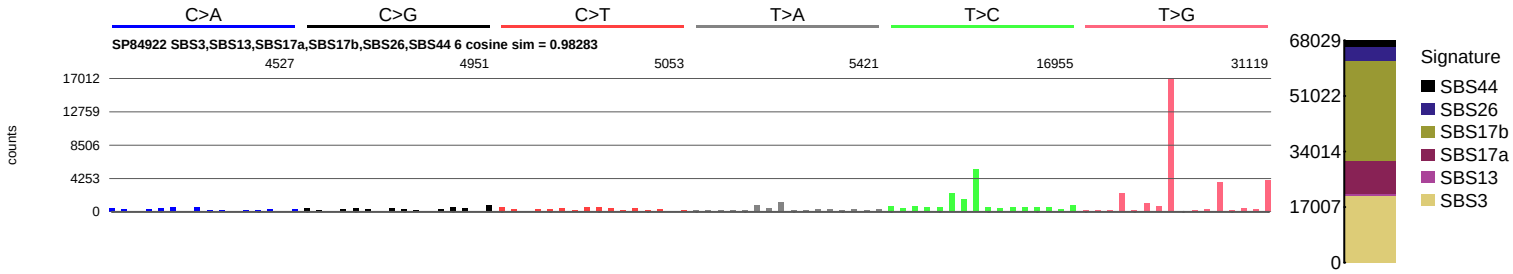
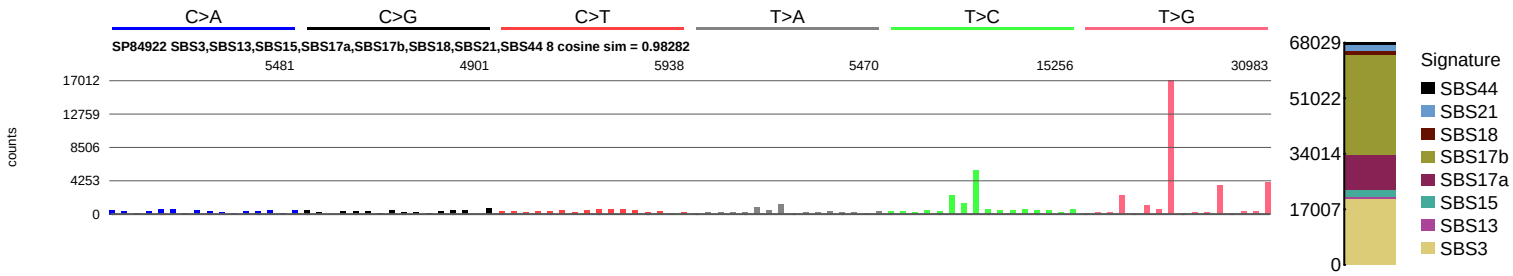
Signature
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS44 6 cosine sim = 0.98283
4527
4951
5053
5421
16955
31119
counts
17012
12759
8506
4253
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS13
SBS3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C>G
C>T
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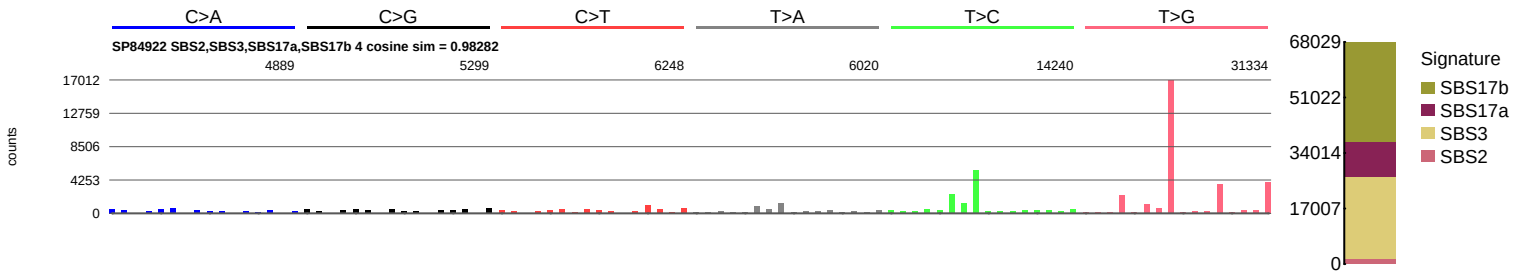
T>G
SP84922 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98282
5481
4901
5938
5470
15256
30983
counts
17012
12759
8506
4253
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3


C>A
C>G
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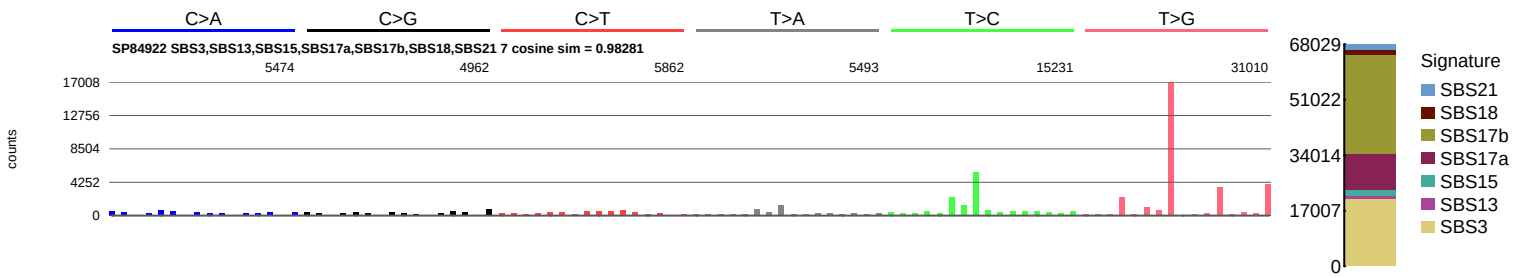
T>C
T>G
SP84922 SBS2,SBS3,SBS17a,SBS17b 4 cosine sim = 0.98282
4889
5299
6248
6020
14240
31334
counts
17012
12759
8506
4253
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
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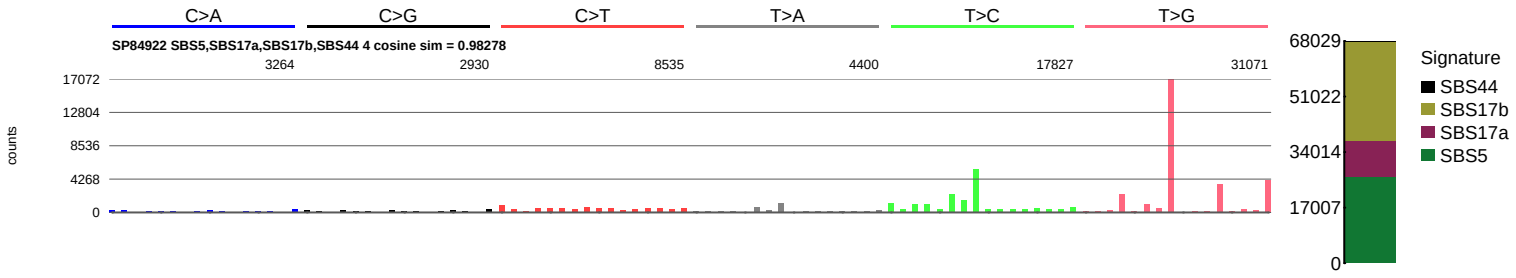
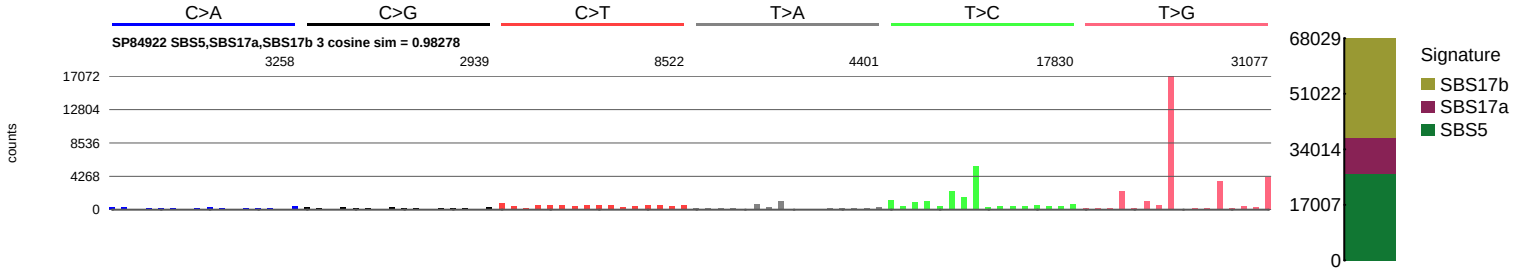
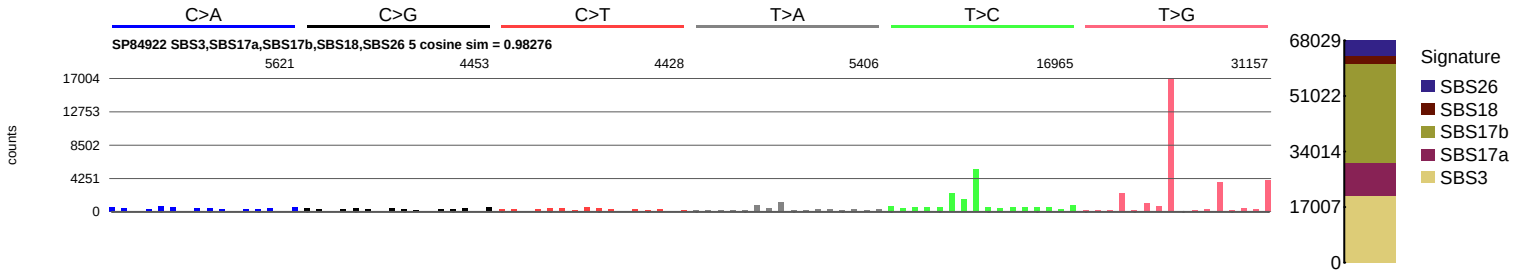
T>C
T>G
SP84922 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98281
5474
4962
5862
5493
15231
31010
counts
17008
12756
8504
4252
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS17a,SBS17b,SBS44 4 cosine sim = 0.98278
3264
2930
8535
4400
17827
31071
counts
17072
12804
8536
4268
0
68029
51022
34014
17007
0
Signature
SBS44
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS5,SBS17a,SBS17b 3 cosine sim = 0.98278
3258
2939
8522
4401
17830
31077
counts
17072
12804
8536
4268
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
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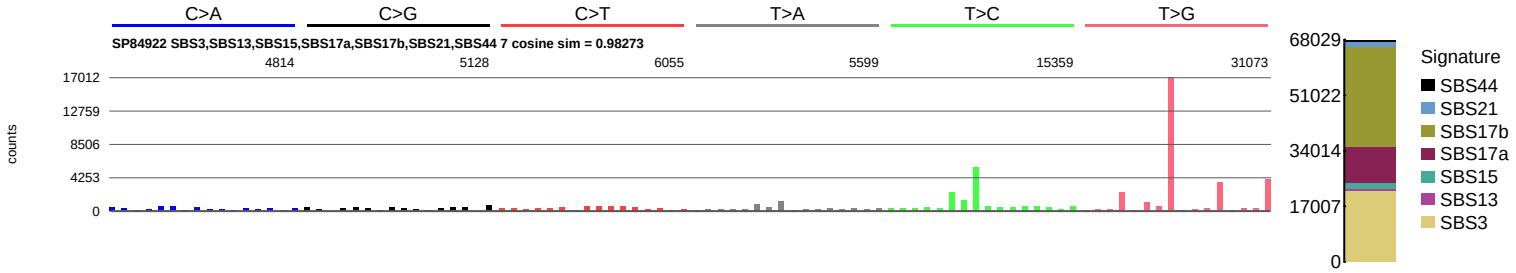
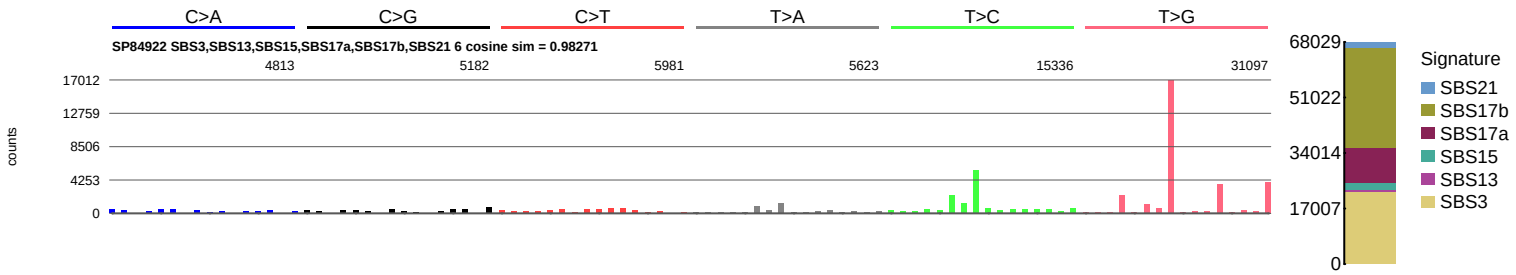
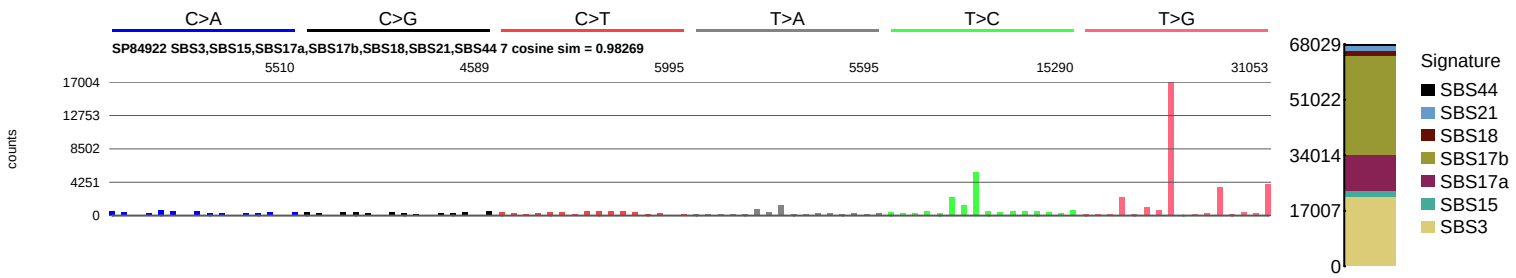
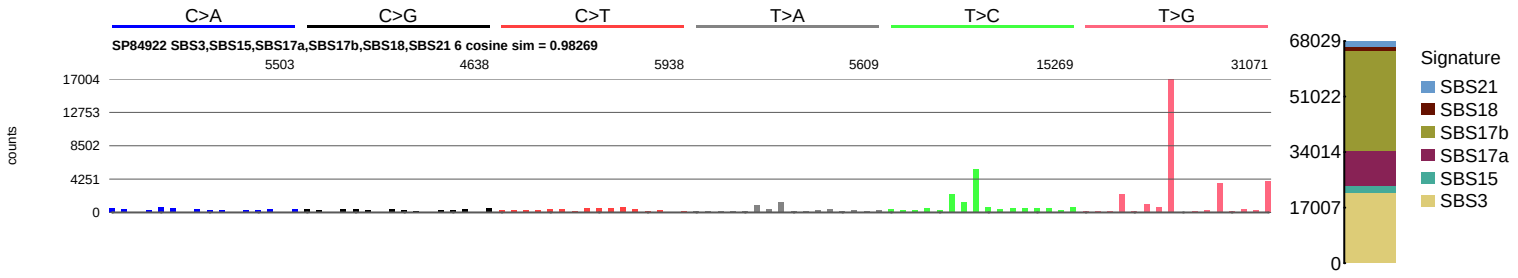
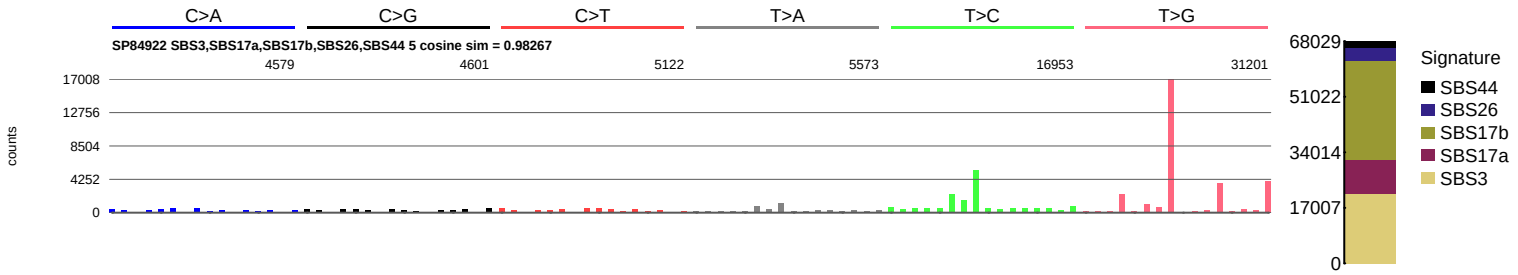
T>C
T>G
SP84922 SBS3,SBS17a,SBS17b,SBS18,SBS26 5 cosine sim = 0.98276
5621
4453
4428
5406
16965
31157
counts
17004
12753
8502
4251
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS3

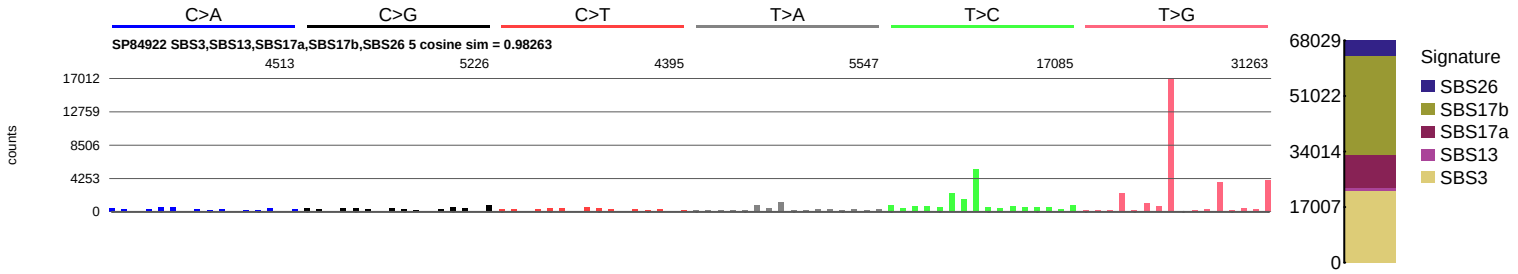

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98273
4814
5128
6055
5599
15359
31073
counts
17012
12759
8506
4253
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3

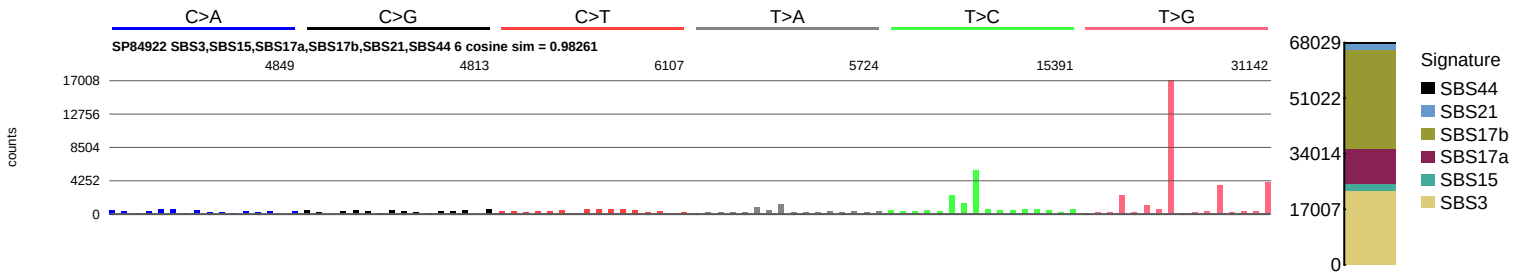

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98271
4813
5182
5981
5623
15336
31097
counts
17012
12759
8506
4253
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3

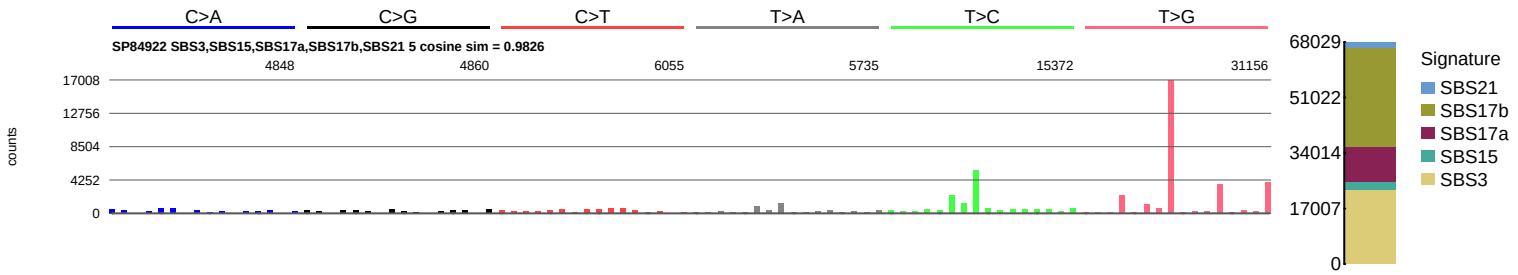

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98269
5510
4589
5995
5595
15290
31053
counts
17004
12753
8502
4251
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3

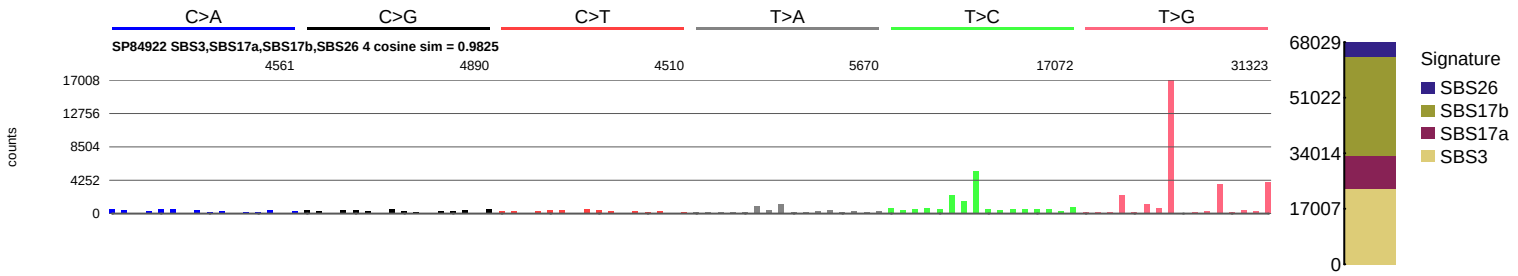

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98269
5503
4638
5938
5609
15269
31071
counts
17004
12753
8502
4251
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS17a,SBS17b,SBS26,SBS44 5 cosine sim = 0.98267
4579
4601
5122
5573
16953
31201
counts
17008
12756
8504
4252
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS13,SBS17a,SBS17b,SBS26 5 cosine sim = 0.98263
4513
5226
4395
5547
17085
31263
counts
17012
12759
8506
4253
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS13
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS44 6 cosine sim = 0.98261
4849
4813
6107
5724
15391
31142
counts
17008
12756
8504
4252
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS15,SBS17a,SBS17b,SBS21 5 cosine sim = 0.9826
4848
4860
6055
5735
15372
31156
counts
17008
12756
8504
4252
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS3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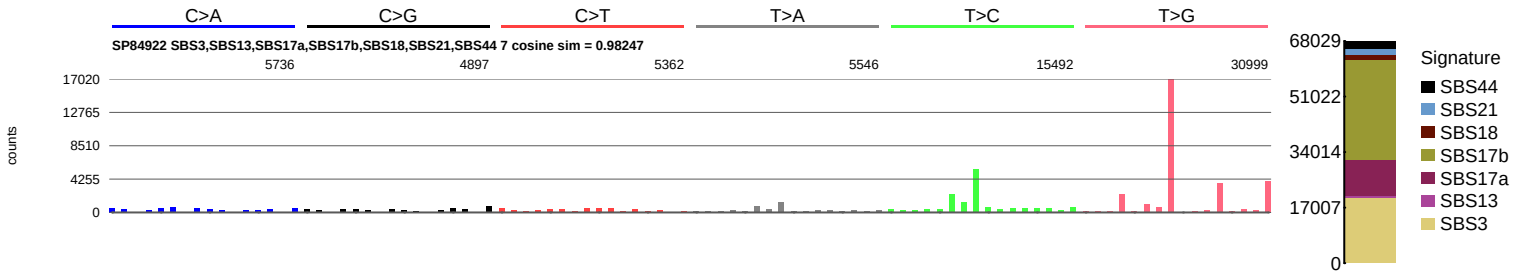
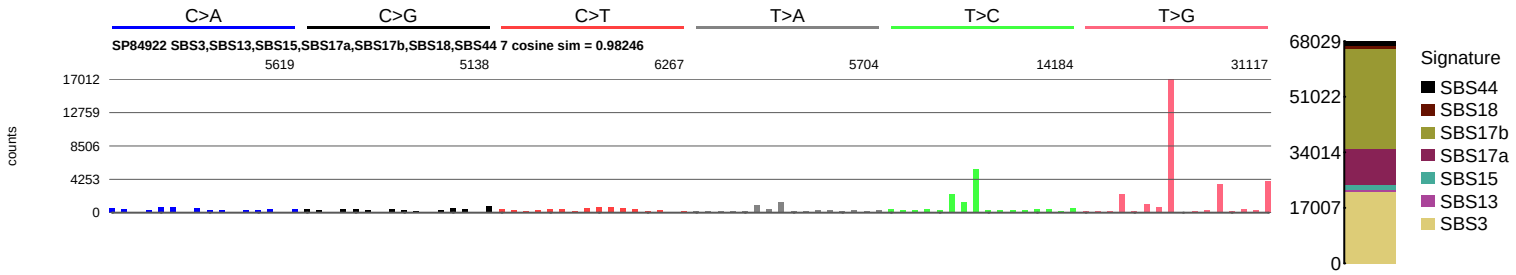
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS17a,SBS17b,SBS26 4 cosine sim = 0.9825
4561
4890
4510
5670
17072
31323
counts
17008
12756
8504
4252
0
68029
51022
34014
17007
0
Signature
SBS26
SBS17b
SBS17a
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98247
5736
4897
5362
5546
15492
30999
counts
17020
12765
8510
4255
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98246
5619
5138
6267
5704
14184
31117
counts
17012
12759
8506
4253
0
68029
51022
34014
17007
0
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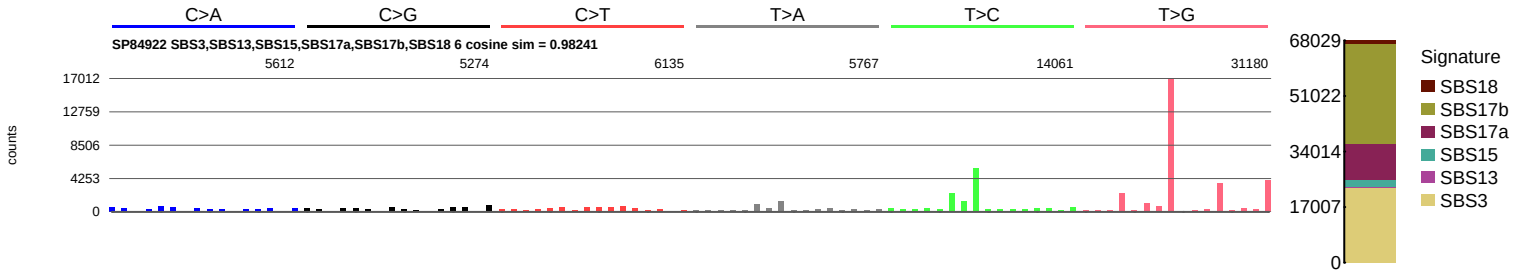
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98241
5612
5274
6135
5767
14061
31180
counts
17012
12759
8506
4253
0
68029
51022
34014
17007
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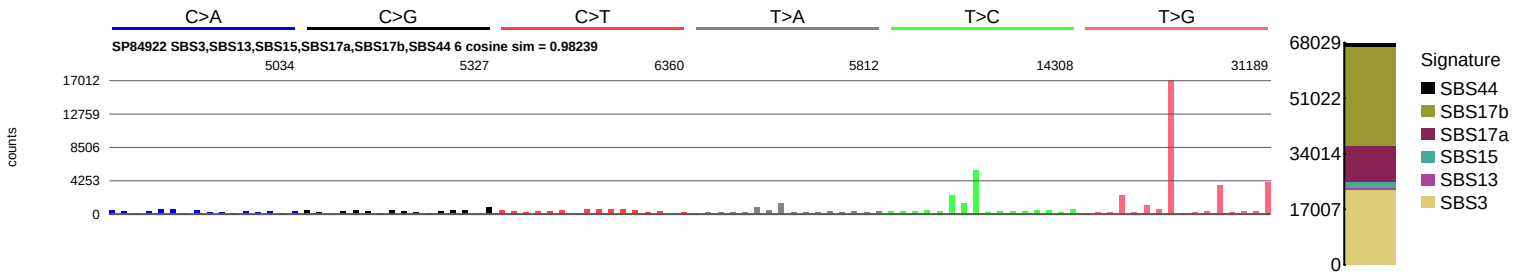
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98239
5034
5327
6360
5812
14308
31189
counts
17012
12759
8506
4253
0
68029
51022
34014
17007
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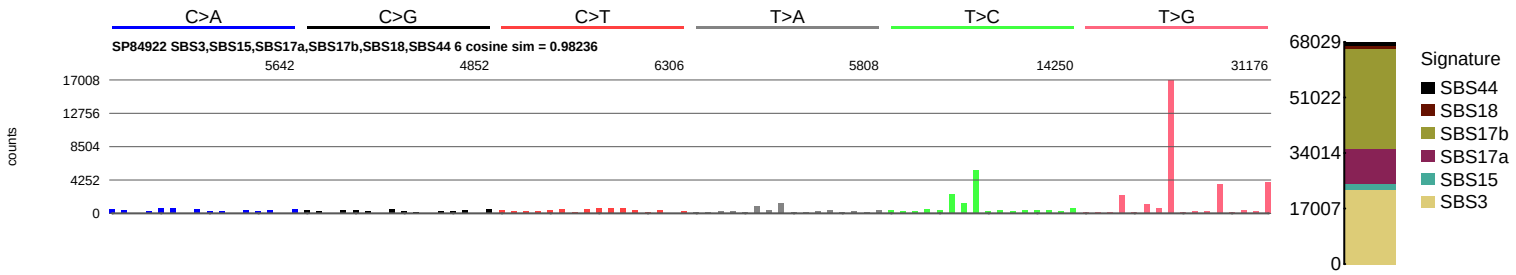
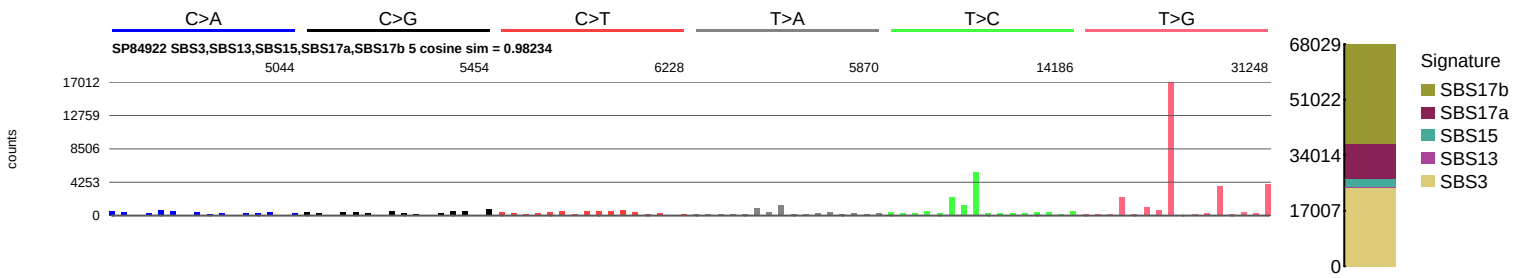
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS13
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98236
5642
4852
6306
5808
14250
31176
counts
17008
12756
8504
4252
0
68029
51022
34014
17007
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS13,SBS15,SBS17a,SBS17b 5 cosine sim = 0.98234
5044
5454
6228
5870
14186
31248
counts
17012
12759
8506
4253
0
68029
51022
34014
17007
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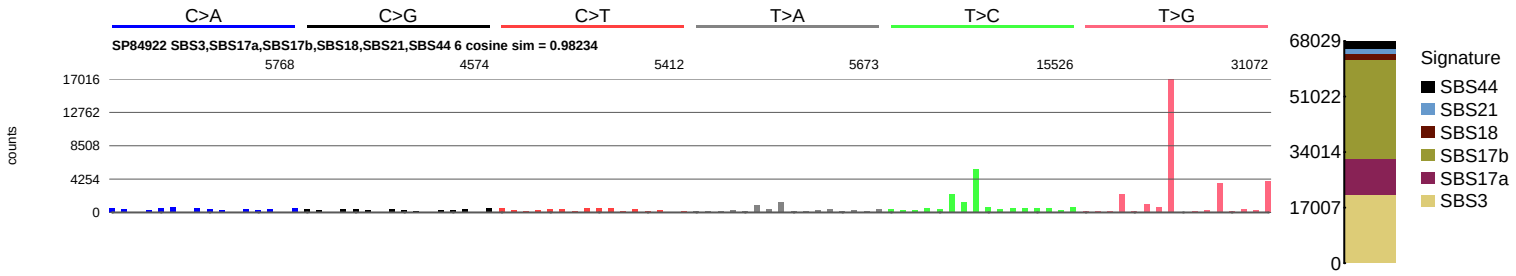
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 6 cosine sim = 0.98234
5768
4574
5412
5673
15526
31072
counts
17016
12762
8508
4254
0
68029
51022
34014
17007
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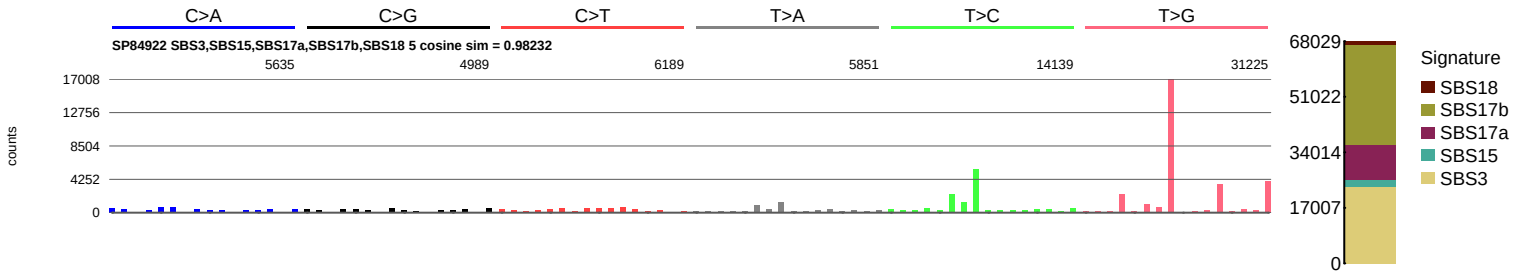
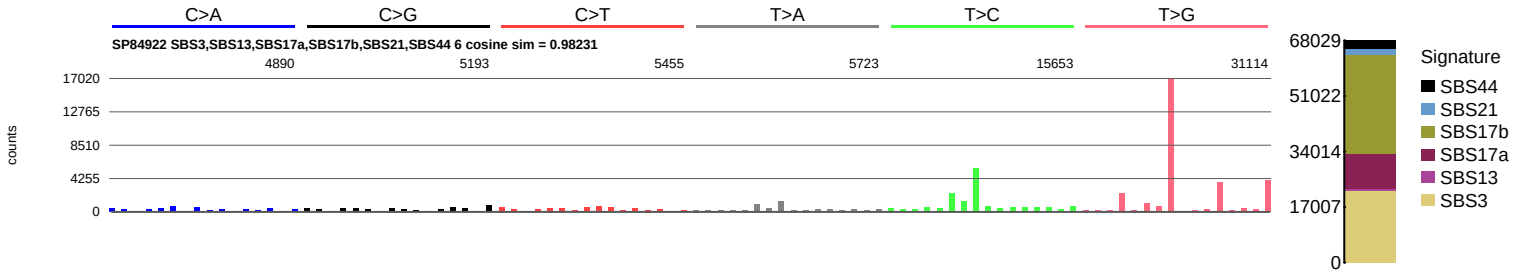
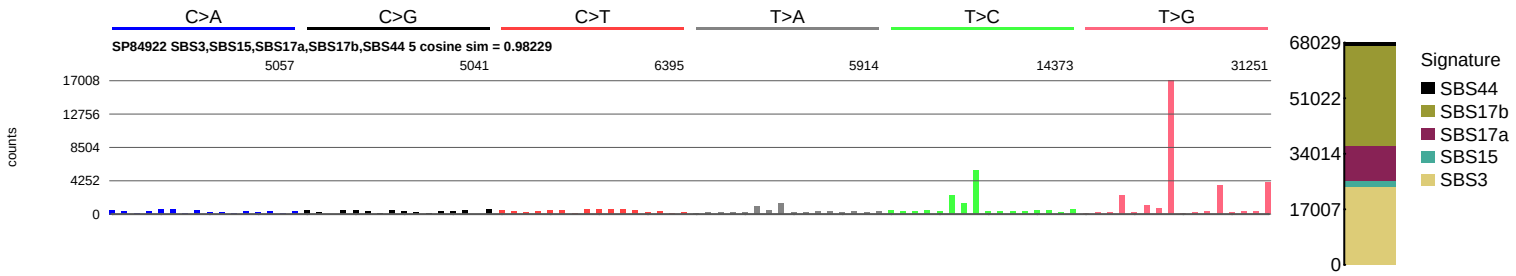
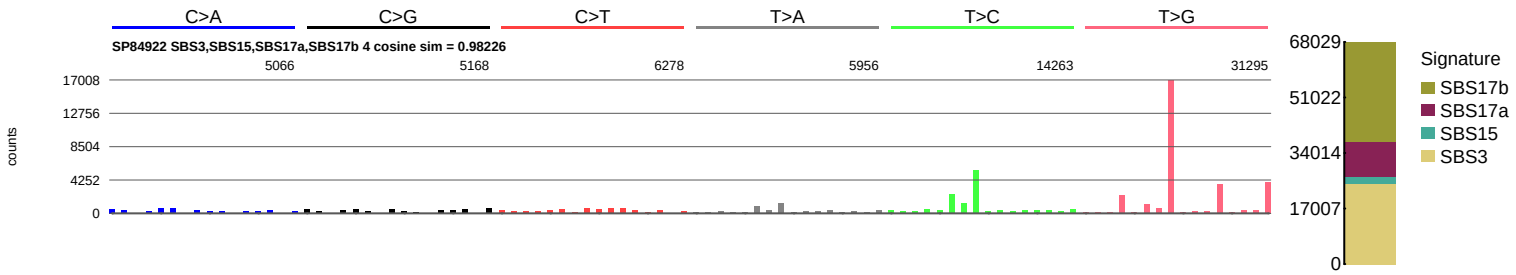
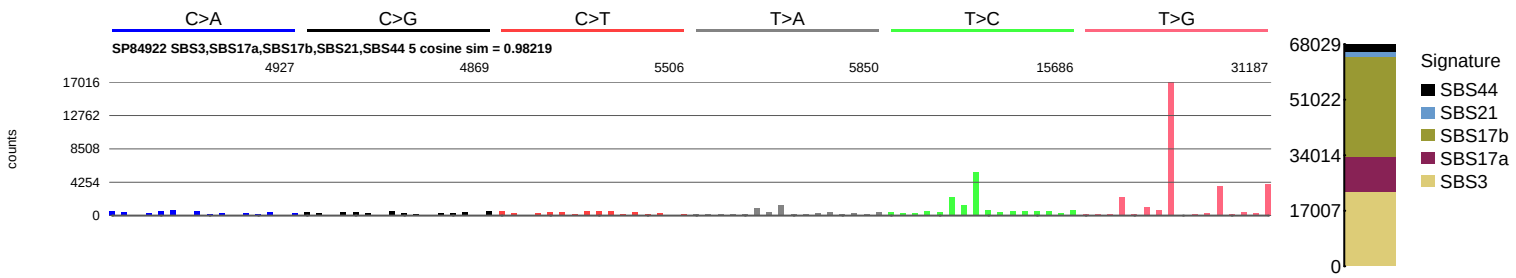
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS15,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98232
5635
4989
6189
5851
14139
31225
counts
17008
12756
8504
4252
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS44 6 cosine sim = 0.98231
4890
5193
5455
5723
15653
31114
counts
17020
12765
8510
4255
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS15,SBS17a,SBS17b,SBS44 5 cosine sim = 0.98229
5057
5041
6395
5914
14373
31251
counts
17008
12756
8504
4252
0
68029
51022
34014
17007
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS15,SBS17a,SBS17b 4 cosine sim = 0.98226
5066
5168
6278
5956
14263
31295
counts
17008
12756
8504
4252
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
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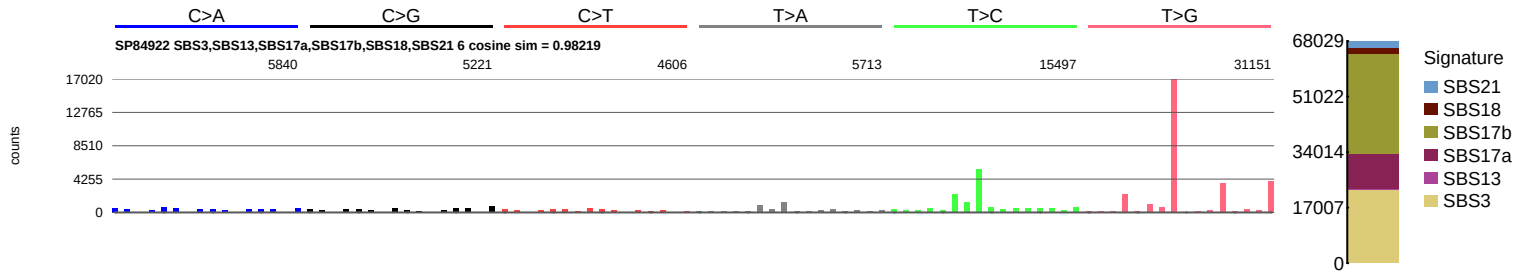
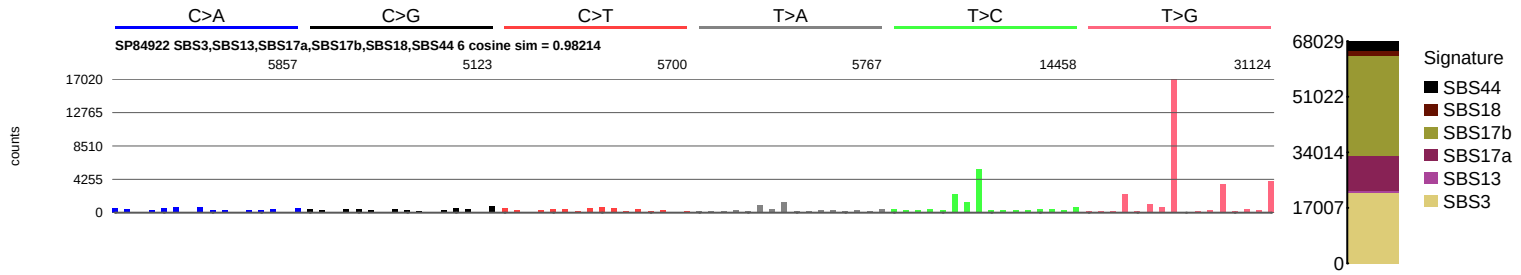
T>C
T>G
SP84922 SBS3,SBS17a,SBS17b,SBS21,SBS44 5 cosine sim = 0.98219
4927
4869
5506
5850
15686
31187
counts
17016
12762
8508
4254
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98219
5840
5221
4606
5713
15497
31151
counts
17020
12765
8510
4255
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3


C>A
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T>A
T>C
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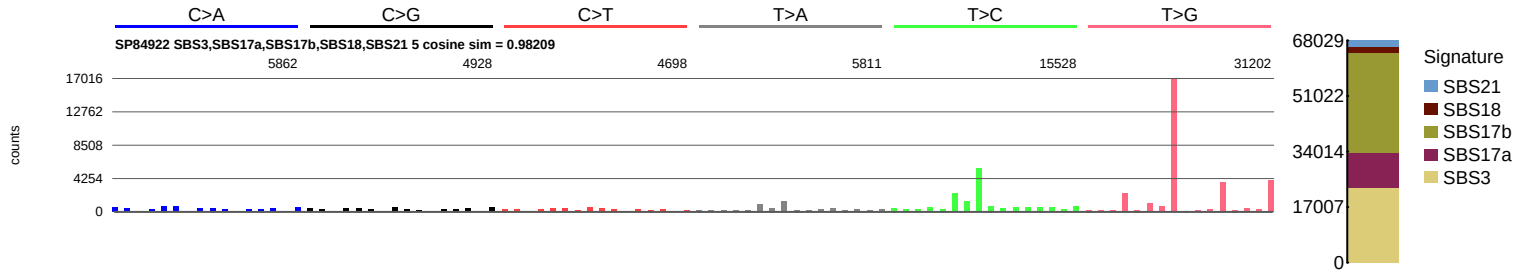
SP84922 SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98214
5857
5123
5700
5767
14458
31124
counts
17020
12765
8510
4255
0
68029
51022
34014
17007
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS3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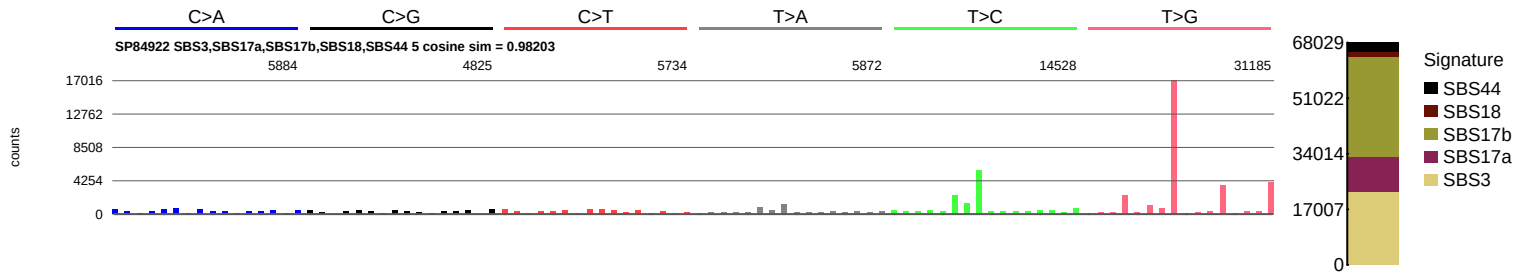
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS17a,SBS17b,SBS18,SBS21 5 cosine sim = 0.98209
5862
4928
4698
5811
15528
31202
counts
17016
12762
8508
4254
0
68029
51022
34014
17007
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS3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C>G
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T>G
SP84922 SBS3,SBS17a,SBS17b,SBS18,SBS44 5 cosine sim = 0.98203
5884
4825
5734
5872
14528
31185
counts
17016
12762
8508
4254
0
68029
51022
34014
17007
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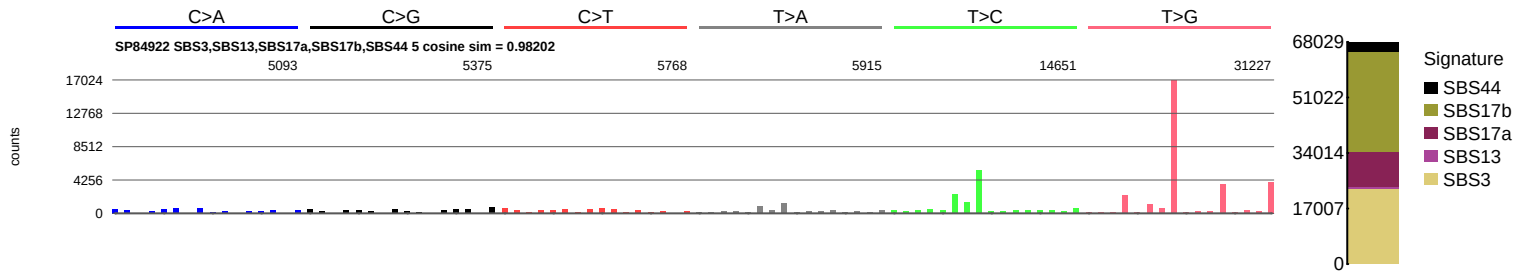
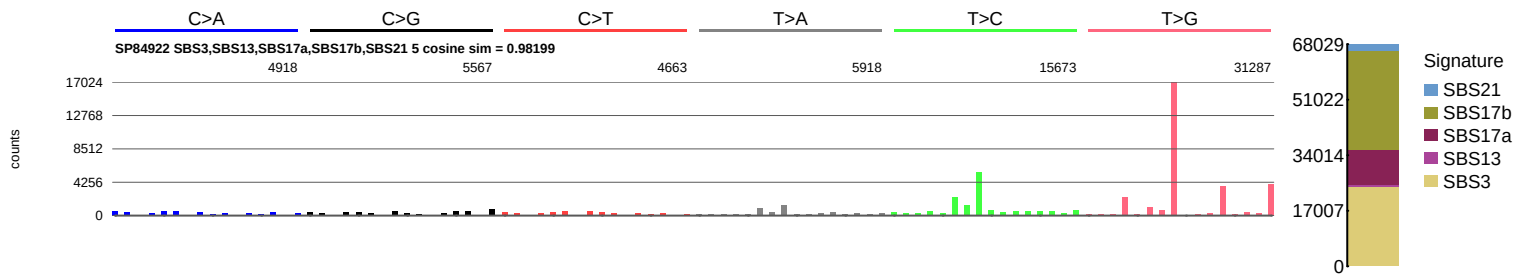
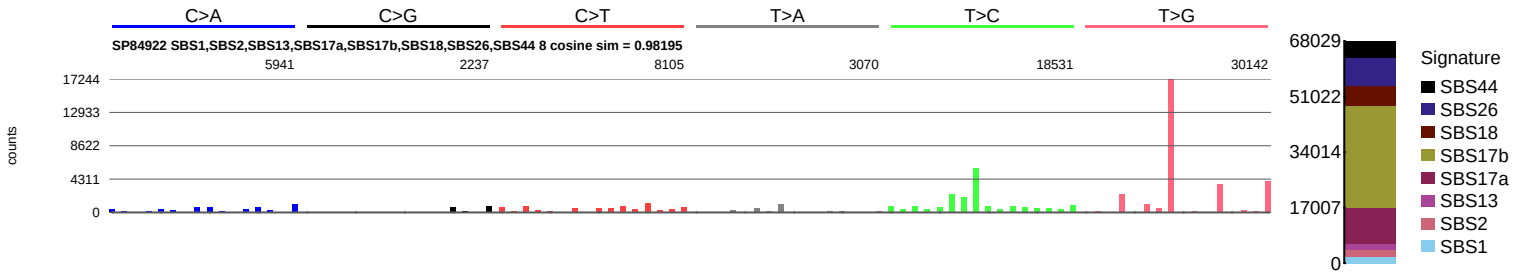
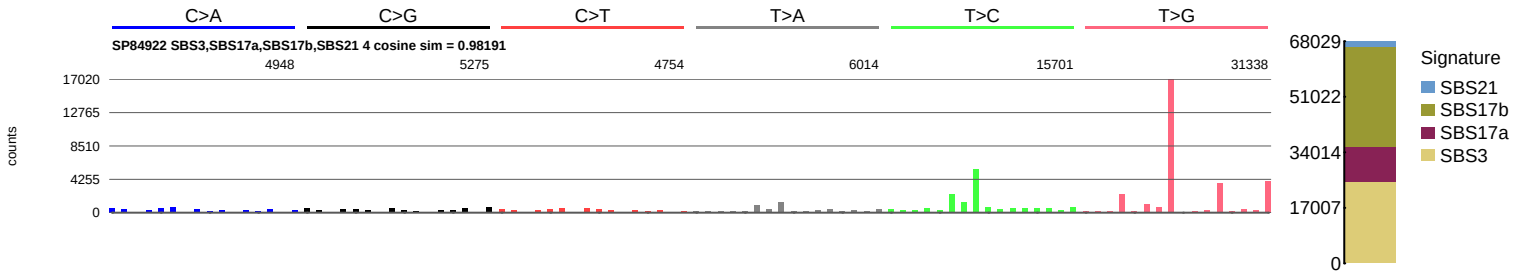
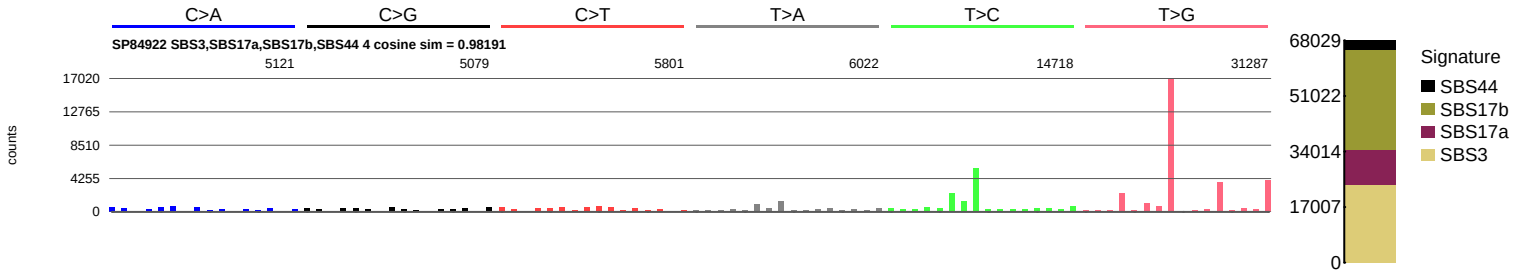
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS3

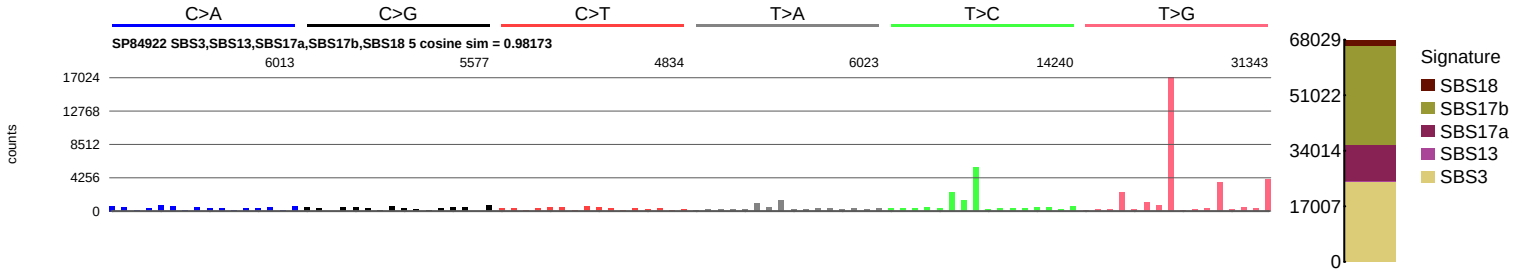

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS13,SBS17a,SBS17b,SBS44 5 cosine sim = 0.98202
5093
5375
5768
5915
14651
31227
counts
17024
12768
8512
4256
0
68029
51022
34014
17007
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS3

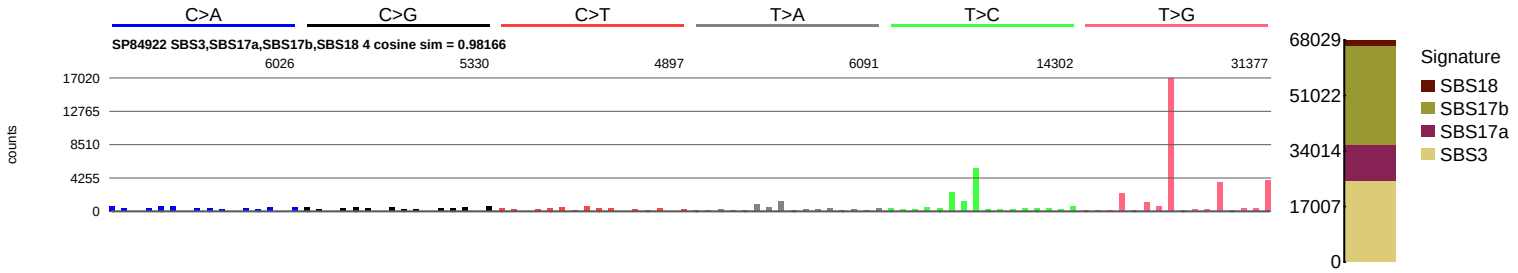

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS13,SBS17a,SBS17b,SBS21 5 cosine sim = 0.98199
4918
5567
4663
5918
15673
31287
counts
17024
12768
8512
4256
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS3

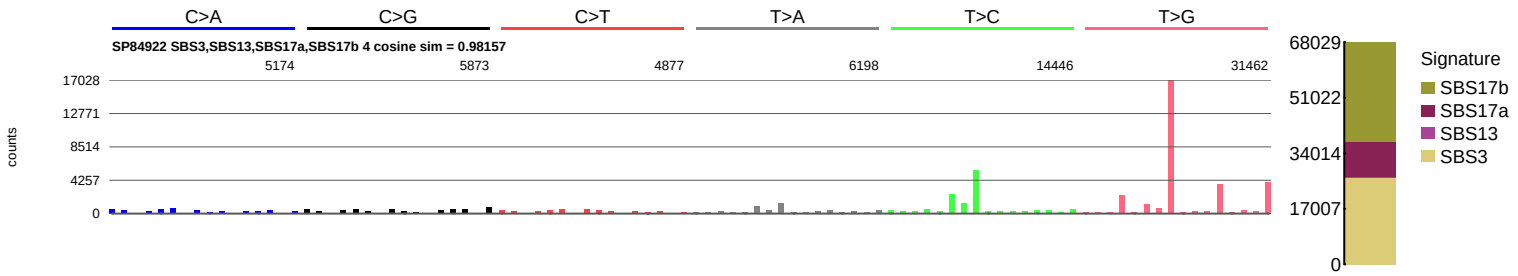

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98195
5941
2237
8105
3070
18531
30142
counts
17244
12933
8622
4311
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

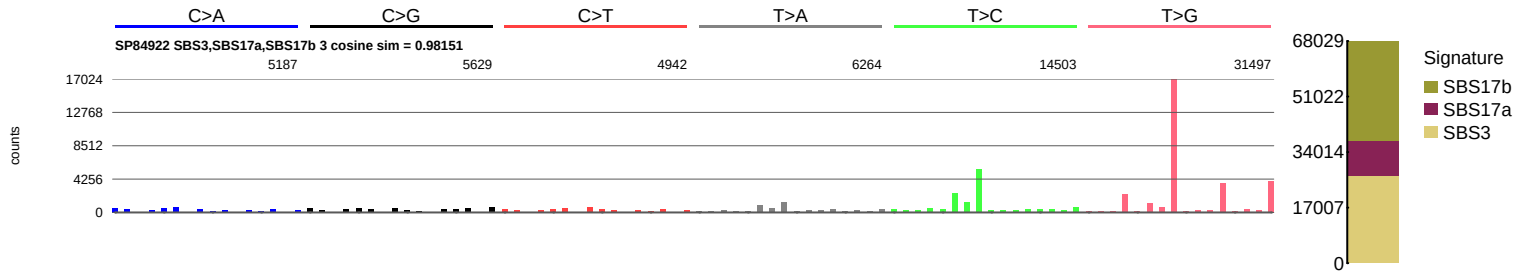

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS17a,SBS17b,SBS21 4 cosine sim = 0.98191
4948
5275
4754
6014
15701
31338
counts
17020
12765
8510
4255
0
68029
51022
34014
17007
0
Signature
SBS21
SBS17b
SBS17a
SBS3

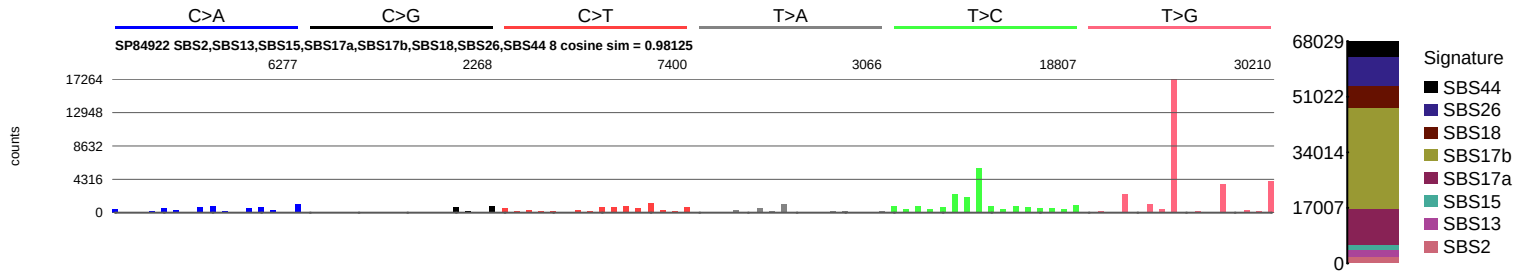

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS17a,SBS17b,SBS44 4 cosine sim = 0.98191
5121
5079
5801
6022
14718
31287
counts
17020
12765
8510
4255
0
68029
51022
34014
17007
0
Signature
SBS44
SBS17b
SBS17a
SBS3

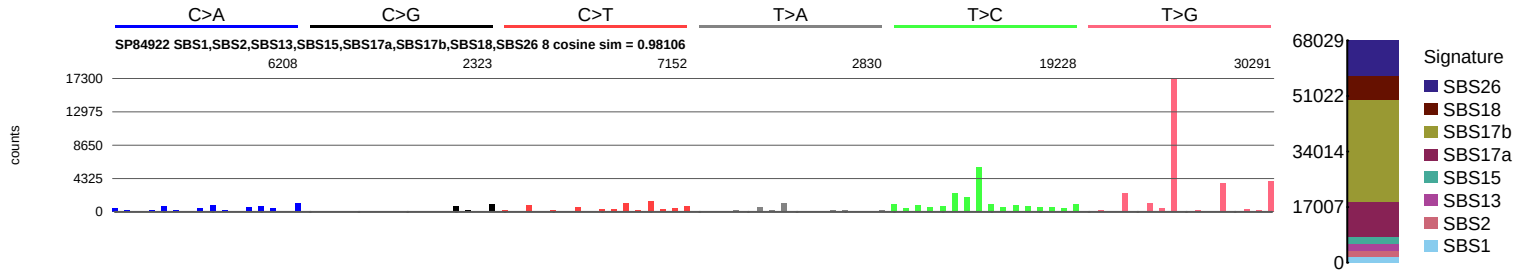

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS13,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98173
6013
5577
4834
6023
14240
31343
counts
17024
12768
8512
4256
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS3

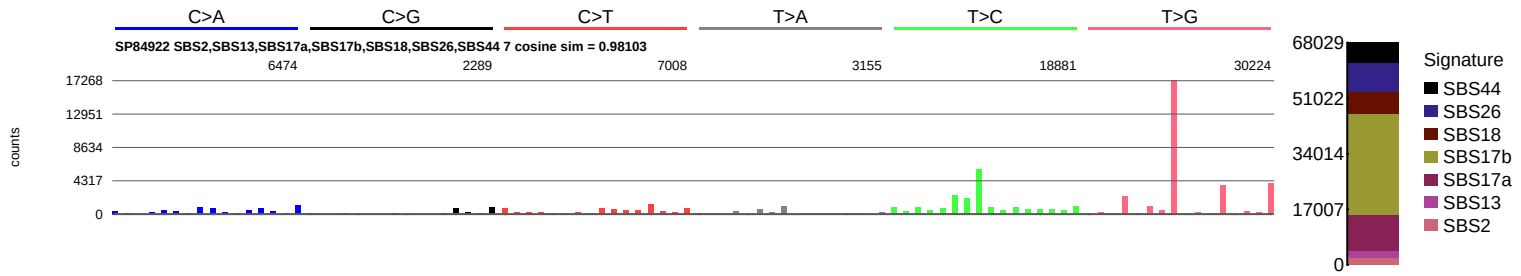

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS17a,SBS17b,SBS18 4 cosine sim = 0.98166
6026
5330
4897
6091
14302
31377
counts
17020
12765
8510
4255
0
68029
51022
34014
17007
0
Signature
SBS18
SBS17b
SBS17a
SBS3

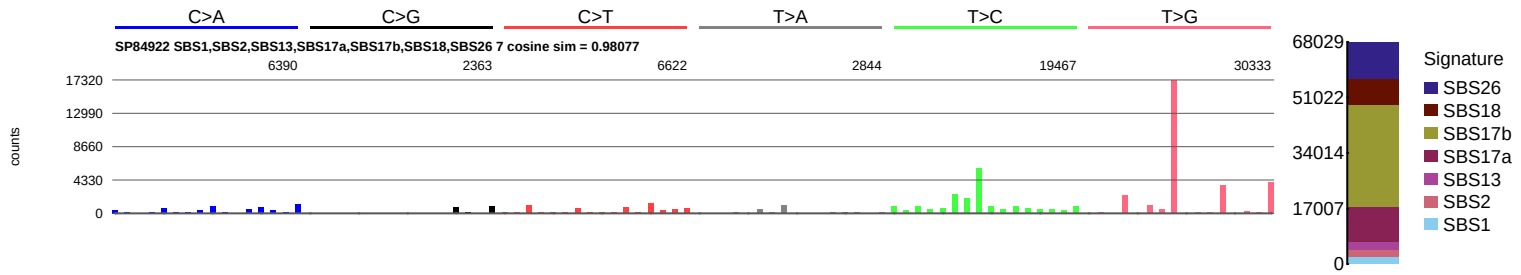

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS13,SBS17a,SBS17b 4 cosine sim = 0.98157
5174
5873
4877
6198
14446
31462
counts
17028
12771
8514
4257
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS13
SBS3

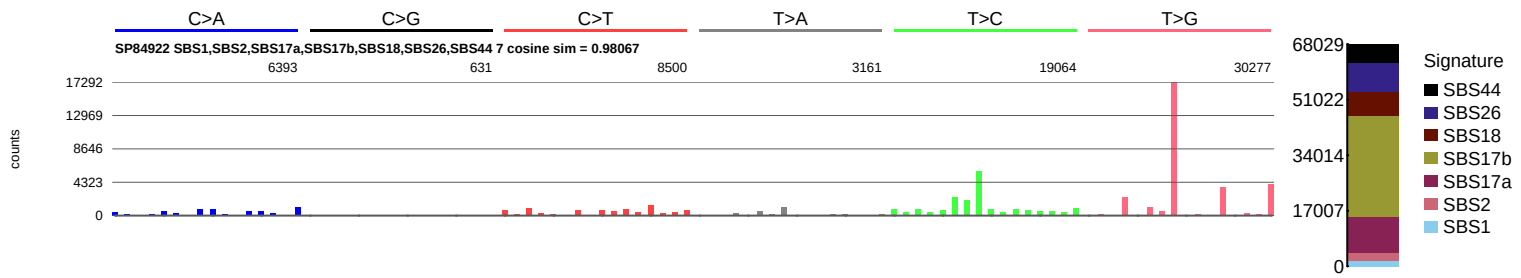

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS3,SBS17a,SBS17b 3 cosine sim = 0.98151
5187
5629
4942
6264
14503
31497
counts
17024
12768
8512
4256
0
68029
51022
34014
17007
0
Signature
SBS17b
SBS17a
SBS3

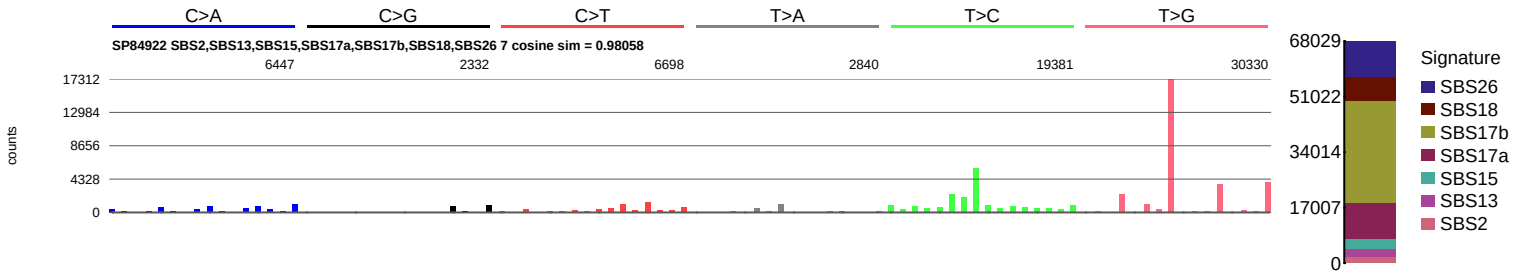

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98125
6277
2268
7400
3066
18807
30210
counts
17264
12948
8632
4316
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2

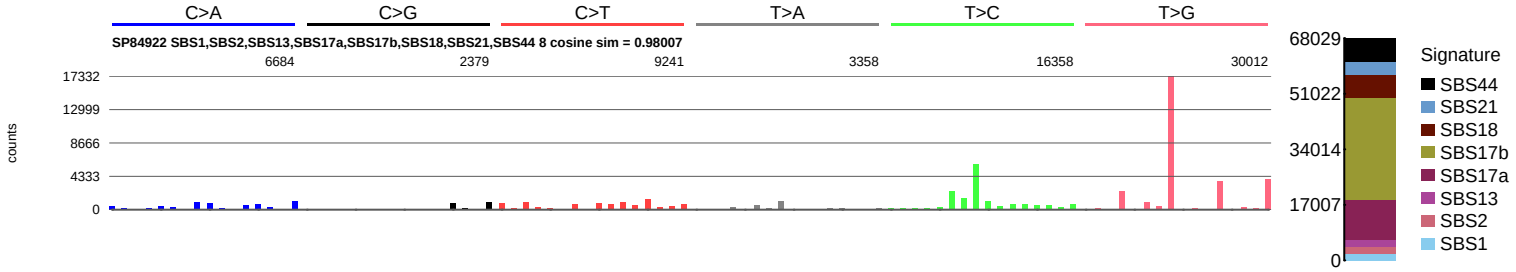

C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98106
6208
2323
7152
2830
19228
30291
counts
17300
12975
8650
4325
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98103
6474
2289
7008
3155
18881
30224
counts
17268
12951
8634
4317
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98077
6390
2363
6622
2844
19467
30333
counts
17320
12990
8660
4330
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98067
6393
631
8500
3161
19064
30277
counts
17292
12969
8646
4323
0
68029
51022
34014
17007
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98058
6447
2332
6698
2840
19381
30330
counts
17312
12984
8656
4328
0
68029
51022
34014
17007
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP84922 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98007
6684
2379
9241
3358
16358
30012
counts
17332
12999
8666
4333
0
68029
51022
34014
17007
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1